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BY PHYSICS WALLAH

UPSC WALLAH

INDIAN GEOGRAPHY

COMPREHENSIVE LEARNING SERIES FOR
PRELIMS AND MAINS



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**COMPREHENSIVE LEARNING SERIES FOR
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A plethora of UPSC Study Material is available in the market but PW ONLY IAS professionals are continuously working to provide the supreme quality study material for our UPSC students.

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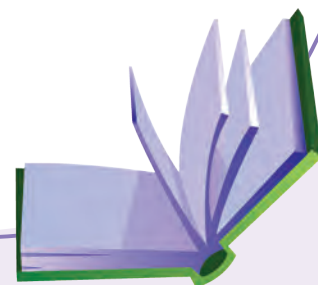
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- Holistic coverage of topics, strictly as per exam syllabus
- One-stop solution for subject wise coverage
- Intensive use of Maps, Diagrams, Flowcharts and Timelines for quick understanding and revision
- Thinking Points in and as 'Ignite Your Mind'
- Updated with latest current affairs and data



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Introduction to Indian Geography

1

India is a country of vast geographical **diversity**. In the north, it is bound by the lofty Himalayas. The Arabian Sea in the West, the Bay of Bengal in the East and the Indian Ocean in the South, wash the shores of the Indian peninsula. The **Tropic of Cancer (23° 30'N)** divides the country into almost two equal parts. To the South-East and South-West of the mainland, lie the Andaman and Nicobar Islands and the Lakshadweep Islands in the Bay of Bengal and Arabian Sea, respectively.

India has an area of **about 3.28 million sq km**, which accounts for about **2.4 per cent** of the total geographical area of the world. India is the **seventh** largest country in the world in terms of **area**. As per United Nations Data, India has surpassed China in 2023 to become the largest country based on population.

The **North to South stretch** of the Indian mainland is from **Kashmir** to **Kanyakumari** respectively and extends between **8°4'N to 37°6'N latitude**. The **East to West stretch** is from **Arunachal Pradesh** to **Gujarat** respectively and extends between **68°7'E to 97°25'E longitude**.

India has a land boundary of about **15,106.7 km** and the total length of the coastline of the mainland, including Andaman and Nicobar and Lakshadweep, is **7,516.6 km**. Further, its territorial limit extends up to **12 nautical miles (about 21.9 km)** towards the sea from the coast.

THE BASICS OF INDIA

The name "India" has its origins in the **River Indus**. The River Indus, known as the **Sindhu** in **Sanskrit**, played a significant role in the early history of the region. The ancient **Persian** and **Greek** civilizations referred to the land beyond the Indus River as "**Hind**" or "**Indoi**," and this term eventually evolved into "**India**."

The ancient Sanskrit name for the country is "**Bharat**," derived from the legendary **king Bharata**, a figure in Hindu mythology. The name Bharat is used as the official Sanskrit name of the country, the "**Republic of India**" or "**Bharat Ganarajya**."

Location

- India lies in the **Northern and Eastern hemispheres** between **8°4'N and 37°6' North latitude** and **68°7' E to 97°25' East longitude**.
- Kibithu** in Arunachal Pradesh is the easternmost point of India. The Westernmost Point is **Guhar Moti** in Gujrat.

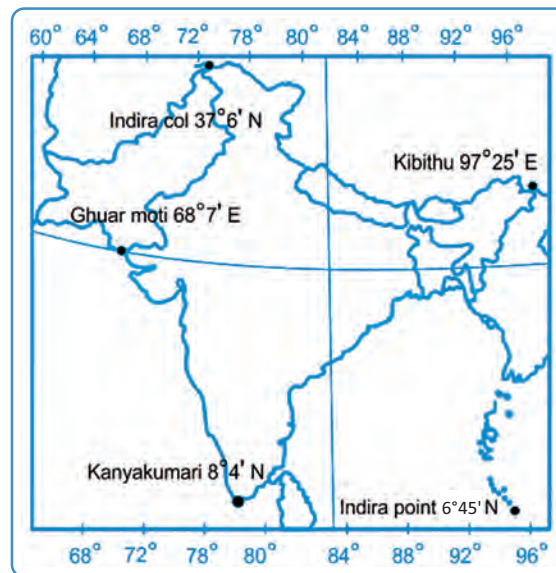


Fig. 1.1: Extreme Points of India

- The Northernmost point is **Indira Col** in the Ladakh region, and the Pygmalion Point, or the Indira Point is the southernmost point of the country at 6° 8' N latitude, located in the Nicobar district of the Andaman and Nicobar Islands.
- The Southernmost point of the Indian mainland is **Cape Camorin** in Kanyakumari of Tamil Nadu.
- The Andaman and Nicobar Islands lie to the Southeast of the Indian mainland in the Bay of Bengal and to the Southwest of the Indian mainland lies the Lakshadweep Islands in the Arabian Sea.

Important Facts

- Bengaluru** lies on a longitude closest to that of **Delhi**.
- The latitude that passes through **Sikkim** also passes through **Rajasthan**.
- Mangalore** and **Chennai** are on the same latitude yet both these cities receive rainfall in different months.
- Amritsar** and **Shimla** have the same latitude yet Shimla is cooler than Amritsar because it is at a higher elevation.

Tropic of Cancer

- Tropic of Cancer is one of the five major circles of latitude that mark maps of the Earth. It is an imaginary line located approximately 23.5 degrees north of the equator.

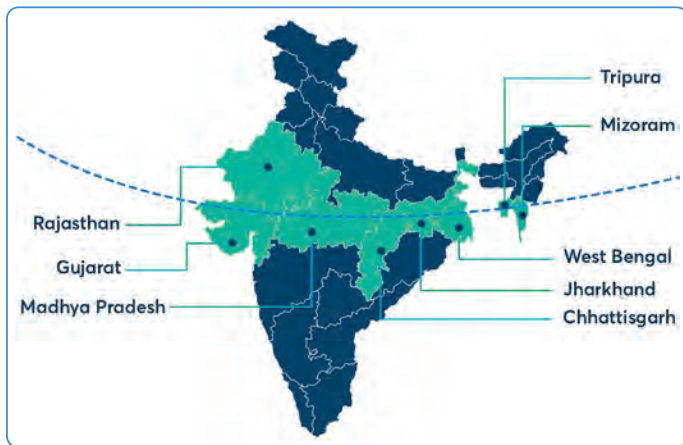


Fig. 1.2: States through which Tropic of Cancer passes

- ❑ This means that it is the northernmost point on Earth where the Sun appears directly overhead at the zenith during the June solstice.
- ❑ The Tropic of Cancer passes through the following 8 Indian states:
 - **Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram**

Standard Meridian of India

- ❑ Indian Standard Time is calculated on the basis of the 82.5°E longitude, roughly located in Mirzapur, Uttar Pradesh. It was chosen because it happens to be in the middle of the eastern and western longitudinal extremities of the country, and this time is observed throughout India and Sri Lanka.
- ❑ Standard Meridian of India passes through **Mirzapur** in the Uttar Pradesh. States along the Standard Meridian include Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha, and Andhra Pradesh.



IGNITE YOUR MIND

Can you analyze the implications of having a single time zone for the entire Indian subcontinent? And also discuss the challenges and benefits of coordinating activities across countries with varying daylight patterns.

Shape and Size

- ❑ With a total geographical area of approximately 32,87,263 square kilometres, India covers **0.57 per cent of the total area of Earth** and **2.4 per cent** of the total land mass on Earth.
- ❑ From North to South i.e. from Indira Col in Kashmir to Kanyakumari, the distance is **3,214 km**. From East to West i.e. from Rann of Kutch to Arunachal Pradesh the distance is **2,933 km**.

- ❑ The difference of **281 km** between the north-south and east-west extent of the country is due to the fact that the distance between two longitudes decreases towards the poles whereas the distance between two latitudes remains the same everywhere.

Historical Land Routes into India

India has several natural routes, including mountain passes and waterways, that have historically served as important entry points into the subcontinent.

Key routes into India

❑ Northwestern Passes:

- **Khyber Pass:** Located in the northwest, it has been a traditional gateway into the Indian subcontinent. It connects Afghanistan and Pakistan to the Indian plains.
- **Bolan Pass:** Situated in southwestern Pakistan, it provides another entry route into India, connecting Afghanistan with the region.

❑ Western Coastal Routes:

- The western coastline of India has served as a maritime route for trade and cultural exchange.
- Ports such as Mumbai, Goa, and Mangalore have historically been important.

❑ Eastern Coastal Routes:

- The eastern coastline also played a crucial role in maritime trade.
- Major ports like Chennai, Visakhapatnam, and Kolkata facilitated sea trade and cultural interactions.

❑ Himalayan Passes:

- **Nathu La and Jelep La:** These mountain passes in the eastern Himalayas connect India with Tibet (China). Historically important for trade, they continue to be strategic routes.
- **Shipki La:** Located in the western Himalayas, this pass connects India and Tibet.

❑ North Indian Plains:

- The vast plains of North India, with the Ganges and its tributaries, provided natural routes for settlement, trade, and invasions.

❑ Indus River Valley:

- The Indus River has historically been a significant waterway, and the plains along the river provided routes for cultural and trade exchanges.

Ancient Silk Route and India

The ancient Silk Route connected India to Central Asia and beyond, facilitating the exchange of goods, ideas, and cultures between the Indian subcontinent and regions along the Silk Road.

Along the ancient Silk Route, several key places played significant roles in trade and cultural exchange.

- ❑ **Taxila (in present-day Pakistan):** An important center of trade and learning, Taxila was a vital junction connecting India to the Silk Route.

- ❑ **Varanasi (Banaras):** A major cultural and religious center in ancient India, Varanasi was linked to the Silk Route, attracting traders and travelers.
- ❑ **Mathura:** Known for its art and commerce, Mathura served as a trading hub and cultural center along the route.
- ❑ **Patliputra (modern-day Patna):** The capital of the Maurya and Gupta empires, Patliputra played a role in facilitating trade and cultural interactions.
- ❑ **Ujjain:** An ancient city and a significant trade center, Ujjain contributed to the economic and cultural exchange along the Silk Route.

These routes have shaped India's historical interactions with neighboring regions and have influenced trade, migrations, and cultural exchanges throughout its history. Additionally, modern transportation infrastructure, such as highways and airports, further connects India with the rest of the world.

Indian States & Union Territories

- ❑ India is politically significant for being the largest democracy that has accommodated and assimilated diversities – ethical, linguistic, cultural, and spiritual within it. Politically India is divided into **28 states and 8 Union Territories**. **Rajasthan is the largest state** in terms of area, followed by **Madhya Pradesh and Maharashtra**. **Goa is the smallest state** in India. The most populous state in India is **Uttar Pradesh**, followed by **Maharashtra** and **Bihar**.

States

Sr.	State	Capital	Area (sq. Km)
1.	Rajasthan	Jaipur	342,239
2.	Madhya Pradesh	Bhopal	308,245
3.	Maharashtra	Mumbai	307,713
4.	Uttar Pradesh	Lucknow	240,928
5.	Gujarat	Gandhi Nagar	196,024
6.	Karnataka	Bengaluru	191,791
7.	Andhra Pradesh	Hyderabad (Proposed-Amravati)	162,970
8.	Odisha	Bhubaneswar	155,707
9.	Chhattisgarh	Raipur	135,191
10.	Tamil Nadu	Chennai	130,058
11.	Telangana	Hyderabad	112,077
12.	Bihar	Patna	94,163
13.	West-Bengal	Kolkata	88,752
14.	Arunachal Pradesh	Itanagar	83,743
15.	Jharkhand	Ranchi	79,714
16.	Assam	Dispur	78,438
17.	Himachal Pradesh	Shimla Dharamshala (Winter Capital)	55,673
18.	Uttarakhand	Dehradun	53,483
19.	Punjab	Chandigarh	50,362
20.	Haryana	Chandigarh	44,212
21.	Kerala	Thiruvananthapuram	38,863
22.	Meghalaya	Shillong	22,429

Sr.	State	Capital	Area (sq. Km)
23.	Manipur	Imphal	22,327
24.	Mizoram	Aizawl	21,081
25.	Nagaland	Kohima	16,579
26.	Tripura	Agartala	10,486
27.	Sikkim	Gangtok	7,096
28.	Goa	Panaji	3,702

Union Territories

Sr.	Union Territories	Capital	Area (sq. Km)
1.	Jammu And Kashmir	Srinagar (May-October), Jammu (November-April)	42,241
2.	Ladakh	Leh	59,146
3.	Andaman and Nicobar Island	Port Blair	8,249
4.	National Capital Territory Of Delhi	New Delhi	1483
5.	Dadar and Nagar Haveli & Daman and Diu	Daman	603
6.	Puducherry	Puducherry	479
7.	Chandigarh	Chandigarh	114
8.	Lakshadweep	Kavaratti	32



Fig. 1.3: India- its States and Union Territories

Boundary Sharing of States

- ❑ **Uttar Pradesh** is the state which shares its boundary with a maximum number of states i.e. eight states and a Union Territory of India including Rajasthan, Haryana, Himachal Pradesh, Delhi, Uttarakhand, Bihar, Madhya Pradesh, Jharkhand, and Chhattisgarh.
- ❑ **Both Assam and Chhattisgarh** share interstate borders with seven states. Assam with West Bengal, Arunachal Pradesh, Meghalaya, Tripura, Nagaland, Manipur and Mizoram. And Chhattisgarh with Uttar Pradesh, Madhya Pradesh, Maharashtra, Jharkhand, Odisha, Telangana and Andhra Pradesh.

- ❑ **Both Meghalaya and Sikkim** are the only states sharing boundary with one state only. Meghalaya has a border with Assam to the north and northeast. Sikkim shares its border with West Bengal.

The minimum number of states you need to travel through by road from Kohima to Kottayam is seven: Nagaland, Assam, West Bengal, Odisha, Andhra Pradesh, Karnataka or Tamil Nadu, Kerala.

THE INDIAN SUBCONTINENT

A subcontinent is part of a continent having a distinguishable cultural, political and geographical identity and also covers a huge area, somewhat less than a continent. That is why India is also considered a Subcontinent.

The Indian Subcontinent includes countries namely **India, Pakistan, Nepal, Bhutan, Bangladesh, Sri Lanka and the Maldives**. These countries were initially part of India but were separated because of political or geographical reasons. Geologically, the landmass which separated from the supercontinent Gondwanaland, is associated with the Indian Subcontinent. The Indian subcontinent is located on the Indian tectonic plate.



Fig. 1.4: Indian Subcontinent Map

The Following Features Distinguish India as a Continent

- ❑ **Distinct Geography:** The Indian subcontinent is separated by the Himalayas in the North, the Indian Ocean in the South, the Arabian Sea in the Southwest and the Bay of Bengal in the Southeast. This geographical feature isolates it from Eurasia.
- ❑ **Huge Population:** The population of India is one of the largest in the world. The combined population of the countries of the Indian subcontinent is much larger than other continents except Asia and Africa.
- ❑ **Diversity of the Region:** The subcontinent is endowed with multiple races, castes and religions. This is mainly due to its physical aspects. However, despite numerous differences, at the root, there are numerous similarities in the sociocultural way of life which are unique and make this region a proper subcontinent.
- ❑ **Diversity of Flora and Fauna:** The subcontinent is endowed with a wide diversity of flora ranging from evergreen forests to alpine meadows. India is one of the biodiversity hotspots of the world. A variety of animals found in the Indian subcontinent are endemic to the region.

All the above characteristics of India, sets it apart from the other regions of Asia, making it a subcontinent in the Asian continent.

IS INDIA A TROPICAL OR A TEMPERATE COUNTRY?

The climate of India resembles the climate of a tropical country although the area north of the tropic of Cancer i.e. the temperate part is twice the area of the tropical part.

- ❑ This is mainly because the lofty Himalayas separates the Indian subcontinent from the rest of Asia and also acts as a barrier to the temperate air masses moving southwards from Central Asia. As a result, compared to other areas on the same latitudes, the northern half of India is warmer by 3°C to 8°C.
- ❑ During summer, due to the overhead position of the sun, the climate of the southern parts of the country is similar to the equatorial dry climate.
- ❑ The northern plains are influenced by “Loo”, a hot dry wind that increases the temperature. Thus India as a whole, which is located south of the Himalayas, has a **tropical climate**.

Overall, India can be considered a country with a predominantly tropical climate, but it also includes temperate and sub-tropical zones due to its diverse geographical features. This diversity in climate has significant implications for agriculture, ecosystems, and the overall way of life across different parts of the country.

THE LONGITUDINAL EXTENT AND ITS IMPLICATIONS

From the longitudinal extent of India, it can be seen that there is a variation of **nearly 30 degrees** between the Easternmost and Westernmost extent of the country. It results in a time difference of **nearly two hours** between the extreme ends of the country. The earth moves around 360° in 24 hours on its axis. Thus, a difference of 1° longitude is equal to 4 minutes in time. Therefore the difference in local time between the western-most point and the eastern-most point is calculated as $30 \times 4 = 120$ minutes or 2 hours.

Despite the difference of two hours, the watches in the Easternmost and the Westernmost parts of India show the same time because Indian Standard Time is followed all over India.

*{In Assam, tea gardens follow a separate time zone, known as the **Chaibagan or Bagan time** ('Tea Garden Time'), which is one hour ahead of IST.}*

Indian Standard Time (IST)

India has adopted the **82.5° East meridian** as its standard time zone, which is **5 hours 30 minutes** ahead of Greenwich Mean Time (GMT). It passes through the city of **Mirzapur** near Allahabad in Uttar Pradesh. The standard meridian of India passes through **5 Indian states** viz. Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Orissa, Andhra Pradesh. Sri Lanka also has adopted Indian Standard Time.

THE FRONTIERS OF INDIA

India is surrounded on three sides by Sea. Its land boundary measures **15106.7 km**, and its shoreline is **7,516.6 Km** including island territories. The coastline of the mainland is **6100 km** long.

All states and territories in India, except for six namely Madhya Pradesh, Chhattisgarh, Jharkhand, Delhi, Haryana and Telangana, have a border with another country or the ocean.

India Shares its Land Boundary

- ❑ In the Northwest with Pakistan and Afghanistan
 - ❑ In the North with China (Tibet), Nepal and Bhutan
 - ❑ In the East with Myanmar and Bangladesh
 - ❑ India has **maritime** borders with Sri Lanka and Maldives.
- From longest to shortest (in length) boundary sharing with India: **Bangladesh (4096 Km) > China > Pakistan > Nepal > Myanmar > Bhutan > Afghanistan (106 Km)**

India-China Border

- ❑ India shares the **second-longest border** with China with a length of **3488 Km**. China is the largest neighbouring

nation of India. Five States namely Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh and Jammu and Kashmir touch the boundary with China.

India's Border with China is Divided into Three Sectors

- ❑ **Western Sector:** This boundary lies between Jammu and Kashmir and the Sinkiang (Xinjiang) province of China. In this region, the dispute is over the Aksai Chin district, the Changmo Valley, Pangong Tso and the Sponggar Tso area of north-east Ladakh.
 - The **Johnson Line**, proposed in 1865, designates Aksai Chin in Jammu and Kashmir under Indian control. In contrast, the **McDonald Line**, proposed in 1893, places Aksai Chin under Chinese control.
 - India asserts that the Johnson Line represents the accurate and rightful national border with China. Conversely, China maintains that the McDonald Line is the correct border with India.
 - Currently, the **Line of Actual Control (LAC)** serves as the boundary, separating Indian areas of Jammu and Kashmir from Aksai Chin.
- ❑ **Middle Sector:** In this area, Uttarakhand and Himachal Pradesh touch the border of the Chinese-occupied region of Tibet.
 - Here India roughly shares about a 545 km long boundary with China, which is considered as least disputed region.
- ❑ **Eastern Sector:** The boundary line in this region is known as the **McMahon Line**. It was the result of the boundary agreement between Tibet and Britain at the **Shimla Accord** in 1913-14. However, China doesn't accept this boundary line. This boundary runs from the easternmost corner of Bhutan to a point near Diphu Pass (Talu Pass) at the trijunction of India, Tibet and Myanmar.
 - The primary dispute revolves around the Tawang Valley of Arunachal Pradesh.
 - Another focal point is the Chumbi Valley, specifically the Doklam Tri-Junction, shared by India and Bhutan.



IGNITE YOUR MIND

Do you know that Yume Samdong, also known as Zero Point, is a remote village near the India-China border in Sikkim and it is one of the highest habitable places in the world and offers breathtaking views of the surrounding mountains. Discuss the challenges faced by indigenous communities in preserving their unique cultural heritage amidst the uncertainties of the India-China border dispute.

India - Nepal Border

- ❑ India and Nepal share a border of **1751 km** length. Five Indian states namely Uttarakhand, Uttar Pradesh, Bihar, West Bengal and Sikkim share the border with Nepal.



Fig. 1.5: Indian States sharing border with Nepal

- ❑ It is a porous border which means the movement of people and goods between India and Nepal is allowed without any restriction.
- ❑ This border runs along the foothill of the Shiwalik Range in the Terai belt.
- ❑ **Raxaul** and **Jogbani** in Nepal are two integrated checkposts on the India-Nepal Border.
- ❑ Currently, India and Nepal are embroiled in a border dispute concerning several regions: Kalapani, Lipulekh Trijunction between India, Nepal, and China, Susta Region of West Champaran (Bihar).

India - Bhutan Border

- ❑ India and Bhutan share a boundary of **699 km** and is an open international border.
- ❑ Four Indian states namely Assam, Sikkim, West Bengal and Arunachal Pradesh share a border with Bhutan.
- ❑ Bhutan serves as a buffer between India and China.



Fig. 1.6: Indian States sharing border with Bhutan

Chicken's Neck' of India

- ❑ The **"Chicken's Neck"** refers to a narrow strip of land in the Indian subcontinent that connects mainland India with its northeastern states.



Fig. 1.7: The Siliguri Corridor

- ❑ Officially known as the **"Siliguri Corridor,"** it is a slender stretch of land, approximately **22 kilometers wide** at its narrowest point, located in the Indian state of West Bengal.
- ❑ The Chicken's Neck is situated in the northern part of West Bengal and serves as the **only land corridor connecting mainland India to the northeastern states.**
- ❑ The narrow width of the Chicken's Neck makes it a **vulnerable point**, and any disruption to this corridor could have significant implications for connectivity to the northeastern region.

India - Pakistan Border

- ❑ India and Pakistan share a border of **3323 km** length. The boundary separating India and Pakistan is known as the **Radcliffe Line** which was demarcated on 17 August 1947.
- ❑ The border dispute is over regions like Saltoro Ridge, Sir Creek, Pakistan-occupied Kashmir, etc.
- ❑ **An integrated check post** has been set up on the India-Pakistan border at Attari (Punjab). Passenger Terminal Building (PTB), Dera Baba Nanak checkpost is operational only for pilgrims to visit the Gurudwara Sri Kartarpur Sahib Ji.
- ❑ **Sir Creek:** It is a disputed 96 km long strip of water, long, in the marshes of Rann of Kutch. The dispute is over the maritime boundary line separating the Sindh province of Pakistan and the Rann of Kutch area. Its importance lies in the fact that it is considered to be the largest fishing ground in Asia.
- ❑ **Line of Control (LoC):**
 - The Line of Control (LoC) is a military control line between Indian-administered territory of Jammu and Kashmir and the Pakistani-administered territory of Azad Jammu and Kashmir and Gilgit-Baltistan.
 - It serves as the de-facto border in the disputed region of Kashmir.

- The Line of Control was established after the first India-Pakistan War in 1947-48 and has witnessed periodic tensions and conflicts between India and Pakistan.
- The Line of Control was originally established as the **Ceasefire Line in 1949** following the **first India-Pakistan War**. It was later renamed as the **Line of Control** after the **Simla Agreement in 1972**.

Siachen Glacier

- The Siachen Glacier is located in the eastern Karakoram Range in the Himalayas and is known as the highest battlefield on Earth. The region has been a source of conflict between India and Pakistan.
- The Siachen Glacier holds strategic significance due to its location near the borders of India, Pakistan, and China. Control over the glacier provides a vantage point in the region.



IGNITE YOUR MIND

There is Baba Chamliyal shrine near the India-Pakistan border in Jammu and Kashmir that hosts an annual fair where people from both countries share the traditional drink "Sharbat." Try to analyze the significance of religious and cultural connectivity along the India-Pakistan border, considering sites like Kartarpur and Chamliyal. How do these cultural linkages contribute to people-to-people diplomacy despite political tensions?

India - Bangladesh Border



Fig. 1.8: India-Bangladesh Map

- ❑ India shares the **longest border** with Bangladesh. They share a **4,096.7 Km** long international border.
- ❑ The 100th Constitutional Amendment Act of 2015 amended the Constitution of India to give effect to the acquisition of territories by India and transfer of certain territories to Bangladesh in pursuance of the agreement

and its protocol entered into between the Governments of India and Bangladesh.

- ❑ The border between India and Bangladesh was created by the **Radcliffe Award**, which divided the former province of Bengal into two parts.
- ❑ Five Indian states namely West Bengal, Mizoram, Meghalaya, Tripura, and Assam. Four integrated check posts namely **Agartala** (Tripura), **Petrapole** (West Bengal), Sutarkandi (Assam) and Srimantapur (Tripura) are set up on the India-Bangladesh border.
- ❑ Three rivers namely Padma (Ganga), Jamuna (Brahmaputra) and Teesta are the important trans-boundary rivers between India and Bangladesh. The presence of these rivers makes boundary management difficult.

India - Myanmar Border

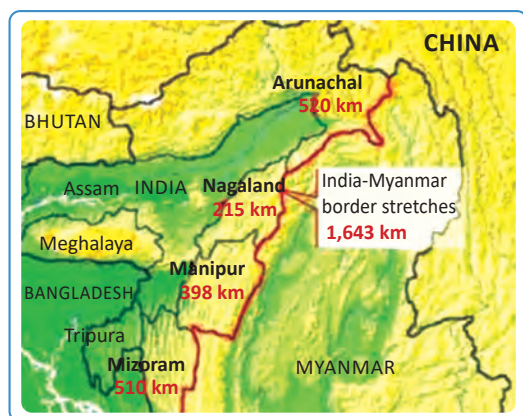


Fig. 1.9: India-Myanmar Border

- ❑ The India and Myanmar border runs **1,643 km** long. India and Myanmar share both land and maritime boundaries.
- ❑ Four North-Eastern States namely Arunachal Pradesh, Nagaland, Manipur and Mizoram share an international boundary with Myanmar.
- ❑ An integrated check post has been set up in **Moreh** (Manipur) on the border of India and Myanmar.
- ❑ There exists a Free Movement Regime (FMR) between India and Myanmar which allows people living in border villages to trade and move freely up to 16 kilometres inside each other's territory.

India - Sri-Lanka Border

- ❑ India and Sri Lanka are separated by a channel of sea called the Palk Strait and the Gulf of Mannar.
- ❑ Dhanushkodi on the Tamil Nadu coast is joined with Talaimannar in the Jaffna peninsula in Sri Lanka by a group of islets forming Adam's Bridge.



Fig. 1.10: India-Sri Lanka

Ram Setu or Adam's Bridge



Fig. 1.11: Dhanushkodi and Talaimannar

- ❑ Ram Setu, also known as Adam's Bridge is a chain of **natural limestone shoals** between **Pamban Island**, also known as Rameswaram Island, off the south-eastern coast of Tamil Nadu, India, and **Mannar Island**, off the north-western coast of Sri Lanka.
- ❑ According to Hindu mythology, this bridge is believed to have been built by Lord Rama to rescue his wife Sita from the demon king Ravana.
- ❑ **Dhanushkodi** and **Talaimannar** are the closest points on either side of the Adam's Bridge.

Dhanushkodi is a town located at the southeastern tip of Pamban Island in Tamil Nadu, India.

- ❑ It is situated near the point where the **Bay of Bengal and the Indian Ocean meet**.
- ❑ Historically, Dhanushkodi was a thriving trade and pilgrimage center, but it was largely destroyed by a cyclone in 1964, leading to the abandonment of the town.

Talaimannar is a town on the northwestern coast of Mannar Island in Sri Lanka.

- ❑ It is known for its strategic location and was historically an important port.

India-Sri Lanka Maritime Border Conflict and Agreements

In the 1970s, India and Sri Lanka faced a maritime boundary dispute over the delineation of their Exclusive Economic Zones (EEZ) in the Gulf of Mannar and the Palk Strait. The disagreement revolved around the interpretation of historical agreements and the maritime boundary. To address this issue:

1974 Agreement:

- ❑ In 1974, India and Sri Lanka signed the Maritime Boundary Agreement, attempting to delimit their maritime boundaries in the Gulf of Mannar and the Palk Strait.
- ❑ The agreement sought to address fishing rights, marine resources, and the demarcation of zones to prevent conflicts over the use of these waters.

1976 Agreement:

- ❑ Despite the 1974 agreement, the maritime boundary dispute persisted, leading to further negotiations.
- ❑ In 1976, India and Sri Lanka signed the Agreement on Maritime Boundary to refine the maritime boundary between the two countries, addressing issues related to the territorial sea and continental shelf.

These agreements aimed to promote cooperation and prevent conflicts over fisheries and other resources in the shared waters.

India - Afghanistan

- ❑ India and Afghanistan have a borderline of **106 km** and the border touches the Union territory of Jammu and Kashmir.
- ❑ **Durand line** touches the union territory of Jammu and Kashmir in India. So, along with a borderline separating Pakistan and Afghanistan, it also acts as a border between India and Afghanistan.
- ❑ Sir Mortimer Durand, a British diplomat, delineated its boundaries in 1896.

India - Maldives

- ❑ India shares a maritime border with Maldives. Maldives lies in the south of the Lakshadweep island.
- ❑ In 1976, India and Maldives officially decided their maritime boundary.
- ❑ **Before 1976**, a dispute between India and the Maldives over the status of Minicoy, an island in the Lakshadweep archipelago, arose from historical claims and linguistic ties.
- ❑ The disagreement was peacefully resolved through diplomatic negotiations, leading to the signing of the **1976 Treaty of Friendship and Cooperation**, which affirmed **Minicoy as part of India's territory**.

- ❑ The **8-degree channel** separates the Indian island of Minicoy and Maldives.



Fig. 1.12: India-Maldives map

INDIA'S POLITICAL DIVISION

- ❑ For administrative convenience, India is divided into **28 states** and **8 Union Territories**. Further, they are divided into districts. **Delhi** is the Capital of India.
- ❑ In terms of area, **Rajasthan** is the largest state and **Goa** is the smallest state. The division of states and UTs is mainly based on languages spoken.
- ❑ These states are categorised into **five zones** according to **States Reorganisation Act, 1956**. These zones are as follows:

Zonal Division of India	
North Zone	Haryana, Himachal Pradesh, Jammu & Kashmir, Punjab, Rajasthan, National Capital Territory of Delhi and Union Territory of Chandigarh
Central Zone	Chhattisgarh, Uttarakhand, Uttar Pradesh and Madhya Pradesh
Eastern Zone	Bihar, Jharkhand, Orissa, Sikkim and West Bengal
Western Zone	Goa, Maharashtra and Gujarat and the Union Territories of Dadra & Nagar Haveli and Daman & Diu
Southern Zone	Kerala, Andhra Pradesh, Karnataka, Tamil Nadu and the Union Territory of Puducherry

The North Eastern states are included in a separate zone, created by the **North Eastern Council Act, 1971**. These states are Assam, Arunachal Pradesh, Manipur, Tripura, Meghalaya, Nagaland and Mizoram. The North Eastern region is also known as the land of seven sisters.

Maritime Boundaries of India

Maritime borders of India are the maritime boundary recognized by the United Nations Convention on the Law of the Sea entails boundaries of territorial waters, contiguous zones, and exclusive economic zones. This creates a maritime border of more than 7,000-kilometre (4,300 mi) shared with **seven nations**.

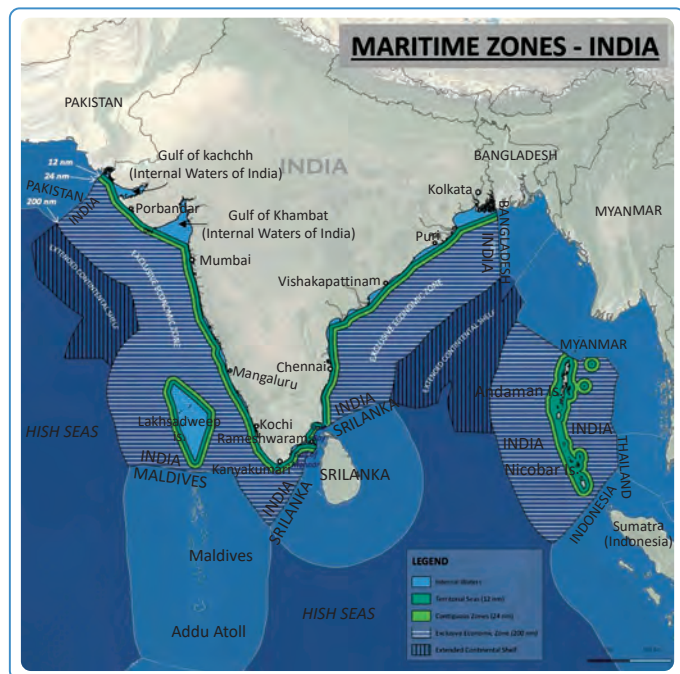


Fig. 1.13: Maritime Boundaries of India on Map

Here are The Main Maritime Boundaries of India

- ❑ **Territorial Sea**—The zone extending 12 nm from the baseline
- ❑ **Contiguous Zone**—The area extending 24 nm from the baseline
- ❑ **Exclusive Economic Zone**—The area extending 200 nm from the baseline except when the space between two countries is less than 400 nm
- ❑ India shares maritime boundaries with several neighboring countries. **The primary ones include:**
 - **Bangladesh:** The maritime boundary with Bangladesh is defined by the Land Boundary Agreement, which was signed in 1974 and later ratified in 2015.
 - **Pakistan:** The maritime boundary with Pakistan is delineated in the Arabian Sea. The Sir Creek area has been a subject of dispute between India and Pakistan.
 - **Sri Lanka:** The International Maritime Boundary Line (IMBL) between India and Sri Lanka is in the Palk Strait and the Gulf of Mannar.

- **Maldives:** India and the Maldives have maritime boundaries in the Indian Ocean.

- ❑ **Andaman and Nicobar Islands,** located in the Bay of Bengal, have their own maritime boundaries, and India exercises sovereignty over the waters surrounding these islands.

Bays, Gulf and Strait in India

Bay

It is a small body of water which is partly surrounded by land on three sides with the fourth side opening towards the ocean. Generally, it is smaller and less enclosed than a gulf. Its mouth, where it meets the ocean, is wider than that of a Gulf.

- ❑ **Bay of Bengal:** The North-eastern portion of the Indian Ocean is the Bay of Bengal. It lies in between the Indian subcontinent and the Indo-Chinese peninsula. It is the largest Bay in the world and is formed due to plate tectonics.



Fig. 1.14: Bay of Bengal on Map

- ❑ **Palk Bay:** It is a shallow water body between India's Southeastern coast and Sri Lanka. It is about 150 kilometres long and 57 to 107 kilometres wide, with an average depth of 13 meters.

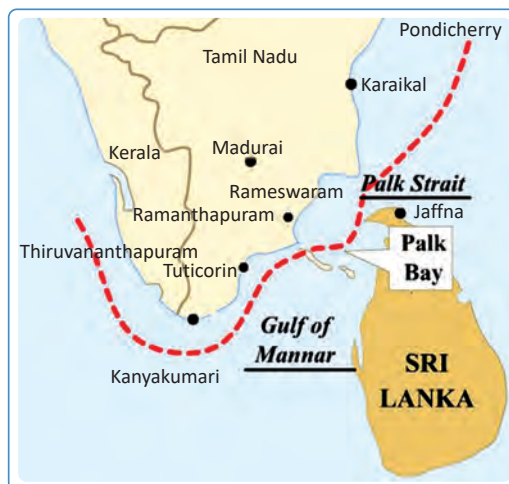


Fig. 1.15: Palk Bay



IGNITE YOUR MIND

Are you aware that Kochi (Cochin), a major port city in Kerala, is known as the “Queen of the Arabian Sea.” In light of this fact discuss the historical importance of Kochi as a major port city. Analyze how historical events and foreign influences have shaped the cultural and economic landscape of Kochi.

Gulf

The Gulf is a larger body of water that penetrates land. The mouth of a Gulf, where it meets the ocean, is narrower than a Bay. Gulfs differ greatly in size, shape, and depth.

- ❑ As compared to bays, they are generally larger and more deeply indented.
- ❑ Like bays, they often make excellent harbours. Many important trading centres are located in gulfs.

Examples of the Gulf in India are as follows:

- ❑ **Gulf of Kutch:** It is an inlet of the Arabian Sea. It is located on the western coast of India, in the Jamnagar district of Gujarat. The Gulf of Khambhat is located in the south of the Gulf of Kutch. It separates the Kutch and Kathiawar peninsulas.
- ❑ **Gulf of Khambhat/Cambay:** The Gulf of Khambhat (also known as the Gulf of Cambay) is situated along the west coast of India in the Arabian Sea, in the state of Gujarat. About 130 kilometres in length, it separates the Kathiawar Peninsula and the southeastern part of Gujarat. The **Narmada**, **Tapti**, **Mahi** and **Sabarmati** rivers evacuate into the Gulf. The depth of the Gulf of Khambhat is less and has abundant shoals and sandbanks.

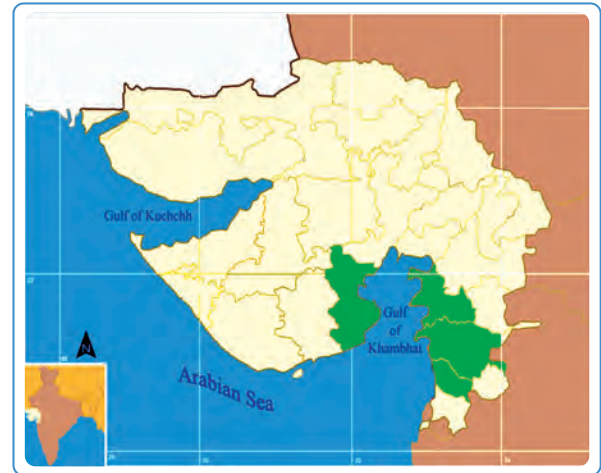


Fig. 1.16: Gulf of Khambhat on Map

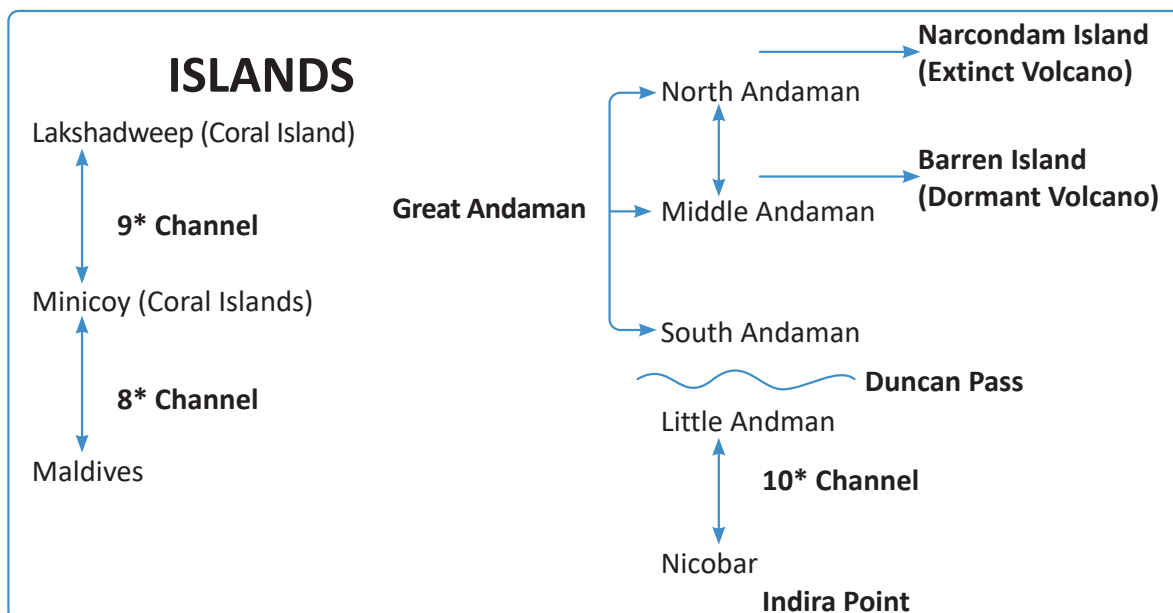
- ❑ **Gulf of Mannar:** It is located in the Indian Ocean.
 - It is bordered by the Southeast coast of Tamil Nadu and the western coast of Sri Lanka.
 - It thus separates India and Sri Lanka. Ram Setu (Adam’s Bridge) which is a chain of low islands and reefs, separates the Gulf of Mannar and Palk Bay.

Strait

A narrow body of water connecting to larger water bodies or continents is called a Strait. An example of a Strait in India is the Palk Strait.

- ❑ **Palk Strait:** It separates India and Sri Lanka. The Palk Strait lies between the Indian state of Tamil Nadu and the Jaffna District of Sri Lanka.
 - It acts as a connecting link between the Bay of Bengal in the northeast and Palk Bay in the southwest.
 - The Vaigai River of Tamil Nadu along with several other rivers drains water into it.

Important Channels



'Channels' refers to natural or artificial waterways that facilitate the movement of water, connecting one body of water to another.

8-Degree Channel

- ❑ The 8-degree channel is located between the **Minicoy Islands (Coral Islands)** and the **Maldives** in the Indian Ocean.

9-Degree Channel

- ❑ The 9-degree channel is situated between the **Lakshadweep archipelago** and **Minicoy Island** in the Arabian Sea.

10-Degree Channel

- ❑ The 10 Degree Channel is a channel that separates the **Little Andaman** and **Car Nicobar Island (northernmost of Nicobar Island group)** in the Bay of Bengal.

Maritime Boundary Issues of India

The Palk Bay Issue

The Palk Bay issue refers to a long-standing maritime dispute between India and Sri Lanka concerning the use of the Palk Bay, a shallow body of water that lies between the southeastern coast of India and the northern coast of Sri Lanka.

Key points:

- ❑ **Fishing Rights:** Local fishermen from both India and Sri Lanka have traditionally relied on the Palk Bay for fishing activities. However, disputes have arisen over the fishing rights in the region. Indian fishermen often venture into the waters claimed by Sri Lanka, leading to tensions between the two sides.
- ❑ **Maritime Boundaries:** The maritime boundaries in the Palk Bay are not clearly defined, leading to disagreements over where the international maritime boundary between India and Sri Lanka should be located.
- ❑ **Narrow Waters:** The Palk Bay is relatively narrow, and its waters are critical for the movement of fishermen and their boats. The close proximity of the coasts of India and Sri Lanka in this region adds complexity to the delineation of boundaries and fishing rights.
- ❑ **Arrests and Conflicts:** Incidents of arrests and conflicts between Indian and Sri Lankan fishermen have been reported over the years. Fishermen from one country inadvertently crossing into the waters claimed by the other often result in arrests and diplomatic tensions.

Efforts have been made by both countries to address the issue through diplomatic channels and negotiations. Talks have been held to find a mutually acceptable solution to the problem, taking into consideration the interests of the fishermen from both sides. However, a definitive resolution remains elusive, and the Palk Bay issue continues to be a source of bilateral concern between India and Sri Lanka.

Sir Creek Issue

The Sir Creek issue is a long-standing territorial dispute between **India** and **Pakistan** over the Sir Creek, a **96-kilometer-long estuary** along the border of the Indian state of Gujarat and the Pakistani province of Sindh. The dispute primarily revolves around the demarcation of the maritime boundary in the Sir Creek region, which opens up into the Arabian Sea.



Fig. 1.17: Sir Creek Dispute

Key Points:

- ❑ **Uncertain Border:** The border in the Sir Creek area is not precisely demarcated, and both India and Pakistan have different interpretations of the alignment of the boundary.
- ❑ **Importance of the Estuary:** The Sir Creek estuary is significant for both countries due to its potential implications for maritime boundaries and exclusive economic zones.
- ❑ **Security Concerns:** The Sir Creek region has gained strategic importance due to security considerations, particularly in the context of the India-Pakistan border.
- ❑ **Negotiations and Talks:** Various rounds of talks have taken place at different levels to address the concerns and reach a mutually acceptable agreement on the maritime boundary.

Efforts have been made to build confidence and enhance bilateral relations through dialogue, but a final resolution to the Sir Creek issue remains pending. The territorial disputes between India and Pakistan, including the Sir Creek issue, are part of broader geopolitical tensions in the region. The challenge lies in finding a compromise that considers the historical context and the interests of both countries.

Maritime Choke Points



Fig. 1.18: Choke Points in the Indian Ocean

- A chokepoint refers to the point of natural congestion along two broader and more critical navigable passages. Maritime choke points are naturally narrow shipping channels with high traffic. They are significant because of their strategic locations.
- Several key maritime choke points are located within or near the Indian Ocean, influencing global shipping routes. Some notable ones include:
 - **Strait of Hormuz:** Connects the Persian Gulf to the Gulf of Oman and the Arabian Sea. It is a critical passage for oil transportation, with a significant portion of the world's oil shipments passing through.
 - **Bab el Mandeb:** Links the Red Sea to the Gulf of Aden and the Arabian Sea. It is important for maritime trade, particularly for vessels traveling to and from the Suez Canal.

- **Strait of Malacca:** Situated between the Malay Peninsula and the Indonesian island of Sumatra, it is one of the most important shipping channels globally, connecting the Indian Ocean to the South China Sea.
- **Lombok Strait:** Located between the islands of Bali and Lombok in Indonesia, it is another key passage for maritime traffic in the Indian Ocean region.

These choke points are critical for global trade and have geo-political implications, as control over them can influence economic and strategic interests for the countries in the region and beyond. The Indian Ocean's significance continues to grow as maritime activities, trade, and geopolitical dynamics evolve.

CONCLUSION

The geography of India is a tapestry of immense diversity, encompassing a myriad of physical features, climates, and ecosystems. From the towering peaks of the Himalayas to the expansive alluvial plains of the Ganges and the arid expanse of the Thar Desert, India's topography is a testament to the country's geological richness. The intricate network of rivers, including the Ganges, Yamuna, Brahmaputra, and Indus, sustains agricultural activities and shapes the landscape.

India's geography is not merely a backdrop but an integral part of its identity. The interplay of mountains, rivers, deserts, and coastlines contributes to the country's ecological richness, making it a captivating and dynamic land. As India continues to navigate the challenges of the modern era, its diverse geography remains a source of resilience, inspiration, and cultural vibrancy.

Geological Structure of India

2

INTRODUCTION

The term “**geological structure**” is primarily used to describe the organization and formation of rocks in the Earth’s crust due to the Earth’s movements. It also pertains to the morphological characteristics of rocks, such as the **Gondwana structure**. The Geological Structure and Rock System of India was analyzed with reference to their geographical locations and their geological history.

Based on this complex and varied geological history, the **Geological Survey of India** has classified the rock systems

of the country into **4 major divisions** as shown in the table below:

Indian Classification	Corresponding Period on Geological Time Scale
Archaean	Early Precambrian Eons
Purana	Late Precambrian (or Proterozoic Eon)
Dravidian	600-400 Mya (largely coinciding with the Palaeozoic era)
Aryan	400 Mya-present

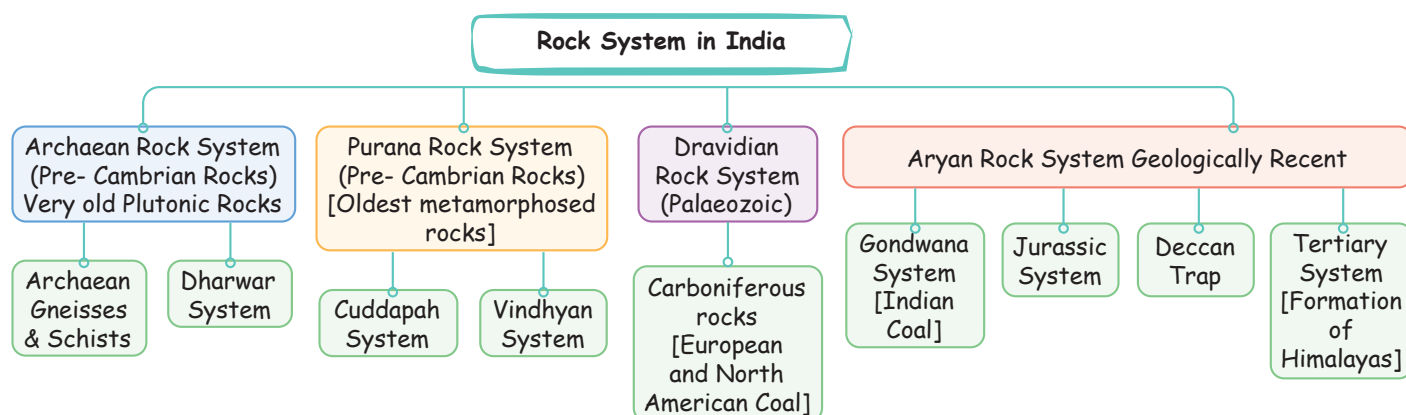


Fig. 2.1: Rock System in India



IGNITE YOUR MIND

You might be interested in knowing that Anthropocene epoch is a proposed epoch that denotes the present geological time interval, in which the Earth’s ecosystem has gone through radical changes due to human impact. In this light can you examine the stratigraphic sequence of major geological formations in India, emphasizing their chronological order and significance.

GEOLOGICAL EVOLUTION OF INDIA

The geological evolution of India spans over billions of years and involves complex processes that have shaped its contemporary landforms.

India’s Geological Evolution with Respect to the Geological Timescale

1. Archean Eon (4.0 - 2.5 billion years ago):

- During the Archean Eon, the Indian subcontinent’s foundation began to form through the accumulation of various rocks, including granite and gneiss.
- These ancient rocks provide the basement upon which later geological processes operated.

2. Proterozoic Eon (2.5 billion - 541 million years ago):

- In the Proterozoic Eon, the Indian subcontinent underwent significant geological events.
- The Dharwar Group, a sequence of metamorphic and sedimentary rocks, was deposited during this time.

3. Cambrian to Ordovician Period (541 - 443 million years ago):

- During the early Paleozoic era, the Indian subcontinent was a part of the supercontinent Gondwana.

- Sedimentary rocks, including limestones and shales, got deposited during this period.

4. Silurian to Devonian Period (443 - 359 million years ago):

- Gondwana continued to evolve, and marine sedimentation continued, leading to the deposition of further sedimentary layers.

5. Carboniferous to Permian Period (359 - 252 million years ago):

- Gondwana continued to experience geological activity, and sediments from this time contributed to the formation of the Gondwana basins.

6. Triassic to Jurassic Period (252 - 145 million years ago):

- During the breakup of Gondwana, India separated from Antarctica and began drifting northwards.
- Sedimentation and volcanic activity characterized this period, leading to the formation of the Deccan Traps.

7. Cretaceous Period (145 - 66 million years ago):

- India continued its northward journey, eventually colliding with the Asian landmass.
- The collision initiated the uplift of the Himalayan mountain range and the Tibetan Plateau.

8. Cenozoic Era (66 million years ago - present):

- The collision with Asia continued, and the ongoing convergence led to further mountain-building processes.

- The Himalayas and associated mountain ranges continue to evolve through tectonic activity.

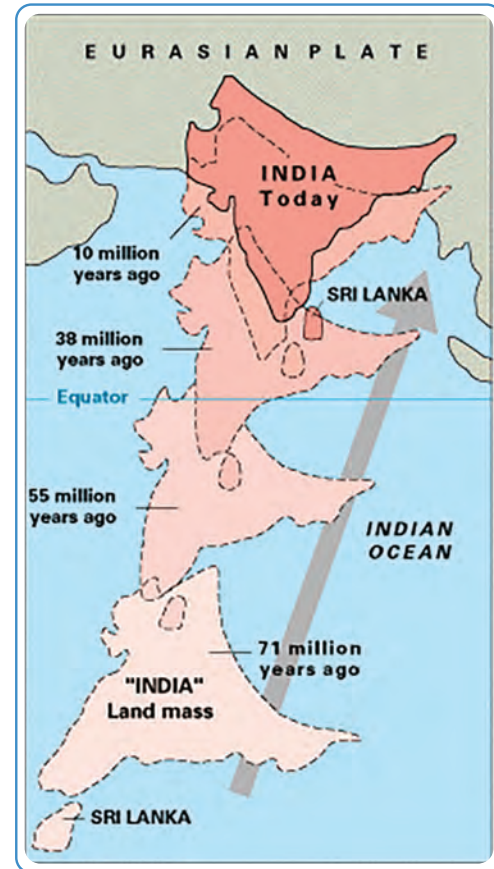


Fig. 2.2: Geological Evolution of India

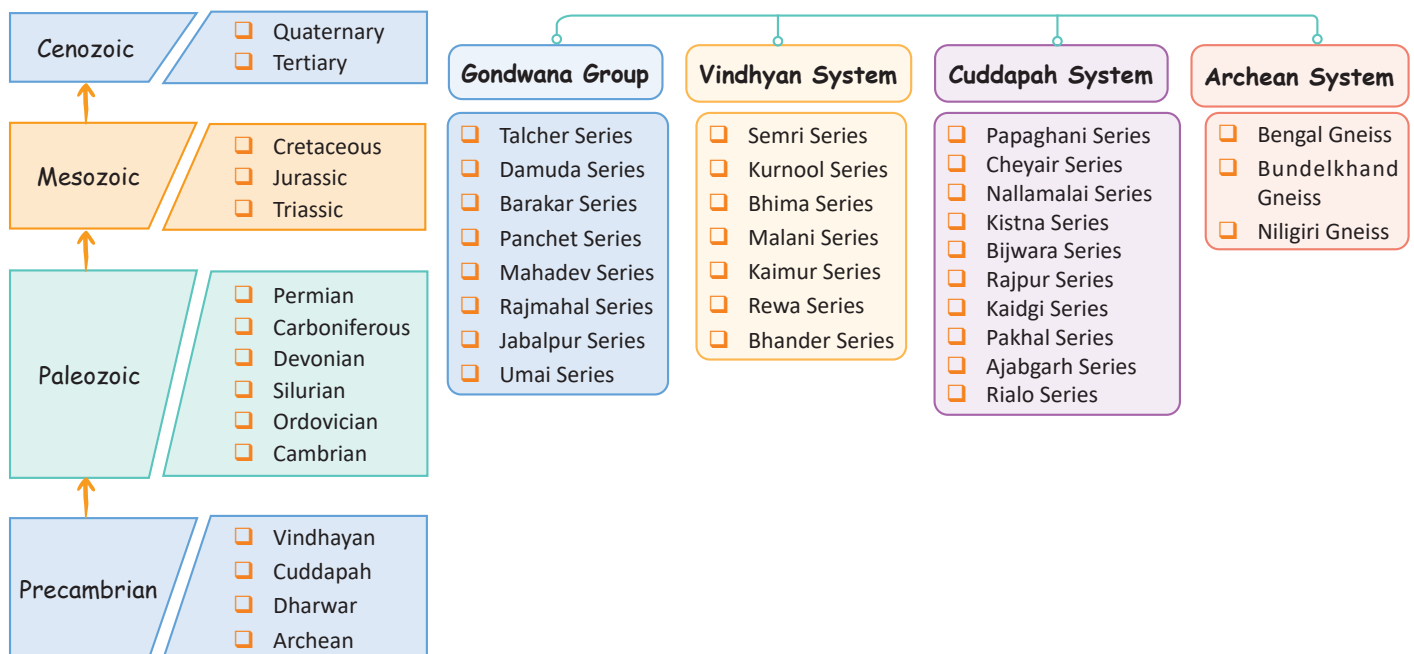


Fig. 2.3: Geological Time Scale and Rock systems found in India

THE ARCHAEN ROCK SYSTEM

- ❑ Prior to **2.5 billion** years ago, during the **Precambrian** Period, the upper crust of the earth's surface began to cool and solidify, which is reflected by the exposure of gneisses and granites, particularly on the Peninsula. This is the first phase of tectonic evolution.
- ❑ The Indian Craton (Block of the Indian Subcontinent of **Gondwanaland**) is composed primarily of these. The **oldest rocks** in the earth's crust are referred to as Archaean a word used by **J.D. Dana** in 1782.
- ❑ There are two systems of rocks in the Archaean group:
 - **Archaean Schists And Gneisses**
 - **Dharwar System**

Archaean Schists and Gneisses

- ❑ **Overview:** The **Archaean** Era, also called the Precambrian Period, represents a significant portion of Earth's geological timeline, stretching from the planet's formation around 4.6 billion years ago to the start of the Cambrian Period, roughly 570 million years ago.

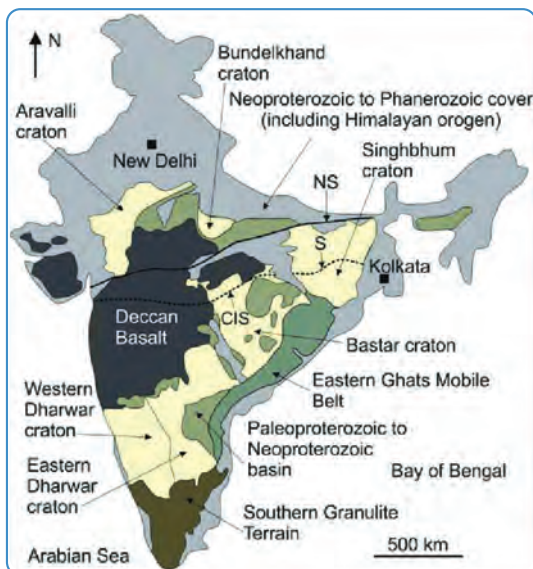


Fig. 2.4: Archaen Rock system in India

- ❑ **Key Developments:** To put it into perspective, the **Precambrian** era accounts for approximately **86.7%** of Earth's history. During the Precambrian era, crucial developments took place, including the formation of Earth's earliest rocks, the evolution of its atmosphere, the emergence of processes like chemosynthesis and photosynthesis, the creation of an environment suitable for life, and the emergence of the planet's current atmosphere. Notably, rocks from the Archaean period do not contain any evidence of ancient life; they are essentially **lifeless**, lacking fossils.
- ❑ **Key Characteristics:** These rocks have a distinct character: they are highly **crystalline**, extensively folded and faulted, and largely devoid of sediment. They often

contain intrusions of **molten rock** (plutonic intrusions) and typically exhibit a well-defined foliated structure.

- ❑ These rocks lack fossils and are commonly referred to as the "Basement" or "Fundamental Complex" due to their foundational role as the base for new geological layers.

The Archean Gneisses Rocks are Categorized into three Types

Bengal Gneiss

- ❑ **Discovery Location:** These were initially found in the Midnapur district, West Bengal.
- ❑ **Characteristics:**
 - Rock with multiple layers.
 - Presence in Odisha's Eastern Ghats (known as Khodolites), Jharkhand, Andhra Pradesh, and Tamil Nadu.
 - Occurrence in Son Valley, Meghalaya Plateau, and Mikir Hills with fewer layers.

Bundelkhand Gneiss

- ❑ **Nature:**
 - Very hard rock since ancient times.
 - It forms a part of the second group of old rocks.
- ❑ **Distribution:** Found in Bundelkhand (Uttar Pradesh), Baghelkhand (Madhya Pradesh), Maharashtra, Rajasthan, Andhra Pradesh, and Tamil Nadu.
- ❑ **Characteristics:**
 - Large, grainy rocks resembling granite.
 - Marked by criss-cross lines and veins of quartz.

Nilgiri Gneiss

- ❑ **Charnockite Series:**
 - Charnockite rocks, often known as the Charnockite series.
 - These are named after Job Charnock's tombstone in Kolkata.
 - **Appearance:** Nilgiri gneiss appears bluish-grey to dark and feels somewhat rough.
 - **Formation:** Formed deep within the Earth and intruded into older rocks.
 - **Geographical Occurrence:** Widespread in Tamil Nadu, Andhra Pradesh, Odisha, Karnataka, Kerala, Jharkhand, Chhattisgarh, and Rajasthan.
 - **Mineral Rich:** Rich in valuable minerals like iron ore, copper, manganese, mica, dolomite, lead, zinc, silver, and gold, contributing to India's wealth.

Dharwar System

- ❑ **Geological Time Period:**
 - It spans from 2,500 million years ago to 1,800 million years ago.
- ❑ **Naming Origin:**
 - Named as the Dharwar System in the Indian Geological Timeline.

- First extensively studied in the Dharwar district of Karnataka.

❑ Rock Composition:

- The rocks mainly consist of metamorphosed or transformed sedimentary materials, including schists and gneisses.
- They contain a significant amount of igneous debris.
- The significant rocks within the Dharwar system include **hornblende, schists, quartzite, phyllite slates, crystalline limestone, and dolomites**.
- Economically, the Dharwar system holds great importance due to the presence of valuable minerals such as **high-grade iron ore, manganese, copper, lead, gold, quartzite, and slates**.
- Notably, these rocks are the **first metamorphic sedimentary rocks** identified in India.

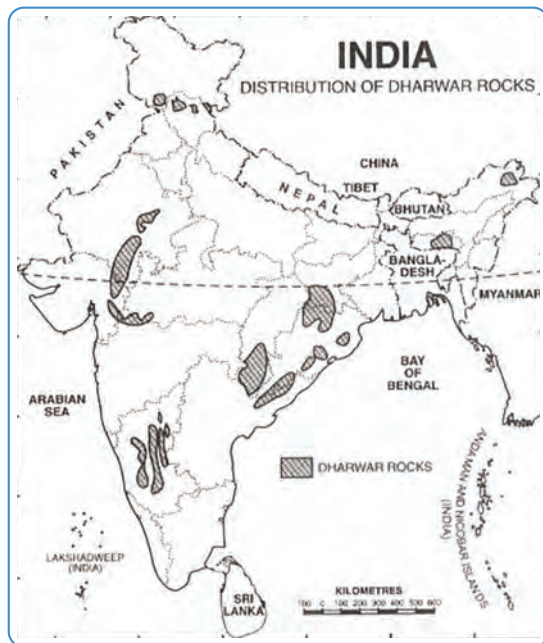


Fig. 2.5: Distribution of Dharwar Rocks

❑ Geographical Distribution:

- ❑ **Places of Discovery:** Aravalli, Tamil Nadu, Chota Nagpur plateau, Meghalaya, Delhi & Himalayas

❑ Series of Dharwar Rock System:

- **Champion Series Rocks:** The Champions series containing gold mines lie within this system. The Champion system is named after the Champion Reef in the Kolar Gold Field (KGF), one of the deepest gold mines in the world.
- **Champenar Series:** Found near Baroda. Sources of lush green variety of marbles. Abundant in quartzite, conglomerates, phyllites, slates, and marbles (green variety).
- **Closepet Series:** Found in Balaghat and Chhindwara region of MP. Rich in copper ore.
- **Chilpi Series:** Found in and around Closepet series of Balaghat and Chhindwada.
- **Iron Ore Series:** Found in Singhbhum and Mayurbhanj and Keojarh ranges.
- **Delhi Series:** The Aravallis, Rialo from Delhi to the south of Alwar and the Himalayan region.

THE PURANA ROCK SYSTEM

The Purana rock system, which includes the **Cuddapah** and **Vindhyan** rock systems, was created through the process of erosion and deposition that occurred roughly between 1,400 to 600 million years ago. These rocks are primarily sedimentary in nature, originating from the wear and tear of older Archean and Dharwar rocks. The Cuddapah system is separated from the Dharwar system by a great unconformity.

The Cuddapah System

- ❑ **Composition:** Cuddapah system consists of rocks like **shales, slates, limestone, and quartzite**. These rocks, generally lack fossils and were formed during a time when the origin of species had not yet occurred.
- ❑ **Name Origin:** Named in honour of the **Andhra Pradesh** region's **Cuddapah** district, these formations are a mix of sedimentary and metamorphic rock types.
- ❑ **Geographical Distribution:** The Cuddapah order is found in various regions, including Cuddapah and Kurnool districts in Andhra Pradesh, Chhattisgarh, the region from Rajasthan to Delhi south of Alwar.

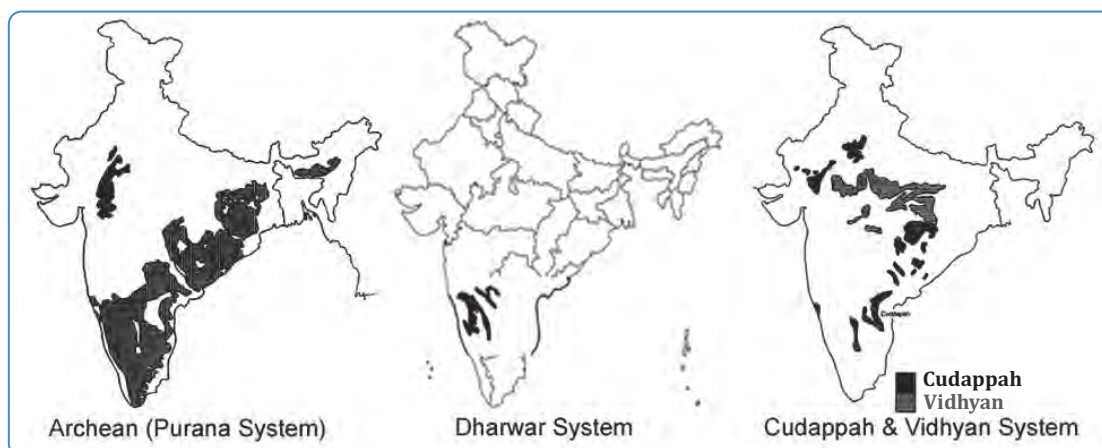


Fig. 2.6: Distribution of Dharwar Rocks

- ❑ **Thickness:** In specific locations, the Cuddapah formations can be as **thick as 6,000 meters**. This significant thickness suggests that over time, the layers of these rocks sank as more sediment accumulated.
- ❑ **Rock System:**
 - These rocks contain ores of iron, manganese, copper, cobalt, nickel, etc. and contain large deposits of cement-grade limestones.

The Vindhyan System

- ❑ **Overview:** The Vindhyan system, named after the **Vindhyan Mountains**, which serve as a natural boundary in between the Ganga Plain and the Deccan Plateau, covers a vast area of 103,600 square kilometres from Chittorgarh in Rajasthan to Sasaram in Bihar.
- ❑ **Geological Characteristics:**
 - This system boasts extensive **sedimentary** deposits, with depths exceeding 4,000 meters in some areas. In certain regions, Vindhyan rocks are concealed beneath Deccan lava.
 - The **Great Boundary Fault (GBF)** stretches for approximately 800 kilometres, separating the Vindhyan System from the Aravalli Range.
 - Mostly unfossiliferous (lacks fossils).
 - Significant Deccan Traps coverage in this belt.
 - Lacks metalliferous minerals but yields durable stones, ornamental stones, limestone, and pure glassmaking sand.
 - Presence of fissile micro-organisms in these structures.
- ❑ **Resources:**
 - The Vindhyan system is renowned for its resources, including red sandstone, which is used for construction, ornamental stone, conglomerates, diamonds, and raw materials for cement, lime, glass, and chemical industries. Some areas yield lower-quality iron ore and manganese.
 - Notably, the famous **diamond mines** of Panna and Golconda are located in the Vindhyan system.
 - The region's red **sandstone** has been employed in the construction of historic landmarks such as Qutab Minar, Humayun's Tomb, Fatehpur Sikri, Agra Fort,

Red Fort, Jama-Masjid, Birla Mandir, and the Buddhist Stupa of Sanchi. Additionally, coarser sandstones from this region have been used as grindstones and millstones.

THE DRAVIDIAN ROCK SYSTEM

- ❑ **Overview:** In India, the Paleozoic rock formation is referred to as the Dravidian system and dates back to the Paleozoic era, approximately 600 to 300 million years ago. It is also called the **Carboniferous rock system** because it is associated with the formation of high-quality coal, a valuable global resource.
- ❑ **Geographical Distribution:**
 - These rocks are predominantly located in the extra-Peninsular regions, including the Himalayas and the Gangetic plain, with restricted occurrences in the Peninsular shield, notably in Umaria, Rewa. (The name 'Dravidian' doesn't mean they are found in South India).
 - Major regions where Dravidian rocks are found include **Pir Panjal, Handwara, Lider Valley**, and Anantnag in Kashmir, additionally Spiti, Kangra, Shimla in Himachal Pradesh, and Garhwal and Kumaon in Uttarakhand.
- ❑ **Rock Varieties:** The Dravidian geological formation includes a range of **rock varieties**, including shales, sandstones, clays, quartzites, slates, salts, talc, dolomite, and marble.
- ❑ **Geological Events:** During this time frame, the supercontinent **Pangaea** began to break apart, and concurrently, the Tethys Sea started to take shape.
- ❑ **Early Life Emergence:** It marked the early emergence of **life on Earth's surface**, with abundant fossil evidence found in rocks from this era, indicating marine conditions prevalent within these ancient Paleozoic rock formations in India.
- ❑ **Onset of Coal Formation:** The Dravidian period also marked the onset of **coal formation**, particularly high-quality Carboniferous coal, although such formations were not widespread in India.

Period	Type of Rocks	Locations/Features
Cambrian (600 Ma)	Slates, Quartzites, Dolomite	North-west Himalayan region
Ordovician (500 Ma)	Quartzites, Grits, Sandstones, Limestones	Overlies Haimanta System in Spiti
Silurian (440 Ma)	Thin continuous band of Silurian strata	Continuation with Ordovician in Spiti valley

Period	Type of Rocks	Locations/Features
Devonian (400 Ma)	Massive White Quartzite (900m thick)	Identified in Muth quartzite of Spiti and Kumaon
Carboniferous (350 Ma)	Limestone, Shale, Quartzite	Upper Carboniferous rocks comprise limestone and dolomite; - Middle Carboniferous rocks found in Spiti, Kashmir, Shimla, and eastern Himalayas; -Lower Carboniferous includes slates, Pir Panjal trap, and rocks in the Kumaon region; - Mount Everest composed of Upper Carboniferous limestones; -Coal formation started in the Carboniferous age

THE ARYAN ROCK SYSTEM

- ❑ **Overview:** The Aryan Group signifies the start of the Upper **Carboniferous** era and stands as a significant milestone in the geological history of our planet. It serves as the gateway to the final, lengthiest, and most event-filled era spanning from the Upper Carboniferous to the Holocene period.
- ❑ **Geological Formations:**
 - The Aryan Group belongs to various geological formations across different eras. It includes the **Upper Paleozoic Era**, which encompasses the Upper Primary Epoch, covering formations from the Upper Carboniferous and Permian periods.
 - Moving forward, the **Mesozoic Era** during the Secondary Epoch, it includes formations from the Triassic, Jurassic, and Cretaceous periods, including the Gondwana Rock system, Deccan trap, and Jurassic system. Transitioning into the Cenozoic Era during the Tertiary Epoch, it encompasses formations from the Paleocene, Eocene, Oligocene, Miocene, and Pliocene periods.
 - Lastly, in the **Neozoic Era** during the Quaternary Epoch, includes formations from the Pleistocene and the present-day Holocene/recent Period. This classification system helps us understand the diverse geological history of our planet.
- ❑ **Components:** Aryan rock system in India comprises the following four systems:
 - Gondwana Rock System
 - Jurassic Rock System
 - Cretaceous System/ Deccan Trap
 - Tertiary Rock System

The Gondwana System

- ❑ **Name & Origin:** The Gondwana System, named after the **Gonds**, the indigenous **tribe** of Telangana and Andhra Pradesh, is a geological formation characterized by deposits that accumulated, is located within folded troughs on ancient elevated plateau regions.
- ❑ **Formation & Evolution:** Over time, as these sediments built up, the troughs gradually sank, creating spaces where freshwater and more sediment gathered,

providing a fertile environment for terrestrial plants and animals to thrive. This process has been ongoing since the **Permian period**, dating back approximately **250 million years**.

- ❑ **Significance:** Notably, the Gondwana **rocks** are of significance due to their association with India's coal reserves, accounting for close to 98% of the nation's coal resources. However, Gondwana coal is comparatively more recent than Carboniferous coal, resulting in lower carbon content.

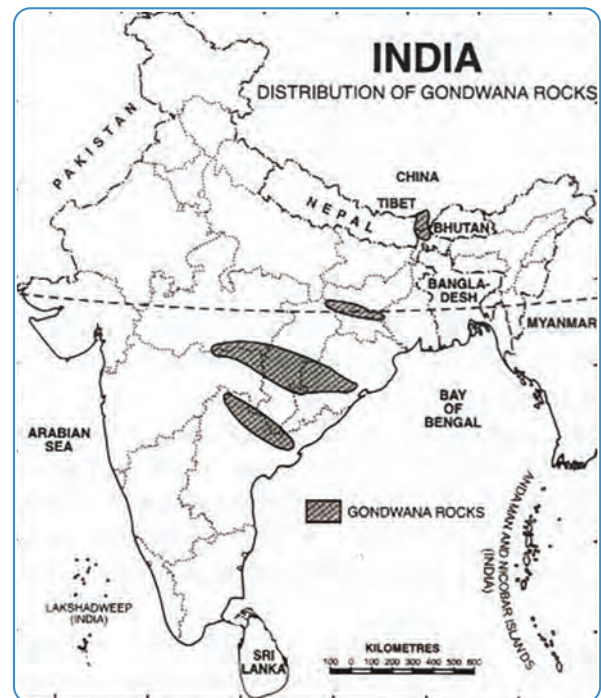


Fig. 2.7: India: Distribution of Gondwana Rocks

- ❑ **Minerals:** These rock formations also contain valuable **deposits of iron ore, copper, uranium**, and antimony. Additionally, sandstones, slates, and conglomerates found within the Gondwana system serve as essential building materials.
- ❑ **Geographical Distribution:**
 - Peninsular areas along Damodar Valley in Jharkhand
 - Mahanadi river valley in Chhattisgarh and Odisha
 - South part of Madhya Pradesh
- ❑ **Gondwana Coal:**
 - Gondwana rocks house nearly 98 per cent of India's coal reserves.

- Gondwana coal, being younger than Carboniferous coal, has lower carbon content.
- Rich deposits of iron ore, copper, uranium, and antimony are found within Gondwana rocks.
- Sandstones, slates, and conglomerates from the Gondwana system serve as essential building materials.

The Jurassic System

- During the later part of the Jurassic period, there was a marine advance that resulted in the creation of extensive **layers of shallow water deposits** in regions like Rajasthan(Jaisalmer) and Kachchh.
- In **Kachchh**, these deposits include materials such as coral limestone, oolitic limestone, sandstone, conglomerates, and shales. Similarly, another marine advance occurred on the eastern coast of the Peninsula, specifically in between Guntur and Rajahmundry.

The Deccan Trap

- **Origin:** A massive volcanic activity that occurred starting after the conclusion of the Cretaceous period to the dawn of the **Eocene period** resulted in the development of the Deccan Traps in Peninsular India.
- **Volcanic Eruption:** This volcanic process involved the eruption of **basaltic lava** through cracks in the Earth's surface, spanning an extensive region of approximately one million square kilometres. These volcanic deposits have distinct features, with flat tops and steep sides, which is why they are referred to as **traps**, a term derived from the Swedish word for 'stair' or 'step.'
- **Geographical Extent:** Over millions of years, **weathering** and erosion processes have significantly reduced the original size of the Deccan Traps. Presently, these

traps encompass roughly 500,000 square kilometres, predominantly in areas such as Kuchchh, Saurashtra, Maharashtra, the Malwa plateau, and northern Karnataka.

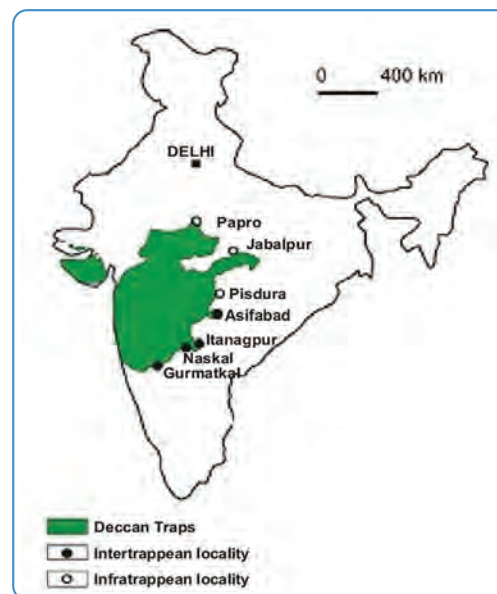


Fig. 2.8: Deccan Traps

- **Thickness Variation:** The thickness of the **Deccan Traps** exhibits variations, ranging from approximately 3,000 meters in the western region to a decrease to 600-800 metres in the south, 800 metres in Kachchh, and a mere 150 metres at the eastern boundary.
- **Soil Formation:** Extensive weathering of these rocks has given rise to the formation of black cotton soil, also referred to as **Regur**.
- The Deccan Trap has been divided into three groups- **Upper, Middle & Lower Trap**.

Group	Found in	Thickness	Age	Inter-trappean beds	Layers of volcanic ash
Upper Trap	Maharashtra and Saurashtra	Thick	Younger	Present	Present
Middle Trap	Central India and Malwa	Moderate	Intermediate	Very rare to absent	Present
Lower Trap	Madhya Pradesh	Thin	Older	Present	Very rare to absent

Note: In basaltic volcanism, which includes the Deccan traps, Siberian shield, and Laurentian shield, sediments settle on the cooled and solidified basaltic layer. Subsequently, these sediments are covered by additional basaltic volcanism, and once again, some sediments settle over this layer. The layers of sediments sandwiched between basaltic flows are referred to as **inter-trappean beds**.



IGNITE YOUR MIND

Examine the geological processes responsible for the formation of the Rann of Kutch and How has the geological history of the Rann of Kutch influenced its unique ecosystem?

THE TERTIARY SYSTEM

- **Overview:** The Tertiary era, which spanned roughly Eocene to Paleocene (66 to 7 million years ago), played a pivotal role in India's geological evolution with two major occurrences. Initially, it marked the conclusive breakup of the ancient Gondwana supercontinent. It is termed the "Age of the Mammals" due to abundant fossil remains.
 - **Eocene System (60 million years ago)**
 - **Distribution:** Jammu and Kashmir, Himachal Pradesh, Rajasthan, Gujarat, and North-East India
 - **Prominent Deposits:** Jaintia series, Barail series, Upper Assam

- ❑ **Notable Locations:** Kachchh, Surat, Broach, Mayurbhanj, Durgapur, Kollam
- ❑ **Recent Discoveries:** Puducherry
- ❑ **Tertiary areas of the extra-Peninsula**
 - ❖ **Outcrop:** Murree series in Jammu hills
 - ❖ **Geological Features:** Dagshai-Kasauli band in Shimla area
 - ❖ **Presence in:** Assam, Gujarat, Odisha, West Bengal, Kerala
- **Oligocene and Lower Miocene System**
 - ❑ **Development:** Poorly developed in India (40-25 million years)
 - ❑ **Rocks:** Oligocene rocks in Barail series of Assam; Lower Miocene rocks in various regions

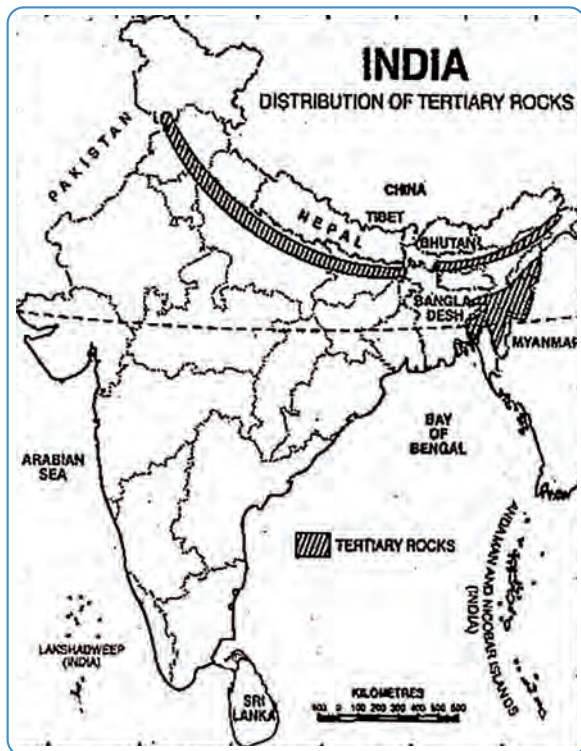


Fig. 2.9: India: Distribution of Tertiary Rocks

- **Shiwalik System (Middle Miocene to Lower Pleistocene)**
 - ❑ **Named After:** Shiwalik hills between Ganga and Yamuna rivers
 - ❑ **Composition:** Sandstones, grits, conglomerates, clays, and silts
 - ❑ **Thickness:** Extends over a width of 10-50 Km
 - ❑ **Fossiliferous Formations:** Notable discoveries near Haridwar in 1839
- ❑ **Formation of Himalayas:** It led to the formation of the Himalayan mountain range, driven by the upliftment of the **Tethys geosynclines**. The Tertiary period holds immense geological importance for India as during this period the Himalayas were formed, and the country's present geography was shaped.
- ❑ **Rock Formations:** Throughout the Tertiary period, geological changes led to the creation of a **series of**

rock formations that extends across various regions, including the Bengal and Ganges Delta, the East Coast, and the Andaman Islands.

- ❑ **Geographical Distribution:** Similar formations can be observed in locations such as the **Salt Range, Potwar Plateau**, the outer Himalayan regions of Jammu and Punjab, Assam, Sind, and Baluchistan. Prominent rock systems from this period encompass the Karewas in Kashmir and the Bhangra and Khadar formations in the Gangetic plains.



IGNITE YOUR MIND

You might know that Meghalaya boasts some of the world's longest and deepest limestone caves, formed through limestone dissolution over millennia, displaying breathtaking stalactite and stalagmite formations. Now can you discuss the unique features of the Meghalaya limestone caves and their significance in the field of speleology. How have these caves contributed to our understanding of geological processes?

THE QUATERNARY SYSTEM (THE PLEISTOCENE AND RECENT FORMATIONS)

The **Quaternary Period** is the most recent period in the geological time scale, spanning approximately the last **2.6 million years to the present**. It is characterized by significant climatic changes, including the onset of glaciations and the presence of Homo Sapiens. The Quaternary is divided into **two epochs: the Pleistocene and the Holocene**.

Pleistocene Epoch (2.6 million to 11,700 years ago)

- ❑ **Ice Ages:** Alternating large ice sheet advancements and retreats.
- ❑ **Hominid Evolution:** Evolution of **Homo Sapiens**.
- ❑ **Cultural Development:** Significant for human culture and diverse faunal/floral assemblages.
- ❑ **Five Major Glacial Events:** Donau, Gunz, Mindel, Riss, Wurm, leaving geological traces.
- ❑ **Holocene Features:** Visible features in various Indian regions, especially along the **Brahmaputra River**.
- ❑ **Alluvial Deposits:** Present in deltaic regions of major rivers like **Cauvery, Krishna, Mahanadi**, and **Ganges**.

Holocene Epoch (11,700 years ago to present)

- ❑ **Stable Climate:** Characterized by relatively stable climatic conditions.

- ❑ **Human Civilization:** Development of civilizations, agriculture, and complex societies.
- ❑ **Recorded History:** Encompasses the entire span of recorded human history.
- ❑ **Present Interglacial Period:** Represents the current interglacial period..

Quaternary System in India

- ❑ The Quaternary period is named after relatively recent deposits that contain fossils of species still in existence today. This period's commencement dates back to the **Pleistocene** epoch, approximately 1 million years ago, and extends into the **Holocene** or recent past.

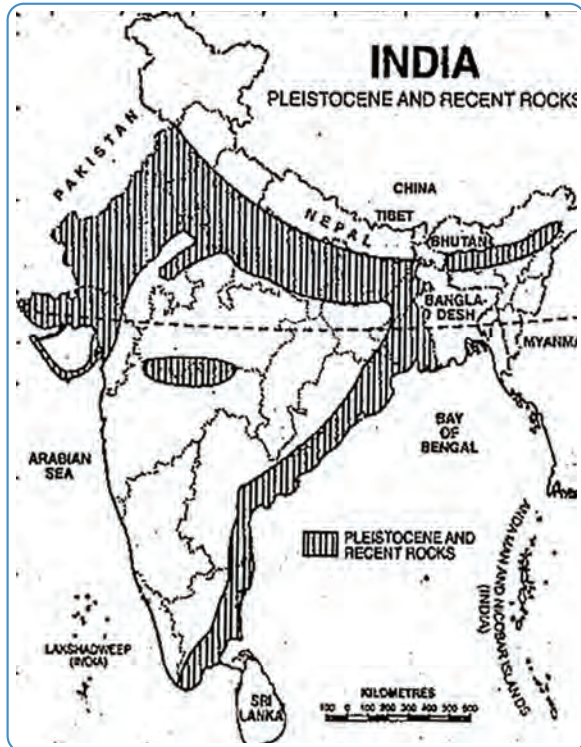


Fig. 2.10: India: Pleistocene and Recent Rocks

- ❑ The Quaternary System encompasses rocks and deposits that can be found in regions like the Indus and Ganga plains. The Quaternary era is divided into two key periods: the **Pleistocene** and the **Holocene**.
- ❑ **Alluvial deposits:**
 - It is characterized by the presence of alluvial deposits in the Indo-Gangetic plain, encompassing the recent alluvial sediments of Northern India's Indo-Gangetic plains.
 - **These sediments are divided into two categories:**
 - ❑ **Bhangar** is composed of old alluvium (located above the flood levels of rivers) of the Upper and Middle Pleistocene age.
 - ❑ **Khadar** is composed of the newer alluvium mainly found within river channels and floodplains, in the **Holocene epoch**. Due to the annual deposition of new silt by the rivers, this region boasts some of the most fertile soil in the country.
- ❑ Originally, this area was a lake, but over time, it evolved into its current shape, known as Karewa due to the continuous deposition of materials. Evidence of **Pleistocene** deposition has been uncovered in unexpected places, including the Thar Desert.
- ❑ Notably, what we now know as the '**Rann of Kachchh**' was once a segment of the ocean during this period. Over the Pleistocene and Holocene epochs, sedimentary materials filled this area, leading to its transformation.
- ❑ **Glacial Activity:**
 - **Pleistocene Impact:** Glaciations shaped northern India with moraines and U-shaped valleys.
 - **Example:** Gangotri Glacier in Uttarakhand shows retreating glaciers, leaving distinct glacial landforms.
- ❑ **Sedimentation in River Valleys:**
 - **Quaternary Period:** Significant sedimentation in river valleys.
 - **Example:** The Indo-Gangetic Plain, formed by the Ganges and Brahmaputra, crucial for agriculture.
- ❑ **Holocene Environmental Changes:**
 - **Holocene Epoch:** Witnessed shifts in vegetation and diverse ecosystems.
 - **Example:** Narmada River Valley showcases vegetation changes and ancient human settlements.

COASTLINE OF INDIA

The coastline of India is a dynamic and diverse region that reflects the geological history and tectonic evolution of the Indian subcontinent. The geological processes have played a significant role in shaping the coastal features, including beaches, estuaries, deltas, and rocky cliffs.

- ❑ India's coastline spans **7,516.6 kilometers (4,671 miles)**, comprising 5,422.6 kilometers of mainland and 2,094 kilometers of island territories..
- ❑ Touching nine states and two union territories, including **Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Odisha, West Bengal, Daman and Diu, and Puducherry**, it passes through a total of 13 states and union territories.
- ❑ The coastline is bordered by the **Bay of Bengal to the east, the Indian Ocean to the south, and the Arabian Sea to the west.**
- ❑ **Western Coast:** The western coast of India, along the Arabian Sea, is characterized by long stretches of sandy beaches, rocky cliffs, and estuaries.
 - Runs straight from Gujarat plains to Kanyakumari.
 - Dotted with coves, creeks, and estuaries (Narmada and Tapi).
 - Malabar coast features lakes, lagoons, and the largest Vembanad Lake.
 - **For Example: The Konkan Coast** in Maharashtra features a mix of rocky cliffs and sandy beaches, influenced by the Western Ghats.

- The elevation and geological structure of the Western Ghats have contributed to the formation of this coastal landscape.

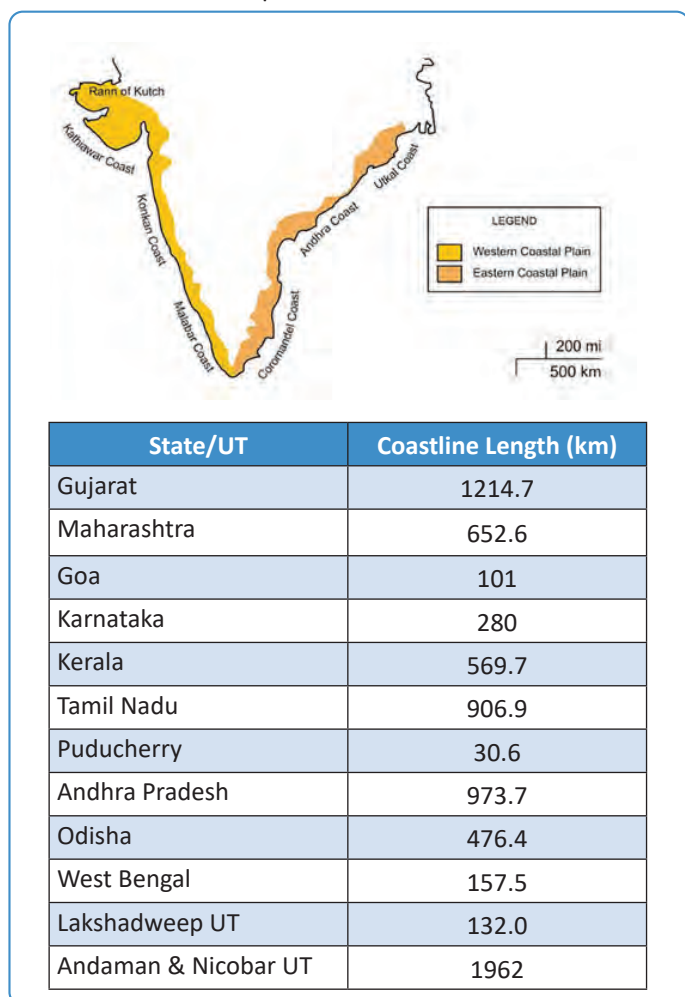


Fig. 2.11: Coastline of India

- The **Western Ghats**, a mountain range parallel to the western coast, has influenced the coastline's formation. The Ghats were formed by tectonic activities, including the rifting of the ancient supercontinent **Gondwana**.
- **Uplift of Western Coast:**
 - Geologists find evidence of uplift in the western coast.
 - Raised beaches and marine shells found at altitudes of 30-45 meters.
 - Steep face of the Sahyadri parallel to the west coast suggests recent elevation from the sea.
 - Coastal plain of Kerala shows multiple phases of uplift.
- **Eastern Coast:** The eastern coast of India, along the Bay of Bengal, is characterized by extensive deltaic regions, mangrove forests, and sandy beaches.
 - Extends from the Ganga delta to Kanniyakumari, facing the Bay of Bengal.

- **Deltas of major rivers:** Mahanadi, Godavari, Krishna, and Cauvery.
- **Notable lagoons:** Chilika lake and Pulicat lagoon.
- **Example: The Sundarbans Delta** in West Bengal is one of the largest deltaic regions in the world. It is formed by the sediment deposition of the Ganges, Brahmaputra, and Meghna rivers.
- **The Eastern Ghats**, another mountain range parallel to the eastern coast, has also influenced the geological features of the coastline.
- Tectonic activities, including the collision between the Indian Plate and the Eurasian Plate, contributed to the formation of the Eastern Ghats.

□ Geological Changes:

- Submergence and emergence observed in Indian coast during geological and historical times.
- Examples include a submerged forest near Mumbai and raised terraces formed by coral reefs.
- Evidence of both submergence and emergence found in the Rann of Kachchh.
- Historical changes in sea level seen in coastal dunes, sea cliffs, sea caves, and raised beaches.

□ Ancient Tectonic Processes:

- The Indian subcontinent's geological history involves ancient tectonic processes, including the breakup of the supercontinent Rodinia, which influenced the configuration of coastlines.
- **For Example: The Narmada River** flows through a rift valley between the Vindhya and Satpura mountain ranges.
- This rift valley is a geological feature formed due to ancient tectonic processes associated with the **rifting of Rodinia**.
- The drifting of the Indian Plate from its position within Gondwana and its subsequent collision with the Eurasian Plate led to the uplift of the Himalayas and the Tibetan Plateau, impacting the hydrography of the region.

□ Sedimentary Processes:

- Sedimentary processes, such as the deposition of river sediments and the formation of deltas, have played a crucial role in shaping the coastal plains.
- **For Example: The Godavari Delta** in Andhra Pradesh is another prominent example of sedimentary processes shaping the coastline.
- Major rivers like the Ganges, Brahmaputra, and Godavari contribute to the deposition of sediments along the eastern and western coasts.

□ Sea Level Changes:

- Fluctuations in sea levels, associated with glacial-interglacial cycles, have influenced the coastal features over geological time.

- Sea level changes have led to the creation of submerged landforms and influenced the extent of coastal erosion and deposition.
- **For Example:** The submerged landforms off the coast of Gujarat, known as the **Gulf of Khambhat**, exhibit evidence of ancient river valleys and archaeological sites. This underwater landscape provides insights into sea level changes and shifts in coastal configurations over time.

ICE AGES IN INDIA

India has experienced multiple ice ages throughout its geological history. The most recent and well-known ice age is the Quaternary glaciation, which began around 2.58 million years ago and continues to the present day. Within the Quaternary, there were several glacial and interglacial cycles, marked by the advance and retreat of ice sheets and glaciers.

❑ Dharwar Ice Age:

- The Dharwar Ice Age is associated with the Archean Eon, which occurred approximately 4.0 to 2.5 billion years ago.
- During this ancient ice age, the Dharwar Craton, a stable geological region in southern India, experienced glacial activity. Glacial deposits, such as tillites (sedimentary rocks formed from glacial debris), are found in the Dharwar Supergroup rocks in **Kaldurg** Karnataka.
- The presence of these glacial deposits indicates that ice sheets or glaciers covered parts of the Dharwar Craton during this period, making it one of the earliest known instances of glaciation in Earth's history.

❑ Gondwana Ice Age:

- The Gondwana Ice Age, also known as the Late Paleozoic Ice Age, occurred during the Carboniferous and Permian periods, roughly 360 to 260 million years ago.
- Glacial deposits, such as tillites and dropstones, are found in sedimentary rocks of the Gondwana Group, indicating widespread glaciation across Gondwana.
- The **Talcher series** of the Gondwana system in **Odisha** provides proof of glaciation during this period.
- Boulders found here bear similarity to those existing in **Shimla, Hazara, and Salt Range**.

❑ Pleistocene Ice Age:

- The Pleistocene period witnessed the effects of an ice age in the Himalayas.
- This period was not a continuous frigid spell but featured alternating cold and warm spells.
- In the Himalayas, extensive glaciation evidence is found up to a height of 1800 m.
- Glacial drifts and terminal moraines covered hill sides and valley floors down to 1,400 m.

- In the Kashmir region, de Terra and Peterson identified four or five periods of glaciation with interglacial periods.
- Large blocks and boulders are commonly seen in different parts of the Himalayas.
- Rock polishing and grooving on the lower steps of the Himalayas, buff-colored sands, and illuminated clays inter-stratified among the Karewa deposits of Kashmir are observed.
- Glaciation led to the formation of several glacial lakes in different parts of the Himalayas, including Kailash Kund, the Sanasar basin near Batoti, the Gulmarg basin, the Sheshnag, and the Kaunsarmag.
- The impact of glaciation also resulted in a sudden and large-scale reduction of Shiwalik mammals.
- The Peninsular part of India has no evidence of glaciation during the Pleistocene period.

❑ Early Pleistocene (2.58 - 0.78 million years ago):

- During this time, there were multiple glacial advances, with ice sheets covering significant portions of the Northern Hemisphere.
- Evidence of glacial deposits, such as moraines and erratic boulders, suggests that high-altitude regions of the Himalayas may have been glaciated.

❑ Middle Pleistocene (0.78 - 0.12 million years ago):

- Glacial-interglacial cycles continued, influencing the climate and landscape of India.
- Evidence of glacial activity is found in regions like Kashmir, where moraines and glacial deposits indicate past ice advances.

❑ Late Pleistocene (0.12 million years ago - 11,700 years ago):

- This period is characterized by more frequent and intense glacial-interglacial cycles.
- The Last Glacial Maximum (LGM), occurring around 20,000 years ago, was a phase of maximum ice sheet expansion globally. Glaciers in the Himalayas reached their maximum extent during this time.

❑ Holocene (11,700 years ago - present):

- The Holocene marks the current epoch, characterized by warmer temperatures and the retreat of ice sheets.
- Glacial retreat in the Himalayas and other high-altitude regions led to the formation of glacial lakes and modification of river courses.

Impact of ice ages on Indian Landscape

- ❑ Glacial activity during ice ages influenced the formation of high-altitude features such as cirques, aretes, and U-shaped valleys in the Himalayan region.
- ❑ Glacial meltwater played a crucial role in shaping river valleys and contributing to the formation of alluvial plains.

{While the Quaternary glaciation had a significant impact on India's landscape, the country did not experience ice sheets or glaciers comparable to those in higher latitudes}

GEODIVERSITY AND GEOTOURISM

Important Terms and their Definitions

□ Geodiversity

- **Definition:** Geodiversity refers to the variety of the geological and physical elements of nature, such as minerals, rocks, soils, fossils and landforms, and active geological and geomorphological processes. Together with biodiversity, geodiversity constitutes the natural diversity of planet Earth.
- **Example:** The Grand Canyon in the United States epitomizes immense geodiversity, revealing layered rock formations that unveil geological history spanning millions of years.

□ Geotourism

- **Definition:** Geotourism is a sustainable tourism form emphasizing the appreciation of a destination's geological heritage, cultural aspects, and local well-being.
- **Example:** The Ring of Fire in the Pacific, notably Yellowstone National Park, magnetizes geotourists with its volcanic landscapes and distinctive geological features.

□ Geoheritage

- **Definition:** Geoheritage involves conserving sites with significant geological features, ensuring their protection for scientific, educational, and cultural purposes.
- **Example:** Giant's Causeway in Northern Ireland, famed for hexagonal basalt columns, is a UNESCO World Heritage Site, showcasing exemplary geoheritage.

□ Geoconservation

- **Definition:** Geoconservation focuses on identifying, managing, and protecting geological sites to maintain scientific value and prevent degradation.
- **Example:** The Jurassic Coast in England, with diverse geological formations, undergoes geoconservation efforts to safeguard its unique features.

□ Geopark

- **Definition:** A geopark is a designated area conserving and enhancing geological heritage, promoting sustainable development, and offering educational opportunities.
- **Example:** Zhangjiajie Geopark in China, renowned for towering sandstone pillars, is a UNESCO Global Geopark, attracting visitors with its unique geological wonders.

□ Geosites

- **Definition:** Geosites are specific locations that hold significant geological importance, providing insights into Earth's history and processes.
- **Example:** Joggins Fossil Cliffs in Nova Scotia, Canada, is a geosite showcasing well-preserved fossils from the Carboniferous period.

□ Geotrail

- **Definition:** A geotrail is a designated route allowing visitors to explore and learn about the geological features and history of a particular area.
- **Example:** Marble Arch Caves Geopark in Ireland offers geotrails, guiding visitors through limestone landscapes and subterranean wonders.

□ Geoconservation:

It involves the identification and preservation of sites that help us learn about the Earth's formation processes. Geoparks promote both geodiversity preservation and public education through guided tours of sites.

□ Understanding the unique nature and culture of islands,

including oceanic aspects, is crucial in geodiversity education. **Gwanmae Island**, Korea's first luxury village recognized by the National Park Service, highlights the **potential** for geotourism based on the island's remarkable geodiversity.

□ Geoheritage:

It focuses on preserving natural features like landforms, exposed rocks, and sites that allow the study and conservation of these features. A geosite is a specific location that forms a component of the geoheritage. **UNESCO's Global Geoparks Network** describes a 'geopark' as an area that includes one or more scientifically significant sites.

□ Geotourism Initiatives in India:

- **National Geotourism Policy (2020):** Aims to identify, develop, and promote geotourism destinations across India.
- **National Geological Monuments:** 34 geological sites declared National Geological Monuments to preserve and promote their geological heritage.
- **Geotourism Certification Program:** Launched by the Ministry of Tourism, certifying geotourism service providers to ensure responsible practices.

National Geological Heritage Monument Sites

National Geological Heritage Sites are geographical areas of national importance and heritage, as notified by the Government of India's **Geological Survey of India (GSI)**, for their maintenance, protection, promotion and enhancement of **geotourism**. There are **34** geoheritage sites in India. These sites showcase various geological phenomena, rock formations, and fossils, providing insights into the Earth's history.

CONCLUSION

India's geological history is an intricate narrative of colliding plates, uplifting mountains, eroding plains, and shaping coastlines. The geological features not only contribute to the country's natural beauty but also play a crucial role in influencing its climate, biodiversity, and socio-economic activities. Understanding and appreciating this geological diversity is essential for managing natural resources, mitigating geological hazards, and fostering sustainable development in this vast and varied landscape.

Physiography of India

3

INTRODUCTION

India's landscape exhibits significant physical diversity. The physiography of an area is the outcome of structure, process and the stage of development. The land of India exhibits great physical variations.

It can be divided into distinct physiographic divisions: the Himalayan Mountains, the Northern Plain, the Peninsular Plateau, the Indian Desert, the Coastal Plains, and the Islands.

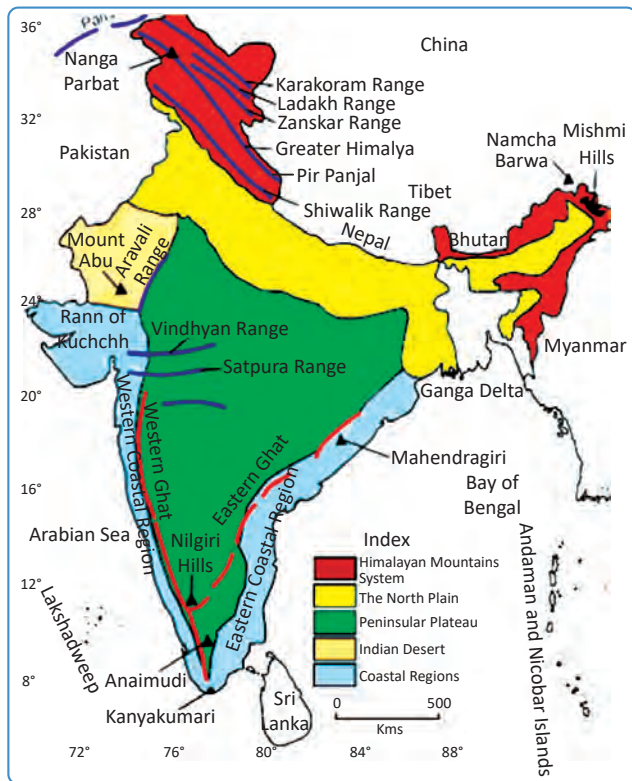


Fig. 3.1: Physiographic divisions of India

In terms of geology, the **Himalayan mountains** exhibit a relatively recent and dynamic topography characterized by towering peaks, deep valleys, and swiftly flowing rivers. This region, along with the Northern Plains, represents some of the most recently formed landforms. In contrast, the **Peninsular Plateau** is an ancient and stable landmass, constituting one of the oldest and most enduring blocks on the Earth's surface. The Northern Plains consist of alluvial deposits, while the Peninsular Plateau is composed of igneous and metamorphic rocks, featuring gently rising hills and expansive valleys.

Additionally, during the Jurassic Period, Gondwanaland underwent further fragmentation, leading to the separation of peninsular India, Madagascar, Australia, Antarctica, and other landmasses. The peninsular part of India, being a component of Gondwanaland, originated from igneous rocks of the Archean era, which underwent transformation into Gneiss and Schist over time.

THE HIMALAYAS

The Himalayan mountain range, a geologically young formation and structurally-fold mountain ranges, that came into existence approximately **50 million years ago**, stretches along India's northern border, running from the Indus River to the Brahmaputra River.

- ❑ These mountains are among the world's tallest and most rugged, forming an arc that spans approximately 2,400 kilometres.
- ❑ Their width varies, being about 400 kilometres in Kashmir and 150 kilometres in Arunachal Pradesh.
- ❑ Altitude changes are more significant in the eastern half of the Himalayas compared to the western half.
- ❑ The **Pamir Knot** connects the Himalayas with the high ranges of central Asia.
- ❑ The Indian Himalayan Region (IHR) is the section of the Himalayas within India, spanning the 13 Indian States/ Union Territories (namely Jammu and Kashmir, Ladakh, Uttarakhand, Himachal Pradesh, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura, Assam and West Bengal).

Major Ranges of the Himalayas

- ❑ The Himalayan range consists of three parallel longitudinal ranges, with valleys lying between them.
 - **Greater Himalayas (Himadri):** The northernmost range is called the Great Himalayas or the 'Himadri.' It is the most continuous range, with the highest peaks averaging 6,000 meters in elevation. This range contains many prominent Himalayan peaks such as **Mount Everest** and **Kanchenjunga**. These peaks are characterized by their **perpetual snow cover and glaciated terrain**. The Greater Himalayas form the **international border between India and Tibet (China)**.

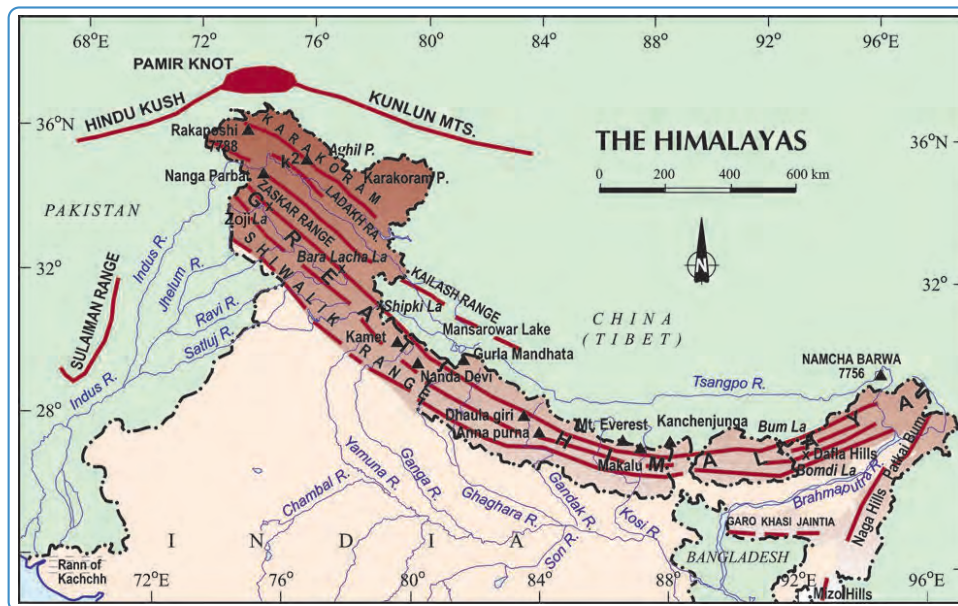


Fig. 3.2: Major Ranges of Himalaya

- **Lesser or Middle Himalayas (Himachal):** South of the Greater Himalayas lies the “Himachal” or “Middle Himalayas.” This range is **lower in elevation** compared to the Greater Himalayas and is known for its diverse landscapes, including **deep valleys, alpine meadows, and dense forests**. Some well-known hill stations like **Shimla** and **Manali** are located in the Lesser Himalayas.
- **Outer Himalayas (Shiwaliks):** The southernmost range is called the “Shiwalik” or “Outer Himalayas.” This range is characterized by relatively lower elevations and consists of **foothills**, which **gradually merge into the Indo-Gangetic Plains**. The Shivalik Hills are known for their picturesque landscapes and are rich in biodiversity. They act as a **transition zone** between the **Himalayas** and the **Indo-Gangetic Plains**.

Himalayan tectonic framework, and it plays a crucial role in the deformation and uplift of the Himalayan range.

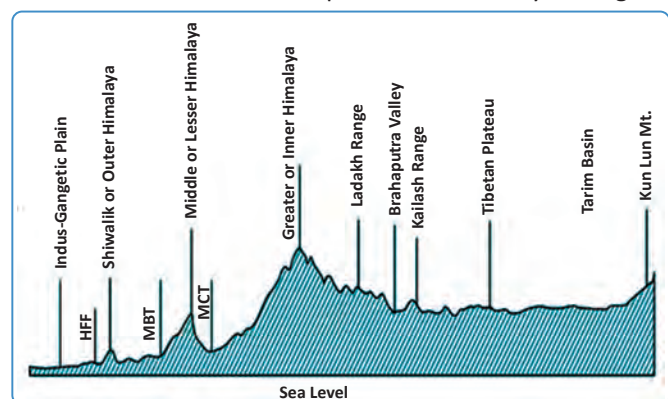


Fig. 3.3: Himalayan Complex

Major Fault Lines in Himalayan Complex

Himalayan Frontal Fault (HFF)

- The Himalayan Frontal Fault is a major fault line that runs along the southern base of the Himalayan mountain range.
- It marks the boundary between the **Indian plate and the Eurasian tectonic plate**. The collision between these two plates is responsible for the upliftment of the Himalayas and the associated seismic activity in the region.

Main Boundary Thrust (MBT)

- The Main Boundary Thrust is a major thrust fault associated with the Himalayan mountain-building process.
- It represents the contact between the Indian plate to the south and the sedimentary rocks of the outer Himalayan range to the north. MBT is a significant structure in

Main Central Thrust (MCT)

- The Main Central Thrust is another important thrust fault in the Himalayan region.
- It is situated **north of the Main Boundary Thrust** and is responsible for the **overthrusting** of the **crystalline rocks of the Greater Himalayas** over the **sedimentary rocks of the Lesser Himalayas**. The MCT marks a major tectonic boundary within the Himalayas and is associated with significant **crustal shortening**.

Origin and Formation of the Himalayas

The origin of the Himalayas has been a subject of debate among geologists and geomorphologists. This mountain system is complex and composed of rocks from different geological periods, including the Precambrian and Eocene periods. It primarily consists of **sedimentary and metamorphic rocks** and has undergone significant **folding and faulting**.

The main **theories** regarding the origin of the Himalayas are as follows:

The Geosynclinal Origin

According to this theory, around 200 million years ago, during the disintegration of **Pangaea**, a vast **Tethys** Sea formed between the Laurentian Shield (Angaraland) in the north and Gondwanaland in the south. This sea occupied the Himalayan region during the Mesozoic Era (180 mya). Sediments eroded from the Eurasian Shield (Angaraland) and **Gondwanaland** were deposited in the Tethys Sea, accumulating considerable thickness due to the sinking of the sea bed.

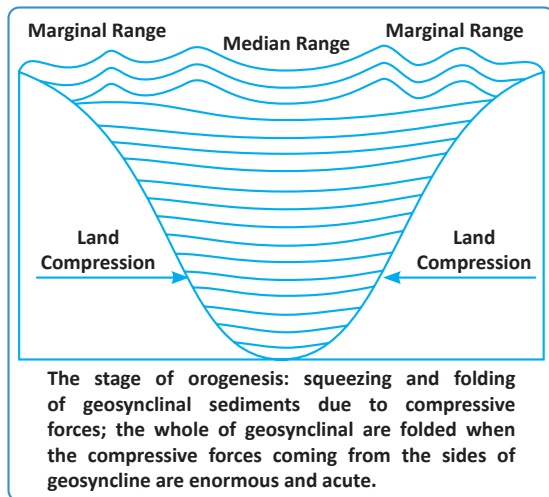


Fig. 3.4: Geosynclinal Origin of Himalayas

- ❑ During the Cretaceous Period, the sea bed began to rise, resulting in the **folding of three successive ranges** of the Himalayas.
- ❑ The first uplift formed the Greater Himalayas during the Eocene Period (about 65 mya), followed by the formation of the Lesser Himalaya during the Miocene Period (about 45 mya), and the Shiwaliks or Outer Himalaya during the Pliocene period (about 1.4 mya).

The Plate Tectonic Origin of the Himalayas

Proposed by **W.J. Morgan** in 1967, theory is based on the concept of **"Sea-Floor Spreading"** by H.H. Hess.

- ❑ Approximately 65 to 30 million years ago, the Indian plate approached the Asian plate, leading to the subduction of the Indian plate beneath the Asian plate.
- ❑ This subduction caused lateral compression, resulting in the squeezing and folding of sediments in the Tethys, forming the three parallel ranges of the Himalayas.
- ❑ The convergence of these plates has caused a crustal shortening of about 500 kilometres in the Himalayan region.

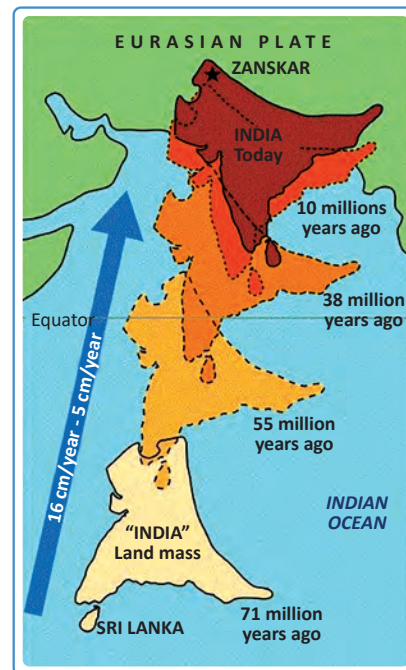


Fig. 3.5: Movement of the Indian Plate

The **northward movement** of the Indian Plate is ongoing, causing the Himalayan peaks to continue rising. The Indian Plate is still moving northward, constantly changing the centre of rotation.

- ❑ Around 50-40 million years ago, the collision between the Indian and Asiatic plates initiated the rise of the Himalayas from the Tethys geosyncline.
- ❑ The first significant uplift led to the formation of the Ladakh and Zaskar ranges in the Trans-Himalaya before the formation of the Great Himalaya.
- ❑ The Main Central Thrust separates the **Greater Himalayas** and the Lesser Himalayas. Around 35 million years ago, the second major uplift occurred, rapidly elevating the southern mountain front of the Lesser Himalaya and forming the Pir-Panjal, Dhauladhar, Karol, and Mahabharat Ranges.
- ❑ These ranges created **intermontane** basins, including **Kashmir, Karol-basin, Dun Valley (Uttarakhand), and the Kathmandu Valley in Nepal.**
- ❑ The **Shiwaliks** rose around 1.4 million years ago as a result of the third major uplift. These sedimentary rocks reflect the Himalayas' ongoing uplift. The Shiwaliks are separated from India's Northern Plains by the active **Himalayan Front Fault (HFF)**, which is characterized by frequent tremors and earthquakes.

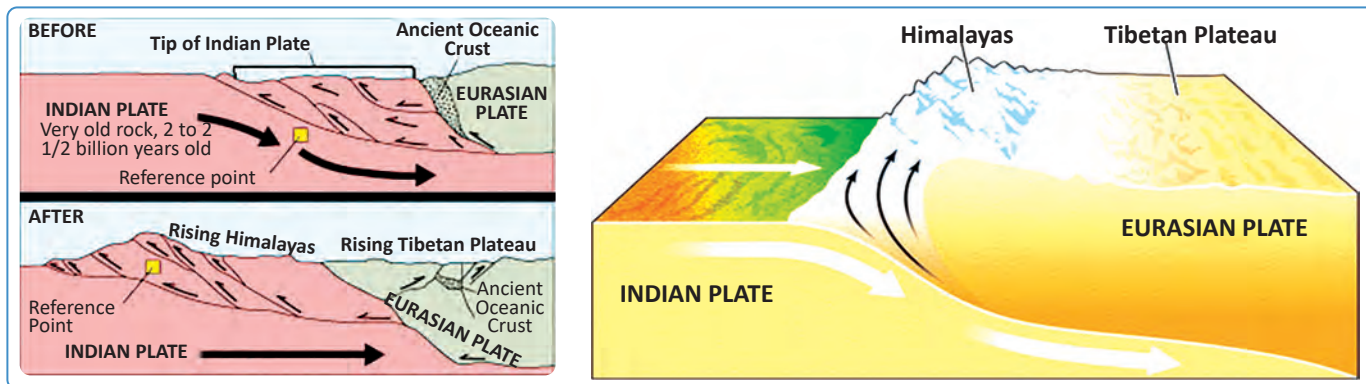


Fig. 3.6: Formation of Himalayas



IGNITE YOUR MIND

What are the key indicators of glacial retreat, and how does this phenomenon affect the local and global environment?

Division of Himalayas: Latitudinal and Longitudinal

The Himalayas, a remarkable mountain system in Asia, stand as a formidable barrier between the Tibetan Plateau to the north and the fertile plains of the Indian subcontinent

to the south. Known for housing the world's tallest peaks, such as Mount Everest and Kanchenjunga, the Himalayas are typically divided based on both latitude and longitude.

Longitudinal Division of the Himalayas

The longitudinal division of the Himalayas, running from **west to east**, categorizes this vast mountain range into four significant segments:

- ❑ The Kashmir/Punjab/Himachal Himalayas
- ❑ The Kumaun Himalayas
- ❑ The Central/Nepal Himalayas
- ❑ The Assam/Eastern Himalayas

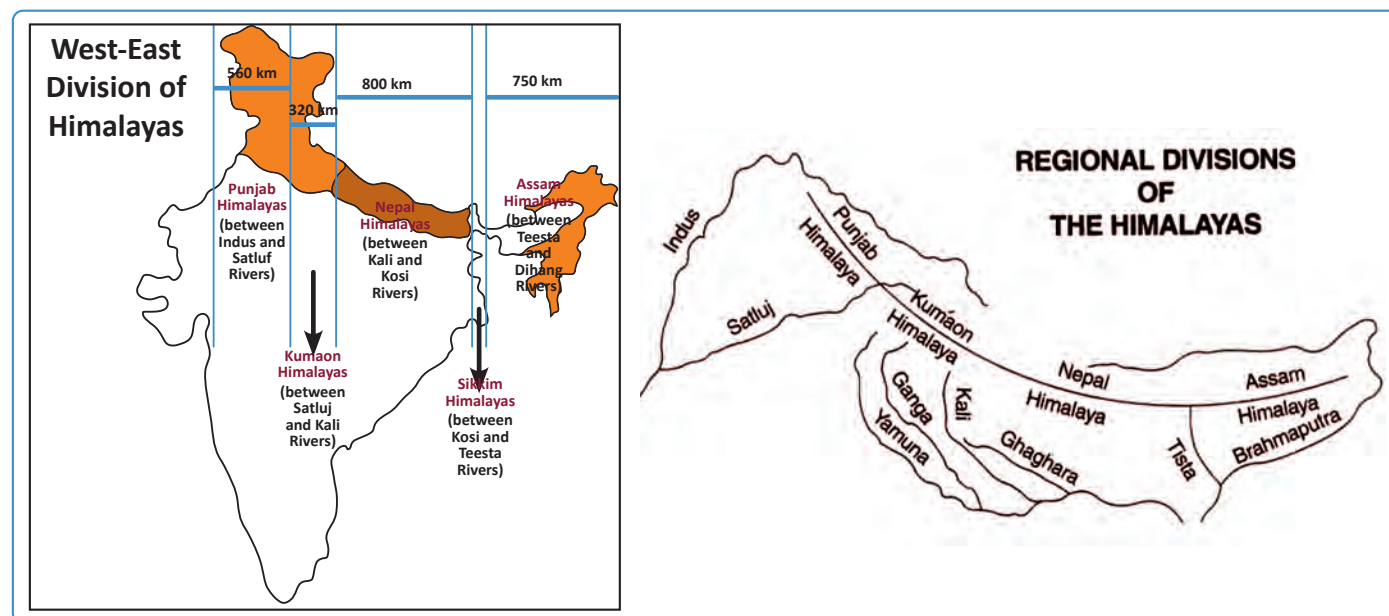


Fig. 3.7: Division of Himalayas

The Kashmir /Punjab/ Himachal Himalayas

- ❑ The Kashmir Himalayas are located between the **Indus and Satluj gorges**. Spans approximately 560 kilometers in length and 320 kilometers in width.
- ❑ They are bound by the **ZASKAR** range to the north and the Shiwaliks to the south.

- ❑ They are characterized by a ridge and valley topography, shaped by the lacustrine deposits of the **Jhelum** River. Notable features include Ox-bow Lakes, Wular Lake, and Dal Lake.
- ❑ The **"Valley of Kashmir"** is known for the presence of Karewas (Lacustrine accumulations) that provide

essential nutrients for **Saffron cultivation**, particularly in the Pulwama to Pampore districts.

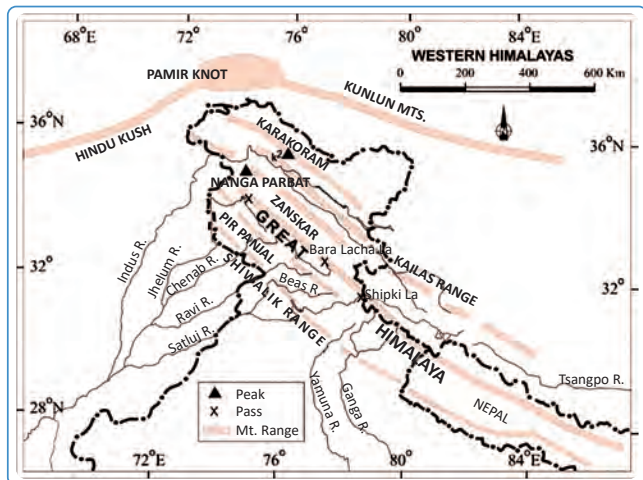


Fig. 3.8: The Kashmir/Punjab/Himachal Himalayas

The Kumaun Himalayas

- ❑ The Kumaun Himalayas extend over a distance of 320 kilometres, spanning from the **Satlu** to the **Kali Gorges**. This region is renowned for its prominent mountain ranges, including Nag Tibba, Mussoorie, and segments of the Greater Himalayas. Among its notable peaks are Nanda Devi, Kamet, Badrinath, and Kedarnath.
- ❑ The presence of numerous **glaciers** and ice caps, especially in Uttarakhand, contributes significantly to the formation of vital rivers such as the Gangotri, Yamunotri, and Pindari.
- ❑ The Kumaun Himalayas experience consistent winter snowfall, and they are characterized by **coniferous forests** that thrive at altitudes exceeding 3200 meters. Furthermore, Deodar forests are found in the elevation range of 1600 to 3200 meters.

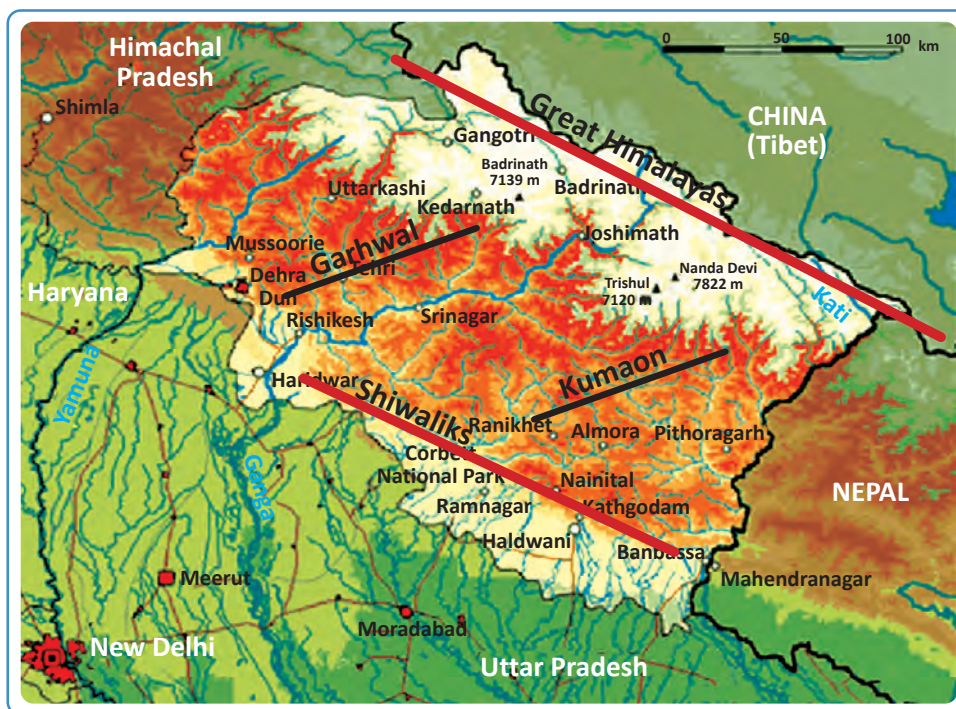


Fig. 3.9: Kumaun Himalaya

The Central/ Nepal Himalayas

- ❑ Spanning a length of 800 kilometres, the Central/Nepal Himalayas are situated between the **river Kali** in the west and the **Tista River** in the east.
- ❑ Notable characteristics of this region include world-renowned peaks, including Mt. Everest (Sagarmatha), Kanchenjunga, Makalu, Annapurna, Gosainthan, and Dhaulagiri.
- ❑ The presence of the Kathmandu valley. The Lesser Himalaya, widely referred to as the Mahabharat range, is within this area.

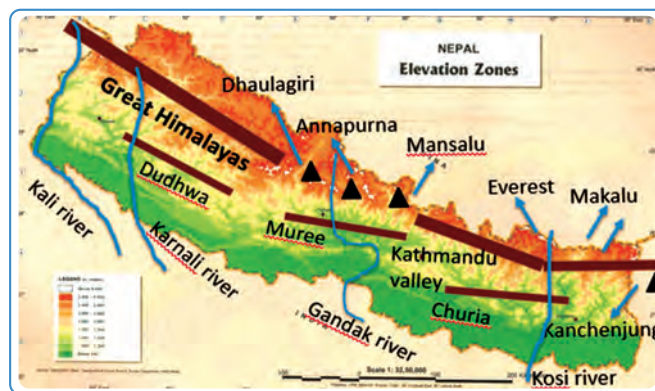


Fig. 3.10: Central/Nepal Himalaya

The Assam/ Eastern Himalayas

- Extending over a length of **750 kilometres**, the Assam/ Eastern Himalayas are situated between the **Tista and Brahmaputra rivers** in the west and east, respectively.
- The salient features of this region are predominant in Arunachal Pradesh and Bhutan among the Eastern Himalayas.
- The occasional merging of the Greater Himalayas and Shiwaliks can be seen at various locations.

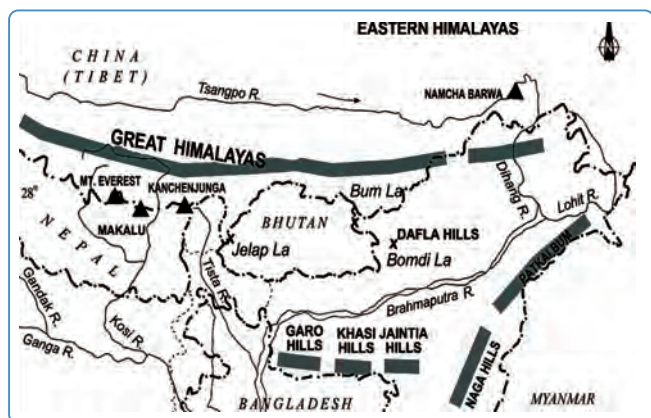


Fig. 3.11: Assam/Eastern Himalaya

- Constrained **longitudinal valleys** called **duars** and high rainfall, exceeding 200 centimeters. Fluvial erosion, primarily driven by heavy rainfall, characterizes the landscape of the Assam Himalayas.

Latitudinal Division of Himalayas

The Himalayas' latitudinal division, also known as the **horizontal division** of the Himalayas, runs from the north to the south of the mountain range. Notably, the northern side of the Great Himalayas is known as the Trans-Himalayas, and it is made up of several sub-ranges, including the Karakoram, Zaskar, Ladakh, and Kailash ranges.

Based on this latitudinal division, the Himalayas can be categorized into the following four segments:

Trans-Himalayan Range

- The Trans-Himalayas encompass several notable ranges, including Ladakh, Karakoram, Kailash, and Zaskar.
- The Karakoram Range marks the northernmost part of the Trans-Himalayas. Key features of this region include **Mount K2** which is also known as Godwin Austen, which stands as the world's second-highest peak, succeeding only Mount Everest. It is situated within the Karakoram Mountains.
- The **Shyok River** flows through the Karakoram and Ladakh regions. The Indus River courses through Ladakh and Zaskar. The Zaskar range is part of the Nanga Parbat region.

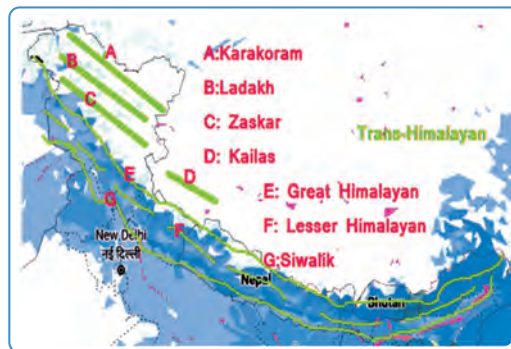


Fig. 3.12: Latitudinal Division of Himalaya

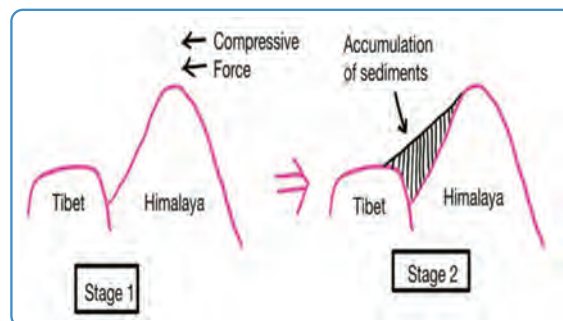


Fig. 3.13: Trans-Himalayan Hogback Structure

The Great Himalayas (Inner Himalayas or Himadri)

- The Great Himalayas, also known as the Inner **Himalayas or Himadri Himalayas**, are situated immediately south of the Trans-Himalayas.
- They are characterized by their average elevation of approximately 6100 meters above sea level and a width ranging about 25 km.
- These mountains are rich in **glaciers** and serve as the source of the Ganga and Yamuna rivers.
- Notably, the southern side of the Greater Himalayas features steeper terrain than the northern side. Within this range, prominent mountain passes include **Shipki-La, Nathu-La, and Zoji-La**, each holding significant strategic and historical value in the Himalayan region.

The Middle Himalayas (Himachal Himalayas or Lesser Himalayas)

- The Himachal Himalayas, often referred to as the **Lesser Himalayas** or Middle Himalayas, encompasses the southern side of the Great Himalayas.
- Key features of this region include, this range extends to a width of nearly 60-80 kilometres. Elevations within this range vary from 3700 to 4500 meters above sea level. Prominent lesser Himalayan **hill stations such as Nainital, Mussoorie, and Shimla** are located here. The renowned valleys of Kullu, Kashmir, and Kangra are also situated between the Great and Lesser Himalayas.

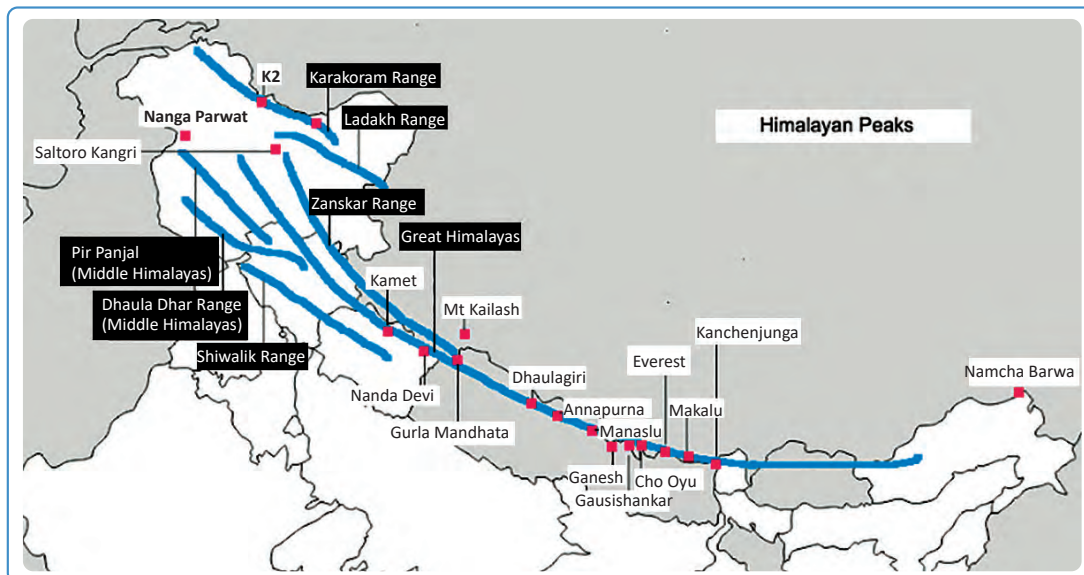


Fig. 3.14: Latitudinal Divisions of Himalaya

Shiwalik Himalayas (Outer Himalayas)

- ❑ The Shiwalik Himalayas, also known as the **Outer Himalayas**, constitute the southernmost mountain range of the Himalayas.
- ❑ Salient features of this region include, these mountains exhibit an average width ranging from 10 to 50 kilometers. Elevations in the Shiwalik mountains fluctuate between **600 to 1500 meters** above sea level.
- ❑ Extensive valleys called **Duns** lie between the Lesser Himalayas and the Shiwalik mountain ranges. Notable and significant Duns in this region include **Dehradun** and **Kotli Dun**.



Fig. 3.15: Shiwalik Range

Region	Name of Shiwaliks
Jammu Region	Jammu Hills
Arunachal Pradesh	Dafla, Miri, Abor, and Mishmi Hills
Uttarakhand	The Dhang Range, Dundwa Range
Nepal	Churia Ghat Hills

REGIONAL DIVISION OF HIMALAYAS

The Western Himalayas

- ❑ Stretch approximately 880 kilometres, the Western Himalayas extend from the Indus River in the west to the Kali River in the east.

- ❑ This region spans **three states** - Jammu and Kashmir, Himachal Pradesh, and Uttarakhand. It comprises three distinct physiographic provinces - Kashmir Himalaya, Himachal Himalaya, and Kumaon Himalaya (Uttarakhand Himalayas).
- Notable areas within the Kashmir Himalayan region includes the **Ladakh plateau and the Kashmir valley**.

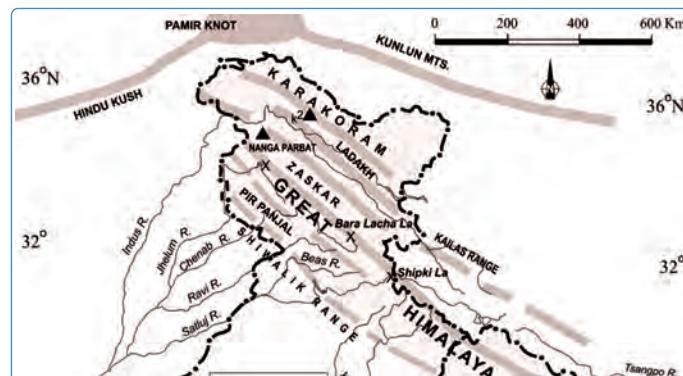


Fig. 3.16: The Western Himalayas

- In the **Himachal** Himalayas, the **Zaskar range** represents the Greater Himalayas, while the **Pir Panjal and Dhauladhar** ranges signify the Lesser Himalayas. The Outer Himalaya is characterized by the Shiwalik range. The southern slopes of this region are rugged, steep, and covered in forests, while the northern slopes are relatively flat and gentle, and feature plains adorned with lakes.
- The **Kumaon** Himalayas are located in Uttarakhand, extending from the Satluj to the Kali River. The **Mussoorie and Nag Tibba** ranges represent the Lesser Himalayas in the Kumaon Himalayas.
- In this area, the Shiwalik range runs south of the Mussoorie range, between the Ganga and Yamuna rivers.

- **Formation of Dun Valleys:** The flat valleys situated between the Lesser Himalaya and the Shiwalik range are commonly referred to as “doons” or “Duns”. Shiwalik hills arose, blocking rivers from the higher Himalayas, are creating temporary lakes. As rivers cut through the Shiwalik range, the lakes drained, forming dun valleys like Dehra Dun, Chumbi, Patli Dun, etc.

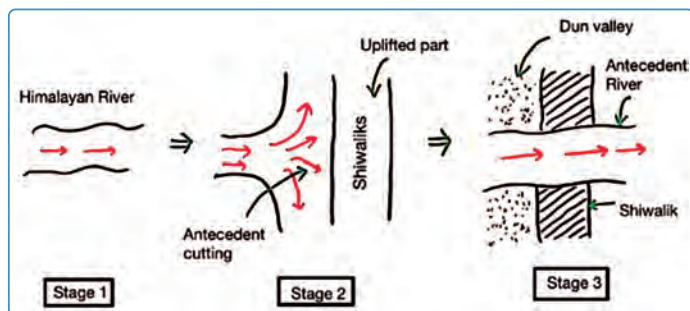


Fig. 3.17: Evolution of Dun Valleys

The Central Himalayas

- The Central Himalayas stretch for 800 kilometres between the Kali River in the west and the Tista River in the east.
- This region contains the highest peaks of the Great Himalaya range, including Mount Everest, Kanchenjunga, Makalu, Annapurna, Gosainthan, and Dhaulagiri.
- The Lesser Himalaya is known as Mahabharat Lekh in this region. This region is traversed by rivers such as the Ghaghara, Gandak, and Kosi.
- Notably, the **lacustrine valleys** of Kathmandu and Pokhara, which were previously lakes, are located between the Great and Lesser Himalayas. The Shiwalik range approaches the Lesser Himalaya to the east and eventually disappears beyond the Narayani (Gandak) River.

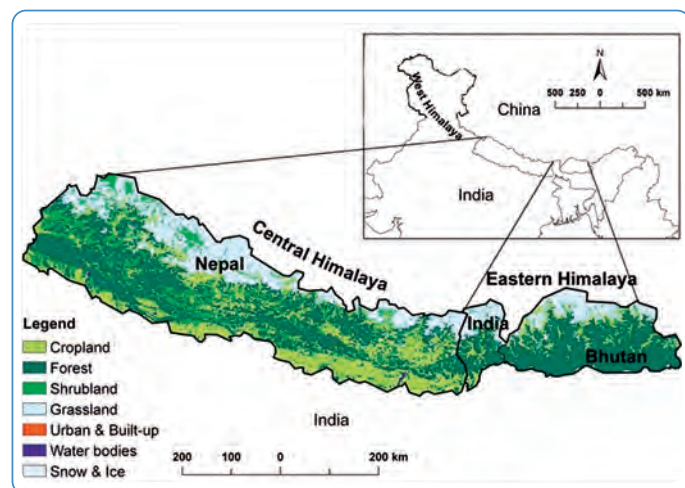


Fig. 3.18: Central Himalayan Region

The Eastern Himalayas (Assam Himalayas)

- The Eastern Himalayas, also known as the Assam Himalayas, stretch for about **720** kilometres between the Tista and Brahmaputra rivers in the west and east, respectively.
- These Himalayas are primarily situated in **Arunachal Pradesh and Bhutan**. Due to heavy rainfall, the Assam Himalayas have significant fluvial erosion.
- The Himalayan range takes a sharp southward turn after the Dihang **Gorge**, running parallel to India's border with Myanmar. Patkai Bum, Naga Hills, Kohima Hills, Manipur Hills, Mizo Hills (formerly the Lushai Hills), Tripura Hills, and Barail Range are some of the local names for these hill ranges.

Purvanchal Himalayas

- The **Patkai Bum** hills are mainly made up of sturdy sandstone, with elevations ranging from 2,000 to 3,000 meters. These hills transition into the Naga Hills, where the highest peak, **Saramati** (3,826 meters), is located.
- **Patkai Bum and Naga** Hills serve as the dividing ridge between India and Myanmar. South of the Naga Hills are the Manipur hills, usually standing at elevations of less than 2,500 meters. These hills are separated from the Naga Hills by the **Barail range**.
- Moving further south, the **Barail Range** extends westward into the Jaintia, Khasi, and Garo hills, representing an eastern continuation of the Indian peninsular block. They are set apart from the main block by the Ganga and Brahmaputra rivers.
- South of the **Manipur Hills** is the **Mizo Hills**, previously known as the Lushai Hills. The highest point in this region is the Blue Mountain (2,157 meters) located in the southern part.



Fig. 3.19: Purvanchal Hills



IGNITE YOUR MIND

The Himalayan region hosts the largest honeybee, *Apis laboriosa*, known for nesting on steep cliffs. Local communities undertake perilous methods to harvest the coveted red honey from these giant bees. Discuss the vulnerabilities of indigenous communities in the Himalayas to climate change and propose community-centric adaptation strategies.

THE SYNTACTICAL BENDS OF THE HIMALAYAS

The Himalayan mountain range, which typically stretches from east to west, experiences a unique change in direction at its western and eastern ends. These **sudden turns**, known as syntactical bends, cause the mountain ranges to sharply curve southward.

- In the western region, near **Nanga Parbat**, the Indus River has carved a deep gorge, resulting in geological

formations that appear as if they have been tightly folded around pivotal points.

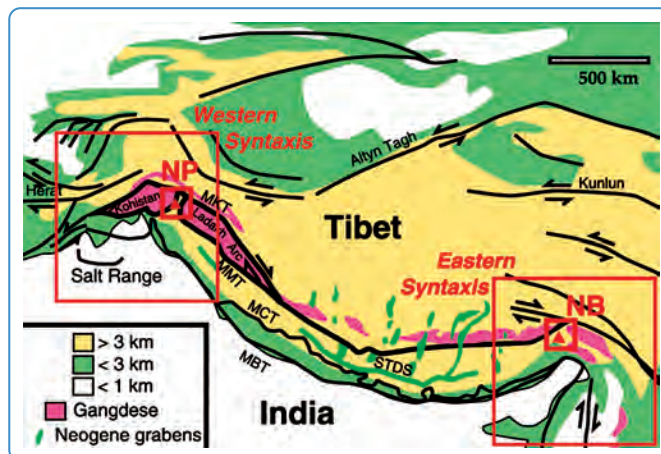


Fig. 3.20: Tectonic sketch map of India-Asia Collision

- Similarly, in **Arunachal Pradesh**, there is a noticeable hairpin bend where the mountains sharply change direction from east-west to north-south after crossing the Brahmaputra River. This shift also led to a significant alteration in the tectonic strike, transitioning from an eastward to a southward trend.

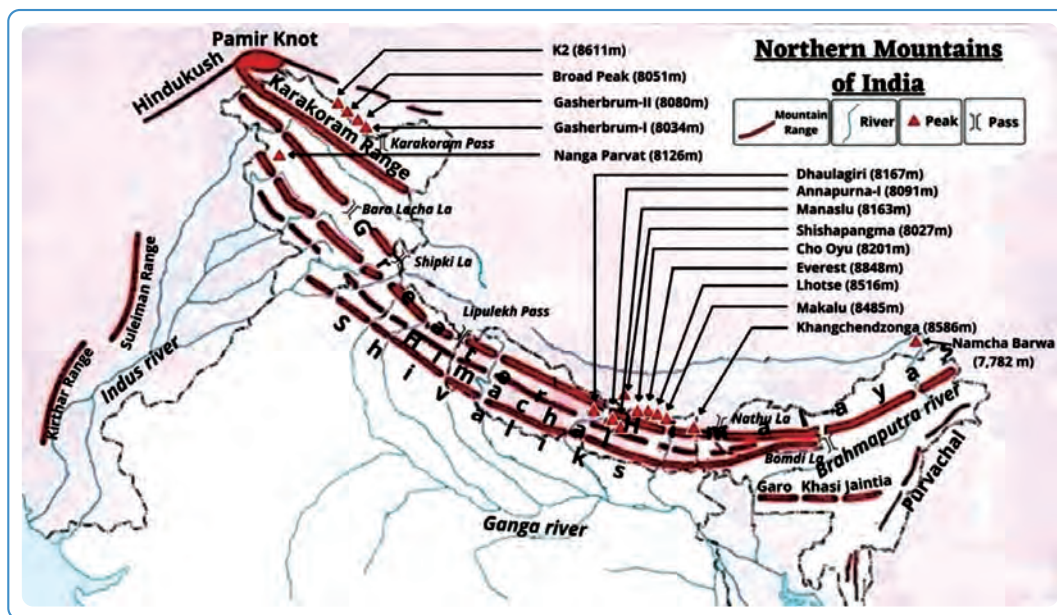


Fig. 3.21: Highest Peaks of the Himalayas

THE SIGNIFICANCE OF THE HIMALAYAN REGION

The Himalayan region holds immense significance due to various factors, encompassing geographical, ecological, cultural, and economic aspects. Here are the key points highlighting the significance of the Himalayan region:

□ Water Tower of Asia:

- The Himalayas act as a colossal water tower, feeding major rivers like the Indus, Ganges, Brahmaputra, and Mekong.

- These rivers sustain vast populations and nourish agriculture across Asia.

□ Biodiversity Hotspot:

- The Himalayas are a biodiversity hotspot, harboring unique flora and fauna adapted to varied altitudes and climates.
- From snow leopards and yaks to Bengal tigers and one-horned rhinos, the region is a biodiverse haven.

- **Climate Regulator:** They play a crucial role in regulating global climate, influencing monsoon patterns and contributing to temperature regulation.

- ❑ **Cradle of Civilizations:** The Himalayas have nurtured ancient civilizations, witnessed rich human history, and hosted spiritual traditions.
- ❑ **Pilgrimage Sites:** Home to sacred sites for Hinduism, Buddhism, and other religions, attracting millions of pilgrims seeking spiritual solace.
- ❑ **Indigenous Communities:** Diverse indigenous communities call the Himalayas home, offering traditional knowledge and practices for sustainable living.
- ❑ **Strategic Location:** Forming a natural border, the Himalayas shape the geopolitical landscape of countries like India, China, Nepal, Bhutan, and Pakistan.
- ❑ **Water Security:** Control and management of Himalayan water resources are crucial, leading to complex water-sharing agreements and geopolitical considerations.
- ❑ **Tourism and Adventure:** A magnet for adventure seekers and tourists, the Himalayas' tourism industry plays a vital role in the region's economy.
- ❑ **Origin of Rivers:** The Himalayas serve as the **origins of India's major rivers**, because of their abundant rainfall, extensive snowfields, and large glaciers. These rivers carry fertile soil down to the plains, making the northern region very fertile.
- ❑ **Key For Energy:** India relies heavily on rivers originating in the Himalayas for energy. About 33% of the country's thermal electricity and 52% of its hydropower depend on these rivers, which receive water from melting glaciers, making them crucial for energy and water security.
- ❑ **Affects the Weather:** The Himalayas have a significant impact on India's climate. Their height and position cause rainfall by intercepting the summer monsoons from the Bay of Bengal and the Arabian Sea. They also serve as a barrier, keeping cold air from Central Asia from entering India.
- ❑ **Rich in Forest:** The Himalayan ranges are rich in forest resources, ranging from tropical to alpine vegetation. These forests supply fuelwood and raw materials to a variety of industries. There are also many valuable medicinal plants in this region.
- ❑ **Popular for Tourism:** The Himalayas attract tourists due to their stunning landscapes and pleasant climate. The hilly areas offer relief from the summer heat making them popular tourist destinations.

The Himalayas are more than just a majestic mountain range; they are a vital life force for the planet. Their environmental,

cultural, and geopolitical significance demands our respect and protection. By understanding and appreciating their multifaceted importance, we can work towards ensuring a sustainable future for this awe-inspiring region and all who call it home.

The Third Pole

- ❑ The "Third Pole" refers to the high mountainous region of the **Hindu Kush Himalayan (HKH) range**, often referred to as the **"water tower of Asia."**



Fig. 3.22: Geographical Location of Third Pole

- ❑ This region encompasses the Himalayas and other associated mountain ranges, including the **Hindu Kush, Karakoram, and Pamir**.
- ❑ It is called the "Third Pole" because, after the North and South Poles, it contains the largest area of frozen water in the form of glaciers and snow outside the polar regions.
- ❑ The Third Pole is a crucial source of freshwater for numerous rivers and supports the livelihoods of millions of people in Asia.

The Hindu Kush Himalayan Assessment Report on Glacier Retreat in the Himalayas

The Hindu Kush Himalayan Assessment Report is a comprehensive study published in 2019 that focuses on the impacts of climate change on the environment, ecosystems, and communities in the Hindu Kush Himalayan region. One significant aspect of the report is its examination of glacial retreat in the Himalayas due to climate change. Some key findings include:

- ❑ **Glacial Retreat:** Rising temperatures, attributed to global climate change, have led to increased melting of glaciers, resulting in their retreat over time. For instance, **Gangotri Glacier**, one of the largest glaciers in the Himalayas and the source of the Ganges River, has been experiencing significant retreat.

- ❑ **Water Resources Impact:** Glaciers act as natural reservoirs, and their meltwater feeds into major rivers, providing a steady source of freshwater. The reduction in glacier size and the alteration of the timing and magnitude of water runoff can impact downstream water availability. For example, in the **Indus River**, originating from the Tibetan Plateau and fed by glacial meltwater from the Himalayas, water availability has reduced for **agriculture** and **hydropower**.
- ❑ **Risk of Glacial Lake Outburst Floods (GLOFs):** The report emphasizes the growing risk of Glacial Lake Outburst Floods (GLOFs) as glacial lakes form due to the accumulation of meltwater. The collapse or overflow of these lakes can result in devastating floods downstream. In February 2021, **Chamoli district** in Uttarakhand witnessed **flash floods** caused by **GLOFs**.

- ❑ **Ecosystem Changes:** The shifting snowlines and vegetation zones impact biodiversity and the livelihoods of communities dependent on natural resources. Warmer temperatures are causing the upward migration of plant species, leading to **shifts in alpine vegetation zones** in the Himalayan ecosystem.
- ❑ **Socioeconomic Impacts:** Changes in water availability, particularly in the context of agriculture and hydropower, can have significant socioeconomic implications for communities in the Hindu Kush Himalayan region. For instance, across the **Hindu Kush Himalayan region**, changes in precipitation patterns, including altered monsoon dynamics and variations in snowfall, impact **crop yields** and **agricultural practices**.

Important Glaciers, Valleys, Passes in Himalaya

Himalayan Glaciers

The Himalayan region is home to around 15,000 glaciers, covering an area of approximately 500,000 square kilometres.

Main Glaciers of the Himalayan Region

Glacier Name	Location	Important Features
Siachen	Karakoram	Largest in Nubra Valley of the Himalayas and second longest glacier outside the Polar region.
Biafo	Karakoram (Baltistan region of Ladakh)	Flows into the Shigar River.
Chogo Lungma	Karakoram	Located in Gilgit Baltistan.
Gangotri	Uttarakhand	Originates below the Chaukhamba Peak and is also known as 'Gomukh.'
Rimo Siachen Area	Ladakh	Drains into the Shyok River.
Hispar	Gilgit-Baltistan	World's longest glacial system.
Zemu	Sikkim/Nepal	Largest glacier in the Eastern Himalaya, feeds River Teesta.
Sonapani	Chandra Valley of Lahaul and Spiti region	Longest glaciers of the Pir Panjal range near the Rohtang Pass.
Milam	Uttarakhand	Major source of River Gori Ganga. Biggest glacier in Kumaon Himalaya.
Rupal	Kashmir	Located in the Greater Himalaya, it flows northeastward.
Gangri Nun Kun Mountains	Himalayan Range	Largest glacier in the Nun Kun peak.
Bara Shigri	Chandra Valley of Lahaul in Himachal Pradesh	Second longest glacier in the Himalayas. Source of water for River Chenab. Deposits of antimony.
Diamir (Nanga Parbat)	Kashmir (Pakistan Occupied)	Diamir mountains are known as the 'King of Mountains'
Pindari	Kumaon Region of Uttarakhand	Gives rise to River Pindari.
Chong Kumdan	Karakoram	Located in Ladakh; they feed and are a threat to Shyok River and might block it.

Karakoram Range glaciers	Pir Panjal Range glaciers	Kumaon-Garhwal glaciers	Central-Nepal glaciers	Kanchenjunga-Everest glaciers
1. Siachen glacier (Nubra Valley) 2. Hispar glacier 3. Biafo glacier 4. Baltoro glacier 5. Godwin Austen glacier 6. Punmah glacier 7. Batura glacier 8. Chogo Lungma glacier 9. Pasu glacier 10. Sasaini/Ghulkin glacier 11. Rimo glacier 12. Chong Kumdan glacier 13. Kichik Kumdan glacier 14. Aktash glacier	1. Sonapani glacier (Chandra Valley) 2. Bara Shigri glacier (Chandra Valley) 3. Gangri glacier 4. Rakhiot glacier 5. Buzhi glacier 6. Tashan glacier	1. Gangotri glacier 2. Milam glacier 3. Mana glacier 4. Bhagirath-Kharak glacier 5. Satopanch glacier 6. Sankalpa glacier 7. Raikane glacier	1. Yepokangara glacier 2. Lidanda Glacier 3. Chhuling glacier 4. Annapurna glacier 5. Mayondi glacier	1. Rongbuk glacier (Northern side of Mt. Everest) 2. Khumbu glacier (Southern side of Mt. Everest) 3. Zemu glacier 4. Kanchenjunga glacier 5. Yalung glacier 6. Talung glacier 7. Alukthang glacier 8. Kangshung glacier 9. Barun glacier

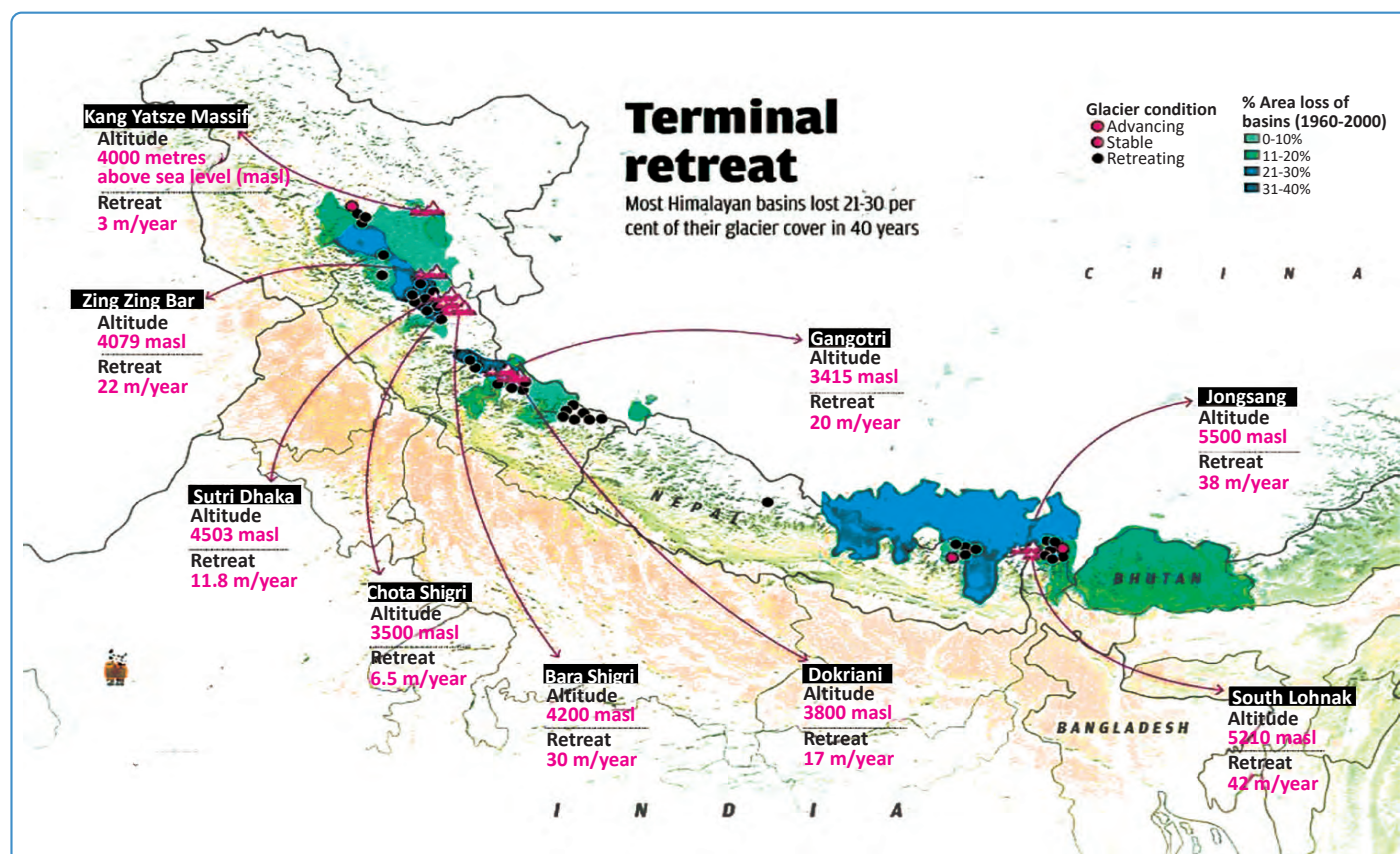


Fig. 3.23: Retreat of Glaciers in Himalayas



IGNITE YOUR MIND

Would you believe me if I told you Kangchenjunga, the world's third-highest mountain, is revered as sacred by Sikkim's people and neighboring regions? Climbing its main summit is deemed forbidden to honor its divine importance. Assess the environmental impact of mountaineering activities on Kangchenjunga and other Himalayan mountains. What measures can be taken to minimize this impact?

Snowline in the Himalayas

The snowline on a mountain demarcates the boundary beyond which perpetual snow is present throughout the year. In the Eastern Himalayas and Kumaon Himalayas, the snowline is situated at an elevation of approximately 3,500 meters above sea level. Conversely, in the Western Himalayas, the snowline is lower, at approximately 2,500 meters above sea level.

Factors Influencing Snowline Variation

Latitude Differences

- ❑ **Eastern and Kumaon Himalayas:** The snowline here is around 3,500 meters above sea level.
- ❑ **Western Himalayas:** Conversely, in the western Himalayas, the snowline sits at about 2,500 meters. This

variance is mainly due to the changing latitude, from about 28° N in Kanchenjunga to 36° N in the Karakoram.

Extent of Glaciers

- ❑ **Glacial Coverage:** The Himalayas encompass a vast area of glaciers and snowfields, totalling approximately 40,000 square kilometres in India. These icy formations stretch from the Karakoram region in Jammu and Kashmir to Arunachal Pradesh.
- ❑ **Numerous Glaciers:** Within this vast region, an estimated 15,000 glaciers are distributed between the bends in the eastern and western Himalayas.

Snowline Heights

Regional Differences: Snowline altitude varies across different parts of the Himalayas:

Influence of the Tibetan Plateau

- ❑ **Tibetan Side:** On the Tibetan side of the Himalayas, the snowline is about 900 meters higher than in neighbouring areas. This elevation difference is due to the arid nature of the region and limited moisture availability.

Moisture-Altitude Relationship

- ❑ **Moisture Levels:** There is a straightforward connection between atmospheric moisture levels and the altitude of the snowline. Typically, regions with higher moisture content experience lower snowline altitudes, while drier areas have higher snowlines.

Mountain Passes in the Himalayas

Pass Name	Location	Significance
Aghil Pass	North of K2 in Karakoram	Connects Ladakh with Xinjiang (Sinkiang) province of China. Closed during winter.
Banihal Pass	Pir Panjal range	Joins Jammu with Srinagar via Jawahar Tunnel.
Bara Lacha	Himachal Pradesh	Connects Manali and Leh. Snow-covered from November to mid-May.
Bomdi La	Arunachal Pradesh (East of Bhutan)	Connects Arunachal Pradesh with Lhasa (capital of Tibet).
Burzail Pass	Srinagar to Gilgit Valley	Connects Kashmir Valley with Deosai Plains of Ladakh. Closed for trade and transport during winter.
Chang-La	Greater Himalayas	Connects Ladakh with Tibet. The road leads to the town of Tangtse. Pass has a temple dedicated to Chang La Baba.
Debsa Pass	Greater Himalayas (Kullu and Spiti districts of Himachal Pradesh)	Provides an easier and shorter alternative to the traditional Pin Parbati Pass route.
Dihang Pass	Arunachal Pradesh	Connects Arunachal Pradesh with Mandalay (Myanmar).
Diphu Pass	Arunachal Pradesh (Eastern part)	Tri-point borders of India, Myanmar and China.
Imis La	Ladakh Union Territory and Tibet in China	Connects Ladakh to Tibet (China). Features difficult terrain and steep slopes.
Khardung La	Near Leh in Ladakh region	Connects Leh with Siachen Glacier. One of the highest motorable pass.
Khunjerab Pass	Karakoram Mountains in Ladakh	Connects Ladakh to Sinkiang Province of China. Snow-covered from November to mid-May.
Likhapani	Arunachal Pradesh	Joins Arunachal Pradesh with Myanmar. Open for trade throughout the year.

Pass Name	Location	Significance
Lipu Lekh	Pithoragarh district, Uttarakhand	Connects Uttarakhand to Tibet. Facilitates pilgrims for Manasarovar Lake. Important border post for trade with China.
Mana Pass	Uttarakhand	Connects Uttarakhand to Tibet. Remains snow-covered for 6 months during the winters.
Mangsha Dhura	Pithoragarh district of Uttarakhand	Connects Uttarakhand to Tibet. Pilgrims for Manasarovar cross this pass. Landslides are a major problem.
Muling La	North of Gangotri, Uttarakhand	Connects Uttarakhand to Tibet.
Nathu La	Sikkim	Forms an offshoot of the ancient Silk Road. One of the three trading border posts between India and China. Reopened in 2006.
Niti Pass	Uttarakhand	Joins Uttarakhand to Tibet.
Pangsau Pass	Arunachal Pradesh	Connects Arunachal Pradesh to Mandalay (Myanmar).
Pensi La	Greater Himalayas	Connects Suru Valley to Zaskar Valley (Ladakh).
Pir Panjal Pass	Mughal Road	Traditional pass between Jammu and Kashmir. It provided the easiest, shortest metalled road access.
Qara Tagh Pass	Karakoram Mountains	The highest pass in the Tarim Basin on the ancient caravan route between Leh and Yarkand. A branch of the Great Silk Road.
Rohtang Pass	Eastern side of Pir Panjal Range (51 km from Manali)	Connects Kullu, the Lahaul and Spiti valleys of Himachal Pradesh. Road access was built by the Border Road Organization (BRO).
Shipki La	Himachal Pradesh	Connects Himachal Pradesh to Tibet (National Highway 5). Important border post at Indo-China border. River Satluj flows through it.
Thang La	Ladakh	One of the highest motorable passes in India.
Traill's Pass	Uttarakhand (Pindari Glacier)	Links Pindari Valley to Milam Valley. Steep and rugged terrain makes it a difficult pass to be crossed.
Zoji La	Kargil District (100 km from Srinagar)	Joins Srinagar with Kargil and Leh. Faces heavy snowfall. Closed from December to mid-May. The beacon force of BRO maintains the road. The centre has approved Rs. 6809 crores to build Asia's longest bi-directional Zoji La Pass Tunnel, which will reduce the travel time to cross the Zoji La pass to 15 min from 3.5 h.

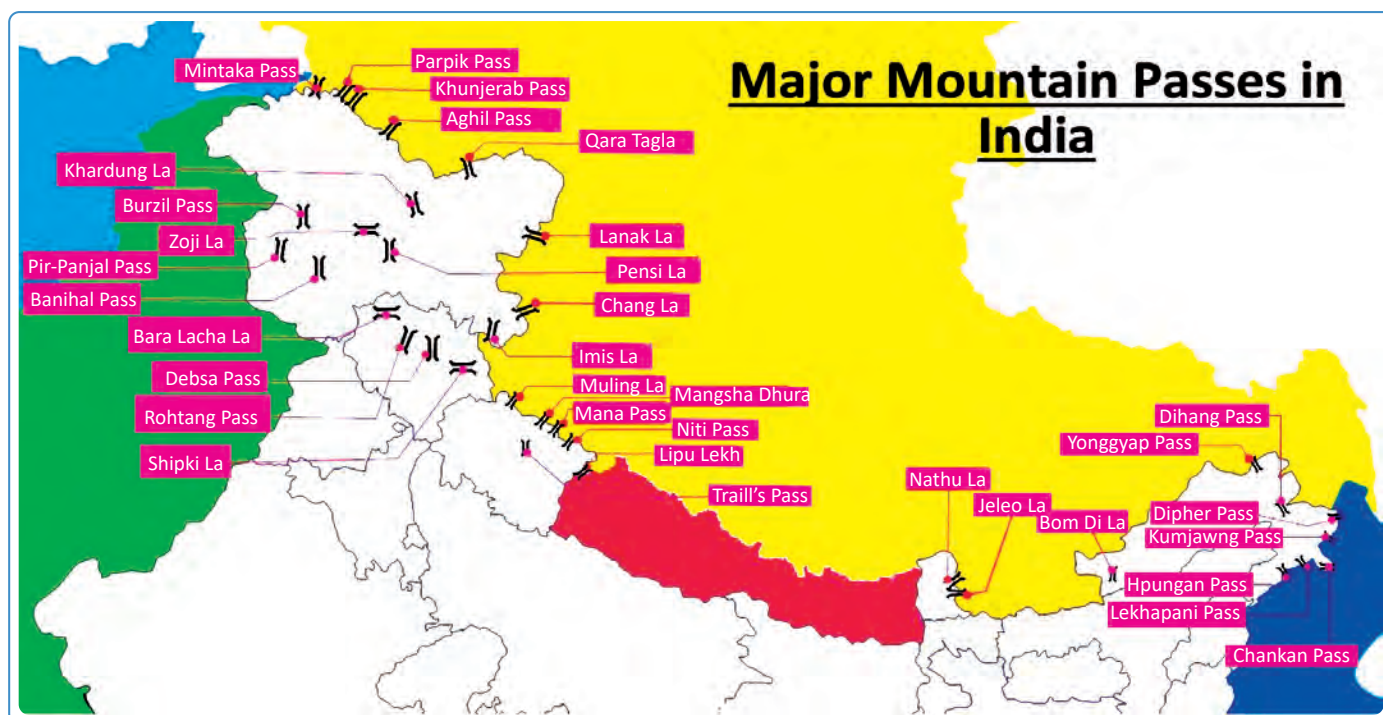


Fig. 3.24: Mountain Passes In Himalayas

Kashmir Valley and Karewas

Kashmir Valley and the surrounding Karewas are extremely important in the Himalayan region. Karewas are **lacustrine deposits** found in the Valley of Kashmir and Bhadarwah Valley of Jammu division. These flat-topped mounds border the Kashmir Valley on all sides and are distinguished by the presence of fossils of mammals and, in some areas, peat.

- Around 1 million years ago, during the **Pleistocene** Period, the entire Valley of Kashmir was submerged underwater. Subsequently, due to internal geological forces, the Baramullah Gorge was formed, allowing the lake to drain through this gorge.
- The deposits left behind in this process are known as Karewas. Karewas has an average thickness of about 1400 meters.
- Over time, the Karewas have been raised, dissected, and relocated due to natural processes like **subaerial denudation** and the influence of the Jhelum River, resulting in their present positions.

Karewas play a **crucial role** in agriculture, primarily devoted to the cultivation of saffron, almonds, walnuts, apples, and orchards. Saffron cultivation, in particular, contributes significantly to the income of local growers.

THE NORTHERN PLAINS

The Northern Plains of India are situated to the south of the Shiwalik mountain range, separated by the **Himalayan Front Fault (HFF)**. This expansive plain serves as a transitional region between the northern Himalayan terrain and the southern Peninsular India.

These plains have been shaped through the deposition of alluvial sediments carried by the Indus, Ganga, and Brahmaputra rivers, and their tributaries. Over time, these

deposits have accumulated, forming the extensive alluvial plain we see today.

The Himalayan Front Fault (HFF)

The Himalayan Front Fault (HFF) is a geological fault that marks the boundary between the Himalayan foothills and the Indo-Gangetic Plain. This fault is the youngest and southernmost thrust structure within the Himalaya deformation front. It's classified as a reverse fault, indicating that older rock layers have moved over younger ones. The HFF is a significant source of earthquakes in the Himalayan region and has contributed to the formation of the Siwalik Hills, a range of foothills along the southern edge of the Himalayas.

Geographical Dimensions of the Northern Plains

The Northern Plains span approximately 2400 kilometres from west to east.

- The **width** of the plain varies significantly along its length. It measures around 90–100 kilometres in Assam, 160 kilometres near Rajmahal in Jharkhand, 200 kilometres in Bihar, 280 kilometres near Allahabad, and expands to 500 kilometres in Punjab. Generally, the width of the plain increases as you move from east to west.
- These plains are primarily composed of **alluvial sediments** transported by rivers originating in both the Himalayan and Peninsular regions.
- The exact **depth** of alluvial deposits remains uncertain; however, recent estimates suggest an average depth of 1300 to 1400 meters in the southern part of the plain (north of Bundelkhand). As one moves towards the Shiwalik region, the depth of alluvium becomes even greater.

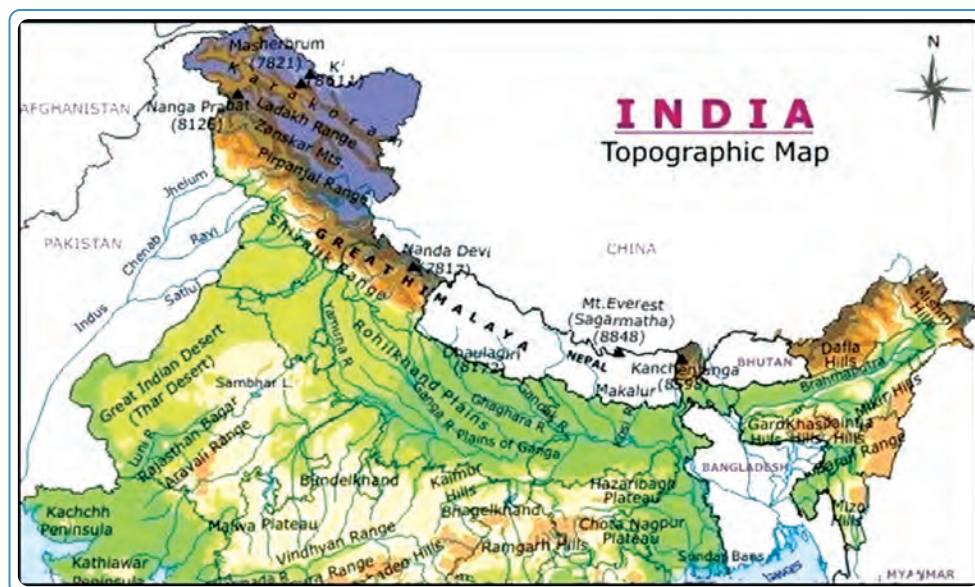


Fig. 3.25: Topographic Map of India

The Northern Plains are remarkably **consistent** in their physical landscape over vast distances. However, at smaller scales, the landscape exhibits variation due to features like river bluffs, Bhurs (natural levees), levees, dead-arms of river channels, ravines, and khols (small gullies).

The plains are characterized by the **shifting courses of rivers**, especially in flood-prone areas. This dynamic process alters the geography of the plains. Despite the damage caused by frequent floods in terms of human life and property, they bring benefits by depositing fresh layers of **nutrient-rich silt** in the floodplains each year, enhancing the fertility of the soils.

EVOLUTION OF NORTHERN PLAINS

Different Theories About Evolution of Northern Plains

❑ Alluviation of the Foredeep:

- Proposed by **Edward Suess**, this theory suggests that a “**foredeep**” formed in front of the rising Himalayas. It created a large synclinorium with an uneven bottom as the Himalayan thrust was stopped by the rigid Peninsular India.
- The Northern Plains were shaped by the alluvial deposits from the Himalayan rivers, with the foredeep’s bed resting on the hard crystalline rocks of the Peninsular region.

❑ Infilling of a Rift Valley:

- Sir G. Burrard proposed that the Northern Plains resulted from filling a rift valley created during the Himalayan formation. This rift valley formed between two parallel faults, one along the southern boundary of the Shiwaliks and the other along the northern boundary of the Peninsular region. Burrard cited similar rift valleys in the Himalayas and the Peninsular region as evidence.

❑ Recession of the Sea:

- Blandford’s theory suggests that during the Eocene Period (66 mya - 23.03 mya), Peninsular India was connected to Africa. Two seas existed during this time, which receded as the Himalayas rose during the

Miocene Period (23.03 mya - 5.333 mya), filled with sediments from Himalayan rivers.

- Evidence includes limestone rocks in the Kumaun-Garhwal region, saline water lakes in Rajasthan, and the joining of islands in the Gulf of Kachchh with the mainland.

❑ Remnant of the Tethys:

- Some geologists propose that the Northern Plains are remnants of the ancient Tethys Sea. After the Shiwalik uplift, the remaining Tethys part formed a large trough connected to the Bay of Bengal in the east and the Arabian Sea in the west. Rivers from the Himalayas deposited sediments in this trough, shaping the Northern Plains.

❑ Recent Views:

- According to recent geological perspectives, the Northern Plains of India are a **sag** in the Earth’s crust created by the northward drift of the Indian Subcontinent.
- This sag formed between the drifting subcontinent and the soft sediments in the Tethyan basin, which crumpled and uplifted into a mountain system. Subsequently, river deposits filled the sag, shaping the Northern Plains.

The Foredeep

A foredeep is a long and narrow depression that forms in front of a mountain range due to the Earth’s crust bending. It gets filled with sediments eroded from the mountains and is valuable in understanding the history of mountain formation and containing oil and gas resources.

The Sag

A sag is a wide and shallow depression in the Earth’s crust, typically resulting from tectonic forces. Sags are frequently found in rift valleys and sedimentary basins, and they can house reserves of oil and gas.

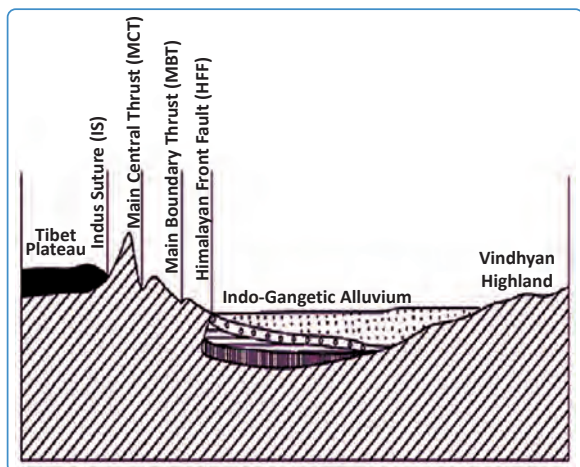


Fig. 3.26: Basement Complex-Hard Crystalline Rocks

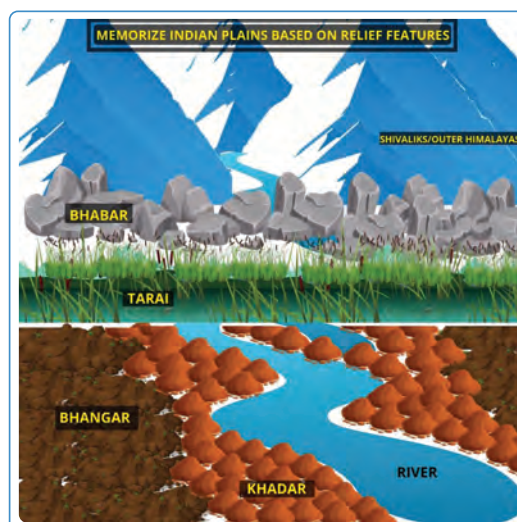


Fig. 3.27: Relief Features of Northern Plains

PHYSIOGRAPHIC DIVISIONS (GENETIC DIVISIONS) OF NORTHERN PLAINS OF INDIA

The Northern Plains of India is a vast, gently sloping region primarily formed by the deposition of sediments carried by the Himalayan and Vindhyan rivers. The Northern Plains of India can be divided into the following sub-regions.

The Bhabar Plain

- ❑ This plain lies south of the **Shiwalik range**, stretching from Jammu Division to Assam. It varies in width, being wider in the western plains and measuring 8 to 15 kilometres.
- ❑ The Bhabar Plain is made up of **gravel and unsorted sediments** deposited by rivers flowing from the Himalayan region and the Shiwaliks.
- ❑ Due to high porosity, smaller streams disappear here, leaving only large rivers to flow on the surface. It's **not suitable for crop cultivation** and is mainly inhabited by **Gujjars**, who are cattle keepers. This area supports the growth of big trees with extensive roots.

The Tarai Tract

- ❑ Located south of the Bhabar tract, the Tarai Tract has a width of 15 to 30 kilometres. It's wider in the eastern parts of the Northern Plains in the Brahmaputra region due to high rainfall.
- ❑ The Tarai is known for **excessive dampness, dense forests, rich wildlife, and mosquito infestations**. Historically, it was marshy with jungle and wild grass along the southern edge of the Shiwaliks. However, it has been largely converted for agriculture in states like Uttarakhand, Uttar Pradesh, Haryana, Punjab, and Jammu Division (J&K).
- ❑ Crops grown here include sugarcane, rice, wheat, maize, oilseeds, pulses, and fodder.

The Bhangar Plains

- ❑ These plains are upland alluvial tracts of India's Northern Plains formed by **older alluvium**. The Bhangar formations were formed during the middle Pleistocene Period and are located above the river's flood limits.
- ❑ The soil here is dark, humus-rich, and productive. It contains concretions and nodules of impure **calcium carbonate**, also known as "**Kankar**." In drier areas, Bhangar may exhibit small areas with saline and alkaline efflorescences known as '**Reh**', '**Kallar**', or '**Thur**'.

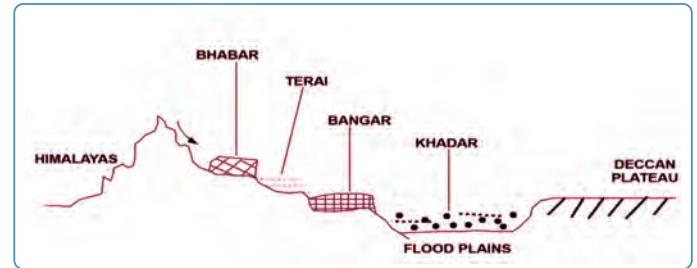


Fig. 3.28: Bhangar Plains

The Khadar Plains

- ❑ The Khadar Plains are **new alluvium tracts** along river courses that are enriched by fresh silt deposits each year during the rainy season. Sand, silt, clay, and mud make up Khadar land.
- ❑ After independence, this region has been cultivated, with crops like sugarcane, rice, wheat, maize, oilseeds, legumes, and fodder crops.

Delta Plains

- ❑ The deltaic plain is an extension of the Khadar land, covering approximately 1.9 lakh square kilometres in Ganga's lower reaches.
- ❑ It forms as the river flows slowly through this area, and it is mostly made up of old mud, new mud, and marsh. In the delta region, the uplands are called '**Chars**', while marshy areas are known as '**Bils**'. The delta of the Ganga is actively spreading towards the Bay of Bengal.

Comparison of Physiographic Divisions of Northern Plains of India

Plain	Location	Characteristics	Land Use and Inhabitants
Bhabar Plain	South of Shiwalik range, Jammu to Assam.	Made up of gravel and unsorted sediments from the Himalayas and Shiwaliks. Smaller streams disappear.	Not suitable for cultivation, inhabited by Gujjars (cattle keepers). Supports big trees.
Tarai Tract	South of Bhabar tract, wider in the east.	Excessively damp, dense forests, rich wildlife, mosquito infestations. Was marshy, now converted for agriculture.	Known for sugarcane, rice, wheat, maize, oilseeds, pulses, and fodder.
Bhangar Plains	Alluvial tracts, formed by older alluvium.	Dark, rich in humus, contains concretions of calcium carbonate. Above flood limits.	Productive land with 'Kankar', sometimes with saline areas ('Reh', 'Kallar', 'Thur').
Khadar Plains	New alluvium tracts along the river courses.	Consists of sand, silt, clay, and mud. Enriched by fresh deposits of silt annually.	Cultivated with sugarcane, rice, wheat, maize, oilseeds, legumes, fodder.
Delta Plains	Lower reaches of Ganga River.	Consists of old mud, new mud, and marsh. Uplands ('Chars') and marshy areas ('Bils').	Actively extending towards the Bay of Bengal.

Regional Division of Northern Plains

The Northern Plains of India can be categorized into four distinct regions based on their geographical and climatic characteristics:

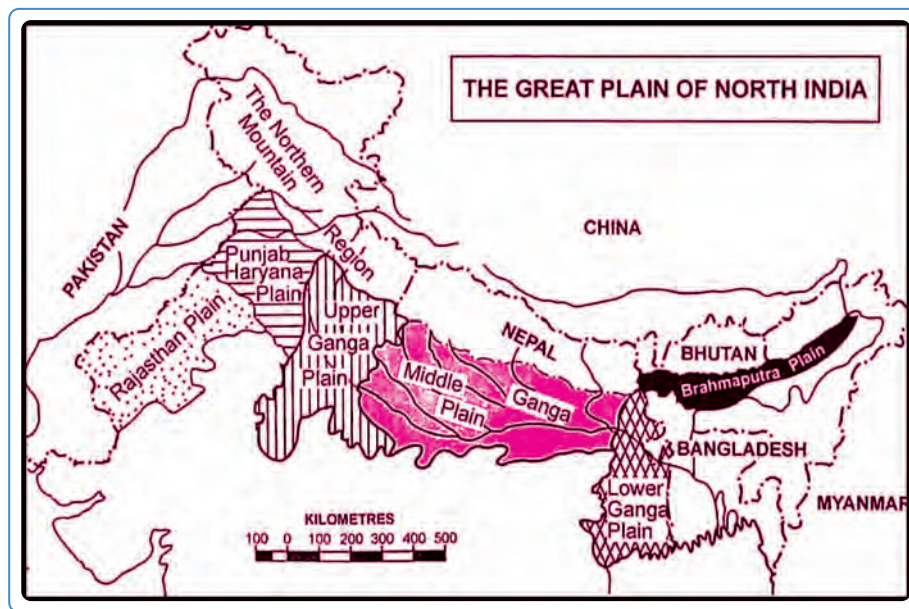


Fig. 3.29: Division of Northern Plains of India

□ The Plains of Rajasthan

- These plains are found to the west of the **Aravalli Range** and cover the Marusthali and Bagar regions of Rajasthan. The Rajasthan plains cover an area of around 175,000 sq km. This plain generally slopes from northeast to southwest, with lower elevations near the **Luni River** in Gujarat.

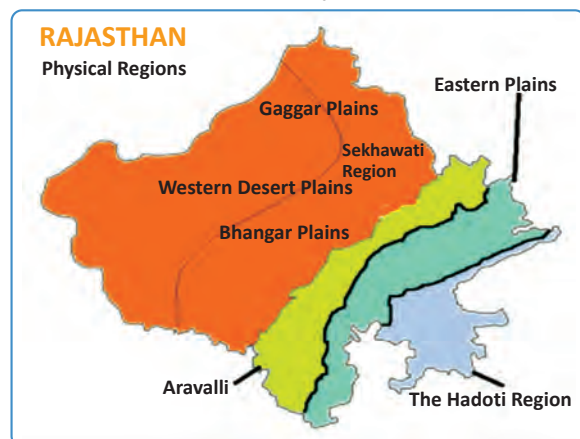


Fig. 3.30: Rajasthan Plains

- Evidence of past **sea recession** is found in the presence of saltwater lakes like Sambhar, Degana, Didwana Kuchaman, Lunkaransar-Tal, and Pachpadra. For instance, Sambhar Lake, covering about 300 sq km during the rainy season, is located approximately 65 km northwest of Jaipur.
- During the Permo-carboniferous Period (300 million years ago), much of the Rajasthan plain was submerged under the sea.

- Dry riverbeds like **Saraswati** and **Drisdavati** indicate that the area was once fertile. Currently, the Luni River is the only one flowing through this region, eventually reaching the Arabian Sea through the Rann of Kachchh.
- Today, most of the Rajasthan Plains are arid deserts characterized by sand dunes and barchans (Barkhans). **Playa lakes** are common in the basins, and some alluvial tracts known as Rohi exist in the southwestern part. The northeastern part features the dry beds of the Ghaggar, known as the Ghaggar Plains.

The Sea Recession

The last time the sea receded in Rajasthan, during the Jurassic period. This retreat of the sea happened because the Indian subcontinent moved northward and eventually collided with Asia.

□ The Punjab-Haryana Plains

- The Punjab-Haryana Plains feature bluffs (steep slopes or rounded cliffs, usually overlooking a shoreline or other body of water), locally known as **Dhaya**, rising as high as three meters or more, and Khadar belts known as Bet. The undulating topography south of the Shiwaliks experiences erosion due to seasonal streams called Chos.
- The southwestern parts, particularly the Hissar District, have sandy terrain with shifting sand dunes. The Satluj, Beas, and Ravi are the only perennial

rivers, while the Ghaggar (the ancient Saraswati) is a seasonal stream that flows through Ambala Cantt, with a course about ten km wide and water only during the rainy season.

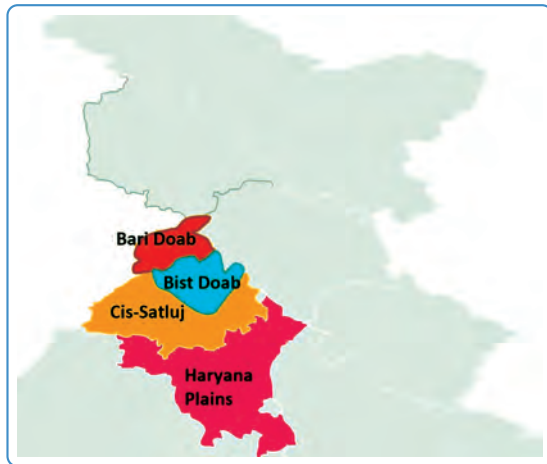


Fig. 3.31: Punjab- Haryana Plains

- The Punjab-Haryana Plains can be further divided into the Bari-Doab (between the Beas and Ravi), Bist Doab (between the Beas and Satluj), Malwa Plain (central part), and Haryana-Bhiwani Bagar (southern and southeastern parts).
- **The Ganga Plains:** The Ganga Plains stretch across a vast area from the Yamuna River basin in the west to the eastern border with Bangladesh. The Ganga Plains display diverse land features, including Bhabar, Tarai, Bhangar, Khadar, river bluffs (levees), abandoned river channels, Kols, dead channels, Bills, Tals, and badlands. These plains can be divided into the following sub-regions:
 - **The Upper Ganga Plain**
 - This region encompasses the **Ganga-Yamuna Doab**, Rohilkhand Division, and parts of the Agra Division. The Upper Ganga Plain features Bhur, which are undulating sandy deposits.
 - It is known for successfully growing sugarcane, rice, wheat, maize, pulses, mustard, vegetables, and orchards.
 - **The Middle Ganga Plain**
 - This region, which encompasses central and eastern **Uttar Pradesh** as well as the Bihar Plains up to Muzaffarpur and Patna, has an area of approximately 144,400 square kilometres.
 - It is bordered to the north by the Shiwaliks and has thick alluvial deposits with less **Kankar** formation. The Middle Ganga Plain is fairly uniform and unremarkable, with a few exceptions such as river levees, bluffs, ox-bow lakes, Dhus, Tals, Jala, and Chauras (marshy lands).
 - Because of the region's low-gradient landscape, river courses change frequently. The Gandak

and Kosi are the main left-hand tributaries of the Ganga in this region, while the Son is an important right-hand tributary.

- **The Lower Ganga Plain**

- This region is bordered by **Assam and Bangladesh** to the east and the Chotanagpur Plateau to the west, extending from Patna in the west to the foot of the Darjeeling Himalaya in the north and the Bay of Bengal in the south.

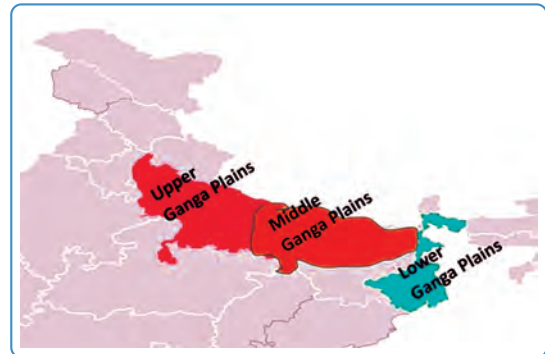


Fig. 3.32: Division of Ganga Plains

- The surface of the Lower Ganga Plain is mostly **uniform**, with minor variations. It is divided into sections by the main streams and their tributaries.
- **Rivers** such as the Tista, Jaldhaka, and Sankosh drain the eastern part of the plain and join the Brahmaputra. Rivers such as the Mahananda, Ajay, and Damodar dominate the western region.
- The **subsidence** of a portion of Peninsular India between the Rajmahal Hills and the Meghalaya Plateau, followed by sedimentation by the Ganga and Brahmaputra rivers, formed this plain. In the southern region of the Lower Ganga Plain lies the Sundarbans Delta, the largest mangrove swamp globally.

The Brahmaputra Plain

- The region is surrounded by high **mountains**, except in the west. It has a gradual **slope** from the Meghalaya Plateau on the southern side. This region is also referred to as the Brahmaputra Valley, Assam Valley, or Assam Plain, primarily because most of the Brahmaputra Valley is located in Assam.

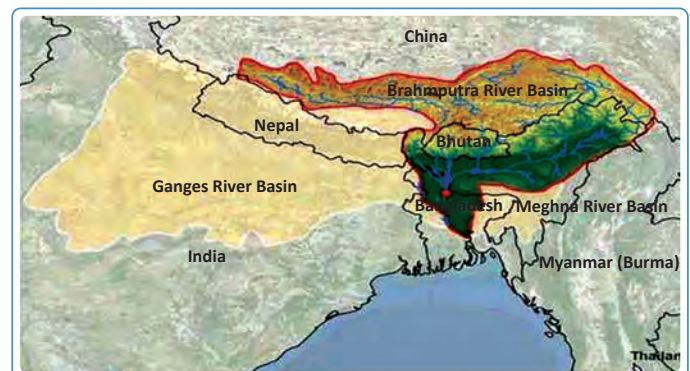


Fig. 3.33: The Brahmaputra Plain

- ❑ The western boundary of this area is marked by the Indo-Bangladesh border and the lower Ganga Plain's boundary. It is a flat region created by the gradual **accumulation of sediment** carried by the Brahmaputra River and its tributaries.
- ❑ Numerous tributaries from the north converge into the Brahmaputra River, creating multiple alluvial fans. This fertile plain is ideal for **cultivating crops** like rice and jute. This branching of tributaries results in river meandering and the formation of oxbow lakes. Large marshy areas are a prominent feature of this region.
- ❑ The coarse **alluvial debris** from the alluvial fans has led to the development of semi-terai conditions.

Significance of the Northern Plains of India

The Northern Plains of India hold immense significance due to their rich cultural heritage and fertile soil. They are known for their ability to support a wide range of crops from both tropical and temperate regions. Here are the key points highlighting their importance:

- ❑ **Fertile Soil for Agriculture:** The soils in these plains are very fertile, making them ideal for growing different crops like Rice (including Basmati in Punjab), Wheat (from Haryana to Uttar Pradesh), Maize, and non-cereal crops like Sugarcane (mainly in Uttar Pradesh and Bihar), Jute (widely grown in West Bengal), and oilseeds. This productive land is often called the "Granary of India."
- ❑ **Continuous Water Supply:** Many rivers in the Northern Plains flow all year round, providing a steady water source for agriculture. For instance, the Ganges, Yamuna, and their tributaries are vital for farming in states like Uttar Pradesh and Bihar. Constructing canals from these rivers, like the Upper Ganga Canal, further improves agricultural productivity.
- ❑ **Abundant Groundwater:** The Northern Plains have a high water table underground, which is used for irrigation, household needs, and industry through tube wells and pumps. States like Punjab and Haryana heavily rely on groundwater for farming.
- ❑ **Easy River Navigation:** The gentle slopes of the rivers in the plains make long-distance navigation possible, supporting trade and transportation. The Ganges, for example, has been a crucial trade route historically.
- ❑ **Infrastructure Development:** The flat landscape of the plains makes it easier to build roads and railways, enhancing connectivity and economic growth. The Grand Trunk Road, linking Kolkata and Delhi, is one of India's oldest and most important roadways.

- ❑ **Energy Resources:** The sedimentary rocks in the plains hold deposits of petroleum and natural gas, significantly contributing to the country's energy resources. Regions like Assam, parts of Bihar, and the Indo-Gangetic plains in Uttar Pradesh have considerable petroleum reserves.
- ❑ **High Population Density:** Even though the plains cover less than one-third of the country's total land area, they are home to over 40% of India's total population. States like Uttar Pradesh and Bihar have some of the highest population densities in the country.
- ❑ **Cultural and Religious Significance:** The Northern Plains have been witness to numerous religious, political, cultural, and social movements throughout history. They are the birthplace of several major world religions, including Hinduism, Buddhism, Jainism, and Sikhism. Additionally, these plains house sacred places and pilgrimage centres like Amritsar, Hardwar, Allahabad, Varanasi, Kushinagar, and Bodh-Gaya.

THE PENINSULAR INDIA

The Peninsular Plateau is the biggest physiographic unit of India, occupying **50% of the country's landmass**. It stretches from the Aravallis in the northwest to the Shillong plateau in the northeast and Kanyakumari in the south. The plateau is triangular in shape and consists of numerous small plateaus and mountains measuring around **16 lakh square km**. The Peninsular Plateau is one of India's oldest and most stable landmasses.

- ❑ The Peninsular Plateau is characterized by a composition of ancient rocks, including the **Dharwar rock system, Archean Gneiss, and the Schist**. Additionally, igneous Basalt rocks are prevalent in the Deccan plateau region. This plateau has experienced minimal structural alterations since its formation and has never been under the sea.
- ❑ The rivers within the Peninsular Plateau have, over time, reached their base level due to their considerable age, resulting in the development of wide and shallow valleys. The plateau generally elevates showing a **west-to-east gradient**, which is evident from the flow of rivers, except in the Narmada-Tapi rift valley.

Physiographic Features of the Peninsular Plateau

The Peninsular Plateau is divided into two broad physiographic regions: the Central Highlands and the Deccan Plateau.

The **Central Highlands** are located in the northern part of the plateau and are wider in the west than in the

east. They include the Vindhya, Satpura, and Aravalli ranges. The **Deccan Plateau**, situated in the southern part of the plateau, is taller in the west and slopes gently eastwards.

Other important physiographic features of the Peninsular Plateau include:

- ❑ **Rift valleys:** The Narmada and Tapi rivers flow through rift valleys in the Peninsular Plateau. Rift valleys are formed when the Earth's crust is pulled apart.
- ❑ **Block mountains:** Block mountains are created when large blocks of the Earth's crust are uplifted. **Examples:** Block mountains in the Peninsular Plateau include the Nilgiris and Western Ghats.
- ❑ **Plateaus:** The Peninsular Plateau is made up of numerous small plateaus, such as the Malwa Plateau, Chota Nagpur Plateau, and Deccan Plateau.
- ❑ **Hills:** The Peninsular Plateau also has a number of hills, such as the Aravalli Hills, Vindhya Hills, and Satpura Hills.

Evolution of the Peninsular Plateau of India

The evolution of Peninsular India has been marked by significant geological changes over time. This region, known for its stability, has undergone several transformations throughout its history.

- ❑ The **Great Plateau**, a remnant of the ancient landmass called **Gondwana**, was formed as a result of the gradual breaking and drifting of Gondwana land. It has remained above sea level since its inception, leading to extensive denudation processes.
- ❑ The plateau's mountains, predominantly relic formations, consist of exceptionally hard rocks that have resiliently withstood the erosive forces better than the surrounding areas.
- ❑ Over time, the rivers here have reached their base levels (after a period of downcutting) due to their ancient age, resulting in the formation of wide and shallow valleys.
- ❑ The dominant rock formations, especially in the southern parts, are of **metamorphic origin**, often featuring granite occurrences.

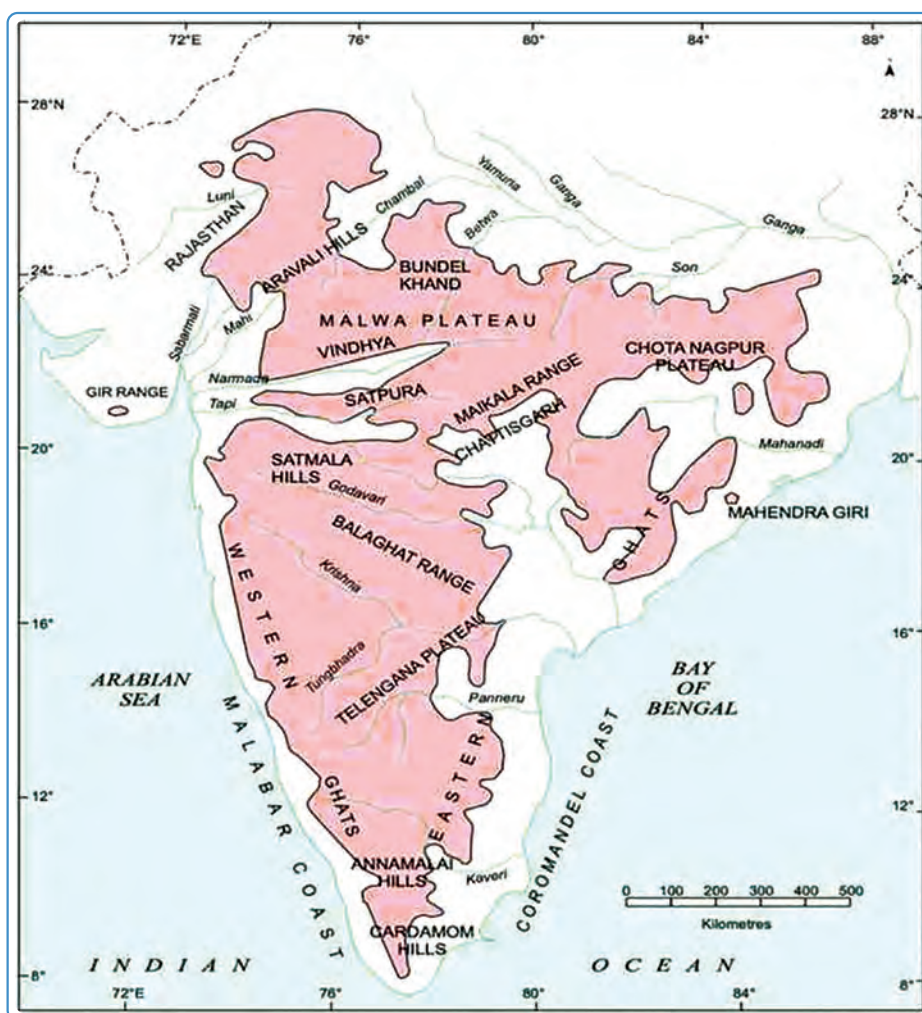


Fig. 3.34: Peninsular Plateau of India

Eastward Tilt of Peninsular India

The **eastward tilt of Peninsular India** can be attributed to several geological factors. Firstly, when India separated from Gondwana, the eastern side of the shield experienced subsidence. Additionally, the Deccan volcanic flow tilted eastward as the western margin underwent uplift. Erosion removed material from the western margin, depositing it in the Arabian Sea. This weight reduction caused the crust to rebound and tilt eastward.

The **oceanic lithosphere** of the Bay of Bengal is colder and denser, resulting in its subduction and dragging the Peninsular region downward. The Bengal sediment fan, comprised of eroded Himalayan sediments deposited in the Bay of Bengal over the Cenozoic era, exerts a downward pull on the Indian Peninsular region.

- **Formation of Rift:** The geological faults or rifts between the Vindhyas and the Satpuras, through which the Narmada and Tapi rivers flow, likely formed during the emergence of the Himalayas from the ancient Tethys Sea.

- The Deccan Traps, a vast volcanic formation, originated around 60 million years ago as the Indian plates moved towards the Asian plates.
- The volcanic hotspot responsible for the Deccan Traps is believed to lie beneath the present-day island of Reunion in the Indian Ocean.

Origin of Western Ghats:

- Western Ghats, a mountainous and faulted edge of the Deccan Plateau, evolved due to several factors. Geological evidence suggests they formed during the breakup of the supercontinent **Gondwana**, approximately 150 million years ago. Crustal doming and rifting preceding the Indian subcontinent's drift led to an initial escarpment of the western margin of India.
- The Ghats also receded eastward due to marine erosion, resulting in the coastal plain and steep Western Ghat **escarpment** (a long, steep slope, especially one at the edge of a plateau or separating areas of land at different heights.).

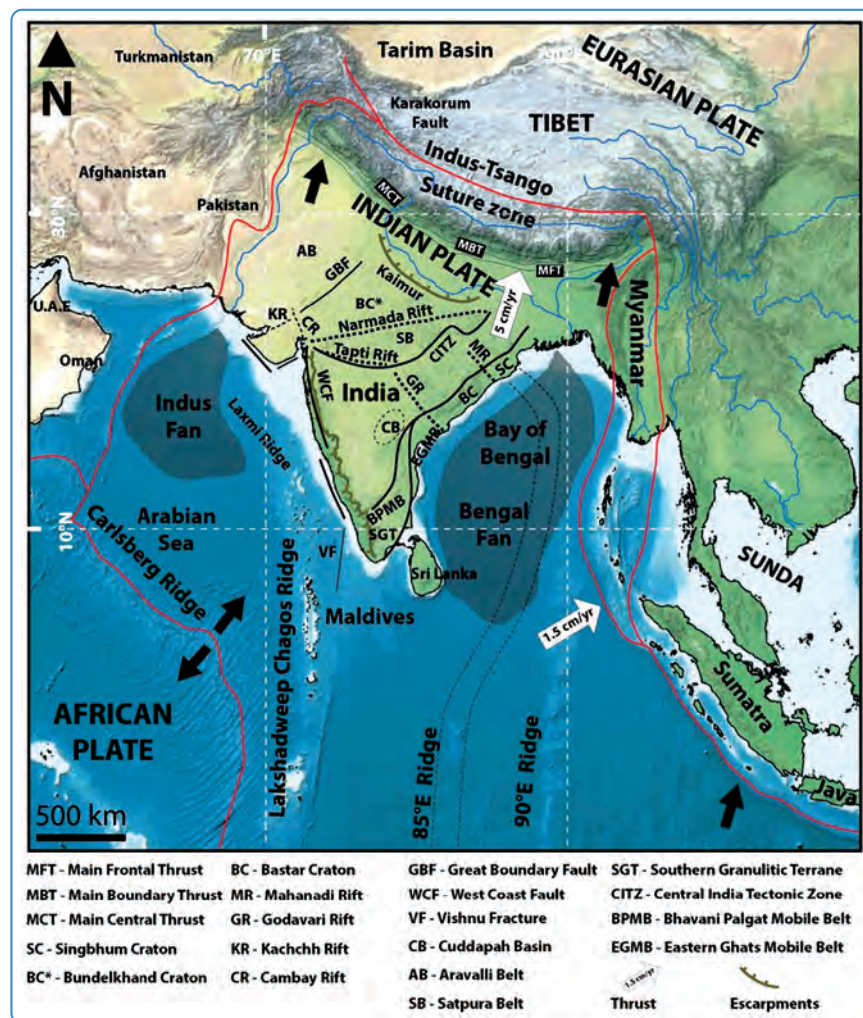


Fig. 3.35: Interplay of Various Tectonic Thrusts

- Other contributing factors include the development of westerly drainage and the gradual **post-glacial rise** in sea levels, leading to the formation of drowned valleys, estuaries, and lagoons.

❑ Origin of Eastern Ghats:

- **Eastern Ghats** are **older** than the Western Ghats and its geological history is complex and linked to the assembling and breaking up of the ancient **supercontinent Rodinia** and the consequent formation of the Gondwana supercontinent.
- The breakup of Gondwanaland led to the creation of Permian-Triassic rift basins along the present eastern margin of India. River deltas have formed numerous passes and broader coastal plains in this region.
- Additionally, lakes like Chilka, Pulicat, and Kolleru have been created by enclosing parts of the Bay of Bengal behind sand bars.

Features of the Peninsular Plateau in India

The Peninsular Plateau, a vast, approximately triangular landmass, encompasses roughly 50% of India's territory. It stands as one of the Earth's most ancient landforms, primarily consisting of **Archaean gneisses and schists**.

- ❑ This plateau maintains an average elevation ranging from **600 to 900 meters** above sea level and gradually inclines from the western to the eastern side.
- ❑ The majority of the peninsular **rivers** follow this eastward course, except for the Narmada and Tapi rivers, which traverse west to east within a rift valley.
- ❑ The Peninsular Plateau is quite a stable block that has seen some structural changes since the time it was formed. It has been a land area for a few hundred million years and has never been submerged under the sea except in a few places.
- ❑ The Peninsular Plateau is a region of rich natural resources, including **minerals, forests**, and agricultural land. It is also home to diverse wildlife, including tigers, elephants, leopards, and sloth bears.
- ❑ The Peninsular India is made up of a series of **patland** plateaus such as the Hazaribagh, the Palamu, the Ranchi, the Malwa, the Coimbatore plateaus etc.
- ❑ Some important physiographic features of this region are tors, block mountains, rift valleys, spurs, bare rocky structures, series of hummocky hills and walls like quartzite dykes offering natural sites for water storage.

The plateau is an aggregation of smaller plateaus, river basins, hill ranges, and valleys. The Peninsular Plateau can be divided into following broad groups:

- ❑ The **Central Highlands** in the north, including the Malwa Plateau and the Bundelkhand Plateau.
- ❑ The **Deccan Plateau** in the south, including the Karnataka Plateau and the Tamil Nadu Plateau.
- ❑ The **Western Ghats and Eastern Ghats** mountain ranges flank the western and eastern edges of the Deccan Plateau, respectively.

- ❑ The **North eastern plateau** is an extension of main peninsular plateau and is separated by the **Malda fault**. It includes Meghalaya and Karbi Anglong plateaus.



IGNITE YOUR MIND

The Peninsular Plateau is one of the oldest landmasses on Earth, believed to have formed around the Precambrian era. Explore the geological processes forming the Peninsular Plateau, emphasizing tectonic activities' impact on its topography.

Minor Plateaus

❑ Marwar or Mewar Plateau:

- The Marwar Plateau can be found in the eastern region of Rajasthan, India. It slopes gently in the eastward direction, boasting an average elevation ranging from **250 to 500 meters** above sea level.
- Composed primarily of shales, sandstone, and limestones from the Vindhyan period, this plateau is characterized by its geological composition. The Banas River, in conjunction with its tributaries, namely the Beach and Khari rivers, originates within the Aravalli Range and meanders in a northwest direction until it merges with the Chambal River.
- The continuous erosional action of these rivers has bestowed upon the plateau a distinctive rolling plain appearance. Rolling plains, in essence, feature undulating terrain, with subtle elevations and depressions in the landscape.

❑ Bundelkhand Upland:

- The region is situated to the north of the Yamuna River, bordered in the west by the Madhya Bharat Pathar, in the east and southeast by the Vindhyan Scarplands, and in the south by Malwa Plateau.

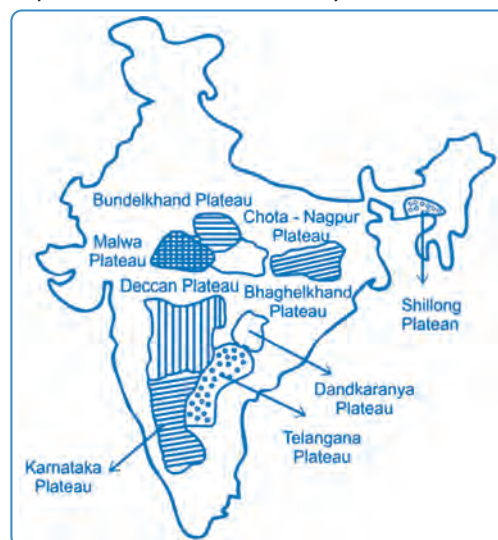


Fig. 3.36: Minor Plateaus of Peninsula

- It represents an ancient dissected upland formed by the **'Bundelkhand Gneiss,'** composed mainly of gneiss and granite. This geographical area spans five districts in Uttar Pradesh and four in Madhya Pradesh, with a mean elevation ranging from **300 to 600 meters** above sea level.
- The terrain gradually slopes from the Vindhyan Scrap towards the Yamuna River, characterized by a series of hillocks crafted from granite and sandstone. Ongoing erosion by local rivers, like the Dhasan, Betwa, and Ken, has given rise to an undulating landscape unsuitable for farming, embodying a **senile topography** indicative of age.

❑ **Meghalaya Plateau:**

- The peninsular plateau, stretching towards the east from the Rajmahal hills, extends into what is known as Meghalaya, also as the **Shillong plateau**. The **Garo-Rajmahal Gap**, a geological feature that was created through down-faulting separates the plateau from the main block. This process, characterized as a normal fault, involved a block of the Earth sliding downwards. Over time, this gap was gradually filled by sediments deposited by the Brahmaputra and Ganga rivers.
- The geological phenomenon of down warping is observed along the Rajmahal-Garo hills, leading to the formation of what is referred to as the **'Malda gap'** Notably, it is through this Malda gap that the Ganga and Brahmaputra rivers flow.
- The Shillong plateau itself is composed primarily of Archaean quartzites, schists and shales. As one moves towards the northern edge of the plateau, it gently slopes down towards the Brahmaputra valley, while in the south, it descends towards the Meghna and Surma valleys.
- Geographically, its western boundary largely aligns with the border of Bangladesh. Within the plateau, three distinct regions are recognized: the Garo Hills, which reach heights of approximately **900 meters**, the Khasi-Jaintia Hills, towering at around 1,500 meters, and the Mikir Hills, with elevations of about **700 meters**.
- Notably, the highest point in the plateau is Shillong, standing at an impressive 1,961 meters above sea level.

❑ **Malwa Plateau:**

- The Malwa Plateau, resembling a triangular shape, is delineated by the Vindhyan Hills to the south, while it is enclosed by the Aravali Range in the western part and in the north by the Madhya Bharat Pathar, and with Bundelkhand bordering it to the east.
- Within this plateau, **two distinct drainage systems** are observed: one flows westward towards the Arabian Sea, encompassing the Narmada, the Mahi,

and the Tapi rivers, while the other system meanders eastward, ultimately leading to the Bay of Bengal, comprising the Chambal and Betwa rivers that converge with the Yamuna.

- In the northern region, the Chambal River serves as the primary drainage, along with numerous right-bank tributaries such as the Kali, Parbati and the Sindh. Additionally, it encompasses the upper reaches of the Sindh, Betwa and Ken rivers.
- The composition of this plateau predominantly consists of extensive lava deposits, which have given rise to the prevalence of black soils.
- Notably, the plateau exhibits a gradual incline from south to north, with elevations decreasing from approximately **600 meters** in the southern region to less than 500 meters in the northern expanse.
- The plateau's topography is characterized by a rolling terrain intersected by various rivers. Particularly in the north, the landscape is distinctive due to the presence of the Chambal ravines.

❑ **Chotanagpur Plateau:**

- The Chotanagpur Plateau constitutes the **northeastern extension** of the Indian Peninsula, primarily encompassing regions within Jharkhand, the northern segment of Chhattisgarh, and the Purulia district of West Bengal.
- Situated to the northwest of the plateau, the Son River eventually merges with the Ganga. This elevated plateau maintains an average height of approximately **700 meters** above sea level and is predominantly characterized by the presence of Gondwana rocks. Its complex drainage system displays a **radial pattern**, with several rivers and streams meandering in various directions.
- **Prominent rivers** including the Damodar, Subarnrekha, South Koel, Barkar, and North Koel have established expansive drainage basins in the region. The Damodar River, which cuts through the plateau from west to east within a rift valley, is host to the Gondwana coal fields, a significant source of coal in India.
- **Hazaribagh plateau** lies to the north of the Damodar River, with an average elevation of approximately 600 meters above sea level. The plateau is characterized by solitary hills and resembles a peneplain, shaped by extensive erosion processes.
- The **Ranchi Plateau** emerges from the southern side of the Damodar Valley reaching an elevation of around 600 meters above sea level. Its terrain is predominantly undulating and encompasses the city of Ranchi, situated at an elevation of 661 meters.
- The **northeastern border** of this plateau is delineated by the **Rajmahal Hills**, predominantly composed of basalt and adorned with lava flows. Stretching from

north to south, these hills attain an average elevation of 400 meters.

- The landscape has undergone extensive dissection, resulting in the formation of separate plateaus within the Rajmahal Hills.

❑ Deccan Plateau:

- The plateau in question spans an area of about **500,000 square kilometres** and boasts a distinctive triangular shape. Its boundaries are demarcated by the Vindhya and Satpura ranges in the northwest, the Maikal and Mahadev ranges in the north, in the east by the Eastern Ghat and in the west by the Western Ghats.
- While the plateau maintains a mean elevation of **600 meters**, it reaches heights of up to 1000 meters in the southern region, gradually descending to 500 meters in the north. This geographical feature exhibits a pronounced west-to-east slope, as evidenced by the directional flow of its major rivers.
- The plateau's topography has been intricately carved by these **rivers**, and results in the formation of numerous smaller plateaus within its expanse.

❑ Baghelkhand:

- Baghelkhand is situated to the north of the **Maikal Range**, a geographical region characterized by its diverse composition. The western expanse comprises sandstones and limestones, whereas the eastern part consists predominantly of granite formations.
- The northern boundary of Baghelkhand is demarcated by the **Son River**. Situated at the heart of this plateau is a crucial hydrological feature, as it functions as a watershed between the Son drainage system towards the north and the Mahanadi river system towards the south.
- The terrain in this region is marked by its undulating topography. Notably, the Bharner and Kaimur regions are in proximity to the trough axis of Baghelkhand.

❑ Dandakaranya Plateau:

- Dandakaranya, a region of historical significance in India according to the **Ramayana**, is located in the central portion of the country within the Bastar region of present-day Chhattisgarh.
- This geographical area covers around **35,600 square miles** and includes the Abujmahad Hills in the west, while its eastern boundaries connect with the Eastern Ghats.
- **Abujmarh**, a densely forested and hilly region in Chhattisgarh, extends through the Bijapur, Narayanpur, and Dantewada districts. It serves as the ancestral homeland for various indigenous tribes of India, like the Gond, Abuj Maria, Muria, and Halbaas.

- Dandakaranya's borders encompass parts of Odisha, Chhattisgarh, Andhra Pradesh, and Telangana states, stretching approximately 200 miles in the north-south direction and 300 miles in the east-west direction.

❑ Central Highlands:

- The region is known alternatively as the **Madhya Bharat Plateau**, or alternatively the Madhya Bharat Pathar is situated east of the Marwar or Mewar Upland.
- This plateau is predominantly characterized by the expansive basin of the Chambal River, which meanders through a rift valley.
- The Chambal River's watershed is nourished by several significant tributaries, including the Kali Sindh, the Banas which flows via the Mewar plateau, and the Parbati and the Parwan, both originating from the state of Madhya Pradesh.
- This plateau is distinguished by its **undulating terrain**, featuring rounded hills primarily composed of sandstone. Within this geographical expanse, dense forests flourish, adding to its natural beauty and ecological significance.
- To the north of the plateau, you will encounter the distinctive **ravines** and badlands characteristic of the Chambal River basin, a unique feature that defines this region's landscape.

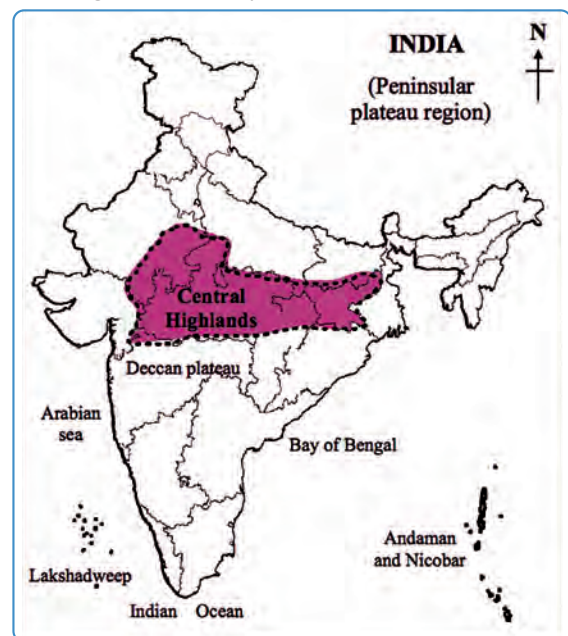


Fig. 3.37: Central Highlands

❑ Bastar Plateau:

- The Bastar Plateau, located in the southernmost region of Chhattisgarh, is a district characterized by its rich forests and abundant mineral resources.

- Compressed between the Godavari and Mahanadi rivers in the southern region of Chhattisgarh, this region is bisected by the Indravati River. Bastar is predominantly inhabited by tribal communities and has long been under the influence of Naxalism.

□ Telangana Plateau:

- The Telangana Plateau, characterized by its unique geological composition, is a remarkable region in India. Comprising primarily **Archaean gneisses and Dharwar rocks**, this plateau also boasts the presence of Gondwana rocks, notably in the Godavari valley, renowned for its abundant coal deposits. The Dharwar rock strata have endowed the plateau with a wealth of mineral resources, contributing significantly to the region's economic importance.
- With an average elevation ranging between **500 to 600 meters**, the Telangana plateau displays an intriguing topographical diversity. The southern part of the plateau stands at a higher elevation compared to its northern counterpart, adding to the region's geographical variation.
- The plateau's hydrology is primarily governed by three major river systems: the Krishna, the Godavari, and the Penner. These rivers play a crucial role in shaping the landscape and sustaining the ecosystem of this plateau region.
- Furthermore, the Telangana plateau is characterized by its division into **Ghats and Peneplains**, the latter being an extensive, featureless, undulating plain formed during the final stages of the deposition process. This unique geological and topographical makeup contributes to the distinctiveness of the Telangana plateau, making it a significant and fascinating part of India's landscape.

□ Karnataka Plateau:

- The Karnataka Plateau, also known by the name Mysore Plateau, is situated to the Maharashtra plateau's south and exhibits the characteristics of a rolling plateau with a mean elevation ranging from **600 to 900 meters**. This region is marked by a complex network of rivers originating from the Western Ghats and extensively dissects the landscape.
- The major streams draining it include the Godavari, Krishna, Kaveri (Cauvery), Tungabhadra, Sharavati, and Bhima. These hills generally align themselves either parallel to or across the Western Ghats. The highest point in this plateau, reaching an elevation of 1930 meters, is found at Mullayanagiri within the Baba Budan Hills, situated in the Chikkamagaluru district.
- Geographically, the plateau can be categorized into two distinct parts: **Malnad and Maidan**. **Malnad**, which translates to "hill country" in Kannada, features deep valleys that are blanketed by dense

forests, creating a picturesque landscape. In contrast, **Maidan** comprises a rolling plain adorned with low granite hills, offering a unique contrast to the rugged terrain of Malnad.

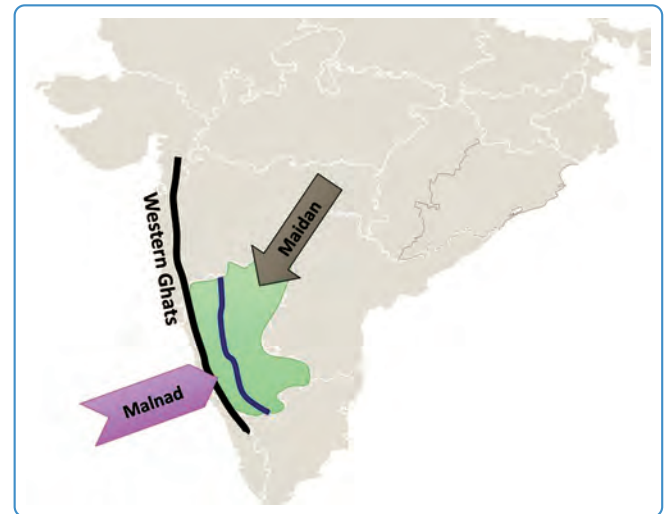


Fig. 3.38: Karnataka Plateau

- As one progresses **southward**, the Karnataka Plateau gradually narrows down between the Eastern Ghats and the Western Ghats, ultimately merging with the Nilgiri hills in the south. This region presents a diverse and captivating topography that encompasses both lush, forested valleys and open, rolling plains.

□ Maharashtra Plateau:

- The Maharashtra Plateau, situated within the state of Maharashtra, constitutes the northern portion of the expansive Deccan Plateau. Predominantly composed of **basaltic rocks** originating from volcanic lava, with a significant portion of the Deccan Traps found in this region, the plateau exhibits an undulating landscape shaped by weathering processes.

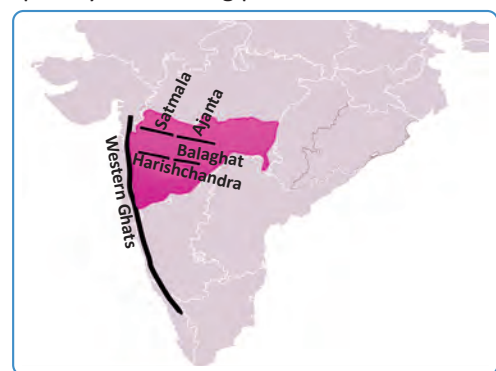


Fig. 3.39: Maharashtra Plateau

- This distinctive Deccan Trap topography, characterized by **step-like formations**, results from the horizontal spread of lava sheets. Alongside, the Bhima, Godavari, and Krishna rivers carve out shallow and broad valleys, with flat-top steep-sided hills and ridges adorning their perimeters. Enveloping the entire area is the renowned black cotton soil, referred to as regur.

Hill Ranges of the Peninsular Plateau

- ❑ Peninsular India exhibits distinctive geographical and climatic characteristics that are synonymous with South India. It takes the shape of an **inverted triangle**, bordered by the Arabian Sea in the west, the Bay of Bengal in the east, and flanked by the Satpura and Vindhya mountain ranges to the north.
- ❑ The majority of the **elevated terrain** within the peninsular region comprises **residual** mountains, which stand as enduring remnants of ancient horsts and hills formed over millions of years. These plateaus within the peninsula are interspersed with various mountain ranges and diverse river basins.
- ❑ Significantly, the historical demarcation between northern and southern India was established by the **Narmada** and **Mahanadi** rivers.
- ❑ To the north, it is shielded by the **Aravallis**, while the northeastern boundary is defined by the Hazaribagh and Rajmahal Hills.
- ❑ The western perimeter is marked by the imposing **Western Ghats**, also known as the Sahyadri Mountains, while the eastern limit is delineated by the **Eastern Ghats**. Notably, the highest peak in Peninsular India, **Anamudi**, rises to an impressive altitude of **2695 meters** above sea level, adding a prominent feature to this diverse and geographically unique region.
- ❑ **Aravali Range:**
 - The Aravali mountain range, which stretches from the northeast to the southwest direction, covers a distance of approximately **670 kilometres** between Delhi and Palanpur in Gujarat.
 - This mountain range is renowned as one of the world's **oldest fold mountains** and is considered the most ancient in India, also known as Fold Mountains or Block Mountains.



Fig. 3.40: Aravali Range

- Originating in the Archaean Era, **4,000 million years ago - 2,500 million years ago**, the Aravallis once boasted towering peaks that nurtured glaciers, with some summits potentially surpassing the height of today's Himalayas.

- Today, the Aravali mountains stand as relict remnants, having endured severe weathering and erosion over millions of years, resulting from the folding process during the **Archaean Era**. Their geographical presence continues beneath the alluvial deposits of the Ganga Plains, extending up to Haridwar. While the Aravali range is notably prominent in Rajasthan, forming a continuous range to the south of Ajmer, where elevations reach up to **900 meters**, its distinctiveness diminishes as it extends into Haryana and Delhi, manifesting as a series of isolated and fragmented ridges beyond Ajmer.
- Notably, certain geographers propose that a section of the Aravali mountain range stretches from the **Gulf of Khambhat to the Lakshadweep Archipelago**, and concurrently, another branch extends into the regions of Andhra Pradesh and Karnataka. The general elevation of this range ranges from 400 to 600 meters, with a few peaks rising above 1,000 meters.
- At the southwestern tip of the range, elevations exceed 1,000 meters, most notably at **Mt. Abu**, which stands at 1,158 meters. The highest peak in the Aravali range, Guru Shikhar, rises to an impressive 1,722 meters and is located in Mt. Abu. Facilitating movement and connectivity, the Dewair, Pipli Ghat, and Desuri passes provide vital transportation links for both roads and railways in the region.

❑ Maikal Range:

- The Maikal Hills constitute a hilly region situated in the eastern section of the Satpuras range, encompassing the Kawardha District in **Chhattisgarh** and the Anuppur district in Madhya Pradesh. This range serves as the source of three significant rivers: **the Narmada, Mahanadi, and Son**.
- Mount Amarkantak stands as the highest peak within the Maikal Hills. This distinctive topographical feature defines a substantial part of Chhattisgarh and also provides a habitat for the renowned Kanha National Park.

❑ Vindhyan Range:

- The Vindhya Range, with a commanding view of the Narmada Valley, forms a prominent escarpment along the northern border of the Narmada-Son Trough—a narrow depression that serves as the conduit for the Narmada River's flow. Stretching approximately **1,200 kilometres** from **Sasaram in Bihar to Jobat in Gujarat**, this range generally maintains an elevation ranging from **300 to 650 meters** above sea level.
- Predominantly, the Vindhyan Range comprises horizontally bedded **sedimentary** rocks of considerable antiquity, constituting a distinct geological formation. To the east, it extends as the Kaimur and Bharner hills.

- Notably, this range plays a pivotal role as a **natural divide**, separating the Ganga river system from the river systems in southern India. Within close proximity, the rivers Chambal, Betwa, and Ken originate, all within a 30-kilometer radius of the Narmada.

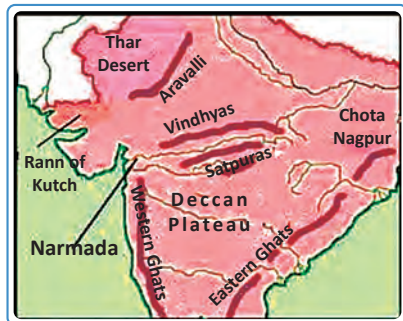


Fig. 3.41: Hills of Peninsular India

□ Satpura Range:

- The Satpura Range comprises seven peaks, where 'Sat' signifies seven and 'pura' denotes mountains. Stretching from east to west, this mountain range lies to the Vindhya's south and is situated between the Narmada and Tapi rivers, running in a generally parallel alignment to these river systems.
- Stretching over a distance of approximately **900 kilometres**, the Satpura Range exhibits notable geological features, including areas that have experienced folding and upheaval, categorizing them as structural uplifts known as 'horst'.
- The highest peak within the Satpura Range is **Dhupgarh**, standing at an elevation of 1,350 meters, and it is situated near Pachmarhi on the Mahadev Hills. Another significant peak in this range is Amarkantak, which reaches an elevation of 1067 meters.

□ Western Ghats (or The Sahyadris):

- The Western Ghats, form the western boundary of the Deccan plateau, extending from the Tapi valley (21° N latitude) to just Kanniyakumari's north (11° N latitude), covering a distance of approximately **1,600 km**.
- The Western Ghats traverse the States of Kerala, Tamil Nadu, Karnataka, Goa, Maharashtra and Gujarat.
- These mountains showcase a distinctive topography characterised by terraced, steep-sided, flat-topped hills facing the Arabian Sea coast. This unique feature results from horizontally layered lavas that have weathered over time, creating a distinctive '**landing stair aspect**.'
- As one moves westward towards the Arabian Sea coast, the Western Ghats rise abruptly to an average elevation of **1,000 meters** from the Western Coastal Plain.
- However, when seen from the Deccan plateau to the east, they appear to slope gently, hardly resembling

a mountain range. South of Malabar, in areas like the Nilgiris and Anamalai, the landscape offers a striking contrast due to distinct geological formations.

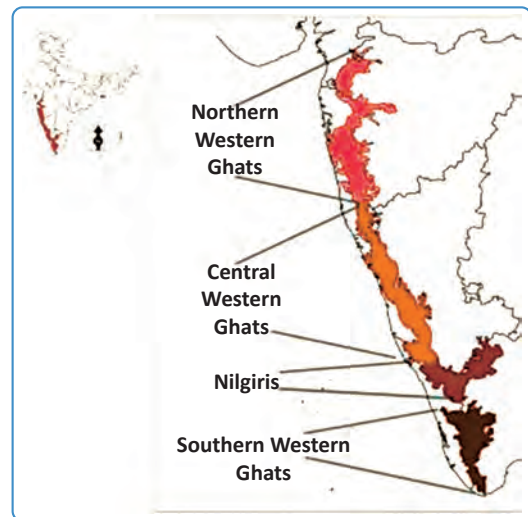


Fig. 3.42: Western Ghats

- The **northern part of the Western Ghats**, stretching from the Tapi Valley to just north of Goa, is primarily composed of horizontal sheets of Deccan lavas, known as the Deccan Traps.
- This region averages an elevation of **1,200 meters** above sea level, with notable peaks such as Salher (1,567 m), Kalasubai (1,646 m), Harishchandragarh (1,424 m), and Mahabaleshwar (1,438 m). Essential passes like Bhore Ghat and Thal Ghat serve as crucial transportation routes, linking the Konkan Plains in the western part to the Deccan Plateau in the eastern part.
- Moving to the **middle section**, spanning from 16°N latitude to the Nilgiri hills, it consists mainly of granites and gneisses, covered densely with forests. The western scarp of this region has been significantly shaped by west-flowing streams through headward erosion. The average elevation here is approximately 1,200 meters, with numerous peaks surpassing 1,500 meters.
- Notable **peaks** include **Kudremukh** (1,892 m), Vavul Mala (2,339 m), and Pushpagiri (1,714 m). Near the junction of Kerala, Karnataka, and Tamil Nadu, the Nilgiri Hills dramatically rise to over 2,000 meters, marking the junction of the Western Ghats with the Eastern Ghats. Prominent peaks in this region include Doda Betta (2,637 m) and Mukurti (2,554 m).
- The **southern section of the Western Ghats** is separated from the main Sahyadri range through the Palakkad Gap, a rift valley. High ranges on either side of this gap terminate abruptly, serving as a vital

transportation corridor connecting the plains of Tamil Nadu and the coastal plains of Kerala.

- The **Palakkad Gap** also facilitates the inland penetration of moisture-laden clouds from the southwest monsoon, providing essential rainfall to the Mysore region. South of the Palakkad Gap, a complex system of rugged and steep slopes characterizes both the western and eastern sides of the Ghats.
- **Anai Mudi** (2,695 m) stands as the highest peak in all of southern India, with three ranges extending in different directions from its base: the Anamalai range (1,800-2,000 m) to the north, the Palani range (900-1,200 m) to the northeast, and the Cardamom Hills or the Yela Mala Hills to the south.

□ Eastern Ghats:

- The Eastern Ghats stretch almost parallel to India's east coast, creating vast plains between their foothills and the shoreline.
- This chain consists of **fragmented** and **discontinuous** hills, extending from Odisha's Mahanadi River to the Vaigai River in Tamil Nadu. They largely fade away between the Krishna and the Godavari Rivers, lacking both structural cohesion and geographical continuity. Consequently, these hill formations are typically treated as separate entities.

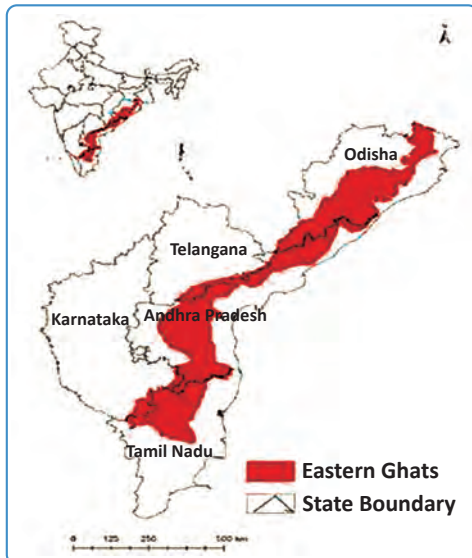


Fig. 3.43: The Eastern Ghats

- In the **northern region**, situated between the Mahanadi and Godavari rivers, the Eastern Ghats exhibit their genuine mountainous character, which is exemplified by the Maliya and Madugula Konda ranges.
- The Maliya range showcases a variety of peaks and ridges, with altitudes ranging between 900 to 1,200 meters. The highest summit within this range is Mahendra Giri, towering at 1,501 meters above sea level.

- On the other hand, the **Madugula Konda** range boasts even greater elevations, with heights ranging from 1,100 to 1,400 meters. Among its notable peaks are Arma Konda (1,680 m), Jindhagada Peak (1,690 m), Gali Konda (1,643 m), and Sinkram Gutta (1,620 m), all of which surpass 1,600 meters in height.
- Between the Godavari and Krishna Rivers, the **middle section** of Eastern Ghats loses its mountainous characteristics and is occupied by Gondwana formations, notably the Krishna Godavari (KG) Basin.
- The Eastern Ghats reemerge as a more or less continuous hill range in the Kurnool and Cuddapah districts of Andhra Pradesh, where they are known as the Nallamala Range, reaching elevations of 900-1100 meters. The southern part of the range is referred to as the Palkonda range.
- Moving into the **southern region**, the hills and plateaus gradually descend in altitude, featuring only the Shevroy-Kalrayan Hills and the Javadi Hills as prominent elevations, both reaching approximately 1,000 meters high. In Karnataka, near the Tamil Nadu border, the Biligiri Rangan Hills soar to a peak of 1551 meters.
- Continuing southward, the Eastern Ghats seamlessly blend with the Western Ghats. It's near the junction of Karnataka, Kerala, and Tamil Nadu that the Nilgiris connect the Western Ghats with the Eastern Ghats, serving as a vital link between these two mountain ranges.

Other Hill Ranges in Peninsular Plateau

- **Satmala Range:** Located in Maharashtra, the Satmala Range is known for its scenic landscapes. It extends from the Ajanta Range in the north to the Harishchandra Range in the south.
- **Ajanta Range:** Also situated in Maharashtra, the Ajanta Range is famous for the Ajanta Caves, a UNESCO World Heritage Site known for its ancient Buddhist cave art.
- **Mahadeo Hills:** These hills are part of the Satpura Range and are located in Madhya Pradesh. They are characterized by dense forests and provide a habitat for diverse flora and fauna.
- **Balaghat Range:** Situated in Maharashtra, the Balaghat Range is part of the Western Ghat and is known for its rich biodiversity and wildlife.

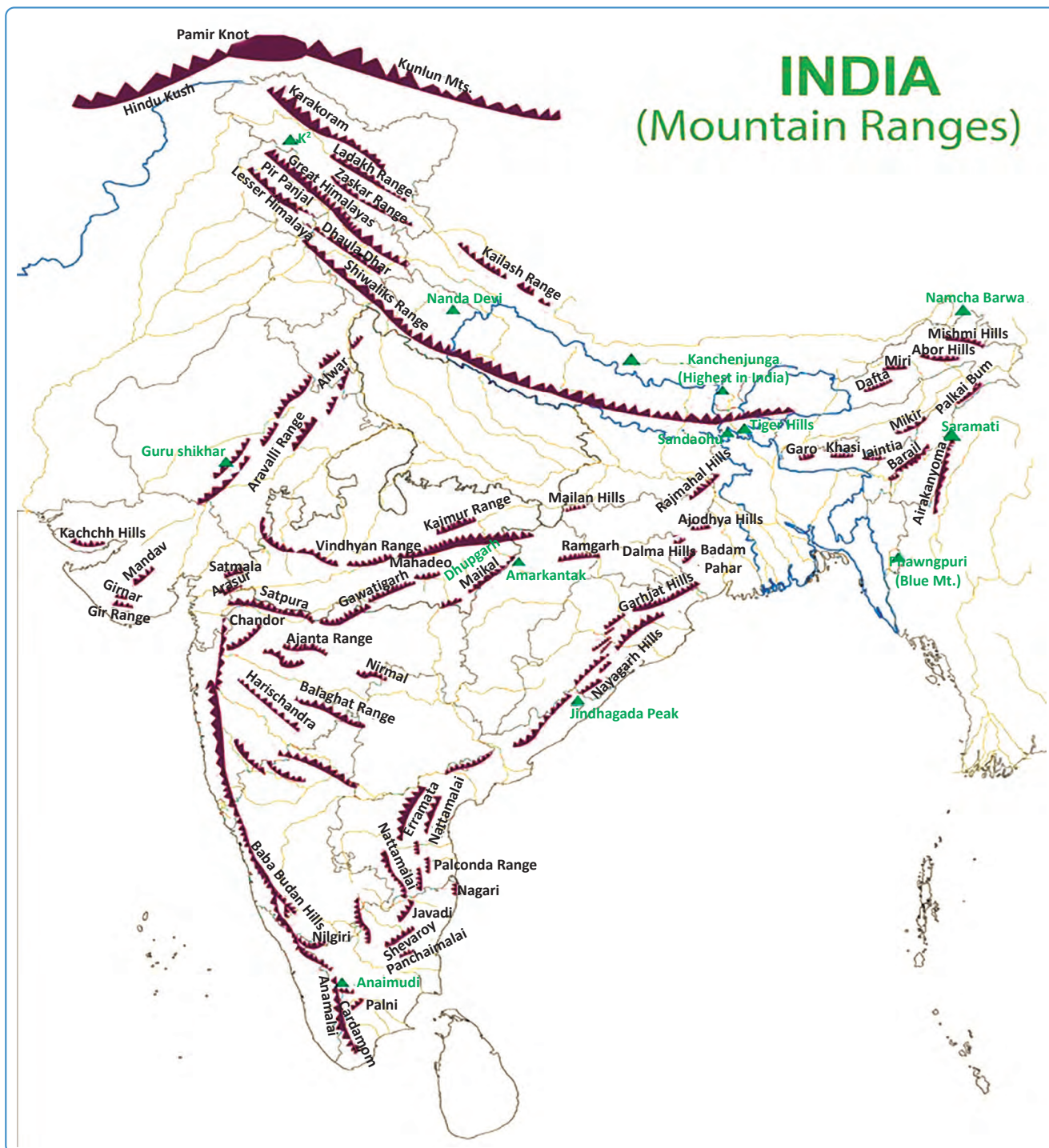


Fig. 3.44: Mountain Ranges of India

- ❑ **Harishchandra Range:** Found in Maharashtra, the Harishchandra Range is part of the Western Ghats. It is named after the legendary sage Harishchandra and is known for its natural beauty.
- ❑ **Palkonda Hills:** Located in Andhra Pradesh, the Palkonda Hills are part of the Eastern Ghats. They are known for their picturesque landscapes and play a crucial role in the regional ecology.
- ❑ **Javadi Hills:** Situated in Tamil Nadu, the Javadi Hills are part of the Eastern Ghats. They are characterized by

dense forests and provide a habitat for various wildlife species.

- ❑ **Shevaroy Hills:** Found in Tamil Nadu, the Shevaroy Hills are part of the Eastern Ghats. They are known for their cool climate and are a popular tourist destination.
- ❑ **Eramala Hills:** Situated in Andhra Pradesh, the Eramala Hills are part of the Eastern Ghats. They are known for their lush greenery and contribute to the biodiversity of the region.

- ❑ **Nilgiri Hills:** Spanning across the states of Tamil Nadu, Karnataka, and Kerala, the Nilgiri Hills are part of the Western Ghats. They are famous for their tea plantations, diverse flora and fauna, and scenic beauty. Ooty and Coonoor are popular hill stations in the Nilgiris.
- ❑ **Biligiri Rangan Hills (B.R. Hills):** Located at the confluence of the Eastern and Western Ghats in Karnataka, the B.R. Hills are known for their biodiversity and the Biligiri Rangaswamy Temple Wildlife Sanctuary.
- ❑ **Palani Hills:** Situated in Tamil Nadu, the Palani Hills are an extension of the Western Ghats. Palani, a prominent pilgrimage town, is located in these hills.
- ❑ **Cardamom Hills:** Also known as the Yela Mala Range, these hills are part of the Western Ghats and are located in Kerala and Tamil Nadu. They are known for their spice plantations, including cardamom.

Differences Between Eastern Ghats and Western Ghats

Western Ghat	Eastern Ghat
❑ Located along the western coast of India, the Western Ghats mark the western boundary of the Deccan Plateau, running parallel to the Arabian Sea. ❑ These mountains are continuous and can only be traversed through designated passes. ❑ The Western Ghats are characterized by higher elevations compared to the Eastern Ghats, with an average elevation ranging from 900 to 1600 meters. ❑ Elevation increases progressively from north to south, with notable peaks like Anai Mudi and Dodabetta. ❑ The Western Ghats encircle a narrow strip between their western slopes and the Arabian Sea, known as the Western Coastal Plain, with a maximum width of 64 km. ❑ This region experiences orographic rainfall, primarily during the summer monsoon season, resulting in a hot and moist climate. ❑ The fertile soil here supports the cultivation of rice, spices, rubber, and various fruits, including coconuts and cashew nuts.	❑ Situated along the eastern coast of India, the Eastern Ghats mark the eastern boundary of the Deccan Plateau, running parallel to the Bay of Bengal. ❑ The Eastern Ghats are characterised by discontinuous, irregular, and intersected rivers that flow into the Bay of Bengal. ❑ The average elevation here is approximately 600 meters. ❑ Prominent peaks in the Eastern Ghats include Mahendragiri and the Javadi Hills. ❑ Similar to the Western Ghats, the Eastern Ghats enclose a strip of land between their eastern slopes and the Bay of Bengal, referred to as the Eastern Coastal Plain. This plain is wider than the Western Coastal Plain, with a maximum width of 120 km. ❑ The Eastern Ghats receive rainfall during both summer and winter, especially through winter monsoons. However, the amount of rain is lower compared to the western strip. ❑ Soil fertility in the Eastern Ghats is not as high as in the Western strip, but it still supports the cultivation of crops such as rice, groundnuts, cotton, tobacco, and coconuts

Significance of the Peninsular Plateau

- ❑ **Relics of Geological History:** The majority of hills in the Peninsular region are remnants from a distant geological past. These hills and horsts formed millions of years ago (horst referring to uplifted blocks and graben to subsided blocks), have endured over time.
- ❑ **Natural Dividers:** These hill ranges, along with various river basins, serve as natural dividers within the Peninsular region. They separate and distinguish the plateaus from each other, contributing to the region's unique topography.
- ❑ **Abundance of Minerals:** The Peninsular Mountains are rich in minerals such as iron, copper, manganese, chromium, mica, bauxite, and gold. These mineral deposits hold significant economic value for the region.
- ❑ **Ideal for Plantation Crops:** Several mountainous areas in southern India, within the Peninsular Mountains, are well-suited for plantation crops. Crops like tea, coffee, and rubber thrive in these elevated terrains, supporting agricultural production.
- ❑ **Diverse Forest Ecosystems:** The highlands of the plateau are adorned with diverse forest ecosystems. These forests yield a wide range of forest products, contributing to the region's ecological and economic diversity.
- ❑ **Hydropower and Irrigation:** The rivers originating in the Western Ghats, a prominent Peninsular mountain range, offer excellent opportunities for hydropower generation. Additionally, they serve as vital sources of irrigation, sustaining agricultural crops across the region.
- ❑ **Hill Resorts:** Several popular hill resorts are nestled within the Peninsular Mountains, including Udagamandalam (Ooty), Kodaikanal, Mahabaleshwar, Khandala, Pachmarhi, Matheran, and Mount Abu. These resorts attract tourists and offer picturesque landscapes and pleasant climates.

Forest Ecosystem of Peninsular India

- ❑ **Tropical Evergreen Forests:** Characterized by tall evergreen trees (>30 m) with buttressed trunks. Predominantly found in the high-rainfall zones in the Western Ghats. Decrease in evergreen-to-deciduous plant ratio with reduced rainfall.
- ❑ **Tropical Semi-Evergreen Forests:** This comprises over 75% of evergreen trees, climbers, and epiphytes. Often located near tropical moist deciduous forests and tropical evergreen.
- ❑ **Tropical Moist Deciduous Forests:** Dominated by deciduous species. Upper canopy irregular, lower storey mainly evergreen trees and shrubs. Predominant in peninsular India with an annual rainfall of 1000-2000 mm. Transition zone to tropical semi-evergreen and dry deciduous forests.
- ❑ **Tropical Dry Deciduous Forests:** Characterized by intermediate-height canopies (<20 m) of deciduous species. Leafless for 1-4 months due to dry spells and frequent fires. Often dominated by teak and sal trees. Now reclassified as savannas in some areas.
- ❑ **Tropical Thorn Forests:** Short canopies (<10 m) primarily consisting of thorny, sun-loving shrubs and grasses. Found in semi-arid regions of peninsular India.
- ❑ **Tropical Dry Evergreen Forests:** With above 75% dominance of hard-leaved evergreen woody plants. Mainly found in the Eastern Ghats of Tamil Nadu and Andhra Pradesh.
- ❑ **Subtropical Broad-Leaf Hill Forests:** Blend of wet evergreen and temperate forest species. Dominated by Lauraceae family species. Located on the lower slopes of the Western Ghats.
- ❑ **Montane Wet Temperate Forests ("Shola"):** These closed-canopied evergreen forests, characterized by their diminutive trees measuring less than 15 meters in height, are adorned with mosses, epiphytes, ferns, and woody climbers. They find their habitat exclusively at elevations exceeding 1500 meters within the Western Ghats.
- ❑ **Littoral and Swamp Forests (Mangroves):** Halophytic evergreen plants featuring stilt roots are a distinctive botanical variety. These plants exhibit a unique characteristic where their seeds begin to germinate while still attached to the parent tree. They are predominantly located in the river deltas situated along the coastal regions of peninsular India.
- ❑ **Scrub Vegetation:** This ecosystem encompasses a blend of both damp and arid open-canopied (<10%) scrub flora. It predominantly comprises deciduous or evergreen shrubs and is found in the arid regions of peninsular India.
- ❑ **Grasslands:** Defined by an abundance of grasses and herbs with limited tree presence, this biome can be found on hilltops and plains, frequently bordering scrub or savanna ecosystems.

DIFFERENT REGIONS OF INDIA

- ❑ **Bundelkhand:**
 - Bundelkhand is a region in central India, spread across parts of the states of Uttar Pradesh and Madhya Pradesh.
 - It is known for its rugged terrain, comprising plateaus and hills, and experiences a semi-arid climate.
 - The region has a rich historical and cultural heritage. Bundelkhand has been home to various ancient kingdoms and is known for its folklore, music, and traditional art forms.
- ❑ **Baghelkhand:**
 - Baghelkhand is a historical and geographical region in central India, primarily in the state of Madhya Pradesh.
 - Similar to Bundelkhand, Baghelkhand features a mix of plateaus, hills, and forests.
 - The region has a diverse cultural landscape with influences from various historical rulers. It is known for its traditional festivals, dance, and music.
- ❑ **Rayalaseema:**
 - Rayalaseema is a region in the southern part of the Indian state of Andhra Pradesh.
 - It is characterized by a semi-arid climate, with rocky terrain and hills.
 - Rayalaseema has a rich cultural heritage, with influences from historical empires. It is known for its distinct cuisine, classical dance forms, and festivals.
- ❑ **Vidarbha:**
 - Vidarbha is a region in the eastern part of the Indian state of Maharashtra.
 - It features a mix of plains and hills. Vidarbha has an agrarian landscape.
 - The region has a diverse cultural heritage with influences from Marathi and Telugu cultures. Vidarbha is known for its traditional handloom and handicraft industries.
- ❑ **Malwa:**
 - Malwa is a historical region in central India, covering parts of Madhya Pradesh and Rajasthan.
 - It features a mix of plateaus, fertile plains, and the Vindhya Range.
 - Malwa has a rich cultural heritage with influences from ancient empires. It is known for its folk music, dance, and traditional art forms.
- ❑ **Konkan:**
 - Konkan is a coastal region along the western coastline of India, spanning parts of Maharashtra, Goa, and Karnataka.
 - It is characterized by lush greenery, beaches, and the Western Ghats.
 - Konkan has a distinct coastal culture with a significant influence on cuisine, festivals, and traditional practices.

□ Mewar:

- Mewar is a historical region in the southern part of Rajasthan.
- It includes the Aravalli Range and is known for its lakes, including Lake Pichola in Udaipur.
- Mewar has a rich Rajput heritage, known for its palaces, forts, and traditional arts.

□ Garhwal:

- Located in the western part of Uttarakhand.
- Predominant language: Garhwali.
- Known for its distinct traditions, folk dances like the “Jhumeila” dance, and festivals such as the “Phool Dei.”

□ Kumaon:

- Located in the eastern part of Uttarakhand.
- Predominant language: Kumaoni.
- Known for its unique culture, traditional music, and festivals like “Harela.”

□ Saurashtra (Gujarat):

- Located in the southwestern part of Gujarat.
- Predominant language: Gujarati.
- Known for its distinct cultural identity, folk traditions, and vibrant festivals like “Navratri.”

□ Mewat (Haryana and Rajasthan):

- Straddles the border regions of Haryana and Rajasthan.

- Significant Muslim population with a distinct cultural identity.
- Known for its cultural richness, traditional crafts, and rural life.

□ Mewar (Rajasthan):

- Encompasses the southern part of Rajasthan, including Udaipur.
- Rich Rajput traditions and historical significance.
- Known for its majestic palaces, lakes, and vibrant festivals like “Gangaur.”

□ Dhundhar (Rajasthan):

- Encompasses the region around Jaipur, the capital city of Rajasthan.
- Rich historical and architectural heritage, including the iconic “Hawa Mahal” and “City Palace.”
- Known for its traditional arts and crafts.

□ Chambal (Madhya Pradesh and Rajasthan):

- Straddles the borders of Madhya Pradesh and Rajasthan.
- Rugged terrain along the Chambal River.
- Known for its wildlife, including the National Chambal Sanctuary.

□ Gird (Madhya Pradesh):

- Located in Madhya Pradesh.
- Diverse historical and cultural influences.
- Known for its archaeological sites, includes Gwalior, Sheopur

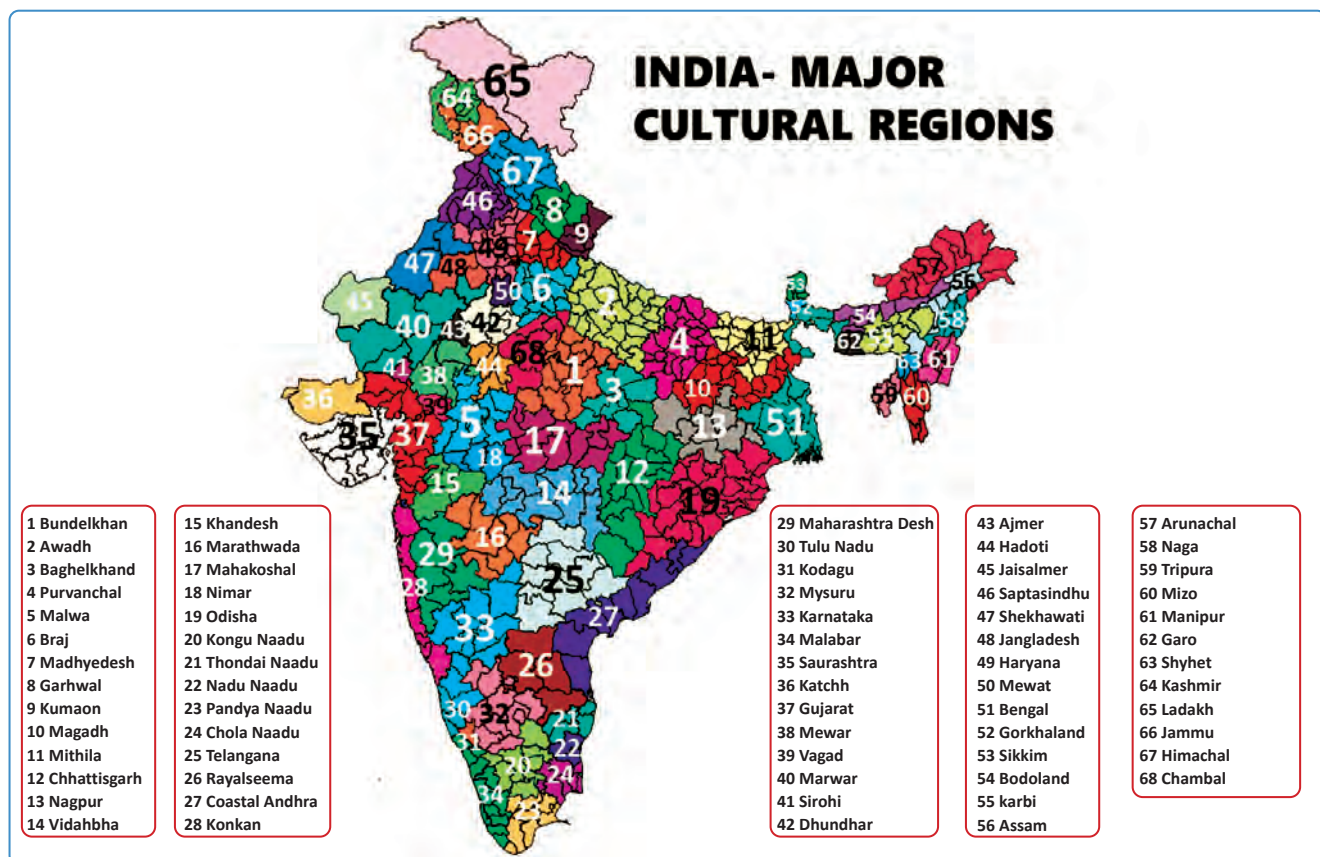


Fig. 3.45: Major Cultural Regions of India

THE DESERT OF INDIA

A desert is a dry area where more water is lost due to evaporation than is received from rainfall. It's a place where evaporation is greater than precipitation. Deserts make up around **33%** of the world's land, with different types like rocky, stony, and sandy.

The vast and arid region known as the **Thar Desert**, often referred to as the Great Indian Desert is situated in the northwestern portion of the Indian subcontinent. It's vast, covering around **200,000 square kilometres**, and acts as a natural border between India and Pakistan. Most of the Thar Desert, about 85%, is in India, while the remaining is in Pakistan. It makes up roughly 4.56% of India's total land area.

Geological History of Desert

- ❑ The Desert landscapes we see today are relatively new in the grand scheme of Earth's history. They resulted from a gradual cooling of the planet and the resulting increase in dryness that took place during a period called the **Cenozoic Era**, which spans from about 65.5 million years ago until now.
- ❑ This cooling process also led to the formation of **savannas** and **scrublands** in regions that were less dry, closer to the tropical and temperate zones near the edges of these emerging deserts.
- ❑ Some scientists believe that many of the plant families we associate with modern deserts, especially those originating from Asia like the Chenipod and Tamarisk families, began to appear during the Miocene period, which occurred between **23 and 5.3 million years ago**.
- ❑ These plant species developed in the saline and arid environments that once existed around the receding Tethys Sea, which is now encompassed by the Mediterranean to Central Asian region.

FEATURES OF DESERTS IN INDIA

The Great Indian Desert, also referred to as the **Thar Desert** is a vast sandy region in India characterized by rolling sand plains covered with sand dunes. These dunes are predominantly crescent-shaped and are known as Barchans

- ❑ The Thar Desert consists mainly of rocks and sand. It has varying elevations, with the eastern side being higher and the western side lower. In some areas, especially near the Aravalli range, the elevation reaches 325 meters, while it's about 150 meters near the western part of Pakistan.
- ❑ You can find salt lakes, locally known as **Rann** in the lower part of the desert and some parts of Jaisalmer, with Sambhar Lake being the largest.
- ❑ Springs exist in certain desert areas, making the land around them somewhat fertile and suitable for palm trees, dates, and grasslands.
- ❑ The Indian Desert can be divided into five regions: **Bagar**, **Rohi**, **Small Desert**, **Marusthali**, and **Desert Region**, each with its unique characteristics and landscape.

THAR DESERT (SANDY DESERT)

The Thar Desert, also known as a tropical desert, is situated at a low latitude. Its name, Thar, is derived from the term **thul**, referring to the region's sand ridges. Covering an expansive area of 200,000 square kilometres, the Thar Desert ranks as the ninth-largest subtropical desert globally.

This place is surrounded by the Indus River plains on one side, the Punjab flatlands to the north and northeast, the Aravalli mountains to the southeast, and the Rann of Kutch to the south. The majority of this desert, over 60%, falls within the state of Rajasthan.

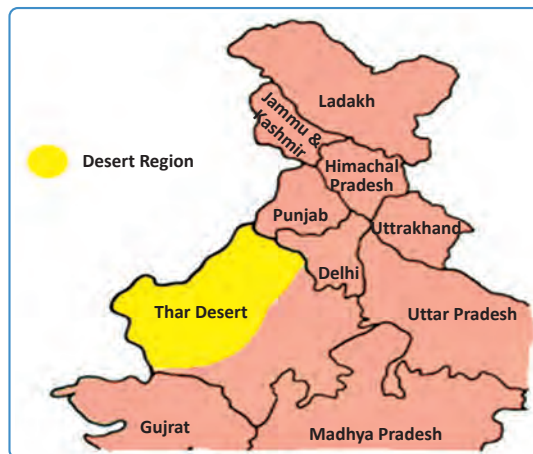


Fig. 3.46: The Thar Desert in India

Landscape and Formation

- ❑ The Thar Desert has a lot of sand that the wind has piled up for about 1.8 million years. This place has big and small sand hills, sandy areas, and low rocky hills called bhakars. The sand dunes in the region are in constant motion, continuously changing in size and shape. Some of the older dunes reach impressive heights of up to 150 meters. The Thar Desert includes playas, locally known as **dhands** scattered throughout. Notable examples include the **Sambhar**, **Kuchaman**, **Didwana**, **Pachpadra**, **Phalodi (Rajasthan)**, **Kharagoda (Gujarat)**, and **Lunkaransar**, which are significant sources of common salt.

Climate and Weather

- ❑ The climate in the Thar Desert is arid and subtropical. It's the 9th largest hot subtropical desert in the world, and its temperatures range from nearly freezing in winter to scorching 50°C in summer. The region experiences very dry weather, with average annual rainfall ranging from 10 cm to 50 cm, primarily during the months of July to September due to the southwest monsoon.

Population and Economic Activities

- ❑ The Thar Desert has a population density of 83 people per square kilometre, making it the most densely populated desert in the world. The region has become a major producer of wool in India. Due to challenging climatic conditions and terrain, animal husbandry has

grown, while kharif crops, primarily bajra, are the main agricultural products.

- ❑ The Thar Desert is increasingly harnessing solar and wind energy to generate electricity. The **Bhadla Solar Park** covers over 5,700 hectares, which is approximately one-third the size of Washington, D.C. With a total capacity of 2,245 megawatts, it ranks among the world's largest solar parks. Its establishment has recently allowed Rajasthan to surpass Karnataka in having the most solar capacity installed among Indian states.



IGNITE YOUR MIND

The Luni River, amid the Thar Desert, is dubbed the “**river of tears**” for its historical importance and past battles on its banks, challenging the perception of deserts as solely arid landscapes. Assess the impact of climate change and human activities on the hydrology of the Luni River and its implications for water resources in the region.

THE COLD DESERT OF INDIA (LADAKH)

Ladakh is India's cold desert, and it is located in the trans Himalayan region on the eastern side of the Jammu and Kashmir region. It is surrounded by the **Karakoram range to the north** and the **Zaskar mountains to the south**. The Indus River flows through Ladakh. The altitude here varies from 3,000 meters in Kargil to over 8,000 meters in the Karakoram.

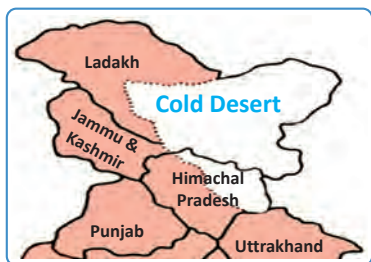


Fig. 3.47: Cold Desert Areas of India

- ❑ **Ladakh** has very thin air, which makes the sun's heat feel quite strong. In the summer, daytime temperatures are just above 0°C, while at night, it can drop as low as -40°C. This region doesn't receive much rainfall, only about 10 centimetres per year, because it's in the rain shadow area of the Himalayan mountains.
- ❑ The area has limited vegetation with small areas of shrubs and grass. In the summer, you can find walnuts, apples, and apricots as the main fruits. The wildlife in Ladakh includes **yaks, wild goats and sheep, and special kinds of dogs.**

Characteristics of Cold Desert of India

- ❑ **Ladakh's Diverse Altitude Range:** In Ladakh, the elevation ranges from roughly 3,000 meters to over 8,000 meters, with the Karakoram Mountains reaching heights of about 8,000 meters. Kargil, located at around 3,000 meters, falls within this altitude range.
- ❑ **Ladakh's Harsh High-Altitude Climate:** Ladakh's climate is extremely dry and cold due to its exceptionally high altitude. During nighttime, temperatures in Ladakh often drop well below -30°C, and throughout the majority of the winter, they remain below -40°C.
 - Because Ladakh is situated within an area protected from rainfall by the mountains of the Himalayas, it gets minimal precipitation, with an average annual precipitation of only 10 centimetres.
 - At these high altitudes, the air is thin, making the sun's heat feel especially intense, resulting in scorching sunlight during the day and chilly winds.



Fig. 3.48: Cold Desert

VEGETATION SYSTEM

- ❑ In both hot and mid-latitude deserts, the most common type of vegetation is xerophytic, which means it's adapted to withstand drought. This includes plants like cacti, thorny bushes, tough and deep-rooted wiry grasses, and scattered dwarf acacias. Trees are quite rare, except in places where there's enough groundwater to support clusters of date palms.
- ❑ Across the western coastal deserts, which are influenced by cold ocean currents like the **Atacama Desert**, there's a limited amount of vegetation. The intense evaporation in deserts leads to increased soil salinity, causing salts to build up on the exterior and form hard layers known as **bajada or palaya**. Because there's very little moisture, the breakdown of organic matter is slow, resulting in desert soils being very low in humus.
- ❑ Most desert shrubs have long roots that are well-spaced out to collect moisture and search for groundwater. These plants typically have few or no leaves, and if they do, the foliage is often waxy, leathery, hairy, or needle-shaped.
- ❑ This helps reduce water loss through transpiration, which is the process of water evaporating from plant leaves. Additionally, the seeds of many grasses and herbs in deserts have tough, thick skins to protect them while they remain dormant.

DRAINAGE SYSTEM

Desert Stream Behavior

- Streams in deserts often dry up quickly, especially before they reach the sea. They remain dry for extended periods but can experience sudden and intense floods during brief heavy rains. These flash floods carry away most of the sediment in desert areas and create fan-shaped deposits of sand, gravel, and boulders at the bases of mountains in deserts. They also carve deep, steep-walled canyons in the mountain regions upstream, the source of the rocks and stones found located at the base of these mountains. In the areas between mountains within deserts, finer-grained materials are deposited by slower-moving currents that result from receding floodwaters.



Fig. 3.49: Sand Dunes in Thar Desert

Risks of Desert Flash Floods

- Flash floods pose significant dangers in desert environments, especially when caused by rainfall in distant areas. More people in deserts die from drowning in flash floods than from thirst or dehydration.
- Often, rain falls in remote mountain regions while downstream residents remain unaware of the rainfall. Desert rainstorms are usually brief but intense, delivering several inches of rain in a short time. This water rapidly flows downstream through mountain canyons, carrying off loose materials within its trajectory. People or automobiles trapped in such a deluge face a significant danger of being swept away by the fast-flowing water.

Dry Lake Beds and Playas

- Many deserts contain arid lake beds located in low-lying flat regions, which may only fill with water sporadically, sometimes taking several years between occurrences. These areas, known as playas or hardpans, often hold deposits of white salts that are created when rainwater evaporates, leaving behind dry lakebeds.



Fig. 3.50: Dry Lake Beds

- These lakes are in large numbers near **Jaisalmer, Rajasthan**. There are small playa lakes near **Khatu, Pokaran, Sujargarh and Kuchaman** in **Rajasthan**.

Badland Topography

- The term badlands was initially used to describe a dry area in South Dakota, USA, where the hills had suffered severe erosion due to occasional rainstorms, forming gullies and ravines.



Fig. 3.51: Playa lake

- The extent of water's impact on the hill slopes and rock surfaces was so significant that this suggests that the whole area was deserted by its inhabitants.
- Deserts exhibiting similar erosion features are now commonly known as **badlands**.
- An example of such badlands is the Painted Desert in Arizona, situated to the southeast of the Grand Canyon of the Colorado River.

Badland Topography in India

- In India, badland topography is commonly found in regions with specific geological formations, such as the **Vindhyan** and **Gondwana** rock formations.
- The **Chambal** and **Son** River basins in central India are well-known for exhibiting **badland topography**.

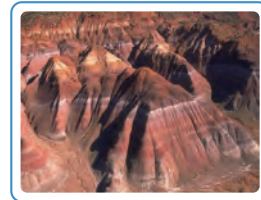


Fig. 3.52: Chambal Badland

- The **Chambal Badlands** are characterized by a maze of deeply eroded gullies, ravines, and rugged terrain.
 - This region spans parts of the states of **Madhya Pradesh and Rajasthan**.
- The dry and arid climate in these areas, coupled with the relatively soft and easily erodible **sedimentary rock** formations, contributes to the development of these unique landscapes.
- Badland topography poses **challenges for agriculture and land use**, as the eroded terrain is often unsuitable for cultivation and may lead to **soil erosion and water runoff** issues.



Fig. 3.53: Regional spread of Chambal Ravines

- However, these areas also hold ecological significance, providing habitats for certain flora and fauna adapted to such harsh conditions.

Alluvial Fans, Pediments, and Inselbergs

- Alluvial fans, composed of coarse-grained deposits, accumulate at the front of mountain canyons in desert regions. These fans, which also contain debris-flow deposits, are prevalent in deserts and hold significant groundwater reserves due to their porous nature.



Fig. 3.54: Ravines of Chambal-Gwalior Region

- In many desert areas, these alluvial fans merge to create an unbroken alluvial apron called a **Bajada**, sloping adjacent to the foundation of mountain ranges as a result of fan coalescence.
- Pediments** are flat expanses in desert areas that incline downward from elevated regions, typically covered by a thin veneer of alluvial deposits and rock fragments. These formations result from the erosion caused by flowing water and shallow channels. Pediments expand as mountains erode.



Fig. 3.55: Pediments

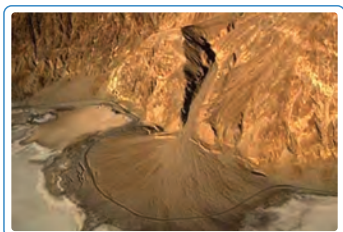


Fig. 3.56: Alluvial Fan

- Inselbergs** are mountains or ridges with sharp, vertical sides that abruptly rise from the flat plains of deserts. Ayres Rock in central Australia is a famous example. They form due to differential erosion, where certain rocks exhibit a higher resistance to weathering and erosion, leaving behind these distinctive desert features.



Fig. 3.57: Inselberg

Agriculture System in Desert

- Agriculture and animal husbandry are the main occupations of the people living in the desert. However, due to irregular rainfall and drought-prone conditions, agriculture is challenging. Land degradation is a significant issue, caused by factors like high animal populations, wind and water erosion, and illegal mining.
- Bajra** is the primary **Kharif crop**, but other crops like **guar**, **jowar**, **maize**, **sesame**, and **groundnuts** are also grown. With the development of irrigation through canals and tube wells, the crop pattern has expanded to include **Rabi crops** like **mustard**, **wheat**, and **cumin seeds**, along with cash crops.
- The government of India, in coordination with international organizations and trusts, has initiated various efforts to promote agricultural activities in the Thar desert, addressing the challenges faced by the region's farmers. Deserts are incredibly dry places where there is very little plant life. You won't typically find trees there, and even shrubs or small plants that grow close to the ground are quite scarce under normal weather conditions.

THE COASTAL PLAINS

India's Coastal Plains are found along the eastern and western shores of the country, flanking the Deccan Plateau. Stretching over approximately **6,150 km**. **Stretching from the Rann of Kutch in the western region to West Bengal in the eastern region**.

- The two primary divisions are the **Western Coastal Plains along the Arabian Sea** and the **Eastern Coastal Plains along the Bay of Bengal**.
- These coastal plains extend from Gujarat's Rann of Kutch to Kanyakumari in the south, both on the western and eastern sides of peninsular India.

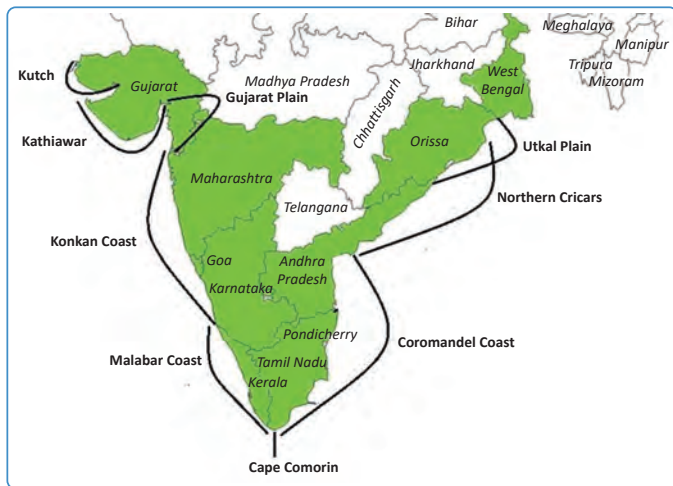


Fig. 3.58: Divisions of Coastal Plains

Formation and Geological Origins

Geologists believe that the origins of India's western and eastern coasts can be attributed to faulting and subsidence processes that occurred in the Arabian Sea and the Bay of Bengal towards the end of the Eocene Period. Consequently, the alluvial deposits found along these coasts are relatively recent, dating from the Pliocene to recent times. These coastal plains provide evidence of both submergence (going underwater) and emergence (rising above water).

THE WESTERN COASTAL PLAINS

Geographical Features of the Western Coastal Plain

- The Western Coastal Plain is located between the Sahyadri Mountains and the Arabian Sea. It stretches for approximately **1600 km.** and varies in width from **10 to 80 km.** The elevation in this region can reach up to 150 meters above sea level and even surpass 300 meters in certain areas.
- The plain is characterized by a variety of features, including sandy beaches, coastal sand dunes, mud flats, lagoons, alluvial areas along rivers, estuaries, laterite platforms, and residual hills.

Western Coastal Plains of India

From the Rann of Kachchh in the northern region to Kanyakumari in the southern region, these plains are relatively narrow, typically measuring around 65 kilometres in width on average.

Kutch and Kathiawar Region

- Kutch and Kathiawar regions, while technically connected to the Peninsular plateau due to their geological composition (with Kathiawar having Deccan Lava and Kutch having tertiary rocks), are typically considered part of the Western Coastal Plains because they have

been largely flattened over time. Historically, the Kutch Peninsula existed as an island enclosed by seas and lagoons.

- Over time, these aquatic regions were progressively filled with sediment transported by the Indus River, which previously coursed through this area. However, due to the recent scarcity of rainfall, this area has transformed into an arid and semi-arid landscape.
- To the **north of Kutch**, there's a salt-soaked plain called the Great Rann, and its southern extension, referred to as the Little Rann, is situated along the coastline and southeast of Kachchh.
- Moving **south** to the Kathiawar Peninsula, its central part consists of the Mandav Hills, which serve as a highland. From this highland, small streams flow outward in various directions, a drainage pattern known as radial drainage.
- The **highest peak** in this region is Mt. Girnar, standing at 1,117 meters, and it has volcanic origins. In the southern region of the Kathiawar Peninsula, one can encounter the Gir Range, characterized by thick forests and renowned as the natural habitat of the Gir lion.

Gujarat Plain

- Situated to the east of Kachchh and Kathiawar, the Gujarat Plain inclines towards the west and southwest. It is formed by the **Narmada, Tapi, Mahi, and Sabarmati** rivers and encompasses the southern territory of Gujarat as well as the coastal areas along the Gulf of Khambhat.
- The eastern segment of this plain is conducive to farming due to its rich soil, while a significant portion of the coastal region is covered with sand deposits called loess, which have been carried by the wind.
- Part of this plan also results from wind-driven deposits and the sea's recession. It contains **Gondwana** rocks known as the Umia Series, which overlay marine Jurassic rocks and are topped by Lower Cretaceous (Apatian) beds. The Deccan lava rests above the Umia series.

Rann of Kutch and Surroundings

The Rann of Kutch is an expansive area characterized by tidal mudflats, crisscrossed by abandoned and active creeks. It is isolated from the Kathiawar Peninsula by the **Gulf of Kutch**. Due to the high salt content in the soil, this low-lying marshy region is almost barren and unproductive.

- The **whitish salt deposits** create the appearance of white bony structures within the dried creek beds. The live creeks form a dendritic drainage pattern, which has been further accentuated by earthquakes. South of the Rann lies Kutch, which used to be an island and is almost entirely Encircled by the Rann except in the southwestern area.

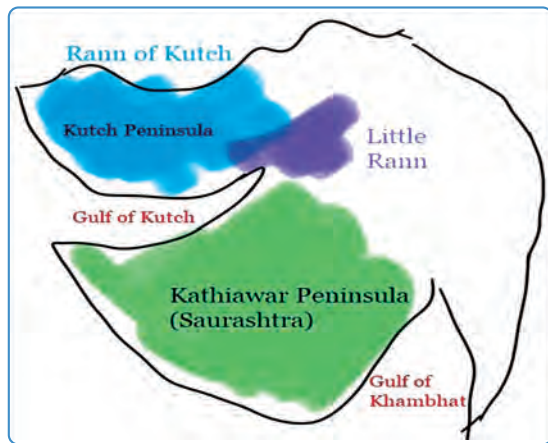


Fig. 3.59: Rann of Kutch and Surroundings

□ Konkan Plain:

- The Konkan Plain, located below the Gujarat Plain, extends from Daman to Goa and is approximately 50 to 80 kilometres wide.
- This region displays signs of marine erosion, such as cliffs, shoals, reefs, and islands in the Arabian Sea.
- One notable geographical feature in this area is the Thane Creek, near Mumbai, which serves as a significant natural bay, offering an excellent harbour for maritime activities.

□ Karnataka Coastal Plain:

- The coastal plain of Karnataka stretches from Goa to Mangalore and is relatively narrow, averaging around 30 to 50 kilometres in width, with the widest section reaching up to 70 kilometres near Mangalore.

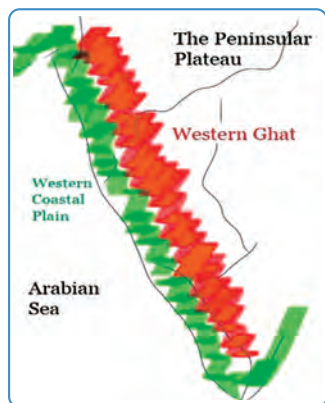


Fig. 3.60: Western Ghat and Coastal Plain

- In certain areas of this region, you can observe that rivers have their source in the Western Ghats descending along steep slopes and forming impressive waterfalls.
- One notable example is the **Sharavati** River, which, as it flows down a steep slope, creates the remarkable Gersoppa (Jog) Falls, measuring a height of 253 meters. (For context, Angel Falls in Venezuela is the tallest waterfall on Earth at 979 meters, and Tugela Falls in South Africa's Drakensberg Mountains is the second highest at 948 meters.) The coastal landscape

in this area also exhibits distinct marine features along the shoreline.

□ Kerala Plain:

- The geographic area recognized as the Kerala Plain or **Malabar** Plain is positioned between Mangalore and Kanyakumari. In contrast to the Karnataka plain, it exhibits notable characteristics of greater width and lower elevation.
- A distinctive characteristic of the Kerala coast is the presence of various water features, including lakes, lagoons, backwaters, and spits. Among these, the backwaters, locally referred to as **kayals**, are shallow coastal inlets running parallel to the shoreline.
- The largest of these backwaters is Vembanad Lake, stretching approximately 75 kilometres in length and 5-10 kilometres in width. This lake also gives rise to a 55-kilometre-long strip of land called a spit.

THE EASTERN COASTAL PLAIN

- The eastern coastal plain extends from the Eastern Ghats to the Bay of Bengal, spanning the shorelines of Odisha, Andhra Pradesh, and Tamil Nadu. These plains have formed over time through the accumulation of alluvial deposits in the coastal region, which includes some of the world's largest deltas.
- The Eastern Coastal Plains mainly consist of recent and tertiary alluvial deposits. These plains are relatively flat and gently slope westward toward the foundation of the Eastern Ghats, with occasional hills breaking the monotonous landscape. The coastline of this plain is straight and features well-defined sandy and shingle beaches, with Marina Beach in Chennai being particularly famous.
- Eastern Coastal Plains, span from the Subarnarekha River in West Bengal to Kanyakumari, primarily owe their formation to the alluvial sediments carried by significant rivers such as the Mahanadi, Godavari, Krishna, and Cauvery.
- Compared to the western coastal plains, the eastern coastal plains are extensive with an **average width of 120km**.
 - These plains can be as wide as **220km** in the **deltaic regions** and as narrow as **35km** in **between the deltas**.
- The eastern coastal plain is known as:
 - **Northern Circars** between the **Mahanadi** and the **Krishna** rivers
 - The term "Circar" is derived from the Persian word "sarkar," meaning district or region.
 - **Carnatic plains** between the **Krishna** and **Cauvery** rivers.
 - This region is often referred to as the **Coromandel Coast**.

Physiographic Divisions of Eastern Coastal Plains

❑ Utkal Plain:

- The Utkal Plain encompasses the coastal regions of Odisha and contains the Mahanadi River delta. The key geographical highlight of this plain is **Chilka Lake**, the largest brackish water lake in India, with its size fluctuating from 906 square kilometers in winter to 1165 square kilometers during the monsoon season. In the southern segment of this plain, you'll find small hills scattered across the landscape.

❑ Andhra Plain:

- The Andhra Plain lies to the south of the **Utkal** Plain and stretches down to Pulicat Lake, which is separated by a long sand spit called Sriharikota Island, home to the ISRO launch facility.
- The most notable character of this plain is the formation of river deltas by the **Godavari and Krishna rivers**. These two deltas have combined over time to create a single landform.
- Remarkably, this merged delta has advanced about 35 kilometres towards the sea in recent years, evident from the current location of **Kolleru** Lake, which used to be a coastal lagoon but is now situated farther inland. In terms of the coastline, this area of the plain exhibits a linear configuration shoreline but lacks good natural harbours, with only Vishakhapatnam and Machilipatnam being notable exceptions.

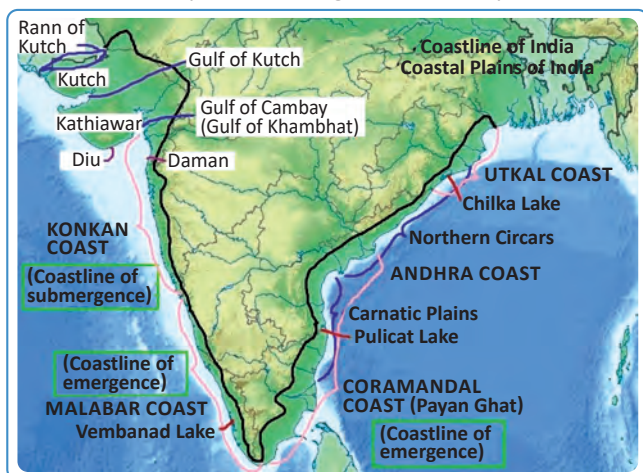


Fig. 3.61: Coastal Plains of India

❑ Tamil Nadu Plain:

- The Tamil Nadu Plain stretches for 675 kilometres along the Tamil Nadu coastline, spanning from Pulicat Lake to Kanyakumari, with an average width of about 100 kilometres.
- Notably, the Cauvery Delta, a prominent feature of this plain, expands to a width of 130 kilometres. Thanks to its fertile soil and extensive irrigation systems, the Cauvery Delta is referred to as the breadbasket of South India.

Important Lagoons and Lakes at Eastern Coast

- The Eastern coast is home to several significant lagoons in India. Among them, **Chilka**, situated southwest of the **Mahanadi delta**, measuring 65 kilometres in length and 8 kilometres in width. **Kulleru Lake** lies in between the deltas of the **Godavari and Krishna rivers**, while the Pulicat Lake is located further south, straddling the border between Andhra Pradesh and Tamil Nadu.

COASTLINES IN INDIA

- India boasts a coastline that spans **7,516.6 kilometres**, comprising 6,100 kilometres of mainland coast and an additional coastline formed by 1,197 Indian islands. This coastal region touches 13 different States and Union Territories.
- The relatively straight and uniform shape of India's coastline is a result of geological events that took place during the Cretaceous period, specifically the faulting of Gondwanaland. This geological process, known as continental drift, shaped India's present coastline. Due to this unique geological history, India's coastline lacks numerous natural harbours suitable for maritime activities.

East Coast of India

- The eastern part of India, known as the East Coast, lies between the **Eastern Ghats** and the Bay of Bengal. It stretches all the way from the Ganga Delta up to Kanyakumari. This coastal region is characterized by the presence of river deltas like the Mahanadi, Godavari, Krishna, and Cauvery.
- Notable geographical features on the East Coast include **Chilka** Lake and Pulicat Lake, which are both important lagoons.
- Different regions along the East Coast have distinct names. In Orissa (Odisha), it is known as the **Utkal** Coast. Moving southward from the Utkal plain, we have the Andhra Coast. Further down, we reach the Tamil Nadu Coast.
- The **Coromandel** Coast, also referred to as Payan Ghat encompasses the collective expanse of the Tamil Nadu Coast and portions of the Andhra Coast. Starting from False Divi Point in Andhra Pradesh, near the Krishna River Delta in the northern region, this coastal stretch continues southward until it reaches Kanyakumari.

Chandipur Beach in Odisha is famously referred to as the "hide-and-seek" beach, where the sea undergoes a disappearing and reappearing phenomenon twice a day. This natural occurrence is attributed to the ebb and flow of tides, causing the sea to withdraw 2–5 kilometers during the ebb tide and then return to the shore during high tide. The receding water leaves behind sand dunes, and upon its return, it brings horseshoe crabs and red crabs from farther reaches of the beach.

West Coast of India

- ❑ The western coastline of India stretches from the northern Gulf of Cambay, alternatively referred to as the Gulf of Khambhat, to the southern Cape Comorin, also known as Kanyakumari. This coastal region is divided into three primary segments, progressing from north to south: the Konkan Coast, the Karnataka Coast, and the Kerala Coast.
- ❑ The shaping of this coastline is primarily attributed to the accumulation of alluvial sediments carried by short rivers originating from the Western Ghats.
- ❑ Along this coastal expanse, one can discover numerous small bays referred to as coves, narrow protected water passages termed creeks, and a handful of estuaries, including the prominent ones like the Narmada and Tapi estuaries.
- ❑ The Kerala Coast, usually known as the Malabar Coast, is distinct in that it features lakes, lagoons, and backwaters, with the most substantial of them being Vembanad Lake.
- ❑ In terms of regional names, the Konkan Coast includes the coasts of Maharashtra and Goa, while the Malabar Coast encompasses the coastal regions of Kerala and Karnataka.

Emergence and Submergence Coastline

- ❑ Coastlines can be classified into, emergence and submergence. An emerging coastline forms when the land either rises up or when the sea level drops. Conversely, a submergence coastline occurs when the land sinks or when the sea level rises.
- ❑ Emergence coastlines are characterized by specific features such as bars, spits, lagoons, salt marshes, beaches, sea cliffs, and arches, which are created due to the land rising or the sea level falling. In the context of India's coastlines, the eastern coast, particularly the southeastern part known as the coast of Tamil Nadu represents a significant example of an emerging coastline.
- ❑ Conversely, the west coast of India exhibits characteristics of both emergence and submergence. The northern section of this coast is submerged due to geological faulting, while the southern portion, referred to as the Kerala coast, is an instance of an emerging coastline.
- ❑ Regional examples include, The Coromandel coast in Tamil Nadu is a representation of an emerging coastline. The Malabar coast in Kerala is also an example of an emerging coastline. However, the Konkan coast, encompassing Maharashtra and Goa, demonstrates features of submergence.

SIGNIFICANCE OF THE COASTAL PLAIN REGION

- ❑ The coastal plains in India are known for their high agricultural productivity. While the western coast specializes in growing specific tropical crops, the eastern coast has experienced a significant agricultural transformation, particularly in rice production, thanks to the green revolution.
- ❑ The delta areas along the eastern coastal plains have a well-developed network of canals that traverse the river tributaries, facilitating agriculture.
- ❑ These coastal plains are a source of valuable resources, providing salt, monazite (used in nuclear power), mineral oil, and gas, and serving as crucial fishing centres.
- ❑ These regions boast a multitude of major and minor ports, making them hubs for trade and leading to the establishment of dense human settlements.
- ❑ The coastal parts of India are renowned for their tourist destinations, thriving fishing activities, and salt production.

THE ISLANDS

India is home to more than 615 islands, with the majority, 572, located in the Bay of Bengal, and the remaining 43 situated in the Arabian Sea. Among the **572 islands in the Andaman and Nicobar group**, only 36 are inhabited.

- ❑ Notably, the islands in the Bay of Bengal, like the Andaman and Nicobar Islands, have primarily formed due to tectonic and volcanic processes, while those in the Arabian Sea are primarily composed of coral formations.
- ❑ Additionally, there are several offshore islands found at the mouth of the Ganga River, along the eastern and western coasts, and in the Gulfs of Khambhat, Kachchh, and Mannar.

Generally, Islands are Divided into two Categories of Islands

- ❑ Continental Islands
- ❑ Oceanic Islands

Continental Islands

At one time, these islands were linked to the mainland but are now separated from it. This separation can occur due to factors such as a shallow lagoon or a deep channel, which may result from land sinking or a rise in sea levels, causing the lowland connections to become submerged by the sea. Evidence of their past connection to the adjacent mainland has the potential to be seen in their similar physical structure, and plant and animal life found on either side of the channel. Over time, human activity and natural forces may lead to changes in their surface features, but the fundamental structural characteristics will remain unchanged. These

types of islands that were once a segment of the continent are known as continental islands.

These islands can be further segregated into three types:

- ❑ **Individual Islands:** Islands that are separate from the mainland but still closely resemble the characteristic features of the mainland they were once connected to are called individual islands.



Fig. 3.62: Individual Island

- A few notable examples include Newfoundland, which is divided from the mainland by the Strait of Belle Isle; Madagascar, separated by the Mozambique Channel; Ceylon, separated by the Palk Strait; Tasmania, separated by the Bass Strait; and Taiwan, separated by the Formosa Strait.
- **Examples (India):** Sriharikota Island, Elephanta Island, Havelock Island (Andaman and Nicobar Islands) and St. Mary's Island (Karnataka)
- ❑ **Archipelagoes or Island Groups:** Archipelagoes, also known as island groups, consist of collections of islands that come in different sizes and shapes.



Fig. 3.63: Archipelagoes or Island Group

- For example, we have the British Isles, the Balearic Islands found in the Mediterranean, and others located in the Aegean Sea.
- **Examples (India): The Andaman and Nicobar Islands** form a group of islands in the Bay of Bengal. This archipelago consists of around 572 islands, and notable ones include North Andaman, South Andaman, and Little Andaman in the Andaman group, and Great Nicobar in the Nicobar group.
- ❑ **Festoons or Island Arcs:** These are groups of islands that create a loop-like shape along the coastline of a continent. These island arcs often follow the pattern of underlying mountain ranges on the mainland.
- Examples include the East Indies, the Aleutian Islands, the Ryukyu Islands, the Kurile Islands, and other island arcs found along the Pacific coast.

- **Examples (India): The Lakshadweep Islands**, situated in the Arabian Sea, can be considered a festoon or island arc. It is a group of coral atolls and islands formed in an arc-like structure. Major islands include Minicoy, Agatti, and Kavaratti.

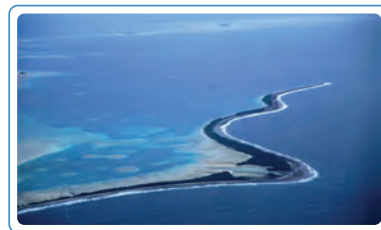


Fig. 3.64: Festoons or Island Arcs

Kuril Islands

A group of four islands lies in the **Sea of Okhotsk**, near Japan's Hokkaido prefecture. **Japan** calls them the Northern Territories, Russia names them the **Kuril Islands**. These volcanic islands, part of the **Pacific Ring of Fire**, boast several volcanoes and numerous hot springs. Russia has controlled them since World War II's end, although both Russia and Japan assert sovereignty, with Japan claiming ownership since the early 19th century.

Oceanic Islands

These are typically small and are situated in the middle of oceans, far away from the mainland, which can be hundreds or even thousands of miles away. They have their own distinct plant and animal life that is different from that of the continents.

For **example**, the **Galapagos** Islands are home to many unique animal species. Because of their remote location, far from major trade hubs, most oceanic islands have very few people living on them. However, some of these islands serve as important stopping points for aeroplanes and ocean-going ships that travel among continents across large expanses of water. These islands can be further segregated into two categories:

❑ Volcanic Islands:

- These are a common feature in our oceans, often being the highest points of undersea volcano cones. While most of these volcanic islands are no longer active, there are still some that remain active.
 - ❑ For example, Mauna Loa in Hawaii, the most famous volcanic peak in the Pacific Ocean, stands at an impressive 13,680 feet above sea level. It's worth noting that when we investigate its beginnings beneath the sea, we discover that Mauna Loa actually commenced its formation on the ocean floor, situated 18,000 feet deep below the water's surface.
 - ❑ **Narcondam Island in the Andaman and Nicobar group** is considered a volcanic island. It features a stratovolcano, an active volcano with the last eruption recorded in November 2022.



Fig. 3.65: Volcanic Island

❑ Coral Island:

- Coral islands are quite different from volcanic islands in that they are much lower in elevation and only rise slightly above the water's surface. These islands are constructed by various species of coral animals and can be located close to the mainland's shores or in the middle of oceans.



Fig. 3.66: Coral Island

- ❑ Examples of coral islands include the Marshall Islands and Gilbert and Ellice Islands in the Pacific Ocean, Bermuda in the Atlantic Ocean, and the Laccadives and Maldives in the Indian Ocean.
- ❑ **Agatti Island, Bangaram Island and Bitra Island in Lakshadweep** are coral islands characterized by its coral reefs.

Katchatheevu Island

Katchatheevu, a uninhabited island in the **Palk Strait**, shifted ownership from India to **Sri Lanka in 1974** through an agreement signed by Prime Ministers Indira Gandhi and Sirima **R.D. Bandaranaike**. This accord granted fishing rights, net drying, and pilgrimage access to Indian nationals.

ISLANDS OF THE BAY OF BENGAL

Many islands are found in the bay, such as the Andaman Islands, Nicobar Islands, and India's Mergui Archipelago, as well as Myanmar. To the northeast, near the Burmese coastline, the Cheduba island cluster is notable for its mud volcanoes, which occasionally exhibit signs of activity.

Andaman and Nicobar Islands

The Andaman and Nicobar Islands comprise two island groups that are divided by the Ten Degree Channel. They are located about 900-2200 km away from the mainland, with the southernmost point being the Indira Point on Great Nicobar Island.

- ❑ The islands have a warm tropical climate throughout the year, characterized by two monsoon seasons. The average temperature is around 25°C, and the sea breeze provides a cooling effect. Rainfall is heavy, with monsoons from mid-May to mid-September and November to mid-December.
- ❑ Various water bodies separate these islands, such as the Duncan Passage between Little Andaman and South Andaman, the **Ten Degree Channel** separating the Great Andaman group from the Nicobar group, and the Grand Channel between Great Nicobar and Sumatra in Indonesia.
- ❑ Additionally, the **Coco Strait** lies between the North Andaman Islands and the Coco Islands of Myanmar. The capital of the Andaman Nicobar Islands is **Port Blair**, situated in South Andaman.
- ❑ Among the Nicobar islands, Great Nicobar is the largest, located at the southernmost tip, while Car Nicobar is the northernmost.
- ❑ Many of these islands are surrounded by coral reefs and are covered by dense forests. The region is famous for its mountainous terrain, with **Saddle Peak** in North Andaman being the highest point at 737 meters.
- ❑ The Andaman and Nicobar Islands experience a tropical marine climate influenced by seasonal monsoon winds. The area is characterized by dense tropical rainforests, and mangrove forests are found in coastal regions.
- ❑ The Andaman and Nicobar Islands are usually called the **Emerald Islands**. The state animal of Andaman is the **dugong**, a herbivorous marine mammal endemic to the Indo-Pacific coastal areas, especially the Andaman Islands.



Fig. 3.67: Andaman And Nicobar Islands

Geology and Biodiversity

- ❑ The islands are densely forested and have diverse marine life around their reefs. They are a haven for birdwatchers, boasting 246 recorded species. The entire region falls within a major earthquake zone, and **Barren Island** in the Andamans has an active volcano. Within the Bay of Bengal, you can find two volcanic islands, namely Barren and Narcondam, positioned approximately 80 kilometres to the east of the Andaman Islands.
- ❑ The Andaman Islands were formed as an extension of the Tertiary mountain chain of **Arakanyoma**, primarily consisting of rocks like sandstone, limestone, and shale similar to the Himalayas. The Nicobar Islands, on the other hand, are mainly of coral origin.

Agriculture and Indigenous Tribes

- ❑ Rice is the primary crop in the Andaman and Nicobar Islands, while cash crops like coconut and arecanut thrive in Nicobar. Tropical fruits like pineapple, various bananas, sweet papaya, and mango are grown on a smaller scale in the Andaman group.
- ❑ Unfortunately, the tribal population in the Andaman Islands is diminishing, with most current inhabitants being migrants from Bangladesh, Myanmar, India, and Tamils from Sri Lanka. Some surviving tribes in the Andamans and Nicobar include the **Onges, Jarawas, and Sentinelese**.

Unique Fauna

- ❑ The Wandoor Marine National Park in South Andaman and Great Nicobar Islands is home to one of the world's largest and rarest crabs, the Giant Robber Crab. These crabs have powerful claws that enable them to climb coconut trees and break open the hard shells of coconuts for food.
- ❑ **Dugong (Dugong dugon)**: Also known as the **sea cow**, the dugong is a marine mammal found in the coastal waters of the Andaman and Nicobar Islands.
 - It is listed as "**Vulnerable**" on the IUCN Red List.



IGNITE YOUR MIND

Dhanushkodi, a former vibrant trade hub on Pamban Island, Tamil Nadu, was devastated by a cyclone in 1964. Now a ghost town, remnants of the railway station and church remain along the coastline. Discuss the factors contributing to the vulnerability of the Indian coastline and assess the impact of anthropogenic activities on coastal ecosystems and suggest sustainable measures for coastal management.

- ❑ **Narcondam Hornbill (Rhyticeros narcondami)**: The Narcondam Hornbill is a species of hornbill found exclusively on Narcondam Island, which is part of the Andaman and Nicobar Islands.

- It is recognized for its distinct appearance and is classified as a '**vulnerable**' species according to IUCN.

- ❑ **Nicobar Tree Shrew (Tupaia nicobarica)**: The Nicobar Tree Shrew is an endemic species found in the Nicobar Islands, including some of the southernmost islands of the Andaman and Nicobar archipelago.
 - It is a small mammal belonging to the family Tupaiidae.
- ❑ **Andaman Day Gecko (Phelsuma andamanensis)**: This is a species of day gecko that is native to the Andaman Islands. Day geckos are known for their vibrant colors, and the Andaman Day Gecko is no exception.
- ❑ **Andaman Horseshoe Bat (Rhinolophus cognatus)**: Endemic to the Andaman Islands, this species of horseshoe bat is adapted to the island's unique environment. It is known for its distinctive horseshoe-shaped nose leaf.

The Bay of Bengal island groups consist of about 572 islands/islets. These are situated roughly between 6°N - 14°N and 92°E - 94°E. The two principal groups of islets include the Ritchie's archipelago and the Labyrinth Island.

Ritchie's Archipelago

- ❑ **Ritchie's Archipelago** is a group of smaller islands located 20 kilometres to the east of the primary Andaman Islands, known as the Great Andaman. Within Ritchie's Archipelago, you can find islands like Neil Island and Havelock Island.
- ❑ Recently, the names of some of these islands were changed: Ross Island is now called Netaji Subhas Chandra Bose Dweep, Neil Island is known as **Shaheed Dweep**, and Havelock Island has been renamed as **Swaraj Dweep**. Ross Island, situated in the South Andaman region, is located just 3 kilometres to the east of Port Blair.

Labyrinth Archipelago

- ❑ The Labyrinth Archipelago has **15 small islands** and is situated **south west of South Andaman**. They are now part of **Mahatma Gandhi Marine National Park**.
- ❑ Major islands include **Boat Island, Jolly Buoy Island** and **Red Skin Island**.

THE ARABIAN SEA ISLANDS

Arabian Sea is home to many islands in which some are habited and some are not. Kavaratti, located on an island bearing the same name, serves as the capital of Lakshadweep. These islands are separated from the Maldives by the Eight Degree Channel and lack significant hills and streams.

Lakshadweep Islands

- ❑ The Lakshadweep Islands are situated in the Arabian Sea, to the southwest of the Indian subcontinent.
- ❑ The latitudinal extent is between **8° N and 12° N**, and longitudinal extent is between **71° E and 74° E**.

- ❑ These islands were formerly referred to as **Laccadive, Minicoy, and Amindivi Islands**, with the name “Lakshadweep” officially adopted on November 1, 1973.
 - The name “Lakshadweep” in Malayalam, the official language, means **one lakh islands**.
 - The main languages spoken are **Malayalam, Jeseri, and Mahl**.
- ❑ **Fishing** is the primary industry in this tiny Union Territory, covering just **32 square kilometres** in total.
 - **Kavaratti** is the capital, and it falls under the authority of the **Kerala High Court**.

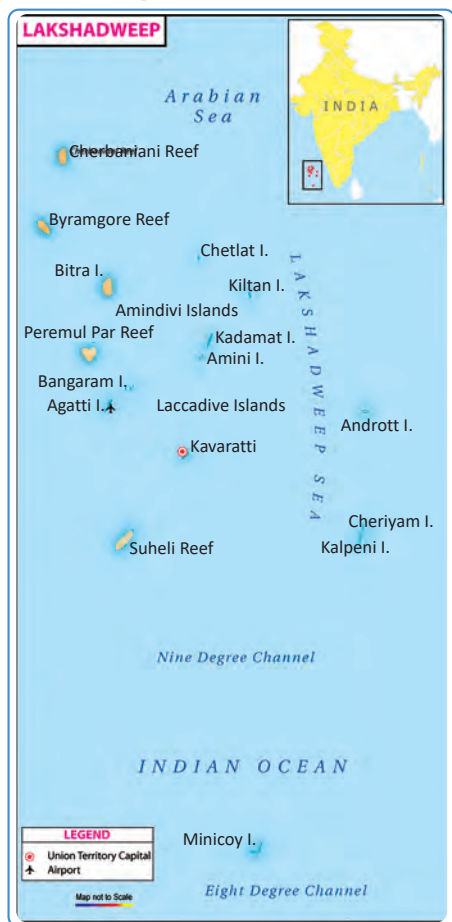


Fig. 3.68: Lakshadweep Islands

Evolution and Formation of Lakshadweep Islands

- ❑ **Lakshadweep** is a tropical archipelago of **36 atolls** and **coral reefs** located in the **Laccadive Sea**, situated 200 to 440 km away from the Kerala coast.
 - Among these, 10 are inhabited, but one island, **Parali 1**, has **submerged** due to sea erosion.
- ❑ The islands are the result of the accumulation of coral debris and the growth of coral organisms in response to favorable environmental conditions, including warm sea temperatures, clear waters, and low wave energy.
 - It is believed that these coral formations developed atop submarine banks and began to evolve around 7,000 to 5,000 years ago.

- ❑ The islands are situated on the **Chagos-Laccadive Ridge**, a submarine feature in the Indian Ocean.

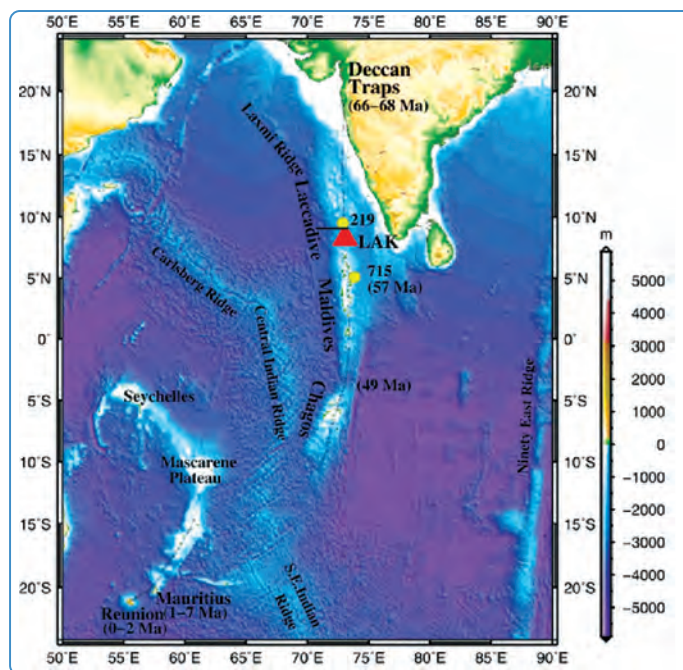


Fig. 3.69: Chagos-Laccadive Ridge

Unique Flora and Fauna

The flora and fauna of Lakshadweep are adapted to the island's unique ecosystem, which includes coral reefs and lagoons.

Flora

- ❑ The vegetation is limited due to the low elevation and sandy nature of the islands. Some common plant species include **coconut palms, screw pines, and various shrubs**.
- ❑ Mangroves are found in some areas, especially along the shores and lagoons.

Fauna

- ❑ The coral reefs around Lakshadweep are rich in marine life. The coral formations support a diverse range of fish species, including **butterflyfish, parrotfish, and angelfish**.
- ❑ **Sea turtles**, such as the **green sea turtle** and **hawksbill turtle**, nest on the beaches of the islands.
- ❑ The surrounding waters are frequented by **dolphins** and various species of **whales**.
- ❑ Bird species include **seabirds** like **terns, gulls, and herons**.

Minicoy - The Largest Island

- ❑ **Minicoy** covers an area of 4.8 sq km. It features a lighthouse and a weather observatory. Fishing is the primary occupation of the local population in Lakshadweep, and coconut is the predominant crop, although they also cultivate pulses and vegetables. The surrounding sea is abundant in marine life.

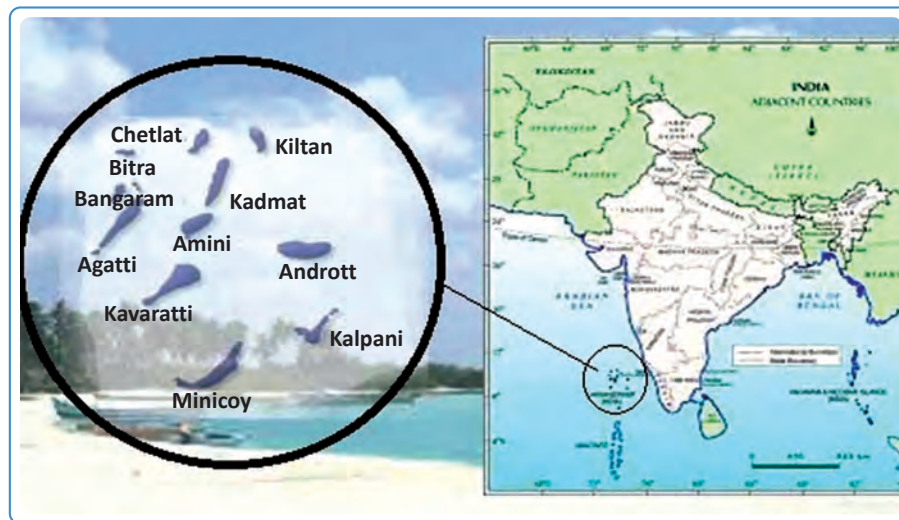


Fig. 3.70: Some Lakshadweep Islands

Lakshadweep's Economic Activities

Lakshadweep's economy is largely reliant on fishing, given its coastal location. Coconut cultivation is the primary agricultural activity, with additional crops like pulses and vegetables being grown. The region's proximity to the rich marine ecosystem of the Arabian Sea contributes to its livelihood and sustenance.

Offshore Islands

The coastal and delta regions of India are home to several islands.



Fig. 3.71: Sagar Island

- Along the western coast, there are islands such as Piram, Bhaisala (Kathiawar), Vaida, Nora, Pirotan (Pirothan), Karunbhar (Kachchh coast), Khadiabet, Aliabet (Narmada-Tapi mouths), Butchers, Elephanta, Karanja, Cross (near Mumbai), Bhatkal, Pegioncock, St. Mary (Mangalore

coast), Anjidiv (Goa coast), Vypin near Kochi, Pamban, Crocodile, Adunda (Gulf of Mannar), Sri Harikota (mouth of Pulicat Lake), Paikud (mouth of Chilika Lake), Short, Wheeler (Mahanadi-Brahmani mouth).

- Additionally, there is Ganga-Sagar in the Ganga Delta. It's important to mention that numerous of these islands lack human inhabitants and are governed by the neighbouring states.

Other Important Islands

□ New Moore Island

- These islands are also known as **South Talpatti** and **Purbasha Island**, it is a small, unoccupied sandy landform in the Bay of Bengal, situated near the Ganges-Brahmaputra Delta area. It first emerged in the Bay of Bengal back in November 1970, following the Bhola cyclone, and it periodically appears and disappears due to natural forces.
- Interestingly, even though this island had no permanent residents or structures, both India and Bangladesh declared ownership of it. The reason behind this dispute was related to rumours suggesting the presence of oil and natural gas resources in the vicinity.



Fig. 3.72: New Moore Island

- The disagreement between India and Bangladesh extended beyond the issue of maritime border determination, involving questions of sovereignty as well, particularly regarding the Radcliffe Award method used for settling such matters.

❑ **Diu Island**

- This island is a petite offshore landmass situated along the southern shoreline of **Kathiawar**, India. Located in the Junagadh district within the **Gulf of Cambay**, it is separated from the Gujarat Coast by a Tidal Brook. The island boasts limestone cliffs, rugged inlets, and sandy shorelines, with **Nagoa Beach** being one of the most renowned. Diu Island is famed for its historic Diu Fort and scenic beaches, and its most notable feature is the expansive fort built by the Portuguese.

❑ **Majuli Island**

- This is a significant river island situated on the **Brahmaputra River** in Assam came into existence due to alterations in the path of the Brahmaputra River and its associated tributaries, especially the Lohit River.
- Historically, it was land situated amid the Brahmaputra River to the north and the Burhidihing River to the south.
- The transformation of Majuli into an island was a consequence of shifts in the Brahmaputra River's flow caused by earthquakes during medieval times. Additionally, Majuli holds deep spiritual significance for Assamese **Neo-Vaishnavites**.



IGNITE YOUR MIND

The Thar Desert's land surface has undergone Aeolian deposition, resulting in the buildup of sand over a period of approximately 1.8 million years. But the Thar Desert is dominated by saline lakes. In Your opinion, what are the causes of the presence of saline lakes in desert regions of India? Do you think that the salt composition of these lakes is different from the composition of seawater?

CONCLUSION

From the towering peaks of the Himalayas in the north to the vast plains of the Indo-Gangetic region, and from the rocky plateaus of the Deccan to the coastal plains and deserts, India boasts a myriad of terrains that have been shaped by geological processes over millions of years. This diverse physiography is not only aesthetically captivating but also has significant implications for India's climate, water resources, biodiversity, and socio-economic activities. The interaction between these varied landscapes has shaped the country's history, culture, and way of life. As India continues to develop, understanding and managing its diverse physiography will be crucial for sustainable growth and environmental conservation.

Drainage System of India

4

During the rainy season, excess water flows through rivers, ponds and nalas. Had it not been for these channels, floods would cause devastating impact. In fact, flooding is a common phenomenon in all those channels that are either poorly defined or blocked and choked due to debris.

This regular flow of water through well-defined channels may be defined as the **Drainage**. Thus, the drainage helps in the removal of the excess surface water. Such a **network of tributaries** and trunk streams that collect and channel surface water to seas and oceans is collectively referred as the **Drainage system**.

'**Drainage basin**' is a spatial **geomorphic** unit of a river system, distinguished by **ridges** and **highlands**. This makes river basins natural land units. The **drainage basin** includes water collected from snowfall, melting of ice, rainfall, streams, floods etc. and diverted to a **single point** where it flows in the direction of the gradient.

Drainage Pattern is the **geometric** design through which the river and its tributaries combine as a unit from source or

origin of the river to the mouth. Any drainage pattern in a given area is dependent on a number of physical elements, including topography, slope, water flow, and rock structure.

Terminologies: Important characteristics associated with the drainage system are:

- ❑ **Source:** A place on the highland where the river begins or originates is called the Source.
- ❑ **Mouth:** The end of the river where the river flows and merges into the sea is called the Mouth.
- ❑ **Tributary:** It is a small river that eventually joins a larger river.
- ❑ **Confluence:** The meeting point at which two rivers join.
- ❑ **Watershed:** The boundary line separating one drainage basin from the other is known as watershed. It is generally found along highlands and ridges.
- ❑ **Catchment Area:** The area within the drainage basin is referred to as the Catchment Area.

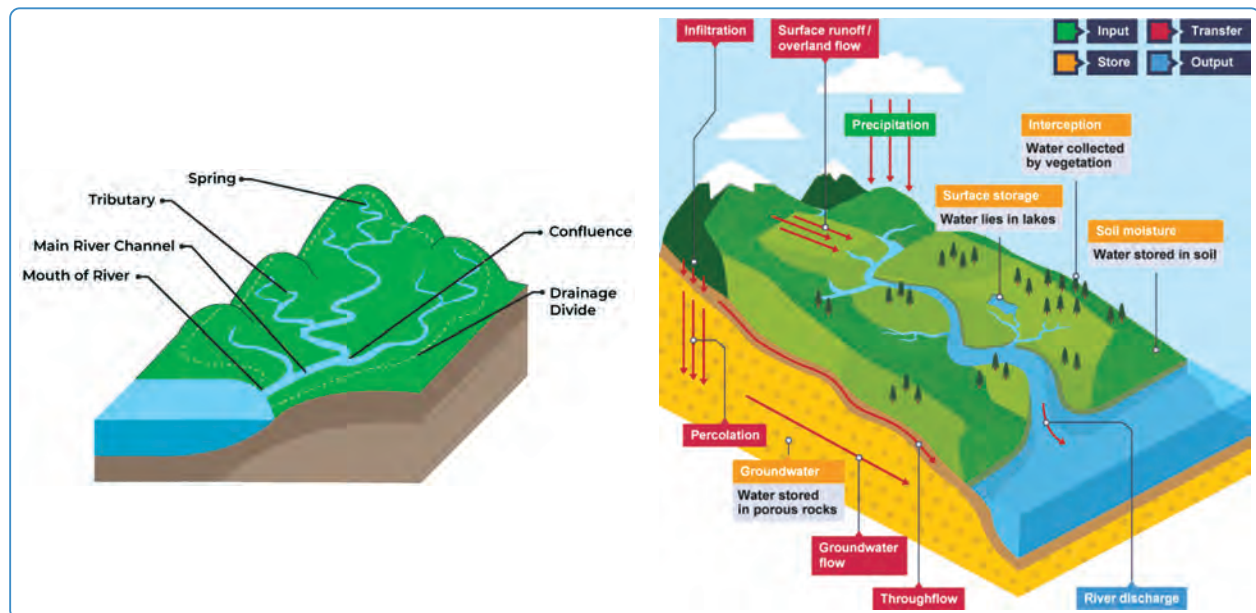


Fig. 4.1: Drainage System

Drainage Basin System

The **drainage basin system** is a crucial element of the water cycle, encompassing various interconnected processes.

❑ Inputs: Introducing Water into the System

- **Precipitation:** The primary input, referring to the introduction of water into the drainage basin system through rainfall or snow.

❑ Stores: Holding Water Within the System

- **Interception:** Water temporarily held by vegetation before reaching the ground.
- **Soil Moisture:** Water retained in the soil.
- **Surface Storage:** Water accumulation in lakes.
- **Groundwater:** Water stored beneath the surface.

❑ Transfers: Movement of Water Within the System

- **Surface Run-off/Overland Flow:** Water flowing over the land surface.
- **Infiltration:** Water penetrating the soil surface.
- **Percolation:** Downward movement of water through soil.
- **Through-flow:** Lateral movement of water within the soil.
- **Groundwater Flow:** Movement of water through underground aquifers.

❑ Outputs: Releasing Water from the System

- **River Discharge:** Water released into rivers.
- **Evapotranspiration:** Water released back into the atmosphere through the combined processes of evaporation and transpiration.

River Basins Vs Watersheds

The catchments created by big rivers are referred to as river basins, while those of small rivulets and rills are often called Watersheds. Thus, **watersheds are small in areas whereas basins occupy huge areas**. This is the main distinction between river basins and watersheds.

Examples of River Basins: The Ganges basin, the Indus basin.

Examples of Watersheds: Watershed of Yamuna, watershed of Chambal, watershed of Gandak etc.

Interconnection between Basins and Watersheds: They are marked by cohesion. Any action or effect in one part of the basin or watershed has a corresponding reaction on the other parts and the system as a whole. Hence, they are usually the most appropriate choices for micro, meso or macro planning regions.

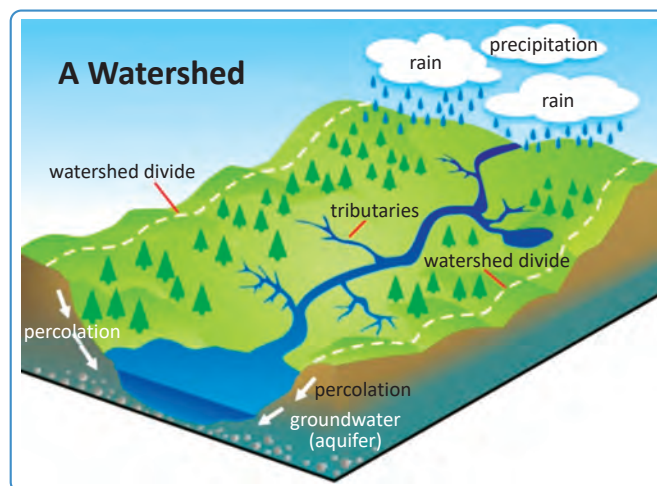


Fig. 4.2: A watershed

Feature	River Basins	Watersheds (or Catchments)
Definition	A region drained by a river and its tributaries.	An area where all the water drains to a common point, typically a river or a lake.
Scale	Larger geographical scale, often involving multiple rivers.	Smaller geographical scale, focusing on the drainage area of a single river or stream.
Boundaries	Defined by the topography and flow of rivers.	Defined by the topography; divides the land into areas where water drains in a specific direction.
Interconnectedness	Multiple river basins may be interconnected within a larger drainage network.	Watersheds can be part of a larger river basin but are more localized in terms of drainage.
Management	It often involves coordinated management across political boundaries.	Management is generally more localized and may involve smaller administrative units.
Example	Amazon River Basin, Mississippi River Basin.	Chesapeake Bay Watershed, Upper Colorado River Watershed.

ORIGIN AND EVOLUTION OF DRAINAGE SYSTEM

The origin and evolution of a drainage system in a region are influenced by the **initial surface, the nature of the slope**, in addition to the **geological structure of the region**.

Factors Influencing Drainage System Evolution

The development of drainage systems is shaped by a few key factors:

❑ Initial Surface and Slope:

- **Flow Directions:** It determines water flow directions.

- **Stream Course Impact:** It influences stream course development.

❑ Geological Structure:

- **Alignment and Behavior:** Folds, faults, and joints guide stream alignment and behaviour.
- **Shaping Drainage Patterns:** Crucial role in forming distinct drainage patterns.

❑ Climate and Weather Patterns:

- **Precipitation and Runoff:** Impact water volume and drainage system resilience.
- **Erosional Processes:** Influence landscape shape through erosion and deposition.

❑ Human Activities and Land Use:

- **Anthropogenic Modifications:** Alters drainage patterns through urbanization and agriculture.
- **Stream Channelization Impact:** Changes affect flow patterns and water quality.

Based on the adjustment of the streams to the initial surface and geological structures, drainage systems are classified into two categories i.e., **Sequent Streams (concordant)** and **Insequent (discordant) Streams**. These can further be subdivided.

MAJOR DRAINAGE SYSTEMS

Sequent Drainage System or Concordant Drainage System

They follow regional slopes and are well adjusted to the local geological structures. They can be further divided into **obsequent streams, subsequent streams, consequent streams and resequent streams**.

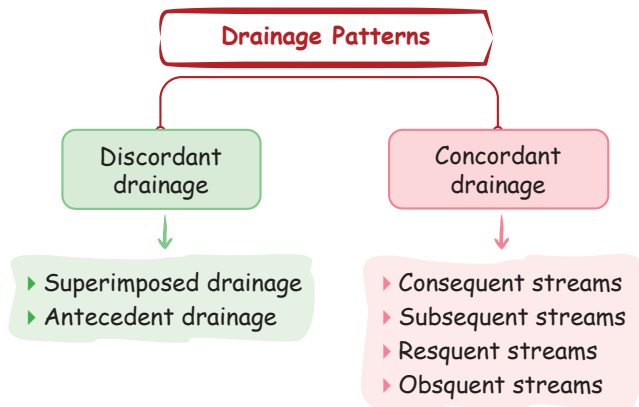


Fig. 4.3: Categorisation of Drainage Patterns

Consequent Streams

The first streams to emerge in a location are known as subsequent streams, also referred to as dip streams. They move in the general slope-side direction. The term “**master consequent**” refers to the longest and principal stream within a drainage system or river network.

The first streams to be initiated on a newly emerged coastal plain are consequent streams which are parallel to each other and thus form a parallel drainage pattern. For **example**, peninsular rivers like **Godavari, Krishna, and Cauvery** that descend from the Western Ghats and flow into the Bay of Bengal.

Subsequent Streams

It is a tributary stream that forms after the main drainage pattern is formed along a non-resistant rock strip. They follow the axis of the anticlines or ridges. They exhibit a transverse orientation, forming a right angle or approximately 90-degree angle with the master stream.

- ❑ Due to the Peninsula's slope towards the Great Plains, the Vindhyan and Satpura ranges flow northwards into the Ganges system. The Chambal, Sind, Ken, Betwa, Tons and Son meet the Yamuna and the Ganga at right angles.

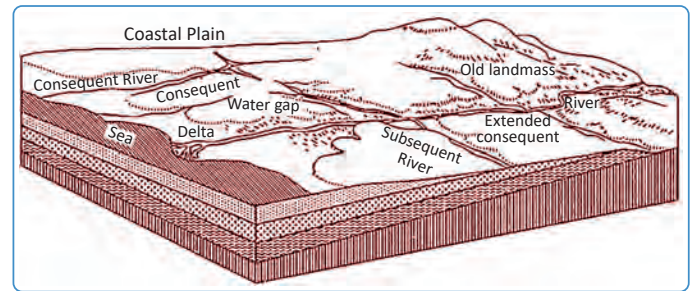


Fig. 4.4: Sequent Drainage System

Resequent Streams

They are the tributary streams which are developed later and flow in the same direction as the original consequent streams, typically as a result of **uplift** or base-level changes. They frequently serve as tributaries to succeeding streams, which flow **perpendicular** to structural features like folds and faults. They are of **recent origin**. Examples include the Gomati river.

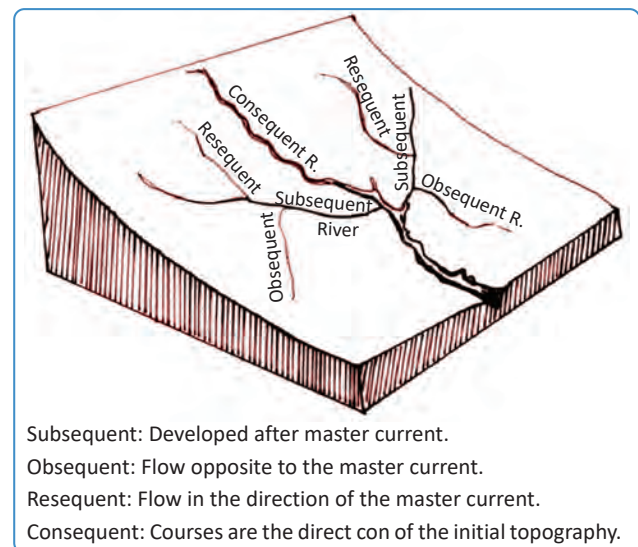


Fig. 4.5: Representation of Sequent Drainage System

Obsequent Streams

They form when a stream adjusts to a rock structure and flows opposite to the original stream (also called master consequent stream).

- ❑ For **example**, the lesser Himalayan streams join **Sun Kosi** as obsequent streams, flowing opposite to the Ganga and Yamuna rivers.

Insequent Drainage System or Discordant Drainage System

Those streams which do not follow regional slopes and are not adjusted to the geological structures. The two major types of insequent streams are antecedent stream and superimposed stream.

Superimposed Drainage

It also known as **epigenetic** or **superinduced** drainage. It is a drainage pattern that evolves on rock strata that have been subsequently removed by erosion. This process, termed superimposition, leads to the formation of river **networks**

that are unrelated to the present geological structure. **Examples** of rivers exhibiting superimposed drainage include the **Damodar, the Subarnarekha, the Chambal, the Banas,** and those flowing through the Rewa Plateau, as well as rivers in eastern USA and southern France.

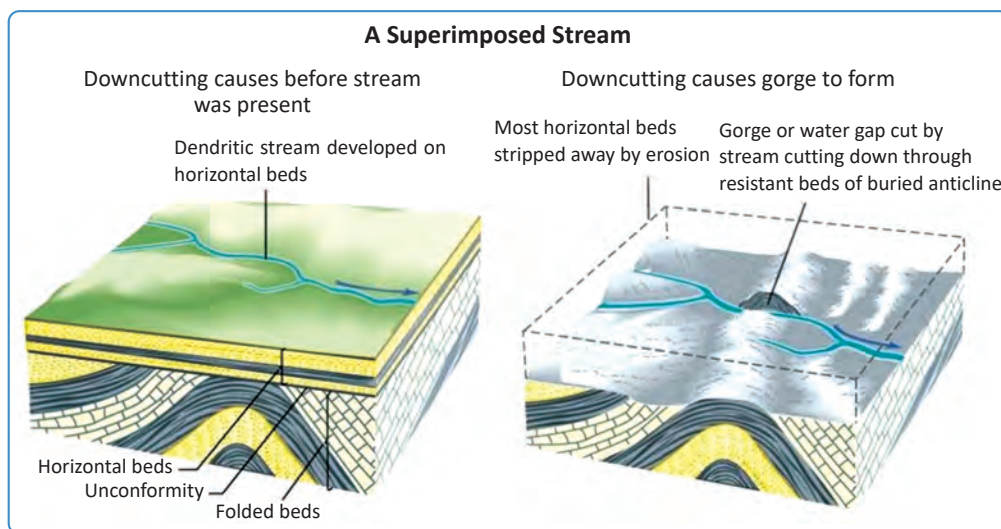


Fig. 4.6: Superimposed River

Antecedent Drainage

- It describes a drainage pattern that predates a period of uplift and folding that gave rise to the current **geological** structure. In this process, known as **antecedence**, the drainage pattern was already established before the uplift occurred.

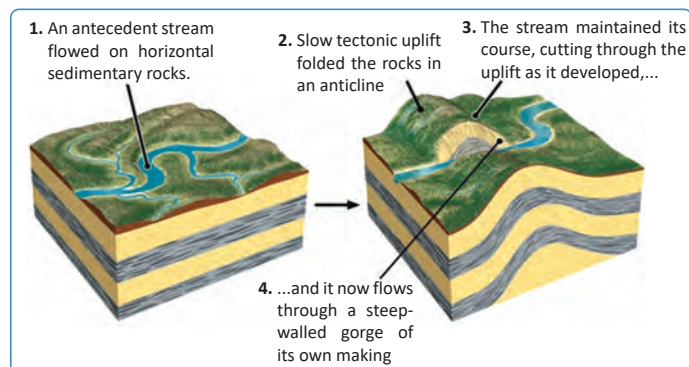


Fig. 4.7: Antecedent Drainage System

- As the uplift and folding processes unfolded, the rivers were able to cause **vertical erosion** and maintain their courses at the same rate approximately.
- This phenomenon allowed the rivers to cut down and persist in their pre-existing paths despite the geological changes.
- Examples** of rivers with antecedent drainage systems are found in various regions where uplift and folding have occurred. Most of the Himalayan Rivers like Indus, Sutlej, Kosi, Gandak have antecedent origins.

FACTORS INFLUENCING DRAINAGE PATTERNS

A '**drainage pattern**' is a geometric arrangement of streams in a region from the source of the river to its mouth. It is the resultant of various factors like:

- The nature and structure of rocks
- The geological time period
- Topography of the region
- Slope
- The volume of the water flowing
- The periodicity and the velocity of the flow.

These factors in turn give rise to a variety of patterns as discussed below.

TYPES OF DRAINAGE PATTERNS

Some of the drainage patterns occurring include **Trellised pattern, Dendritic pattern, Rectangular pattern, Radial pattern, Parallel pattern, Deranged pattern, Annular pattern**, among others. A detailed analysis of each of these is presented below:

- Trellis Drainage Pattern:** The trellis drainage pattern is characterized by the parallel flow of primary tributaries, while secondary tributaries intersect them at right angles.
 - Description:** It resembles a garden trellis, forming in folded topography with parallel hard and soft rocks.
 - Formation:** Primary tributaries follow down-turned folds (synclines) forming valleys.

- **Areas of Formation:** This pattern typically emerges when there is a **parallel** arrangement of hard and soft rock formations.
- **For example,** Rivers in the upper Himalayan region; Indus, Ganga, Brahmaputra; the old folded mountains of the Singhbhum (Chotanagpur Plateau).

❑ **Rectangular Pattern:** In a rectangular drainage pattern, the main stream bends at right angles and the tributaries join at right angles creating rectangular patterns.

- **Areas of formation:** It is found in regions that have undergone faulting. This pattern typically forms on terrains with pronounced jointing in the rock structure. It distinguishes itself from the trellis drainage pattern by its greater irregularity, with tributary streams being shorter and less parallel.
- **Geological Conditions:** It develops in areas with limited topography and a network of bedding planes, fractures, or faults.
- **Examples:** Streams in the Vindhya mountain range like Tapi, Chambal, Betwa, Ken; and Colorado River in the USA.

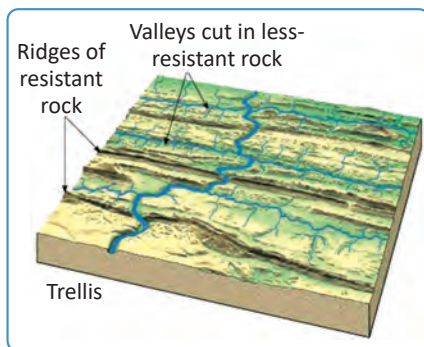


Fig. 4.8: Trellised Pattern

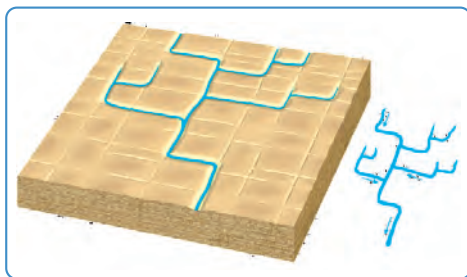


Fig. 4.9: Rectangular Pattern



Fig. 4.10: Dendritic drainage pattern

❑ **Dendritic Pattern:** It is the most common form that resembles tree roots. The dendritic pattern emerges

when the river channel aligns with the slope of the terrain, resembling the branching structure of a tree.

- **Name Origin:** This configuration is aptly named dendritic due to its resemblance to the branches of a tree.
- **Geological Conditions:** It occurs in regions with rock beneath the stream lacking a specific structure, allowing for equal erosion in all directions.
- **Tributaries:** The tributaries join larger streams at acute angles (less than 90°).
- **Examples:** Indus, Ganga, Brahmaputra, Godavari, Mahanadi etc.

❑ **Radial Pattern:**

- **Description:** It develops around a central elevated point, streams radiate outward, are known as radial or centrifugal patterns like lava erupting from a volcano.
- **Formation:** Common around conically shaped features like volcanoes.
- **Examples:** Rivers originating from the Amarkantak Hills (Narmada, Son and Johila river), Girnar Hills, Gujarat (Sabarmati & others) and Mikir Hills, Assam (Dhansiri, Kopili Rivers).

❑ **Centripetal Pattern:** Rivers' discharge of water into a lake or depression. Similar to a radial pattern, but with the distinction that centripetal passages flow inward as opposed to outward like radial passages do. For instance, the streams of Ladakh and the Loktak Lake in Manipur, Bagmati and its tributaries in Nepal.

❑ **Parallel Pattern:** It is found in regions of long parallel landforms with slopes. Tributaries flow parallel to natural downfalls, such as rivers in the Western Ghats flowing into the Arabian sea, Godavari, Kaveri, Krishna, & Tungabhadra.

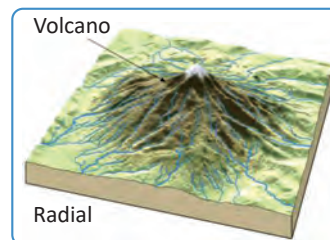


Fig. 4.11: Radial Pattern

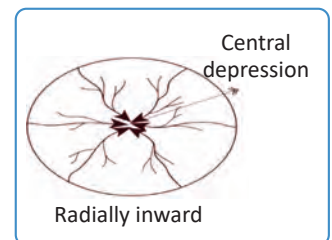


Fig. 4.12: Centripetal Pattern

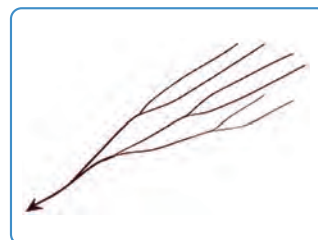


Fig. 4.13: Parallel Pattern



Fig. 4.14: Annular Pattern

❑ **Annular Pattern:** An annular pattern characterizes a segment of a drainage pattern where the successive streams adopt curving or arcuate courses before converging with the main consequent stream.

- This phenomenon arises from a partial adjustment to an underground circular structure, such as batholiths.
- This pattern is **not common** but is found in Pithoragarh (Uttarakhand), Nilgiri Hills, and Kerala, with easier erosion of the concentric strata.

❑ **Barbed Pattern:** In this drainage pattern, the point where a tributary meets the main river is marked by a discordant junction, creating the impression that the tributary seeks to flow upstream rather than downstream.

- This distinctive pattern arises from the main river being captured, leading to a complete reversal in its flow direction, while the tributaries maintain their orientation in the direction of the original flow.
- **For instance**, the Kosi River has a tributary in Nepal called the Arun River.

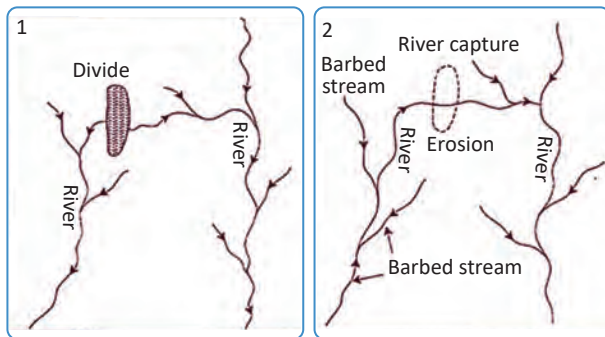


Fig. 4.15: Barbed Pattern

❑ **Pinnate Pattern:** It is formed in the narrow valleys flanked by steep mountain ranges. The tributaries from the steep sides of the parallel ridges join the longitudinal main river like veins joining the nerves **at acute angles**. **For example**, the Son, and Narmada rivers.



Fig. 4.16: Pinnate Pattern

❑ **Herringbone or Rib Pattern:** Herringbone drainage pattern is identified by a series of streams that flow in a series of **V-shaped** patterns. They are typically found in areas with a regular pattern of faults or joints.

- **Example:** Patterns linked to alternating layers of hard and soft rock or fault lines in the landscape. The upper reaches of the Jhelum River in the Vale of Kashmir are fed by numerous tributaries on both its flanks.

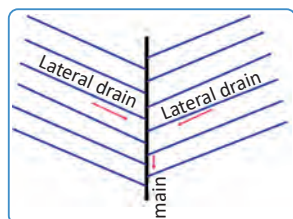


Fig. 4.17: Herringbone Pattern

❑ **Deranged Pattern:** Inconsistencies in materials deposited by glaciers and a lack of time for drainage to adapt to solid rock structures are likely to be to blame for the disorderly drainage pattern in a region that has recently been cleared by an ice sheet.

- This type of drainage, which includes numerous waterways, lakes, and marshes, is present in the glaciated valleys of the **Karakoram**.



Fig. 4.18: Deranged Pattern

❑ **Angular Pattern:** It is characterized by straight stretches connected by sharp, often right-angled bends.

- **Formation:** It is associated with bedrock joints and faults that intersect at acute angles.
- **Examples:** Common in the Himalayan Foothill region.

OTHER IMPORTANT TERMS RELATED TO DRAINAGE

❑ **Avulsion:** It refers to the sudden and rapid change in the course of a river, typically caused by the erosion of its banks. Unlike gradual changes in a river's course, avulsion involves a more abrupt and often unpredictable shift in the direction of the river channel. This can result from various factors, including changes in sedimentation, tectonic activity, or human-induced alterations to the landscape. As the sediments deposit, it may eventually reach a point where the river overflows its banks, especially during periods of high flow (e.g., floods). The overflow can create a new channel that diverts the river's flow away from its previous course.

❑ **River Channel Migration:** The gradual and natural movement of a river channel over time.

❑ **Stream Capture:** The diversion of one stream's flow into another, often occurring due to changes in topography or erosion.

❑ **Riverbank Erosion:** The gradual wearing away of riverbanks due to natural processes, sometimes leads to changes in the river's course.

❑ **Wind Gap:** A gap through which a waterway once flowed but which is now dry as a result of stream capture is referred to as a wind gap, also known as an **Air Gap**.

❑ **Stream/River Capture:** When a stream or river drainage is diverted from its own bed and flows down the bed of a neighbouring stream, this is referred to as **stream capture, river capture, or river piracy**.

- ❑ **Water Gap:** A pass or gorge actively used by a river, allowing its flow through mountainous terrain.
- ❑ **Defile:** A narrow gorge or passage, often through mountains, shaped by the erosive action of rivers or streams.
- ❑ **Nickpoint:** A location with a sharp change in river channel slope, such as a waterfall or rapid.

Let us now examine the drainage system of the Indian subcontinent.

CLASSIFICATION OF DRAINAGE SYSTEM OF INDIA

The Indian drainage system has been categorized into many types on the basis of different criteria under consideration like mode of origin, water discharge and the catchment area size. The following are the details based on these criteria.

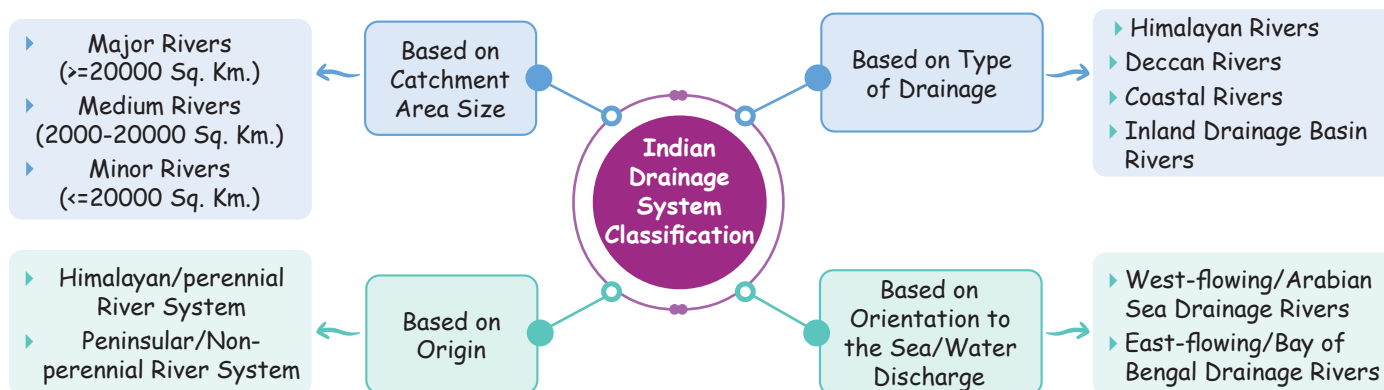


Fig. 4.19: Classification of Drainage system of India

❑ Size of the Catchment Area:

The drainage basins of India are divided into three categories based on the size of the catchment area:

- **Major river basins:** They have a catchment area of more than **20,000** square kilometers and include **14** drainage basins like the Ganga, the Barak and the Godavari.
- **Medium river basins:** The river basins with catchment areas of between **2,000 and 20,000** square kilometers. They consist of 44 river basins like the Kalindi, the Periyar, and the Meghna
- **Minor river basins:** The river basins with catchment areas of less than **2,000** square kilometers. A number of rivers flowing in the areas of low rainfall come under this. These include Sharavati, Vaigai etc.

Catchment Areas of Major Indian Rivers

Major Rivers of India	Catchment Area (in sq. km.)
Ganga	861404
Godavari	312812
Krishna	258948
Brahmaputra	194413
Mahanadi	141589
Narmada	98796
Cauvery	81155
Tapi	65145
Pennar	55213
Brahmani and Baitarani basin	51822
Barak	41723

Mahi	34842
Subarnarekha	29196
Sabarmati	21674

❑ Mode of Origin:

Depending on the mode of origin, the drainage of India can be divided into:

- **The Himalayan Drainage (Perennial River System):** It consists of relatively younger rivers like the Ganges, the Brahmaputra and the Indus.

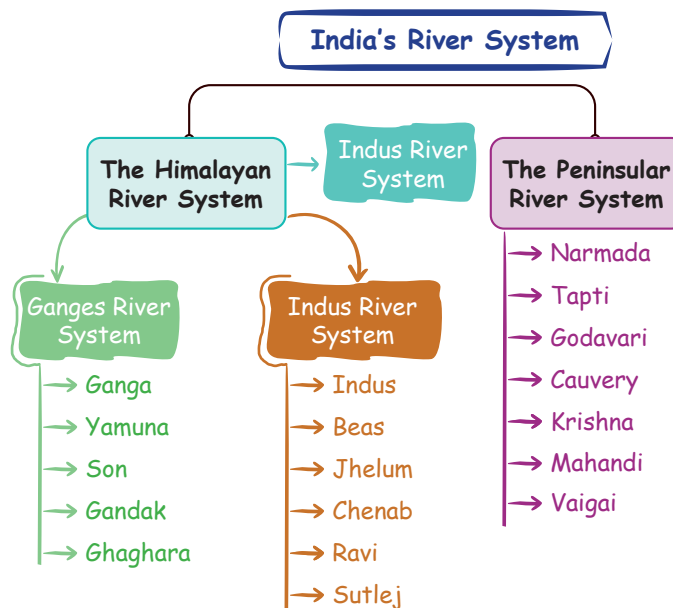


Fig. 4.20: Classification of the Himalayan and Peninsular river system

- **The Peninsular Drainage (Non-Perennial River System):** It consists of comparatively older rivers like the Godavari, the Krishna and the Penna
- Although many Peninsular rivers, like the Chambal and Betwa, are older in age and origin than the Himalayan rivers, there is no obvious distinction between these two drainage systems.

Himalayan Rivers Vs Peninsular Rivers

Aspects	Himalayan River	Peninsular River
Location	Flow through the Himalayan mountain range in the northern part of India.	Flow through the Peninsular Plateau in the southern part of India.
Origin	Generally originates from glaciers or high mountain peaks in the Himalayas.	Originate from various sources within the Peninsular plateau, including hills (Western & Eastern Ghats) and plateaus.
Flow Direction	Flow towards the northern plains and eventually into the Bay of Bengal or the Arabian Sea.	Flow towards the east or west into the Bay of Bengal or the Arabian Sea.
Perennial vs. Seasonal Flow	Mostly perennial rivers with continuous flow throughout the year due to glacier-fed sources.	Many are seasonal , with intermittent flow as they depend on rainfall.
Water Volume	Generally has a higher water volume and discharge compared to Peninsular rivers, especially during the monsoon season.	Generally have a lower water volume and discharge, with significant variations in flow throughout the year.
Gradient and Rapids	Characterized by steep gradients and fast-flowing rapids in their upper reaches due to their mountainous origins.	Tend to have gentler gradients and slower-moving waters, with occasional rapids in hilly areas.
Tributaries	Receive water from numerous tributaries and sub-tributaries along their course.	Have fewer tributaries and sub-tributaries in comparison.
Erosion and Sedimentation	Carry a significant amount of sediment eroded from the mountains, leading to alluvial deposits in the plains.	Carry comparatively less sediment and have a lesser impact on alluvial deposits.
Examples	Gangas, Brahmaputra, Yamuna, Indus, Sutlej, Chenab, and others.	Godavari, Krishna, Cauvery, Narmada, Tapi, Mahanadi, and others.

Drainage Type

India's river systems can be categorized into four groups:

- ☐ Himalayan rivers
- ☐ Deccan rivers
- ☐ Coastal rivers that drain into the sea
- ☐ Rivers of the inland drainage basin (endorheric basin)

Category	Geographical Region	Origin and Source	River Characteristics	Hydroelectric Potential	Examples of Rivers
Himalayan Rivers	Northern India	Himalayan mountains, glaciers, snowmelt	Perennial, fast-flowing, high sediment load	High due to steep gradients and elevation drop	Ganga, Yamuna, Brahmaputra
Deccan Rivers	Plateau region of central and southern India	Plateau regions, rainfall, local runoff	Variable flow, some seasonal rivers	Moderate due to plateau terrain	Godavari, Krishna, Narmada, Tungabhadra
Coastal Rivers	Coastal regions and Western Ghats	Coastal areas, Western/Eastern Ghats, rainfall	Variable flow, moderate sediment load	Limited due to variable flow and low gradients	Mahanadi, Krishna, Cauvery
Inland Drainage Basin Rivers (Endorheic Basin)	Arid or semi-arid regions, often interior	Dependent on local rainfall and runoff	Variable flow, often ephemeral, minimal sediment	Limited due to arid or semiarid conditions	Luni, Ghaggar, many small seasonal rivers

- ❑ In the Rann of Kutch, the Luni is the sole river that flows through the salt desert.
- ❑ Each river category possesses distinctive features, influencing agriculture, human settlements, ecosystems, and navigation opportunities.

Water Discharge/Orientation to the Sea

Based on the direction of water-discharge, drainage system can be categorized as:

- ❑ **East-flowing Rivers:** These flow into the **Bay of Bengal** and account for about **77%** of the total drainage. They

include the Ganga, the Brahmaputra, the Godavari, the Krishna, the Mahanadi etc.

- ❑ **West-flowing rivers:** They flow into the **Arabian Sea** and comprise **23%** of the total drainage. Some of them are the Indus, the Narmada, the Tapi etc.

In spite of varying drainages, about **90%** of water drains into the Bay of Bengal. The East and the West flowing rivers are separated from each other through the **Delhi Ridge**, the **Aravallis** and the **Sahyadris** and are collectively referred to as the **Water Divide**.

East Flowing River System Vs. West Flowing River System

Aspect	Bay of Bengal Drainage (East Flowing River System)	Arabian Sea Drainage (West Flowing River System)
Location	Eastern side of the Indian subcontinent	Western side of the Indian subcontinent
Major Rivers	Ganges, Brahmaputra, Godavari, Krishna, Mahanadi	Indus, Narmada, Tapi, Sabarmati
River System	More extensive and complex river system	Less extensive river system
River Discharge	Generally higher discharge due to abundant rainfall and snowmelt in the Himalayas	Lower discharge compared to Bay of Bengal drainage due to less rainfall in the western part of India
Rainfall Pattern	Experiences heavy rainfall during the monsoon season	Receives less rainfall due to the rain shadow effect from the Western Ghats, leading to arid regions in some areas
Coastal States	India (eastern coast) and Bangladesh	India (western coast), Pakistan, and parts of Oman
Major Ports	Chennai, Kolkata, Vishakhapatnam, Chittagong	Mumbai, Karachi, Kandla, Kochi
Ocean Currents	Affected by the North Equatorial Current and the East India Coastal Current	Affected by the Somali Current and the West India Coastal Current
Sea Trade Routes	Connected to major trade routes with Southeast Asian countries and beyond	Historically linked with trade routes to the Persian Gulf, Europe, and Africa



Fig. 4.21: Water Divide in India

THE HIMALAYAN DRAINAGE

The drainage system of the Himalayas primarily includes the **Indus, Ganga, and Brahmaputra** rivers, which are **perennial** due to continuous feed through **snow melt** and **precipitation**. Both the southern and northern Tibetan slopes of the Himalayas are drained by the *great Himalayan rivers, which are older than the Greater Himalayas*.

- ❑ They act as eroding agents and create **erosional landforms** like waterfalls, cataracts, rapids, gorges, steep slopes, V-shaped valleys and river terraces during the course.
- ❑ They also create **depositional landforms** like flat valleys, flood plains, ox-bow lakes, deltas near the river mouth and braided channels. Their courses are **tortuous**, with **meandering** tendencies in plains.
- ❑ These rivers exhibit wide **seasonal fluctuations**, causing devastating floods during rainy seasons and shrinking to the bottom during the dry season. They continue to cause intensive erosion and transport heavy loads of sand and silt annually.
- ❑ For **example**, the River **Kosi** brings huge amounts of sediments from the upper reaches and deposits in the

plain areas. As a consequence, the natural course is blocked and the river changes its course. For this reason, the Kosi is called the **Sorrow of Bihar**.

The rivers have great socio-economic and cultural importance in Indian life, used for irrigation, industries, hydel-power generation, navigation, and domestic purposes.

Evolution of the Himalayan Drainage System

Geologists believe that a mighty river called **Shiwalik or Indo-Brahma** spanned the entire length of the Himalayas from Assam to Punjab, eventually flowing into Sind, formed around **5-24** million years ago during the **Miocene** period. The river's continuity and **lacustrine** (relating to lakes) origin, along with alluvial deposits of silt, clay, sand, support this view. (**Indo-Brahma or Shiwalik River Theory**)

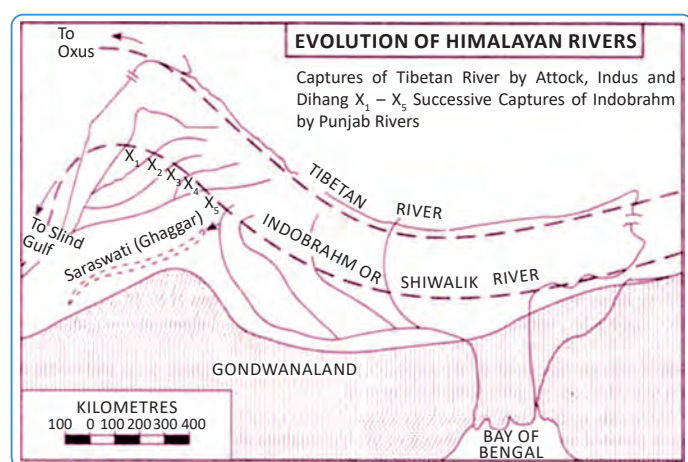


Fig. 4.22

The Indo-Brahma River system was divided over time into:

- ❑ The Indus and its five tributaries in the western portion
- ❑ The Ganga and its Himalayan rivers in the central region and
- ❑ The Brahmaputra stretch of the Eastern Indian parts of Assam and Himalayan rivers.

The Ganga and Brahmaputra systems were diverted towards the Bay of Bengal during the **Pleistocene upheaval** in the western Himalayas, which included the uplift of the **Potwar Plateau (Delhi ridge)** and the downthrusting of the **Malda gap** region between the **Meghalaya plateau** and the **Rajmahal hills**.

Another theory, namely, **Multiple River Theory** proposes that the intricate network of Himalayan rivers evolved through the integration of multiple ancient river systems. Geological processes, tectonic activity, and river capture have shaped the diverse drainage patterns. Continuous uplift and erosion contribute to the dynamic evolution of the Himalayan river system.

The **Persian word "doab"** refers to the region that lies between two rivers. Between the **Indus** River and the **Jhelum** River is where the **Sindh Sagar Doab** is produced. Despite having the largest land area of the Punjab doabs, it is also the poorest because so little of the land is used for agriculture.

Indus River System

The **Indus**, also known as Sindhu, a significant drainage system in India, spans 2880 km, with 1114 km in India. Its catchment area is 1,165,000 sq km, with 3,21,289 sq km in India. The famous Indus Valley Civilisation is named after this river.

- ❑ Originating from a glacier near **Bokhar Chu Glacier (4,164 m)** in **Mt. Kailash**, it drains the largest number of glaciers and mountain slopes in the **Karakoram** region.
- ❑ It flows in the North-west direction passing through **Ladakh, Gilgit and Baltistan**.
- ❑ Major **tributaries** of the Indus River can be classified as right-bank rivers and left-bank rivers, based on where they meet the river.
 - **Right-bank tributaries:** The Shyok, Gilgit, Zhoob, Hunza, Nubra, Shigar, Gilgit, Kabul River, Kurram, Gomal, Tochi, Viboa, Sangar.
 - **Left-bank tributaries:** The Zaskar, the Sutlej, the Beas, the Chenab, the Ravi, the Jhelum.
 - ❑ **Doabs of Indus River System:** A "doab" refers to a region of land lying between two rivers. The word "doab" is derived from Persian, where "do" means "two," and "ab" means "water" or "river." These regions are typically fertile and well-suited for agriculture. The 5 doabs of this system are:
 - ❑ **Sindh Sagar Doab:** This doab is situated between the Indus River and the Jhelum River. The region is named after the historic Sindh province and the Sagar Lake.
 - ❑ **Jech Doab:** The Jech Doab is formed between the Jhelum River and the Chenab River. The name is derived from the combination of the two rivers.
 - ❑ **Rechna Doab:** This doab lies between the Chenab River and the Ravi River.
 - ❑ **Bari Doab:** The Bari Doab is located between the Ravi River and the Beas River. It is one of the most fertile regions in the Indian subcontinent.
 - ❑ **Bist Doab:** The Bist Doab is formed between the Beas River and the Sutlej River.
- ❑ The Indus River is conjoined by the **Dhar** River near the **Indo-China border**.
- ❑ Notably, it is joined by the **Zaskar** River at **Leh**. Proceeding towards **Skardu**, the **Shyok** River converges with the Indus.
- ❑ The Indus concludes its **mountainous** journey by traversing a deep **gorge** near **Attock**, reaching a depth

of 5181 m, north of the **Nanga Parbat**, and takes a sharp southerly bend (**syntaxial bend**).

- ❑ The **Kabul** River from Afghanistan joins the Indus near Attock. Subsequently, the river flows through the **Potwar** plateau, crossing the Salt Range at the southeastern edge of the Potwar Plateau.
- ❑ Notably, just upstream of **Mithankot**, the Indus receives the combined waters of the five eastern tributaries—the

Jhelum, Chenab, Ravi, Beas, and Satluj—flowing from the Panjnad (**Panchnad**). Finally, the river empties into the Arabian Sea south of Karachi, forming a vast delta.

In Tibet, the Indus River is referred to as **Singi Khamban** or the **Lion's Mouth**.

The **blind Indus River Dolphin**, a sub-species of dolphin, is found only in the Indus River.

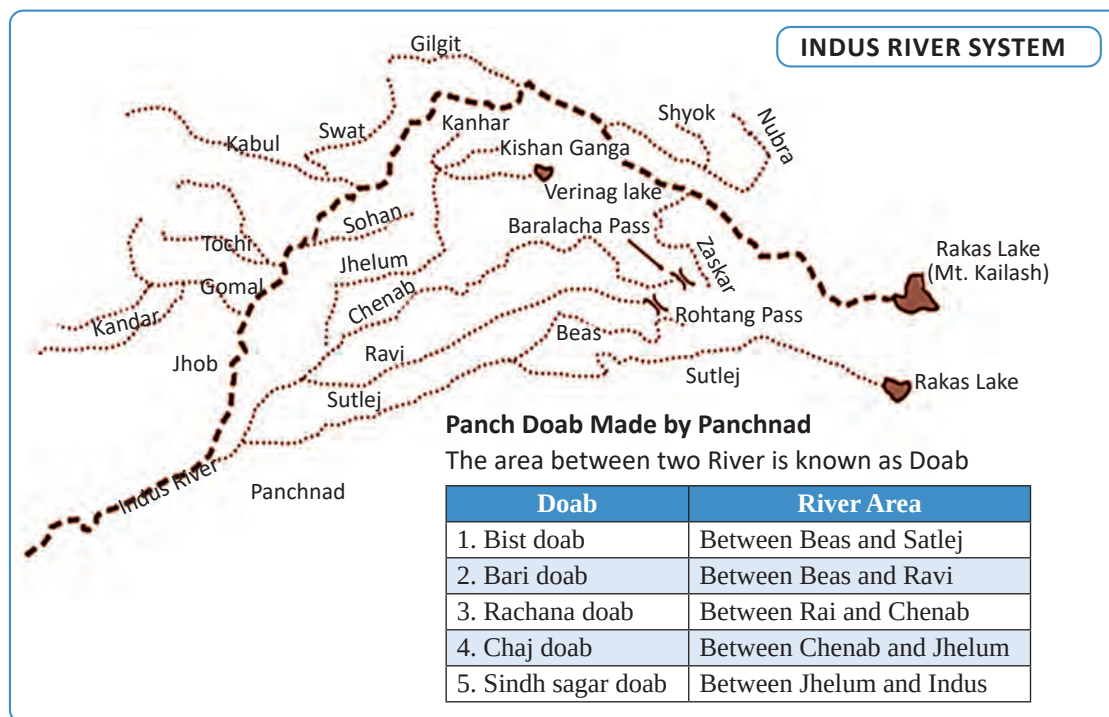


Fig. 4.23

Indus Water Treaty (1960)

An important water distribution deal between India and Pakistan was the Indus Water Treaty (1960), which was signed in that year. The agreement created the **Permanent Indus Commission** and mandated that Pakistan and India hold biannual meetings and schedule technical site visits.

- ❑ In the final reckoning of the agreement, Pakistan received 80% of the IRS waters while India got 20%. The **World Bank** served as the treaty's intermediary.
- ❑ **Rights over Rivers:** The Treaty allocates the Western Rivers (Indus, Jhelum, Chenab) to Pakistan and the Eastern Rivers (Ravi, Beas, Sutlej) to India. However, according to Annexure C of the IWT, India is granted specific agricultural usage rights. Additionally, Annexure D permits India to construct '**run of the river**' hydropower projects, which refers to projects that do not necessitate the retention of water for storage.
- ❑ **Upper and Lower Riparian System:** The terms "upper riparian" and "lower riparian" refer to the geographical positioning of states or regions along a river.

- The upper riparian state is located upstream, while the lower riparian state is situated downstream. These terms are particularly relevant in the context of river water sharing and international water laws like Indus. In the case of the Indus River, which flows through China, India, and Pakistan, India is considered an upper riparian state with respect to Pakistan.
- **India's strategic advantage in the Indus River basin:**
 - ❑ **Geographical Advantage:** India's **upper riparian status** facilitates control over water resources.
 - ❑ **Control Over Flow:** Essential for managing resources, including **agriculture, hydroelectric power, and domestic supply**.
 - ❑ **Strategic Infrastructure:** Significant projects, like **Bhakra-Nangal Dam**, contribute to effective water resource management.
 - ❑ **Negotiation Leverage:** India's upper riparian position provides **negotiation leverage** in discussions, including those concerning the **Indus Waters Treaty** and other agreements.

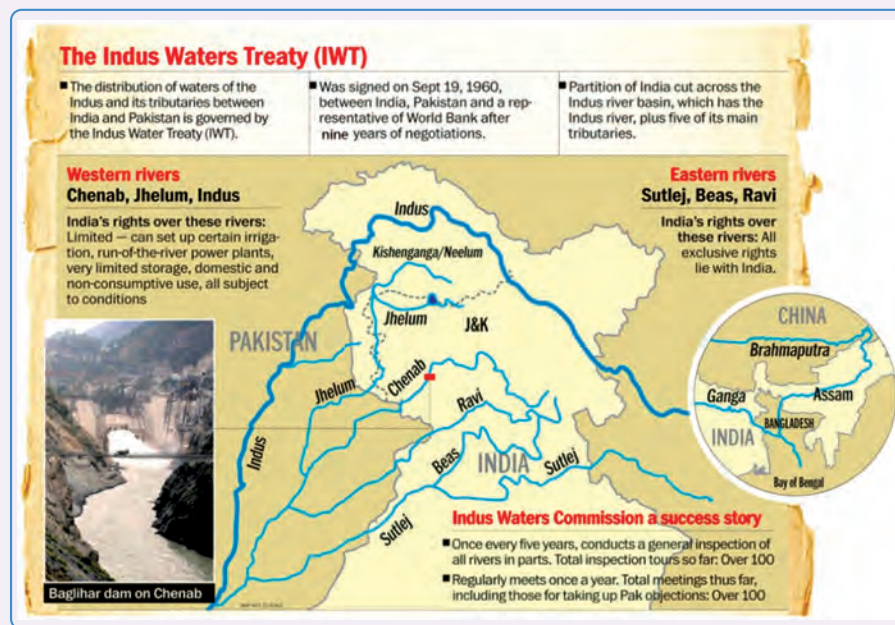


Fig. 4.24: Indus Water Treaty

- **Important projects on the Indus River:** Bhakra Nangal, Indira Gandhi Project, Pong Project, Chamera Project, Thein Project, Nathpa Jhakri Project, Salal, Baglihar Project, Dulhasti Project, Tulbul Project, and Uri Project.

❑ **Geostrategic concerns of India over Indus river:** India has several geo-strategic concerns related to the Indus River system, which plays a crucial role in the country's water security, energy generation, and overall development.

- **Water Security:** India aims to ensure a reliable and sustainable water supply from the Indus River system to meet the diverse needs of its population, including agriculture, domestic consumption, and industrial use.
- **Hydroelectric Power Generation:** India strategically focuses on harnessing hydropower potential from the Indus River system to enhance energy security and support economic development.
- **Transboundary Water Sharing:** India seeks a fair and effective mechanism for water allocation that aligns with its developmental requirements.
- **Environmental Impact and Climate Change:** India is attentive to the potential environmental consequences of climate change on the Indus River system, necessitating adaptive strategies to ensure sustainable water management.
- **Geopolitical Tensions:** The geo-political context between India and Pakistan adds complexity to the management of the Indus River system. India aims to navigate this context while maintaining adherence to international agreements.

❑ **Dispute Resolution Mechanism:** The Indus Waters Treaty (IWT) incorporates a three-step mechanism for dispute resolution, as outlined in Article IX.

- Issues or questions on either side can first be addressed at the Permanent Commission or elevated to the inter-government level.
- In instances where questions remain unresolved, particularly those involving water-sharing disagreements such as technical differences, either nation has the option to engage the World Bank. The World Bank can then appoint a Neutral Expert (NE) to render a decision.
- If either party remains dissatisfied with the NE's decision or if disputes arise regarding the interpretation and scope of the treaty, the matter can be escalated to a Court of Arbitration for resolution.
- **Current Disputes (January 2023):**
 - ❑ India seeks IWT amendments; notice issued to Pakistan on January 25.
 - ❑ Pakistan announces Court of Arbitration in The Hague on January 27.
 - ❑ Hydroelectric power projects on Kishanganga and Chenab rivers under dispute.
 - ❑ Pakistan initially sought World Bank's involvement in August 2016.
 - ❑ India proposes Neutral Expert; World Bank appoints Michel Eno and Sean Murphy for Arbitration on October 17, 2022.

Now let us understand the major tributaries of the river Indus in detail.

Jhelum (Ancient Name: Vitasta)

It is an important tributary which rises from a spring at **Verinag** at the foot of the **Pir Panjal** (part of Kashmir valley).

- ❑ It flows northwards through Srinagar and from the **Wular lake** changes its course southwards. The beautiful city of **Srinagar** is situated on the bank of this river.
- ❑ The largest river flowing into Jhelum is the **Kishanganga River**.
- ❑ Upon reaching **Baramulla**, the river enters a gorge nestled in the hills. Below Baramulla, the Indus creates a narrow gorge with steep sides as it courses through the **Pir Panjal Range**.
- ❑ Moving southward from Muzaffarabad, the river takes a distinctive **hairpin bend**. Subsequently, it delineates the **India-Pakistan boundary** for a stretch of 170 km and surfaces on the **Potwar Plateau** near Mirpur.
- ❑ Continuing its course through the spurs of the **Salt Range**, the Indus then emerges onto the plains in the vicinity of the city of Jhelum after debouching from a confined space into a wide and open area.
- ❑ Uri, Kishanganga Project are important hydroelectricity projects.

Chenab (Ancient Name: Asikni)

The Chenab is the largest tributary of the Indus with 960 km of length in India with a catchment area of **29,050 sq km, upto the international border**. It is created in Himachal by the fusion of two streams, the **Chandra** and the **Bhaga**, and is referred to as **Chandrabhaga**.

- ❑ It originates in **Lahaul-Spiti (Zaskar range)**. It cuts a deep gorge near **Kishtwar**. It flows through the plains of Pakistan Punjab to reach **Panchnad** where it **joins the Satluj** after receiving the waters of Jhelum and Ravi.
- ❑ **Salal, Dulhasti** hydel project lies on Chenab and **Baglihar Dam** is constructed on the river and is popular for Chenab's bridge which is one of the world's highest railways.

Ravi (Ancient Name: Parushni or Iravati)

Originating in the **Kullu** hills near the **Rohtang Pass** in Himachal Pradesh, the Ravi River flows through the region between the **Pir Panjal** and **Dhauladhar** ranges.

- ❑ Following its course beyond **Chamba**, the river adopts a south-westerly direction and carves a profound gorge through the **Dhaola Dhar** range.
- ❑ Upon reaching the Punjab Plains near **Madhopur**, it eventually crosses into Pakistan below Amritsar. The river eventually **merges into the Chenab**, slightly upstream from Rangpur in the Pakistani Punjab, after debouching from its course.

- ❑ **Ranjit Sagar Dam and Shahpur Kandi** dam project are major projects built on the river.

Beas (Ancient Name: Vipasa or Argikiya)

It originates from the **Beas kund** near the Rohtang pass at a height of 4000 metres above sea level. At **Harike**, it merges with the Satluj after passing through the **Kullu** valley and the Punjab plains.

- ❑ Compared to other Indus tributaries, it is a short river (470 km in length), and it only flows through Indian territory.
- ❑ The river crosses the Dhauladhar range and follows a south-westerly direction and then river eventually confluences with the Sutlej River at Harike in Punjab.

Satluj (Ancient Name: Satadru or Satudri)

Originating from the **Mansarovar-Rakas** Lakes (where it is referred to as **Langchen Khmabab**) in western Tibet at an elevation of 4,572 m, the Satluj River lies within 80 km of the Indus River's source.

- ❑ Mirroring the course of the Indus, the Satluj initially flows in a north-westerly direction until it reaches the **Shipki La** on the Tibet-Himachal Pradesh boundary. As it penetrates the Great Himalaya and other Himalayan ranges, the river carves deep gorges.
- ❑ Before entering the Punjab plain, it cuts a gorge in Naina Devi Dhar, where the famous Bhakra dam has been constructed. Upon entering the plain at Rupnagar (**Ropar**), the river takes a westward turn and meets the **Beas at Harike**.
- ❑ Serving as the boundary between India and Pakistan for approximately 120 km, from near Ferozepur to Fazilka, the Satluj receives the **combined drainage of the Ravi, Chenab, and Jhelum rivers** during its onward journey.
- ❑ It eventually joins the Indus a few kilometers above **Mithankot**. Out of its total length of 1,450 km, the Sutlej flows for 1,050 km within Indian territory.
- ❑ Among the five tributaries of the Indus River, the Satluj is distinguished as the **longest**.

Ghaggar, often known as the **fabled Saraswati**, is an Indian **inland drainage** that rises in the Shiwaliks of Sirmour, close to **Ambala**. It vanishes in the plains before resurfacing in the **Karnal** District.

- ❑ In **Bikaner**, the stream, also known as **Hakra**, is lost close to **Hanumangarh**.
- ❑ The Satluj was thought to flow south through this **Ghaggar-Hakra** river rather than southwestward due to the broad river bed and loamy terrain.
- ❑ The river feeds into the Rann of Kutch and is identified as **Eastern Nara**, an ancient Indus channel in Sindh. The Ghaggar-Hakra streams are now **ephemeral**, and the land is **desert**.

Saraswati River - The Mystery of the Lost River

The Saraswati is a significant river in the Vedic Period, part of the **Sapta Sindhu** rivers. It was mentioned in **Rig Veda** and other Vedic texts. It flows between Satluj and Yamuna and into the Samudra.

- ❑ The river is believed to have originated from the **Har-ki-Dun glacier** in **Uttarakhand**. It is 1500 km long, 5 m deep, and 3-15 km wide.
- ❑ Later Vedic text and Mahabharata mentioned that the river dried up in a desert.
- ❑ It flowed between 6000 and 4000 BCE, and geological upheavals caused it to dry up.
- ❑ According to contemporary experts, the **Ghaggar-Hakra** river system in northwest India and eastern Pakistan is where the Saraswati originates.

Recent efforts by the government for revival of Saraswati river:

- ❑ **Adi Badri Dam:** On January 21, 2022, a Memorandum of Understanding (MoU) was executed between the Governments of **Haryana and Himachal Pradesh** for the collaborative construction of the Adi Badri Dam on the **River Somb** in Himachal Pradesh, including its associated linkages with the **Saraswati River**.
 - The construction of the Adi Badri Dam aims primarily to revive the Saraswati River and promote the development of **Saraswati heritage**.
 - Additionally, the project is designed to allocate **61.88 hectare-metres** per annum of water for drinking and irrigation purposes in the state of **Himachal Pradesh**.

A **glacial lake** is a body of water that forms in a depression on the surface of a glacier or in a glacially carved basin. These lakes originate through various glacial processes, such as the melting of ice, the damming of glacial meltwater, or the deposition of sediments by the glacier. For instance, the **Garhwal** Himalayas are home to **Devtal Glacial Lake**.



IGNITE YOUR MIND

What evidence supports the existence of the Saraswati River, and what are the leading theories about its origin? How could its potential revival impact water management and agriculture in arid regions of India?

Now, let us turn to the Ganga River System, which is considered to be of the highest prominence both on account of the huge area it drains as well as the **cultural** significance.

Ganga River System

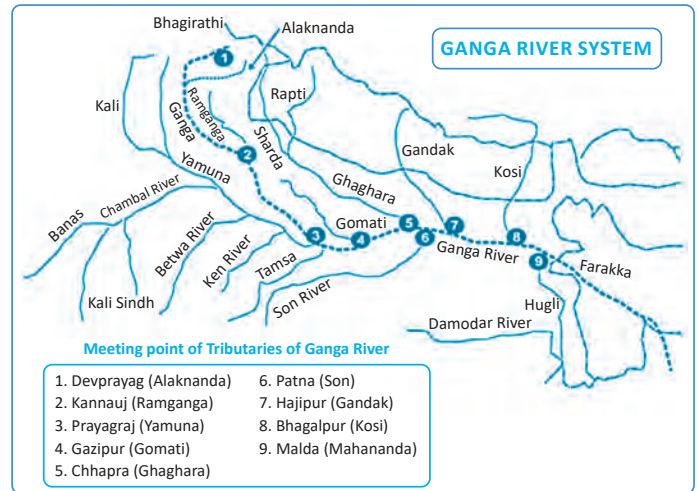


Fig. 4.25: Ganga River System

The 861452 sq km Ganga basin, **India's largest river basin**, has a length of 2525 km. It is India's most revered and significant river, flowing across **Bangladesh** before joining the Bay of Bengal.

- ❑ The river is associated with **fertility** and **spiritual nourishment** and originates as **Bhagirathi** from the **Gaumukh** glacier (3900 m) in Uttarkashi District.
- ❑ It carves its way through the narrow gorges of the Central and Lesser Himalayas. The confluence of the Bhagirathi and Alaknanda at **Devprayag** marks the commencement of the Ganga. (*Alaknanda has its source in the Satopanth glacier above Badrinath.*)
- ❑ Originating from the Satopanth glacier above Badrinath, the Alaknanda comprises the Dhauli and the Vishnu Ganga, which meet at Joshimath or Vishnu Prayag.
- ❑ Upon reaching **Haridwar**, the **Ganga enters the plains**. It initially flows southward, then changes course to the southeast and east before bifurcating into two **distributaries**, namely the **Bhagirathi** and the **Padma**.
- ❑ The river spans a length of 2,525 km, traversing through Uttarakhand (110 km), Uttar Pradesh (1,450 km), Bihar (445 km), and West Bengal (520 km). The Ganga basin covers an extensive area of about 8.6 lakh sq. km in India alone.
- ❑ The Ganga river system stands as the **largest** in India, featuring numerous perennial and non-perennial rivers that originate from the Himalayas in the north and the Peninsula in the south, respectively.
- ❑ **Left-bank** tributaries are Ramganga, Gomti, Gandak, Burhi Gandak, Kali, Kosi, Ghaghara, Mahananda etc
- ❑ **Right-bank** tributaries include Yamuna, Damodar and Son.

The Prayags of the Ganga

Confluence of rivers is known as **Prayag**. There are Five sacred river confluences in the Himalayas in Uttarakhand.

- ❑ **Vishnuprayag**: It is a confluence of the **Alaknanda** River (Source: **Satopanth** Glacier) and the Dhauliganga River (Source: **Mana** Glacier).
- ❑ **Nandaprayag**: Nandaprayag is a confluence of the **Alaknanda** and **Nandakini** Rivers (Source: **Trishul** Glacier).
- ❑ **Karnaprayag**: The **Alaknanda** and **Pindar** Rivers merge here. (Source: **Pindari** Glacier).
- ❑ **Rudraprayag**: The **Alaknanda** and **Mandakini** rivers merge here. (Source: **Chorabari** Glacier). **Kedarnath**, a Hindu holy town is located 86 km from Rudraprayag.
- ❑ **Devprayag**: The **Alaknanda** River and the **Bhagirathi** River meet at this point. (Source: **Gaumukh** or Gangotri Glacier). Both rivers flow on as the Ganga River.

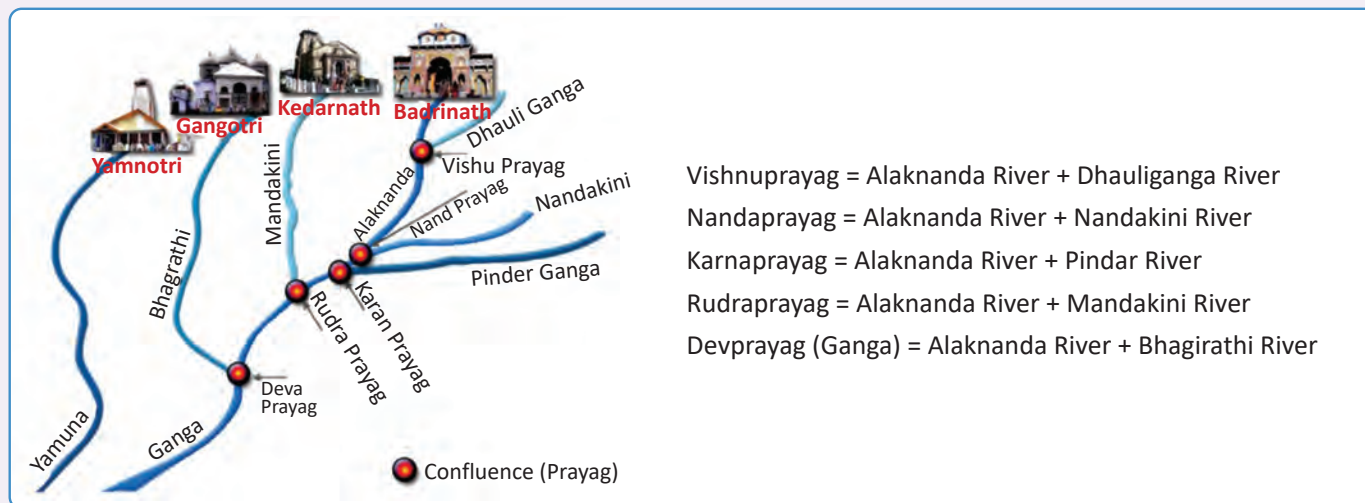


Fig. 4.26: The Panch Prayag

Ganga-Brahmaputra Delta: It forms the **largest delta** in the world between the **Hooghly** River and the Meghna River.

- ❑ The delta's coastline exhibits significant **indentations**, forming a complex network of distributaries and islands, all enveloped by dense forests known as the deltaic region.
- ❑ A substantial portion of the delta comprises **low-lying swamps**, subject to **inundation** by marine water during high tide.
- ❑ Upon entering Bangladesh, the Ganges River is referred to as the **Padma** River.
- ❑ It continues its southward course and ultimately converges with the **Brahmaputra** River near the town of **Chandpur** in Bangladesh. The amalgamated river is subsequently recognized as the **Meghna** River. Ultimately, the river discharges into the Bay of Bengal near Sagar Island.

Major rivers of Ganga river system and their source is summed up in the following table.

River	Source
Bhagirathi	Gangotri glacier
Yamuna	Yamunotri glacier (Banderpoonch)

Chambal	Janpao hills in the Vindhyas
Banas	Aravalli range
Betwa	Vindhyas, north of Narmadapuram
Ken	Barner range
Son	Amarkantak plateau (Radial Drainage)
Ramganga	Garhwal (Uttarakhand)
Ghagra	Gurla Mandhata peak, south of Mansarovar in Tibet (river of the trans-Himalayan origin)
The Kali River (border between Nepal and Uttarakhand)	Glaciers of trans-Himalayas
Gandak River	Tibet-Nepal border
Burhi Gandak	Sumesar hills near the India-Nepal border
Kosi ('Sorrow of Bihar')	Tumar, Arun and Sun Kosi unite at Triveni north of the Mahabharata Range to form the Kosi
Damodar ('Sorrow of Bengal')	Chotanagpur plateau

Left Bank Tributaries

Ramganga River

It rises near river rising in the Garhwal hills near Gairsain and joins the Ganga near Kannauj. It is a tiny river in the **Kumaon** Himalayas. The Shiwalik redirects it to the southwest. In **Nainital**, Uttarakhand, the river flows through the **Jim Corbett National Park**.

Sharda River

It originates from the rises in the **Milam glacier** in the Nepal Himalayas (where it is known as the Goriganga) at **Kalpiani** in the Pithoragarh district of Uttarakhand near **Lipu-Lekh** pass. The river, known as **Kali or Mahakali or Chauk (184 km)**, before joining the right bank of **Ghaghara** near Barabanki.

- ❑ The Kali River forms the **Indo-Nepal border**. In 1992, NHPC commissioned **Tanakpur** Hydroelectric Power in Champawat, demonstrating the potential of the river for hydroelectric power generation.

Sarayu River

It is a significant river in the Indian state of Uttarakhand that drains the **Central Kumaon** region. Before entering **Mahakali** at Pancheshwar, it runs from Sarmul through the cities of Kapkot, Bageshwar, and Seraghat.

- ❑ The **Sharda** River's major tributary is the Sarayu. The contemporary Ghaghara River is said to be an exact replica of the Sarayu River.
- ❑ The city of Ayodhya is situated on the banks of river Sarayu.

Ghaghara River

It originates in the **Mapchachungo** glacier in Tibet's northern Himalayan highlands (near Gurla Mandhata Peak). The Ghaghara River is also known as the **Karnali River** in Nepal, **Mapcha Tsangpo** in Tibet, and **Sarayu River** in Awadh.

- ❑ After collecting the waters of its tributaries – **Tila, Seti and Beri**, it comes out of the mountain, cutting a deep gorge at Shishapani.
- ❑ It is a perennial **transnational** river. This river carves a deep valley through the **Mahabharata Range** in western Nepal.
- ❑ It merges with the **Ganga** close to **Chhapra**, Bihar. With a total length of 1080 km, it is the **second-longest** tributary of the Ganges after the Yamuna in terms of volume.
- ❑ The flood frequency of the river is high and is infamous for shifting courses.

Gomti River

It is a tributary of the Sarayu, originating in the **Madho Tanda** region near Mainkot from a lake called **Fulhar Jheel**. It flows through various districts before meeting the Ganga at Kaithi in Ghazipur.

- ❑ The river is a **perennial**, flat, and sloping river with a drainage area of 30,437 sq km, covering about 960 km. Its major tributary is the **Sai** River.

The Gomti Valley is also known as '**Katyuri Valley**' and is the major **agricultural** region of **Kumaon**. However, **pollution** from industries from Sitapur to Sultanpur is affecting the river, as it passes through sugar processing, paper, and plywood industries.

Gandak River

Originating in Nepal's central Himalayas, it flows between **Dhaulagiri, Annapurna, and Mt. Everest**. It is formed by the **Kaligandak** and the **Trishul Ganga**.

- ❑ It flows into the Indian Great Plains in the Champaran District of Bihar before joining the Ganga River across from Patna at Sonpur.

Its major tributaries are Kali Gandak, the Bari and Trishuli. It is also renowned for its frequent route alterations.

Burhi Gandak

- ❑ This river originates from the western slopes of Sumeswar hills near the India-Nepal border.
- ❑ It joins the Ganga near Monghyr town.

Kosi River

Commonly referred to as the "**Sorrow of Bihar**," is a predecessor river that rises on the Tibetan side of Mount Everest. Kosi is an **antecedent river** with its source to the north of Mount Everest in Tibet, where its main stream Arun rises. It is approximately 729 kilometers long overall.

- ❑ Running parallel to the **Mahabharata Range** and joining the **Arun** to form the **Sapta-Kosi**, it connects the **Son-Kosi** and **Tamur-Kosi** in Nepal.
- ❑ The river emerges into the Bihar Plains close to Chatra in the Saharsa District after cutting through the Shivalik Hills and the Mahabharata Range. 30 km west of Manihari, it merges with the Ganga River.
- ❑ In the Bihar Plains, it divides into a great number of channels (**braided**). It carries a huge amount of water at great speeds. Large scale deposition of sediments occurs on the plains.
- ❑ In 2008, the river Kosi completely changed its course inundating areas which were supposed to be safe from floods. The breach of the Kosi River is now called one of India's worst floods in recent memory and has been classified as a national calamity.
- ❑ To control the river, **embankments** (joint ventures of India and Nepal) have been built on both of its banks since 1962.

Mahananda River

Originating in West Bengal's **Darjeeling Hills**, it merges with the duars near **Siliguri**. Its total length is about 360 km. It forms sharp curves before joining the Ganga River, making it the **last north-bank** tributary.



IGNITE YOUR MIND

What roles do the Ganga floodplains, including the unique floating islands called “phumdis,” play as critical habitats for migratory birds? How do these floodplains contribute to ecological services, and what measures can be taken to protect and enhance their biodiversity?

Right Bank Tributaries

Son River

It rises from the **Amarkantak** Plateau not far from the Narmada. It traverses through waterfalls and the **Kaimur Range** and approaches the Ganga Plain before emptying into the Ganga at **Bankipora**, close to Aara.

- ❑ **Rihand, Johilla, Gopat and North Koel** are the main tributaries of the Son River. Rihand rises in the Matiranga hills of Chattisgarh and meets in Uttar Pradesh to help transform into a mighty stream.

Damodar River

It flows from west to east for about 592 km, drains the eastern **Chotanagpur** Plateau and emerges on Bengal's deltaic plains.

- ❑ **Barakar** is its largest feeder, meeting the main river above **Asansol**. It bends below Bardhaman and meets the Hugli River at Falta.
- ❑ The Damodar River is a heavily **contaminated** river due to the dumping of industrial waste, eutrophication and deposition of **coal** in the region has increased the waste from coal industries.
- ❑ The Damodar valley is famous for hydroelectric power generation and '**The Ruhr of India**'.

Yamuna

Yamuna originates in the **Yamunotri** Glacier on Banderpunch's western slopes, is the Ganga's **longest** and western-most tributary. It flows for about 1,376 km and drains about 366,223 sq. km. area.

- ❑ It traverses through the Nag Tibba, Mussoorie, and Shiwalik ranges, cutting across these geographical features.
- ❑ The **primary** tributary in its upper reaches is the **Tons**, originating from the Bandarpunch glacier. Below Kalsi, where the Yamuna departs from the hills, the Tons river merges with the Yamuna. Notably, at this confluence, the volume of water carried by the Tons is twice that of the Yamuna.
- ❑ It then flows southward, making a **boundary** between Haryana and Uttar Pradesh. It passes through Delhi,

Mathura, and Agra, and joins the **Ganga at Triveni Sangamam in Prayagraj** (Allahabad).

- ❑ The Yamuna River spans a total length of 1,376 km from its source to Allahabad. It gives rise to the incredibly **fertile alluvial** region known as the **Yamuna-Ganges Doab**, situated between the Yamuna and the Ganges in the vast **Indo-Gangetic plain**.
- ❑ Important right bank tributaries mostly originate from the Aravallis, Vindhya Range, and Malwa Plateau.
- ❑ The Yamuna and the mythological River **Saraswati** may have shared water while the **Vedic** era was in effect in Bikaner.

Yamuna River System

Non- Peninsular Tributaries of Yamuna	Peninsular Tributaries of Yamuna
❑ Rishiganga ❑ Uma ❑ Hanuman Ganga ❑ Tons join it in the mountains. ❑ Hindon joins at Ghaziabad in the plain area	❑ Chambal ❑ Sind ❑ Betwa ❑ Ken ❑ Most of the Peninsular rivers flow into the Yamuna between Agra and Allahabad.

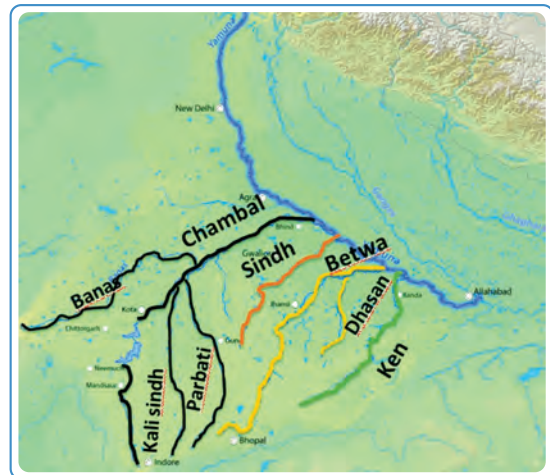


Fig. 4.27: Yamuna River System

Major Tributaries of Yamuna

Chambal River

It is a 1024 km long river, originating from **Singar Chouri peak** in the **Vindhya** mountains near Mhow and flows northwards through a gorge to Kota, Rajasthan.

- ❑ It is a **perennial** river and tributary of Yamuna, surrounded by Luni, Yamuna, Vindhya, and Aravalli ranges.
- ❑ It forms the boundary between Madhya Pradesh and Rajasthan. The river is **rainfed**, and is relatively **pollution-free**.
- ❑ **Tributaries** include Banas, Kalisindh, and Parbati. The river is known for its **badland topography (Chambal Ravines)**, crocodile species, and **Keoladeo** National Park.

❑ **Dams on the Chambal river:** Many dams have been constructed on the perennial chambal, to harness its water both for electricity and irrigation. They include:

- The **Gandhi Sagar dam** marks the initial construction among the four dams established on the Chambal River, situated on the **border** of Rajasthan and Madhya Pradesh.
- Positioned 48 km downstream from the Gandhi Sagar dam, the **Rana Pratap Sagar** dam spans the Chambal River in the **Chittorgarh** district of Rajasthan.
- The **Jawahar Sagar Dam**, the third installment in the Chambal Valley Projects, is located 29 km upstream of Kota city and 26 km downstream of the Rana Pratap Sagar dam.
- Subsequently, the **Kota Barrage**, the fourth structure in the Chambal Valley Projects, is positioned approximately 0.8 km upstream of Kota City in Rajasthan.
- Water released after electricity generation at the Gandhi Sagar dam, Rana Pratap Sagar dam, and Jawahar Sagar Dams is redirected by the Kota Barrage for irrigation purposes in Rajasthan and Madhya Pradesh through an extensive **canal system**.

Chambal Ravines

The Chambal River in Uttar Pradesh, Madhya Pradesh, and Rajasthan is home to a maze of ravines, valleys, and **saw-toothed ridges**. These ravines are being reclaimed for agriculture, pastures, social forestry, and wildlife sanctuaries.

- ❑ The **National Chambal Sanctuary**, established in 1979 is home to the endangered **Gangetic Dolphin**, eight turtle species, Chinkara, Sambar, Blue Bull, wolf, and wild boar.
- ❑ The river's large sediment load causes floods in flood-affected areas, leading to erosion and the creation of **badlands**. The river incision may have resulted from tectonic uplift and tilting.

Betwa River

A Yamuna tributary, which rises in the Vindhya mountains not far from Hoshangabad (**Narmadapuram**), Madhya Pradesh, and flows for 590 kilometers. It confluences with Yamuna near **Hamirpur**. Major tributaries include **Halali** and **Dhasan**, with Halali being the longest.

Sind River

- ❑ The Sind river originates in Vidisha plateau of Madhya Pradesh.
- ❑ It flows for a distance of 415 km before it joins the Yamuna.

Ken River

It begins in Madhya Pradesh and merges with the Yamuna in the Uttar Pradesh district of **Banda**. **Rajghat** and **Paricha** are its two principal tributaries, and it flows through the **Panna Tiger Reserve**.

Brahmaputra River System

Brahmaputra River originates from the **Chemayungdung glacier** of the **Kailash range**, which is close to **Mansarovar Lake**. It is one of the biggest rivers in the world. The Brahmaputra is also known as **Tsangpo** in Tibet, runs about 1,200 kilometers eastward in southern Tibet.

Region	River Name
Tibet	Tsangpo (meaning 'The Purifier')
China	Yarlung Zangbo
Assam Valley	Dihang or Siang, South of Sadiya: Brahmaputra
Bangladesh	Jamuna River
	Padma River: Combined Waters of Ganga and Brahmaputra
	Meghna: From the confluence of Padma and Meghna

One significant right bank tributary is the **Rango Tsangpo**. The river carves out a deep valley in the **Central Himalayas** and close to **Namcha Barwa** takes a **syntaxial bend** before emerging as a turbulent and active river.

- ❑ It enters India as **Dibang** or **Sikang** and gets its principal left bank tributary **Lohit** west of Sadiya town in Arunachal Pradesh. It thereafter acquired the name **Brahmaputra**.
- ❑ **Right bank** tributaries include Subansiri, Bhareli, Kameng, Jaldhaka, Manas, Sankos, Tista, and Raidak, while **left-bank** tributaries include Dihang, Lohit, and Burhi Dihang Dhasiri, Kalang, and Kapili.
- ❑ The **Teesta** was a tributary of the Ganga prior to the floods of **1787** after which it diverted its course eastwards to join the Brahmaputra.
- ❑ As the Brahmaputra courses through Assam, it adopts a braided channel, characterized by the flow into shallow, interconnected channels separated by deposited earth. In this region, the channels continually shift, leading to considerable meandering, and the river carries a substantial amount of silt. This is why it is called the Sorrow of Assam.
- ❑ It enters Bangladesh near **Dhubri** known by the name **Jamuna**, joined by Ganga and flows southward joining **Teesta** on its right bank, known as the river **Padma** in Bangladesh. It finally falls into the Bay of Bengal.
- ❑ The river in its course forms a large number of Islands. The most important riverine Island of them is "Majuli" island.

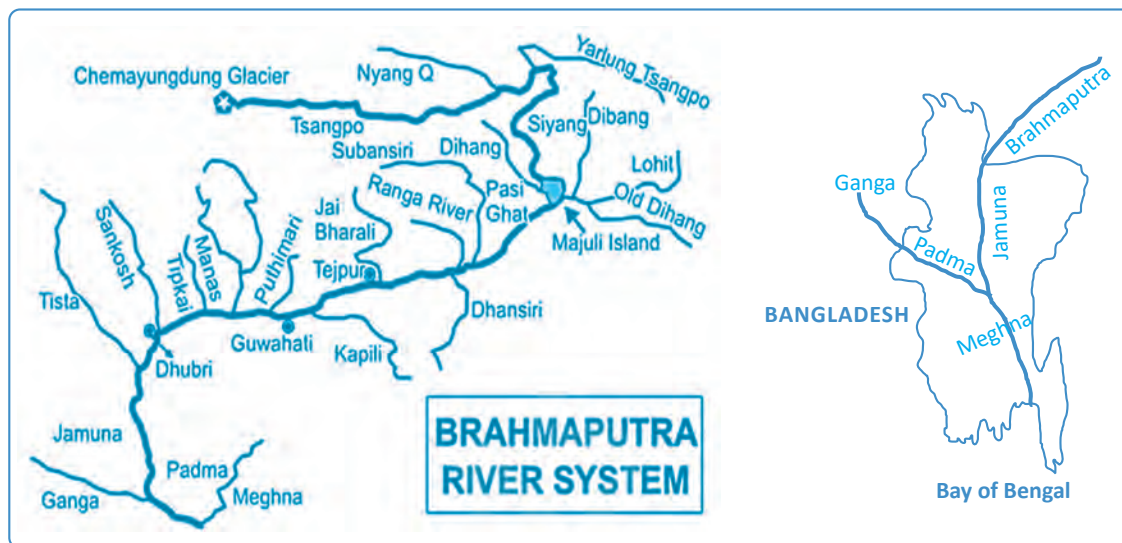


Fig. 4.28: Brahmaputra River System & Its Confluence

Let us now examine the tributaries of Brahmaputra in detail.

Lohit River

Originating in Tibet as the **Zayu**, flows through **Mishmi** Hills, Arunachal Pradesh. Its total length is about 2677 km. It joins the **Siang** River to form the Brahmaputra. It is characterized by a **reddish** color as it carries **lateritic** soil.

Barak River

Originating in Manipur hills, it is part of the **Surma-Meghna** River System. It flows about 564 km through the four states of **Manipur, Nagaland, Mizoram, and Assam** before reaching the Bay of Bengal.

- ❑ The Barak river enters Bangladesh as it traverses **Surma and Kushiara**, both of which reunite into **Meghna river**.
- ❑ The river Tiulu is the most important tributary of the river **Barak**. Apart from the Tiulu River, the Barak River is joined by various tributaries such as the **Jiri, Irang, and Makru in Manipur, and the Tuivawl and Koladyne in Mizoram**.
- ❑ The river is very much prone to floods, particularly during the monsoon season.
- ❑ The Barak basin has **Mawsynram** and **Cherrapunji** which receive the highest rainfall in the world.

Kameng River

- ❑ It originates near Arunachal Pradesh's **Gorichen** mountains. Its total length is about 264 km. Its drainage basin is about 11,843 sq. km.
- ❑ It flows between the **Eagle Nest Sanctuary** and **Pakke Tiger Reserve**. It helps in ensuring ecological balance.

Rangit River

- ❑ Originating from **Sikkim**, is renowned for its numerous **rapids** and is known for **rafting sports** worldwide, offering excellent **camping grounds** on its banks. Its total length is about 80 km.

Sankosh River

- ❑ It is also known as '**Puna Tsang Chu**' in Bhutan, forms the boundary between Assam and Arunachal Pradesh, meeting the Brahmaputra River below Dhubri.

Manas River

- ❑ Originating from Tibet, it flows through the **Greater Himalayas**, collecting tributaries in the Lesser Himalayas. It flows for a total length of about 400 km. It joins the Brahmaputra in the plains. It is a **transnational** river between **Bhutan and India**.
- ❑ It flows through two crucial protected areas: The **Royal Manas National Park** in Bhutan and the **Manas Wildlife Sanctuary**.

Subansiri River

- ❑ It is the greatest tributary of the Brahmaputra River that travels 442 km across the Himalayas and Upper Assam Plain. It is a trans-Himalayan river predecessor. It divides the **Miri** Hills and the **Abor** Hills.

Dhansiri River

- ❑ Originating from the Naga Hills, it flows approximately 350 km through Nagaon before joining the Brahmaputra River. It flows through **Itanki National Park**, Nagaland.

Manipur River

- ❑ It originates in northern Manipur, flows southwards. It drains **Loktak Lake** and joins the **Chindwin** Valley, a tributary of the River **Irrawaddy** in Myanmar.

Teesta River

It originates in **Kanchenjunga**, flows South-west through West Bengal, joining the Brahmaputra in Bangladesh.

- ❑ It merges with the **Padma** River in Bangladesh, ending in the Bay of Bengal. The river forms a deep gorge in **Darjeeling hills**, dividing them into **Tiger Hill and Kalimpong Hill** Ranges.

- ❑ The **Teesta Barrage Dam** aids in irrigation for **tea** and **jute** production.
- ❑ The **Rangit** River (originating in Sikkim), a major tributary, is a contentious issue between India and Bangladesh.

Impact of Melting of Glaciers on the Himalayan Drainage System

The melting of glaciers is the primary source of the Himalayan drainage system. The Himalayas are referred to as the **third pole**. Glaciers are **lifeline** to the population of major sub-continent countries of Pakistan, India, Nepal, Bhutan and Bangladesh. The melting of these glaciers is causing problems and putting pressure on the resources of these countries. Himalayan glaciers are **receding** and **global warming** is also contributing to it. Some of the key impacts are:

- ❑ **Increased Runoff:** Melting glaciers contribute to an increased volume of water entering river systems. This influx of water can lead to higher levels of runoff, affecting river discharge and potentially causing floods during peak melting periods.
- ❑ **Shortage in the long term:** Compromise the Indian drainage system's water supply, leading to decreased river flow and potential **imbalances** in seasonal flow.
- ❑ **Altered River Flow:** Glacier meltwater serves as a crucial component of river flow, especially in regions where glaciers are a primary water source. Changes in the **timing** and **magnitude** of this meltwater can lead to alterations in the overall flow patterns of rivers.
- ❑ **Shifts in Hydrological Regimes:** The melting of glaciers can result in shifts in hydrological regimes, affecting the **seasonality** of river flows. This can impact water availability for both ecosystems and human activities, with potential consequences for agriculture, industry, and water supply.
- ❑ **Sediment Transport:** Glacial meltwater carries a significant amount of sediment, including rocks and debris eroded by the glacier. As glaciers melt, there can be changes in sediment transport patterns, affecting **riverbed morphology** and potentially leading to increased sedimentation in downstream areas.
- ❑ **Glacial lake outburst floods:** Melting glaciers can give rise to the formation of glacial lakes. If the **moraine dams** of these lakes fail, it can result in Glacial Lake Outburst Floods (**GLOFs**), posing significant threats to downstream communities, infrastructure, and the overall drainage system.
 - In June 2013, Uttarakhand experienced an uncommonly high level of precipitation, resulting in the **Chorabari glacier melting** and triggering the surge of the Mandakini river.
 - In August 2014, the village of **Gya** in Ladakh was struck by a flood resulting from the outburst of a glacial lake.
 - In February 2021, flash floods occurred in **Chamoli** district, Uttarakhand, suspected to be associated with GLOFs.
 - Sikkim recently witnessed a GLOF when the South **Lhonak Lake**, situated at an elevation of 17,000 feet in the northwest of the state, underwent a breach due to persistent rainfall. This event further led to the breach of the **Chungthang Hydro-Dam** on the **Teesta** River in Sikkim, exacerbating the overall situation.
 - The subsidence of **Joshimath** is thought to be influenced by the cascading streams originating from Vishnuprayag and coursing along natural channels, with their augmented flow attributed to glacier melting.
 - **Changes in water quality:** The quality of river water may deteriorate due to increased sedimentation. Increased melting may lead to changes in **nutrient levels**, affecting aquatic ecosystems and water quality downstream.
 - The **long-term** consequences of glacier melt in the Himalayas could affect **agriculture, hydroelectric power generation, and the ecological balance**.
 - **Mitigation** strategies include reducing greenhouse gas emissions and investing in water management infrastructure like dams and irrigation systems.

Important Hydro-Power Projects in the Himalayan Region

- ❑ **Tehri Stage 2:** 1000 MW on Bhagirathi River
- ❑ **Tapovan Vishnugadh:** 520 MW on Dhauliganga River
- ❑ **Vishnugadh Pipalkoti:** 444 MW on Alaknanda River
- ❑ **Singoli Bhatwari:** 99 MW on Mandakini River
- ❑ **Phata Bhuyan:** 76 MW on Mandakini River
- ❑ **Madhyamaheshwar:** 15 MW on Madhyamaheshwar Ganga River
- ❑ **Kaliganga 2:** 6 MW on Kaliganga River



IGNITE YOUR MIND

In recent time there have been concerns about China's plans for dam construction and potential diversion of the Brahmaputra (Yarlung Tsangpo) River, which could impact water flow downstream in India and Bangladesh. Examine the measures that downstream countries can take to mitigate the adverse effects of upstream water projects.

Let us now move on to the peninsular drainage system that is markedly different from the Himalayan drainage system.

PENINSULAR DRAINAGE SYSTEM

With its wide, shallow valleys and established rivers, the Peninsular drainage system is **older than the Himalayan drainage** system.

- ❑ Major Peninsular rivers are split by the **Western Ghats**, which run from north to South and run close to the Western Coast. Due to them, the discharge is split into the Bay of Bengal and the Arabian Sea.
- ❑ Except for Narmada and Tapi, most significant rivers go from **west to east**.
- ❑ The Mahanadi, Godavari, Krishna, and Kaveri are some of the principal river systems.
- ❑ Peninsular rivers run seasonally (**non-perennial**), have well-defined channels, and are characterized by lack of meandering.
- ❑ The slopes are largely gentle or present a low gradient. Hence, the velocity of the stream is lesser than the young Himalayan rivers. However, steep slopes can be found

in the areas of faulting causing sudden changes in the speed of flow.

- ❑ The **east-flowing** peninsular rivers form **deltas** at their mouths. These are very fertile lands and water-intensive crops are grown here.
- ❑ The **west-flowing** rivers form **estuaries** in place of deltas.

Evolution of Peninsular River System

The current drainage networks in Peninsular India can be explained by using the two following significant geological occurrences:

- ❑ **Theory around Sahyadri-Aravali axis:** Geologists propose that in the past, the Sahyadri-Aravali axis served as the primary **water divide**.
 - According to the hypothesis, the current peninsula is the remnant of a larger landmass, with the **Western Ghats** positioned centrally.
 - This setup resulted in two drainage systems—one flowing eastwards into the Bay of Bengal and the other flowing westwards into the Arabian Sea.



Fig. 4.29: The Peninsular River System

- During the early **Tertiary period**, corresponding with the Himalayan formation, the western part of the Peninsula experienced **cracking** and **submersion** in the **Arabian Sea**.
- The collision of the Indian plate with the **Eurasian plate** led to subsidence in certain regions of the Peninsular block, creating a series of rifts and faults. (It coincides with the formation of Himalayas)
- The **Narmada** and **Tapi** rivers, now west-flowing, follow these rifts.
- The plausibility of this theory is supported by the **straight coastline**, the steep western slope of the Western Ghats, and the **absence of delta** formations along the western coast.

□ **Theory around bending of Northern part of Peninsula at the time of upheaval of Himalayas:** It is theorized that the west-flowing rivers in the peninsula don't follow valleys carved by the rivers themselves. Instead, they have situated themselves within **two fault rifts** running **parallel** to the Vindhyas.

- These faults are believed to have originated during the bending of the **northern** part of the Peninsula at the time of the **Himalayan upheaval**.
- The Peninsular block, located south of these cracks, underwent a slight **eastward tilt** during this geological event, establishing the overall drainage orientation towards the Bay of Bengal.
- **Critics** argue that such tilting should have led to an increase in the gradient of the river valleys and triggered **rejuvenation** of the rivers.
- However, this phenomenon is notably absent in the Peninsula, with only a few exceptions such as the presence of **waterfalls**.

East Flowing River System

The East-flowing rivers of the Peninsular system flow into the Bay of Bengal. Most of these rivers form deltas which are some of the most fertile regions and hence rich in agriculture. Let us now examine the major rivers of the system in detail.

Mahanadi River

The Mahanadi River has a length of 851 km, making it the third longest river in peninsular India. Before reaching the Bay of Bengal, it originates in the **Sihawa Mountain** in the Dhamtari region of the Amarkantak Plateau (Bastar highlands of Dandakaranya near Raipur) and travels through Madhya Pradesh, Odisha, Jharkhand, and Maharashtra.

□ It is surrounded by the **Central India hills** to the north, the **Eastern Ghats** to the south and east, and the **Maikala range** to the west.

- The fertile soil and stream of silt deposition in the river valley are well known. The 15 km long artificial Hirakud Dam was built in Sambalpur on a river to aid in agriculture. Other dams include Tikarpara and Naronj dams, Jobra Barrage near Cuttack, Amritdhara Waterfall on the Hasdeo River and Chandrasasini Devi on the confluence of Mand and Mahanadi.

□ It encompasses **Chilka Lake** and is the biggest river in the state of Odisha. The well-known pilgrimage site **Puri** is situated along the river. At the tip of the Mahanadi Delta is **Cuttack City**.

□ At **False Point**, the river enters Bay of Bengal through many channels.

□ **Tributaries:** The initial stretch of the river is situated within the basin known as the '**Chhattisgarh Plain**,' which has a **saucer-shaped topography**.

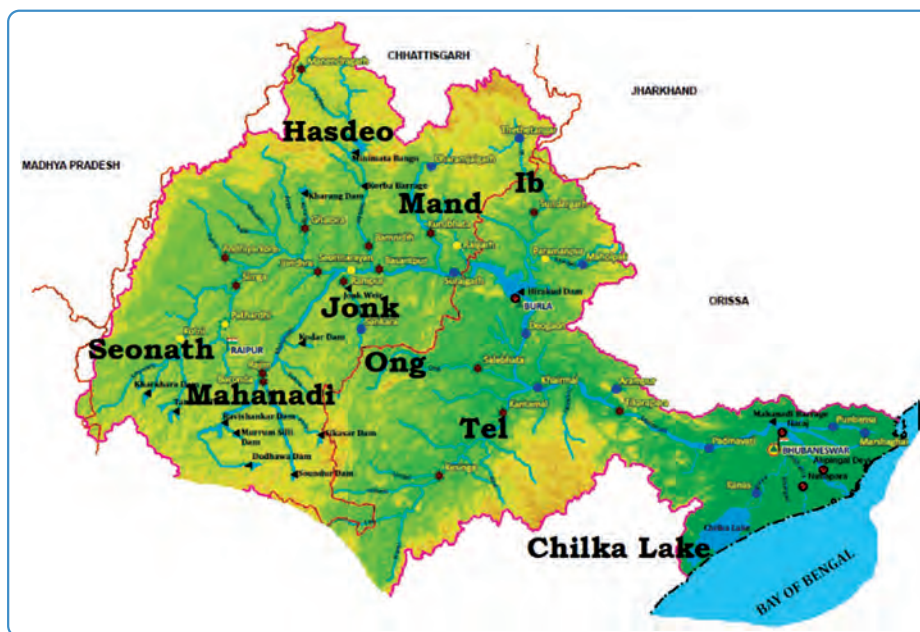


Fig. 4.30: Mahandi River System

- Encircled by hills on the north, west, and south, this basin witnesses the **convergence** of numerous tributaries into the main river from these directions.
- Tributaries from the left bank include the Seonath, Hasdeo, Mand, and Ib. The Ong, Tel, and Jonk are right bank tributaries.
- **Industries around Mahanadi basin:** Three important urban centers in the basin are **Raipur, Durg and Cuttack**. Mahanadi basin, because of its rich mineral resource and adequate power resource, has a favorable industrial climate.
 - The Important industries presently existing in the basin are the **Iron and Steel** plant at Bhilai, aluminum factories at Hirakud and Korba,

paper mill near Cuttack and cement factory at Sundargarh.

- Other industries based primarily on agricultural produce are sugar and textile mills. **Mining** of coal, iron and manganese are other industrial activities.

Godavari River

It is the **largest river in peninsular India**, spanning 1465 km and covering 312812 sq km and it has the largest basin area among Peninsular rivers, constituting one-third of the Indus Basin area. It originates at Triambakeshwar of Nashik district in the Western Ghats and drains eastern and southeastern Maharashtra, Bastar plateau, and Telangana and Andhra regions.

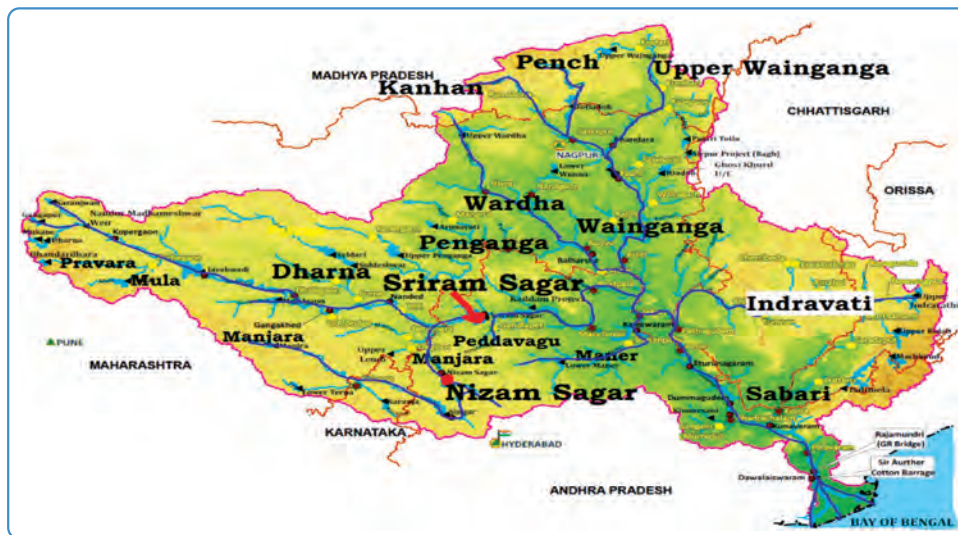


Fig. 4.31: Godavari Basin

- It is regarded as '**Dakshin Ganga**'. The river is bound by the Eastern Ghats on the south and the east, by Satmala hills, the Ajanta range and the Mahadeo hills on the north, and by the Western Ghats on the west.
- **Tributaries:** The left bank tributaries surpass the right bank tributaries both in number and size.
 - **Right Bank tributaries:** **Manjira** (724 km) joins the Godavari after passing through the Nizam Sagar. Others include Pravara, Mula, Peddavagu, Maner.
 - **Left bank tributaries:** The Dharna, Penganga, Wainganga, Wardha, **Pranahita** (conveying the combined waters of Penganga, Wardha, and Wainganga), Pench, Kanhan, Sabari, Indravati, and others.
- The river is subjected to heavy floods south of **Polavaram**, where it flows through a picturesque **gorge**.
- Beyond **Rajahmundry**, the river bifurcates into two main streams—
 - The **Gautami** Godavari on the east and
 - The **Vasishta** Godavari on the west.
- It forms a vast delta before flowing into the Bay of Bengal, which is characterized by a **lobate** type, featuring a **round bulge** and numerous **distributaries**.
- It forms an **inland lagoon**, the '**Kolleru Lake**', lying to the south-east of **Kakinada**. Important cities located on the bank of the Godavari River are Nashik, Triambakeshwar, Nizamabad, Seoni, etc.
- **Projects on the Godavari River:** Significant projects accomplished during the planning period include Godavari Barrage, Upper Penganga, Jayakwadi, Upper Wainganga, Upper Indravati, Upper Wardha, **Pranahita-Chevala**, **Polavaram**, **Srirama Sagar**, **Nizam Sagar in Telangana** (on Manjira), **Kaleshwaram project in Telangana** (off shoot of Chevala(Pranahita)), **Polavaram in Andhra** (Pattiseema Project interlink Godavari and Krishna rivers); **Bheemgarh dam**(on Wainganga in MP), Sir Arthur cotton Barrage and Pochampad project.
- **Mineral Wealth of the Godavari basin:** The upper reaches of the Godavari drainage basin are characterized by the presence of the **Deccan Traps**, which contain

metallic minerals such as magnetite, epidote, biotite, zircon, and chlorite.

- In the middle part of the basin, the predominant geological composition includes phyllites, quartzites, amphiboles, and granites.
- Moving downstream in the middle basin, the landscape is primarily occupied by sediments and rocks of the **Gondwana** group. The Gondwanas consist mainly of **detritals**, featuring some substantial coal seams, including the **Singareni Coal Seam**.
- The lower part of the drainage basin is prominently influenced by the Eastern Ghats, primarily formed from the **Khondalites**.

❑ **Industrial richness of the Godavari basin:** Key urban centres within the basin include Nagpur, Aurangabad, Nashik, and Rajahmundry. Nashik and Aurangabad are notable for hosting a substantial number of industries, particularly in the **automobile** sector.

- Apart from these, industries in the basin are predominantly oriented around agricultural products, encompassing activities such as **rice milling**, **cotton spinning** and weaving, and the extraction of sugar and oil.
- Additionally, the basin houses **cement** production and a few small engineering industries.

❑ **Drought and Floods in the Godavari basin:** The lower reaches of the Godavari basin grapple with **flood** issues. The coastal regions are susceptible to **cyclones**.

- **Drainage congestion** is a challenge in the delta areas due to the flat topography.
- A significant portion of Maharashtra within the basin, particularly **Marathwada**, is prone to **drought**.

The Godavari River has a cultural significance due to the **Kumbh Mela** which takes place on its banks. Other locations of Kumbh Mela are the Shipra River (Ujjain), Ganga (Haridwar), and Saraswati River (Prayag).

Krishna River

It is the **second largest** river in the peninsula and rises near **Jor village of Satara district** of Maharashtra at an altitude of 1,337 m just north of Mahabaleshwar in the **Sahyadri** hills.

- ❑ It flows for 1300 km through Maharashtra, Karnataka, Andhra Pradesh, and Telangana. It is bounded by the Eastern Ghats on the south and the east, the Balaghat range on the north, by and by the Western Ghats on the west.
- ❑ The predominant portion of the basin is characterized by agricultural land, constituting **75.86%** of the total area.
- ❑ The Krishna River forms an extensive **delta** with a shoreline stretching approximately 120 km. The Krishna delta seems to blend with the delta formed by the Godavari and extends around 35 km into the sea.

❑ The river is fed by seasonal monsoon rains, causing land erosion and deposition of productive soil. It is used for agricultural purposes but fluctuating flow limits its irrigation use.

❑ **Tributaries:** The river has a large number of tributaries.

- **Right bank tributaries:** Koyna, Ghataprabha (Hiranyakashi + Markandeya), Malprabha and Tungabhadra (Unification of Tunga + Bhadra originating from Gangamula in the Central Sahyadri. Total length is 531 km).

- **Left bank tributaries:** Bhima (longest of length 861 km, originates from the Matheron Hills and joins the Krishna near Raichur), Dindi, Musi (Origin in Anantgiri Hills, joins it at Wazirabad) and Munneru.

- **The Koyna:** Although a small tributary, it gained prominence due to the **Koyna Dam**. This dam is believed to have been a primary factor in the destructive **earthquake** of 1967, measuring 7.5 on the Richter scale, resulting in the loss of about 200 lives.

- **Bhima River:**

❑ **Originates** in the Matheron Hills, near **Bhimashankar Temple** in the Bhimashankar hills, **Western Ghats, Sahyadri Hills, Maharashtra**. The Bhima covers a distance of 861 km before joining the Krishna near **Raichur**.

❑ **Left Bank tributaries:** Ghod, Sina, Kahini

❑ **Right Bank tributaries:** Bhama, Indrayani, Mulamutha, Nira

❑ **Flows through** Maharashtra, Karnataka, Telangana, then joins **Krishna near Raichur**.

❑ Also known as **Chandrabhaga River**, particularly at **Pandharpur, Maharashtra**, due to its moon-like shape.

❑ **Important locations:** **Bhimashankar Temple (Jyotirlinga shrine)**, **Vitthal Temple of Pandharpur**, **Kanganahalli Buddhist site** in Karnataka.

❑ **Dam:** Ujani Dam

- **Malaprabha River:**

❑ **It originated** in **Kanakumbi** in the Belgaum district.

❑ **Flows through cities:** **Kanakumbi, Khanapur, Soundatti (Malaprabha Dam), Nargund, Pattadakal, Kudalasangam**.

❑ **Important towns** on its banks: **Aihole, Pattadakal, and Badami**.

❑ Joins **Krishna at Kudalasangama** in Bagalkot district.

- **The Tungabhadra:** It is created by the confluence of the **Tunga** and the **Bhadra**, both originating from **Gangamoola** in the Central Sahyadri. It has a total length of 531 km. Its origin Gangamula in Varaha Parvatha, Central Sahyadri; **Famous City:** Hampi; Tungabhadra also known as Pampa during epic period.



Fig. 4.32: Krishna River Basin

- **The Musi:** At Wazirabad, the Krishna River receives its final significant tributary, the Musi, with the city of **Hyderabad** situated along its banks
- ❑ **Drought in the Krishna basin:** Certain regions within the basin, notably the **Royalaseema** area of Andhra Pradesh, along with Bellary, Raichur, Dharwar, Chitradurga, Belgaum, and Bijapur districts of Karnataka, as well as Pune, Sholapur, Osmanabad, and Ahmednagar districts of Maharashtra, are prone to drought.
- ❑ **Flood Prone areas of the Krishna:** The **deltaic** areas of the basin face the risk of flooding. There is a consistent observation of the riverbed in the deltaic region being elevated due to continuous **silt** deposition, resulting in a decrease in the channel's carrying capacity. The flood issue is exacerbated by the intense and short-duration rainfall from coastal cyclones.
- ❑ **Projects on the Krishna river:** Key hydro power stations within the basin encompass Koyna, Tungabhadra, SriSailam, Nagarjuna Sagar, Almatti, Naryanpur, and Bhadra.
 - The **Tungabhadra** is a significant **inter-state** project within the basin. Its operation is crucial for regulating flows among the beneficiary states of Karnataka and Andhra Pradesh.
- ❑ **Mining and Industry:** The basin boasts abundant mineral deposits, presenting a promising potential for industrial development.
 - Presently, key industrial activities in the basin include iron and steel production, cement manufacturing, sugar cane processing, vegetable oil extraction, and rice milling.
 - Recent discoveries of oil in the basin are anticipated to influence the future industrial landscape significantly.
 - Major urban centres in the basin include **Pune** and **Hyderabad**. Hyderabad, the capital of Telangana, has evolved into a major **IT** hub. Pune, situated in Maharashtra, is home to numerous **automobile** and **IT** industries and is a prominent education center.

Kaveri River

The **Talakaveri** on the **Brahmagiri Hills** in the Karnataka district of **Kodagu** are the source of the 805 km long **Kaveri (Cauvery) River**, also known as “the Ganga of the South.”

The river spans a total length of 805 km from its origin to its outfall. The Cauvery basin covers the states of Tamil Nadu, Karnataka, Kerala, and the Union Territory of Puducherry, encompassing a drainage area of 81155 square kilometers.

- ❑ It is surrounded by the Western Ghats to the west, the Eastern Ghats to the east and south, and ridges that serve as boundaries separating it from the Krishna basin and the Pennar basin to the north.
- ❑ The **Nilgiris**, an extension of the Western Ghats, reach eastward to the Eastern Ghats, creating a **natural** and **political** division in the basin. This division consists of the **Karnataka plateau** in the north and the **Tamil Nadu plateau** in the south.
- ❑ The **upper catchment** receives rain during the summer **southwest** monsoon and the **lower catchment** winter **retreating** monsoon, therefore making it a **perennial river** that carries water all year. It travels across Tamil Nadu, Kerala, and Karnataka.
- ❑ **Physiographic division:** The basin can be categorized into three main parts: The Western Ghats, the Plateau of Mysore, and the Delta. The delta region stands out as the most fertile tract in the basin.

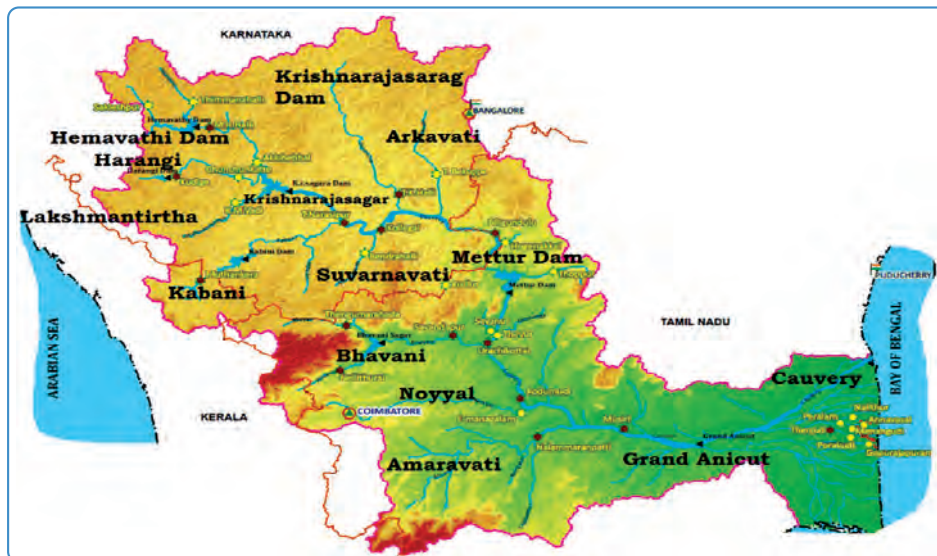


Fig. 4.33: Cauvery River System

- Various soil types prevail in the basin, including black soils, red soils, laterites, alluvial soils, forest soils, and mixed soils.
- Red soils dominate extensive areas in the basin, while alluvial soils are prevalent in the delta regions.
- ❑ The Cauvery River is exceptionally **well-regulated**, with **90 to 95** percent of its irrigation and power production capabilities effectively harnessed. A significant portion of the basin is dedicated to **agricultural land**, constituting **66.21** percent of the total area.
- ❑ The Cauvery is known for its relatively less fluctuation of water level among the peninsular rivers
- ❑ **Tributaries:**
 - **Left Bank tributaries:** The Harangi, Hemavati, Shimsha, and Arkavati.
 - **Right Bank tributaries:** Lakshmantirtha, Kabbani, Suvarnavati, Bhavani, Noyil, and Amaravati.
 - The **Kabini River**, which serves as a border between **Bandipur and Nagarhole National Parks**, meets the Nagarhole River as it passes through the Nagarhole National Park. The major river that drains the park is the Kabini, a branch of the Cauvery River.
 - The **Sathyamangalam Tiger Reserve (STR)** is situated where the **Western and Eastern Ghats converge**.
 - The **Palar River** empties into the Cauvery River and flows into it. This region is home to a sizable **tiger** population and is adjacent to **Bandipur, Mudumalai, and Nagarhole**, three Tiger conservation landscapes.
- ❑ The river descends from the South Karnataka Plateau to the Tamil Nadu Plains, passing through the **Shivasamudram** waterfalls (101 m high).
- ❑ At Shivasamudram, the river bifurcates, creating a series of falls and rapids with a height of 91 m, utilized for power generation.
- ❑ The two branches reunite after the falls, flowing through a wide gorge known as **'Mekedatu'** (Goat's Leap) and forming the **boundary** between **Karnataka** and **Tamil Nadu**.
- ❑ Heading south after **Hogenakkal** Falls, the river enters the **Mettur Reservoir**.
- ❑ The tributary **Bhavani**, joins about 45 km below the reservoir, and the river widens with a sandy bed, known as **'Akhandu Cauvery'**, after the addition of **Noyil** and **Amaravathi** on the right bank.
- ❑ Beyond **Tiruchirapalli** district, the river divides into the **Northern** branch, known as **'The Coleron'**, and the Southern branch remains as Cauvery, marking the beginning of the Cauvery Delta.
- ❑ After 16 km, the two branches reunite to form **'Srirangam Island'**. Along the Cauvery branch lies the **"Grand Anicut,"** believed to be constructed by a **Chola King** in the 1st Century A.D.
- ❑ Below the Grand Anicut, the Cauvery branch bifurcates into **Cauvery** and **Vennar**, with these branches further dividing into smaller networks all over the delta.
- ❑ **Floods in the Cauvery basin:** The Cauvery basin is **leaf** shaped in **Tamil Nadu** and **fan** shaped in **Karnataka**. Due to the shape, the run-off does not drain off quickly because of and therefore **no fast rising** floods occur in the basin.
- ❑ **Industry of the Cauvery basin:** The city of **Bangalore** lies just beyond the boundaries of this basin. Key industries within the basin encompass.
 - the **cotton** textile industry in **Coimbatore** and **Mysore**,
 - **Cement** factories in **Coimbatore** and **Trichinapally**, and industries associated with minerals and metals.
 - Additionally, the **Salem steel** plant and numerous engineering industries in **Coimbatore** and **Trichinapally** are located within this basin.
- ❑ **Important projects on the Cauvery:** In the pre-plan era, several projects were successfully executed in this basin,

including **Krishnarajasagar** in Karnataka (Mysore), **Mettur** dam, and the **Cauvery delta system** in Tamil Nadu.

- Notably, lower Bhavani, Hemavati, Harangi, and Kabini are significant projects that were completed during the plan period.

Other East Flowing Rivers

Apart from the major rivers, peninsular region is traversed by the following rivers:

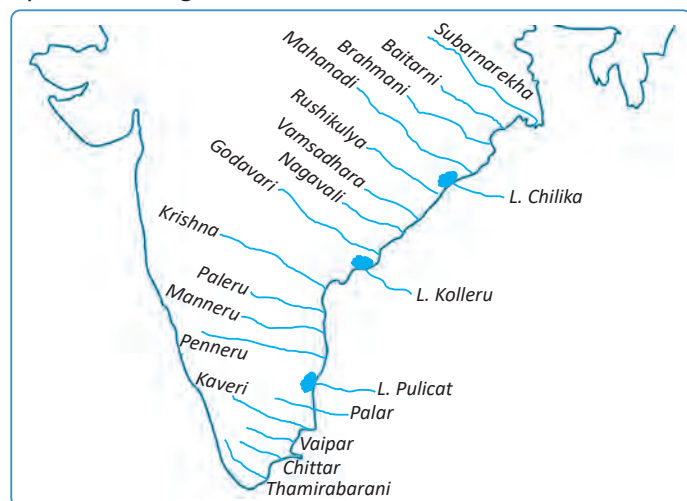


Fig. 4.34: East flowing rivers

Pennar River

The Pennar, also known as **Uttara Pinakini**, **Penneru** stands as one of the prominent rivers in the peninsula. Originating in the **Chennakesava** hill of the **Nandidurg** range within Karnataka's **Chikkaballapur** district, the river flows eastward, eventually emptying into the Bay of Bengal.

- Covering a total length of 597 km from its origin to the Bay of Bengal outfall, the Pennar basin extends across the states of Andhra Pradesh and Karnataka, encompassing an area of approximately 55 thousand sq. km.
- The **fan-shaped basin** is demarcated by the **Erramala range** to the north, the **Nallamala** and **Velikonda ranges** of the Eastern Ghats to the east, the Nandidurg hills to the south, and a narrow ridge separating it from the **Vedavati valley** of the Krishna Basin to the west. Additional hill ranges to the south of the river include the **Seshachalam** (renowned for **Red Sanders**) and **Paliconda ranges**.
- A significant portion of the basin is dedicated to agriculture, constituting **58.64%** of the total area.
- **Left Bank** tributaries include the Jayamangali, the Kunderu and the **Right bank** tributaries are the Chiravati, the Papagni etc.
- Tungabhadra high level canal in the Krishna basin continues to irrigate areas in Pennar basin. The **Somasila project** is the only major project in the basin.
- **Industry:** The scope for industrial development is limited due to scarcity of water, power potential and mineral resources. There are by and large, no major industries.

The existing small industries are mostly based on agricultural produce such as cotton weaving, sugar mills, oil mills, rice mills etc.

The **Gandikota Canyon**, situated in the Kadapa district of Andhra Pradesh, is a mesmerizing labyrinth of rugged rocks adorned in various shades of red.

This breathtaking gorge has been shaped by the flowing waters of the renowned **Pennar** River, originating from the Erramala hills.

Often referred to as the **Grand Canyon of India** due to its striking resemblance to the iconic U.S. landmark, the area also features two ancient temples and a 12th-century fort perched atop the hills encircling the gorge.

The Pennar River gracefully descends from the Erramala hills, contributing to the splendor of this remarkable canyon, acknowledged as the Grand Canyon of India.

Ponnaiyar River

- Also called **Dakshina Pinakini**, Ponnaiyar is a modest watercourse primarily restricted to the coastal region. It spans a limited area across the states of Tamil Nadu, Karnataka, and Andhra Pradesh.
- The basin is bordered to the northwest and south by diverse ranges of the Eastern Ghats, including the **Velikonda** Range, the **Nagari** Hills, the **Javadi** Hills, the **Shevaroy** Hills, the **Chitteri** Hills, and the **Kalrayan** Hills. To the east, it is demarcated by the Bay of Bengal.
- Kelavarapalli and Krishnagiri dams; Sathanur Dam near Tiruvannamalai

Vaigai River

To the south of the **Cauvery delta**, numerous streams traverse the region, with the Vaigai emerging as the longest among them.

- The Vaigai basin holds significance among the 12 basins situated between the Cauvery and Kanyakumari. Vattaparai Falls are located in it.
- Bounded by the **Varushanadu** hills, the **Andipatti** hills, the **Cardamom** hills, and the **Palani** hills to the west, and the **Palk Strait** and **Palk Bay** to the east, this basin covers an area of 7,741 sq. km, entirely situated within the state of Tamil Nadu.

Subarnarekha River

- It originates from Jharkhand's **Ranchi Plateau** and, in its lower course, divides West Bengal from Odisha.
- Between the **Ganga** and **Mahanadi deltas**, it joins the Bay of Bengal to create an estuary. The river provides water to **TISCO Jamshedpur**.
- Spanning a total length of 395 km, the river features **Hundru Falls** along its course.

Brahmani River

- It is 420 km long, formed by the confluence of the **Koel** and **Sankh** rivers at **Rourkela**.
- It drains the western **Garhjat Hills**, flows through the Bonai, Talcher, and Balasore districts, merges into the Bay of Bengal, and forms a **delta complex** below Bhadrak.
- Rengali Dam, Jokadia & Jenapur Anicuts are notable dams here.

Nagavali River

- It is one of the principal rivers in **Southern Odisha** and **Northern Andhra Pradesh** is the 256 km long Nagavali River, often referred to as **Langulya**.
- The Eastern Ghats' **Kalahandi** District is where it first appears.

Sarada River

- A medium-sized east-flowing river, it is situated in the Visakhapatnam district of Andhra Pradesh.
- Bound by the Nagavali River to the north, the Gosthani and Gambiramgedda rivers to the east, the Bay of Bengal to the south, and the Machhkund sub-basin of the Godavari River to the west, the river basin is well-defined in its geographical context.

Vamsadhara River

- It flows through the states of **Odisha** and **Andhra Pradesh**, between **Rushikulya** and **Godavari**, the Vamsadhara River is a significant east-flowing river.
- The river's source is in the Odisha districts of Rayagada and Thuamul Rampur, respectively, in the **Kalahandi** and **Kalyansingpur** regions.
- It travels for around 254 kilometers until ending up in the Bay of Bengal at **Kalingapatnam**, Andhra Pradesh.
- Left-bank** tributaries include Harangi, Hemavati, Shimsha, and Arkavati, while
- Right-bank** tributaries include Lakshmantirtha, Kabbani, Suvarnavati, Bhavani, Noyil, and Amaravati.

West Flowing River System

The rivers of Peninsular India that flow westwards are limited in number and smaller compared to their eastward counterparts.

- The primary westward-flowing rivers are the **Narmada** and the **Tapi**, exhibiting unique behavior as they traverse through faults rather than forming valleys.
- These **faults**, aligned parallel to the **Vindhya** and the **Satpuras**, result from the bending of the northern peninsula during the Himalayan formation process.
- Additional westward-flowing rivers include the **Sabarmati**, **Mahi**, and **Luni**. Numerous small streams

originating in the Western Ghats swiftly flow westward to join the Arabian Sea.

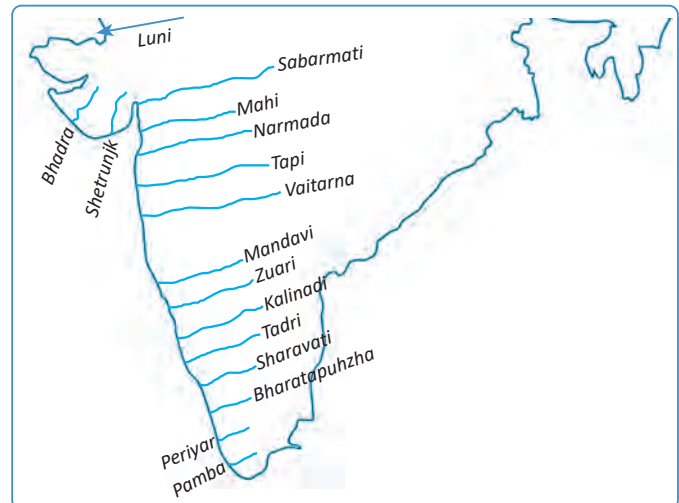


Fig. 4.35: West flowing rivers

- Notably, these westward-flowing rivers do not create deltas but **estuaries**, as the Narmada and Tapi, flowing through hard rocks, carry **minimal silt** due to their small tributaries, preventing the formation of distributaries or a delta before reaching the sea.
- Approximately **six hundred** small streams originate from the Western Ghats and flow westwards, eventually emptying into the Arabian Sea. The **western slopes** of the Western Ghats receive substantial rainfall from the **south-west monsoons**, enabling the generation of this multitude of streams.
- While only around **3%** of the total area experiences **swift downhill** flow on the steep slopes, some of these streams form waterfalls.
- The **Jog** or **Gersoppa** Falls, created by the **Sharavati** River and towering at 253 meters, stands out as the most renowned waterfall in India.

Narmada River

The Narmada River is the largest west-flowing river in peninsular India, courses westward through a **rift valley** nestled between the **Vindhya** Range to the north and the **Satpura** Range to the south.

- Originating from the **Maikala** Range near **Amarkantak** in Madhya Pradesh, at an elevation of approximately 1057 meters, it traverses a basin spanning the states of Madhya Pradesh, Gujarat, Maharashtra, and Chhattisgarh, covering an expansive area of about 1 Lakh sq. km.
- Bounded by the **Vindhyas** to the north, **Maikala** Range to the east, **Satpuras** to the south, and the **Arabian** Sea to the west, the Narmada River's total length extends to 1,312 km from its source in Amarkantak to its estuary in the Gulf of **Khambhat**.
- The upper regions of the basin are hilly, while the lower and middle reaches are wide and fertile, ideal for agriculture. Notably, **Jabalpur** stands as the **sole significant urban center** within the basin.

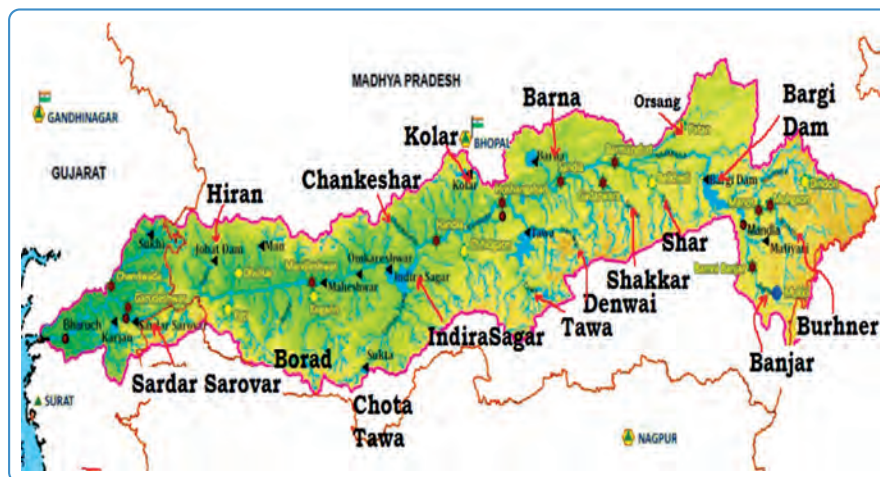


Fig. 4.36: Narmada River System

- ❑ As the river slopes down near Jabalpur, it forms the **DhuanDhar Falls** (meaning “Cloud of Mist”), a 15-meter cascade into a **gorge**, known as the **Marble Rocks** due to its composition of marble. Two additional waterfalls of 12 meters each are present at **Mandhar** and **Dardi**.
- ❑ Moving downstream, near **Maheshwar**, the river descends from another small fall of 8 metres, called the **Sahastradhara Falls**. **Kapildhara falls** is also notable here.
- ❑ The Narmada River’s estuary features several islands, with **Aliabet** being the largest among them. Notably, the river remains navigable up to 112 km from its mouth.
- ❑ **Tributaries of the Narmada:** Due to its course through a narrow valley hemmed in by steep hills, the Narmada River exhibits a **scarcity** of tributaries.
 - **Right Bank:** **Hiran**, the Orsang, the Barna, and the Kolar.
 - **Left bank:** **Shakkar**, **Sher**, **Tawa** (172km, longest, origin in Satpura), **Burhner**, **Kaveri**, and **Banjar**.
- ❑ The basin hosts several major **hydroelectric projects**, including **Indira Sagar**, **Sardar Sarovar**, **Omkareshwar**, **Bargi**, **Tawa Dam** (on Tawa river) and **Maheshwar**.

Kevadia, nestled amidst the verdant hills of **Gujarat**, **India**, cradles the iconic **Statue of Unity**, a tribute to **Sardar Vallabhbhai Patel** towering over the mighty **Narmada River**.

River island haven: Kevadia sits on **Sadhu Bet**, a picturesque island amidst the Narmada’s serpentine embrace, 3.2 km downstream from the colossal **Sardar Sarovar Dam**.

Dramatic backdrop: Framed by the majestic **Vindhyachal and Satpura ranges**, Kevadia’s landscape blends verdant valleys with rugged outcrops, offering breathtaking panoramas.

Narmada’s embrace: The lifeblood of the region, the **Narmada River** carves a sinuous path through Kevadia, its glistening waters reflecting the sky and the colossal statue.

Tribute in steel and bronze: The **Statue of Unity**, a 182-meter marvel of engineering, dominates Kevadia’s skyline, symbolizing India’s unity and Patel’s vision.

Emerging tourism hub: Kevadia, once a sleepy village, has blossomed into a thriving tourist destination, offering cultural experiences, adventure activities, and serene natural beauty.

Tapti River

The **Tapti**, also known as the **Tapi**, stands as the second-largest west-flowing river in Peninsular India and is often referred to as the ‘**twin**’ or ‘**handmaid**’ of the **Narmada**.

- ❑ Originating near the **Multai reserve forest** in Madhya Pradesh at an elevation of 752 meters, it **meanders** for approximately 724 km before finally emptying into the Arabian Sea through the Gulf of Cambay (Gulf of Khambhat).
- ❑ The Tapti River, along with its tributaries, courses through the plains of **Vidarbha**, **Khandesh**, and **Gujarat**, covering extensive areas in the states of Maharashtra, as well as smaller portions in Madhya Pradesh and Gujarat.
- ❑ The basin encompasses states such as Madhya Pradesh, Maharashtra, and Gujarat, spanning an area of approximately 65,000 sq. km.
- ❑ Situated in the Deccan plateau, the basin is encircled by natural features, including the **Satpura** range to the north, **Mahadev** hills to the east, **Ajanta** Range and the **Satmala** hills to the south, and the Arabian Sea to the west.
- ❑ The basin delineates two distinct physical regions:
 - The **well-forested hilly** areas encompassing Satpura, Satmalas, Mahadeo, Ajanta, and Gawilgarh hills, and the broad, fertile **plains** predominantly in the **Khandesh** region
- ❑ **Tributaries of Tapti:**
 - **Right Bank:** The Suki, the Gomai, the Arunavati and the Aner.

- **Left Bank:** The Vaghur, the Amravati, the Buray, the Panjhra, the Bori, the Girna, the Purna, the Mona and the Sipna.

- ❑ The Narmada and Tapi are in close proximity to one another at the **Khandwa-Burhanpur Gap**.
- ❑ **Projects on Tapi:** Hathnur Dam in the Upper Tapi Project (Maharashtra); Kakrapar weir and Ukai Dam in the Ukai Project (Gujarat) and Girna Dam and Dahigam Weir in the Girna Project (Maharashtra)

Sabarmati River

The Sabarmati River is formed by the confluence of the **Sabar** and **Hathmati** streams. Its basin spans the states of Rajasthan and Gujarat, covering an area of 21,895 sq km.

- ❑ **Boundaries:** It is bounded by the **Aravalli** hills to the north and northeast, the **Rann of Kutch** to the west, and the Gulf of **Khambhat** to the south, the basin takes on a roughly triangular shape, with the Sabarmati River as the base and the source of the Vatrak River as the apex.
- ❑ **Originating** from the **Aravalli hills** near the village of Tepur in the Udaipur district of Rajasthan at an elevation of 762 meters, the Sabarmati River flows for a total length of 371 km before reaching its outfall into the Arabian Sea.
- ❑ Approximately **74.68%** of the basin is dedicated to **agriculture**, making it a significant contributor to the region.
- ❑ **Rainfall** within the basin varies, ranging from minimal amounts in **Saurashtra** to over 1000 mm in the southern part.
- ❑ **Tributaries:** Noteworthy **left** bank tributaries include the Wakal, Hathmati, and Vatrak, while the **Sei** is a significant **right** bank tributary.
- ❑ Several **key projects** have been completed during the plan period, including the **Sabarmati Reservoir (Dharoi)**, **Hathmati Reservoir**, and **Meshwo Reservoir Project**.
- ❑ **Industry:** Significant urban hubs within the basin include **Gandhinagar** and **Ahmedabad**.
 - Ahmedabad, an industrial city located on the banks of the Sabarmati, is notable for its diverse industrial sectors such as textiles, leather and leather goods, plastic, rubber goods, paper, newsprint, automobile, machine tools, drugs, and pharmaceuticals, among others.
 - The industrial activities in Ahmedabad raise concerns about potential **water pollution** in the area.

Mahi River

The Mahi basin spans across the states of Madhya Pradesh, Rajasthan, and Gujarat, encompassing a total area of 34,842 sq km.

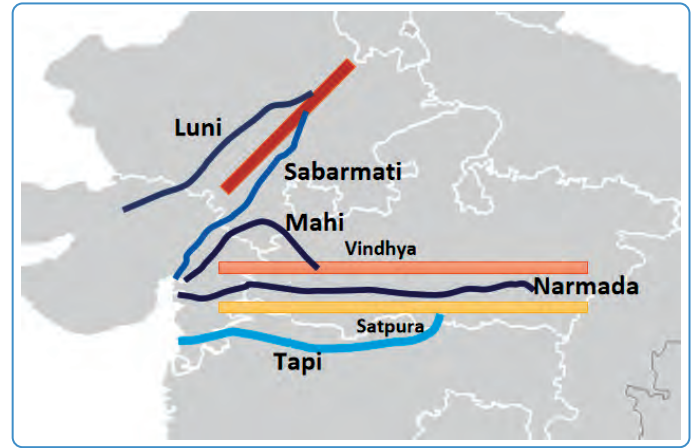


Fig. 4.37: West Flowing Peninsular rivers

- ❑ **Boundaries:** It is bordered by the **Aravalli** hills to the north and northwest, the **Malwa Plateau** to the east, the **Vindhyas** to the south, and the Gulf of **Khambhat** to the west.
- ❑ The Mahi River stands as a significant **interstate** west-flowing river in India.
- ❑ **Originating** from the northern slopes of the Vindhyas at an altitude of 500 meters in the **Dhar** district of Madhya Pradesh, the Mahi River extends for a total length of 583 km, ultimately draining into the Arabian Sea through the Gulf of **Khambhat**.
- ❑ **Agricultural** land dominates the major portion of the basin, constituting **63.63%** of the total area.
- ❑ **Hydro power** stations are strategically positioned at the **Mahi Bajaj Sagar Dam** and the **Kadana Dam**.
- ❑ **Vadodara** emerges as the sole important urban center within the basin, and while industrial presence is limited, some industries, including cotton textile, paper, newsprint, drugs, and pharmaceuticals, are found in **Tatlam**.

NOTE: The Mahi River uniquely intersects the **Tropic of Cancer twice** within India. Its initial crossing occurs in Madhya Pradesh, from where it flows southward into Rajasthan before ultimately entering the state of Gujarat.

Luni River

The Luni, also known as the **Salt River (Lonari or Lavanavari** in Sanskrit), earns its name from the **brackish** quality of its water below **Balotra**.

- ❑ This river basin holds singular significance in **Western Rajasthan**, a region predominantly characterized by aridity.
- ❑ **Originating** from the western slopes of the **Aravalli** ranges at an elevation of 772 meters near **Ajmer**, the Luni flows in a south-west direction, covering a distance of 511 km in Rajasthan before ultimately dissipating into the **Rann of Kachchh**, where it **gets lost in the marshes**.

- ❑ The majority of its tributaries stem from the steep north-west slopes of the Aravalli hills, joining the Luni on its left side. The **entire catchment** area of the Luni is situated within the state of **Rajasthan**.
- ❑ One **distinctive** feature of this river is its tendency to **widen rather than deepen** its bed.
- ❑ This is attributed to the **easily erodible** soils comprising the banks, contrasted with the sandy composition of the riverbed.
- ❑ **Floods** in the Luni develop and recede so rapidly that there is insufficient time for the water to scour the riverbed.

INLAND DRAINAGE SYSTEM

An inland drainage system refers to a network of rivers, streams, lakes, and other water bodies that do not flow into the ocean or any external body of water. Instead, the drainage is internal, and water accumulates in closed basins or endorheic regions. Inland drainage systems are characteristic of certain geographical areas where the terrain and topography prevent the natural outflow of water to the seas or oceans.

Key features of inland drainage systems include:

- ❑ **Closed Basins:** Inland drainage systems are often associated with closed basins or endorheic basins, where water does not have a natural outlet to the ocean. Instead, water accumulates within the basin and may form lakes or **salt pans**.
 - ❑ **Lack of River Outflow:** Unlike rivers in exorheic or open drainage systems, rivers in inland drainage systems terminate within the closed basin, and their waters do not reach the ocean.
 - ❑ **Evaporation and Salt Accumulation:** In arid or semi-arid regions with inland drainage, high evaporation rates can lead to the concentration of salts in the remaining water. This can result in the formation of saline or brackish lakes.
 - ❑ **Salt Pans:** Some inland drainage areas may have salt pans or playas where water evaporates, leaving behind salt deposits. These areas are common in **deserts** or regions with high evaporation rates.
 - ❑ **Examples:** The **Great Basin** in the United States, the **Qaidam Basin** in China, and certain regions in Australia, such as **Lake Eyre** Basin, are examples of areas with inland drainage systems. **Luni** and the **Ghaggar** are a great example of the Inland River System in India.
 - ❑ **Climate Influence:** Inland drainage systems are often associated with **arid or semi-arid** climates where **evaporation** rates are **high**, and **precipitation** is **insufficient** to sustain a continuous outflow of water.
 - ❑ **Endorheic Lakes:** Inland drainage systems may include endorheic lakes, which are lakes that have no outflow and retain water within the closed basin.
- Understanding the characteristics of inland drainage systems is crucial for managing water resources and addressing potential challenges, such as salinization and ecological impacts in these unique hydrological environments.

West Flowing Rivers of the Sahyadris (Western Ghats)

- ❑ **Pamba:** It flows in **Kerala** and falls in the **Vembanad** lake.
 - The Pamba River plays a crucial role in the Sabarimala pilgrimage, and the **Pamba Triveni Sangamam**, the confluence of the Pamba, Achankovil, and Manimala rivers, is a sacred site.
 - Pilgrims take a ritual dip in the Pamba River before ascending to the temple. It is venerated as the **Ganga of Kerala**.
- ❑ **Periyar:** The Periyar originates at the **Sivagiri Hills** in Western Ghats. It is an important river for Kerala and Tamil Nadu.
 - It flows through the Periyar National Park, which is home to diverse flora and fauna, including elephants, tigers, and various bird species.
 - The **Mullaperiyar dam** located on this river is crucial for providing water to arid regions of Tamil Nadu.
- ❑ **Bharathapuzha:** Originating near **Anamalai hills**, it drains into the Arabian sea. The river is also known as **Ponnani**.
 - The **Thrissur Pooram**, one of the largest temple festivals in Kerala, takes place on the banks of the Bharathapuzha River in Thrissur. It involves grand processions and cultural events.
- ❑ **Mandovi:** Originates in the **Bhimgad Wildlife Sanctuary in the Western Ghats of Karnataka**. Boat cruises and river tours on the Mandovi are popular activities in Goa.
 - The Mandovi River played a significant role during the **Portuguese colonial era** in Goa. It was an essential waterway for trade and transportation.
- ❑ **Zuari:** It is the **longest** river of Goa and is a Tidal river. **Vasco da Gama, the port city** of Goa lies on the mouth of this river.
 - Mandovi and Zuari are the backbone of the economy and they together form an estuarine system.
- ❑ **Sharavati:** Rises from the **Shimoga** district of Karnataka. The **Jog or the Gersoppa falls**, one of the highest plunge waterfalls in the country, is located on this river and is a famous tourist destination.
 - The river also traverses through the **Sharavathi wildlife sanctuary, which is home to Lion-tailed Macaque and Malabar Giant Squirrels**.
- ❑ **Bedti:** The river traces its source to **Hubli Dharwad** (Karnataka).
 - Recently, a river linking project has been envisaged to connect the river **Bedti to Varada**, a tributary of the river **Tungabhadra**. It has been criticized by both environmental experts and economists on

the grounds that the cost outweighs the projected benefits.

- ❑ **Kalinadi:** Begins from the **Belgaum** district and drains in **Karwar Bay**.
 - The river is known to flow through the **Dandeli wildlife sanctuary**, known for its diverse ecosystem and rare species of birds.
 - The river forms the natural boundary between the states of Goa and Karnataka.
- ❑ **Vaitarna:** Originates near the **Trimbak Hills** in **Nasik District** (Maharashtra).
 - The Vaitarna Dam, also known as the **Modak Sagar Dam**, is constructed on the Vaitarna River. It is a significant reservoir that provides water to the city of Mumbai. The reservoir helps in storing water during the monsoon for later use.
- ❑ **Dhadhar:** It starts from the **Ghantar** village in **PanchMahal** district (Gujarat) and drains into the **Gulf of Khambhat**.
- ❑ **Shetrunji:** The river originates near **Dalkahwa** in **Amreli** district (Gujarat). It flows through the Gir forest or the **Gir National Park** and the **Gir wildlife sanctuary**.
 - The Shetrunji River basin attracts bird watchers, as it is home to a variety of bird species. The diverse ecosystem supports both **resident and migratory** birds.
- ❑ **Bhadar:** The river begins in the **Aniali** village in **Rajkot** district (Gujarat).
 - It is one of the most important rivers of the Kathiawar (Saurashtra) peninsula of Gujarat. Bhadar dam is the second largest dam in Saurashtra after the Shetrunji dam.

Why Don't Peninsular Rivers that Flow into the Arabian Sea Form Deltas?

Peninsular rivers that flow over **hard rock** and lack silt buildup create estuaries rather than deltas when they enter the Arabian Sea. Because they travel only a **short distance** before emptying into the sea, west-flowing rivers do not develop deltas, which inhibit depositions.

- ❑ The tributaries of Narmada, Tapi, and Luni are small and do not contribute to silt.
- ❑ Western ghats, due to rising sea levels, are **submerged coasts**, preventing the formation of distributaries or deltas.
- ❑ **High gradients** of west-flowing rivers carry sediments along with them to sea instead of depositing them at the mouth of the river.
- ❑ West-flowing rivers do **not meander** due to fixed courses. Narmada and Tapi flow through faults, rift valleys, or troughs.

ESTUARY

An estuary is a partially enclosed coastal water body where the convergence of freshwater from rivers and streams with saltwater from the ocean occurs.

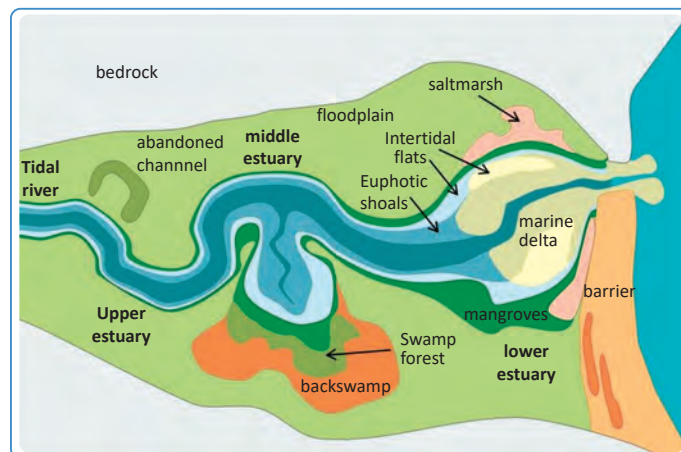


Fig. 4.38: Estuary

- ❑ The **primary productivity** within estuaries is notably **high**, with **fishing** emerging as a **dominant** occupation in these regions. Many estuaries also serve as **bird sanctuaries**.
- ❑ Estuaries, along with the surrounding lands, represent **transitional** zones from land to sea and from freshwater to saltwater.
- ❑ Despite being influenced by tides, estuaries are **shielded** from the full impact of ocean waves, winds, and storms by features like **barrier islands** or peninsulas, making them suitable for various purposes, including the establishment of **ports**.
- ❑ These environments are among the **most productive** on Earth, generating more organic matter annually than comparable forested, grassland, or agricultural areas.
- ❑ The sheltered tidal waters of estuaries support unique communities of plants and animals specially adapted to life at the ocean's edge.
- ❑ Estuaries hold significant commercial value, contributing to the economies through tourism, fisheries, and recreational activities. They serve as crucial public infrastructure, functioning as **harbors and ports** vital for shipping and transportation.
- ❑ Estuaries offer additional valuable services. Water draining from uplands carries sediments, nutrients, and pollutants to estuaries, where wetlands, such as swamps and salt marshes, act as natural filters, removing much of the sediments and pollutants.
- ❑ Estuarine plants, including salt marsh grasses and **mangroves**, play a role in erosion prevention and **shoreline stabilization**.

Why do Rivers Flow from West to East?

Indian rivers mostly flow eastward and drain into the Bay of Bengal, except Narmada & Tapi. The formation of the Western Ghats caused a **slight tilt** of the **Peninsular Plateau** towards the East which looks like the **upliftment** of landmass on the **Western side**.

- ❑ Himalayan rivers follow the **natural cycle** of water flow through the Plateaus depression, falling from high altitude to low altitude.
- ❑ **Gravity** is the most determining force for water flow towards the south.
- ❑ The **west-to-east gradient** of the Deccan plateau also influences river flow.

River Regime

River regime denotes the typical fluctuations in a river's flow, encompassing variations in discharge, velocity, and sediment transport throughout the year. River regimes encompass seasonal variations, flashiness, baseflow and stormflow dynamics, snowmelt contributions, human impacts, and sediment transport variations.

Factors affecting/encompassing river regimes:

River regimes encompass various factors:

- ❑ **Seasonal Variations:** Temperate rivers peak in spring due to snowmelt, while tropical rivers maintain consistent flow.
- ❑ **Flashiness:** Rivers are classified as flashy (rapid changes) or stable (gradual changes) in response to precipitation.
- ❑ **Baseflow vs. Stormflow:** Groundwater sustains baseflow, while stormflow increases during precipitation events.
- ❑ **Snowmelt Impact:** Spring snowmelt in snowy regions significantly affects river flow and sediment transport.
- ❑ **Human Impacts:** Dams, irrigation, and water extraction alter natural river regimes, affecting flow and sediment transport.
- ❑ **Sediment Transport:** Flow variations influence sediment transport, shaping channel morphology.

Understanding these factors helps differentiate peninsular and Himalayan river regimes, crucial for water management.

NOTE: The **discharge** is the volume of water flowing in a river measured over time. It is measured either in Cusecs (cubic feet per second) or cumecs (cubic meters per second)

Himalayan River Regimes

- ❑ **Origin:** Begin at high elevations, fed by glacier melt and rainfall.
- ❑ **Catchment Area:** Wide catchment ensures consistent and abundant flow.
- ❑ **Seasonal Variation:** Ganga, for example, peaks in August-September after a minimum flow from January to June.

- ❑ **Monsoon Influence:** Ganga shows a monsoon regime, with significant flow before and during the monsoon season.
- ❑ **Discharge Levels:** Ganga's highest discharge at Farakka Barrage is 55,000 cusecs, lowest is 1,300 cusecs.
- ❑ **Groundwater Source:** Flows through permeable alluvial soils, contributing to groundwater.

Peninsular River Regimes

- ❑ **Origin:** Begin at low elevations in the Western Ghats.
- ❑ **Dependency:** Non-perennial due to reliance on monsoonal rainfall.
- ❑ **Catchment Area:** Narrow catchment leads to reduced flow.
- ❑ **Seasonal Variation:** Narmada, for instance, peaks in August after low discharge from January to July.
- ❑ **Discharge Levels:** Narmada's maximum flow at Garudeshwar is about 2,300 cusecs, minimum is 15 cusecs.
- ❑ **Rock Composition:** Flows through impermeable rocks, limiting groundwater percolation and recharge.

Major Indian Cities and their River Banks

Rivers forge essential associations with cities, shaping their identity and sustenance. They serve as lifelines, providing water for drinking, agriculture, and industry. Rivers facilitate trade, offering transportation routes crucial for economic growth. Cities often flourish along riverbanks, benefiting from fertile soil and strategic locations. The cultural and historical significance of rivers is woven into the fabric of urban life, witnessed in rituals, festivals, and folklore.

- ❑ **Delhi - Yamuna River:** Provides water for drinking, irrigation, and industry. Offers recreational spaces and enhances city aesthetics.
- ❑ **Mumbai - Mithi River and Arabian Sea:** Arabian Sea supports maritime trade. Mithi River regulates city climate; facing pollution challenges.
- ❑ **Kolkata - Hooghly River:** Vital waterway for transportation and trade. Ports contribute significantly to city economy.
- ❑ **Chennai - Cooum and Adyar Rivers:** Historically provided water for various uses. Estuarine region holds economic importance.
- ❑ **Ahmedabad - Sabarmati River:** Contributes to city aesthetics; site of Sabarmati Riverfront Development.
- ❑ **Kanpur - Ganga River:** Historically facilitated trade; facing pollution issues.
- ❑ **Varanasi - Ganga River:** Sacred river central to religious activities; focal points for rituals and ghats.
- ❑ **Patna - Ganga River:** Transportation route; city's economy linked to river activities.
- ❑ **Pune - Mutha River:** Contributes to water supply and irrigation; offers recreational spaces.

- ❑ **Hyderabad - Musi River:** Despite pollution challenges, used for recreation and city development.
- ❑ **Lucknow - Gomti River:** Historically provided water; riverbanks offer green spaces.
- ❑ **Jaipur - Yamuna River:** Historically benefited from river for trade and water supply.
- ❑ **Bengaluru - Vrishabhavathi River:** Historical water supply; facing pollution challenges.
- ❑ **Bhopal - Upper Lake (Bada Talab):** Acts as water source; contributes to city aesthetics.
- ❑ **Indore - Saraswati River:** Historical role in water supply; altered due to urbanization.
- ❑ **Chandigarh - Sukhna Lake:** Reservoir adds to city beauty and offers recreation.
- ❑ **Agra - Yamuna River:** Facilitated trade; significant for city's cultural heritage.
- ❑ **Surat - Tapi River:** Historical trade role; riverbanks contribute to greenery.
- ❑ **Visakhapatnam - Meghadri Gedda Reservoir:** Contributes to water supply and recreation.
- ❑ **Guwahati - Brahmaputra River:** Vital for transportation; enhances city's scenic beauty.

Having examined the Indian context, let us now turn to the cities of the world and their river banks.

Famous Cities of the World and their River Banks

- ❑ **Paris, France (Seine River):** Iconic landmarks like the Eiffel Tower along the Seine River offer scenic river cruises.
- ❑ **London, United Kingdom (Thames River):** Historical architecture, including the Tower of London, lines the Thames River.
- ❑ **Rome, Italy (Tiber River):** Connects ancient sites like the Colosseum and Vatican City, shaping the city's history.
- ❑ **New York City, USA (Hudson River):** Integral to trade and transportation, providing a stunning skyline backdrop.
- ❑ **Cairo, Egypt (Nile River):** Central to Egypt's civilization, history, and culture, influencing the capital city's development.
- ❑ **Shanghai, China (Huangpu River):** Divides modern skyscrapers and historic architecture, showcasing Shanghai's diversity.
- ❑ **Buenos Aires, Argentina (Río de la Plata):** Known for vibrant culture and tango music along the Rio de la Plata.
- ❑ **Cape Town, South Africa (Atlantic Ocean):** Coastal city influenced by the Atlantic Ocean, offering stunning views.
- ❑ **Baghdad, Iraq (Tigris River):** Part of the fertile crescent, contributing to the city's rich history.
- ❑ **Sydney, Australia (Sydney Harbour):** Iconic harbor hub for tourism and maritime activities, featuring landmarks like the Sydney Opera House.

- ❑ **Moscow, Russia (Moskva River):** Enhances scenic beauty, winding through the city and playing a role in its history.
- ❑ **Bangkok, Thailand (Chao Phraya River):** Connects historical temples with modern developments and transportation.
- ❑ **Seoul, South Korea (Han River):** Scenic backdrop to skyscrapers, parks, and cultural attractions.
- ❑ **Berlin, Germany (Spree River):** Meanders through a mix of historic and contemporary architecture, adding to Berlin's charm.
- ❑ **Lisbon, Portugal (Tagus River):** Estuary offering panoramic views of the city, with maritime significance.
- ❑ **Dublin, Ireland (River Liffey):** Contributes to the city's charm, providing a focal point for locals and tourists.
- ❑ **Vancouver, Canada (Fraser River):** Waterfront featuring urban development, parks, and scenic mountain views.

However, urbanization poses challenges, contributing to pollution and encroachment. Sustainable river management is paramount, ensuring cities continue to thrive in harmony with these vital waterways, fostering ecological balance and supporting diverse aspects of human existence.

LAKES OF INDIA

Lakes play a crucial role in drainage systems as they are often integral components of a watershed. Lakes act as natural reservoirs, storing water during periods of excess rainfall or snowmelt. They help regulate the flow of water, preventing floods during heavy precipitation events and releasing water during drier periods to maintain a consistent flow in downstream rivers.

Lakes provide diverse habitats for various aquatic species. They contribute to the overall biodiversity of a region and support a wide range of flora and fauna. Healthy ecosystems in and around lakes contribute to the resilience of the drainage system.

Significant Lakes of India

- ❑ **Vembanad Lake:** Vembanad Lake, the **longest** lake in India and the **largest** in Kerala along the Malabar Coast, is renowned for its intricate backwater ecosystem, weaving through interconnected rivers, lakes, and estuaries.
 - Vembanad Lake, central to Kerala Backwaters and **Kuttanad's "Rice Bowl,"** aids **below-sea-level farming** with crucial irrigation.
 - The renowned **Nehru Trophy Boat Race**, held on **Punnamada Lake** within Vembanad in Alappuzha (Alleppey), features the iconic **Chundan Vallam (snake boats)**, highlighting boatmen's skill and Kerala's rich heritage.
- ❑ **Chilika Lake:** Chilika Lake, located predominantly in Odisha on India's eastern coast, holds the distinction of being the country's **largest coastal lagoon** and **Asia's largest brackish water lagoon**.

- This habitat is crucial for the endangered **Irrawaddy dolphins**.
 - **Nalabana Island (Forest of Reeds)**, serves as a vital refuge for winter migratory birds, including flamingos, sandpipers, and herons.
 - The **2022 water bird status survey** at documented nearly **1.07 million water birds**
 - It is the **first Indian wetland under the Ramsar Convention**, declared in 1981.
- ❑ **Sambhar Lake:** Sambhar Lake, the **largest inland saltwater** lake in India, holds significance as a major hub for salt production, with its expansive **salt pans** tracing their historical roots back to the **Mauryan period**.
- Its salt reserves were historically utilized by the **Mughal dynasty**.
 - Later, it was jointly owned by the princely states of **Jaipur** and **Jodhpur**.
 - The **salt industry** has played a crucial role in contributing to the region's economic prosperity.
 - **Migratory birds** make the region their temporary home, leading to its designation as an **Important Bird Area (IBA)**.
 - In 2019, around 22,000 migratory birds died due to avian botulism. In response, the Rajasthan government proactively constructed temporary shelters for migratory birds in 2020.
- ❑ **Kolleru Lake:** Situated in Andhra Pradesh, nestled between the **Krishna and Godavari deltas**, this expansive water body spans an area of 308 km², is one of the largest freshwater lakes in India.
- Acts as a **natural flood-balancing reservoir** for the Krishna and Godavari rivers.
 - It is fed by the **Budameru and Tammileru streams** during their respective seasons.
 - The lake connects to the Krishna and Godavari systems through a network of over **68 inflowing drains and channels**.
 - Recognised as **Kolleru wildlife sanctuary** in November 1999 under India's **WildLife (Protection) Act, 1972** as well as a wetland of importance under the **Ramsar Convention** in November 2002.
- ❑ **Pulicat Lake:** Pulicat Lagoon, ranking as the **second-largest brackish water lagoon** in India, comes after Chilika Lake in this distinction.
- The vast expanse of the lagoon is predominantly distributed, with approximately **96%** residing in **Andhra Pradesh** (Nellore district) and the remaining **3%** in **Tamil Nadu**.
 - The two contributing rivers are the **Arani river** and the **Kalangi River**.
 - On the **western fringe** of the lagoon, the **Buckingham Canal** serves as a navigation channel.
- The **barrier island of Sriharikota** acts as a natural boundary, separating the lagoon from the Bay of Bengal and hosting the **Satish Dhawan Space Centre**.
 - The geographical composition includes **Venadu Reserve Forest** and **Pernadu Reserve Forest** (both situated in Andhra Pradesh).
- ❑ **Wular Lake:** Wular Lake, a **Ramsar site** located in the Kashmir Valley and spanning parts of **Bandipora** and **Baramulla** districts in Jammu and Kashmir, stands as the largest freshwater lake in Asia. It serves as a vital water body within the **Jhelum River** basin, being fed by the Jhelum River.
- Legend attributes the creation of the **artificial island of Zaina Lank** at the lake's center to the directive of **Kashmiri Sultan Zain-ul-Abidin** in 1444, adding a historical and cultural dimension to the lake's narrative.
 - The **Tulbul Project**, described as a "**navigation lock cum control structure**" situated at the lake's mouth, has been a source of contention between India and Pakistan since 1987.
 - ❑ The disagreement stems from Pakistan's claim that the project violates the terms of the **1960 Indus Waters Treaty (IWT)**. This unresolved issue has persisted, contributing to diplomatic discussions between the two nations.
- ❑ **Dal Lake:** Situated in Srinagar, the summer capital of Jammu and Kashmir, the lake is enveloped by the **Pir Panjal** mountains, creating a picturesque setting.
- Dal Lake is known as the "**Jewel in the crown of Kashmir**" or "Srinagar's Jewel,"
 - Ranking as the second-largest lake in Jammu & Kashmir, its shoreline is adorned with Mughal-era gardens, parks, houseboats, and hotels.
 - The lake freezes during winter, with temperatures dropping to **-11 °C (12 °F)**. Dal Lake is part of a 21.1-square-kilometer natural wetland featuring **floating gardens** called "**RAAD**" that bloom with lotus flowers in July and August.
 - It is renowned for a floating market where vendors on **Shikaras** offer a unique shopping experience to tourists.
 - In May 2023, an ecological incident resulted in the loss of thousands of fish, this was attributed to a sudden temperature fluctuation.
- ❑ **Loktak Lake:** Loktak Lake stands as the **largest freshwater lake in Northeast India**, captivating visitors with its pristine beauty. Renowned for its unique floating circular swamps locally known as "**phumdis**," the lake attracts tourists from distant places.
- These phumdis, resembling islands, consist of a composite of soil, organic matter, and vegetation, creating a distinctive landscape on the lake.

- Notably, Loktak Lake is home to the **world's only floating national park, the Keibul Lamjao National Park**, serving as the last refuge for the endangered **brow-antlered deer**, also known as **sangai**, which holds the status of Manipur's state animal.
- It was also included in both the Ramsar sites and the **Montreux Record**.

❑ **Pangong Tso Lake:** Nestled in the enchanting landscape of Leh Ladakh, this renowned lake draws its name, "Pangong Tso," from the Tibetan language, translating to "**high grassland lake**." Commonly referred to as Pangong Lake, it is a lengthy and slender **endorheic (landlocked)** lake nestled at an altitude exceeding 14,000 feet (4,350 meters) amid the Ladakh Himalayas.

- It is the **world's highest saltwater lake**, in a distinctive **boomerang shape**.
- Notably, **one-third** of this extraordinary lake is under the control of **India**, while the remaining **two-thirds** is under **Chinese** control.
- It is renowned for its mesmerizing **color transformations**, exhibiting hues of **blue, green, and red** that captivate observers at different times.
- It was one of the **friction points** of the 2020 violent **Galwan clash** between our Indian soldiers and the Chinese soldiers.

❑ **Bhojtal Lake:** Also recognized as **Upper Lake**, stands as a prominent water reservoir situated in **Bhopal**. Holding the title of the **largest artificial lake** in Asia, this water body owes its name to **Raja Bhoj, a Paramara king** credited with both the lake's construction and the renowned Bhojeshwar Temple on its eastern shore.

- Originating in the 11th century, the lake came into existence through the construction of an **earthen dam**, strategically designed to retain the waters of the **Kolans River**.
- Beyond its historical significance, Bhojtal plays a vital role as a primary water source for the residents of Bhopal and the neighboring areas. It is also a recognised **Ramsar site**.

❑ **Deepor Beel:** It is a **freshwater** lake and serves as a crucial biodiversity hotspot. It has been declared as an **Important Bird Area (IBA)** and also included in the list of **Ramsar** sites.

- Situated in the southwestern region of **Guwahati** city, Assam, Deepor Beel is the former watercourse of the **Brahmaputra** River.
- Recently, the Ministry of Environment, Forest and Climate Change (MoEFCC) has notified the **Eco-sensitive zone** of the Deepor Beel Wildlife Sanctuary (Assam).

❑ **Maha Rana Pratap Sagar lake:** The lake is fed by **Beas River** and its numerous perennial tributaries such as Gaj, Neogal, Binwa, Uhl, Bangana, and Baner.

- It has been declared "a wetland of national importance," and also as a **Ramsar** site.
- It is also known as **Pong Dam** lake.
- More than one lakh migratory water birds arrived at the Pong Dam Lake Wildlife Sanctuary in Himachal Pradesh in winter 2020-21.

❑ **Ashtamudi Kayal:** It means 8 braids (branches) in Malayalam, due to its **octopus** shape.

- It has a unique wetland ecosystem and is a **Ramsar** site.
- It is a gateway to the Kerala Backwaters and is famous for houseboat tourism.
- It is the **second largest estuarine lake** of Kerala after Vembanad
- Ashtamudi Lake is a major contributor to Kerala's extensive network of interconnected canals, rivers, and lakes.

❑ **Kanwar lake:** Also known as **Kanwar Jheel**, it is a significant freshwater lake situated in the Begusarai district of Bihar.

- It is recognized as the **largest oxbow lake (crescent-shaped body of water)** in Asia. This was formed due to the meandering of **Gandak river**,
- Despite being designated as a Ramsar Wetland of International Importance, it lies neglected and is on the brink of drying up.
- **Five critically endangered species** inhabit the site, including three vultures – the red-headed vulture (*Sarcogyps calvus*), white-rumped vulture (*Gyps bengalensis*) and Indian vulture (*Gyps indicus*) – and two waterbirds, the sociable lapwing (*Vanellus gregarius*) and Baer's pochard (*Aythya baeri*).

State	Lakes	Type of Lake	Comments
Andhra Pradesh	Kolleru, Nagarjuna Sagar	Fresh water	Kolleru lake is located between the Godavari and Krishna river deltas.
	Pulicat	Brackish water	It is the second largest brackish water lagoon lake in India.
Assam	Haflong lake, Deepor beel, Son beel	Fresh water	Deepor Beel is a riverine wetland of Brahmaputra
	Chandubi	Oxbow lake	It is a natural lagoon lake.
Bihar	Kanwar	Oxbow lake	Asia's largest freshwater oxbow lake.

State	Lakes	Type of Lake	Comments
Gujarat	Hamirsar, Kankaria, Narayan Sarovar	Artificial Lake	Narayan Sarovar is a pilgrimage site on the Kori Creek.
	Thol, Vastrapur	Freshwater lake	Thol was constructed for irrigation purpose
Haryana	Badkhal, Blue Bird, Brahma Sarovar, Karna, Tilyar, Damdama	Freshwater lake	Karna lake is connected to the epic Mahabharata.
Himachal Pradesh	Chandra Taal, Suraj Taal, Maharana Pratap Sagar(Pong Dam), Prashar Lake	Freshwater lake	Pong dam is the highest earthfill dam.
Jammu and Kashmir	Wular, Dal lake	Freshwater lake	Wular is the second largest freshwater lake of Asia.
Karnataka	Agara, Ulsoor, Kukkarahalli, Honnamana, Pampa Sarovar	Freshwater lake	Pampa Sarovar is to the south of Tungabhadra and is sacred to the Hindus.
Kerala	Ashtamudi, Kuttanad, Vembanad	Brackish water	Ashtamudi is an estuarine lake.
	Sasthamkotta	Freshwater lake	It is called Queen of lakes.
Madhya Pradesh	Bhojtal	Freshwater lake	It was constructed by King Bhoj of Paramara dynasty.
Maharashtra	Salim Ali, Shivsagar	Freshwater lake	Shivsagar lake was formed after construction of Koyna Dam.
	Lonar	Saline and Alkaline	It is a national geo-heritage monument.
Manipur	Loktak	Lenticular	Large number of floating islands.
Meghalaya	Umiam	Freshwater lake	Created for hydropower.
Mizoram	Tam Dil	Freshwater lake	Also called Lake of Mustard.
Odisha	Chilika	Brackish water	India's largest brackish water and lagoon lake.
Punjab	Harike, Kanjili	Freshwater lake	Both are Ramsar sites
Rajasthan	Sambhar Lake	Saltwater lake	India's largest inland saline lake.
	Rajsamand	Freshwater lake	The lake is fed by the Gomti river.
Sikkim	Tsomgo, Khecheoplari	Freshwater lake	Khecheoplari is sacred for Hindus and Buddhists
Tamil Nadu	Ooty, Chembarambakkam	Freshwater lake	Chembarambakkam is crucial for Chennai's water supply and is an artificial lake.
	Kaliveli	Freshwater and brackish water	Coastal Lagoon lake
Telangana	Hussain Sagar, Bhadrakali	Freshwater lake	Hussain Sagar is an artificially constructed lake
Uttar Pradesh	Gobind Ballabh Pant Sagar, Bela Sagar	Freshwater lake	GB Pant Sagar is an artificially constructed lake on Rihand river.
Uttarakhand	Bhimtal	Freshwater lake	Bhimtal is a lake formed by constructing a masonry dam.



IGNITE YOUR MIND

Discuss the success stories of community-led conservation efforts around Loktak Lake. How can similar community-based approaches be applied to other Indian lakes for sustainable management and biodiversity conservation?

Usability of Rivers

Rivers have numerous practical applications and contribute significantly to the environment, society, and economy. Here are some key aspects of the usability of rivers, with examples from the Indian context:

❑ Water Supply:

- **Domestic Use:** In India, the **Ganges** River is a vital source of freshwater for millions, providing water for drinking, cooking, and sanitation.

- **Industrial Use:** The **Yamuna** River, flowing through **Delhi**, plays a crucial role in supplying water to industries in the region.

❑ **Agriculture:**

- **Irrigation:** The **Godavari** River is essential for irrigating extensive agricultural lands in states like Andhra Pradesh and Telangana.
- **Fertile Soil Deposition:** The Brahmaputra River deposits **nutrient-rich** sediments in Assam, enhancing soil fertility for **tea and rice** cultivation.

❑ **Navigation and Transportation:**

- **Trade and Commerce:** The **Hooghly** River, a distributary of the Ganges, facilitates trade and transportation in the **Kolkata port**, a major commercial hub in India.

❑ **Hydropower Generation:** The Bhakra Nangal Dam on the Sutlej River in northern India is a prime example of hydropower generation, supplying electricity to multiple states.

❑ **Recreation and Tourism:** The **backwaters of Kerala**, formed by rivers like the **Periyar** and **Pamba**, attract tourists for **houseboat** cruises and cultural experiences.

❑ **Wildlife Habitat:** The Ganges River is home to the endangered **Gangetic dolphin**, drawing eco-tourists interested in wildlife conservation.

❑ **Flood Control:** The **Brahmaputra** River in Assam plays a critical role in natural flood control by distributing excess water across the floodplains, thus acting as natural buffer.

❑ **Biodiversity and Ecology:**

- **Habitat:** The Western Ghats rivers, like the Cauvery and Krishna, harbor diverse ecosystems and are vital for the rich biodiversity in the region.
- **Migration:** The **Hilsa fish** in the Ganges-Brahmaputra delta undertake long migrations for spawning, showcasing the importance of river systems.

❑ **Cultural and Historical Significance:**

- **Cultural Heritage:** The **Saraswati** River, mentioned in ancient texts, holds cultural importance despite being mostly perceived to be mythical, reflecting India's cultural heritage.
- **Religious Practices:** Rivers like Ganga, Yamuna, Godavari and Krishna are considered sacred in Hinduism, and millions participate in religious rituals along its banks.

❑ **Waste Disposal:** The **Teesta** River in Sikkim acts as a natural filter, purifying water as it flows downstream and supporting local ecosystems.

❑ **Climate Regulation:** The **Narmada** River in central India contributes to the regulation of temperatures in the surrounding areas, providing a **cooling effect**.

The usability of rivers in India is multifaceted, influencing various aspects of life, culture, and the environment.

Sustainable management and conservation efforts are crucial to preserve and enhance the benefits derived from these vital water bodies.

Contradiction in India: Floods and Droughts

India's climatic contradiction is marked by its **dual struggle** with both floods and droughts, showcasing the country's vulnerability to **extreme** weather events. These opposing challenges present a complex and delicate balancing act for India's agriculture, infrastructure, and water management systems.

Floods, primarily driven by monsoon rains and river overflows, can wreak havoc. The annual monsoon season brings relief but also the risk of devastating floods. For instance, the **2013 North India** floods caused widespread destruction in **Uttarakhand**, claiming thousands of lives and causing substantial economic losses.

On the flip side, **droughts** are a recurring menace in many regions, particularly during weak monsoons or extended dry spells. In **2019, Maharashtra** faced one of its worst droughts, with depleted water reservoirs, parched fields, and severe water scarcity for millions of people.

Agriculture: Both drought and floods have adverse impacts, but it is more pronounced in agriculture.

❑ **Drought:** In times of drought, the scarcity of water leads to a reduction in crop productivity. Insufficient water availability hampers the normal growth and development of crops, resulting in yield losses and diminished quality. Farmers face challenges in sustaining their livelihoods, and communities may grapple with food shortages due to decreased agricultural output.

❑ **Floods:** Excessive water absorption by crops during floods damages their structural integrity and compromises their ability to thrive. Submergence, soil erosion, and waterlogging contribute to crop losses, impacting the overall agricultural yield.

Thus, both droughts and floods inflict severe consequences, exerting profound impacts on food security of the nation.

The contradiction lies in the fact that India must simultaneously prepare for both these extremes. Adequate infrastructure for flood control, such as **embankments and reservoirs**, is necessary, but these structures should also allow water storage for times of drought. Implementing efficient water management practices, promoting rainwater harvesting, and modernizing irrigation systems are crucial.

RIVER INTERLINKING PROJECT - NATIONAL WATER GRID PROGRAM

Interlinking rivers involves transferring water through channels, dams, reservoirs, or upliftment to address water deficit, drought conditions, and rainfall scarcity.

- ❑ This problem is primarily experienced in peninsular regions due to their seasonal nature and lack of perennial water sources like northern rivers.
- ❑ A **few examples** of river interlinking projects are
 - The **Ken-Betwa** River link project,
 - The **Godavari** River interlinked with the **Krishna** River near Vijayawada and
 - The **Ravi-Beas-Sutlej** interlinking project.
- ❑ The **National Interlinking River Project (NIRP)** was proposed during **British Rule** to connect rivers for navigation purposes. It is currently **managed** by the **National Water Development Agency** under the Ministry of **Jal Shakti**.
- ❑ The project aims to connect Himalayan rivers through reservoir construction over the Ganga and Brahmaputra rivers, along with dams, canals, and reservoirs over peninsular rivers.

Ken-Betwa River Linking Project

The Ken-Betwa Link Project (KBLP), the inaugural project between the states of MP and UP under the National Perspective Plan (NPP), received government approval in December 2021. The estimated cost, set at ₹ 44,605 crore (2020-21 price level), includes a central support of ₹ 39,317 crore. Implemented through the Ken-Betwa Link Project Authority (KBLPA), the initiative focuses on connecting the Ken and Betwa rivers in Madhya Pradesh, which later join the Yamuna near Banda and Hamirpur. KBLP will annually irrigate 10.62 lakh hectares (8.11 lakh hectares in MP and 2.51 lakh hectares in UP).

- ❑ The project aims to provide irrigation facilities, drinking water supply, hydropower generation, and solar power projects.

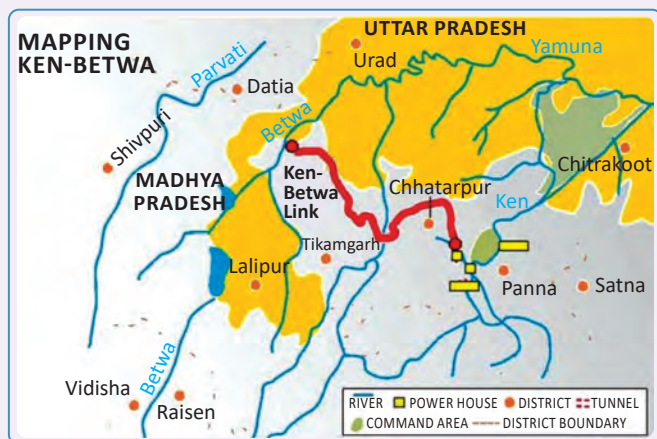


Fig. 4.39: Ken-Betwa riverlinking project

- ❑ The project will help address water scarcity in the water deficit region of both states due to delayed rainfall.
- ❑ The **Center** will bear **90%** cost by constructing a **Daudhan dam** and canal to transfer water. However,

the project faces concerns including the **submergence** of the **Panna Tiger Reserve** due to the **Daudhan dam**, impacting wildlife sanctuaries like **Nauradehi**, **Rani Durgawati**, and **Ranipur**, migration of locals, potential poverty, and environmental clearance obstacles.

Benefits of Interlinking of Rivers

The numerous advantages of interlinking of rivers include, but are not limited to:

- ❑ **Hydropower** projects and hydroelectricity will generate **green electricity**, **reduce coal** usage, address water crisis, provide transportation, reduce traffic congestion, and address air pollution.
- ❑ They will also improve irrigation in regions with water scarcity due to less rainfall during monsoon. It will provide another source of transportation and connectivity, reduce traffic congestion on roads and highways, and reduce air pollution from vehicles.
- ❑ This will also address **inland navigation** issues in South India.
- ❑ River-linking projects can **convert floods into irrigation sources** by transferring water between basins.
- ❑ **Floods** are caused by excessive rainfall, **surface runoff**, and over-storage of reservoirs and dams. **Rainwater harvesting** helps store and use water for drinking and irrigation, reducing water peak levels before rainfall and releasing it regulated to manage floods in southern India.

Concerns with the Interlinking of Rivers

In spite of the many advantages interlinking of rivers offers, it has many downsides like:

- ❑ **Inter-state disputes:** River **disputes** among states, such as Kaveri and Godavari, often **overlook human nature**.
- ❑ **High Costs:** Interlinking rivers is **costly** due to **environmental** impacts, high **economic** costs, and humanitarian issues. It can cause **displacement**, **migration**, and **rehabilitation** issues.
- ❑ **Transnational Rivers:** Conflicts with neighboring countries, such as Bangladesh, Nepal, and China, also exist. For **instance**,
 - Disputes over the **Teesta** River and its tributaries with **Bangladesh** and
 - Over the **Kali** River with **Nepal**.
- ❑ **Ecological Impact:** The project will adversely affect various ecological aspects such as delta formation, mangrove growth, and aquatic life.
- ❑ **River Course Alteration:** Rivers may alter their courses approximately every century, rendering the interlinking project potentially obsolete after a century.
- ❑ **Loss of Forest Cover:** Implementing river interlinking will necessitate clearing vast swathes of land, leading to deforestation.

- ❑ **Water Management Challenges:** Challenges like water logging, heightened salinity levels, and potential water pollution may arise from inland waterway navigation resulting from the project.
- ❑ **Impact on society:** constructing dams and reservoirs will cause the displacement of many local communities.

Way Forward

- ❑ **Prioritize efficient utilization** of existing water resources through integrated water resource management before investing in large-scale river interlinking projects.
- ❑ **Focus on groundwater management**, including identifying and protecting recharge mechanisms, enhancing recharge where possible, and regulating groundwater use at the aquifer level.
- ❑ **Promote the concept of virtual water**, where importing goods rather than producing them domestically conserves local water for other uses.
- ❑ **Consider the National Waterways Project (NWP)** as an alternative to the National River Linking Project (NRLP). NWP operates as a water grid, transferring water from flooded rivers to others, requiring less land, offering year-round navigation, and requiring no pumping. Additionally, it can irrigate almost double the land and generate 76% more power compared to river interlinking projects.

Thus, the interlinking of rivers necessitates careful planning and planning for long-term projects.

ISSUES RELATED TO DRAINAGE SYSTEM

River Pollution

It is estimated that approximately 70% of surface water is unfit for human consumption in India. CPCB identified Ganga, Yamuna, Godavari, Ghaggar and Gomti as the five most polluted rivers in India.

Causes of River Pollution

The major causes of river pollution are dealt in detail below:

- ❑ **Industrial Discharges:** The discharge of untreated industrial effluents into rivers, such as the **Ganga and Yamuna**, contains heavy metals, chemicals, and toxins, impacting water quality and aquatic life.
- ❑ **Agricultural Runoff:** Excessive use of fertilizers and pesticides in agriculture leads to runoff containing pollutants like nitrogen and phosphates, contributing to **eutrophication** in rivers like the **Godavari and Krishna**.
- ❑ **Domestic Sewage:** Inadequate sewage treatment facilities in urban areas result in the direct release of untreated domestic wastewater into rivers, affecting the water quality of rivers like the **Sabarmati and Mithi**.
- ❑ **Solid Waste Dumping:** Improper disposal of solid waste, including plastics and other non-biodegradable materials,

in rivers like the **Brahmaputra and Hooghly**, exacerbates pollution and poses a threat to aquatic ecosystems.

- ❑ **Mining Activities:** Unregulated **sand mining** in rivers like the **Yamuna and Chambal** leads to erosion, sedimentation, and habitat destruction, degrading the overall health of the rivers.
- ❑ **Urbanization and Encroachment:** Rapid urbanization along riverbanks, as seen in cities like **Mumbai** along the **Mithi** River, results in increased impervious surfaces, leading to higher runoff and pollution.
- ❑ **Disposal of Religious Offerings:** Immersion of religious idols and offerings in rivers like the **Ganga** during festivals introduces pollutants such as **heavy metals** and non-biodegradable materials.
 - **Dead bodies** are cremated on the river banks and **partially burnt bodies** are also flung into the river.
- ❑ **Deforestation and Soil Erosion:** Deforestation in catchment areas, particularly along the Western Ghats affecting rivers like the **Kaveri and Periyar**, contributes to soil erosion and increased sedimentation.
- ❑ **Hydropower Projects:** Construction and operation of hydropower projects, such as the **Tehri Dam** on the **Bhagirathi River**, can alter the natural flow regime, affecting water quality and aquatic habitats.
- ❑ **Lack of Regulatory Enforcement:** Weak enforcement of environmental regulations allows industries and individuals to continue polluting rivers without facing significant consequences, contributing to the deterioration of rivers across the country.

Understanding these specific causes in the Indian context is crucial for developing effective policies and implementing sustainable practices to address river pollution and ensure the long-term health of India's river ecosystems.

CPCB 2021 Study Findings

The Central Pollution Control Board (CPCB) identified 351 polluted river stretches in India, with untreated wastewater discharge being a major contributor.

- ❑ Approximately **60%** of these polluted stretches are concentrated in Maharashtra, Assam, Madhya Pradesh, Kerala, Gujarat, Odisha, West Bengal, and Karnataka. Maharashtra has the highest number of polluted river stretches.
- ❑ According to the CPCB's 2021 report on the National inventory of sewage treatment plants, the country generated 72,368 million liters per day (MLD) of sewage, while the operational treatment capacity was only 26,869 MLD.
- ❑ This **untreated or partially treated sewage**, when discharged into rivers, elevates **biological oxygen demand**, contributing to river pollution.

Methods to Tackle River Pollution in the Indian Context

The below are some of the Indian efforts in combating pollution:

- ❑ **Wastewater Treatment Plants:** Implementation and upgradation of sewage treatment plants, like the **Yamuna Action Plan** in Delhi, to treat domestic and industrial wastewater before discharge into rivers.
- ❑ **Effluent Standards and Regulation:** Enforcing stringent effluent discharge standards, as seen in the case of the Ganga Action Plan, to control industrial pollutants and maintain water quality.
- ❑ **Riverfront Development and Restoration:** Riverfront development projects, such as the **Sabarmati Riverfront in Ahmedabad**, focusing on restoring river ecosystems and improving water quality.
- ❑ **Community Participation and Awareness:** Initiatives like the **Namami Gange** program involving local communities in cleaning and rejuvenating the Ganga, fostering awareness and active participation.
- ❑ **Green Belts and Riparian Vegetation:** Planting and maintaining green belts along riverbanks, exemplified by the **Green Wall of Aravalli** along the Sabarmati River, to reduce runoff and improve water quality.
- ❑ **Regulation of Sand Mining:** Implementing regulations on sand mining, as witnessed in the case of the **Kerala government's** measures to control illegal sand mining in rivers like the **Periyar**.
- ❑ **Floating Treatment Wetlands:** Utilizing floating treatment wetlands, demonstrated in Dal Lake, Srinagar, to naturally treat pollutants and improve water quality.
- ❑ **Bioremediation Techniques:** Applying bioremediation techniques, such as the use of microbes, to clean polluted stretches, as experimented in the Mula-Mutha River in Pune.
- ❑ **Strict Enforcement of Anti-Pollution Laws:** Implementing and strictly enforcing laws like the **Water (Prevention and Control of Pollution) Act** and the **Environment Protection Act** to hold polluters accountable.
- ❑ **River Monitoring and Data Collection:** Establishing comprehensive river monitoring systems, like the Real-Time Water Quality Monitoring on the **Ganga**, to collect data and promptly address pollution incidents.
- ❑ **Restoration of Riparian Habitats:** Focusing on restoring and protecting riparian habitats, as seen in the case of the Kali River Conservation Project, to enhance biodiversity and ecosystem health.
- ❑ **Sustainable Agricultural Practices:** Promoting sustainable agricultural practices, exemplified by initiatives in the **Cauvery basin**, to reduce chemical runoff and enhance water quality.

Implementing a combination of these methods, tailored to the specific challenges of each river, is crucial for effectively tackling river pollution in the Indian context.

SAND MINING

Sand mining involves the extraction of primary natural sand and sand resources, including mineral sands and aggregates, from terrestrial, riverine, coastal, or marine environments. This process aims to extract valuable minerals, metals, crushed stone, sand, and gravel for subsequent processing. Driven by various factors, this activity poses significant threats to ecosystems and communities.

Sources of Sand in India:

Sustainable Sand Mining Management Guidelines (SSMMG) 2016 identify sources:

- ❑ River (riverbed and flood plain)
- ❑ Lakes and reservoirs
- ❑ Agricultural fields
- ❑ Coastal/marine sand
- ❑ Palaeo-channels
- ❑ Manufactured Sand (M-Sand)

Factors Driving Illegal Sand Mining:

- ❑ **Lack of Regulation and Enforcement:** Weak regulatory frameworks and enforcement contribute to illegal mining proliferation.
- ❑ **High Demand for Construction Materials:** Construction industry demand intensifies illegal extraction, stressing riverbeds and coastal areas.
- ❑ **Corruption and Mafia Influence:** Organized sand mafias and corruption perpetuate illegal mining, undermining regulation.
- ❑ **Lack of Sustainable Alternatives:** Limited adoption of alternatives like **M-sand** leads to overreliance on riverbed sand.
- ❑ **Weak EIA Implementation:** Ineffective EIA implementation allows unauthorized sand extraction, unnoticed due to insufficient awareness.

Consequences of Sand Mining:

- ❑ **Erosion and Habitat Disruption:** Unregulated mining alters riverbeds, causing erosion and disrupting aquatic habitats.
- ❑ **Flooding and Sedimentation:** Sand depletion contributes to increased flooding and altered flow patterns, impacting ecosystems.
- ❑ **Groundwater Depletion:** **Deep pits** from mining lower the groundwater table, leading to local water scarcity.
- ❑ **Biodiversity Loss:** Habitat disruption results in significant biodiversity loss, affecting aquatic and riparian species.

Initiatives to Prevent Sand Mining in India:

- ❑ **Mines and Mineral Development and Regulation Act, 1957 (MMDR Act):** Under the Mines and Minerals

(Development and Regulations) Act, 1957 (MMDR Act), sand is categorized as a “minor mineral.”

- The **authority** over minor minerals is delegated to the **State Governments**. Section 3(e) of the MMDR Act is designed to counteract illegal mining by empowering the government to enact laws aimed at curbing unlawful practices.
- ❑ **MMDR Amendment Act, 2023:** Recent parliamentary amendment to the MMDR Act, strengthening regulations.
- ❑ **2006 Environment Impact Assessment (EIA) Mandate:** Supreme Court mandates approval for all sand mining, addressing ecological impacts, even for areas less than 5 hectares.
- ❑ **Sustainable Sand Management Guidelines 2016:** Issued by MoEFCC, focusing on environmentally sustainable and socially responsible mining.
- ❑ **Enforcement and Monitoring Guidelines for sand mining 2020:** Uniform protocol for monitoring sand mining, covering identification, dispatch, and end-use.

Way Forward

- ❑ **Sand Alternatives:** Prioritize alternatives like manufactured sand, artificial sand, fly ash, and crushed rock to reduce sand extraction. For example, Singapore successfully replaced sand with copper slag in concrete production.
- ❑ **Regulatory Strengthening:** Classify sand as a major mineral and centralize regulation to uniformly manage sand mining across states.
- ❑ **Digital Technology Implementation:** Utilize satellite imagery, drones, GPS tracking, and remote sensing for real-time monitoring of mining activities. For instance, Odisha’s integrated mines and mineral management system portal facilitates electronic regulation of mining activities.
- ❑ **Enforcement Enhancement:** Stricter enforcement of existing mining regulations and higher penalties for illegal sand mining. Bihar has established permanent checkpoints at sensitive locations to combat illegal sand mining.
- ❑ **Establishing Basic Standards:** Implement a global program to monitor and standardize the sand mining industry to prevent illegal activities and ensure compliance.

Thus, reforming regulations, promoting sustainable alternatives, and enforcing guidelines are crucial in combating the detrimental impacts of sand mining in India.

INTER STATE RIVER DISPUTES

One of the most controversial topics today flagging the Indian states is the inter-state river conflicts. The **Satluj Yamuna Link Canal** and the **Cauvery Water Dispute** are two recent, well-known, and contentious river disputes.

- ❑ For the country’s many rivers, a number of **Inter-State Water Disputes Tribunals** have been established.
- ❑ **Composition of Inter-state river disputes tribunal:** Include the Chief Justice of India, a serving Supreme Court justice, and two additional judges who are either Supreme Court or High Court judges make up the tribunal.
- ❑ **The Inter-State Water Disputes Act of 1956** governs the resolution of water disputes.
- ❑ According to the law, state governments are free to go to the center and request a tribunal to settle any water disputes they may have with other states. The tribunal’s judgment is conclusive.
- ❑ The Inter-State Water Disputes Legislation of 1956 was **amended** by the **Inter-State River Water Disputes (Amendment) Bill, 2017**, which was introduced in Lok Sabha.
 - A modernized Tribunal with a permanent establishment, a permanent office, and infrastructure is introduced by the bill.
 - The bill also includes a provision for the Central Government to establish a **Dispute Resolution Committee (DRC)** to settle disputes amicably within a six- to one-year time frame.
- ❑ **Other well known River Disputes**
 - **Krishna Water Disputes Tribunal II (2004)** – Karnataka, Telangana, Andhra Pradesh, Maharashtra
 - **Mahanadi Water Disputes Tribunal (2018)** – Odisha & Chhattisgarh
 - **Mahadayi Water Disputes Tribunal (2010)** – Goa, Karnataka, Maharashtra
 - **Ravi & Beas Water Tribunal (1986)** – Punjab, Haryana, Rajasthan
 - **Vansadhara Water Disputes Tribunal (2010)** – Andhra Pradesh & Odisha.

GOVERNMENT INITIATIVES

New National Water Policy

Until recently, the National Water Policy 2012 was in practice. To address the recent and emerging challenges of the water sector, the new National Water Policy calls for a multi-stakeholder and multi-disciplinary approach. This was the **first time** that the government asked a committee of independent experts to draft the policy in 2019.

Need for a new national policy:

To overcome the following Drawbacks of 2012 policy:

- ❑ **Overemphasis on Interlinking Rivers:**
 - The ambitious proposal to link major rivers across India has encountered technical, environmental, and social concerns, with limited progress made so far.

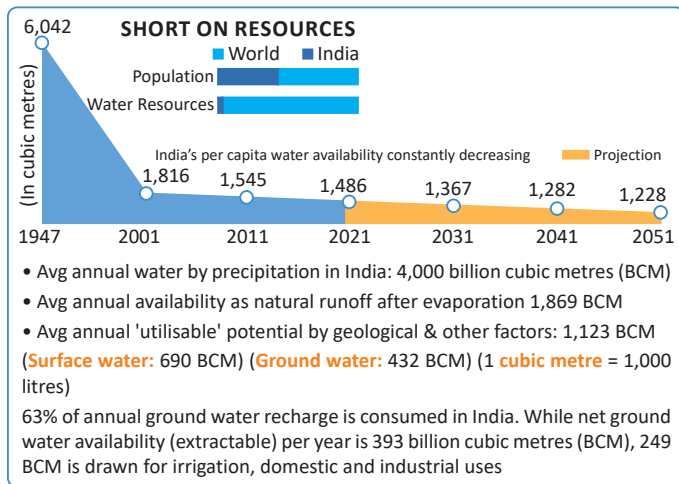


Fig. 4.40: Important facts on shortage of water resources

❑ Inadequate Focus on Groundwater Management:

- Overexploitation of groundwater resources continues to be a major challenge, and the policy's emphasis on surface water hasn't adequately addressed this issue.

❑ Neglected Inter-State Water Disputes:

- Complex water conflicts between states remain unresolved, hindering effective water management at a national level.

❑ Limited Public Participation:

- The policy formulation process lacked sufficient public consultation and stakeholder engagement, leading to concerns about its inclusivity and effectiveness.

For catering the following New Challenges:

❑ Addressing Climate Change Impacts:

- The policy must include strategies to manage water resources amidst changing climatic conditions, including heightened variability and extreme weather events.

❑ Promoting Integrated Water Management:

- An integrated approach is crucial, encompassing rainwater harvesting, inter-basin transfers, and the conjunctive use of surface and groundwater.

❑ Focus on Water Conservation and Efficiency:

- Prioritize water conservation efforts in agriculture, industry, and domestic sectors to maximize the utilization of existing resources.

❑ Strengthening Institutional Framework:

- Establish robust institutions with clearly defined roles and responsibilities to ensure effective policy implementation and water governance.

❑ Enhancing Public Participation:

- Formulate the new policy through a transparent and inclusive process, incorporating diverse stakeholder perspectives to foster public engagement.

Issues Dealt with in National Water Policy

❑ **Demand Management:** The National Water Policy recognizes the limitations of endlessly increasing water supply and advocates a shift towards demand management to ensure sustainable water use.

❑ **Irrigation:** A staggering 80-90 percent of India's water is consumed by irrigation, predominantly for rice, wheat, and sugarcane. Without a substantial shift in this water demand pattern, meeting the basic water needs of millions remains a challenge.

Groundwater: The NWP prioritizes sustainable and equitable management of groundwater, emphasizing participatory groundwater management. Stakeholders, designated as custodians of their aquifers, would be provided user-friendly information on aquifer boundaries, water storage capacities, and flows to enable effective groundwater management.

River: While historically seen as primarily serving economic purposes, the NWP accords rivers protection and revitalization as a top priority. It outlines the development of a Rights of Rivers Act, recognizing their right to flow, meander, and meet the sea.

Water Quality: Addressing water quality emerges as a critical, unaddressed issue in India. The NWP suggests the incorporation of water quality departments in every water ministry at the central and state levels.

Hydro Schizophrenia: The policy proposes radical governance reforms to address three forms of "hydro-schizophrenia" prevalent in water management:

- **Irrigation vs. Drinking Water:** The policy aims to bridge the gap between irrigation and drinking water management.
- **Surface vs. Groundwater:** It seeks to harmonize management strategies for surface and groundwater resources.
- **Water vs. Wastewater:** By avoiding the separation of water and wastewater in planning, the policy aims to prevent a decline in water quality.

The over-extraction of groundwater, for instance, leads to rivers drying up, affecting base-flows necessary for sustained river water levels post-monsoon. Integrating drinking water and irrigation management prevents aquifers from drying up due to conflicting uses. Similarly, avoiding the separation of water and wastewater planning prevents a decline in overall water quality.

Two Major Recommendations of the Proposed NWP:

❑ Shift the focus away from ceaselessly increasing water supply, instead prioritize measures for **demand management**.

- This entails diversifying crop patterns to incorporate less water-intensive varieties, aligning with regional agroecology.

- Additionally, it necessitates reducing the industrial water footprint, currently one of the highest globally, by minimizing freshwater consumption and transitioning to recycled water.
- Cities should mandate the redirection of all non-potable uses, such as flushing, fire protection, vehicle washing, landscaping, and horticulture, to treated wastewater.
- ❑ A necessary shift in focus is imperative within the supply-side, given the dwindling availability of sites for the construction of additional large dams in the country.
 - Simultaneously, water tables and groundwater quality are diminishing in many areas.
 - Despite trillions of liters of water stored in large dams, a significant portion fails to reach the intended farmers.
 - The policy outlines a solution involving the deployment of pressurized closed conveyance pipelines, integrated with **Supervisory Control and Data Acquisition (SCADA)** systems and pressurized micro-irrigation.

The draft policy also states that a global consensus is emerging in favor of “**nature-based solutions**” for water storage and supply. Consequently, the National Water Policy (NWP) places a major emphasis on the supply of water through the revitalization of catchment areas. This rejuvenation needs to be incentivized through compensation for ecosystem services, particularly directed towards vulnerable communities in upstream, mountainous regions.

Namami Devi Narmade

The Namami Devi Narmade was a programme aimed at conservation of Narmada river through **social mobilization**, started by the Madhya Pradesh government. It was called the **Namami Devi Narmade Sewa Yatra**

- ❑ It lasted for about 118 days, beginning on December 11, 2016. Throughout the Yatra, many **cultural and religious events** were conducted.
- ❑ The campaigns aimed at increasing public knowledge, participation and support for organic farming, soil conservation, afforestation and pollution abatement.

Namami Gange Programme

The Namami Gange Programme, also known as the **National Mission for Clean Ganga (NMCG)**, was launched by the Government of India in 2014 to address the pressing issue of pollution and degradation in the Ganges River. This iconic river, revered by millions, has suffered from environmental challenges due to human activities.

Key objectives and components of the Namami Gange Programme include:

- ❑ **Sewage Treatment Plants (STPs):** Constructing and upgrading sewage treatment infrastructure along the Ganges and its tributaries to effectively treat domestic and industrial wastewater.

- ❑ **Industrial Effluent Monitoring:** Monitoring and regulating industrial discharges into the Ganges to reduce pollution from industries.
- ❑ **River Surface Cleaning:** Initiatives to cleanse the Ganges surface from floating **solid waste** and debris.
- ❑ **Bioremediation of Rivers:** Implementing bio-remediation techniques to enhance the water quality of the Ganges and its tributaries.
- ❑ **Afforestation:** Planting trees and promoting afforestation along riverbanks to prevent soil erosion, boost groundwater recharge, and enhance overall **riverine ecosystems**.
- ❑ **Public Awareness and Participation:** Engaging communities, religious institutions, and the public to raise awareness about the importance of a clean Ganges and encouraging active participation in its conservation.
- ❑ **Biodiversity Conservation:** Implementing measures to safeguard and conserve the biodiversity of the Ganges River and its associated ecosystems.
- ❑ **Riverfront Development:** Developing riverfront areas for recreational and cultural activities while ensuring sustainability and eco-friendly practices.
- ❑ **Research and Development:** Promoting research and development activities to find innovative and sustainable solutions for Ganges rejuvenation.
- ❑ **Cultural and Spiritual Heritage:** Integrating cultural and spiritual aspects to emphasize the historical and religious significance of the Ganges in Indian heritage.

The Namami Gange Programme adopts an integrated and holistic approach to Ganges conservation, involving collaboration among government agencies, Non-governmental organizations (NGOs), and local communities. Its successful implementation is vital for ensuring the long-term health and sustainability of the Ganges River ecosystem and fulfilling the cultural and spiritual aspirations of millions who consider the river sacred.

National Hydrology Project (NHP)

It is a comprehensive initiative aimed at enhancing India’s **hydrological information systems** and water resource management. It was launched with the support of the **World Bank**.

- ❑ The project seeks to strengthen the country’s **capacity** for hydrological monitoring, data management, and decision-making processes.
- ❑ The NHP focuses on the **modernization of infrastructure**, including the establishment of hydromet stations, river gauging stations, and data centers to facilitate the collection and analysis of hydrological data.
- ❑ By promoting the use of advanced technology and improving **institutional capabilities**, the project aims to enhance flood forecasting, reservoir operations, and overall water resource planning.

- ❑ The NHP plays a pivotal role in promoting **sustainable water management** practices and addressing the challenges posed by climate change, ensuring efficient utilization of water resources for the benefit of both rural and urban communities.

CONTEMPORARY ISSUES

❑ Dam Safety Bill

The Dam Safety Bill, initially presented in the Lok Sabha in July 2019, successfully traversed the legislative process and was approved by the Rajya Sabha in December 2021, emerging as the Dam Safety Act of 2021.

Rationale for Dam Safety:

- **Ageing Dams:** India stands as the third-largest builder of large dams globally, with over 5,200 already in existence. Of these, approximately **1,100** have **surpassed** the **50-year mark**, with some exceeding 120 years.
 - ❑ Signs of aging dams, including increased instances of failures, escalating costs for repair and maintenance, growing reservoir sedimentation, and diminished functionality, underscore the imperative for dam safety measures.
- **Lower Shelf Life:** Medium and minor dams, numbering in the hundreds of thousands, pose an even more precarious situation due to their inherently shorter shelf life compared to large dams.
 - ❑ This vulnerability jeopardizes water security, impacts farmers' income, and heightens the risk of flooding, necessitating urgent attention.
- **Structurally Vulnerable Dams:** Scholarly studies consistently reveal that Indian reservoirs lack a robust understanding of sedimentation science during the design phase.
 - ❑ Any large storage structure, irrespective of its composition (concrete, masonry, or earth), can weaken structurally over time, emphasizing the need to address these vulnerabilities.
- **Mitigating Climate Change Impacts:** Increasing instances of severe flooding and other extreme environmental events, influenced by climate change, can surpass a dam's design limits. This necessitates proactive measures to mitigate the impacts of climate change on dam infrastructure.
- **Hazardous Infrastructure:** Despite structural integrity, large dams are categorized as **"high hazard" infrastructure** due to the potential for catastrophic loss of human lives, livelihoods, and widespread destruction in the event of failure.
 - ❑ The recent **Uttarakhand flash floods** highlight the compounding effect of global warming-induced glacier melting exacerbated by the construction

of infrastructure such as Hydroelectric Power Projects (HEPs).

Objective: The Dam Safety Act mandates the supervision, inspection, and maintenance of specified dams nationwide, encompassing those over 15 meters in height or 10-15 meters with specific design conditions.

Salient Features:

- It establishes a **two-tiered** institutional mechanism, comprising the National Committee on Dam Safety (NCDS) and the National Dam Safety Authority (NDSA), along with State Level Bodies and obligations for dam owners.
- **Institutional Mechanism:**
 - ❑ **National Bodies:**
 - ◇ NCDS formulates policies and recommends regulations.
 - ◇ NDSA implements NCDA policies, aids State Dam Safety Organisations (SDSOs), and resolves disputes.
 - ❑ **State Bodies:**
 - ◇ SDSOs conduct perpetual surveillance and inspection.
 - ◇ The State Committee on Dam Safety oversees rehabilitation programmes and reviews SDSO work.
- **Obligations of Dam Owners:** Owners are responsible for safe dam construction, operation, and maintenance. They must establish dam safety units, conduct inspections pre and post-monsoon, after seismic events, floods, or signs of distress.
 - ❑ **Functions** include emergency action plans, regular risk assessments, and comprehensive safety evaluations.
- **Offenses and Penalties:** Obstruction or non-compliance may result in imprisonment for up to two years in case of loss of life.
- **Issues Associated:** Debates arise on Parliament's jurisdiction over intrastate river dams. Initiatives like the Dam Rehabilitation and Improvement Project (DRIP) complement the Dam Safety Act.

Recommendations for Improved Dam Safety:

- Environmental considerations in inspections.
 - Recognizing local factors and climate variations.
 - Prioritizing dam decommissioning.
 - Establishing an overarching authority to address interstate disputes, exemplified by the Mullaperiyar Dam issue between Kerala and Tamil Nadu.
- ❑ **Dam Rehabilitation and Improvement Project:**
- Recently, the Government of India inked a \$250 million loan pact with the World Bank for the Second Phase of Dam Rehabilitation and Improvement Project (DRIP Phase II).

Key aspects of this project are

- **Phase-I:** The Government, aided by the World Bank, initiated DRIP in April 2012, aiming to enhance safety and performance of selected dams and strengthen dam safety institutions.
- It covered 223 dams in seven states, with the **Central Water Commission** coordinating.
- A web-based tool, Dam Health and Rehabilitation Monitoring Application (**DHARMA**), was developed for data capture and AI-driven dam safety management.
- DRIP Phase-II and Phase III: Based on DRIP's success, the Ministry of Jal Shakti launched DRIP Phase II and Phase III in October 2020.
 - It spans 10 years with 19 states and 3 central agencies, budgeting Rs 10,211 Cr for 736 dams.
 - DRIP Phase-II, **co-financed** by the **World Bank and AIIB**, focuses on dam safety, institutional strengthening, and revenue generation from selected dams.

Objectives:

- Enhance safety and performance sustainably.
- Strengthen dam safety institutions.
- Explore alternative revenue generation methods for dam operation and maintenance.

Significance:

- India ranks third globally with 5334 large dams after the China and USA.
- Addresses the ageing infrastructure risk with over 1,115 large dams reaching 50 years by 2025.
- Mitigates flood risks and reduces greenhouse gas emissions.
- Focuses on dam safety legislation and employment generation.

DRIP Phase II advances dam safety, incorporates AI, and aligns with climate resilience, flood mitigation, and employment generation, crucial for India's water security and sustainable development.

- **TamiraSES project:** The TamiraSES project is a comprehensive initiative at the district level, with the primary goal of restoring the Social Ecological Systems within the Tamiraparani river landscape.
 - This restoration effort spans from the headwaters to the estuary, aiming to create favorable conditions for the flourishing of native biodiversity. The project further seeks to safeguard and enhance multiple ecosystem services for the benefit of local stakeholders.
 - As an initial step, five social ecological observatories will be established in the first phase of the project. These observatories will function as pilot sites, allowing for the extrapolation of insights gained from their experiences to scale up the initiative effectively.

- The overarching vision of the TamiraSES project extends beyond the rejuvenation of the Tamirabarani River alone, encompassing the revitalization of all water bodies within the riverscape of Tirunelveli.

□ Rejuvenation Project of 13 Major Rivers:

River rejuvenation is the process of revitalizing a river's natural flow and well-being, which may have been compromised by human activities like waste disposal, encroachment, and pollution. This restoration process entails the removal of pollutants, cleaning of riverbeds, and the rehabilitation of the surrounding natural environment.

Impacts: The following are the consequences of river rejuvenation:

- **Habitat Restoration:** River rejuvenation contributes to the restoration of natural habitats for plants and animals, fostering biodiversity and creating a healthier ecosystem.
- **Cleaner Water:** The process results in cleaner water, suitable for both drinking and irrigation purposes, thus addressing water quality concerns.
- **Tourism and Livelihood Improvement:** Reviving rivers has the potential to boost tourism in the region, attracting visitors interested in the restored natural beauty. Additionally, it positively affects the livelihoods of communities dependent on the river.
- **Flood Control:** River rejuvenation plays a crucial role in controlling floods by allowing a healthy river to absorb excess water during heavy rains. This helps prevent downstream flooding, saving lives, protecting property, and mitigating the economic impact of floods.

In the year 2022, the Ministry of Environment, Forest and Climate Change unveiled Detailed Project Reports (DPRs) amounting to Rs. 19,342.62 crore for the rejuvenation of 13 major rivers. These initiatives focus on implementing forestry interventions to enhance the ecological health of the following rivers: **Jhelum, Chenab, Ravi, Beas, Sutlej, Yamuna, Brahmaputra, Narmada, Godavari, Mahanadi, Krishna, Cauvery, and Luni.**

Examples of River Rejuvenation:

- **The Rhine River** in Europe, once heavily contaminated, has experienced a substantial cleanup initiative, leading to its transformation into a dynamic and flourishing ecosystem.
- **The Kuttamperoor River**, spanning 7.2 km in **Alappuzha, Kerala**, has undergone a revival, achieved through a collaborative effort involving the **public and government interventions** over a span of six years.
 - For more than a decade, the river remained lifeless, grappling with issues such as waste dumping, encroachments, and various other human-induced activities. The rejuvenation process involved deepening the river channel,

eliminating encroachments, and constructing embankments on both sides.

- ❑ The revitalization of the **Kuttamperoor** River is anticipated to play a significant role in flood control within the region.

❑ **Rights of Rivers:**

Rights of Rivers encompass various aspects, emphasizing a holistic view of rivers as ecosystems. These include:

- **Flow:** Instead of merely quantifying water volume, considering a river as an ecosystem broadens the scope of its rights.
- **Flora and Fauna:** Encompassing aquatic life, biodiversity in catchment areas, forests, tributaries, groundwater, rocks, and soil in its bed and banks.
- **Human Settlements:** Recognizing the ecological causes and conditions supporting natural habitats, with consideration for human settlements dependent on the river.
- **Economy:** Balancing rights to accommodate fishing and local, subsistence-based needs, fostering a respectful relationship with the river as an ecosystem.

The **Universal Declaration of the Rights of Rivers** is a global civil society initiative acknowledging basic entitlements for all rivers. Originating in 2008 with Ecuador, this movement sees **rivers as living entities**, marking a shift from an extractive mindset to conservation safeguards.

- Recent campaigns globally, including in the US, Canada, Ecuador, and Argentina, have recognized river rights.
- The passing of **legislation** in the parliament of **New Zealand** in 2017 that recognised the country's third-longest river, the **Whanganui river**, as a legal person with the same set of rights as citizens of the country would have for themselves, is a significant boost to the campaign.

Indian scenario: In 2017, the **Uttarakhand HC** declared **Ganga and Yamuna**, along with glaciers, as legal persons, emphasizing corresponding rights.

Challenges include the need for cultural change, questioning current development paradigms, addressing cross-boundary issues, and overcoming a cooperation deficit in implementing laws safeguarding river rights.

❑ **Punjab Government has declared the Indus river dolphin as Punjab's state aquatic animal:**

Indus river dolphins represent one of only four species and subspecies of river dolphins globally that exclusively inhabit **freshwater** throughout their entire lives.

- **Distribution:** The Indus river dolphin stands as the second most endangered freshwater river dolphin. Presently, there are only about 1,800 individuals found in the Indus River in Pakistan, with a population ranging from eight to 10 in the Beas River.
- **IUCN Status:** Designated as "**Endangered**" by the International Union for Conservation of Nature (IUCN).

- **Characteristics:** Indus Dolphins display a **brown/gray** coloration.
 - ❑ **Functionally blind**, they heavily rely on **echolocation** for navigation, communication, and hunting prey such as prawns, catfish, and carp.
 - ❑ With a lifespan of at least 30 years, they can grow over 2 meters in length.
- **Threats:** The primary threat faced by these dolphins is the construction of numerous **dams and barrages**, which not only **fragment** the population into smaller groups but also degrade their habitat and hinder migration.
 - ❑ Other factors that contribute include **poaching**, **entanglement** in fishing nets, and river pollution.

NATIONAL WATERWAYS

National waterways play a crucial role in a country's transportation and economic development. Water transport offers numerous benefits, including cost-effectiveness, reduced congestion, eco-friendliness, accessibility, economic development, trade facilitation, energy efficiency, disaster relief, tourism, infrastructure development, and connectivity.

- ❑ It can carry large quantities of goods over long distances, reducing transportation costs for businesses and consumers.
- ❑ In 2022 FY, the **total cargo traffic** on Indian national waterways amounted to almost 11 million metric tons.
- ❑ India has notified **111 National Waterways** under the **National Waterways Act, of 2016**, out of which 13 are operational for shipping or navigation of cargo/passenger vessels.
- ❑ The government created the **Inland Waterways Authority of India (IWAI)** in **1986** for the purpose of **regulating** and developing inland waterways.
- ❑ Waterways also help alleviate congestion on roads and railways, reducing traffic jams and infrastructure wear.
- ❑ They produce **fewer greenhouse gas emissions** per unit of cargo transported, contributing to cleaner air and lower carbon footprints.
- ❑ National waterways provide access to remote regions, promoting economic development and connecting them to major markets and trade routes.
- ❑ They also facilitate trade by facilitating the import and export of goods. Water transport is energy efficient, as ships and barges can carry large volumes of cargo with low fuel consumption per ton-mile, contributing to energy conservation.
- ❑ Waterways also serve as critical routes for **disaster relief** and **emergency response** efforts, attracting visitors for activities like boating, fishing, and sightseeing.
- ❑ Investment in national waterways can lead to the construction and maintenance of infrastructure, creation of jobs and a major support in economic growth.

National Waterway	River System	Route	Location
NW-1	Ganga-Bhagirathi-Hooghly (Longest waterways 1620 km)	Prayagraj – Haldia	Uttar Pradesh, Bihar, Jharkhand, West Bengal
NW-2	Brahmaputra	Sadiya-Dhubri	Assam
NW-3	West Coast Canal, Champakara Canal, and Udyogamandal Canal	Kottapuram – Kollam	Kerala
NW-4	Krishna and Godavari	Kakinada–Puducherry stretch of canals, Kaluvelly Tank, Bhadrachalam – Rajahmundry, Waziraba–Vijayawada	Andhra Pradesh, Tamil Nadu, and Puducherry
NW-73	Narmada River	–	Gujarat and Maharashtra

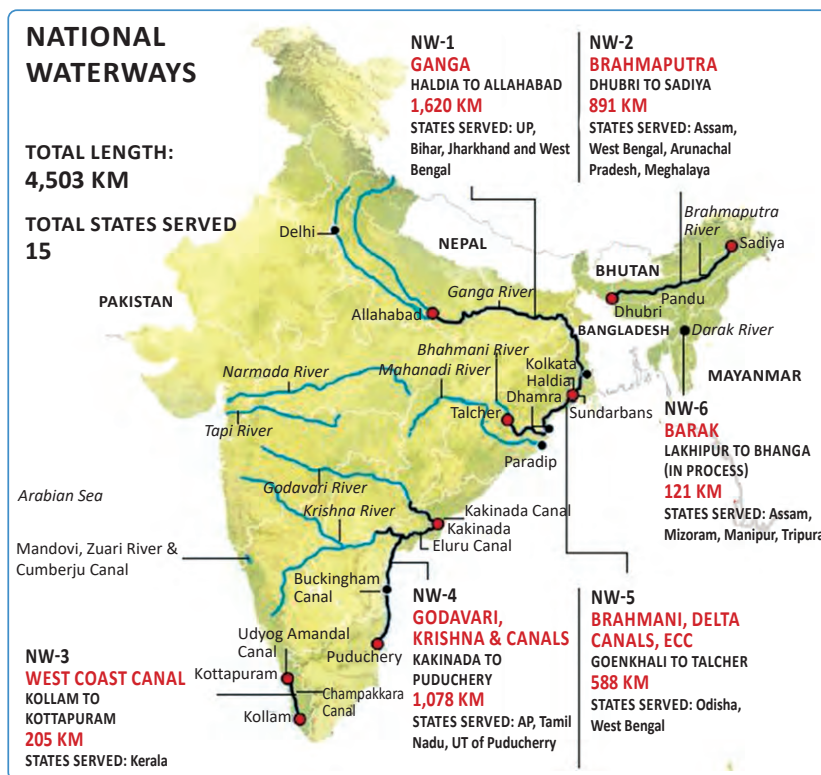


Fig. 4.41: National Waterways of India



IGNITE YOUR MIND

The spatial and temporal changes in precipitation across India frequently cause human suffering such as a lack of drinking water, inundation of agricultural lands, crop failure, and so on. In your opinion would river interlinking provide a permanent solution to the problem?

CONCLUSION

India's drainage system, comprising rivers like the Ganges, Brahmaputra, and Indus, supports diverse ecosystems and agricultural livelihoods but faces challenges from pollution, urbanization, and climate change. To sustain it, effective waste treatment, conservation practices, and resilient infrastructure are crucial. Climate change exacerbates risks like floods, necessitating adaptive measures. Integrated water resource management is key, involving flood control, watershed development, and sustainable agriculture. Preserving river health is essential for biodiversity and national well-being, requiring a united effort towards sustainable practices and responsible governance amid evolving environmental dynamics.

Climate of India

5

INTRODUCTION

Climate describes the average weather conditions for a given place across time. According to the Köppen classification, India has **six** main climatic groups, including deserts in the west, alpine tundra, the northern glaciers, and humid tropical areas with rain forests in the southwest and on the islands. With several distinct micro-climates, India is one of the most climatically diverse countries.

Weather and Climate

Climate is the average of weather conditions and fluctuations across a vast area over a long period of time (more than 30 years), whereas **weather** is the state of the atmosphere over a specific region at a certain moment in time. While the condition of the atmosphere may change rapidly (within a day or a week), climatic shifts are more gradual and may take 50 years or more to become noticeable.

SALIENT FEATURES OF INDIAN CLIMATE

The diversified climate of India, which spans tropical, subtropical, and alpine regions, is influenced by a unique monsoon season, seasonal fluctuations, regional temperatures and rainfall inequalities. These characteristics have a substantial influence on agriculture, economics, and lifestyle, making understanding these factors critical for analysing India's environmental and geographical dynamics.

- ❑ **Reversal of Winds:** With the passage of each season, the Indian climate is characterised by a full reversal of the wind system.
 - Over the winter, winds blow from north-east to south-west in the direction of trade winds. These winds are known for their dryness, lack of moisture, low temperatures, and high-pressure systems throughout the country.
 - The winds completely reverse their course during the summer, mostly blowing from the south-west to the north-east.
- ❑ **Formation of Alternatively High and Low Pressure Areas Over the Land:** With each season, there is a shift in the atmospheric pressure conditions.
 - High-pressure regions form on the northern side of India during the winter due to low-temperature conditions.
 - On the other side, the severe land heating throughout the summer causes a thermally induced low-pressure cell to develop over the country's northwest.
 - The wind's direction and strength are governed by these pressure zones.
- ❑ **Seasonal Variability in Rainfall:** In India, the end of the summer, which may last anywhere from 1 to 5 months depending on where you are in the country, receives more than 80% of the country's annual rainfall.
 - Flooding and soil erosion are difficulties brought on by rainfall since they come in the form of a heavy downpour.
 - Sometimes there is a long stretch of dry weather followed by days of continuous rain. The rainfall at Cherrapunji in a single day is equal to ten years' rainfall in Jaisalmer, Rajasthan.
- ❑ **Unification of Indian Climate:** The Himalayas and surrounding mountain ranges expand into northern India from east to west.
 - These towering mountain ranges obstruct India's access to the extremely cold northerly winds from Central Asia.
 - Consequently, India's northernmost regions, which extend beyond the Tropic of Cancer, experience tropical weather.
 - The monsoon winds are driven by these mountains to deliver rain to India, and their effects are felt across the country. As a result, the entire country is affected by monsoon-like weather.
- ❑ **Climatic Variation in India:** Due to its large geographical size and diversified terrain, India experiences a wide range of climate variations. These variations may be seen in the winds, temperature, rainfall, humidity, aridity, and other factors. The following are key aspects of climatic variation across different regions of India:
 - **Northwestern Plains:** Temperatures in the northwest plains, which include states such as Punjab, Haryana, and parts of Rajasthan, vary dramatically. Summers are hot, with temperatures frequently topping 40°C (104°F), while winters may be extremely cold, with at night temperatures falling below freezing point.
 - **Himalayan Region:** The Himalayan region in northern India features alpine and subarctic climates, with high-altitude areas experiencing heavy snowfall during winter. Throughout the summer, this location provides relief from the heat.

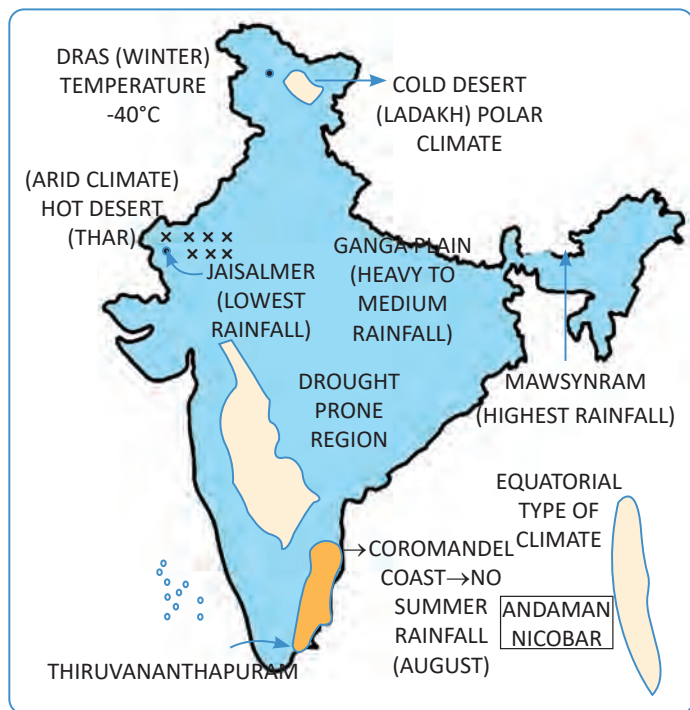


Fig. 5.1: Climatic Variation across India

- **Western Coast and Eastern Coast:** The Western Ghats and other western coastal regions benefit from a tropical wet climate. These areas receive a lot of rain, which keeps them lush and green all year long. This zone includes large cities like Kochi and Mumbai. The Bay of Bengal has an impact on the eastern coastal regions, including parts of Andhra Pradesh and Odisha. They have a tropical wet and dry climate, with the monsoon season bringing a lot of rain.
- **Desert Zone:** The Thar Desert in the northwestern state of Rajasthan is arid and has a desert climate. It experiences extremely hot temperatures during the day and cooler nights, with very little annual rainfall.
- **Deccan Plateau:** The Deccan Plateau, comprising states like Karnataka and Maharashtra, has a semi-arid to tropical climate. It experiences seasonal variations with hot, dry summers and colder winters.
- **Islands:** India has two island groups - The Andaman and Nicobar Islands are located in the Bay of Bengal, and the Lakshadweep Islands, situated in the Arabian Sea. These regions have a tropical climate with high humidity and receive significant rainfall.
- **High Rainfall Zones:** Some parts of India, such as the Western Ghats and the northeastern states, are known for receiving exceptionally high annual rainfall, contributing to their rich biodiversity and fertile soil.
- **Northeastern Region:** Because of their closeness to the Bay of Bengal, the northeastern states, especially Assam and Meghalaya, have some of the highest rainfall levels in the nation. This leads to a subtropical environment that is humid and has abundant vegetation.

FACTORS DETERMINING THE CLIMATE OF INDIA

The climate of India is influenced by several factors that interact to create India's diverse climate zones, ranging from the warm and humid climate in the south to the cold and dry climate in the northern Himalayan regions. It's important to note that India experiences three main seasons: summer, monsoon, and winter, each with its climate characteristics.

Factors Related to Location and Relief

- **Location:** The temperature rises as one goes closer to the equator. When one travels closer to the poles, the temperature drops.
 - India is located in the Northern Hemisphere, $8^{\circ}4'$ closer to the Equator, and the $23\frac{1}{2}^{\circ}$ Tropic of Cancer runs through the centre of the country.
 - As a result, a tropical climate can be found to the south and a subtropical climate to the north of this latitude. **Andhra Pradesh**, for example, would be warmer than Haryana. In general, regions south of the Tropic of Cancer receive more solar radiation than regions north of it.
- **Altitude:** It is the height above sea level. As we move higher above the earth's surface, the atmosphere thins and we experience shortness of breath, causing the temperature to drop.
 - **For example**, towns on the hills, such as Shimla, have a milder climate than towns on the plains, such as Ludhiana.
- **Relief:** Temperature, air pressure, wind direction and speed, and the amount and distribution of rainfall are all influenced by India's physiography or relief.
 - The windward portions of the Western Ghats and Assam receive heavy rainfall from June to September, while the southern plateau remains dry due to its leeward position along the Western Ghats.
- **Distance from the Sea:** The southern region of India is geographically bounded by the Arabian Sea to the west, the Bay of Bengal to the east, and the Indian Ocean to the south.
 - Due to the cooling impact of the sea, neither the winters nor the summers in this area are very cold or extremely hot.
 - As an example, the northern part of India situated at a considerable distance from the coastline, has an exceptionally severe climatic pattern, whereas a region of south India near the sea has an identical climate.
- **Distribution of Land and Water:** India is surrounded by the Indian Ocean to the south and the Himalayas to the north.

- Water, unlike land, takes longer to warm up or cool down. This temperature difference between the land and the sea causes varying air pressure regions around the Indian subcontinent during different seasons.
- These differences in air pressure cause the monsoon winds to shift direction.

- ❑ **The Himalayan Mountains:** The rising Himalayas in the north, along with their subsequent expansions, act as an effective climate boundary.
 - The Himalayas mountain range shields the subcontinent from the harsh northern winds.
 - The Arctic Circle is where these very cold winds come from, and they blow across central and eastern Asia. The Himalayas also act as a barrier to the monsoon winds, causing them to spread rainfall over the subcontinent.

Causes of the Formation of Thar Desert and Least Rainfall in Gujarat and Rajasthan

- ❑ Monsoon winds flow parallel to Aravallis in Rajasthan and Gujarat and are not obstructed by any orographic barrier, hence there is no orographic rainfall occurring in these regions.
- ❑ There are no vertical wind patterns or convection cells in Rajasthan or Gujarat. Horizontal wind movements are the only type of wind movement prevalent in Gujarat and Rajasthan because the monsoon winds are directed towards low-pressure cells in Tibet. The STJ, or Subtropical Jet, causes a strong divergence in the area during the winter.

Reasons why Haryana and Punjab are not Deserts like Rajasthan Despite Locational Proximity

Despite their northern proximity, Haryana and Punjab boast fertile landscapes unlike Rajasthan's desert expanses. Three key factors drive this contrast:

- ❑ **Himalayan Embrace:** Haryana and Punjab are closer to the Himalayas, benefiting from moisture-laden winds that bring life-giving rain. Rainfall is significantly lower in distant Rajasthan, particularly the Thar Desert.
- ❑ **Rainfall Pattern:** Haryana and Punjab receive comparatively higher annual rainfall than Rajasthan.
 - The precipitation brought by western disturbances and monsoon along with the irrigation infrastructure fed by rivers, supports agriculture and prevents desert like conditions in Haryana and Punjab.
 - Rajasthan experiences low and erratic rainfall, with high rainfall variability, making it arid. The limited water availability restricts agricultural activities in this region.

- ❑ **Geographical Features:** Abundant rivers like the Indus and extensive canal networks nourish Haryana and Punjab's plains, fostering agriculture and vibrant ecosystems. Rajasthan's arid regions, lacking such water resources, struggle with desertification.

Factors Related to Air Pressure and Winds

To properly understand the variations in local climates across India, we must first understand the workings of the following three components:

1. **Wind and Air Pressure Distribution on the Earth's Surface:** During the winter, a high-pressure system exists north of the Himalayas.

- Dry and cold winds travel from this region to low-pressure zones over the southern oceans.
- A low-pressure area could form over the middle of Asia and in the northwest of India during the summer.

2. **Impact related to upper Air Circulation:** Regional and global weather control elements, as well as the entry of numerous jet streams and air masses, all have an impact on upper air circulation.

- Atmospheric circulation transfers heat throughout the Earth's surface, impacting the water cycle, including cloud formation and precipitation events. The flow of air masses determines our everyday weather, while long-term circulation patterns impact regional climate and ecosystems.
- At the equator, air rises, resulting in low pressure and rain. As the air hits the boundary of the atmosphere, it can no longer move any farther, thus it travels north and south. The air cools, grows denser, and descends, resulting in high pressure and dry conditions roughly 30 degrees north and south of the equator.
- Different weather patterns, jet streams, and prevailing winds are a result of the global weather systems caused by the Earth's rotation and the differing amounts of heat received in various regions.

3. **Impact of Tropical Depressions and Western Cyclones:** Tropical depressions and western cyclones arriving in India during the southwest monsoon and winter seasons, respectively, create favorable conditions for rainfall.

- Western cyclonic disturbances are low-pressure systems that originate in the Mediterranean Sea region.
- These systems travel towards the Indian subcontinent under the influence of westerly jet streams.
- As a result, they disrupt the weather, particularly in the northwestern states of Jammu and Kashmir, Punjab, Haryana, and Himachal Pradesh.

Funnelling Effect

- ❑ When clouds are channelled into a narrow region between mountains and hence the cloud density increases.
- ❑ Mountains often cause winds to flatten out and pass over or around them. If a mountain range is broken by a pass or valley, the air is forced through the gap at a high rate of speed. The funnelling effect is a term used to describe wind that is driven through small valleys.
- ❑ For example, the Funnelling effect occurs in mountain ranges of western ghats, Eastern ghats and northeastern regions of the Himalayas such as Cherrapunji and Mawsynram in Meghalaya.

Funnelling Effect in Cherrapunji and Mawsynram

❑ Cherrapunji is situated on the windward side of the Khasi Hills, facing the Bay of Bengal. ❑ The funnelling effect occurs as moist air from the Bay of Bengal is forced to ascend the windward slopes of the Khasi Hills. ❑ As the air rises, it cools and condenses, leading to heavy rainfall in Cherrapunji. ❑ The funnel-like topography enhances the lifting of moist air, contributing to the extraordinary rainfall levels experienced in Cherrapunji.	❑ Mawsynram, located not far from Cherrapunji, also experiences a similar funnelling effect. ❑ It is situated on the eastern slopes of the Khasi Hills, facing the Bay of Bengal. ❑ Moisture-laden air is funneled and forced to ascend these slopes, leading to cooling and condensation of the air, resulting in significant rainfall. ❑ The unique topography enhances the precipitation, making Mawsynram one of the wettest places globally.
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INDIAN MONSOON

The term “monsoon” comes from the Arabic word “**Mausim**,” which means “season.” Several centuries ago, Arab navigators used the term monsoon to describe a seasonal reversal of winds off the coasts of the Indian Ocean, particularly the Arabian Sea. In the summer, it blows from the southwest to the northeast, and in the winter, it blows from the northeast to the southwest. Arab traders conducted the first scientific study of monsoon winds. Ten centuries ago, an Arab traveller named Al Masudi described how the monsoon winds and ocean currents across the northern Indian Ocean reversed.

MECHANISM AND THEORIES OF INDIAN MONSOON

The monsoons are a complicated meteorological phenomenon. The genesis of monsoons has given rise to a

variety of theories, according to meteorology experts. The following list includes some key ideas on the monsoon’s beginnings:

Classical Theory (Halley’s Theory)

- ❑ **Explanation of Monsoon:** Sir Edmund Halley in the 17th century attributed the monsoon to temperature variations between continents and seas.
- ❑ **Solar heating:** Halley suggested that the monsoon winds were driven by differential solar heating of land and sea. During summer, the Sun is directly overhead or slightly north of the equator, leading to more direct sunlight in the northern hemisphere, including India.
- ❑ **Wind Reversal:** The theory proposed that the air over the cooler ocean would then move toward the warmer, low-pressure land, resulting in a seasonal reversal of the prevailing winds.
- ❑ **Summer Conditions:** Intense sunlight causes rapid heating of the land, creating low-pressure areas over the Indian subcontinent.
- ❑ **Winter Conditions:** In winter, the Sun is over the southern hemisphere, causing the northern hemisphere, including India, to cool. And cooler land results in a high-pressure zone.
- ❑ **Limitations of Halley’s Theory:** It fails to explain variations in monsoon development, such as severe rainfall. additionally, it was unable to account for irregular delays in the onset of rain and other observed variations.

Air Mass Theory

- ❑ **Focus:** The air mass theory of monsoon centers around the seasonal changes in air masses and their characteristics, particularly over large land masses and adjacent oceans.
- ❑ **Key Features:**
 - It emphasizes the role of temperature and humidity differences between the land and sea, leading to the development of specific air masses.
 - During summer, the land heats up more quickly than the ocean, creating a low-pressure area over the land. This low-pressure area draws in moist air from the ocean, resulting in the monsoon rains.
 - In winter, the land cools faster than the ocean, leading to a reversal of wind patterns.
- ❑ **Frontal Boundaries:** Air mass theory often considers the presence of frontal boundaries, such as the monsoon front, where different air masses meet and interact.
- ❑ **Local and Regional Scale:** This theory tends to explain monsoon phenomena at a more local or regional scale, focusing on the characteristics and interactions of air masses over specific land and sea areas.

The Intertropical Convergence Zone (ITCZ) is a belt of low pressure near the equator where trade winds from the Northern and Southern Hemispheres converge and rise. The ITCZ is also known as the monsoon trough.

In July, the ITCZ is centered over the Gangetic plain at a latitude of about 20°N-25°N. This helps to form thermal lows over north and northwest India. Due to the movement of the ITCZ, the trade winds from the Southern Hemisphere cross the equator between 40° and 60° E longitude and start blowing from southwest to northeast. This is called the southwest monsoon. During the winter, the ITCZ moves southward, and the northeasterly winds become southerly winds. These are called the northeast monsoons.

Dynamic Theory

- ❑ **Focus:** The dynamic theory emphasizes the role of large-scale atmospheric circulation patterns and global pressure systems in driving the monsoon.
- ❑ **Key Features:**
 - It involves the seasonal shift of the Intertropical Convergence Zone (ITCZ), which is the region near the equator where trade winds converge.

- The migration of the ITCZ is influenced by the changing position of the sun, particularly during the summer and winter solstices.
- The monsoon winds are seen as a result of the differential heating between land and sea, leading to the establishment of a low-pressure system over land during the summer, drawing in moist air from the ocean.

- ❑ **Meteorological Phenomena:** The theory considers large-scale phenomena such as the Tibetan Plateau heat low and the development of the Asian low-pressure system.
- ❑ **Synoptic Scale:** Dynamic theory often operates on a synoptic scale, analyzing the overall atmospheric circulation and pressure patterns.
- ❑ **Comparison with Airmass theory:** In summary, while the dynamic theory of monsoon emphasizes the large-scale atmospheric circulation patterns, including the movement of the ITCZ and the influence of pressure systems, the air mass theory of monsoon focuses on the seasonal changes in air masses over land and sea, driven by differential heating. Both perspectives contribute to a holistic understanding of the monsoon phenomenon.

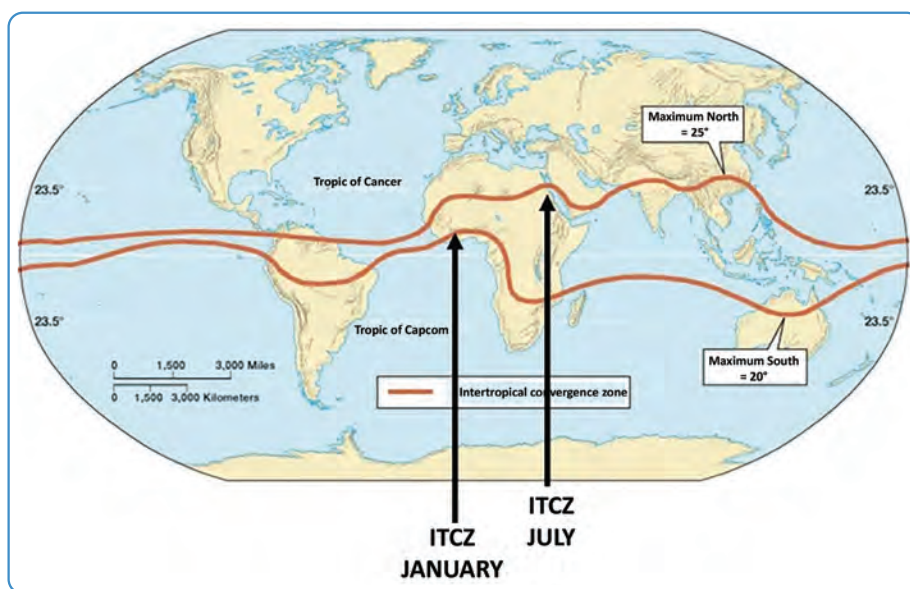


Fig. 5.2: Movement of ITCZ

The Jet Stream Theory (Modern Theory of Indian Monsoon)

- ❑ The Jet Stream Theory of Monsoon is inspired by ideas of M.T. Yin and Pierre Pedelaborde explains the influence of the jet stream on the onset, progress, and withdrawal of the Indian monsoon.
- ❑ The jet stream is a high-altitude, upper air circulation of fast-flowing air current in the atmosphere, and its position plays a crucial role in the dynamics of the monsoon in the Indian subcontinent.

Role of Sub-Tropical Jet Stream (STJ)

- ❑ Sub-Tropical Jet stream is a narrow band of fast moving air flowing from west to east [Westerlies].
- ❑ STJ in northern hemisphere flows between 25° to 35° N in the upper troposphere at a height of about 12-14 km.
- ❑ The Subtropical Jet Stream (STJ) is important for both lowering monsoon winds and hastening their onset.
- ❑ The burst of monsoons depends upon the upper air circulation which is dominated by STJ.

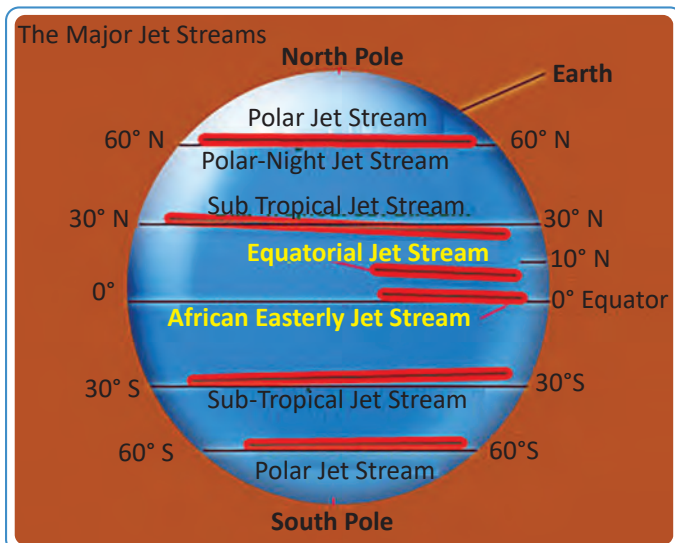


Fig. 5.3: The Major Jet Streams

Seasonal Migration of Subtropical Jet Stream

- ❑ During the **summer** months, the **jet stream over the Indian subcontinent** is positioned to the **north of the Himalayas**, around the **Tibetan Plateau**.
- ❑ This **northward shift** of the jet stream creates a **low-pressure** area over the Indian subcontinent.

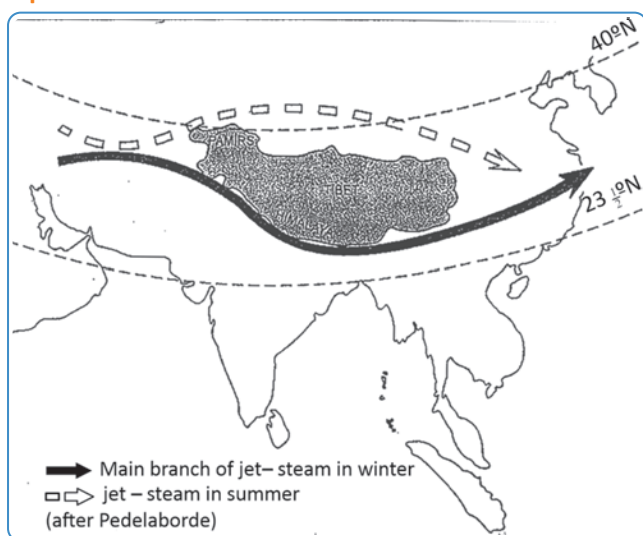


Fig. 5.4: Jet stream during summer

- ❑ **Warm air from the Indian Ocean** is drawn towards this low-pressure area, resulting in the **onset of the monsoon winds**.
 - The southwesterly winds carrying moisture-laden air from the Indian Ocean are known as the **southwest monsoon**.
 - These winds bring heavy rainfall to various parts of India, marking the onset of the monsoon season.
- ❑ As the **monsoon progresses**, the jet stream continues to influence the weather patterns.
- ❑ Its position and intensity **impact the distribution of rainfall** across different regions of the Indian subcontinent.

- ❑ Towards the **end of the monsoon season**, the jet stream starts to **shift southward**.
- ❑ This southward shift **weakens the low-pressure gradient** over the Indian subcontinent, leading to the **withdrawal of the monsoon**.
- ❑ In the **post-monsoon period**, the jet stream is positioned to the **south of the Himalayas**.
- ❑ This **southward shift** allows for the establishment of a **high-pressure area** over the Indian subcontinent.

Jet Stream in Winter

- ❑ **Winter Jet Stream and Subtropical Region:** Winter sees the presence of a rapid westerly Subtropical Jet (STJ) stream over the subtropical region.
- ❑ **Himalayan Influence on Jet Stream:** Unfortunately, the Himalayan peaks and Tibetan Plateau divide this jet stream into two distinct branches.

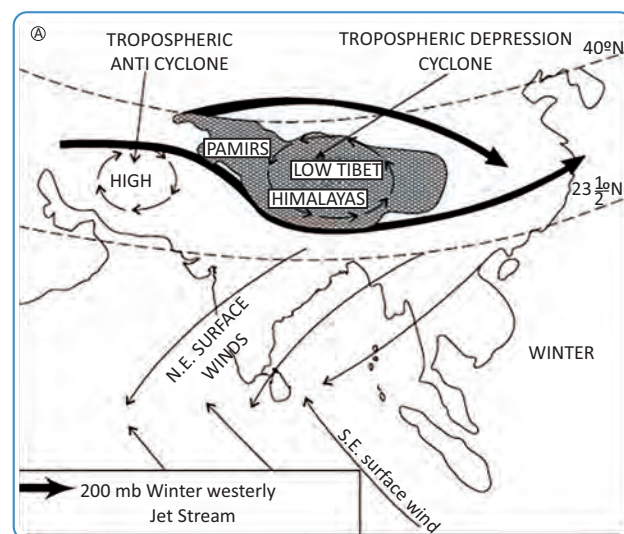


Fig. 5.5: Jet stream during winter

- ❑ **Northern Branch Impact on Tibetan Plateau:** The northern branch of the jet stream affects the northern border of the Tibetan Plateau.
- ❑ **Southern Branch Movement:** The southern branch of the jet stream moves towards the southern side of the Himalayan peaks.
- ❑ **Winter Weather Influence on India:** India's winter weather is significantly shaped by the southern branch of the jet stream.
- ❑ **Control over Western Disturbances:** This jet stream governs Western Disturbances, originating in the Mediterranean Sea and reaching North-West India.
- ❑ **High-Pressure Zone and Northeast Trade Winds:** The southern branch creates a high-pressure zone over North India.
 - **Result:** Strengthening of the northeast trade winds, sometimes termed as northeast monsoon winds.

The Jet Stream in Summer

- ❑ **Summer Jet Stream Dynamics:** In late May, the southern jet stream undergoes a split, moving north of Tibet.

❑ Monsoon Activation and STJ Weakening:

- Result: Significant increase in monsoon activity.
- Weakening of the southern branch of the subtropical jet stream (STJ).

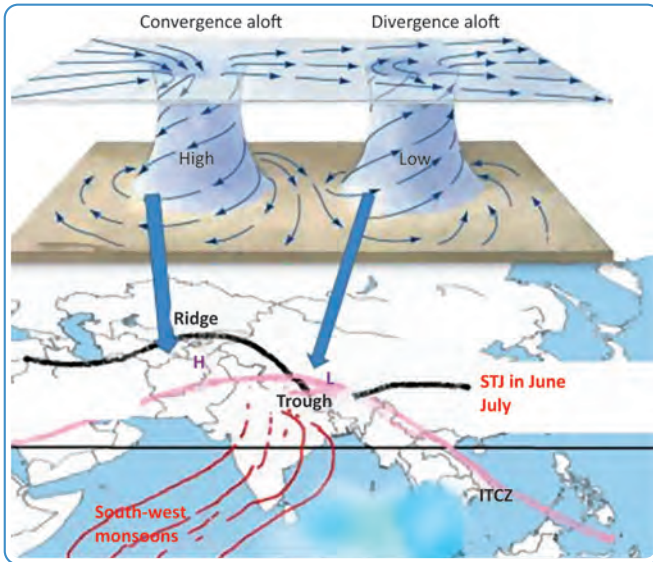


Fig. 5.6: Sub-tropical jet stream in summer

❑ ITCZ Movement and WJ Stream Departure:

By mid-July, the Intertropical Convergence Zone (ITCZ) moves northward, running parallel to the Himalayas (20° to 25° N).

- The westerly jet stream (WJ stream) or subtropical jet stream exits the Indian region during this period.

❑ Correlation with ITCZ and Jet Stream Withdrawal:

Meteorologists observe a correlation between the northward movement of the ITCZ and the withdrawal of the westerly jet stream from the North Indian Plain.

❑ STJ North Migration and TEJ (tropical easterly jet) Appearance:

During STJ's northward migration, the Tropical Easterly Jet (TEJ) emerges over peninsular India. It results in a burst of Indian monsoon.

❑ Anticyclonic Divergence and TEJ Formation:

Partially caused by anticyclonic divergence in the high atmosphere.

- Originates from a thermally produced low-pressure area above the Tibetan Plateau during summer.

❑ Westerly and Easterly Wind Relationship:

In the lower atmosphere, westerly winds are linked to easterly winds in the upper atmosphere.

❑ SW Monsoon Connection:

Across the Indian subcontinent, these westerly winds contribute to the south-west monsoon.

Role of Somali Jet

❑ Somali Jet is a transitory jet stream

that coexists with Tropical Easterly Jet. The Somali Jet is seen to originate from Mauritius in the summer and travels over Kenya, Somalia, and the Sahel before arriving on the coast of Kenya at roughly 3° S.

❑ It also aids in accelerating and intensifying the S-W monsoons

as they move towards India by strengthening the high-pressure area around Madagascar.

❑ It creates a break in the monsoon called the "lull"

when it is weak, and when it is strong, the monsoon will also be strong. It is much easier for the southwest monsoon to move towards India when the Somali jet moves through Kenya, Somalia, and the Sahel.

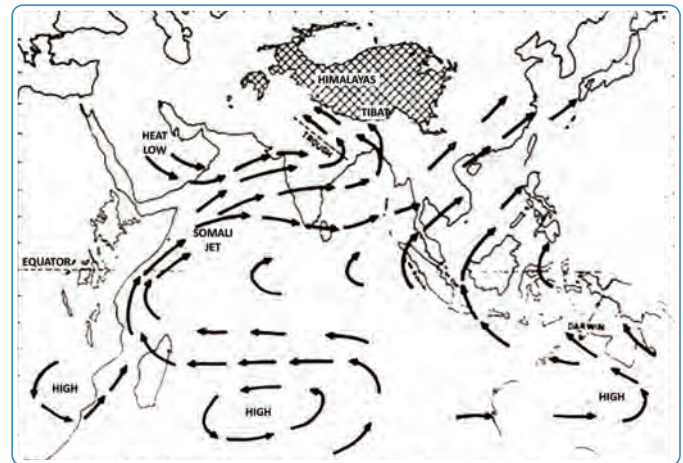


Fig. 5.7: The Summer Monsoon. Low level winds

❑ It makes the stable high near Madagascar stronger

and speeds up and intensifies the S-W monsoons as they move towards India.



IGNITE YOUR MIND

It would be interesting to note that the existence of Jet streams was confirmed during World War II when high-altitude pilots noticed strong tailwinds, revolutionizing aviation by enabling more efficient travel through the use of these high-speed air currents. Assess the importance of understanding jet streams in improving the accuracy of monsoon forecasting and also Discuss the challenges faced by meteorologists in predicting monsoon variability due to the influence of jet streams.

Tibet

❑ Tibetan Plateau Characteristics:

- **Extensive Dimensions:** Measures over 3000 km east-west and 1500 km north-south.
- **Considerable Altitude:** Typically stands at an elevation of 4,500 meters.

❑ Insolation Advantage:

Receives more insolation due to its substantial height.

❑ Influence on Atmospheric Circulation:

It is the key geographical factor shaping the overall atmospheric circulation in the region.

❑ **Dual Role as a Barrier and Heat Source:**

- Acts as a substantial physical barrier.
- Serves as a source of intense heat, impacting the atmosphere.

❑ **Jet Stream Movement:** Subtropical jet stream completely disappears across northern India in early June. It shifts to approximately 40° N, just north of the Himalayas and Tibet.

❑ **Seasonal Influence on Jet Stream:**

- In winter, extreme coldness of the Tibetan Plateau prompts the southward movement of the jet stream by mid-October.

❑ **Heat Engine Effect:** Tibet plateau functions as a high-level heat source during summer. This “Heat Engine” mechanism generates a thermal anticyclone over the plateau.

❑ **Anticyclone Formation:** During summer monsoon, a heated core anticyclone (high pressure) forms at 500 mb level in the troposphere and the Formation process is termed Anti-cyclogenesis.

Role of Walker Cell

The Walker circulation, commonly referred to as the Walker cell, is a theoretical representation of how air moves through the troposphere in the tropics. This model predicts that air packets move in a closed circulation both horizontally and vertically.

This circulation, which broadly agrees with observations, is brought on by variations in heat distribution between the ocean and the land. Gilbert Walker was the one who found it.

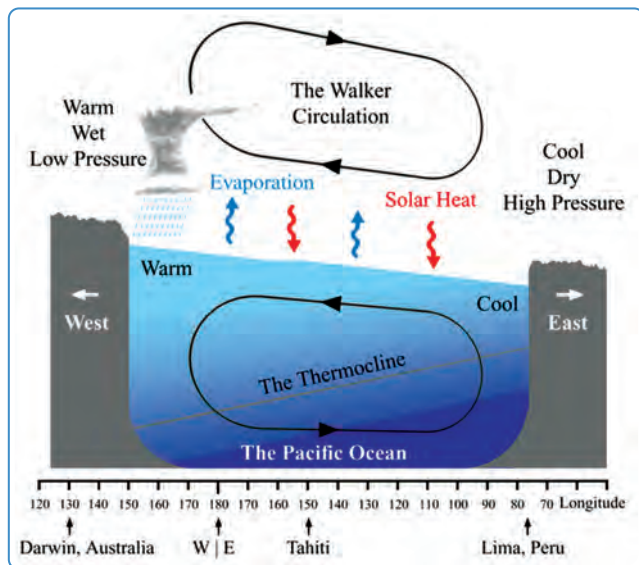


Fig. 5.8: Walker cycle

Impact on Indian Monsoon

- ❑ The Walker Cell is an atmospheric circulation pattern that plays a crucial role in influencing the Indian monsoon.
- Under normal conditions, the Walker Cell consists of two main branches—the east-to-west trade winds in

the lower atmosphere and the west-to-east return flow in the upper atmosphere.

❑ The **Walker Cell** is associated with **variations in sea surface temperatures (SSTs)** in the **tropical Pacific ocean**.

- In the **western Pacific**, warm SSTs create a **low-pressure area**, leading to **ascending air**.
- In the **eastern Pacific**, cooler SSTs result in a **high-pressure area**, causing **descending air**.

❑ The Walker Cell's influence on SSTs affects the Indian monsoon by modulating the atmospheric pressure and wind patterns.

❑ During **normal conditions**, the Walker Cell helps maintain a **low-pressure area over the Indian subcontinent**, promoting the inflow of moist air from the Indian Ocean.

❑ **La Niña**, the cold phase of the El Niño-Southern Oscillation (ENSO), **enhances the Walker Cell**. Cooler-than-average SSTs in the central and eastern Pacific lead to a **stronger Walker Cell**, contributing to **above-average rainfall in the Indian subcontinent**.

❑ **El Niño**, the warm phase of ENSO, **weakens the Walker Cell**. Warmer-than-average SSTs in the central and eastern Pacific **suppress the Walker Cell**, leading to **drier conditions in the Indian subcontinent**.

❑ The **Walker Cell's variations** influence the **onset and withdrawal** of the monsoon over India. During La Niña, the strengthened Walker Cell promotes an early and robust monsoon onset, while El Niño tends to delay the onset and weaken the monsoon.

El Nino

El Nino is a climate pattern that describes the unusual warming of surface waters in the eastern tropical Pacific Ocean. It is the “warm phase” of a larger phenomenon called the El Niño-Southern Oscillation (ENSO) climate pattern, which includes both El Niño and its counterpart, La Niña. El Niño events typically occur every 2 to 7 years and can have significant impacts on weather patterns around the world.

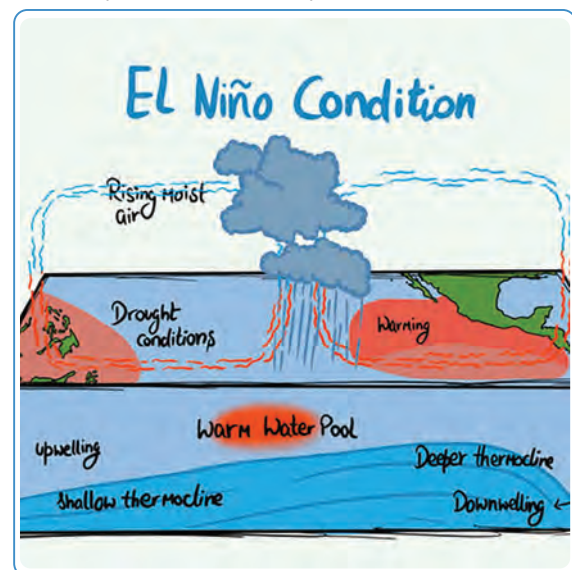


Fig. 5.9: El-Nino

Impact on Indian Monsoon

- ❑ El Niño is associated with a **delayed onset of the monsoon in India**.
- ❑ El Niño tends to **weaken** the normal southwest monsoon over the Indian subcontinent.
 - The warming of the central and eastern Pacific leads to changes in **Walker Cell circulation**, weakening the monsoon winds.
- ❑ The weakening of the monsoon winds results in **reduced rainfall** over many parts of the country.
- Parts of North India, including states like **Punjab** and **Haryana**, experienced **below-normal rainfall**, affecting the sowing and growth of crops.
- ❑ While El Niño typically leads to drier-than-average conditions in India, the impact can vary across regions.
- ❑ Some areas may experience more intense **drought** conditions, affecting agriculture and water resources.
 - Regions like **Marathwada** faced acute **water scarcity**, impacting agriculture and leading to distress conditions for farmers.

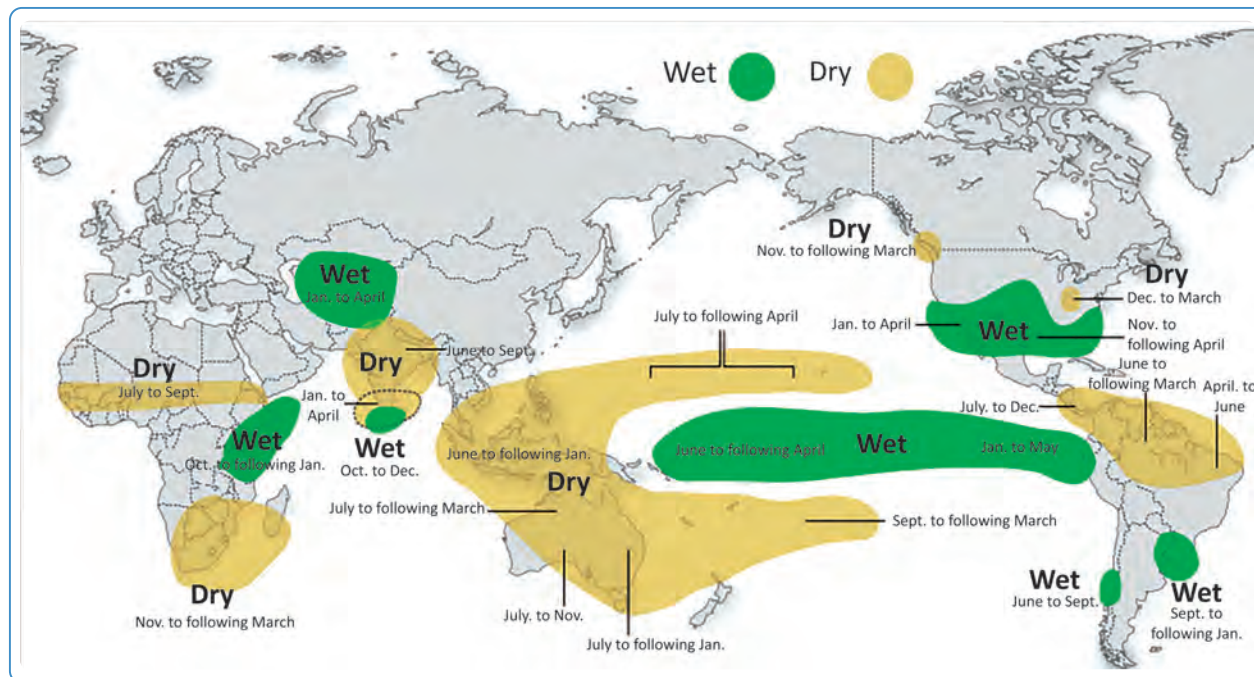


Fig. 5.10: El Niño and Rainfall

La-Nina

- ❑ In Spanish, La Nina means "**The Little Girl**." La Nina events is the period when the sea surface temperature in the central and western Pacific Oceans falls below normal.
- ❑ La Nina event is observed when the **water temperature in the Eastern Pacific gets comparatively colder than normal**, as a consequence of which, there is a **strong high pressure over the eastern equatorial Pacific**.
- ❑ La Niña often results in above-average sea surface temperatures in the western Indian Ocean and cooler temperatures in the eastern Indian Ocean.
- ❑ This configuration encourages the inflow of moist air over the Indian subcontinent, leading to enhanced monsoon rainfall.
- ❑ La Niña is associated with an early onset of the monsoon over India.

Impact on Indian Monsoon

- ❑ **La Niña** is associated with the **strengthening of the Walker Cell**, a part of the atmospheric circulation over the tropical Pacific Ocean.
- ❑ The Walker Cell involves the east-to-west trade winds in the lower atmosphere and the west-to-east return flow in the upper atmosphere.
- ❑ La Niña tends to enhance the positive phase of the Indian Ocean Dipole (IOD), a climate pattern characterized by temperature anomalies in the Indian Ocean.
- ❑ The combined influence of La Niña and a positive IOD usually leads to favorable conditions for the Indian monsoon.
- ❑ The strengthened Walker Cell and favorable conditions in the Indian Ocean contribute to a prolonged and robust monsoon season.
- ❑ The increased rainfall during La Niña events is generally beneficial for agriculture, promoting good soil moisture and supporting crop growth.

ENSO

- ❑ **ENSO's most influential factor is the Indian Monsoon**, the large-scale circulation pattern responsible for the majority of the Indian subcontinent's annual precipitation. India receives the majority of their annual precipitation from June to September.

- ❑ **The El Niño–Southern Oscillation (ENSO)** is an unpredictable variation in winds and sea surface temperatures in the tropical eastern Pacific Ocean. Every three to seven years, the surface waters of the tropical Pacific Ocean warm or chill by 1°C to 3°C.
- ❑ When the sea temperature rises, it is referred to as El Niño; when it declines, it is referred to as La Niña. As a result, El Niño and La Niña are opposite phases of the El Niño–Southern Oscillation (ENSO) cycle.
- ❑ These variations from normal surface temperatures can have a significant impact on global weather and climate, as well as oceanic circulation.

Effect of El-Niño and La-Niña

El Niño Effect on Indian Monsoon	La Niña Effect on Indian Monsoon
It is associated with weak monsoons and drought-like conditions.	It is associated with above-normal rainfall and flooding.
The equatorial Pacific Ocean's sea surface temperature will rise above typical levels.	The equatorial Pacific Ocean's sea surface temperature falls below average.
Modifications in atmospheric circulation patterns could be seen.	Modification in the patterns of atmospheric circulation could be seen.
The position of the jet stream has shifted, impacting the strength and direction of the monsoon winds.	Monsoon winds get stronger, bringing additional moisture and rains to India.
Less rainfall, dry spells, and heatwaves result in crop failures and water scarcity.	Heavy rainfall can also cause floods and landslides, inflicting agriculture and infrastructural damage.

Tele-Connections

- ❑ Variations in the manner in which air travels through the atmosphere produce teleconnection patterns. The duration of the changes could range from a few weeks to several months. Patterns of teleconnection are inherent. These may, however, change when global temperatures rise. El Niño, for instance, is a significant teleconnection pattern induced by atmospheric and oceanic El Niño–Southern Oscillation changes (ENSO).
- ❑ **During El Niño** events, cold and dry air is blown into normally warm and moist areas. As a result, the weather changes dramatically. For example, the amount of cloud cover in one region may impact precipitation elsewhere. Temperature changes at the ocean's surface have an effect on the weather in the tropical Pacific Ocean.



IGNITE YOUR MIND

Can you believe that scientists can trace the occurrence of El Niño events back thousands of years by studying natural records such as tree rings, ice cores, and sediment layers? Explore historical instances of significant El Niño events and their consequences on global weather patterns, including the Indian monsoon.

Indian Ocean Dipole

The Indian Ocean Dipole (IOD), sometimes known as the Indian Niño, is a phenomenon similar to El Niño that occurs in the Indian Ocean between the Indonesian and Malaysian coastlines in the east and the African coastline around Somalia in the west. In the Indian Ocean Dipole (IOD), a sea surface temperature fluctuation, rarely affects the northern or equatorial Indian Ocean Area.

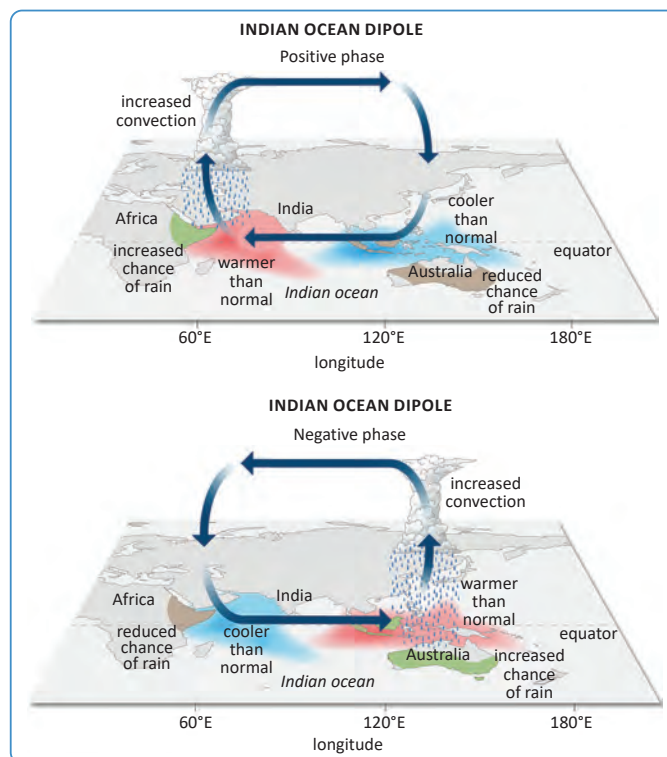


Fig. 5.11: The Indian Ocean Dipole (IOD)

Phases of IOD

- ❑ **Positive IOD:** When the western side of the Indian Ocean, along the Somalian Coast, becomes warmer than the eastern side of the Indian Ocean, along the Malaysian coastline.
 - Positive phase of the IOD alters the wind, temperature, and rainfall patterns in the region, typically bringing floods to eastern Africa and droughts and bushfires to eastern Asia and Australia.

- ❑ **Negative IOD:** The eastern side of the Indian Ocean gets warmer than the western side of the Indian Ocean.
 - Negative IOD events can also influence weather patterns, typically resulting in increased rainfall in regions that are usually dry and drier conditions in regions that are typically wet.

Impact on Indian Monsoon	
Positive IOD	Negative IOD
❑ Typically enhances the Indian monsoon. ❑ Warm waters in the western Indian Ocean lead to increased evaporation, providing additional moisture to the atmosphere. ❑ This moisture is transported to the Indian subcontinent, contributing to above-average rainfall during the monsoon season. ❑ A positive IOD is often associated with increased rainfall in central and northern India. ❑ Central and northern regions of India, including states like Maharashtra, Madhya Pradesh, and parts of North India, experienced above-average rainfall due to increased moisture transport from positive IOD during the monsoon season.	❑ Can have a suppressing effect on the Indian monsoon. ❑ Cooler waters in the western Indian Ocean reduce evaporation and moisture availability. ❑ This may lead to drier conditions and below-average rainfall in parts of the Indian subcontinent. ❑ A negative IOD may result in drier conditions, particularly in parts of southern India. ❑ Southern parts of India, including states like Karnataka, Tamil Nadu, and Kerala, faced drier-than-average conditions due to reduced moisture availability, from negative IOD during the monsoon.

Azores High

The Azores High, also referred to as the North Atlantic Subtropical High, is a subtropical high-pressure system that spans over western Europe and Eastern North Atlantic Ocean in the winter. It is a subtropical high-pressure cell, during summer it moves westward. When it falls, it is known as Bermuda High. It is created when dry air ascends from the subtropics and it occurs along the Hadley Circulation's downstream branch.

Recent Concept About the Origin of Indian Monsoon

- ❑ Over the past 50 years, extensive research on higher atmospheric circulation has challenged the traditional Indian monsoon origin theory. Recent hypotheses focus on the Tibetan Plateau, jet streams, and El-Nino, along with the upper atmospheric circulation, indicating that the monsoon circulation cannot be influenced by sea and land warming differently.
- ❑ The information obtained by meteorologists following World War II has shown that the following events are connected to the genesis and mechanism of monsoons:
 - The Himalayas and Tibetan Plateau function as a natural barrier and a source of intense heat.
 - The tropospheric circulation of higher air jet streams.
 - The presence of an upper air circum-polar swirl in the troposphere above the north and south poles.
 - The vast landmass of Asia, as well as the Indian and Pacific Oceans, are differentially heated and cooled.
 - El Nino's occurrence in the Indian and South Pacific Oceans.

FACTORS INFLUENCING INDIAN MONSOON

The Indian monsoon is a complex weather system influenced by a combination of oceanic, atmospheric, and geographical factors. These factors interact to determine the timing, intensity, and distribution of rainfall across the Indian subcontinent, making the monsoon a vital and dynamic aspect of India's climate. The Indian monsoon is influenced by the following factors:

- ❑ **Latitudinal Location:** The Tropic of Cancer passes through India, from the Rann of Kutch in the west to Mizoram in the east.
 - This means that about half of the country is in the tropical zone, south of the Tropic of Cancer, and the other half is in the subtropics, north of the Tropic of Cancer. As a result, the monsoon in India is both tropical and subtropical.
- ❑ **Distance From Sea:** Coastal areas of India have a temperate climate because of their long coastline. However, areas of India's interior are far from the sea's moderating influence, and therefore experience climate extremes.
 - As a result, residents of Mumbai and the Konkan coast have little understanding of temperature extremes and seasonal weather patterns, while weather patterns in the country's interior, such as Kanpur, Delhi, and Amritsar, have a significant impact on all aspects of life.
- ❑ **The Himalayas:** The Himalayas and their northern extensions function as an efficient climatic division. The high mountain range protects the subcontinent by acting as an unbreakable barrier against the cold northern winds.

- Monsoon winds are also trapped by the Himalayas, forcing them to do rainfall across the Indian subcontinent.
- ❑ **Physiography:** Temperature, air pressure, wind speed and direction, as well as the amount and distribution of precipitation, are all affected by India's geography or relief.
 - From June to September, the windward regions of the Western Ghats and Assam receive substantial rainfall, while the leeward side of the Western Ghats keeps the southern plateau barren.
- ❑ **Seasonal transition:** The sudden change in seasons is caused by a complete reversal of the monsoon winds, which bring most of the rainfall to the Indian subcontinent.
- ❑ **Upper Air Circulation:** The Subtropical Jetstream's southern branch is responsible for bringing western disturbances from the Mediterranean region into the Indian subcontinent.
 - The formation of the Tropical Easterly Jet Stream aids in the rapid onset of the southwest monsoon.

Western Disturbances

Western Disturbances (WDs) are **extratropical weather systems** originating in the **Mediterranean** region and traveling eastwards across northern regions of the Indian subcontinent. These disturbances play a crucial role in influencing weather patterns, particularly during the winter months.



Fig. 5.12: Western Disturbances

Key Features

- ❑ Western Disturbances typically form over the Mediterranean region and move eastward. They are associated with mid-latitude westerly winds and can extend into the upper troposphere.
- ❑ As WDs approach the Indian subcontinent, they bring changes in weather patterns. The most notable impact is seen during the winter months, contributing to precipitation, cloud cover, and variations in temperature.

- ❑ WDs are a significant source of winter rainfall in parts of northern India, particularly in the Himalayan region and the adjacent plains. This precipitation is important for winter crops and the overall water balance in the region.
- ❑ In mountainous areas like the Himalayas, Western Disturbances often bring snowfall. This is crucial for the replenishment of glaciers, rivers, and contributes to the availability of water resources during the dry season.
- ❑ WDs can lead to fluctuations in temperatures, especially when they interact with pre-existing air masses. The warm air ahead of the disturbance and the cold air behind it can result in varying temperatures in affected regions.
- ❑ Western Disturbances are associated with increased cloud cover, which can influence both daytime and nighttime temperatures. Cloud cover can also have implications for solar radiation reaching the surface.
 - **On Daytime Temperatures:** The reduction in solar radiation reaching the Earth's surface due to cloud cover results in a cooling effect leading to lower daytime temperatures compared to clear-sky conditions.
 - **On Nighttime Temperatures:** Increased cloud cover due to western disturbances act as a barrier to outgoing longwave radiation, trapping some of the Earth's heat, and preventing rapid cooling leading to higher nighttime temperatures.
- ❑ In the presence of clouds, the Earth's surface radiates less heat into space, contributing to relatively warmer nighttime temperatures compared to clear nights.
- ❑ WDs often induce cyclonic circulation in the region where they influence the weather. This circulation contributes to the development of weather systems and precipitation.
- ❑ The interaction of Western Disturbances with the Indian monsoon and other atmospheric features can lead to complex weather scenarios, including the formation of cyclonic storms and the modulation of monsoonal patterns.

Monsoon Expedition

- ❑ **Monex (Monsoon Expedition)** was founded in **1973** as a collaboration between India and the Soviet Union. It was organised to conduct research during the Indian monsoons. In 1973, an Indo-Soviet joint expedition was conducted to better understand the causes and mechanisms of the monsoon. Four Soviet and two Indian ships were outfitted with scientific

tools for meteorological observations in the Indian Ocean, Bay of Bengal, and Arabian Sea. According to Dr. P. Koteswaram and Professor H. Flohn, the Tibetan Plateau is a source of heat for the upper atmosphere.

- ❑ Rising air motion is caused by rising temperatures at higher altitudes. During the summer, the Tibetan plateau is 20 - 30 degrees Celsius warmer than the surrounding areas. The low pressure created on the Tibetan Plateau pushes the westerly jet northward. It corresponds to the arrival of the Indian summer monsoon. The prevailing easterly winds over south India are strengthened by the southward blow of air from Tibet. It explains why the easterly jet appears around the same time as the Indian monsoon.
- ❑ There is a correlation between the shifting of the jet and the slowing of the westerly jet across Eurasia. In contrast, the Tibetan Plateau becomes very cold during the winter and is the most important factor in advancing the westerly jet to the south by mid-October. It contributes to the formation of cold waves and western disturbances over the north Indian plains, resulting in high pressure over the land surface and low pressure over the Indian Ocean. These meteorological conditions contribute to a favourable monsoon.

India Meteorological Department (IMD)

The India Meteorological Department was established in 1875 which is the country's main meteorological service and the primary government agency responsible for dealing with meteorology, seismology, and related topics. The Ministry of Earth Sciences of the Indian Government is in charge of the Department's administrative tasks. The IMD's main office is in New Delhi. Categories of Rainfall according to IMD:

- ❑ **Large Excess:** $\geq 60\%$ of long period average (LPA).
- ❑ **Excess:** 20% to 59% of long period average (LPA).
- ❑ **Normal:** -19% to +19% of the long-period average (LPA).
- ❑ **Deficient:** -59% to -20% of long period average (LPA).
- ❑ **Large Deficient:** -99% to -60% of long period average (LPA).

Long Period Average (LPA): LPA of rainfall is the rainfall recorded over a particular region for a given interval (like month or season) average over a long period like 30 years, 50 years etc. September. When predicting the quantitative rainfall during the monsoon season each year, it serves as a standard.

NATURE AND IMPORTANT ASPECTS OF MONSOON

Monsoon is a complex climatic phenomenon whereby a single theory hasn't been able to explain the exact nature and causation of monsoon fully. However, systematic studies of the causes of rainfall in the South Asian region help to understand the causes and salient features of the monsoon, particularly some of its important aspects, such as:

- ❑ The onset of the monsoon.
- ❑ Rain-bearing systems (e.g. tropical cyclones) and the relationship between their frequency and distribution of monsoon rainfall.
- ❑ Break in the monsoon.

The Onset of the Monsoon

The onset of monsoon rains is referred to in monsoon literature as the monsoon burst, which usually takes the form of a large thunderstorm.

Mechanism of Onset

- ❑ During April and May, there is an intense **low pressure** in the northwestern part of the subcontinent caused by the vertical heating of the Sun over the **Tropic of Cancer**.
- ❑ This low pressure cell attracts the south-east trade winds from the high pressure region of Indian Ocean across the equator.
 - These winds cross the Equator between **40°E and 60°E** longitudes.
- ❑ After crossing the equator these south-east trade winds are deflected as **south-west monsoon winds** towards the Indian subcontinent due to the **Coriolis effect**.
- ❑ These conditions help in the **northward shift in the position of the ITCZ**.
- ❑ The shift in the position of the ITCZ is also related to the phenomenon of the **withdrawal of the westerly jet stream** from its position over the north Indian plain, south of the Himalayas (**Jet Stream Theory**).
- ❑ After the withdrawal of the westerly jet stream, the **easterly jet stream** sets in along **15°N latitude**.
 - This easterly jet stream is responsible for the **onset of monsoon**, also called the '**monsoon burst**' in India.

{The Southwest monsoon normally sets in over Kerala around 1st June. It advances northwards, usually in surges, and covers the entire country around 15th July.}

Rain Bearing Systems and Rainfall Distribution

- ❑ The monsoon approaches the landmass through two channels:
 - the **Arabian Sea channel and the Bay of Bengal channel**.

- The rains begin in the Bay of Bengal and spread across the plains of northern India. The second is the Arabian Sea current of the southwest monsoon, which brings rain to India's west coast.

❑ **Monsoon Winds of the Arabian Sea:** The monsoon winds that originate over the Arabian Sea are further subdivided into three branches:

- **Western Ghats:** The Western Ghats mountain range blocks one of the branches of the monsoon winds. As the winds ascend the 900-1200 meter slopes of the Western Ghats, they cool rapidly, leading to extremely heavy rainfall (250-400 cm) on the windward side of the mountains and the Western Coastal Plain. After crossing the Western Ghats, the winds descend and

heat up, resulting in much less rainfall on the leeward side of the mountains.

- **Arabian Sea Branch:** Another branch of the Arabian Sea monsoon strikes the coast north of Mumbai. These winds, which blow along the Narmada and Tapi River basins, provide rain to most of central India. This segment of the branch brings 15 cm of rain to the Chotanagpur plateau. They subsequently enter the Ganga plains and join with a branch of the Gulf of Bengal.
- **Third Branch:** The third branch of the monsoon wind strikes the Saurashtra Peninsula and the Kutch region. It then joins a branch of the Bay of Bengal in Punjab and Haryana. When these two branches merge, they bring rain to the western Himalayas.

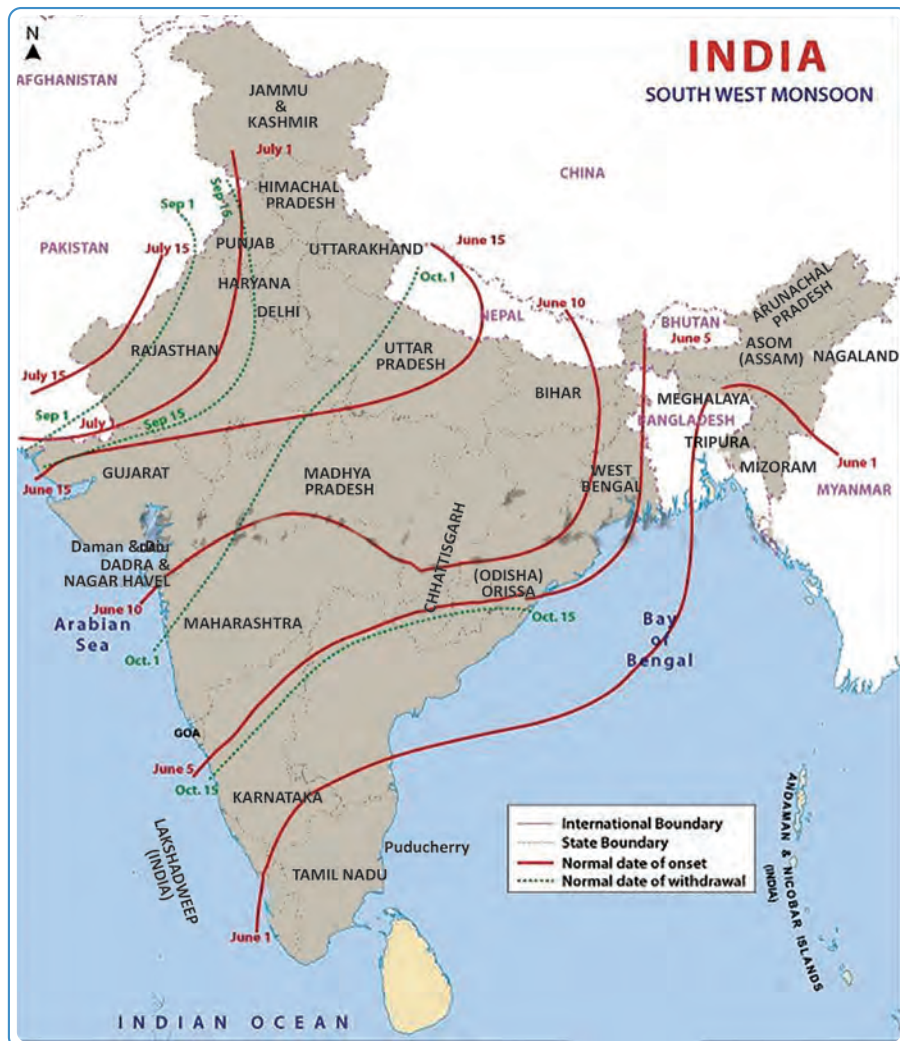


Fig. 5.13: India South West Monsoon

❑ **Monsoon Winds of the Bay of Bengal:** The Arakan Hills, a mountain range along the coast of Myanmar, deflect a large portion of the Bay of Bengal branch of the monsoon winds towards the Indian subcontinent.

- As a result, the monsoon winds enter West Bengal and Bangladesh from the south and southeast, instead of from the southwest.

- Under the influence of the Himalayas and a thermal low in northwest India, this branch of the monsoon winds splits into two.
- The first branch continues to follow the coasts of Myanmar and southeast Bangladesh, while the second branch travels westward across the Gangetic Plain to the Punjab region of India.

Break in the Monsoon (July to August)

- ❑ After a few days of rain during the southwest monsoon, a break in the monsoon occurs when it does not rain for one or more weeks. During the rainy season, these dry spells are quite common. These regional breaks are caused by a variety of factors, including:-
 - Rainfall in northern India is likely to fail if rain-bearing storms do not occur frequently along the monsoon trough or the ITCZ over this region.
 - Days when winds blow parallel to the coast along the west coast are connected with dry spells.

Retreat in Monsoon

- ❑ Retreating monsoon season commences with the beginning of the withdrawal of the south-west monsoon [mid-September – November] and lasts till early January.
- ❑ It is a 3-month long process where it starts from the peninsula in October and from the extreme south-eastern tip by December. The south-west monsoons withdraw from the Coromandel coast in mid-December. In Punjab, the south-west monsoons withdraw from there in the second week of September.
- ❑ The withdrawal of the retreating monsoon is gradual and takes a longer duration compared to the advancing one.
- ❑ The end of the southwest monsoon season is indicated by rising temperatures and clear sky. The earth is still moist. The weather may be extremely uncomfortable due to the high temperature and humidity. This is usually referred to as the “**October heat.**” Temperatures begin to fall sharply in the second half of October, particularly in northern India.

Weather During the Monsoon’s Retreat

- ❑ The weather during the monsoon’s retreat is arid in northern India but rainy in the eastern peninsula. In this region, the wettest months are October and November.
- ❑ This season’s extensive rainfall is caused by cyclonic depressions that originate in the Andaman Sea and traverse the southern Peninsula’s eastern coastline. These tropical cyclones pose a grave threat. Their preferred targets are the heavily populated deltas of the Godavari, Krishna, and Kaveri rivers. Each year, cyclones wreak devastation on this region. Several cyclones also strike Bangladesh, West Bengal, and Myanmar. These depressions and cyclones are responsible for the preponderance of Coromandel’s rainfall.



IGNITE YOUR MIND

The break in the monsoon, with its interplay of light and shadow, has inspired poets and writers. It often serves as a metaphor in literature, symbolizing moments of reflection, change, or respite amidst life’s challenges. Compare and contrast the break in the monsoon in the Indian subcontinent with similar phenomena in other parts of the world. Consider regional variations and their implications.

SIGNIFICANCE OF MONSOONAL CLIMATE

The monsoonal climate is of significant importance for regions that experience it, such as South Asia and parts of Southeast Asia.

- ❑ **Agriculture and Food Security:** The monsoon is a lifeline for agriculture in many regions.
 - The seasonal reversal of winds and the accompanying rainfall provide much-needed moisture for crops.
 - Farmers rely on the monsoon to irrigate their fields and cultivate various crops like rice, wheat, and sugarcane.
 - The timely arrival and distribution of monsoon rains are crucial for food production and, consequently, food security for millions of people.
- ❑ **Economic Impact:** The monsoon has a significant economic impact in affected areas. Agriculture forms a substantial part of the economy in countries like India, and a successful monsoon season can lead to higher agricultural output, increased income for farmers, and overall economic growth. Conversely, a weak or delayed monsoon can lead to crop failures, food shortages, and economic challenges.
- ❑ **Water Resources:** Water resources such as rivers, lakes, and reservoirs are restored by monsoon rainfall.
 - This water is not only vital for agricultural irrigation but also for drinking water supply and industrial use.
 - Adequate monsoon rainfall ensures the availability of water resources throughout the year.
- ❑ **Biodiversity and Ecosystems:** Monsoons play a crucial role in maintaining biodiversity and supporting ecosystems.
 - They provide the necessary moisture for forests, wetlands, and other natural habitats.
 - Many species of plants and animals have adapted their life cycles to the monsoonal patterns, and the monsoon contributes to the overall health and balance of these ecosystems.
- ❑ **Hydroelectric Power Generation:** The monsoon’s contribution to river flow is essential for hydroelectric power generation.
 - The increased water flow during the monsoon season allows for the efficient generation of electricity through hydroelectric dams.
 - This contributes to the energy needs of the region and reduces reliance on fossil fuels.

MONSOON PREDICTION AND FORECASTING IN INDIA

The monsoon 2023 forecast for the month predicts that 32 of the 36 meteorological subdivisions will receive less rain than usual. The rainfall total was expected to be 20 millimetres less than the 1996-2013 average. Prediction of Monsoon in India is very difficult for the meteorological department.

Here are some factors which are responsible for predicting the monsoon in India.

- ❑ **Complex Weather Systems:** The Indian monsoon is impacted by a number of atmospheric and oceanic elements, including the Indian Ocean, El Nino/La Nina occurrences, the Tibetan Plateau, and the Himalayan mountain range. These interrelated processes make precise forecasting of monsoon patterns challenging.
- ❑ **Variability:** The Indian monsoon differs greatly from year to year. Natural climate variability, such as the El Nino Southern Oscillation (ENSO), can create droughts or heavy rains in certain parts of India. Because of this unpredictability, making long-term projections is difficult.
- ❑ **Spatial Variability:** Because of India's unique geography, the monsoon does not come consistently across the nation. Various regions have diverse rainfall patterns and timings, making it difficult to correctly forecast local implications.
- ❑ **Timing:** Forecasting the start and end of the monsoon is a huge task. The monsoon season usually starts in June and ends in September, however, the precise time varies from year to year and area to region.
- ❑ **Complicated Modelling:** Monsoon forecast requires complex computer models that replicate the interplay of various factors. These models need a massive quantity of data and computational power, and even minor inaccuracies in input data might result in incorrect predictions.
- ❑ **Climate Change:** Climate change is changing traditional monsoon patterns, making forecasting even more difficult. Rising global temperatures have the potential to destabilise existing climate systems and cause unanticipated monsoon behaviour.
- ❑ **Local Factors:** Land-use changes, deforestation, and urbanisation may all have an impact on local weather patterns, making it difficult to forecast how these variables will interact with the monsoon.

Despite these challenges, meteorologists and scientists continue to improve their understanding of the Indian monsoon and develop better prediction models. Advances in technology and increased data collection are helping to enhance the accuracy of monsoon forecasts, **The prediction is:**

- ❑ **Normal to below normal rainfall:** Rainfall is expected to be 96% of the long-term average from June to September. Normal to below-normal rainfall in parts of northwestern India, west-central India, and pockets of northeast India.
- ❑ **El Nino Effect:** El Nino is the primary cause of this year's poor rainfall. Due to the effect of La Nina, India has been receiving heavy monsoon rains since 2019. El Nino, caused by warm water in the equatorial Pacific, has the potential to affect worldwide weather patterns. El Nino has been associated with a drier climate as well as less rainfall in India.

- ❑ **Current La Nina:** Over the equatorial Pacific, the current La Nina conditions, which are typically favourable for the monsoon, have become neutral.
- ❑ **Predicting the Monsoon:** The monsoon season in India runs from June to September, with an LPA of 88 cm and a standard deviation of 9 cm (approximately 10% of the mean value). When the average rainfall for the entire nation falls within 10% of its LPA or between 90% and 110% of its LPA, it is considered "normal," when it falls below 90% of its LPA, it is considered "below normal," and when it exceeds 110% of its LPA, it is considered "above normal."

Karakoram Anomaly

The Karakoram Anomaly is a phenomenon in which the glaciers in the central Karakoram mountains have remained stable or even grown, while glaciers in other nearby mountain ranges, such as the Himalayas and other mountains around the world, have been retreating. Himalayan glaciers are extremely important in India, particularly for people living downstream. They rely on perennial rivers for their daily water supply. However, as a result of global warming and climate change, the bodies of water are rapidly disappearing. Glaciers in the central Karakoram, on the other hand, have not changed. They have increased slightly over the last few decades. This phenomenon is limited to a small geographical area. Aside from the Karakoram range, similar or comparable phenomena can be observed in the Kunlun ranges.

INITIATIVES TAKEN FOR WEATHER FORECASTING

- ❑ Forecasts and warnings are sent via email, WhatsApp groups, social media, and websites.
- ❑ Registered users are also sent SMS alerts, and press releases are issued as needed.
- ❑ In 2020, IMD launched seven services via the '**UMANG**' mobile app, providing the public with information on various weather aspects.
- ❑ For weather forecasting, agromet advisory, and lightning alerts, mobile apps such as '**MAUSAM**,' '**Meghdoot**,' and '**Damini**' were developed.
- ❑ To address meteorological events that cause widespread damage, IMD developed a web-based Climate Hazard & Vulnerability Atlas.
- ❑ Through the Central Sector Scheme "**atmosphere & Climate Research-modelling Observing Systems & Services (ACROSS)**," IMD is modernising and expanding its services.
- ❑ ACROSS sub-schemes include **Atmospheric Observation Network (AON)**, Forecast System Upgrade (UFS), Weather & Climate Services (WCS), and Polarimetric Doppler Weather Radar Commissioning (PDWR).

- ❑ Commissioning of **Automated Weather Observing Systems (AWOS/HAWOS)** or Heliport Automated Weather Observing Systems (HAWOS), establishment and improvement of Meteorological Centres (MCs), and other facilities throughout the region.

Long Range Forecast (LRF)

The India Meteorological Department (IMD) issues the operational long-range forecast (LRF) for the southwest monsoon seasonal (June-September) rainfall averaged across the entire country in two stages. In its most recent Long Range Forecast (LRF), the India Meteorological Department (IMD) forecasted that the monsoon will be ordinary this year. The Long Period Average (LPA) of Rainfall is the Amount of Rainfall Recorded Across a Particular Region for a Particular Period (such as a Month or Season) Averaged over a Long Period (like 30 Years, 50 Years, etc.). It functions as a benchmark for predicting the amount of precipitation for that region during a particular month or season. It is customary to validate the seasonal rainfall every ten years.

SEASONS OF INDIA

Seasons are periods of the year that have unique weather patterns and day lengths. Seasons happen because of the Earth's axis is tilted at an angle of around 23.4 degrees away from vertical, and different portions of the Earth receive more solar radiation than others. A distinct seasonal pattern distinguishes the monsoon climate. The weather varies significantly from season to season. These changes are most visible in the country's interior. The temperature does not vary much along the coast, but the rainfall pattern does. The Indian Meteorological Department divides India's seasons into four categories:

A. Seasons of North-East Monsoon

1. Cold Weather Season (Winter season), mid-December to mid-March
2. Hot weather season, mid-March to May

B. Seasons of South-West Monsoon

3. Rainy season, June to September
4. Season of retreating monsoon, October to mid-December

COLD WEATHER SEASON

Duration

- ❑ The cold weather season in the majority of India begins in late November in the north and by early December in the rest of the country. December and January are the coldest months in northern India. The days are warm during the cold season, but the nights are cold.
- ❑ The flow of cyclonic disturbances from the west and northwest defines characteristics of the cold weather

season in the northern plains. With the westward flow, these low-pressure systems form over the Mediterranean Sea and western Asia and then move to India.

- ❑ They bring freezing rain to the mountains and much-needed rain to the plains in the winter. Even though the total proportion of winter rain, known locally as "**Mahawat**," is small, it is essential for growing "rabi" crops.
- ❑ The peninsular area is devoid of a clear cold season. Because the sea has a calming effect, winter temperatures don't change much from one season to the next.

Temperature

- ❑ From south to north, the temperature decreases. The average temperature in Chennai, which is on the east coast, is between 24° and 25° Celsius, while the average temperature in the northern plains is between 10° and 15° Celsius.
- ❑ During the day, it's warm, but at night, it's cold. Frost happens often in the north, and it snows on the higher Himalayan slopes.
- ❑ In January, the mean temperature in the northwestern Great Plains of India (Punjab, Haryana, western Uttar Pradesh, and Rajasthan) is less than 15°C.
- ❑ The temperature occasionally drops below freezing at night in the plains of Punjab, Haryana, and Rajasthan (Amritsar, Hissar, and Jodhpur), which causes ground frost.

Rainfall

- ❑ The cold weather precipitation, while minor, is extremely beneficial to Rabi crops. Furthermore, snowfall from western disturbances feeds the Western Himalayan glaciers. During the winter season, some rain falls in the northeastern parts of India.
- ❑ During these months, Arunachal Pradesh, Nagaland, and Assam may receive up to 20 cm of rain. India's rainfall is unevenly distributed.
- ❑ The heaviest rainfall occurs along the southeastern coast of Tamil Nadu and adjacent areas of Andhra Pradesh and gradually decreases from there.
- ❑ Northwest India also receives rain from western disturbances, but the amount decreases gradually from north and northwest to east. During the winter, rain falls in the northeastern part of India.

Pressure in the Winter Season

- ❑ Due to low temperatures and the STJ (Sub Tropical Jet Streams) ridge, which causes divergence, the air pressure is high over a large part of northwest India.
- ❑ The pressure in South India is lower than in the rest of the country.
- ❑ The winds start to move from a high-pressure area in the northwest to a low-pressure area in the southeast.

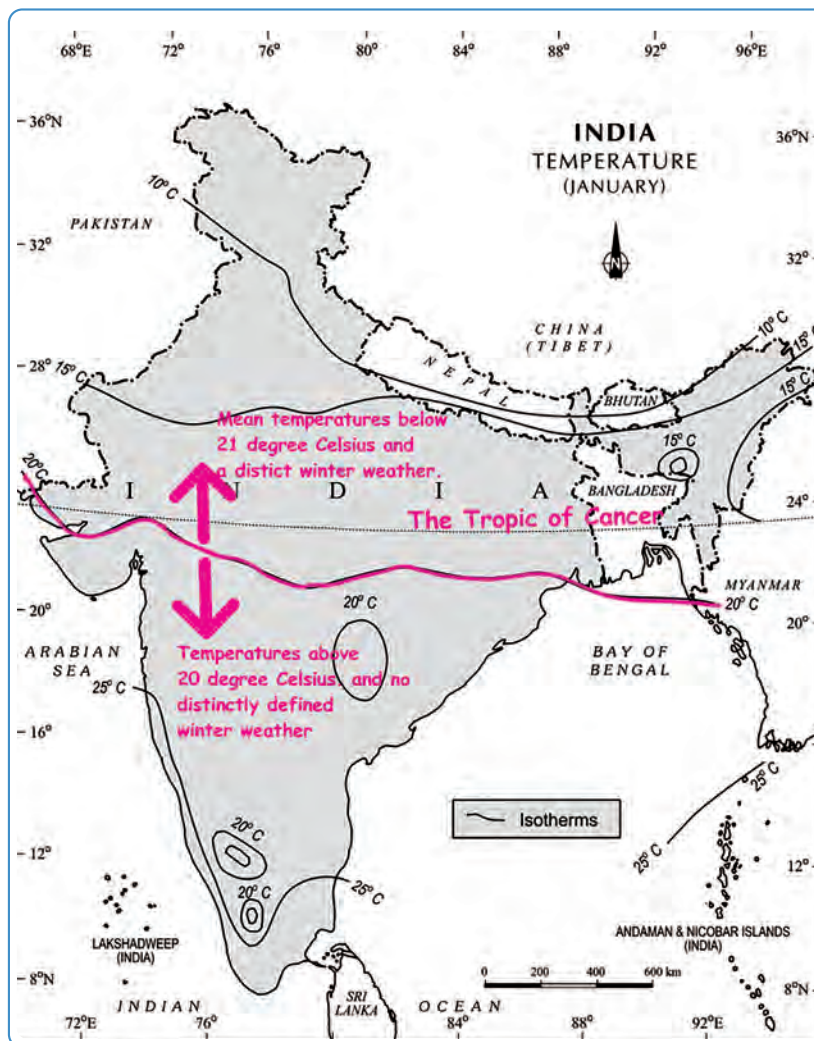


Fig. 5.14: India Temperature (January)

Reasons for no South-West Monsoons During Winter

❑ Reversal of Wind Direction:

- During winter, there is a reversal in the wind direction.
- Instead of the southwest, the prevailing wind pattern becomes northeast, commonly known as the northeast monsoon.
- Hence the north-east monsoon becomes **onshore winds** that carry less or no moisture.

❑ Shift in the Inter-Tropical Convergence Zone (ITCZ):

- The Inter-Tropical Convergence Zone (ITCZ) is a belt of low pressure near the equator, where the northeast and southeast trade winds converge.
- In winter, the ITCZ and monsoon trough shifts southward, causing the retreat of the southwest monsoon and the onset of the northeast monsoon.

❑ Cooling of the Indian Ocean and high-pressure system over Indian subcontinent:

- During winter, the northern hemisphere, including the Indian subcontinent, experiences cooling temperatures.

- This cooling **reduces the temperature contrast** between the land and the ocean, **diminishing the driving force** for the southwest monsoon winds.
- This leads to development of a **high-pressure system** over the Indian subcontinent.
- High-pressure systems are associated with descending air, which inhibits the ascent of moist air and the development of monsoon rainfall.



IGNITE YOUR MIND

As the severe grasp of winter tightens its hold on Kashmir, the region prepares to enter the formidable 40-day phase known as Chilla-i-Kalan, which commenced recently. In this light can you discuss the specific impacts of climate change on the winter conditions in the Himalayan region and the consequences for the ecosystem and communities?

HOT WEATHER SEASON

Duration

- ❑ The north Indian region experiences a distinct hot weather season from mid-March to mid-June.
- ❑ The hot season in southern and western India will last from April until the start of May. In the northern parts of India, May is the warmest month.

General Characteristics of Weather Conditions

- ❑ It is characterised by unstable air pressure and wind circulation as a transitional season between the northeast and southwest monsoons (rainy season).
- ❑ The low-pressure area moves from southeast to northwest as the sun moves northward. It eventually settles over northwestern India at the end of May or the beginning of June.
- ❑ In the neighbouring sea, pressure generally rises towards the south. Winds blow from the northwest and west in northwestern India and from the southwest in the Arabian Sea and its surrounding coasts.
- ❑ In Punjab, Haryana, and western part of Uttar Pradesh, there are dust storms that look like tornadoes. In western India, there are hot winds called “**Loo**,” and in West Bengal, they are called “**Kalbaisakhis**”.
- ❑ The sun is scorching during the summer, and the relative humidity is generally less than 30%, occasionally falling below 10%.

Temperature

- ❑ The temperature starts rising after the Vernal Equinox as the sun moves northward towards the Tropic of Cancer.
- ❑ The temperature starts to rise sharply as March approaches. By April, the Peninsular regions south of the Vindhyan Range have reached 40°C on average.
- ❑ In May, the average maximum temperature in Rajasthan, Delhi, west Uttar Pradesh, south Punjab, Jammu and Haryana is 42°C. Daytime temperatures in some parts of India, particularly in northwestern India, can reach 45°C or 47°C. In May, most of the interior is around 32–40 °C (90–104 °F) on average.
- ❑ The average minimum daily temperature in May remains quite high, at around 26°C in Delhi and Jaipur.

Vernal Equinox

The phenomenon occurs when the Sun is directly above the Equator and day and night are the same length, this is called the vernal equinox. The Vernal equinox is one of the two times each year when the Sun crosses the celestial equator, which is the imaginary line in the sky that divides the Northern and Southern Hemispheres. In the

and March 21. In the Southern Hemisphere, it happens between September 22 and September 23. Astronomically, the Vernal equinox marks the start of spring. This is because the Sun is now shining directly on the equator, and the days and nights are of equal length. After the vernal equinox, the days start to get longer in the Northern Hemisphere and shorter in the Southern Hemisphere.

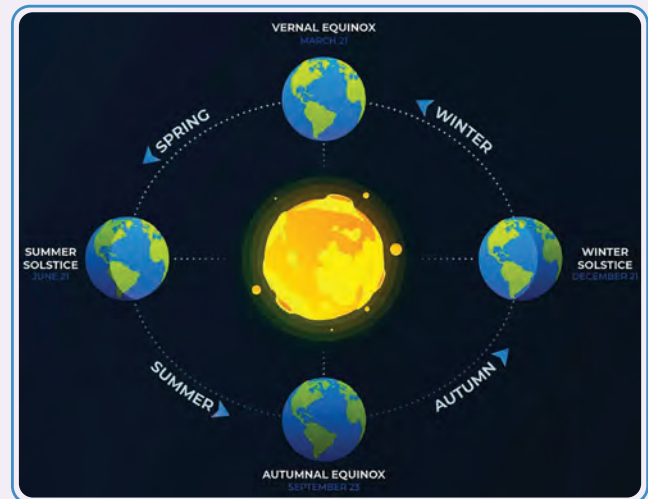


Fig. 5.15: Equinox and Solstice

Heat Wave

In India, heat waves generally occur from March to June, but they can sometimes last until July. Every year, five to six heat wave events occur in the northern areas of the nation.

Criteria for Declaring Heat Wave in India

A heat wave is declared when the highest temperature at a station is 40 degrees Celsius or higher for Plains and 30 degrees Celsius or higher for Hilly areas.

❑ Based on Departure from Normal

- **Heat Wave:** 4.5°C to 6.4°C above or below normal.
- **Severe Heat Wave:** Temperature is more than 6.4°C above normal.

❑ Based on Actual Maximum Temperature:

- **Heat Wave:** When actual maximum temperature $\geq 45^{\circ}\text{C}$.
- **Severe Heat Wave:** When actual maximum temperature $\geq 47^{\circ}\text{C}$.

If the above requirements are fulfilled at least in two stations in a Meteorological subdivision for at least two days in a row, and it is declared on the second day.

Rainfall

- ❑ The season's total rainfall is less than 2 cm in Rajasthan, Gujarat, and Madhya Pradesh; between 5 and 15 cm in the sub-montane region of Uttarakhand, Himachal Pradesh, Punjab, Uttar Pradesh, Bihar, and Odisha; between 15 and 25 cm along the Malabar coast; and more than 50 cm in Assam, Meghalaya, Mizoram, and Nagaland.
- ❑ Tropical depressions are directed towards India by the easterly jet stream. These depressions have an impact on the monsoon rainfall distribution over the Indian subcontinent.

Pressure in Summer Season

- ❑ All over the country, the air pressure is low due to the hot weather.
- ❑ Strong disparity caused by changes in the atmosphere, on the other hand, stops the southwest monsoons from starting over northwest India.

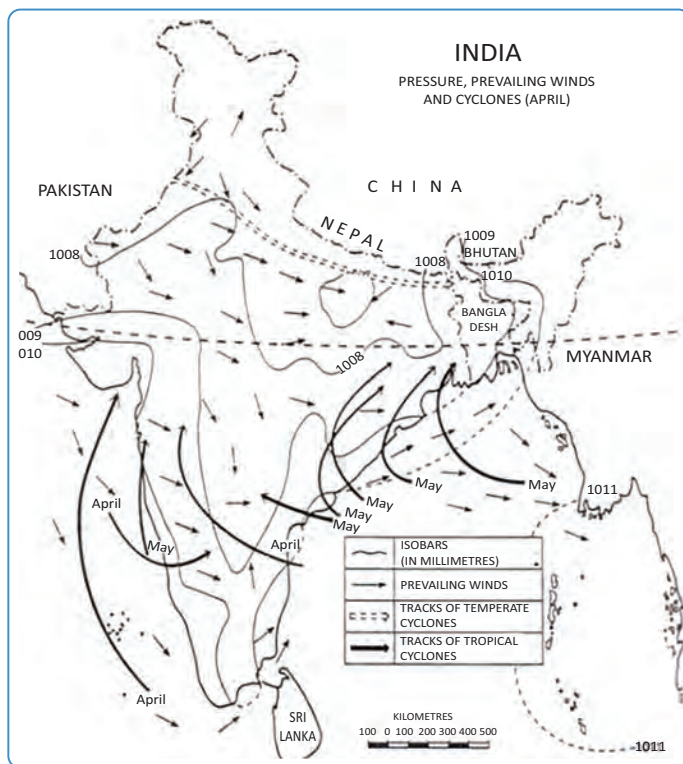


Fig. 5.16: India: Pressure, Prevailing Winds and Cyclones in January

Winds Distribution

- ❑ Because of the high temperatures, atmospheric pressure is low throughout the country. The Inter Tropical Convergence Zone had formed by the middle of July (ITCZ) which moved northward, nearly parallel to the Himalayas between 20° N and 25° N.
- ❑ At this period, the westerly jet stream had left the Indian subcontinent. The ITCZ pulls winds from all directions since it is a low-pressure region.

- ❑ The maritime tropical airmass (mT) from the southern hemisphere flows south-westerly to the low-pressure region after crossing from the equator. People often call this flow of moist air the “southwest monsoon.”

Frontal Thunderstorms in Summer Season

- ❑ Thunderstorms occur in the east and north-east of the country since the westerly jet stream causes strong convectational movements.
- ❑ Most of the time, they start over the Chota Nagpur plateau and move east on winds that come from the west.
- ❑ Thunderstorms are most common in Assam, Nagaland, Mizoram, Arunachal Pradesh, Manipur, Tripura, West Bengal, Meghalaya, and adjoining Odisha and Jharkhand.

Summer Season is also the season for isolated thunderstorms with strong winds and torrential downpours, which are frequently accompanied by hail. These storms are known as the ‘Kaal Baisakhi’ in West Bengal.

Some Famous Local Storms of Hot Weather Season

- ❑ **Mango Shower:** Pre-monsoon showers are short-lived thunderstorms that occur in the Indian subcontinent before the start of the monsoon season. They typically occur between April and May and are most common in Kerala and coastal Karnataka. Mango showers get their name because they help mangoes ripen early.
- ❑ **Blossom Shower:** Coffee flowers bloom in Kerala and surrounding areas as a result of this rain.
- ❑ **Nor Westers:** In Bengal and Assam, evening thunderstorms are expected. Its popularity comes from the local name ‘Kalbaisakhi,’ which refers to a calamity that occurs during the month of Baisakh. These rains are beneficial to tea, jute, and rice growth. In Assam, these storms are known as “Bardoisila.”
- ❑ **Loo:** Winds are hot, dry, and oppressive in the Northern Plains from Punjab to Bihar, with greater intensity between Patna and Delhi.

RAINY SEASON (SOUTHWEST AND NORTH-EAST MONSOON SEASON)

South-West Monsoon Season (Monsoonal Rainfall)

Duration

- ❑ It will last from June to September. The majority of the rain falls during this season, but monsoons are frequently delayed. The distribution of rainfall varies spatially on a large scale.

- ❑ The duration of the monsoon decreases from southern India to northern India.

General Characteristics

- ❑ Monsoonal rainfall is influenced heavily by relief or terrain. For example, The windward of the Western Ghats mountains receives more than 250 cm of rainfall during the monsoon season.
- ❑ Similarly, the northeastern states of India also receive a lot of rainfall, due to the presence of the Eastern Himalayas. The mountains force the monsoon winds to rise, just as the Western Ghats do.
- ❑ The amount of rainfall decreases as you move away from the sea. This is because the monsoon winds lose moisture as they travel over land. Kolkata, which is located on the coast, receives 119 cm of rain during the monsoon season. Patna, which is located further inland, receives 105 cm of rain. Prayagraj, which is even further inland, receives 76 cm of rain. Delhi, which is located in the far north of India, receives the least rainfall, at 56 cm.
- ❑ Since more than three-quarters of the rain in the country falls during the southwest monsoon season, the monsoons are very important to India's agricultural economy. In some parts of the country, it can take a long time before it starts to rain. Sometimes the rain stops much earlier than usual, which hurts crops that are already growing and makes it hard to plant winter crops.

Temperature

- ❑ With the arrival of the monsoons, temperatures plummet and humidity levels rise. With the end of the southwest monsoons in September, the temperature rises. Whenever there is a break in the monsoons, the temperature rises.
- ❑ Because of clouds and rain, the diurnal temperature range is small. During the break, it rains a lot in the sub-Himalayan areas and on the southern sides of the Himalayas.

Wind Distribution

- ❑ By early June, the low-pressure system across the northern plains has become more intense. It attracts the southern hemisphere's trade winds.
- ❑ The warm subtropical areas of the southern oceans give rise to these south-east trade winds. They go across the equator and blow into the Indian peninsula from the southwest.
- ❑ This is called the southwest monsoon. These winds bring a lot of water to the subcontinent because they blow over warm oceans. The average speed of these winds is 30 km/h, which means they are strong.
- ❑ All but the far northwest of the country are covered by monsoon winds for about a month.

The Monsoon Comes to the Landmass from two Directions:

1. The Monsoon Winds of the Arabian Sea branch
2. The Monsoon Winds of Bay of Bengal branch

1. Monsoon Winds of the Arabian Sea: The monsoon winds that originate over the Arabian Sea are further subdivided into three branches:

- **First Branch:** First Branch is halted by the **Western Ghats**. These winds go up the 900–1200 m Western Ghats slopes. Because of this, the windward side of the Sahyadris and the Western Coastal Plain gets between 250 cm and 400 cm of rain every year. After going over the Western Ghats, these winds come down and get warmer. This makes the winds have less moisture. Because of this, it doesn't rain much east of the Western Ghats. The rain-shadow area is a place where it doesn't rain as much. Take note of how Kozhikode, Mangalore, Pune, and Bengaluru get different amounts of rain.
- **Second Branch:** The coast north of Mumbai is hit by another branch of the Arabian Sea. When these winds move along the valleys of the Narmada and Tapi rivers, they bring rain to a lot of central India. From this part of the branch, 15 cm of rain falls on the Chotanagpur plateau. Afterwards, they flow into the Ganga plains and mix with the branch that flows into the Bay of Bengal.
- **Third Branch:** A third branch of the Arabian Sea monsoon wind strikes the Saurashtra Peninsula and the Kachchh. It then travels over western Rajasthan and along the Aravallis, where it brings only light rainfall. In Punjab and Haryana, it joins the Bay of Bengal branch of the monsoon wind. When these two branches come together, they bring rain to the western Himalayas.

2. Monsoon Winds of the Bay of Bengal: The Bay of Bengal division goes along the coasts of Myanmar and southeast Bangladesh.

- But the Arakan Hills on the coast of Myanmar send a large part of this branch into the Indian subcontinent. So, the monsoon comes into West Bengal and Bangladesh from the south and southeast instead of the southwest.
- This branch is divided in two because of the Himalayas and the thermal low in northwest India. One of its branches runs west along the Ganga plains all the way to the Punjab plains. The other branch moves north and northeast up the Brahmaputra valley, bringing rain to a wide area. Its branch goes through the Garo and Khasi hills in Meghalaya. On top of the Khasi hills is the town of Mawsynram, which receives the highest rainfall in the world.

Why the Tamil Nadu Coast Remain Dry During this Season?

- ❑ The branch of the southwest monsoon that comes from the Bay of Bengal and goes along the coast of Tamil Nadu.
- ❑ It is in the rain shadow area of the Arabian Sea branch of the southwest monsoon.

Rainfall

- ❑ The weather in India changes completely when the southwest monsoon comes. Early in the season, more than 250 cm of rain falls on the side of the Western Ghats that faces the wind.
- ❑ Even though the Deccan Plateau and parts of Madhya Pradesh are in the rain shadow, they still get some rain. The most rain falls in the northeastern part of the country during this season.

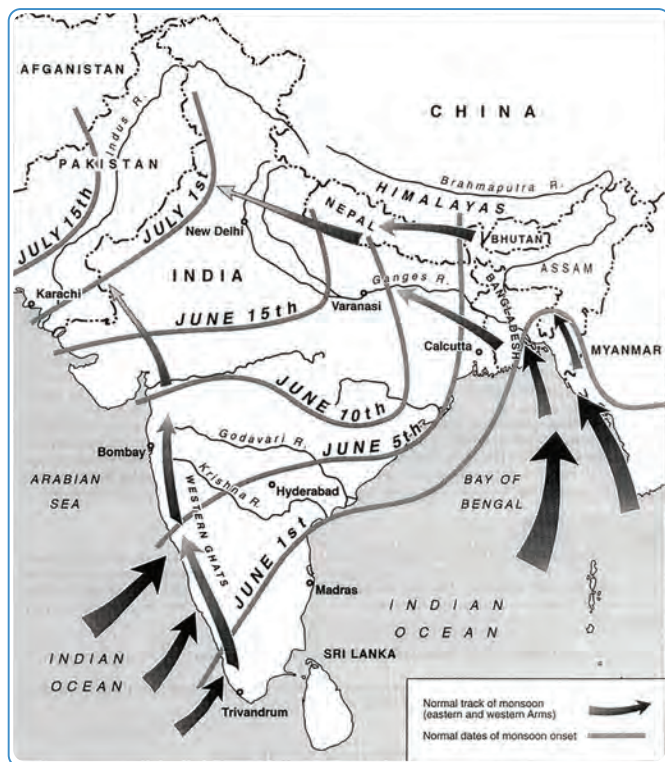


Fig. 5.17: Rainfall - South West Monsoon

- ❑ The most rain falls on average in Mawsynram, which is in the southern ranges of the Khasi Hills. From east to west, it rains less in the Ganga Valley. Rajasthan and some parts of Gujarat don't get much rain.
- ❑ The amount of annual rainfall in the northern plains of India decreases from east to west.
- ❑ Another point about the monsoon is that it tends to stop raining for a while. Because of this, it has both wet and dry times. This means that monsoon rains only last a few days at a time. There are times when it doesn't rain between them.
- ❑ The monsoon trough moves, which makes these breaks in the rainy season. The trough and its axis keep moving southward or northward for different reasons. This affects where the rain falls. When the monsoon trough's centre passes over the plains, there is a lot of rain.

- ❑ The plains undergo longer dry spells and extended rainfall in the mountainous catchment regions of the Himalayan rivers as the centre shifts closer to the Himalayas. These heavy rains cause devastating floods in the plains, causing loss of life and property.
- ❑ The amount and duration of monsoon rains are also determined by the frequency and intensity of tropical depressions.
- ❑ These depressions start at the mouth of the Bay of Bengal and move onto the mainland. The depressions are aligned along with the **"monsoon trough of low pressure."**
- ❑ The monsoon is known for being hard to predict. Dry and wet spells vary in how bad they are, how often they happen, and how long they last. In some places, it causes heavy flooding, but not in others.

Mawsynram (a town in Meghalaya state's East Khasi Hills district), the world's wettest place, is also known for its stalagmite and stalactite caves. The warm moist air arising from the Bay of Bengal during the monsoon season is forced down as it approaches Khasi Hills, which results in heavy rainfall over Mawsynram. Mawsynram is a hill station that is known for being the wettest place in India. The region receives annual rainfall of almost 12 metres.

Monsoon Trough

- ❑ Monsoon Trough is a long area of low pressure that stretches from Pakistan to the Bay of Bengal. This is one of the monsoon's more permanent features. Monsoon trough may be a trait of Himalayan ranges that run east to west and Khasi-Jaintia Hills that run north to south.
- ❑ With the progress of the monsoon, this heat low slowly moves east until it forms an elongated low-pressure zone that runs west to east along the Himalayas. This is called the monsoon trough by Indian meteorologists. Its axis runs about parallel to the foothills of the Himalayas.
- ❑ This monsoon trough moves to the north and south of its normal position from time to time. When it moves north and gets close to the foothills of the Himalayas, it changes the way it rains in India in a big way. Over the plains of northern India, the rain stops quickly, but over the foothills of northeast India, the rain gets stronger quickly.

North-East Monsoon Season (Retreating Monsoon)

The Northeast Monsoon, also known as the retreating monsoon or post-monsoon season, is a three-month period that begins in mid-September and ends in early January.



Fig. 5.18: North East Monsoon in India

- ❑ With the Sun's southward migration, the south-west monsoon begins to retreat from northern India by the third week of September.
- ❑ Starts with the beginning of the withdrawal of southwest monsoon (middle of September – November).
- ❑ Monsoons withdraw from the extreme north-west end of the country in September, from the peninsula by October and from the extreme south-eastern tip by December.
- ❑ In Punjab, the south-west monsoons reach in the first week of July and withdraw from there in the second week of September.
- ❑ South-west monsoons reach Coromandel coast in the first week of June and withdraw from there only in the middle of December.
- ❑ Unlike the sudden burst of the advancing monsoons, the withdrawal is rather gradual and takes about three months.

Temperature During Retreating Monsoon Season

- ❑ In November, the temperature starts to drop, and by December, it's cold. The average temperature in the north and northwest of the Indian plains is 15°C, in the centre it's 20°C, and along the coasts it's 25°C.
- ❑ When October comes, the low-pressure area over the northwestern parts of India goes away and is replaced by a low-pressure cell above the northern parts of Bengal. By the beginning of December, it moves south.

While the days are hot, the nights are cool and pleasant. The ground is still having moisture. During the day, the weather becomes oppressive due to the high temperatures and humidity. This is commonly referred to as **October Heat**.

Pressure and Winds During Retreating Monsoon Season

Pressure

- ❑ Over the Indian subcontinent, the pressure conditions start to change during the retreating monsoon season.
- ❑ The large-scale **low-pressure area** that dominated the Indian subcontinent during the peak monsoon season begins to **weaken and shift**.
- ❑ With the retreat of the southwest monsoon winds, **high-pressure areas start forming over northern India** and the adjoining regions.
 - The formation of these high-pressure systems is influenced by the **cooling temperatures** over the landmass.

Winds

- ❑ The wind direction undergoes a noticeable shift from the southwest to the northeast.
- ❑ Over northern India, the **prevailing winds** become **northeasterly**, bringing in **cooler and drier air**.
- ❑ As the **monsoon trough shifts southward**, the northeast monsoon begins to establish itself over the Indian subcontinent.

Rainfall

- ❑ As the southwest monsoons move away, humidity and cloud cover go down, and most of the country stays dry. October and November are the main rainy season in **Tamil Nadu** and nearby parts of Andhra Pradesh to the south of the Krishna delta.
- ❑ This is also a secondary rainy season in Kerala. The rain is caused by the monsoons taking in water as they move over the Bay of Bengal on their way out.

Cyclones During Retreating Monsoon Season

- ❑ By the beginning of November, the low pressure that was in northwest India had moved to the Bay of Bengal. **Cyclonic depressions** that form over the Andaman Sea are linked to this change.
- ❑ Most of the time, these cyclones pass along India's eastern coasts, bringing heavy rain to a wide area. Most of the time, these tropical cyclones do a lot of damage.
- ❑ Cyclones often hit the heavily populated deltas of the Godavari, Krishna, and Kaveri rivers, doing a lot of damage to people and things.
- ❑ The coasts of West Bengal, Odisha, and Bangladesh are sometimes hit by these cyclones. The majority of the Coromandel Coast's rainfall is caused by depressions and cyclones.
- ❑ The months of October and November are when the frequency of cyclones is highest during the retreating monsoon season.

Reasons for Increase in Frequency of Cyclones in October and November

- ❑ October and November represent the transition period between the southwest and northeast monsoons.
- ❑ Warm sea surface temperatures in the Bay of Bengal and Arabian Sea provide energy for cyclone intensification.
- ❑ Favorable upper-level atmospheric conditions with reduced wind shear support cyclone organization and strengthening.
- ❑ The active monsoon trough contributes to moist air convergence, facilitating cyclone formation.
- ❑ Interaction between northeast and southwest monsoon winds aids in air mass convergence, conducive to cyclone development.
- ❑ Late withdrawal of the southwest monsoon may lead to additional rainfall, providing moisture for cyclone development.

TRADITIONAL INDIAN SEASONS

A year is divided into six two-monthly seasons in Indian tradition. The common people of central India and the north follow a seasonal cycle based on practical experience and age-old perceptions of weather phenomena. This system, however, does not correspond to the seasons in southern India, where the seasons vary little.

Seasons	Months (Indian Calendar)	Months (Gregorian Calendar)
Vasanta	Chaitra-Vaisakha	March-April
Grishma	Jyaistha-Asadha	May-June
Varsha	Sravana-Bhadra	July-August
Sharada	Asvina-Kartika	September-October
Hemanta	Margashirsa-Pausa	November-December
Shishira	Magha-Phalguna	January-February

RAINFALL VARIABILITY IN INDIA

Rainfall variability refers to the amount of fluctuation from the mean values in rainfall recorded over a long period of time. It is calculated using the Coefficient of Variability (C.V.) of rainfall using the following formula:

{The Coefficient of Variability varies between 15-35 per cent for India as a whole}

- ❑ Rainfall variability is inversely proportional to the total amount of rainfall. This means areas of low annual rainfall have high variability, while areas of high annual rainfall have low variability.

Distribution of Annual Rainfall in India

The average annual rainfall in India is roughly 125 cm, but it varies greatly across the country.

- ❑ **Areas of High Rainfall:** The heaviest rainfall in India occurs on the Western Ghats mountains, the western coast, the sub-Himalayan regions of the northeast, and the hills of Meghalaya. This area receives more than 200 cm of rain per year. Some parts of the Khasi and Jaintia hills receive more than 1,000 cm of rain per year. The Brahmaputra valley and the surrounding hills receive less than 200 cm of rain per year.
- ❑ **Areas of Medium Rainfall:** Between 100 and 200 cm of rainfalls in the southern parts of Gujarat, the east of Tamil Nadu, the northeastern peninsula, which includes Jharkhand, Bihar, Odisha, and eastern Madhya Pradesh, the northern Ganga plain along the sub-Himalayas, the Cachar Valley, and Manipur.
- ❑ **Areas of Low Rainfall:** Rainfall in western Uttar Pradesh, Delhi, Punjab, Haryana, Jammu and Kashmir, Gujarat, eastern Rajasthan, and the Deccan Plateau ranges between 50 and 100 cm.
- ❑ **Areas of Inadequate Rainfall:** Parts of the Peninsula, particularly Andhra Pradesh, Karnataka, and Maharashtra, as well as Ladakh and most of western Rajasthan, receive rainfall of less than 50 cm. Snowfall is limited to the Himalayan range.

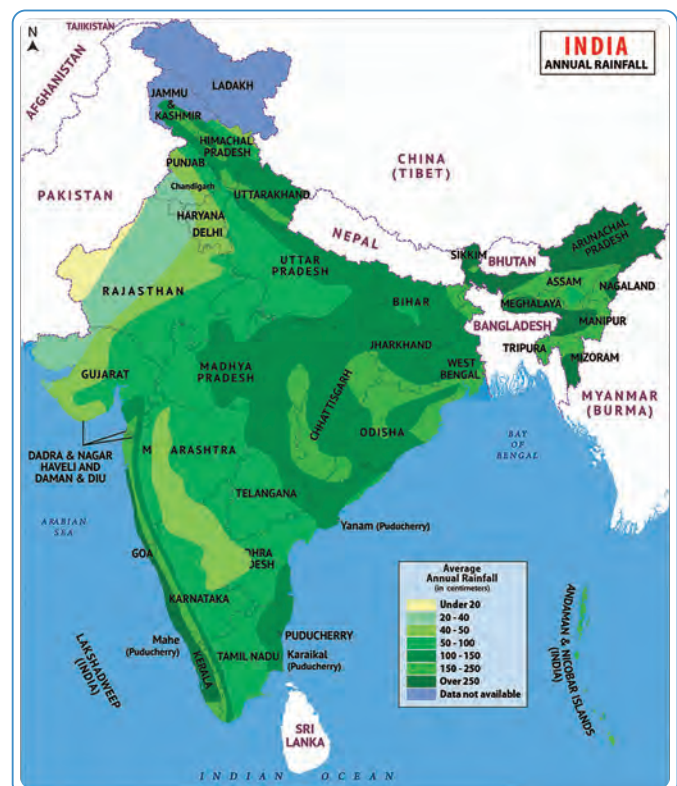


Fig. 5.19: India Annual Rainfall

Isohyet

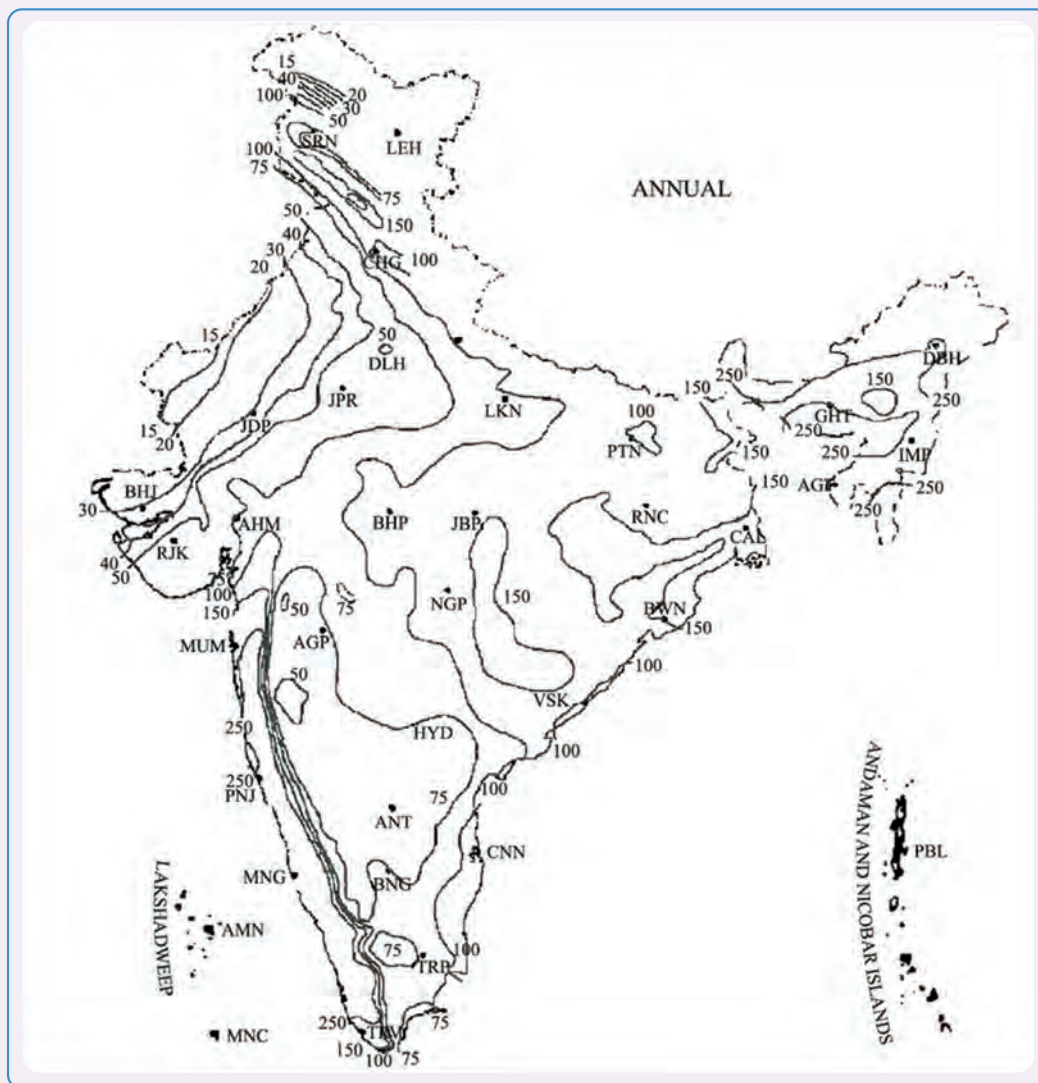


Fig. 5.20: Isohyets - India

- ❑ An **isohyet** is a line on a map connecting points that receive the **same amount of precipitation** within a specified time period.
- ❑ In other words, it is a contour line that joins **locations with equal rainfall** or precipitation values.
- ❑ Isohyets help meteorologists, climatologists, and geographers analyze and understand the spatial patterns of precipitation, which is crucial for various purposes, including agriculture, water resource management, and disaster preparedness in India.
 - The information derived from isohyets aids in planning and decision-making related to water availability and agricultural practices in different parts of the country.

HUMANIZING FACTORS AFFECTING THE INDIAN MONSOON

- ❑ **Disturbance in natural balance:** The natural balance has been disturbed as a result of the humanization of landscapes. As a result, there has been excessive deforestation, an increase in pollution and pollutants, an increase in built-up areas, global warming, and so on.

Humanizing Landscape

Humanizing the landscape involves altering the natural architecture of the landscape to fit the needs of people, like "Urbanization or Industrialization." Features of human landscapes include elements of infrastructure such as buildings, roads, transport, energy, sewerage and telecommunication systems.

- ❑ **Rainfall in Urban Areas:** Because of the increased availability of pollutants and dust particles, the number of condensation nuclei has increased, resulting in greater rainfall in urban areas than in rural areas. The increased built-up area has allowed for more surface heating. This increased temperature has increased evaporation and influenced rain patterns.
- ❑ **Deforestation:** Monsoon distribution has suffered as a result of deforestation. Increased concretization causes more runoff and less water infiltration, affecting the water cycle and disrupting the monsoon cycle.
- ❑ **Pressure differences:** Pressure differences, which are necessary for monsoonal precipitation, have been unstable and unpredictable.
- ❑ **High-Rise Buildings:** High-rise buildings have changed local wind patterns and precipitation patterns.
- ❑ **Wetlands:** A wetland is a land area that is permanently or seasonally saturated with water. Wetlands play a variety of environmental roles, including water purification, flood control, carbon sequestration, and shoreline stability. The destruction of wetlands in urban areas, such as lakes and ponds, is a major cause of irregular rain patterns in major cities such as Chennai and Srinagar.
- ❑ **Materials used for Infrastructure:** The widespread use of concrete in urban areas reduces groundwater recharge and alters the hydrological cycle of the environment, ultimately disrupting India's Monsoon system.
- ❑ **Use of Chemicals in Agriculture:** Another example of humanising the landscape is the widespread use of chemicals and borewell irrigation in agriculture.
- ❑ **Dams:** Humanizing the landscape also includes the construction of large dams in hilly areas that alter the ecological flow of rivers.

In the Bhojpur region, the South-West monsoon is known as "**Purvaiya**," denoting the easterly direction from which the winds enter the area.

- ❑ The South-West monsoon winds bifurcate into the Arabian Sea and Bay of Bengal branches.
- ❑ The Bay of Bengal branch takes a path towards the Purvanchal Himalayas, making a northwest turn and entering the Bhojpur region of Bihar from the east.

The term "**Purvaiya**" specifically signifies the **easterly winds associated with the South-West Monsoon** in the Bhojpur region, with the local word "**Purva**" meaning east.

These Purvaiya winds play a crucial role in shaping the culture of Bhojpur, influencing traditions, rituals, and daily life.

- ❑ Many folk songs and dances in the region highlight the importance of the Purvaiya winds, such as Kajari songs and Jhijhia dance.
- ❑ The agricultural practices driven by these winds also impact the types of crops grown, subsequently influencing the local cuisine.

CLIMATE CHANGE AND ITS IMPACT ON INDIA & INDIAN MONSOON

Climate Change

- ❑ **Definition:** Climate change refers to long-term alterations in temperature, precipitation, and other atmospheric conditions on Earth. It is primarily driven by human activities, such as the burning of fossil fuels, deforestation, and industrial processes, leading to an increase in greenhouse gas concentrations in the atmosphere.
- ❑ Climate change encompasses both global warming, the increase in Earth's average surface temperature, and broader changes in weather patterns and climate-related phenomena. It involves shifts in temperature, precipitation, sea levels, and the frequency and intensity of extreme weather events.

India and Climate Change

- ❑ India, being a large and diverse country, is highly vulnerable to the impacts of climate change. The country experiences changes in monsoon patterns, rising temperatures, and increased frequency of extreme weather events. These changes have far-reaching consequences for agriculture, water resources, and vulnerable communities.

Consequences of Climate Change

- ❑ **Impact on Agriculture:** Changes in temperature and precipitation patterns affect crop yields. For example, variations in monsoon timings and intensity can impact rice and wheat production, key staples in India.
- ❑ **Water Scarcity:** Altered precipitation patterns contribute to water scarcity. Glacial melt in the Himalayas, crucial for river systems like the Ganges, is accelerating, affecting water availability for millions of people.
- ❑ **Extreme Weather Events:** Increased frequency of extreme events like cyclones, floods, and droughts poses a threat to infrastructure and human lives. The 2018 Kerala floods, exacerbated by intense rainfall, are an example of climate change impacts.

- ❑ **Heatwaves in India:** Heatwaves during the monsoon season are extremely uncommon in India. Due to the delayed arrival of the monsoon in the country's east, A rare heatwave caused damage to the country.
- ❑ **Effect on Rice Producing Areas:** Rice-producing states have been severely deficient due to the southward movement of majors, all major monsoon low-pressure areas, and depressions. Despite an active monsoon current in July and August, Bihar, West Bengal, and Uttar Pradesh, which account for one-third of the country's total rice production, have been severely deficient. This would immediately have an impact on the country's food security.
- ❑ **Pests and diseases impact:** Pests and diseases are exacerbated by the uneven distribution of rain, as well as rising temperatures and humidity.
- ❑ **Heat Stress:** Extremely high temperatures (above 35°C) cause heat stress and disrupt plant physiological processes, resulting in spikelet sterility, non-viable pollen, and poor grain quality.
- ❑ **Effect of drought:** Drought reduces plant transpiration rates, which can lead to leaf rolling and drying, reduced leaf expansion rates, and plant death.
- ❑ **Rising sea levels:** When the Earth's temperature goes up due to climate change, the sea level rises because warm water takes up more space than cooler water.
 - The melting of glaciers makes this problem worse. People who live in low-lying places, on islands, or near the coast are in danger because the sea is rising.
 - It damages property and kills habitats like mangroves and marshes that protect coastlines from storms.

Mitigation

- ❑ **Renewable Energy Adoption:** Shifting to renewable energy sources, such as solar and wind power, helps reduce dependence on fossil fuels and lowers greenhouse gas emissions. India's National Solar Mission aims to increase solar energy capacity.
- ❑ **Afforestation and Reforestation:** Planting trees and restoring forest ecosystems act as carbon sinks, absorbing CO₂ from the atmosphere. India's Green India Mission focuses on increasing forest and tree cover.
- ❑ **Policy Initiatives:** Implementing and adhering to policies and international agreements aimed at reducing emissions and promoting sustainable practices. India is a signatory to the Paris Agreement, committing to reduce its carbon intensity.

Climate change is a global challenge with severe implications for India's diverse ecosystems and communities. Mitigation efforts, along with global cooperation, are essential to address the root causes and reduce the impact of climate change on India and the world.

Relationship Between the Shrinking Himalayan Glaciers and the Symptoms of Climate Change

- ❑ **Global Warming:** The primary driver behind the retreat of Himalayan glaciers is the increase in global temperatures due to climate change.
 - According to IPCC Assessment Report 6, anthropogenic global warming of 1.1 degrees Celsius has spurred changes to the Earth's climate that are unprecedented in recent human history.
- ❑ **Altered precipitation patterns:** In some regions, there may be shifts from snowfall to rainfall, reducing the accumulation of snow on glaciers. This change in the form of precipitation affects the mass balance of glaciers, contributing to their shrinkage.
- ❑ **Glacial Retreat:** Higher temperatures cause the glaciers to melt at their lower ends faster than the rate at which they gain new ice from snowfall at higher elevations. This leads to a net loss of glacial mass, resulting in the retreat of glaciers.
- ❑ **Glacial Lake Outburst Floods:** As glaciers retreat, meltwater accumulates in glacial lakes formed at their snouts. These lakes can pose a risk of glacial lake outburst floods (GLOFs) if the natural moraine dams holding them back fail.
- ❑ **Impacts on Water Resources:** The Himalayan glaciers serve as a crucial source of freshwater for major rivers, including the **Ganges, Brahmaputra, and Indus**. The shrinking glaciers can affect the timing and quantity of water flow in these rivers, impacting **downstream water availability** for agriculture, drinking water, and other purposes.
- ❑ **Ecosystem Changes:** The retreat of glaciers affects the ecosystems that depend on glacial meltwater. Changes in water availability can impact biodiversity, vegetation, and the livelihoods of communities relying on these ecosystems.
- ❑ **Global Sea Level Rise:** While the Himalayas are landlocked, the melting of glaciers contributes to rising sea levels globally. This is because the water released from melting glaciers eventually flows into the oceans, adding to the volume of seawater.

Relevant Examples from India

- ❑ **Gangotri Glacier Retreat:** The Gangotri Glacier, one of the largest glaciers in the Himalayas and the source of the Ganges River has been experiencing significant retreat. This retreat has a direct implications for the Ganges River's flow and the millions of people who depend on it for water.
- ❑ **Glacial Lake Outburst Floods (GLOFs):** Sikkim recently experienced a Glacial Lake Outburst Flood (GLOF).
- ❑ The **South Lhonak Lake**, a glacial lake located at an altitude of 17,000 feet in the state's northwest, experienced a rupture as a result of continuous rainfall.

Waiting to Happen! What was the Trigger?

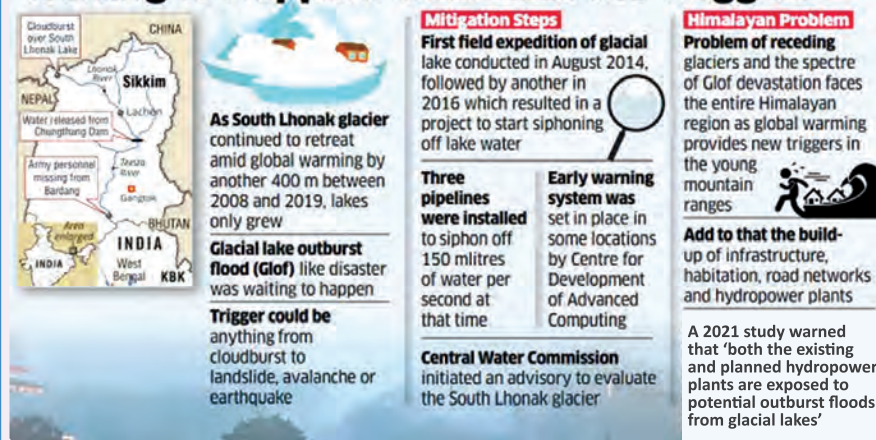


Fig. 5.21: Lhonak Lake Induced GLOF disaster

CLIMATIC REGIONS OF INDIA

Monsoon weather prevails throughout India. However, the combination of weather elements reveals many regional variations. These variations represent monsoon climate subtypes. A climatic region has consistent climatic conditions as a result of a variety of factors. Temperature and rainfall are two important variables that are used in all climatic classification systems.

There are two important classifications of climatic regions of India:

- ❑ Stamp's Classification
- ❑ Koeppen's Classifications

Koeppen's Classifications

A Koppen classification is an empirical classification based on climatic data. Koppen considered climatic regions when defining them like

- ❑ The Mean Monthly Temperature
- ❑ The Mean Monthly Rainfall
- ❑ The Mean Annual Rainfall

Koppen's Classification Criteria

Humid Tropical Climate: Humid Tropical climates make up half of the earth's surface (over 40 per cent of the sea surface and 20 per cent of the land). Due to how dry they are, The hot deserts of Arabia and North Africa are not suitable for this type. The mean annual isotherm of 20°C is used to describe it. Most of the time, you can tell what kind of climate is most common by how the rain falls at different times of year. In this belt, the temperature is mostly the same.

Koppen Climate Classification Types	Criteria
Tropical Humid Climate (A)	The lowest average temperature for a month is more than 18°C.
Tropical rainforest (Af)	The driest month gets at least 6 cm or more of rain or snow.
Tropical monsoon (Am)	Even though it rains less than 6 cm in the driest month, that's still more than 4% of the year's rain.
Tropical wet and dry (Aw)	In the driest month, it starts raining less than 10 cm, which is less than 4% of the total amount of rainfall recorded in a year.

Dry Climates: These have been divided into desert or arid types and steppe or semi-arid types. There are many things that the two parts of climate have in common. Koeppen's system for classifying climates uses a formula with three variables to show where wet and dry climates meet.

Koppen Climate Classification Type	Criteria
Arid desert (BW)	The amount of rain that falls each year is less than 50% of the rain threshold.
Semi-arid steppe (BS)	The amount of rain that falls each year is more than 50% of the rain threshold.

Warm temperate Climates: There are China, West European, and Mediterranean climates within this group. In the month with the highest temperature, it was over 10°C. The temperature in the coldest month is between 18° and 0°C.

Koeppen's Climate Classification Type	Criteria
Mild temperate dry summer (Cs)	The driest month of summer gets less than 1/3 as much rain as the wettest month of winter.
Mild temperate dry winter (Cw)	The driest month of winter receives less than 1/10 as much rain as the wettest month of summer.
Mild temperate humid (Cf)	Doesn't fit the Cs or Cw types of weather.

Cold Snow Forest Climates: Mid-latitude humid climates bearing lower temperatures have been subdivided into the following types

Koppen Climate Classification Type	Criteria
The Taiga Climate (Dwd/Dwe/Dfe)	The average summer temperature is 15°C, and the summers last only a few weeks. Winter lasts a long time and is very cold. The temperature can drop as low as -50°C.
The Cool East Coast Climate (Dwa)	Summers are hot, sticky, and last a long time. The average temperature in the winter ranges from 0°C to -40°C. Rainfall comes and goes.
The Continental Climate (Dfa)	In the summer, it is warm, and the temperature stays between 10°C and 21°C for less time. During the winter, the temperature can drop below freezing.

The Cold/Polar Climate: The average temperature of the warmest month in a polar climate is less than 10°C. These are different because they don't have a warm period and stay cold for a long time. This group of weather has been split into two types: Tundra climate, where the warmest month is above freezing but below 10°C and Frozen climate, where even the warmest month is below 0°C.

Koppen Divided the Country into three Broad Climatic Zones

1. Humid (A)
2. Arid (B)
3. Semi-Arid (C and D)

Koppen's Classification of Indian Climate

Code	Climate Type	Features	Temperature	Rainfall	Regions of India
Aw	Tropical savanna	Tropical Savanna grassland, monsoon deciduous vegetation	Hottest month— May (max. temp.: 40°C and min. temp.: 27°C) Coldest month temperature is greater than 18°C High ranges of annual and diurnal temperatures	Occurs during the season of the southwest monsoon—July to September Dry winters	Peninsular India, Jharkhand, Chattisgarh, Odisha, Andhra Pradesh, Telangana, Maharashtra, Karnataka, Purulia district of West Bengal

These three broad climatic divisions were further subdivided into sub-types based on seasonal variations in precipitation and temperature distribution patterns, for which the symbols **S, W, m, f, w, s, c, and h** were used. Koeppen identified five major climatic groups, four of which are temperature-based and one of which is precipitation-based.

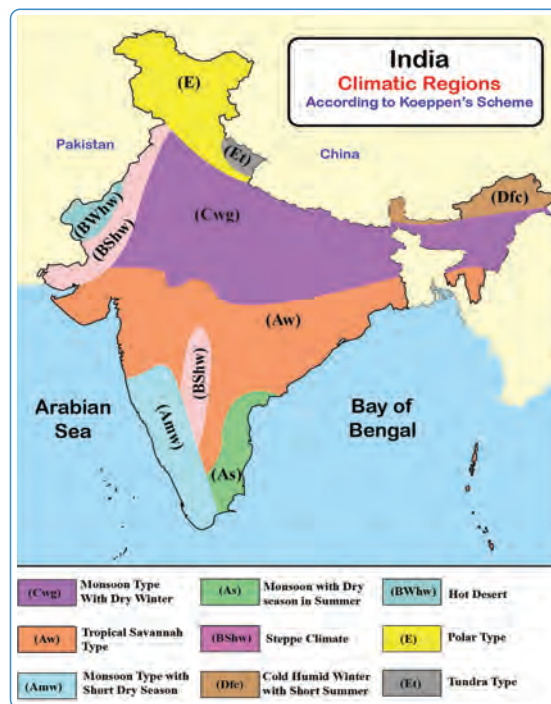


Fig. 5.22: India Climatic Regions as per Koeppen

Groups	Characteristics
A-Tropical	The average temperature in the coldest month is at least 18° C.
B-Dry Climates	The amount of water that could evaporate is greater than the amount that could fall as rain or snow.
C- Warm Temperate	In (Mid-latitude) climates, the average temperature of the coldest month of the year is more than -3°C but less than 18°C.
D-Cold Snow forest	The average temperature in the coldest month is at least -3° C.
E-Cold Climates	The average temperature each month is less than 10° C.
H-Highlands	Cold due to the altitude.

Amw	Tropical monsoon	Short winter season luxurious evergreen rain forests	Annual temperature ~24°C to 27°C	Heavy rainfall over 200 cm during the season of southwest monsoon	Konkan, Malabar Coast, adjoining areas of the Western Ghats, Plateaus of Tamil Nadu, southern areas of Tripura and Mizoram
As	Tropical moist	Dry summers	Mean monthly temperature >18°C	Average rainfall varies between 75 and 100 cm 75% of total rainfall occurs between September and December	Narrow zone along the Coromandel Coast
BShw	Semi-arid steppe	Hot and dry summers from March to May	Maximum temperature as high as 40°C	Seasonal rainfall (in summer)	Stretches over rain shadow zone of Karnataka, Tamil Nadu, eastern Rajasthan, Gujarat, south-western Haryana
Cwg	Mesothermal climate; Gangetic Plain Type	Hot desert type of climate; dry winters	Average temperature <18°C and temperature of coldest month >15°C Max. temperature is recorded in May or the first half of June	Below 40 cm	Thar desert
Dfc	Cold humid winter	Short and humid summer and cold humid winter	Winter temperatures are about 10°C and summer temperature 30 to 35°C	Over 200 cm	Sikkim, Arunachal Pradesh, Assam, Tripura, Mizoram
E	Polar	Areas remain under the ice during the greater part of the year	Temperature of warmest month <10°C Temperature falls by 0.6°C for every 100 m rise in altitude in the Himalayas	Lesser rain on the leeward side of the mountains and heavy rainfall on the well-exposed slopes	Higher mountainous areas of Jammu and Kashmir, Himachal Pradesh, Uttarakhand
ET	Tundra	Polar ice caps, the soil is frozen to hundreds of metre depth, vegetation constitutes of mosses, lichens, dwarf trees, scattered woody shrubs	Average temperature of warmest month is between 0°C and 10°C		Higher altitudes of Ladakh, Kashmir, Himachal Pradesh, Uttarakhand

CYCLONES

- ❑ Cyclones are caused by atmospheric disturbances in the proximity of a low-pressure area, which are distinguished by quick and often disturbing air circulation. Henry Piddington coined the term cyclone, which is derived from the Greek word Cyclos, which means snake coils. In meteorology, a cyclone is defined as a system of winds rotating inwards to an area of low atmospheric composition, with anticlockwise circulation in the Northern Hemisphere and clockwise circulation in the Southern Hemisphere.
- ❑ When a huge amount of energy moves from the ocean to the air, this is called a cyclone. Studies show that 70 to 90 cyclonic systems form around the world every year. The surface winds spiral towards the low-pressure area

because of the Coriolis force. Because the Coriolis force isn't very strong between 5 degrees north and 5 degrees south, cyclonic systems don't form in the equatorial regions.

❑ Cyclone Prone Areas in India

- Cyclones typically strike the East Coast of India, along the Bay of Bengal, that is, the states of West Bengal, Odisha, Andhra Pradesh and Tamil Nadu, but also parts of Maharashtra and Gujarat on the West Coast in front of the Arabian Sea.
- **Example:** A very severe cyclonic storm Phailin in 2013 and another severe cyclonic storm Hudhud in 2014 killed hundreds of people, cattle, and wild animals, and damaged property, transportation, and communication along the coasts of Andhra Pradesh, Odisha, and West Bengal.

- The IMD (Indian Meteorological Department) classified Cyclone Mocha, which made landfall in Myanmar in 2023, as an Extremely Severe Cyclonic Storm and a 'Super Cyclone' that originated in the Bay of Bengal. Amphan had a very high wind speeds ranging up to 155-165 kmph gusting to 185 kmph in 2020, Tauktae had a surface wind speed of 160-170 kmph gusting to 185 kmph at the time of landfall in 2021, and Gonu had a speed of 210 km/hr in 2007.

Correlation Between Cyclones and Indian Climate

- ❑ **Monsoon Influence:** Cyclones play a crucial role in the Indian summer monsoon. The formation and movement of cyclones in the Bay of Bengal and the Arabian Sea can impact the distribution of rainfall during the monsoon season. Cyclones bring heavy rainfall to coastal regions and, in some cases, can enhance the overall monsoon precipitation.
- ❑ **Cyclone Tracks:** The paths that cyclones take can affect different parts of India differently. For example, cyclones that move northward from the Bay of Bengal can bring heavy rainfall to the eastern and northeastern parts of India, while those moving westward from the Arabian Sea can impact the western coast.
- ❑ **Flooding and Storm Surge:** Cyclones often bring heavy rainfall, leading to flooding in coastal and low-lying areas.
- ❑ **Tropical Cyclones:** India experiences tropical cyclones, which are low-pressure systems characterized by strong winds and heavy rain. The frequency and intensity of tropical cyclones in the region can vary from year to year, influenced by factors such as sea surface temperatures, atmospheric conditions, and climate phenomena like El Niño and La Niña.
- ❑ **Impact on Agriculture:** Cyclones can have both positive and negative effects on agriculture. While they bring much-needed rainfall, excessive rainfall during a short period can lead to waterlogging and damage crops. On the other hand, regions experiencing a deficit in monsoon rainfall may benefit from cyclonic activity.
- ❑ **Cyclone Seasons:** The primary seasons for cyclones in the North Indian Ocean are the pre-monsoon (April to June) and post-monsoon (October to December) periods. The Bay of Bengal is more active during the pre-monsoon season, while the Arabian Sea is more active during the post-monsoon season.
- ❑ **General Mitigation and Preparedness Measures Against Cyclone**
 - **Mapping Hazards:** Displaying cyclone hazards on a map to show the frequency and probability of different intensities or durations.
 - **Regulating Land Use:** Implementing policies to control land use and enforce building codes. Designating

vulnerable areas for parks, grazing grounds, or flood diversion instead of human settlements.

- **Engineered Structures:** Adopting good construction practices, such as building on stilts or earthen mounds and ensuring buildings are wind and water-resistant.
- **Cyclone Shelters:** Establishing cyclone shelters in areas prone to cyclones for the protection of residents.
- **Flood Management:** Building barriers along rivers and coasts to prevent flooding in vulnerable areas.
- **Plantation of Mangroves:** Involving communities in mangrove plantation efforts organized by local authorities, NGOs, or the community itself.



IGNITE YOUR MIND

IMD stops recording monsoon data after September 30 but certain parts of India now receive high rainfall in October month. Do you think that the monsoon period in India should be redefined? In your opinion what weather phenomenon can be attributed to this unusual rainfall?

Diverse Impacts of the Indian Monsoon

The Indian monsoon, a unique climatic phenomenon, plays a pivotal role in shaping the diverse climate of India.

This Diversity is Characterized by Various Factors Associated with the Monsoon System

- ❑ **Spatial Variability:**
 - The Indian subcontinent experiences spatial variations in monsoon dynamics, resulting in different regional climates.
 - Coastal regions, such as the west coast, receive heavy rainfall due to orographic effects, while inland areas like the Thar Desert may be arid.
- ❑ **Seasonal Shifts:**
 - The Indian monsoon follows a distinct seasonal pattern.
 - The southwest monsoon, occurring from June to September, brings the majority of the annual rainfall. In contrast, the post-monsoon and winter months experience drier conditions.
- ❑ **Temperature Modulation:**
 - The monsoon significantly influences temperature patterns.
 - The arrival of the southwest monsoon brings relief from high temperatures, cooling the subcontinent. Conversely, the post-monsoon and winter seasons witness lower temperatures.

❑ **Agricultural Impact:**

- The diversity in monsoonal rainfall profoundly affects agriculture.
- Regions receiving abundant rainfall during the monsoon season are suitable for cultivating crops, while areas with less rainfall may face agricultural challenges, necessitating water management strategies.

❑ **Economic Implications:**

- The diversity in monsoonal patterns influences economic activities.
- Sectors like agriculture, which heavily depend on seasonal rainfall, are directly impacted.
 - ❑ Adequate monsoonal rainfall is crucial for crop yields and food production.

❑ **Ecological Variations:**

- The monsoon shapes the ecological diversity across India.
- Regions with heavy rainfall support lush vegetation, diverse ecosystems, and biodiversity.
- In contrast, arid regions may have unique flora and fauna adapted to water scarcity.

❑ **Climatic Extremes:**

- The diversity in monsoonal behavior contributes to climatic extremes.
- Excessive rainfall during the monsoon can lead to floods, while deficient rainfall may result in drought conditions.
- These extremes impact water availability, infrastructure, and livelihoods.

❑ **Hydrological Dynamics:**

- The monsoon significantly influences hydrological cycles, replenishing rivers, reservoirs, and groundwater.
- Regions with abundant monsoonal rainfall contribute to the water resources that sustain various human activities.

❑ **Cultural Significance:**

- The monsoon holds cultural significance in India, influencing festivals, traditions, and daily life.
- Seasonal shifts are often celebrated, and the cultural practices of different regions reflect the dependence on monsoonal cycles.

Understanding the diversity in the Indian monsoon is crucial for adapting to climatic variations, managing water resources, and developing strategies for sustainable agriculture and economic activities. The dynamic interplay between the monsoon and the climate underscores the complexity of India's weather patterns and the need for diverse approaches to climate resilience.

Extreme Weather Events

- ❑ **Droughts:** Periods of deficient rainfall can lead to droughts, impacting water availability, agriculture, and overall socio-economic conditions.

- ❑ **Floods:** Excessive rainfall, especially during the monsoon, can lead to floods, causing damage to infrastructure, displacing communities, and affecting agriculture.

❑ **Indian Ocean Dipole (IOD):**

- **Positive IOD:** A positive IOD phase, characterized by warmer sea surface temperatures in the western Indian Ocean, can enhance monsoonal rainfall in India.

- ❑ **El Niño:** El Niño events, characterized by warmer-than-average sea surface temperatures in the central and eastern Pacific Ocean, can lead to reduced monsoonal rainfall and drought conditions in India.

Impact on Agriculture

- ❑ **Water Availability for Crops:** Adequate monsoon rains are crucial for providing sufficient water to crops. Insufficient rainfall can lead to drought conditions, affecting crop yields and water availability for irrigation.

- ❑ **Crop Planting and Growth:** Monsoons play a vital role in determining the timing and success of crop planting. The timely onset of monsoons is essential for farmers to initiate sowing, and a well-distributed rainfall pattern is crucial for the proper growth of crops.

- ❑ **Crop Yields:** Favorable monsoon conditions contribute to higher crop yields. Conversely, erratic or deficient rainfall can lead to crop failure, impacting food production and the livelihoods of farmers.

- ❑ **Crops Vulnerable to Excess Rainfall:** Excessive rainfall during the monsoon season can lead to waterlogging and flooding in certain areas, particularly low-lying regions. This can adversely affect crops like rice and pulses that are vulnerable to waterlogged conditions.

- ❑ **Disease Outbreaks:** Monsoons create a conducive environment for the spread of diseases affecting crops. Excessive moisture can lead to the outbreak of fungal and bacterial diseases, impacting the health of plants and reducing yields.

- ❑ **Impact on Cash Crops:** Cash crops, such as cotton and sugarcane, are significantly influenced by monsoon patterns. Adequate rainfall is essential for these crops, and variations in monsoon conditions can directly affect the economic well-being of farmers involved in cash crop cultivation.

- ❑ **Shifts in Crop Patterns:** Unpredictable monsoons may lead to shifts in crop patterns. Farmers might alter their choice of crops based on rainfall expectations, affecting the overall agricultural landscape.

- ❑ **Food Security:** The monsoon's impact on agriculture directly influences food production and, subsequently, food security. A poor monsoon season can lead to shortages in staple food items, affecting both rural and urban populations.

- ❑ **Livestock and Pasture Availability:** Monsoon conditions also influence pasture availability for livestock. Inadequate rainfall can lead to a shortage of fodder, affecting the health and productivity of livestock.
- ❑ **Economic Impact on Farmers:** Monsoon variability can lead to economic stress for farmers. Crop failures or lower yields reduce income, affecting the financial stability of agricultural communities.



IGNITE YOUR MIND

As you might know Makar Sankranti, Pongal, and Lohri are festivals celebrated during the winter season and they mark the harvest of crops like sugarcane, rice, and wheat. Can you analyze the role of Winter Harvest Festivals in promoting agricultural sustainability and rural development. How can these festivals be leveraged to enhance farmers' income and address agrarian challenges?

Government Monitoring and Management

- ❑ **India Meteorological Department (IMD):** The primary agency responsible for monitoring and forecasting monsoons in India is the India Meteorological Department (IMD). IMD employs advanced technologies, satellite imagery, and weather models to track the progression of the monsoon across the country.
- ❑ **Monsoon Mission:** The Ministry of Earth Sciences in India launched the Monsoon Mission, an initiative aimed at improving the prediction of the monsoon. It involves research and the use of advanced technology, including high-performance computing, to enhance the accuracy of monsoon forecasts.
- ❑ **Seasonal Climate Outlook:** IMD releases seasonal climate outlooks, providing predictions for the upcoming monsoon season. These outlooks are crucial for farmers, policymakers, and various sectors to plan and make informed decisions based on expected rainfall patterns.
- ❑ **Hydro-Meteorological Services:** Government agencies, including the Central Water Commission (CWC) and the National Disaster Management Authority (NDMA), work in coordination to provide hydro-meteorological services. These services include monitoring river levels, water storage in reservoirs, and issuing flood alerts based on IMD forecasts.
- ❑ **Early Warning Systems:** The government has implemented early warning systems to alert communities about extreme weather events associated with the monsoon, such as heavy rainfall, cyclones, and floods. These warnings are crucial for facilitating timely evacuations and disaster preparedness.
- ❑ **Agricultural Advisory Services:** The government provides agricultural advisory services to farmers based on monsoon forecasts. These services offer guidance on optimal sowing times, crop selection, and water management practices, enabling farmers to make informed decisions.
- ❑ **Drought Management Strategies:** In anticipation of deficient monsoons and drought conditions, the government develops and implements drought management strategies. These strategies involve water conservation measures, provision of relief to affected areas, and support for alternative livelihoods during periods of water scarcity.
- ❑ **National Disaster Response Force (NDRF):** The NDRF plays a crucial role in disaster response during extreme weather events linked to the monsoon. NDRF teams are deployed in vulnerable areas for search and rescue operations, evacuation, and providing humanitarian assistance.
- ❑ **Research and Development:** The government invests in research and development initiatives to improve the understanding of monsoon dynamics. This includes collaborations with scientific institutions and international organizations to enhance the country's capability to predict and manage monsoons effectively.
- ❑ **Climate Adaptation and Resilience Building:** Government policies and programs emphasize climate adaptation and resilience building in the face of changing monsoon patterns. This involves incorporating climate-smart practices in various sectors, such as agriculture, water management, and infrastructure development.

CONCLUSION

India's climate is diverse and influenced by various geographical features, including its vast size, topography, and proximity to the equator. India's climatic diversity influences various aspects of life, including agriculture, water management, and regional ecosystems. The country's climate resilience is challenged by factors such as climate change, leading to shifts in weather patterns and increasing frequency of extreme events. Sustainable climate policies and adaptation strategies are crucial for addressing these challenges and ensuring the well-being of the population.

INTRODUCTION

Pedology is the scientific study of soil, which focuses on the formation, classification, and mapping of soils, as well as their physical, chemical, and biological properties. It examines the factors that influence soil development, such as climate, topography, organisms, and parent material.

Biogeography is the interdisciplinary study of the distribution of species and ecosystems across geographical space and through geological time. It explores the interactions between living organisms and their environments, considering factors like climate, landforms, and historical events that shape the spatial patterns of biodiversity.

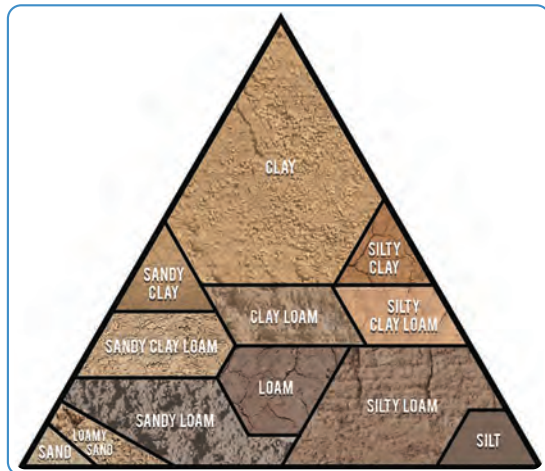


Fig. 6.1: Different Soil Textures

Soil is a combination of **rock debris and organic materials** which develop on the earth's surface. Relief, parent material, climate, vegetation, other life forms, and time are the major factors influencing soil formation.

Aside from these, human activities have a significant impact on it. Mineral particles, humus, water, and air are soil components. The exact amount of each of these determines the type of soil. Some soils are deficient in one or more of these, while others have varying combinations.

Composition: Soil is mostly made-up of **mineral or rock** particles, a small amount of **decomposed organic** matter, soil water, soil air, and living organisms. These things live together in a complex and changing way. Natural soil is affected by its **parent material, topography, its climatic condition, the physical, chemical, and biological agents** (microorganisms) in it, its usage, and the passage of time.

Components: In general, soil is made up of **four things:** Inorganic or mineral matter (which comes from the parent

material), organic matter (decayed plants and animals), air, and water.

Pedogenesis: Soil is made under certain natural conditions, and each part of the natural environment plays a role in this complicated process, which soil scientists call **pedogenesis** or soil formation. It is the natural process of soil formation that includes weathering, leaching, calcification, and other processes.

Soil is a product of several factors: the influence of climate, relief (elevation, orientation, and slope of terrain), organisms, natural vegetation, land use practices, microorganisms, and the soil's parent materials (original minerals) interacting over time. These factors do not act on soil independently or in isolation, but rather in close association, resulting in a complex network of interrelationships.

FACTORS THAT INFLUENCE SOIL FORMATION

- ❑ **Active factors:** **Climate and biotic agents** serve as the sources of energy that operate on the material to build soil.
- ❑ **Passive factors:** These include **parent material, relief, and time**, which stand for the origin of the soil-forming mass and the factors that influence it.
- ❑ **Climate:** Temperature and rainfall affect the formation of soil as they influence how well weathering, water seepage, and microorganisms work. In different climates, different types of soil can be made from different kinds of parent materials. In wet places, crystalline granites make laterite soil, and in some parts of Tamil Nadu, we can find black soil. Due to evaporation and the build-up of lime, the soil in dry and semi-dry areas is light, while the soil in the Himalayas is acidic and the decay of plants is slow.
- ❑ **Parent Rock:** The materials from which soils originate are referred to as parent materials, and, in most cases, they dictate the soil's coloration, mineral composition, and texture.
 - The physical properties of the soil that forms may or may not align with those of the parent rock in certain cases.
 - Chemical changes induced by climatic factors also impact the physical properties of the soil.
 - Surface rocks undergo weathering processes, transforming them into fine grains that form the basis for soil development.

- Within Indian conditions, the parent material is generally classified based on the types of rocks present.
- Red soils, derived from ancient crystalline and metamorphic rocks, exhibit their characteristic color due to the presence of iron oxides.
- **Calcareous and argillaceous soils** result from Cuddapah and Vindhyan rocks, which are ancient sedimentary rocks lacking metalliferous minerals.
- **Gondwana rocks** contribute to soils with low fertility and less maturity, given their status as younger sedimentary rocks.
- Deccan basalts give rise to regur or black soils, the dark colour of which is attributed to the richness of titanium, magnetite, aluminum, and magnesium in the basalts.

- ❑ **Biotic organisms:** Natural vegetation in India plays a crucial role in soil formation and development, with decayed leaf material adding humus and increasing fertility. Densely forested areas contain some of the best soils in India.
 - Vegetation helps control soil erosion, facilitates percolation and drainage, and increases mineral dissolution through carbon dioxide and acidic substances. Burrowing animals and microorganisms play a significant role in soil formation, affecting humification and mineralization.
- ❑ **Time:** Soils that have been subjected to soil formation processes over a longer length of time are deeper, more developed, and comparatively more fertile.
- ❑ **Topography or Relief:** Relief is the most important component in soil formation in areas with steep slopes, such as mountainous regions, plateau boundaries, and so on. Soil formation is hampered by rampant soil erosion on bare hillsides. **Example:** Chambal ravines, upper parts of the Himalayas with less forest cover, and so forth.

SOIL DEVELOPMENT: GENESIS OF SOIL-STRUCTURE

Processes Contributing to Soil Formation

Weathering

- ❑ **Weathering** is the process of disintegration leading to the breakdown and weakening of solid rock. It involves the transformation of larger rocks into smaller ones.
- ❑ Primary cause of **residual soil formation** in India, contributing to red and black soils.

Leaching

- ❑ Process of water percolating through soil, which carries away soluble minerals.
- ❑ In regions with heavy rainfall, like the Western Ghats, leaching forms **lateritic soils** rich in iron and aluminium oxides.

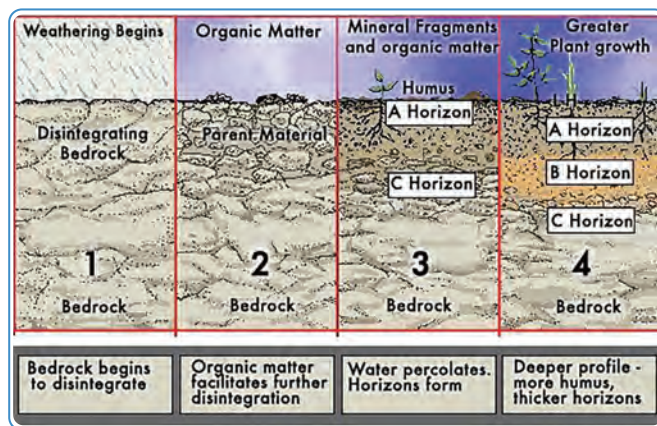


Fig. 6.2: Genesis of Soil Structure

Biological Activity

- ❑ Microorganisms and plants significantly contribute to soil formation.
- ❑ Decay of organic matter by microbes produces **humus**, enhancing soil structure and fertility.

Initiating Soil Formation

- ❑ The **genesis of soil structure** begins with the **physical and chemical disintegration** of rocks.
- ❑ Rocks exposed to the atmosphere and water percolation initiate this process.

Result of Weathering

- ❑ The fundamental outcome is the **weakening and breakdown** of solid rocks.
- ❑ Results in the creation of **smaller rock particles** from larger ones.

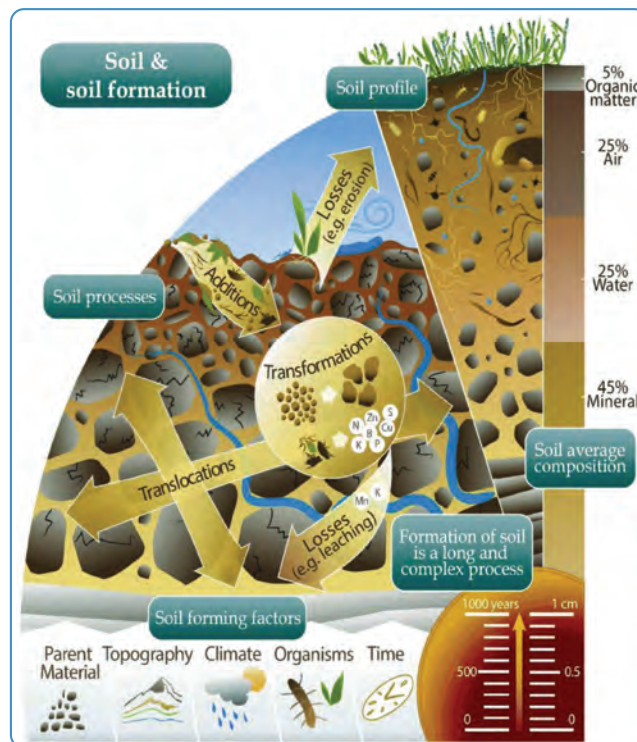


Fig. 6.3: Soil and Soil Formation

Regolith Formation

- ❑ **Regolith** ("blanket rock") covers intact rock beneath in a blanket-like manner.
- ❑ **Primary Outcome:** The primary outcome of weathering is the formation of a **layer of loose inorganic material** known as **regolith**.
- ❑ **Composition of Regolith:**
 - Comprises material weathered from underlying rock.
 - Exhibits a gradual variation in particle sizes, with larger fragments near the bedrock.
- ❑ **Particle Size Gradation:** Within the regolith, there is a gradual change in particle sizes, with the largest fragments near the bedrock.
- ❑ **Regolith Composition Variation:** Occasionally, regolith may comprise material transported from other locations, leading to significant **composition variations**.
- ❑ **Upper Regolith Differentiation:** The upper half-metre of the regolith displays distinctions, particularly in **biological and chemical processes**.

Formation of Soil

- ❑ This upper portion is recognized as **soil**, consisting of finely fragmented mineral particles.
- ❑ It represents the final outcome of weathering processes.
- ❑ **Components of Soil:** Soil comprises various components: root systems, decaying plant matter, microscopic flora and fauna, air, and moisture.

Continual Evolution

- ❑ Soil is not the end product but a stage in an ongoing sequence of **physical, chemical, and biological processes**.
- ❑ In forested regions, abundant organic matter accumulates, leading to the formation of rich, dark **forest soils**.

SOIL PROFILE

The soil profile is a **vertical section** through different layers of soil. The texture, colour, depth, and chemical composition of each layer vary.

Soil Horizon

A layer that is generally **parallel to the soil surface** and has physical properties that differ from the layers above and beneath it. In most cases, these are distinguished by obvious physical characteristics, most **notably colour and texture**.

O-Horizon

- ❑ **Composition:**
 - **Organic Material:** It is primarily composed of organic material.
 - **Components:** It includes partially decomposed or undecomposed litter such as leaves, needles, twigs, moss, and lichens.

- ❑ **Decayed Plant Matter:** Some O layers consist of partially decayed plant matter accumulating on the soil surface.
- ❑ **Location:** Found on both mineral and organic soils.
- ❑ **Hydration:** Unlike other horizons, O horizons are not consistently saturated with water for prolonged periods.
- ❑ **Topsoil Enrichment:**
 - **Rich in Humus:** Part of the top soil which is rich in organic material, especially humus.
 - **Dark Color:** Imparts a dark color to the soil.

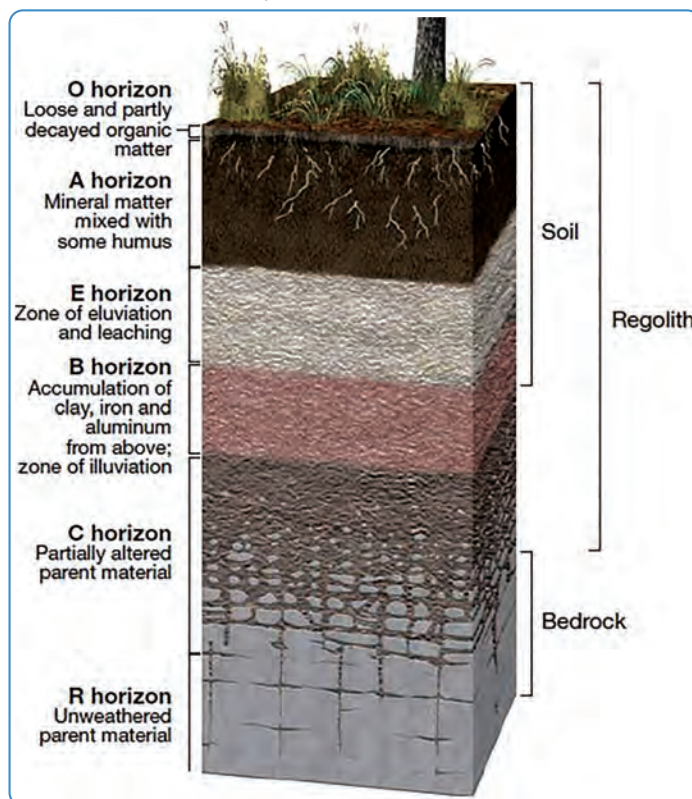


Fig. 6.4: Soil Profile

- ❑ **Importance:**
 - **Nutrient Source:** It serves as a source of nutrients for plants.
 - **Biological Activity:** It supports various biological activities in the soil.
- ❑ **Decay Variation:**
 - **Decomposition Levels:** It varies from partially decomposed to undecomposed litter.
 - **Organic Enrichment:** It contributes to the organic enrichment of the soil profile.

A-Horizon

- ❑ **Position:** The A horizon is a mineral horizon. This horizon always forms at the surface and is what many people refer to as topsoil.
- ❑ **Composition:** This layer is composed of a **mixture** of both organic and mineral matter.
- ❑ **Organic Richness:**
 - The A horizon displays a **significant accumulation** of organic matter.
 - It supports a **diverse population** of soil organisms.

- ❑ **Leaching Effects:** Within the A horizon, there is a **reduction** of substances like iron, clay, aluminum, and soluble components due to **leaching**.
- ❑ **Subsurface Variation:** In cases of pronounced depletion, a lighter-colored “**E**” **subsurface horizon** can be observed just beneath the A horizon.
- ❑ **Characteristics of A Horizon:**
 - **Texture:** The layer is soft enough to accommodate both **water and air**.
 - **Seed Germination:** This layer is crucial for the **germination of seeds**.
 - **Organism Support:** It provides **food for organisms** such as earthworms and fungi.
 - **Depleted Constituents:** Iron, clay, aluminum, organic compounds, and other soluble constituents are **depleted** in this layer.
 - **Eluviation Layer:** The A horizon is categorized as an **eluviation layer**.

Eluviation: It is the procedure by which soil constituents are washed out from the higher levels to the bottom layer by percolating water.

- ❑ **Factors Influencing Eluviation:**
 - **Rainfall:** Intensity and frequency of rainfall affect the rate of eluviation.
 - **High Temperatures:** Temperature variations contribute to the eluviation process.
 - **Vegetation Removal:** The removal of protective vegetation influences eluviation rates.
- ❑ **Material Removal in Eluviation:** In the worst cases, eluviation can remove materials that don't dissolve, like quartz, iron and aluminium hydroxides, and iron oxides. A good example of this kind of soil is laterite.
- ❑ **Application:** Eluviation is a common process in forest soils, where water percolating from the top to the bottom of the soil moves nutrients. In this way, eluviation helps the cycle of materials needed to keep forest plants alive and healthy.

E Horizon

- ❑ **E-Horizon** is characterized by the loss of a considerable amount of clay, iron oxides, and aluminum oxides, leaving behind resistant minerals like quartz in the sand and silt sizes.
- ❑ **Significance of “E”:** The term “**E**” **signifies the eluviated layer**, indicating a specific soil horizon.
- ❑ **Leaching Process:** This horizon has undergone significant **leaching of substances** like clay, iron, and aluminum oxides.
- ❑ **Resistant Minerals:** Due to leaching, there is a concentration of durable minerals, such as **quartz**, found in the sand and silt particles.

- ❑ **Occurrence:** The eluviated horizons are typically found in **well-developed, older soils**.
- ❑ **Position:** They generally occur **between the A and B horizons**, contributing to the vertical arrangement of soil layers.
- ❑ **Characteristics of E Horizon:**
 - **Residue of Minerals:** The resistant minerals like quartz are the **residue** left after leaching.
 - **Soil Development Indicator:** Presence of E horizon is an **indicator of advanced soil development**.
 - **Vertical Arrangement:** It plays a crucial role in the **vertical stratification** of soil layers.
 - **Age of Soil:** Commonly found in soils that have undergone **extensive weathering and development**.

B-Horizon

- ❑ The **B-Horizon** is a sub-surface layer that indicates the chemical or physical modifications of the original parent material.
- ❑ **Characteristics of B Horizon:**
 - **Mineral Retention:** This horizon retains minerals leached from upper horizons, contributing to its distinct composition.
 - **Subsurface Accumulation:** It is characterized by the **accumulation of various components** beneath the soil surface.
 - **Leaching Impact:** It reflects the **impact of leaching processes** from the overlying horizons.
 - **Diverse Composition:** The composition can vary, and it plays a crucial role in **soil fertility and structure**.
 - **Parent Material Transformation:** Indicates the **transformation of the original parent material** over time.
 - **Alteration of Parent Material:** It represents a layer where the **original parent material has undergone changes** either chemically or physically.
 - **Mineral Accumulation:** The B horizon **accumulates minerals** that were leached from the A and E horizons above.
 - **Accumulated Components:** This horizon accumulates **iron, clay, aluminum, and organic compounds** due to the leaching process.
 - **Subsurface Nature:** All B horizons are, or initially were, **situated beneath the surface** of the soil profile.

Illuviation is the process of water movement causing dissolved substances or particles in water to settle into lower soil layers.

- ❑ **Soil Horizon Dynamics:**
 - **Surface Horizons:** Near the surface, soil horizons are full of life, organic matter, and change rapidly.
 - **Deep Horizons:** Deeper soil horizons tend to be more stable.

❑ Influence of Rainfall:

- **Rainfall** significantly impacts soil changes over time.
- B horizons, influenced by clay minerals, iron, humus, and calcium carbonate buildup, often undergo substantial changes.

❑ Factors Leading to Illuviation:

- **Clay Minerals:** Accumulation of clay minerals in B horizons.
- **Iron:** Buildup of iron content.
- **Humus:** Presence and buildup of organic matter.
- **Calcium Carbonate:** Accumulation of calcium carbonate.

❑ Material Seepage:

- Illuviation is primarily driven by **material seepage** from the upper layers.
- Substances and particles are transported downwards by the movement of water.

❑ Overall Impact:

- Illuviation contributes to the development and transformation of soil horizons.
- It plays a crucial role in the differentiation and characteristics of various soil layers.

C-Horizon

The **C-Horizon** represents a layer where weathered parent material, particularly in sedimentary deposits, accumulates.

❑ Characteristics:

- **Accumulated Weathered Material:** This horizon **accumulates weathered parent material**, especially in sedimentary deposits.
- **Layer of Large Rocks:** Characterized by the **presence of larger, unbroken rocks**, which contributes to its distinct composition.
- **Accumulation of Soluble Compounds:** It might accumulate more **soluble compounds (inorganic substances)**.
- **Potential Modification:** The C horizon can **undergo modification**, even without clear signs of soil formation (pedogenesis).
- **Plant Growth Medium:** Despite being less developed, plant roots can penetrate into the C horizon, making it important as a **medium for plant growth**.

❑ Significance:

- **Weathered Parent Material:** It acts as a repository for weathered parent material.
- **Geological Impact:** It reflects geological processes in forming sedimentary deposits.
- **Root Penetration:** It is important for plant root penetration despite larger rocks.
- **Early Soil Formation:** It represents an early stage in soil profile development.

- **Sub-surface Composition:** It contributes to the overall composition of the soil profile.

R-Horizon

- ❑ The **R-Horizon** represents the layer of partially weathered bedrock found at the bottom of the soil profile.

❑ Characteristics:

- ❑ **Continuous Masses of Hard Rock:** Unlike preceding layers, R-Horizon mainly consists of continuous masses of hard rock.

- ❑ **Similarities to Bedrock:** Soils formed in situ will share significant similarities with this bedrock layer.

- ❑ **Underlying Depth:** Typically located underneath other profiles, often at depths exceeding 50 feet.

- ❑ **Examples of Bedrock:** Designations include rocks like granite, basalt, quartzite, and indurated limestone or sandstone.

❑ Significance:

- **Striking Similarities:** Soils formed in situ will closely resemble this layer, indicating a connection to the underlying bedrock.
- **Geological Context:** It provides insights into the geological history and composition of the area.
- **Depth Indicator:** It is positioned at the bottom, indicating underlying depths beyond 50 feet.
- **Bedrock Identification:** It helps identify bedrock types such as granite, basalt, quartzite, and indurated limestone or sandstone.

GENERAL CHARACTERISTIC FEATURES OF INDIAN SOILS

1. Formation and Distribution:

- Diverse landscape influences soil formation, with factors like rock type, climate, and vegetation.
- Alluvial soils dominate northern plains and river valleys, known for fertility.
- Black soils, rich in organic matter, found in parts of the Deccan Plateau, ideal for cotton cultivation.
- Red and yellow soils in peninsular India, often deficient in nutrients.

2. Texture and Structure:

- Vary in texture, ranging from sandy to loamy to clayey.
- Loamy soils considered ideal for agriculture due to good drainage and water-holding capacity.
- Sandy soils drain quickly with low water-holding capacity; clayey soils prone to waterlogging.

3. Chemical Properties:

- Many soils naturally acidic, requiring lime addition for optimal crop growth.
- Common nutrient deficiencies in nitrogen, phosphorus, and potassium, necessitating fertilizer use.

- Low organic matter content, emphasizing the need for organic farming practices.

4. Salinity and Alkalinity:

- Coastal areas and arid regions may have saline or alkaline soils, limiting agricultural productivity.
- Crucial to use land reclamation techniques and salt-tolerant crops in such areas.

5. Erosion and Degradation:

- Soil erosion, driven by deforestation, intensive agriculture, and overgrazing, poses a major threat to fertility.
- Essential to implement sustainable land management practices and soil conservation efforts.

CLASSIFICATION OF SOIL IN INDIA

The first scientific classification of Indian soils was made by **Voeleker (1893) and Leather (1898)**. According to them, the Indian soils may be classified into four categories, namely **alluvial, regur (black-earth), red soil, and lateritic soil**.

- ❑ **Urvara and Usara:** India has varied relief features, landforms, climatic realms and vegetation types. India has a lot of different kinds of soil because of these things. In the past, soils were put into two main groups: **Urvara and Usara**. **Urvara**, represented fertile soils and was good for growing plants, and **Usara** represented sterile soils and did not support vegetative growth.
- ❑ **Characteristics-Based Groups:** In the 16th century A.D., soils were put into groups based on their internal and external characteristics, such as texture, colour, land slope, and the amount of water in the soil.

ICAR has classified the soils of India into the following order as per the USDA soil taxonomy

S. No.	Order	Area (in Thousand Hectares)	Percentage
(i)	Inceptisols	130372.90	39.74
(ii)	Entisols	92131.71	28.08
(iii)	Alfisols	44448.68	13.55
(iv)	Vertisols	27960.00	8.52
(v)	Aridisols	14069.00	4.28
(vi)	Ultisols	8250.00	2.51
(vii)	Mollisols	1320.00	0.40
(viii)	Others	9503.10	2.92
Total			100

Fig. 6.8: ICAR classification - The Soil of India

- ❑ **Texture-Color Classification:** The main types of soil were found to be sandy, clayey, silty loam, and so on. They were red, yellow, black, etc., based on their colour. Since

the country got its independence, different agencies have done scientific surveys of the soil.

- ❑ The **Soil Survey of India** was set up in 1956, it did detailed studies of the soil in places like the Damodar Valley. The **Indian Council of Agricultural Research (ICAR)**, which is in charge of agricultural research, had the **National Bureau of Soil Survey** and the **Land Use Planning Institute** do much research on Indian soils.
- ❑ **ICAR's Soil Classification:** In order to study soil and make it comparable internationally, the **Indian Council of Agricultural Research (ICAR)** has put Indian soils into groups based on their nature and characteristics. These groups are based on the **Soil Taxonomy by U.S. Department of Agriculture (USDA)**.

The **Indian Council of Agricultural Research (ICAR)** and the **National Bureau of Soil Survey (NBSS)** have put Indian soils into four groups based on where they came from, what they are made of, what colour they are, and where they are.

1. Alluvial soils
2. Black soils
3. Red and Yellow soils
4. Laterite soils
5. Arid soils
6. Saline soils
7. Peaty soils
8. Forest soils

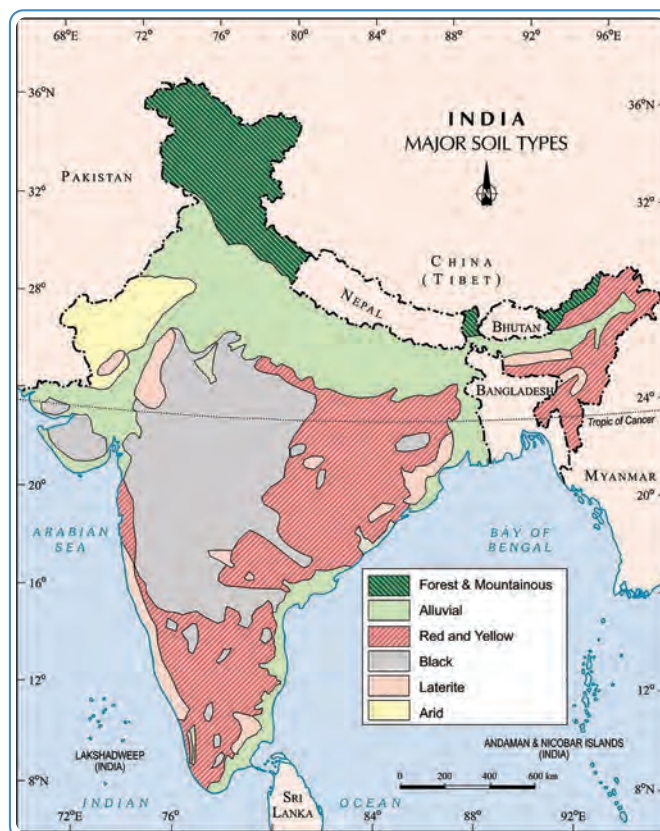


Fig. 6.5: Major soil Types

Alluvial Soil

The alluvial soil covers approximately 143.1 million square kilometres, accounting for approximately **43.4%** of the total reporting area.

- ❑ **Origin:** It originates primarily from silt deposits carried by the Indo-Gangetic-Brahmaputra Rivers and coastal regions through wave actions.
- ❑ **Source Material:** **Himalayan rocks** act as the primary source material for alluvial soils.
- ❑ **Extensive Coverage:** The alluvial soil constitutes the largest soil group, covering approximately **45.6%** of India's total area.
- ❑ **Population Support:** It is pivotal for sustaining over **40%** of India's population.
- ❑ **Immature Profile:** It generally possesses weak, immature profiles.
- ❑ **Texture:** It is pre-dominantly **loamy**, with variations of sandy and clayey compositions. Pebbly and gravelly compositions are rare.
- ❑ **Porosity and Drainage:** It is notably porous due to loamy nature, aiding in efficient drainage.
- ❑ **Kankar Beds:** Some regions harbor kankar beds along river terraces.
- ❑ **Recurrent Replenishment:** Regular replenishment results from periodic floods.
- ❑ **Chemical Traits:**
 - **Nitrogen Proportion:** Nitrogen content is typically low.
 - **Nutrient Content:** Satisfactory levels of **potash, phosphoric acid (phosphate), and alkalis (lime)**. Variable iron and lime levels.
- ❑ **Agriculture and Crops:**
 - **Suitability:** Highly suitable for agriculture due to flat and uniform characteristics.
 - **Irrigation:** Optimal suitability for both canal and well/tube-well irrigation methods.
 - **Rich Crop Varieties:** Yields abundant harvests of crops like **Rice, Wheat, Sugarcane, Tobacco, Cotton, Jute, Maize, and Oilseeds**.

- ❑ **Characteristics:** Most of the things in these soils came from the Himalayan region or were left behind by the sea when it went away. In the **Peninsular** region, they live along the east coast in the river valleys and deltas. The texture of these soils is more like sand and clay.
- ❑ **Distribution:** In the Brahmaputra valley and the middle and lower Ganga plains. There are a lot of alluvial soils in the northern plains and river valleys. Through a narrow corridor, they reach from Rajasthan to the plains of Gujarat. In the Peninsular region, they live along the east coast in the river valleys and deltas.
- ❑ **Minerals:** There are various types of alluvial soils, from sandy loam to clay. They are rich in potash but poor in phosphorus.
- ❑ **Colour:** The alluvial soils range from light grey to ash grey in colour and from sandy to silty loam in texture. Both good and bad drainage can be found in these soils.
- ❑ **Profile:** They generally have an immature profile in undulating areas and a well-developed and mature profile in levelled areas. In the lower and middle Ganga plains as well as the Brahmaputra valley, the soils exhibit a higher proportion of loam and clay.

These Soils May be Divided Into

- ❑ The **Khadar** soils are low, so they often get flooded when it rains. Because of this, the khadar is in the flood plains of the rivers and gets new silt every year. It also has long stretches of salty and alkaline blooms called "**Reh**," "**Kallar**," and "**Thur**" by the people who live there.
- ❑ The **Bhangar** soil is higher than the flood level. It is generally well-drained but contains impure calcium carbonate concretion (kankars). The texture of the soil ranges from loamy to clayey-loam. It has a good drainage system.
 - **Nature:** It is made from older alluvial soil, generally **clayey**, with characteristic kankars.
 - **Features:** It forms terrace-like structures above the flood plain.

Characteristic	Bhabar	Terai	Khadar	Bhangar
Area	Northernmost part of Gangetic plain, 8-16 km wide	Narrow track of swampy land, south of Bhabar	Riverbanks and floodplains	Made from older alluvial soil
Location	Along the Shivalik foothills	South of Bhabar	Extends along riverbanks and floodplains	Generally clayey, found above floodplain
Deposition	Alluvial fans deposited by rivers from hills	Lost rivers of Bhabar make reappearance	Regular deposition of silt and fine particles	Formed from older alluvial soil
Vegetation	Large trees with big roots due to porosity	Suitable for large trees, not ideal for agriculture	Enriched by new silt annually, suitable for crops	Terrace-like structures, suitable for agriculture

Black Soil

- ❑ Black soil covers about **15% of the total land area** of the country.
- ❑ It encompasses most of the **Deccan Plateau**.
- ❑ It is also known as **“Regur Soil”** or **“Black Cotton Soil.”**
- ❑ It retains moisture for an extended period.
- ❑ **Characteristics of Black Soils:**
 - **Clay Dominance:** **Clay-rich** (62% or more).
 - **Fertility Variability:** Upland soils are less fertile, while valley soils are more fertile.
 - **Moisture Retention:** Retains moisture effectively, aiding in **irrigation**
 - **Expansion and Contraction:** Swells significantly when moist and develops deep cracks in dry seasons.
- ❑ **Color Diversity:**
 - Black color comes from iron-rich parent rock.
 - Presence of **magnetite or iron** and black constituents.
 - Various tints of the black color.
- ❑ **Chemical Composition:**
 - Its constituents include **10% alumina, 9-10% iron oxide, 6-8% lime and magnesium carbonates**, variable potash (less than 0.5%), and low levels of phosphates, nitrogen, and organic matter.
- ❑ **Crops:**
 - It is suited for **cotton cultivation** and various crops like **wheat, jowar, linseed, virginia tobacco, castor, sunflower, and millet**.
 - **Rice and sugarcane** flourish with irrigation facilities.
 - It is utilised for centuries without substantial fertilizers or manures, exhibiting limited signs of depletion.
- ❑ **Self-ploughing:** These soils are also referred to as **“Regur Soil”** or **“Black Cotton Soil.”**
 - The black soils are typically clayey, deep, and impervious. When wet, they swell and become sticky, and when dry, they shrink. As a result, during the dry season, these soils develop large cracks.
 - As a result, there is some **self-ploughing**. Because of its slow absorption and loss of moisture, black soil retains moisture for an extended period, which aids in irrigation.
- ❑ **Distribution:** It covers most of the Deccan Plateau and parts of the states of **Maharashtra, Andhra Pradesh, Telangana, Madhya Pradesh, Gujarat, and Tamil Nadu**, covering 5.46 lakh sq km (16.6% of total area). In the upper parts of the Godavari and Krishna rivers and in the northwestern part of the Deccan Plateau, the black soil is very deep.

Red and Yellow Soil

Red soil forms on crystalline igneous rocks in low-rainfall areas of the Deccan Plateau's eastern and southern reaches. It covers **about 18.5 %** of the total land area of the country.

- ❑ **Prominence:** Red soils, along with their minor variations, constitute a significant soil group in India.
- ❑ **Parent Rock:** Major parent rocks are crystalline and metamorphic, including acid granites, gneisses, and quartzite.
- ❑ **Distinctive Color:**
 - These soils appear red due to the presence of **iron oxides**.
 - Exhibit a yellow hue when found in a hydrated state.
- ❑ **Chemical Composition of Red Soils:**
 - **Acidic Nature:** Red soils are generally **acidic** due to the characteristics of their parent rocks. They contain a fair amount of **alkali content**.
 - **Deficiencies:** These soils are deficient in **lime, magnesia, phosphates, nitrogen, and humus**.
 - **Potassium Content:** Red soils are fairly rich in **potash and potassium**.
- ❑ **Texture Variation:** Red soils can exhibit a range of textures from **sand to clay**, with a majority being loams.
- ❑ **Topographical Influence:** On uplands, red soils tend to be **poor, gravelly, and porous**, while in lower areas, they are **rich, deep, dark, and fertile**.
- ❑ **Distribution of Red Soils:**
 - Red soils are commonly found in regions with **low rainfall**.
 - They cover an area of about **3.5 lakh sq km (10.6%)** in India.
 - They have an extensive presence in states like **Tamil Nadu, Karnataka, Maharashtra, Telangana, Andhra Pradesh, Madhya Pradesh, Chhattisgarh, Odisha, Bihar, West Bengal, Uttar Pradesh, Rajasthan, and parts of Northeastern states** as well as in the southern reaches of the Ganga plain.
- ❑ **Crops in Red Soils:**
 - Due to their mostly **loamy nature**, red soils have lesser water retention compared to black soils.
 - With the appropriate use of fertilizers and irrigation, red soils can yield crops like **cotton, wheat, rice, pulses, millets, tobacco, oilseeds, potatoes, and fruits**.

Laterite Soil

Laterite was named after the Latin word **“later”** which means **“brick”**. Laterite soils form in areas with high temperatures and rainfall. These are the results of heavy leaching from tropical rains. Laterite soils are commonly cut into bricks

for use in building houses. It accounts for **about 3.7%** of the total area of the country.

❑ **Formation:**

- This soil is the end product of **weathering** under high temperature and heavy rainfall conditions.
- It results from alternating wet and dry periods.

❑ **Leaching Effect:**

- Heavy rainfall leads to **leaching**, washing away nutrients like **lime and silica**.
- Leaves behind a soil rich in **iron and aluminum oxide compounds**.

❑ **Latin Origin:**

- Term 'Laterite' comes from Latin, meaning **"brick."**
- It hardens significantly when it loses moisture.

❑ **Red Color:**

- Due to higher content of **iron and aluminum compounds** and less clay.
- Contains more gravel from red sandstones.

❑ **Chemical Composition:**

- Rich in **bauxite or ferric oxides**.
- Low in organic matter, nitrogen, phosphate, and calcium.
- Excess amounts of **iron oxide and potash**.
- Higher **humus content** in wetter regions.
- Lime and silica are washed away by rain, leaving behind soils that are full of iron oxide and aluminium compounds. Bacteria that do well in hot places quickly eat up the humus in the soil. The dirt is rougher and more crumbly, and it tastes sour.
- **Minerals:** Laterites are most often made up of quartz, zircon, and oxides of titanium, iron, tin, aluminium, and manganese that are left after the weathering process. Quartz is the mineral that is found most often in the parent rock.

❑ **Distribution of Laterite – Lateritic Soils:**

- It covers around **2.48 lakh sq km** in India.
- It is commonly found in Karnataka, Kerala, Tamil Nadu, Madhya Pradesh, and hilly Odisha and Assam.
- It is predominant on **Western Ghats' summits, Eastern Ghats, Rajmahal Hills, Vindhyan, Satpuras, and Malwa Plateau**.
- Also found at lower levels and in valleys in various parts of the country.
- Well-developed in **south Maharashtra, parts of Karnataka**, and scattered in other regions.

❑ **Crops in Laterite – Lateritic Soils:**

- Due to intensive leaching, these soils lack fertility.
- It supports **plantation crops** like tea, coffee, rubber, cinchona, coconut, areca nut, etc.
- It is suitable for **grazing grounds and scrub forests**.

- Laterites are thus unsuitable for cultivation; however, manure and fertiliser application is required to make the soils fertile for cultivation.
- Red laterite soils in Tamil Nadu, Andhra Pradesh, and Kerala are better suited for cashew nut cultivation. Tapioca and cashew nuts, tea, coffee etc are other examples.

❑ **Economic Value of Laterite – Lateritic Soils:**

- The soil serves as valuable **building materials**.
- Easily cut into bricks or blocks, but **harden significantly** when exposed to air.
- Durable due to substantial weathering.

Arid Soils

Soils in dry areas are made up of sand. Some places contain so much salt in the water that it has to be evaporated to get regular salt. These come from nearby rocks trying to break apart, and most of them are blown from the coast and the Indus Valley. These are **made of salt**. In some places, there is so much salt in the water that **common salt can be made by letting the salty water evaporate**.

❑ **Composition:** It is primarily composed of **Aeolian sand (90 to 95%)** and a smaller proportion of clay (5 to 10%).

❑ **Coverage:** It encompasses an area of **1.42 lakh sq km (4.32%)**, desert soils are a notable presence.

❑ **Inhibiting Soil Growth:** High sand content hampers soil growth, leading to **desertification** in nearby soils due to the intrusion of desert sand carried by the wind.

❑ **Fertility and Limitations:**

- Despite sandy nature, these soils are fertile, with **water scarcity** being the primary constraint.
- Reclamation is possible through effective **irrigation measures**.

❑ **Distribution of Arid – Desert Soils:**

- **Geographical Presence:** Found in arid and semi-arid regions encompassing **Rajasthan, Punjab, and Haryana,,** extending to the **Rann of Kutch in Gujarat**.
- Sand transported by southwest monsoon winds from the **Indus basin and the coast**.
- Coastal areas of **Odisha, Tamil Nadu, and Kerala** also have sandy soils without clay.

❑ **Chemical Properties of Arid–Desert Soils:**

- **Organic Matter:** Typically low, exhibiting varying degrees of **alkalinity and soluble salts** such as calcium carbonate.
- **Calcium Distribution:** Increases downward, with subsoil containing significantly higher calcium levels.
- **Phosphate and Nitrogen:** Despite low original nitrogen levels, these soils often have high phosphate content similar to normal alluvial soils. Some nitrogen is available in the form of **nitrates**.

- Kankar layers are in the lower horizons of the soil because the amount of calcium increases as you go down. When the “**Kankar**” layer forms in the bottom horizons, it makes it harder for water to get into the soil.

❑ **Crops of Arid – Desert Soils:**

- Usually not good for growing many crops due to low moisture content and buildup of soluble salts.
- Management and sufficient water can make these soils productive.
- **Fertility and Irrigation:** Enhanced by phosphates and nitrates, given sufficient **moisture availability**.
- Proper irrigation can potentially **reclaim these soils**.
- Support crops that are **drought-resistant and salt-tolerant**, including barley, cotton, millets, maize, and pulses.



IGNITE YOUR MIND

Fun fact: Rajasthan is recognized for its distinctive terracotta-colored soils, resulting from the erosion of volcanic rocks, contributing a unique reddish tint to the terrain. Assess the challenges faced by farmers cultivating crops in terracotta soil regions. What strategies can be employed to enhance agricultural productivity in these areas?

Saline and Alkaline Soils

They are also referred to as Usara soils. These soils are top soil impregnated with saline and alkaline efflorescence, indicating the presence of salts, infertile and incapable of supporting vegetative growth. They have more salts, owing to the dry climate and poor drainage. They can be found in arid and semi-arid regions, as well as waterlogged and swampy areas. These soils have developed in places that are dry and don't have good drainage. These are places that get a little more rain than desert soils, but not much.

- ❑ **Composition:** Their composition varies from sandy to loamy. The fertile alluvial soils are becoming saline in areas of intensive cultivation with excessive irrigation, particularly in green revolution areas. In coastal areas, these soils are also made when salty water spreads out on the land during high tide. Also, seawater seeping into the deltas helps salty soils grow there. Un-decomposed rock fragments undergoing weathering contribute to the formation of **sodium, magnesium, calcium salts, and sulfurous acid**.
- ❑ **Distribution:** Saline soils are more common in western Gujarat, eastern coast deltas, and Sunderban areas of

West Bengal. They live in dry and semi-dry areas, as well as in areas with a lot of water and swamps especially in parts of Andhra Pradesh, Telangana, Karnataka, Bihar, Uttar Pradesh, Haryana, Punjab, Rajasthan, and Maharashtra. Found in canal-irrigated areas and regions with high sub-soil water tables.

- ❑ **Minerals:** They are deficient in nitrogen and calcium. Saline soils have a higher proportion of **sodium, potassium, and magnesium**.
- ❑ **Salt Percolation:**
 - In regions with a low water table, salts percolate into the subsoil.
 - In well-drained areas, salts are carried away by flowing water.
- ❑ **High Sub-soil Water Table:**
 - Regions with a **high sub-soil water table** experience capillary action transporting injurious salts from below due to evaporation during dry seasons.
- ❑ **Colour:** Saline soils are easy to spot in the field because crops grow in spots and there are often white salt breads on the surface. When the salt problem isn't too bad, plants that are going to grow often have a **blue-green tint**. In saline areas, cereal or forage crops may grow with barren spots and plants that are too short.
- ❑ **Cultivation:** Near the coast, there are a lot of **coconut trees**. As was already stated, farming **salt-tolerant crops like dhaincha, berseem**, and other grain legumes may also help bring these soils back to life.
- ❑ **Coastal Impact:** **Gulf of Khambhat** has been affected by tides carrying salt-laden deposits. Saline seawater infiltration degrades soil quality along coastlines.
- ❑ **Coastal Degradation:** Low-lying areas of coastal Andhra Pradesh and Tamil Nadu face this kind of soil degradation.

Peaty Soils

These are found in places with lots of rain and high humidity, where plants grow well. This kind of soil is found in places with the most water.

- ❑ **Characteristics:** As a result, a large amount of dead organic matter accumulates in these areas, providing the soil with rich humus and organic content. Organic matter in these soils can reach 40-50 per cent. Characterized by being black, heavy, and highly acidic in nature and formed under conditions of submergence, resulting in **heavy, dark, and acidic soils**.
- ❑ **Distribution:** In many places, they are also acidic. It is common in the coastal parts of West Bengal, Orissa, Tamil Nadu and Kottayam and Alappuzha districts of Kerala (referred to as “**Kari**”), as well as in the northern part of Bihar and the southern part of Uttaranchal. Affected

areas also include the coasts of Odisha and Tamil Nadu, the Sunderbans in West Bengal, Bihar, and the Almora district in Uttarakhand.

- ❑ **Minerals:** They are low in potash and phosphate.
- ❑ **Colour:** These soils are typically dense and dark in colour.
- ❑ **Cultivation:** During the rainy season, most peaty soils are saturated, but as soon as the rains stop, they are converted over to paddy farming.

Forest Soils

As the title indicates, forest soils form in forest areas where there is enough rain. Soils are different in how they are put together and how they feel depending on where they form in the mountains.



Fig. 6.6: Forest Soils

- ❑ **Heterogeneous Composition:**
 - Influenced by factors like **parent rocks, ground configuration, and climate**, resulting in diverse characteristics.
 - Formation governed by the **deposition of organic matter from forests**, changing with parent rocks, ground configuration, and climate.
 - Significant differences even in close proximity.
- ❑ **Characteristics:** On the sides of valleys, the soil is sandy and muddy, while on the upper slopes, it is coarse-grained.
 - They are acidic and do not have much humus. In the snow-covered parts of the Himalayas, they get worn down.
 - The lower valleys have nutrient-rich soils. These have serious consequences for other components of the system, of which they are important parts.
- ❑ **Distribution:** These soils cover around **2.85 lakh square kilometres or 8.67% of India's total land area**. Such soils are usually found in valleys, less steep slopes, and slopes facing north in the Himalayan area. The south-facing slopes are pretty steep and vulnerable to denudation, and so do not sustain soil formation. Forest soils can also be discovered in the **Western and Eastern Ghats**.
- ❑ **Minerals:** Humus is present in forest soils. They lack potassium, phosphorus, and lime and they need fertilisation.

- ❑ **Colour:** The colour of the soil changes with the vegetation. The soil in some sections is light grey (dry), whereas in others it is darker and damp.
- ❑ **Cultivation:** In the peninsular forest area, they are good for farms that grow **tea, coffee, spices, and tropical fruits**. In the Himalayan forest, **wheat, maize, barley, and temperate fruits** are grown.

Other Types of Soils Classified by ICAR

ICAR has classified mountain and sub montane soil as other types:

- ❑ **Mountain Soils:**
 - **Formation:** Found in the Himalayas and other mountainous regions, these soils are formed by the weathering of rocks and the accumulation of organic matter from forest vegetation.
 - **Properties:** Texture varies depending on altitude and slope. Upper slopes typically have coarse-grained, sandy soil, while valleys and lower slopes have loamy and silty soil. In snowbound areas, the soil is acidic and has low humus content due to denudation.
 - **Distribution:** Covers a vast area including the Himalayas, Eastern Ghats, Western Ghats, and other mountain ranges.
 - **Suitability:** Generally not very fertile due to leaching and erosion on steep slopes. However, valleys and lower slopes can support agriculture with proper management. Suitable for crops like apples, pears, peaches, and potatoes.
- ❑ **Sub-Montane Soils:**
 - **Formation:** These soils are formed by the deposition of eroded material from the Shiwaliks and lower Himalayas in the Terai region, stretching from Jammu and Kashmir to Assam.
 - **Properties:** Rich in nutrients and well-drained due to their alluvial origin. However, prone to soil erosion due to their hilly terrain.
 - **Distribution:** Found in the foothills of the Himalayas and other mountain ranges, particularly in the northern and eastern parts of India.
 - **Suitability:** Highly fertile and support a variety of crops like rice, sugarcane, jute, tea, and vegetables. However, erosion control measures are crucial for sustainable agriculture.

Karewa soils are the **lacustrine deposits** in the **Kashmir Valley** and the **Bhadarwah Valley**. Fine silt, clay, and boulder gravel are the composition of Karewa soil. They are characterized by **fossils**. These soils are mainly devoted to the cultivation of saffron, almonds, apple, walnut, etc.

SUMMARY: SOILS IN INDIA

Soil type	Area (Million Hectares)	Coverage (%)	Composition/ Characteristics	Regions	Crops
Alluvial Khadar	143.1	43.36	Rich in phosphoric acid, poor in potash	Indus Ganga– Brahmaputra plains, Narmada– Tapi Plains	Wheat, Maize, Sugarcane, Pulses, Oilseeds, Berseem (Fodder)
Red (Omnibus group)	61.0	18.49	Ferric oxide gives it red colour; deficient in lime, phosphate, manganese	Western Tamil Nadu, Karnataka, southern Maharashtra, Andhra Pradesh, Chattisgarh, Jharkhand, Odisha, some areas of Bundelkhand, Mirzapur, Sonbhadra, Banswara, Bhilwara, Udaipur	Wheat, Cotton, Pulses, Tobacco, Millets, Oilseeds, Potato, Orchards
Black (Regur); internally known as Tropical Chernozems	49.8	15.09	Iron, lime, calcium, potassium, aluminium, magnesium; deficient in nitrogen, phosphorous, organic matter	Gujarat, Maharashtra, western Madhya Pradesh, north-western Andhra Pradesh, Karnataka, Tamil Nadu, Rajasthan, Chattisgarh, Jharkhand up to Rajmahal	Cotton, Pulses, Millets, Linseed, Castor, Tobacco, Sugarcane, Citrus Fruits And Vegetables
Mountain	18.2	5.51	Acidic and low humus content	Mid-latitude zone of Himalaya, regions of Assam, Darjeeling, Uttarakhand, Himachal Pradesh, Jammu and Kashmir	Maize, Rice, Legumes, Fodder, Orchards
Desert	14.6	4.42	High percentage of soluble salts, low organic matter, varying percentage of calcium carbonate	Rajasthan, West of Aravallis, northern Gujarat, Saurashtra, Kutch, western parts of Haryana, southwestern parts of Punjab	Bajra, Pulses, Guar, Fodder And Lesser Water Requiring Crops Like Aloe Vera.
Laterite	12.2	3.70	Rich in iron and aluminium; poor in nitrogen, potash, potassium, lime, organic matter	Western Ghats, Eastern Ghats, Rajmahal Hills, Satpura, Vindhya, Odisha, Chattisgarh, Jharkhand, West Bengal, North Cachar Hills of Assam, Garo Hills of Meghalaya	Rice, Ragi, Sugarcane, Cashew Nuts
Submontane	5.7	1.73	Neutral to alkaline	Jhelum and Indus Valley of Tarai region of submontane from J & K to Assam	Rice
Snowfields	4.0	1.21	Covered with ice and glacier	Greater Himalayan Region, Karakoram, Ladakh, Zaskar	Unsuitable For Cultivation of Crop

Soil type	Area (Million Hectares)	Coverage (%)	Composition/ Characteristics	Regions	Crops
Grey and Brown	3.6	1.09	Iron oxide	Rajasthan and Gujarat	Cotton And Oilseeds
Peaty and marshy	–	–	Rich in organic matter, highly saline, deficient in phosphate and potash	Kottayam, Alappuzha (Kerala), Sundarban Deltas of Mahanadi, Krishna, Kaveri, Rann of Kutch	Rice Cultivation
Karewa	–	–	Characterised with fossils of mammals and peat	Kashmir Valley, Bhadarwah Valley of Doda district of Jammu	Saffron, Almond, Walnut, Apple, Orchards
Red and Black	17.8	–	–	Isolated parts of Bundelkhand, East of Aravallis in Rajasthan, Gujarat	Maize, Bajra, Millets, Pulses, Oilseeds

SIGNIFICANCE OF SOIL

- ❑ **Nutrient Cycling:** Soil provides essential nutrients for plants, such as nitrogen, phosphorus, and potassium, cycle through the soil. The soil-plant system recycles these nutrients, which are vital for plant growth.
- ❑ **Food Security & Agriculture:** Soil is the basis of most food production in the world, as crops grow on it. High agricultural productivity depends on healthy soil.
- ❑ **Biodiversity Support:** Soil hosts a diverse range of organisms, such as bacteria, fungi, insects, and other microorganisms. These organisms are important for the functioning and balance of ecosystems.
- ❑ **Water Filtration:** Soil filters water naturally, removing impurities and pollutants as water passes through it. This helps keep water clean and safe from groundwater contamination.
- ❑ **Carbon Sequestration:** Soil stores carbon in organic matter, which is part of the global carbon cycle. This helps reduce the effects of climate change by lowering the amount of carbon dioxide in the air.
- ❑ **Infrastructure Support:** Soil provides various materials for construction, such as clay, sand, and gravel. These materials are necessary for building infrastructure and fulfilling the needs of construction projects.
- ❑ **Biodiversity Support:** Soil is home to a vast array of micro-organisms, invertebrates, and other organisms, all of which contribute to the health of the ecosystem.
- ❑ **Erosion Prevention:** Soil prevents erosion by stabilizing land and reducing the loss of fertile topsoil.
- ❑ **Plant Disease Regulation:** Healthy soil regulates plant diseases by hosting beneficial microorganisms that suppress harmful pathogens.
- ❑ **Medicinal Resources:** Certain soils contain valuable minerals for medicinal purposes, contributing to traditional medicine and pharmaceuticals.
- ❑ **Waste Decomposition:** Soil decomposes organic matter, including waste materials, recycling nutrients and reducing environmental impact.
- ❑ **Recreation and Aesthetics:** Soil provides the foundation for recreational activities and contributes to landscape aesthetics.
- ❑ **Habitat for Soil Organisms:** Soil serves as a habitat for various organisms, enhancing overall biodiversity in ecosystems.
- ❑ **Natural Reservoir of Water:** Soil acts as a natural water reservoir, regulating availability, and sustaining vegetation and groundwater levels.
- ❑ **Cultural and Historical Significance:** Certain soils hold cultural and historical importance, preserving ancient civilizations and traditional practices.
- ❑ **Bioenergy Production:** Soil contributes to bio-energy production through crops used as feedstocks for biofuels, aligning with sustainable energy practices.
- ❑ **Public Health:** Soil plays a crucial role in public health by regulating pathogens, reducing dust, and preventing waterborne diseases.
- ❑ **Economic Value:** Healthy soil supports productive agriculture, contributing to economic growth and livelihoods. Degraded soil can lead to economic losses.

PROBLEMS RELATED TO THE INDIAN SOILS

Human activities such as deforestation, overgrazing of animals, and unscientific use of agricultural land, on the other hand, damage and destroy soils.

Soil Erosion

- ❑ The **removal of topsoil** is referred to as soil erosion. Soil erosion is becoming more of a problem in many parts of India.

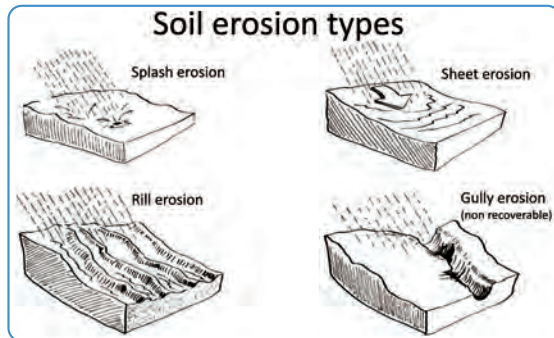


Fig. 6.7: Soil erosion types

- ❑ **Sheet** erosion occurs when the topsoil is removed, and gully erosion occurs when run-off creates gullies. Soil erosion is a widespread issue in India. Water is the primary agent of soil erosion in areas with heavy rainfall, whereas wind is the primary agent of soil erosion in arid and semi-arid areas.
- ❑ The National Bureau of Soil Survey & Land Use Planning (NBSS & LUP) estimates that soil erosion extends to about 119.2 million hectare area in India.
- ❑ The average annual rate of soil erosion in the country is 16.35 tonnes per hectare i.e. 5334 million tonnes per year.
- ❑ Water, wind, sea waves, glaciers, and shifting cultivation are the primary causes of soil erosion.
- ❑ **Gully erosion** is also prevalent in the Shiwalik tracts of Punjab, Haryana, Himachal Pradesh, Jammu and Kashmir, Uttarakhand, Uttar Pradesh, and along the southern Himalayan slopes, as well as the Western and Eastern Ghats.

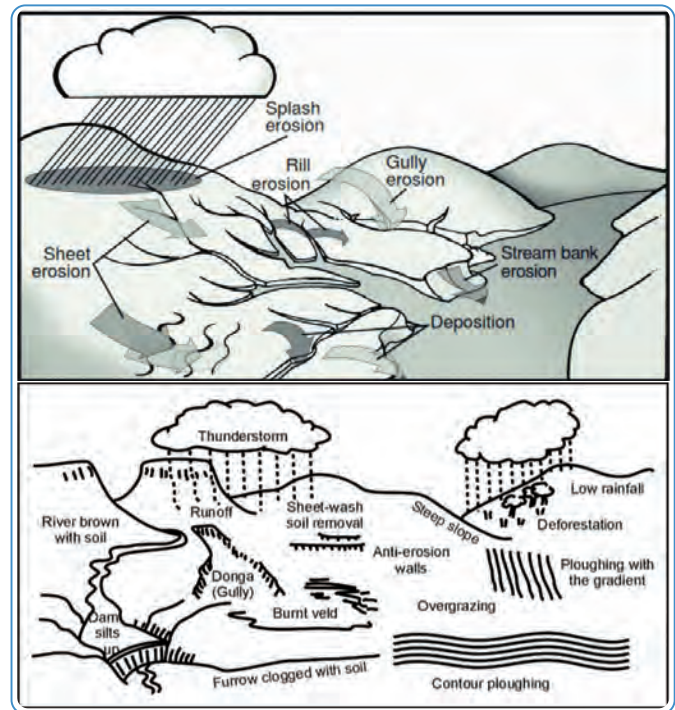


Fig. 6.8: Soil Erosion and Its mitigation

- ❑ **Wind erosion** is prevalent in arid and semi-arid regions of Rajasthan, Haryana, Punjab, western Madhya Pradesh, and Gujarat. Wind erodes soil along Peninsular India's coastal plains.
- ❑ The **tidal waters** of the Arabian Sea and the Bay of Bengal cause significant damage to coastal soils. Kerala, Tamil Nadu, Andhra Pradesh, Odisha, West Bengal, and Gujarat all have very eroded beaches.
- ❑ India loses **5,334 million tonnes** of soil every year because of soil erosion, which is caused by the careless and excessive use of fertilisers, insecticides, and pesticides over the years.

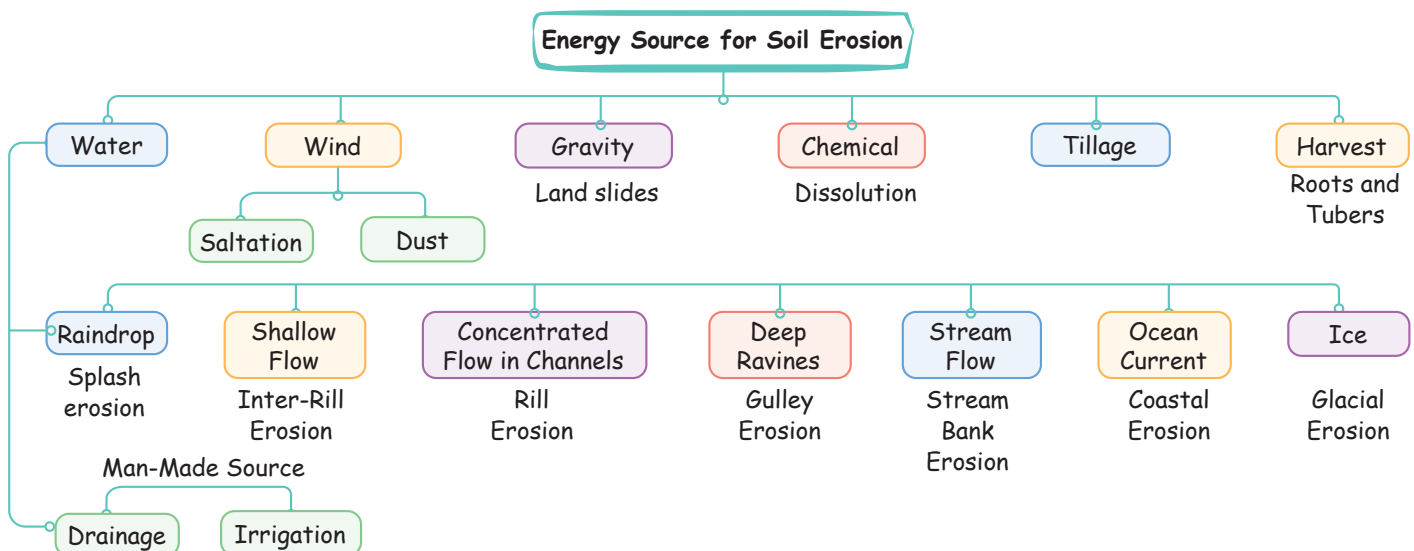


Fig. 6.9: Energy Source for Soil Erosion

Mitigating Soil Erosion

- ❑ **Afforestation:** Planting trees and shrubs on barren lands helps in binding the soil with their roots, reducing erosion by wind and water.
- ❑ **Contour Farming:** This practice involves ploughing and planting crops along the contours of the land, rather than straight up and down the slopes. This helps to break up the flow of water and prevent it from washing away soil.
- ❑ **Terracing:** Creating terraces on steep slopes helps to break down the slope into smaller, more manageable sections, reducing the velocity of runoff water and preventing soil erosion.
- ❑ **Cover Cropping:** Planting cover crops, such as legumes and grasses, between cash crops helps to protect the soil from erosion and improve soil fertility.

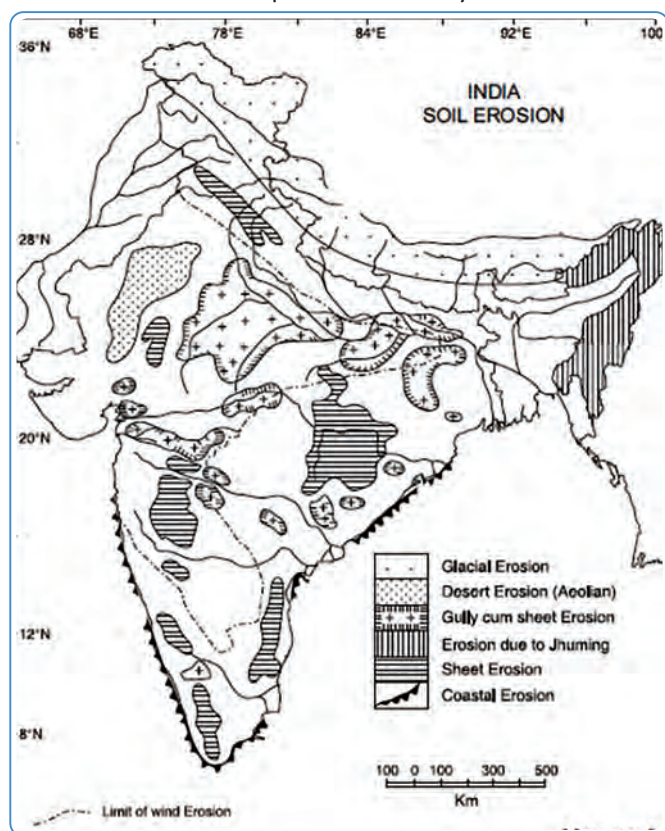


Fig. 6.10: India Soil Erosion

- ❑ **Mulching:** Applying a layer of organic matter, such as straw or wood chips, to the soil surface helps to retain moisture, suppress weeds, and prevent erosion.
- ❑ **Bunding:** Constructing earthen embankments across slopes helps to trap runoff water and prevent it from carrying away soil.
- ❑ **Gully Plugging:** Filling gullies with stones, brushwood, or other materials helps to stabilize the soil and prevent further erosion.
- ❑ **Check Dams:** Building small dams across streams and gullies helps to slow down the flow of water and trap sediment, preventing it from reaching rivers and reservoirs.

- ❑ **Stone Walls:** Constructing stone walls along slopes can provide a physical barrier to erosion and help to stabilize the soil.



Fig. 6.11: Sustainable Soil Management

Water-Logging

Waterlogging occurs when the **water table rises** to the point where soil pores in the root zone of a crop become **saturated**, resulting in a restriction of normal air circulation, a decrease in oxygen levels, and an increase in carbon dioxide levels.

- ❑ It is considered that **average every year, 75 lakh hectares of land is affected** of land will be flooded in total. Around the world, tropical swamps, rice fields, and floodplains make up about 14, 12, and 10% of all waterlogged land, respectively. Odisha, West Bengal, Bihar, and Uttar Pradesh have the most waterlogged soil in India, and the total area is estimated to be one million hectares.

Mitigating Waterlogging

- ❑ **Improve Drainage Systems:** This is the most crucial step. It involves building and maintaining proper drainage channels, culverts, and pumps to remove excess water from low-lying areas.
- ❑ **Desilting of Existing Drains:** Over time, drains get clogged with silt and debris, reducing their effectiveness. Regular cleaning and desilting of existing drains is essential.
- ❑ **Urban Planning:** Urban planning should consider the natural drainage patterns of an area and avoid building in low-lying areas or blocking natural drainage channels.

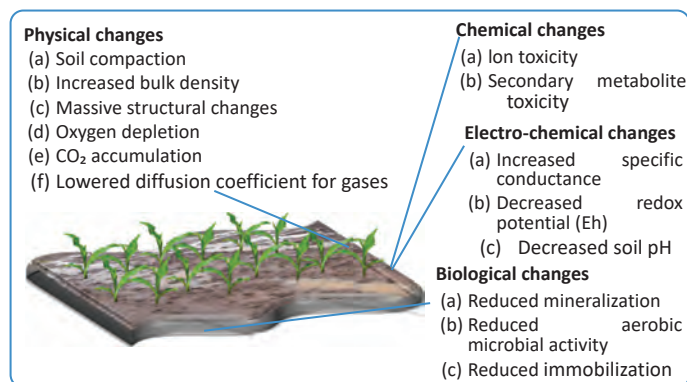


Fig. 6.12: Waterlogging effects on Soil

- ❑ **Increase Green Cover:** Trees and plants help absorb rainwater and improve soil drainage. Planting trees and creating green spaces in urban areas can significantly reduce waterlogging.
- ❑ **Wetland Restoration:** Wetlands act as natural sponges, soaking up excess water and releasing it slowly over time. Restoring degraded wetlands can help mitigate waterlogging.
- ❑ **Real-time Monitoring Systems:** Installing sensors and monitoring systems can provide early warning of flooding and allow authorities to take timely action.
- ❑ **Use of Permeable Pavements:** Replacing traditional pavements with permeable materials like porous concrete or gravel can allow water to drain through into the ground.
- ❑ **Rainwater Harvesting:** Collecting and storing rainwater for later use can reduce the amount of water that runs off and causes flooding.

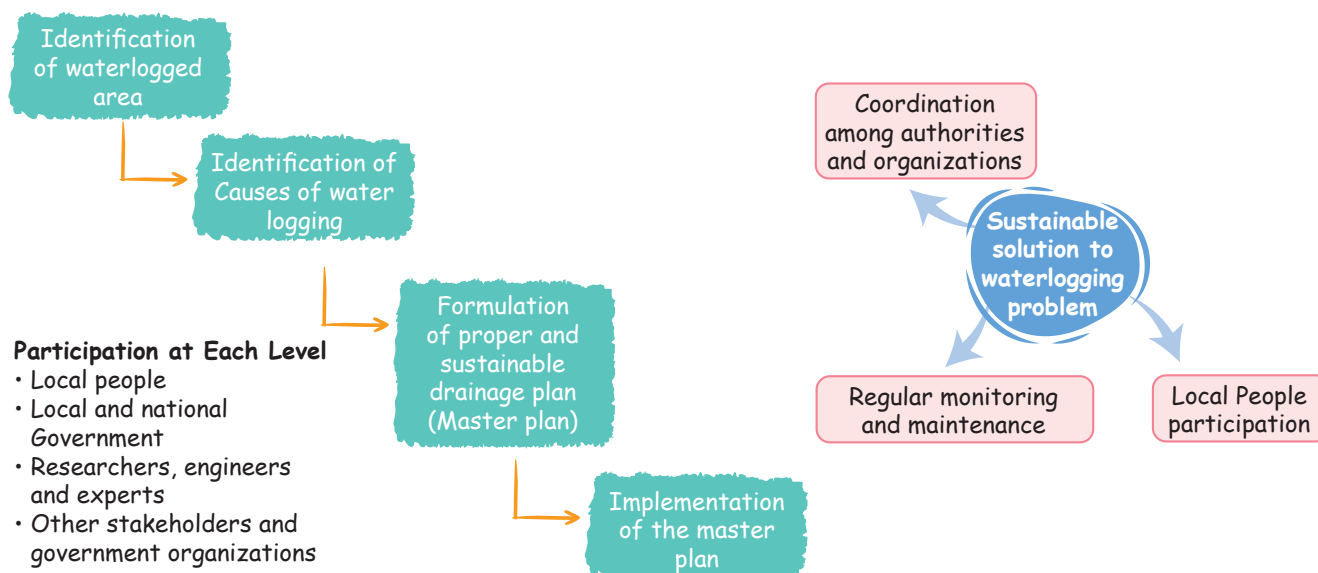


Fig. 6.13: Sustainable solution to waterlogging problem

Declining Soil Fertility

The natural fertility of the soil is rapidly **depleting** due to centuries of use and multiple cropping without following the agricultural land. Farmers in Punjab, Haryana, and western Uttar Pradesh frequently complain about their soils' declining fertility. Indeed, unscientific crop rotation (wheat and rice) has significantly depleted soil fertility in India's Great Plains over several decades.

A National Database on Land Degradation prepared by the Indian Space Research Organisation in 2016 shows that around 120.7 million hectare (mha), or 36.7 per cent of India's total arable and non-arable land, suffers from various forms of degradation with water erosion being its chief contributor.

Improving Soil Fertility

- ❑ **Composting:** Composting kitchen scraps, yard waste, and manure creates nutrient-rich humus that improves soil structure, aeration, and water holding capacity.
- ❑ **Vermicomposting:** Using earthworms to break down organic matter into vermicompost, a potent fertilizer rich in nutrients and beneficial microbes.

- ❑ **Green Manuring:** Growing cover crops like legumes and incorporating them into the soil adds nitrogen, organic matter, and suppresses weeds.
- ❑ **Crop Rotation:** Planting different types of crops in succession helps to break pest and disease cycles and replenish nutrients in the soil.
- ❑ **Precision Agriculture:** Using GPS and sensor technology to apply fertilizers and water only where and when they are needed, reducing waste and environmental impact.
- ❑ **Agro-forestry:** Integrating trees and shrubs into agricultural landscapes provides shade, windbreaks, and habitat for beneficial insects.

Soil Degradation

- ❑ Soil degradation can be defined broadly as a decline in soil fertility caused by **erosion** and **misuse** when the nutritional status of the soil deteriorates and the depth of the soil decreases. Soil degradation is the primary cause of India's depleting soil resource base.

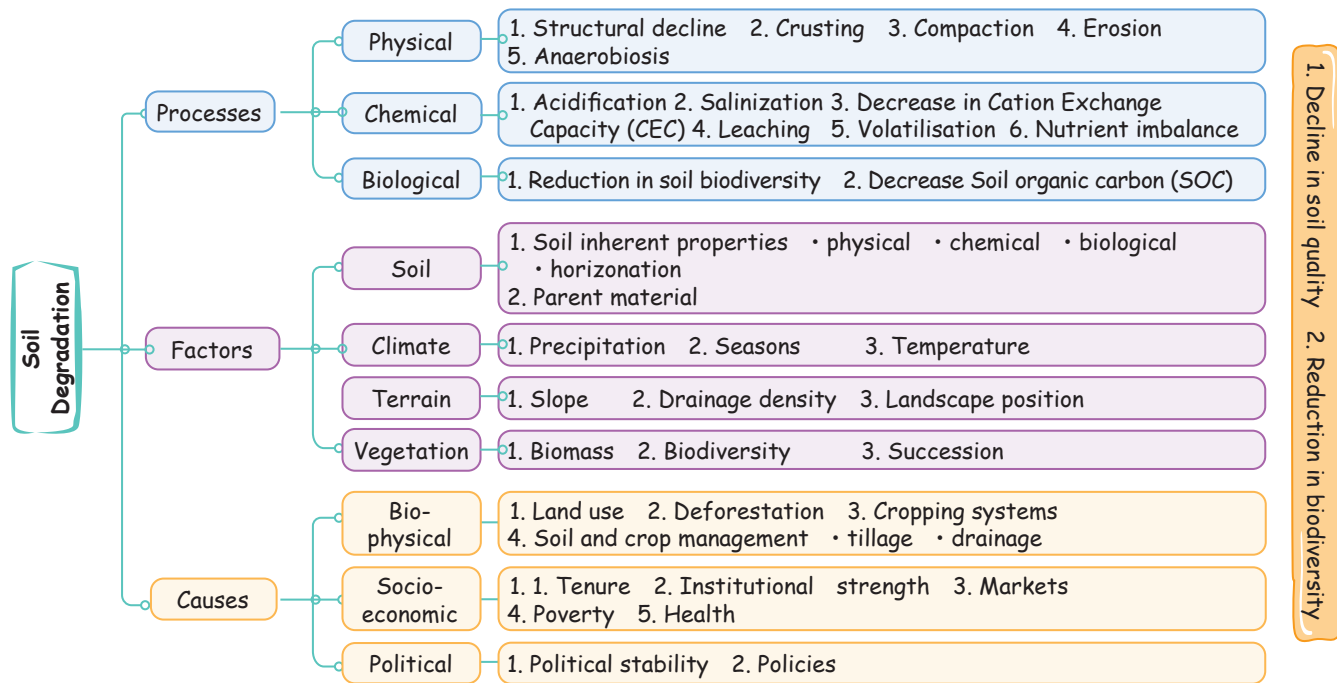


Fig. 6.14: Soil Degradation

- ❑ The degree of soil degradation varies depending on the topography, wind velocity, and amount of rainfall. Every year, India loses millions of tonnes of soil and nutrients to agents of soil degradation, reducing our national productivity. **Soil erosion and salinization** are two major causes of soil degradation.
- ❑ It is assumed that out of India's **329 Mha** total geographical area (TGA), 179.9 M ha (60.47% of TGA) is used for agriculture and 120.4 M ha is degraded by one or more types of degradation, which is hurting the country's productive resource base. It is thought that more than 5,000 tonnes of topsoil are lost every year. 30% of the total area that has been eroded, or about 1600 million tonnes, is always being lost to the sea.

SDG 15

SDG 15 aims at protecting, restoring and promoting sustainable use of terrestrial ecosystems, sustainable manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

Soil Salinity

- ❑ Soil salinity refers to the **presence of excessive amounts of soluble salts in the soil**, primarily sodium and chloride ions. These salts can originate from natural sources like weathering rocks, seawater intrusion, or tidal floods, or from human activities such as irrigation with saline water, excessive fertilization, and deforestation.
 - High salt concentration in the soil disrupts its natural balance and hinders plant growth.
 - India faces a significant challenge with soil salinity, affecting around 6.73 million hectares of land, roughly equivalent to the size of Kerala.

- ❑ In places with little rain, the soil will become salty or alkaline. One such place is **Rajasthan**. When sea water gets into the land, it makes the soil salty. When there is kankar (calcium carbonate) under the soil, water pools. This can be seen in the **Indira Gandhi Canal** areas of Rajasthan. If the soil is full of salt, it will spread to the rest of the land. Leguminous crop cultivation following a soil-exhausting crop can improve soil fertility in affected areas.

Causes of Soil Salinity

❑ Natural Causes:

- **Arid and Semi-Arid Climate:** Low rainfall and high evaporation rates in these regions lead to salt accumulation in the soil.
- **Seawater Intrusion:** In coastal areas, rising sea levels and tidal floods can increase salinity levels in agricultural lands.
- **Saline Groundwater:** Irrigation with naturally saline groundwater can gradually salinize the soil.

❑ Human-induced Causes:

- **Over-Irrigation:** Excessive irrigation without proper drainage can lead to waterlogging and salt buildup in the soil profile.
- **Poor Water Quality:** Using saline water for irrigation directly adds salts to the soil.
- **Deforestation:** Loss of vegetation cover exposes the soil to erosion and accelerates salt accumulation.
- **Excessive use of Fertilizers:** Certain fertilizers, particularly those containing ammonium and chloride ions, can contribute to soil salinization over time.

Mitigating Soil Salinity

- ❑ **Efficient Irrigation Practices:** Using drip irrigation or precision agriculture techniques reduces water wastage and minimizes salt buildup.
- ❑ **Drainage Systems:** Proper drainage channels and subsurface drainage systems can remove excess water and salts from the soil.
- ❑ **Saline Water Management:** Treating saline water for irrigation or using alternative water sources like rainwater harvesting can prevent further salinization.

- ❑ **Use of Salt-Tolerant Crops:** Planting crops like halophytes or varieties with inherent salt tolerance can be beneficial in saline areas.
- ❑ **Soil Amendments:** Applying gypsum or organic matter can improve soil structure and drainage, aiding in salt leaching.
- ❑ **Afforestation and Soil Conservation:** Planting trees and shrubs helps prevent soil erosion and reduces salt accumulation.

Salinity	Alkalinity
❑ Refers to the presence of an excess of soluble salts in the soil, predominantly composed of sodium chloride (NaCl), but also including other salts like sulfates and carbonates. ❑ Typically has a neutral or slightly acidic pH in the range of 7 to 8.5. ❑ Has a low exchangeable sodium percentage (ESP)- Less than 15%. ❑ Excess salts disrupt water uptake by plants, leading to dehydration and stunted growth. Specific ions like sodium can also interfere with nutrient uptake and cause toxicity.	❑ Refers to a high concentration of carbonate and bicarbonate ions in the soil, creating a high pH. ❑ Has a high pH, typically greater than 8.5. ❑ May have a high ESP - Can be high or low (but > 15% if considered sodic soil). ❑ High pH reduces the availability of some essential nutrients like phosphorus and iron, making them less accessible to plants. ❑ Sodium dominance in alkaline soils can also harm plant growth, similar to saline soils.

Deforestation

- ❑ Human settlements are putting more and more pressure on forests, which has caused forest land to be used for other things, like farming, living, logging, etc. The roots of trees and plants hold the soil particles together and control the flow of water, which keeps the soil from washing away.

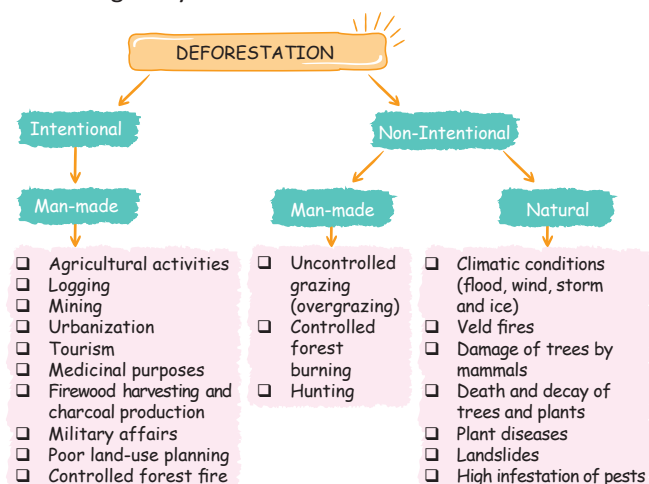


Fig. 6.15: Causes of Deforestation

- ❑ When trees are cut down, the soil is more likely to be eroded by wind and water. Forests also return nutrients to the soil by putting back organic matter that has broken down. Deforestation changes how nutrients are added to the soil, which means the soil doesn't have enough nutrients to support healthy plant life.
- ❑ India's Deforestation Rates and Statistics show that there were 31.3 Mha of natural forests in India in 2010. This was

more than 11% of the land area of the country. It lost 117 Kha of natural forest in 2022, which is like letting out 62.9 Mt of CO₂.

- ❑ As per a report by UK-based Utility Bidder - India **ranked the second highest for the rate of deforestation** after losing 668,400 hectares of forest cover in the last 30 years. In India, the study blamed pressures due to expanding human settlements: "As the country with the largest population in the world, India has had to compensate for the increase in residents – this has come at a cost in the way of deforestation."

Addressing Deforestation

- ❑ **Forest Management and Restoration:** Implement sustainable forest management practices that balance the extraction of forest resources with conservation efforts.
 - Promote reforestation and afforestation projects to restore degraded areas and establish new forests.
- ❑ **Land Use Planning:** Develop and enforce land use planning that designates specific areas for agriculture, urbanization, and conservation, taking into account the ecological value of forests.
 - Implement zoning laws and regulations to prevent unauthorized land clearing and deforestation.
- ❑ **Law Enforcement and Regulations:** Strengthen and enforce laws and regulations that prohibit illegal logging, land clearing, and other activities contributing to deforestation.
 - Implement monitoring systems, satellite technology, and other tools to detect and combat illegal logging and land-use changes.

- ❑ **Economic Incentives:** Develop economic incentives for sustainable forestry practices, such as certification programmes that reward companies for responsible sourcing.
 - Explore mechanisms like payments for ecosystem services (PES) to financially compensate landowners and communities for preserving forests.

Overgrazing

- ❑ When cattle walk over grasslands, again and again, the pressure from their feet crushes the soil into smaller pieces. This makes the soil more likely to be washed away by wind or rain.
- ❑ During the dry season, when there isn't enough grass near the fields to feed the cattle, they are left to graze on the grasslands nearby. This pulls the grass and its roots out of the ground, which makes the ground less stable.
- ❑ When it rains, the loose soil is quickly washed away. More than 70% of the land in states like Madhya Pradesh, Rajasthan, Maharashtra, and Karnataka is used for grazing and the problem of overgrazing can be seen there.

Overgrazing Mitigation

- ❑ **Rotational grazing:** Move livestock between different pastures, allowing vegetation to recover before being grazed again.
- ❑ **Adjust stocking density:** Ensure the number of animals matches the carrying capacity of the land.
- ❑ **Plant diverse forage species:** Promote healthier and more resilient pastures.
- ❑ **Rest periods:** Allow pastures to rest during critical growth periods.
- ❑ **Holistic management:** Integrate grazing with other practices like water management and wildlife conservation.
- ❑ **Stored fodder:** Supplement livestock diets with stored feed to reduce grazing pressure.
- ❑ **Monitor pasture health:** Regularly assess vegetation cover and soil condition to adjust grazing practices.
- ❑ **Controlling Invasive Species:** Preventing the spread of undesirable plants that compete with forage species.



IGNITE YOUR MIND

IoT is enhancing agriculture by using field sensors to gather data on soil conditions, temperature, and more, empowering farmers with valuable insights for informed decision-making. Examine the role of soil testing in precision agriculture and how can farmers benefit from soil testing, and what policy measures can be implemented to encourage widespread adoption of this practice in India?

SOIL CONSERVATION

Soil Conservation is a combination of practices used to protect the soil from **degradation**. First and foremost, soil conservation involves treating the soil as a living ecosystem.

- ❑ This means **returning organic** matter to the soil on a continual basis. Farmers in Gujarat, Haryana, Punjab, Rajasthan, and western Madhya Pradesh have successfully protected their fields from soil erosion by planting rows of trees to reduce the velocity of winds that erode soil cover.

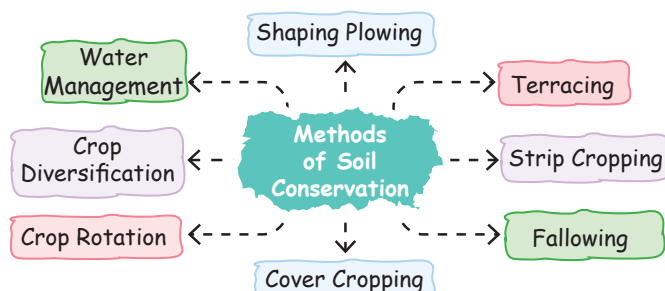


Fig. 6.16: Methods of Soil Conservation

- ❑ Soil conservation entails **reducing soil erosion**, reforestation, rational soil use, and ways to improve soil sustainability.
- ❑ Some of the important steps that can help with soil conservation are as follows:
 - **Afforestation:** Soil erosion can be reduced by planting trees. Trees reduce the intensity of runoff while increasing water seepage to the underground water table. Along the banks of rivers, canals, lakes, roads, and railway tracks, social forestry can be developed.

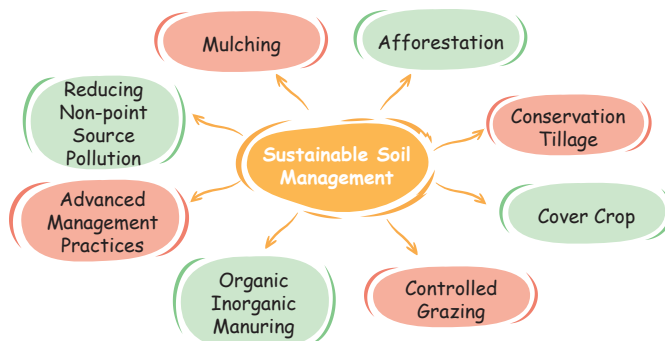


Fig. 6.17: Sustainable Soil Management

- **Restriction on the Felling of Trees:** Aside from afforestation, it is also critical to prevent indiscriminate tree felling. People's awareness, which resulted in the Chipko Movement's launch, can aid in achieving this goal.

- **Contour Ploughing and Strip Cultivation:** Ploughing in hilly and mountainous areas should be done along contours rather than in an up-and-down slope direction. Contour ploughing is an effective method of preventing soil erosion. Similarly, small strips can be developed on gentle slopes for crop planting, assisting in overcoming the threat of soil erosion.

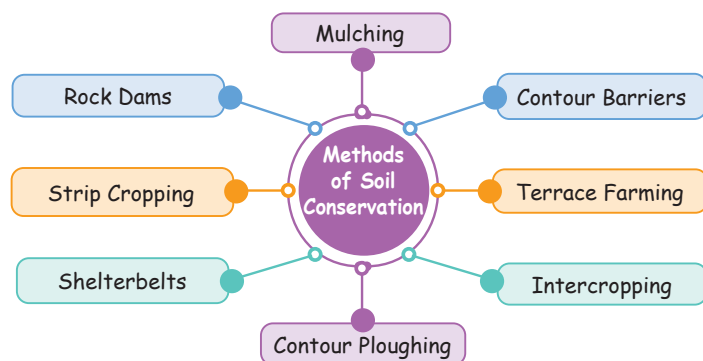


Fig. 6.18: Methods of Soil Conservation

- **Control of Floods:** In India, the issue of soil erosion is closely linked to floods. Floods are most common during the rainy season. Efforts must therefore be made to either store flood water or divert additional rainwater. River interconnections, such as the Garland Canal Project or the Ganga-Kaveri Link Canal Project, can be extremely beneficial in this regard.
- **Reclamation of Ravines and Badlands:** To address the issue of soil erosion, gullies and ravines must be reclaimed. Several such schemes are being implemented in the Chambal ravines of Madhya Pradesh, Uttar Pradesh, and Rajasthan, including the plugging of gully mouths, the construction of bunds across the gullies, the levelling of gullies, afforestation, and grazing restrictions.

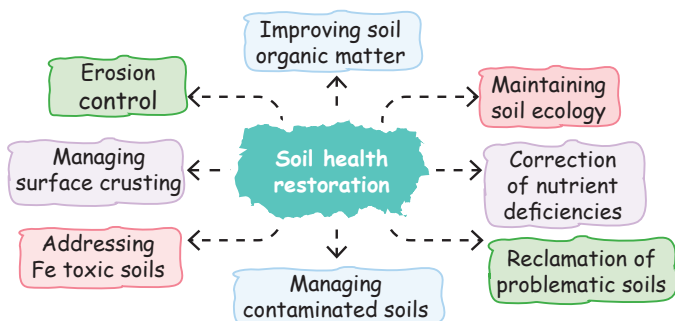


Fig. 6.19: Soil health restoration

- **Restriction on Shifting Cultivation:** In the states of North-east India and the Western and Eastern Ghats, soil erosion is mostly caused by shifting cultivation,

also known as “**slash and burn.**” Shifting cultivators need to be trained and encouraged to switch to terraced farming.

- A plan to stop shifting cultivation has been put in place in the seven North-East Indian states. This is a beneficiary-focused programme aimed at rehabilitating Jhumia families (shifting cultivators). This programme should be expanded to other states of the country, and sedentary farming should gradually replace this agricultural system.
- **Restoration of Long Fallow:** In 2021, fallow land for India was 23,612 thousand hectares. Though India’s fallow land fluctuated substantially in recent years, it tended to decrease through the 2002 - 2021 period ending at 23,612 thousand hectares in 2021.
 - Andhra Pradesh, Telangana, Karnataka, Madhya Pradesh, Maharashtra, Gujarat, Rajasthan, Tamil Nadu, Bihar, Uttarakhand, and Uttar Pradesh have the most old fallow land. This old fallow land can be converted to agriculture, grazing, and orchards, which will help to reduce soil erosion.
- **Five-pronged strategy:** The Indian government has a five-pronged plan to protect the land. This includes getting chemicals out of the soil, saving the biodiversity in the soil, increasing SOM, keeping the soil moist, reducing soil degradation, and stopping soil erosion.
- **Soil Health Card (SHC) scheme:** Earlier farmers had less knowledge about the type, soil deficiencies, and soil moisture levels. In 2015, the government of India started the Soil Health Card (SHC) scheme to deal with these problems.
 - The SHC is used to figure out how healthy the soil is right now and to see how it has changed over time. The SHC shows soil health indicators and the words that describe them, which help farmers know what changes to make to the soil.
- **Pradhan Mantri Krishi Sinchayee Yojana:** The Pradhan Mantri Krishi Sinchayee Yojana is another important programme that aims to stop soil erosion, restore natural vegetation, collect rainwater, and refill the groundwater table.
- **National Mission for Sustainable Agriculture (NMSA):** Under NMSA, Organic farming practises are being spread. NMSA also has programmes that support traditional indigenous practices like organic farming and natural farming. This makes

smallholder farmers less dependent on chemicals and other agri-inputs and lessens their financial burden.

Role of the Central Soil Conservation Board (CSCB) in India:

The CSCB plays a crucial role in promoting sustainable land management and soil conservation practices across India. Established in 1953, it operates under the Ministry of Agriculture and Farmers Welfare.

- ❑ **River Valley Projects (RVPs):** These projects aim to prevent soil erosion and land degradation in the catchment areas of major rivers and reservoirs.
- ❑ **Watershed Development Projects (WDPs):** These projects focus on the integrated development of natural resources such as land, water, vegetation and livestock in a watershed area.
- ❑ **Soil Conservation in the Catchment of River Ganga (SCCRG):** This project aims to reduce the sediment load and improve the water quality of the river Ganga by implementing soil and water conservation measures in its catchment area. The project covers 9.5 million hectares in 11 states.
- ❑ **National Watershed Development Project for Rainfed Areas (NWDPA):** This project aims to enhance the productivity and sustainability of rainfed agriculture by adopting a participatory approach involving farmers and local communities.
- ❑ **Reclamation and Development of Alkali and Acid Soils (RADAS):** This project aims to reclaim and improve the productivity of salt affected and acidic soils by applying suitable amendments and agronomic practices.

Land Degradation Neutrality: It is an ambitious global goal that aims to achieve a balance between the ongoing loss of productive land and efforts to restore degraded land. It's essentially like striving for zero net land degradation, ensuring our planet's valuable land resources remain healthy and productive for future generations.

The United Nations Convention to Combat Desertification (UNCCD) plays a crucial role in this mission. Through its Land Degradation Neutrality (LDN) framework, the UNCCD encourages countries to set voluntary targets to avoid, reduce, and reverse land degradation. These targets can be tailored to each country's specific circumstances and priorities.

Global UNCCD Targets

- ❑ **Avoiding new land degradation:** This involves protecting healthy land from unsustainable practices like deforestation, overgrazing, and soil erosion.

- ❑ **Reducing existing land degradation:** This means implementing sustainable land management practices like crop rotation, cover cropping, and agroforestry to improve soil health and fertility.

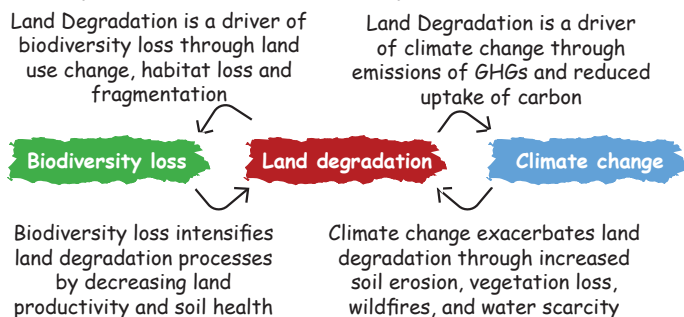


Fig. 6.20: Connections among biodiversity, land degradation and climate change

- ❑ **Restoring degraded land:** This involves actively bringing back degraded land to a more productive state through activities like tree planting, assisted natural regeneration, and biochar application.

India's LDN Targets: India, a country facing significant land degradation challenges, has set ambitious LDN targets for itself. These include:

- ❑ Restoring 26 million hectares of degraded land by 2030
- ❑ Achieving land productivity neutrality by 2030
- ❑ Increasing soil organic carbon content at a rate of 0.4% per annum in the top 1 m soil layer by 2030.

To achieve these goals, India is implementing various initiatives like:

- ❑ **The National Mission on Sustainable Agriculture:** This mission promotes soil health management, water conservation, and climate-resilient agriculture practices.
- ❑ **The Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS):** This scheme provides rural employment opportunities while also supporting land restoration activities like afforestation and bunding.
- ❑ **The Pradhan Mantri Fasal Bima Yojana (PMFBY):** This crop insurance scheme incentivizes farmers to adopt sustainable land management practices.

According to a **Ministry of Agriculture report**, the Government of India aims to achieve land degradation neutrality by 2030. In India, **69%** of the land is dry (about 105 million hectares) and **32%** is desertifying. Every year, approximately **1 mm** of top-soil is lost, and the total amount of land lost due to soil erosion is **5334 million tonnes**. Integrated land use planning has the potential to halt or reverse desertification on Indian land.

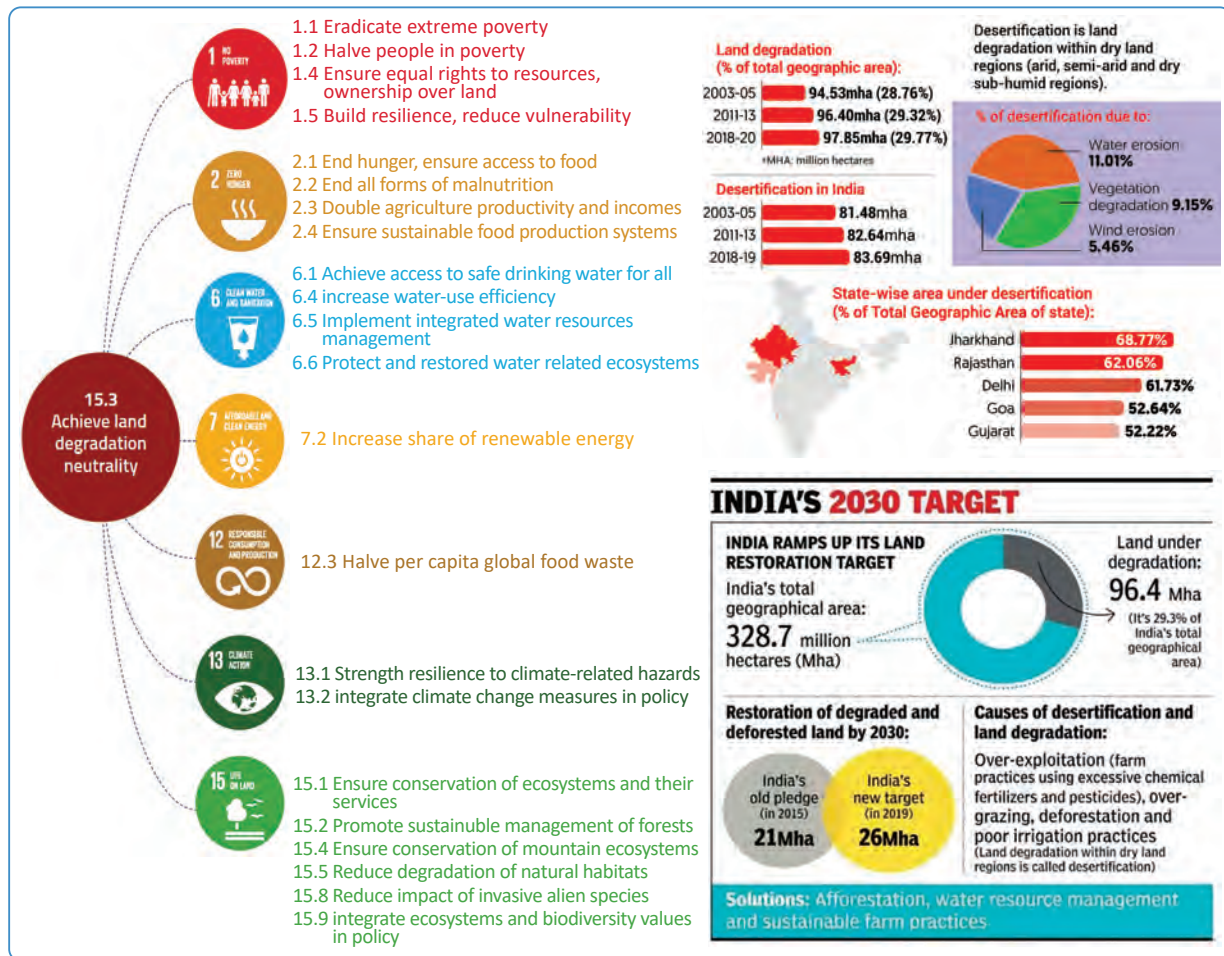


Fig. 6.21: SDG goals and targets

Soil Health and Climate Change Mitigation

Soil health, often overlooked, plays a crucial role in mitigating climate change, acting as a silent guardian influencing the Earth's capacity to absorb and store greenhouse gases.

Carbon Sequestration

- ❑ **Giant Carbon Sink:** Healthy soil serves as a significant carbon sink, storing substantial amounts of carbon dioxide.
- ❑ **Preventing Release:** Managed organic matter ensures that carbon remains locked away in the soil, preventing its release into the atmosphere and contributing to global warming.

Reduced Emissions

- ❑ **Minimizing Nitrous Oxide:** Unhealthy soil, often due to unsustainable agricultural practices, releases nitrous oxide, a potent greenhouse gas.
- ❑ **Promoting Cleaner Air:** Healthy soil with diverse microbial communities minimizes emissions, and contributes to cleaner air.

Enhanced Resilience

- ❑ **Climate Resilience:** Healthy soil, with good water-holding capacity and organic matter, exhibits greater resilience to extreme weather events like floods and droughts.

- ❑ **Preventing Erosion:** Improved soil health reduces erosion, prevents the release of stored carbon and maintains agricultural productivity in challenging conditions.

Improved Food Security

- ❑ **Fostering Healthy Crops:** Healthy soil fosters robust crop growth, resulting in increased food production.
- ❑ **Reducing Deforestation:** Enhanced soil health reduces the need for deforestation, minimizing associated greenhouse gas emissions.
- ❑ **Organic Matter Decomposition:** Soil health influences the decomposition of organic matter, affecting greenhouse gas emissions.
- ❑ **Agroforestry and Cover Crops:** Agroforestry practices and cover cropping contribute to soil health and carbon sequestration.
- ❑ **Reduced Erosion:** Healthy soils are less prone to erosion, maintaining their carbon sequestration capacity.
- ❑ **Water Management:** Proper soil health management enhances water retention, reducing runoff and contributing to climate change mitigation.
- ❑ **Nutrient Cycling:** Healthy soils support efficient nutrient cycling, reducing the need for synthetic fertilizers with associated greenhouse gas emissions.

- ❑ **Biodiversity Support:** Soil health is connected to biodiversity, contributing to ecosystem resilience.
- ❑ **Land Use Planning:** Prioritizing soil health in land use planning aligns with climate-smart strategies for climate change mitigation.

Practices for Healthy Soil

- ❑ **Minimize Tillage:** Opt for no-till farming or minimal tillage practices to preserve soil structure and reduce carbon release.
- ❑ **Cover Cropping:** Plant cover crops between main crops to protect the soil, add organic matter, and suppress weeds.
- ❑ **Composting and Mulching:** Enrich the soil with nutrients, improve water retention, and enhance microbial activity through composting and mulching.
- ❑ **Crop Rotation:** Break disease cycles and replenish soil nutrients by rotating different crops.
- ❑ **Diversification:** Planting diverse species contributes to a healthy soil microbiome and overall ecosystem resilience.

Studies estimate that improving soil organic carbon content by just 0.4% per year could offset up to 12% of global carbon dioxide emissions. By adopting these practices and promoting soil health, we significantly contribute to climate change mitigation. Understanding the vital link between healthy soil and climate mitigation is essential for effectively addressing the climate crisis.

Maintaining and enhancing soil health is a multifaceted approach to climate change mitigation, which involves

practices such as Carbon Sequestration, Reduction of GHGs and promotion of overall ecosystem resilience. Integrating soil health management into broader climate mitigation strategies is essential for building sustainable and resilient agricultural and land-use systems.



IGNITE YOUR MIND

Soil formation occurs within distinct natural settings, whereby several components of the natural environment collectively contribute to this intricate process. In your opinion, why Indian soils are nitrogen deficient? Why do black soils have a clayey texture? Do you think that the nature of the soil is controlled more by climatic factors than the type of parent rock?

CONCLUSION

India's diverse soils are integral to its agricultural prosperity and environmental sustainability. While the country's varied soil types contribute significantly to agricultural productivity, challenges such as soil degradation and improper land use necessitate urgent attention. Sustainable soil management practices are crucial to mitigate these issues. There is a need for a collective effort involving policy-makers, scientists, and farmers to preserve and enhance India's soil health for long-term prosperity and environmental well-being.

Natural Vegetation of India

7

India has a varied natural vegetation influenced by climate and soil variations. The vegetation includes temperate forests in the Himalayas, tropical rainforests in the Western Ghats and Andaman Nicobar Islands, tropical forests and mangroves in deltaic areas, hardy cacti and thorny shrubs in Rajasthan, and other plant communities. The natural vegetation of India is a rich and complex mosaic that illustrates the country's environmental diversity.

FACTORS INFLUENCING NATURAL VEGETATION IN INDIA

The natural vegetation across regions is influenced by many factors, creating a rich diversity of flora and fauna. These factors can be broadly categorised into two main groups:

Relief Factors

- ❑ **Land:** The character of the land directly impacts natural vegetation. Hilly regions differ from plateaus and plains, with diverse ecosystems. Rich, flat lands are often used for agriculture, like great plains of northern India, while uneven terrains support grasslands and forests, providing habitats for various fauna.
- ❑ **Soil:** Different soil types serve as the foundation for various vegetation types. Sandy desert soils support cacti and shrubs giving rise to xerophytic vegetation while marshy deltaic soils foster mangroves and deltaic flora. Hill slopes with adequate soil depth host conical trees.
- ❑ **Altitude:** As the elevation increases, the distribution of flora changes, akin to the variations seen with changes in latitude. There is a strong correlation between latitude and altitude regarding climatic zones and natural vegetation, like the mountainous region of northern India houses alpine vegetation.
- ❑ **Topography:** India's diverse topography, including plains, plateaus, mountains, and coastal regions, contributes to variations in temperature, rainfall, and soil, resulting in a wide range of vegetation.

Climatic Factors

- ❑ **Temperature:** Temperature, along with humidity, precipitation, and soil conditions, determines the nature and extent of vegetation. Lower temperatures, often associated with higher elevations, lead to colder climates and affect the types of vegetation found.
 - **For example,** the slopes of the Himalayas and peninsular hills above 915 metres experience shifts

from tropical to subtropical, temperate, and alpine vegetation.

- ❑ **Photoperiod (Sunlight):** Variations in the duration and amount of sunlight due to factors such as latitude, altitude, season, and day length profoundly impacts plant growth. More extended periods of sunlight, as seen in the Southern slope of the Himalayas, support denser vegetation than the Northern slopes.
- ❑ **Precipitation:** The timing and quantity of rainfall are pivotal in determining vegetation. Regions with heavy rain, such as western slope of western ghats during the advancing Southwest monsoon (June to September), tend to have denser vegetation than areas with lesser rainfall.
 - **For example,** the Western Ghats' western slopes see tropical evergreen forests flourish due to the southwest monsoon, while the eastern slopes have less dense forests due to differences in precipitation.

Other than these factors, the **presence of rivers, lakes, and other water bodies** can foster the growth of specific vegetation types, especially in floodplains and wetlands. Mangroves, for example, thrive in coastal areas where saltwater and freshwater mix.

SPATIAL DISTRIBUTION OF NATURAL VEGETATION IN INDIA

- ❑ The spatial distribution of natural vegetation in India is closely tied to annual and spatial variations in rainfall.
- ❑ Regions receiving over **200 cm** of rain annually, such as the Western Ghats, North-eastern India, and the Andaman and Nicobar Islands, support tropical evergreen rainforests with year-round greenery.
- ❑ Areas with moderate rainfall, **between 70 and 200 cm**, host deciduous forests where trees shed leaves seasonally, including Teak trees. In contrast, regions with **less than 70 cm** of rainfall, like semi-arid areas, feature thorn forests with specialised drought-resistant vegetation.
- ❑ **Coastal regions**, particularly river deltas, harbour mangrove forests adapted to saline and freshwater mixtures, crucial for coastal ecology and biodiversity.

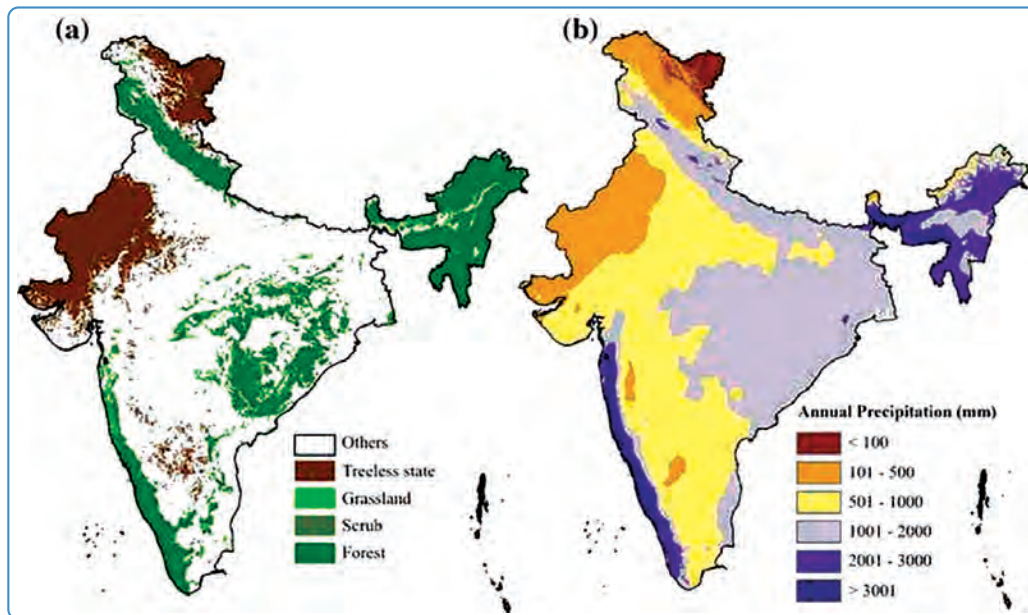


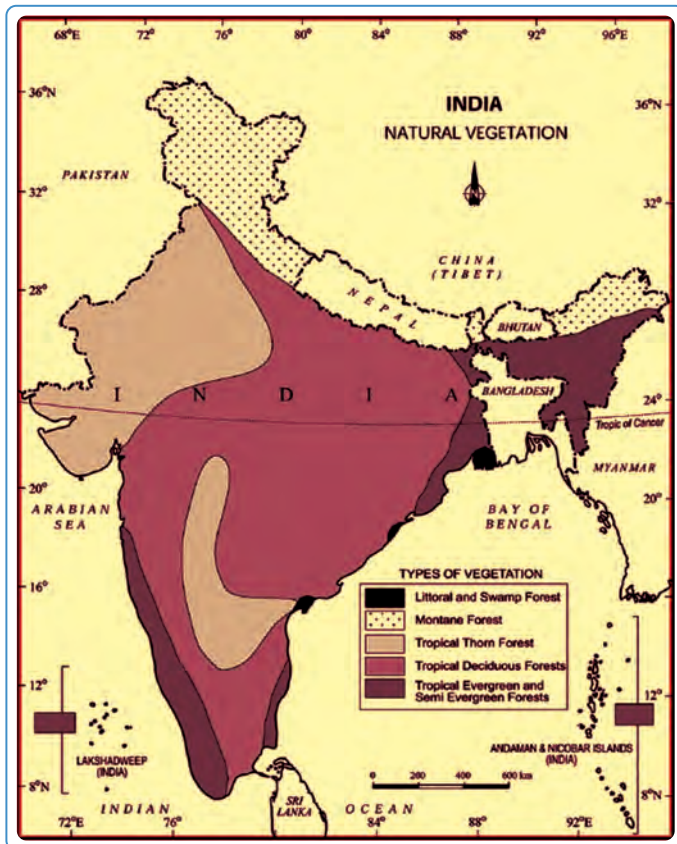
Fig. 7.1: Precipitation and Distribution of Natural Vegetation In India

SIGNIFICANCE OF NATURAL VEGETATION

- ❑ **Habitat Preservation:** Natural vegetation provides diverse habitats for plants, insects, animals, and microorganisms. These habitats support numerous species, including endangered ones, and help preserve genetic diversity.
- ❑ **Ecosystem Preservation:** Different species within these ecosystems interact in complex ways, contributing to ecological stability. For example, predator-prey relationships help control populations and maintain equilibrium.
- ❑ **Soil Erosion Prevention:** The root systems of plants bind soil particles together, preventing erosion caused by wind and water. This is particularly important in regions with steep terrain or heavy rainfall.
- ❑ **Landslide Mitigation:** In hilly or mountainous areas, vegetation helps stabilise slopes. The roots of trees and other plants provide mechanical support to the soil, reducing the risk of landslides.
- ❑ **Riverbank Stability:** Along riverbanks, vegetation protects against erosion by absorbing the energy of flowing water. This maintains the integrity of riverbanks and prevents sedimentation in water bodies.
- ❑ **Carbon Sequestration:** Forests and other natural ecosystems act as carbon sinks. They absorb more carbon dioxide from the atmosphere than they release, which is crucial in mitigating climate change.
- ❑ **Oxygen Production:** Through photosynthesis, plants release oxygen into the atmosphere, which is essential for the respiration of animals, including humans.
- ❑ **Temperature and Humidity Control:** Vegetation provides shade, which lowers temperatures in urban areas. It also releases water vapour through transpiration, contributing to humidity and local climate moderation.
- ❑ **Natural Water Regulators:** Plants absorb rainwater and release it slowly, reducing the risk of floods and droughts. This natural regulation benefits agriculture and reduces the impact of extreme weather events.
- ❑ **Groundwater Recharge:** Vegetation helps replenish groundwater by allowing rainwater to percolate into the soil. This contributes to the sustainability of water resources.
- ❑ **Sustainable Resources:** Forests and natural vegetation offer renewable resources like timber for construction, fruits for nutrition, and medicinal plants for traditional and modern healthcare.
- ❑ **Economic Contribution:** These resources are not only essential for local livelihoods but also contribute to national economies through industries such as forestry and agriculture.
- ❑ **Recreation:** Natural landscapes, parks, and forests provide spaces for outdoor activities, improving physical and mental well-being. They offer opportunities for hiking, camping, and wildlife viewing.
- ❑ **Cultural and Spiritual Significance:** Indigenous cultures connect deeply to their natural surroundings. Forests, mountains, and rivers hold cultural and spiritual significance, often playing roles in traditional ceremonies and beliefs.
- ❑ **Air and Water Filtration:** Trees and vegetation act as natural filters, trapping airborne pollutants and purifying water. This helps improve air and water quality, benefiting human health and the environment.

CLASSIFICATION OF NATURAL VEGETATION IN INDIA

India's natural vegetation, extensively studied and classified by notable scholars like **H.G. Champion** and later simplified by others, encompasses a rich diversity of plant life. This classification system has identified **5 major types** and **16 sub-types of vegetation** across the country. These categories help us understand the intricate tapestry of India's ecosystems, influenced by climate, geography, and rainfall.



Forest Type in India	% of Total Area
Tropical Moist Deciduous	37
Tropical Dry Deciduous	28
Tropical Wet Evergreen	8
Sub-Tropical Moist Hill	6
Tropical Semi-Evergreen	4
Rest below 4 %	

Fig. 7.2: National Vegetation of India

Classification of Natural Vegetation in India

Major Types of Vegetation	Sub-Types
Moist Tropical	☐ Tropical Wet-Evergreen ☐ Tropical Semi-Evergreen ☐ Tropical Moist Deciduous ☐ Littoral and Swamp

Dry Tropical	☐ Tropical Dry Evergreen ☐ Tropical Dry Deciduous ☐ Tropical Thorn
	☐ Sub-Tropical Broad Leaves Hill ☐ Sub-Tropical Moist Hill (Pine) ☐ Sub-Tropical Dry Evergreen
	☐ Montane Wet Temperate ☐ Himalayan Moist Temperate ☐ Himalayan Dry Temperate
Alpine	☐ Sub-Alpine ☐ Moist Alpine Scrub ☐ Dry Alpine Scrub

Moist Tropical Vegetation

Tropical Wet Evergreen

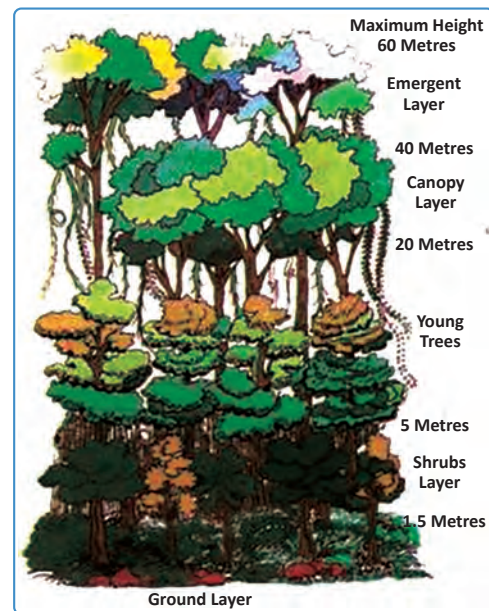


Fig. 7.3: Tropical Evergreen Forest

- ☐ Tropical wet evergreen forests, also known as tropical rainforests, are one of the most biologically diverse ecosystems on Earth and are known for their lush vegetation and high species richness.
- ☐ They are found between **5° North and South** of the equator.
- ☐ **Climatic Conditions:** Tropical Wet Evergreen forests thrive in areas with specific climatic conditions.
 - **Rainfall:** These areas typically receive an annual rainfall of more than 250 centimetres.
 - **Temperature:** The yearly temperature remains around 25°C to 27°C, and the average relative humidity is over 77 percent.
 - **Vegetation:** One of the main distinguishing features of this type of vegetation is the short dry season.
- ☐ **Characteristic:** The name "**Wet Evergreen**" indicates that the trees in this forest do not shed their leaves seasonally.
 - There is **no specific time period** for trees to shed their leaves, flowering and fruition, creating an evergreen appearance.

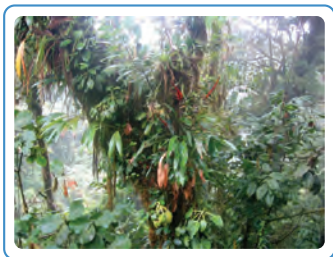


Fig. 7.4: Vegetation in a Tropical Wet Evergreen Forest

- These forests are characterised by their **dense and multi-layered structure**. The trees are typically quite tall, reaching up to 45 metres but not exceeding 60 metres.
 - The **forest canopy is thick** and provides continuous cover, making it difficult for sunlight to reach the forest floor. As a result, the undergrowth is limited, consisting mainly of ferns, climbers, bamboo, and orchids.
 - Leaf litter decomposes faster than in any other biome as a result, the soil surface is often almost bare. Apart from trees, the vegetation is largely composed of plant forms that reach up to the canopy vicariously by climbing trees or growing as epiphytes, rooted on the upper branches of trees.
- ❑ **Distribution:**
- **Western Ghats:** Notable regions with these forests include the Malabar Coast in Kerala, the Agumbe region in Karnataka, and parts of Goa.
 - **Northeastern States:** The heavy rainfall regions of states like Assam, Meghalaya, Nagaland, Mizoram, and Tripura.
 - **Andaman and Nicobar Islands.**
- ❑ **Flora and Forest Resources:** The timber obtained from these forests is highly valued. It is described as fine-grained, hard, and durable.
- Notable tree species in these forests include **Mahogany, Mesua, White Cedar, Jamun, Canes, Dhup, and Bamboo.**
 - Timber
 - ❑ **Hardwood:** The timber of these forests is fine-grained, hard and durable.
 - ❑ It has high commercial value but it is highly challenging to exploit due to dense undergrowth, absence of pure stands and lack of transport facilities.

Reasons for Underdevelopment in Tropical Wet Evergreen Regions

- ❑ **Inaccessibility:** Many tropical wet evergreen regions are geographically remote and have challenging terrains, making them less accessible for infrastructure development.

- **For Example:** The forests of **Parambikulam Wildlife Sanctuary** in **Kerala** are challenging to access due to their remote location and rugged terrain, limiting large-scale development initiatives.

- ❑ **Difficult Topography:** The Western Ghats, where these forests are predominant, is characterized by rugged terrain, steep slopes, and dense vegetation, making it difficult for large-scale development projects.

- **For Example:** The **Anamalai Hills** in Tamil Nadu and Kerala are characterized by steep slopes and dense vegetation, making it difficult for extensive infrastructure development.

- ❑ **Conservation Policies:** Stringent conservation policies and protected area designations limit the extent to which these regions can be exploited for economic development. Regulations are in place to protect the unique biodiversity of these ecosystems.

- **For Example:** The Western Ghats, including the Agasthyamalai Biosphere Reserve in Kerala and Tamil Nadu, is subject to strict conservation policies to protect its unique biodiversity, limiting extensive development activities.

- ❑ **Indigenous Populations:** Many tropical wet evergreen regions in India are home to indigenous tribal communities. Development activities may conflict with their traditional lifestyles, and conservation efforts prioritize the well-being of these communities.

- **For Example:** The tribal communities in the **Wayanad Wildlife Sanctuary** in Kerala have traditional ties to the forest. Development projects need to consider and respect the rights and lifestyles of these indigenous populations.

Tropical Semi-Evergreen Forest

- ❑ Tropical Semi-Evergreen vegetation in India stands as a fascinating ecological **transition**, bridging the gap between the luxuriant tropical wet evergreen forests and the deciduous woodlands.



Fig. 7.5: Vegetation in the Tropical Semi Evergreen forest

- ❑ **Climatic Conditions:** Tropical semi-evergreen forests experience milder moisture conditions compared to their wet evergreen counterparts.

- **Rainfall:** They receive an annual rainfall over 200 centimetres.
- **Temperature:** They have mean annual temperatures above 22°C.
- **Relative humidity** remains around 75 percent, significantly influencing the ecological dynamics of this region.

❑ **Characteristic:** The flora and fauna in Tropical Semi-Evergreen forests are exceptionally diverse.

- Trees often feature **buttressed trunks and host numerous epiphytes**. The bark is characterised by roughness and thickness.
- These forests are home to an array of **heavy climbers** and boast a unique mix of plant species. Bamboo species are comparatively fewer in number but contribute to the rich tapestry of the ecosystem.
- Compared to wet evergreen forests, **semi-evergreen vegetation is less dense**. They often exist in gregarious groupings of various plant species, creating a mosaic of life.
- The undergrowth consists of a rich mix of plants adapted to the transitional climatic conditions, showcasing adaptability and resilience.

❑ **Distribution:**

- **Western Ghats:** Extensive areas in the Western Ghats, particularly in states like Kerala, Karnataka, Goa, Maharashtra, and Tamil Nadu.



IGNITE YOUR MIND

It is interesting to know that in the village of Pimpri in Maharashtra, non-native Baobab trees, introduced centuries ago, stand as a testament to historical trade and cultural exchanges in the region. *Can you evaluate the role of Baobab trees in traditional medicine practices in the Western Ghats and their impact on local healthcare.*

- **Eastern Ghats:** Parts of Odisha, Andhra Pradesh, and Tamil Nadu along the Eastern Ghats may have Tropical Semi-Evergreen vegetation.
- **Northeastern States:** Certain areas in the northeastern states, such as Assam, Arunachal Pradesh, Meghalaya, Nagaland, and Mizoram.
- **Andaman and Nicobar Islands.**

❑ **Flora and Forest Resources:** Prominent tree species in these forests include **Laurel, Rosewood, Mesua, thorny Bamboo, Kenju, Kusum, Mundani, Semul, Kadam, and Irul**.

- In the **Western Ghats**, one can encounter White Cedar, Indian Chestnut, and Champa, while the **Himalayan** region also hosts these intriguing forests, contributing to the region's ecological diversity.

Tropical Moist Deciduous

❑ Tropical Moist Deciduous forests are a distinct type of vegetation characterised by a combination of trees and grasses.

❑ **Climatic Conditions:**

- **Rainfall:** These forests are typically found in regions with an annual rainfall ranging from approximately 100 to 200 centimetres.
- **Temperature:** They have a mean annual temperature of around 27°C.
- **Humidity:** The **average relative humidity** in these areas falls between 60 to 75 percent.

❑ **Characteristic:** Notably, they often feature heavily buttressed tree trunks and possess a fairly dense undergrowth.

- While these forests cover a larger expanse compared to evergreen forests, extensive areas have been cleared for cultivation.
- These forests display an irregular tree canopy with trees reaching heights between 20 to 60 metres.

❑ **Distribution:**

- **Central and eastern India:** These forests are present in the hills of eastern Madhya Pradesh and Chattisgarh, the Chota Nagpur plateau, significant parts of Odisha and portions of West Bengal.
- Tropical Moist Deciduous vegetation is also present in parts of **Karnataka, Andhra Pradesh, and Telangana**.
- The Western Ghats, particularly the rain-shadow regions on the leeward side. These areas experience a distinct wet season during the monsoon, followed by a dry season.
- Parts of Gujarat and Rajasthan along the northern edges of the Western Ghats also exhibit Tropical Moist Deciduous vegetation in areas with sufficient rainfall.
- They are also found along the **Shivalik range**, including the **Terai** and **Bhabar** regions.

❑ **Flora and Forest Resources:** These forests are valuable for the resources they provide, including timber from tree species such as Teak, Sal, Laurel, Lendi, Semul, Irul, Rosewood, Amla, Kusum, Jamun, and Bamboo, among others.

- However, like other forest types, they face challenges due to deforestation and habitat conversion for agriculture.

Littoral and Swamp

❑ "Littoral" and "swamp" are terms used to describe specific types of ecosystems, typically associated with aquatic or semi-aquatic environments.

❑ **Climatic Conditions:** Littoral and Swamp vegetation is a unique type that thrives in areas with both fresh and brackish water, characterised by varying levels of salinity ranging from 0.5 to 35 grams per kilogram.

❑ **Characteristic:** These ecosystems are typically found in and around deltas, estuaries, and creeks that are influenced by tidal flows. They play a crucial role in the transition zones between land and water, providing habitat for specialised plant and animal species.

❑ **Distribution:**

- **Littoral Vegetation:** Littoral vegetation is found along the coastlines of India, in states along the Arabian Sea (west coast) and the Bay of Bengal (east coast), such as **Gujarat, Maharashtra, Karnataka, Kerala, West Bengal, Odisha, and parts of Tamil Nadu.**
- **Swamp vegetation** is found in areas characterized by waterlogged and poorly drained soils, creating swampy conditions. Swamps are distributed across different regions of India, often associated with river deltas, wetlands, and low-lying areas.
 - ❑ Swamp vegetation is observed in various states, including **Assam, West Bengal, Uttar Pradesh, Bihar, Kerala, Tamil Nadu, and the Sundarbans region.**

❑ **Flora and Forest Resources:** The vegetation in these areas consists mainly of mangrove trees, which can reach heights of around 30 metres.

- These mangrove forests are essential for fuel purposes and serve as a protective buffer against coastal erosion.
- Epiphytes, which are plants that grow on other plants without harming them, are abundant in these forests. Additionally, higher ground areas support screw pines, while marshes and creeks are inhabited by nipa palms (*Nypa fruticans*).
- One well-known example of Littoral and swamp vegetation is the Sundarbans delta, which is covered with **Sundari trees** (*Heritiera minor*).
- However, these ecosystems are vulnerable to habitat destruction and pollution, making their conservation crucial for both the environment and local communities.

Differences Between the Littoral Zone and Swamps

Characteristic	Littoral Zone	Swamp
Location	Along the shoreline of bodies of water (lakes, ponds, rivers, oceans).	Low-lying areas with stagnant water or slow-moving water (wetlands).
Water Presence	Subject to periodic submersion by water, especially during high water levels.	Consistently saturated with water, often with standing water year-round.
Hydrology	Water levels can vary seasonally or with changes in the water body's level.	Consistently high water levels with minimal fluctuation.
Common Ecosystem Type	Often found in the transitional zone between aquatic and terrestrial ecosystems.	A type of wetland ecosystem with standing or slow-moving water.



IGNITE YOUR MIND

Did you know that Myristica swamps in the Western Ghats are distinctive ecosystems known for their rich plant endemism, featuring species like *Myristica*, a primitive angiosperm? Try to analyze the role of swamp vegetation in climate change mitigation and adaptation, considering factors such as carbon sequestration and resilience to extreme weather events.

Dry Tropical Vegetation

Tropical Dry Deciduous

❑ Tropical Dry Deciduous forests serve as a transitional vegetation type between moist deciduous and tropical thorn forests.

❑ **Climatic Conditions:**

- **Rainfall:** These forests typically receive an annual rainfall ranging from 70 to 100 centimetres.

- **Temperature:** These regions experience average annual temperatures spanning from 15°C to 22°C. Relative humidity levels in these regions fluctuate between 63 to 77 per cent.

❑ **Characteristics:** These are similar to moist deciduous forests and shed their leaves in dry season. The major difference is that they can grow in areas of comparatively less rainfall.

- They represent a transitional type – moist deciduous on the wetter side and thorn forests on the drier side.
- They have a closed but uneven canopy.
- The forests are composed of a mixture of a few species of deciduous trees rising up to a height of 20 metres.
- Undergrowth: Enough light reaches the ground to permit the growth of grass and climbers.
- These forests have been subject to significant deforestation and clearance for agriculture, leading to habitat loss and environmental challenges.
- Tropical Dry Deciduous forests play an important ecological role and are a source of valuable timber and other forest products.

- Their conservation is essential to maintain biodiversity and protect against soil erosion and desertification in these regions.

❑ **Distribution:**

- **Central India:** Parts of Madhya Pradesh, Chhattisgarh, and parts of Maharashtra.
 - ❑ Notable examples include the forests in Kanha National Park (Madhya Pradesh) and Pench National Park (Madhya Pradesh and Maharashtra).
- **Eastern India:** Particularly in Jharkhand, Odisha, and parts of West Bengal.
 - ❑ The Saranda Forest in Jharkhand is an example of Tropical Dry Deciduous vegetation.
- **Southern India:** Karnataka, Andhra Pradesh, and parts of Tamil Nadu.
 - ❑ Areas like Bandipur National Park (Karnataka), Nagarhole National Park (Karnataka), and Sathyamangalam Wildlife Sanctuary (Tamil Nadu and Karnataka) feature Tropical Dry Deciduous vegetation.
- **Western India:** Parts of Gujarat and Rajasthan also have pockets of Tropical Dry Deciduous forests, where species like teak, sal, and bamboo are common.
 - ❑ Gir National Park in Gujarat, known for its Asiatic lion population, features this type of vegetation.
- **However, they are not found in Rajasthan, the Western Ghats, and Gujarat.**

- ❑ **Flora and Forest Resources:** Key tree species found in tropical dry deciduous vegetation include Teak, Laurel, Khair, Harra, Bel, Rosewood, Satinwood, Anjeer, Palas, Amaltas, and Bilsal, among others.

Tropical Thorn

- ❑ “Tropicathorn” typically refers to plants in tropical regions with thorns or spines as adaptations for defence against herbivores and water conservation in arid environments.
- ❑ **Climatic Conditions:**
 - **Rainfall:** Tropical Thorn vegetation thrives in regions characterised by very low annual rainfall, typically ranging from 50 to 70 centimetres.
 - **Temperature:** The mean annual temperature in these areas falls between 25°C to 27°C.
 - **Humidity:** Relative humidity is quite low, often below 47 percent.
- ❑ **Distribution:**
 - **Northwestern India:**
 - ❑ Rajasthan, Gujarat, parts of Haryana, and Punjab.
 - ❑ The Thar Desert, which spans across northwestern India, is a prominent area with tropical thorn vegetation. The vegetation in this region is adapted to arid conditions, featuring plants such

as acacia species, *Prosopis juliflora* (mesquite), and various cacti.

- **Central India:**

- ❑ Parts of Madhya Pradesh, Maharashtra, and Chhattisgarh.
- ❑ Some areas in central India experience a dry deciduous climate, leading to the presence of tropical thorn vegetation. Common species include *Acacia catechu*, *Anogeissus latifolia*, and *Boswellia serrata*.

- **Southern India:**

- ❑ Karnataka, Andhra Pradesh, Telangana, parts of Tamil Nadu, and Maharashtra.
- ❑ Certain regions in the Deccan Plateau, including parts of the Eastern Ghats, exhibit tropical thorn vegetation. The vegetation is adapted to the semi-arid conditions of these areas and includes thorny shrubs like *Ziziphus mauritiana* (Indian jujube) and species of *Acacia*.

- **Western India:**

- ❑ Parts of Gujarat and Maharashtra.
- ❑ Some areas in the western parts of India experience arid to semi-arid conditions, leading to the presence of tropical thorn vegetation. These areas may include species like *Acacia senegal*, *Anogeissus pendula*, and others adapted to drier climates.

- ❑ **Characteristics:** In these arid and semi-arid regions, trees tend to be relatively small, ranging from 6 to 9 metres in height, and they often appear stunted. Coarse grasses dominate large patches of the landscape.

- ❑ **Flora and Forest Resources:** Common plant species in this vegetation type include Indian Date Palms, Acacias, Cacti, Palash, Wild Palms, Kokko, Kanju, and Khejra. Despite the challenging environmental conditions, these plants have adapted to survive in arid climates, making them an essential part of these ecosystems.

Tropical Dry Evergreen

- ❑ Tropical dry evergreen forests are a type of forest ecosystem found in tropical regions characterised by trees that retain their green leaves throughout the year despite seasonal dry periods.
- ❑ **Climatic Conditions:**
 - **Rainfall:** Tropical Dry Evergreen forests are typically found in areas that receive approximately 100 centimetres of rainfall, primarily occurring from October to December.
 - **Temperature:** The annual mean temperature in these regions is around 28°C, and the relative humidity levels hover at approximately 74 percent.
- ❑ **Characteristics:** These forests are characterised by relatively low-height trees, with canopies ranging from 9 to 12 metres in height.

❑ **Distribution:**

- **Coromandel Coast:** Coastal regions of Tamil Nadu, including areas near Chennai, Puducherry, and the surrounding districts, are known for these forests.
- **Eastern Ghats:** parts of Andhra Pradesh, Odisha, and Karnataka along the Eastern Ghats, Seshachalam Hills in Andhra Pradesh and the Simlipal National Park in Odisha.
- **Western Ghats:** Karnataka and Kerala, especially in the rain shadow regions, may have elements of TDEF.
- **Southern Tip of India:** parts of Kerala and Tamil Nadu, may feature Tropical Dry Evergreen Forests.

❑ **Flora and Forest Resources:** Key tree species that are part of the tropical dry evergreen vegetation include **Toddy Palm, Kokko, Jamun, Ritha, Khirni, Tamarind, and Neem, among others.** These trees have adapted to survive in regions with a distinct dry season.

❑ **Significance:** Tropical Dry Evergreen forests are ecologically significant, providing habitat and resources for various wildlife species. Preserving these ecosystems is essential to maintaining biodiversity in these coastal regions.

Casuarina Plantation

- ❑ It resembles a feathery conifer in general appearance.
- ❑ They are rapid-growing, carefree species for sites and climates as varied as coastal sand dunes, high mountain slopes, hot humid tropics, and semi-arid regions.
- ❑ They have the ability to fix atmospheric nitrogen. It grows 15 to 25 metres in height on average.

Distribution

- ❑ Casuarina is the most popular farm forestry in the states of Andhra Pradesh, Tamil Nadu, West Bengal, Odisha, Maharashtra, Gujarat, and Karnataka.

Benefits

- ❑ Reduces damage in the event of natural calamities.
- ❑ Line planting in the coastal areas helps in controlling the wind force.
- ❑ It is also used for tourism promotion in view of its ornamental appearance.
- ❑ It provides top-quality firewood.
- ❑ The Wood is suitable for paper pulp and useful raw material for the manufacture of paper for writing, printing, and wrapping.
- ❑ It has some serious medicinal value as well.

Montane Sub-Tropical Vegetation

Sub-Tropical Moist (Pine)

- ❑ Subtropical moist pine forests are forest ecosystems located in subtropical regions with moderate moisture levels, typically featuring pine trees adapted to these conditions.

❑ **Climatic Conditions:** These forests are situated at altitudes ranging from 1000 to 1800 metres above sea level.

- **Rainfall:** The annual rainfall in these areas typically falls between 100 to 200 centimetres.
- **Temperature:** Average temperature ranges from 15°C to 22°C.
- **Relative humidity** levels vary from 63 to 77 percent.

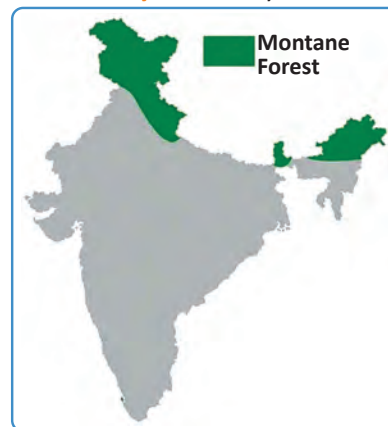


Fig. 7.6: Montane forest

❑ **Characteristics:** During the rainy season, these forests often have a thick grass cover.

- Sub-tropical moist Pine forests are **ecologically significant** and provide valuable habitat for a variety of flora and fauna in these regions.
- Maintaining these ecosystems is crucial for biodiversity conservation and preserving the unique characteristics of these sub-tropical mountainous areas.

❑ **Distribution:** Sub-tropical moist Pine forests are primarily found in specific regions, including the northwestern Himalayas (excluding Kashmir), Himachal Pradesh, Uttarakhand, the Khasi hills, Nagaland, and Manipur.

- This vegetation type is particularly prominent in the **Shivalik Hills**, which are the southernmost range of the Himalayas.

❑ **Flora and Forest Resources:** One of the prominent tree species in these forests is the Chir pine, particularly prevalent in areas with higher humidity. Additionally, you can find other important tree species such as Jamun, Oak, and Rhododendron.



IGNITE YOUR MIND

The Ghost Trees” in Kaladhungi forest, Uttarakhand, are Eucalyptus trees covered in thick cobwebs, creating a ghostly appearance due to millions of silk-spinning caterpillars. Explain the role of traditional knowledge and indigenous practices in the conservation of endangered vegetation in the Himalayan region. How can modern conservation strategies integrate traditional wisdom for effective preservation?

Sub-Tropical Dry Evergreen

- ❑ Subtropical dry evergreen forests are characterised by evergreen vegetation adapted to subtropical regions with pronounced dry seasons.
- ❑ **Climatic Conditions:**
 - **Rainfall:** Rainfall in this ecosystem typically falls within the range of 50 to 100 centimetres annually, with the number of rainy days totalling about 26 to 38 days each year.
 - **Temperature:** The climate in these areas is characterised by an average annual temperature of around 20°C.
- ❑ **Distribution:** Sub-tropical dry Evergreen forests are typically found in specific regions, including the Bhabar, the Shiwaliks, and the western Himalayas, up to an elevation of approximately 1,000 metres above sea level.
 - These forests cover the Piedmont area of the Himalayas, including parts of Punjab, Haryana, and some regions of the Kashmir hills. They are situated at elevations ranging from 450 to 1,500 metres.
- ❑ **Characteristics:** During the rainy season, these forests often feature lush grass growth.
 - Sub-tropical dry Evergreen forests are ecologically important and provide habitat for various plant and animal species, contributing to the biodiversity of these regions.
 - Conservation efforts are essential to preserve these unique ecosystems.
- ❑ **Flora and Forest Resources:** These forests consist of various species of acacia, along with trees like Olive, Acacia, Modesta, and Pistacia, among others.

Sub-Tropical Wet Hill

- ❑ Subtropical wet hill forests refer to forested ecosystems in hilly or mountainous subtropical regions with abundant rainfall, creating unique ecological conditions and supporting diverse plant and animal species.
- ❑ **Climatic Conditions:**
 - **Rainfall:** The annual rainfall in these regions is abundant, exceeding 150 centimetres, and it spans over 78 to 146 rainy days.
 - **Temperature:** The average annual temperature in these areas ranges from 18°C to 24°C, with a relative humidity varying between 51 to 81 per cent.
- ❑ **Distribution:**
 - Sub-tropical Wet Hill forests are typically found in relatively moist areas at lower altitudes on mountain ranges, particularly at elevations ranging from 900 to 1,050 metres above sea level.
 - These forests are prevalent in various regions, including the Eastern Himalayan region around Assam and the Khasi hills, as well as in places like Pachmarhi, Nilgiri, and Mahabaleshwar hills.

- ❑ **Characteristics:** Sub-tropical Wet Hill forests are characterised by a blend of tropical and temperate wet evergreen trees.
 - These trees typically have an average height ranging from 15 to 30 metres, and the canopy is less dense compared to some other forest types.
- ❑ **Flora and Forest Resources:** Major tree species found in the Eastern Himalayan region include Jamun, Machilus, Elaeocarpus, Colitis, Oak, and Chestnut, among others.
 - These forests contribute to the rich biodiversity of the region and are essential for ecological balance and conservation efforts.

Montane Temperate Vegetation

Wet Temperate

- ❑ Montane wet temperate forests are high-altitude forest ecosystems typically found in mountainous regions with temperate climates and ample precipitation, featuring a variety of plant and animal species adapted to these cool and moist conditions.
- ❑ **Climatic Conditions:**
 - **Rainfall:** These forests experience a mean annual rainfall of 150 to 300 centimetres.
 - **Temperature:** These forests have a mean annual temperature that varies between 11 to 18°C. During the winter months (December to February), temperatures can drop to freezing levels, often accompanied by dense fog.
 - **Humidity:** The average relative humidity in these regions is over 80 percent.
- ❑ **Distribution:** Wet Temperate forests thrive at elevations ranging from 1,800 to 3,000 metres above sea level and can be found along the slopes of various mountain ranges, including the **Nilgiris**, **Annamalai**, **Palni hills** (above 1,500 metres), the **Himalayas**, and the **Assam hills** (ranging from 1800 to 2900 metres).
- ❑ **Characteristics:** Trees in Wet Temperate forests typically reach heights of 15 to 18 metres and are known as “**Sholas**” in South India.
 - These forests are characterised by dense undergrowth and host numerous epiphytes, mosses, and ferns.
- ❑ **Flora and Forest Resources:** Key tree species found in these forests include Magnolia, Laurel, Rhododendron, Elm, Prunus, and Plum.
 - In the northern part of India, Oak, Chestnut, and Laurel are common tree species.
- ❑ **Significance:** Wet Temperate forests are ecologically significant and contribute to the rich biodiversity of mountainous regions, playing a crucial role in maintaining ecological balance and providing habitat for various plant and animal species.

Himalayan Moist Temperate

- ❑ Himalayan moist temperate forests are forested ecosystems located in the Himalayan mountain range, characterised by temperate climates and diverse flora and fauna.
- ❑ **Climatic Conditions:** These forests are typically found at altitudes ranging from 1,500 to 3,300 metres above sea level.
 - **Rainfall:** The annual rainfall in these areas varies from 150 to 250 centimetres,
 - **Temperature:** And average annual temperature hovers around 12°C to 13°C.
- ❑ **Characteristics:** These forests form a high but relatively open tree cover, often accompanied by shrubby undergrowth consisting of **Oaks, Rhododendrons, and some Bamboo.**
 - Mainly composed of coniferous species and Species occur in mostly pure strands.
 - Himalayan Moist Temperate forests are ecologically significant, contributing to the rich biodiversity of the Himalayan region.
 - They provide habitat for a variety of plant and animal species, including many endangered and endemic ones. Conservation efforts are essential to protect these vital ecosystems.
- ❑ **Distribution:** The Himalayan Moist Temperate vegetation is distributed across several states along the Himalayan range, including **Kashmir, Himachal Pradesh, Uttarakhand, Darjeeling and Sikkim.**
- ❑ **Flora and Forest Resources:** The vegetation in Himalayan Moist Temperate forests is characterised by a mix of broad-leaf evergreen and conifer species.
 - Notable tree species include **Oak, Fir, Spruce (Picea), Deodar (Cedrus deodara), Celtis, Chestnut, Cedar (Chamaecyparis), Maple, Silver Fir, Kail, Yew, and Birch, among others.**
 - **Timber:** It provides fine wood which is of much use for construction, timber and railway sleepers.

Himalayan Dry Temperate

- ❑ Himalayan dry temperate forests are forest ecosystems situated in the Himalayan region, characterised by temperate climates with lower moisture levels, and they typically support vegetation adapted to these semi-arid or dry conditions in the mountainous terrain.
- ❑ **Climatic Conditions:** These areas receive minimal precipitation, typically below 100 centimetres, and much of it comes in the form of snowfall. These forests are situated at altitudes above 1,500 metres.
- ❑ **Characteristics:** The unique characteristic of Himalayan Dry Temperate forests is their occurrence in the inner dry

ranges of the Himalayas, where the southwest monsoon loses much of its moisture content. Consequently, these forests are adapted to arid conditions and support xerophytic shrubs and trees.

- ❑ **Distribution:** Himalayan Dry Temperate forests are primarily found in regions such as Ladakh, Chamba, Lahaul, Garhwal, and Sikkim.
 - The Himalayan Dry Temperate vegetation is prevalent in parts of Himachal Pradesh, particularly in the lower elevations of the state.
 - Certain areas of Jammu and Kashmir, especially in the transitional zones between the higher elevations and the plains, exhibit characteristics of the Himalayan Dry Temperate vegetation.
- ❑ **Flora and Forest Resources:** Key tree species found in these forests include Deodar, Juniper, Chilgozah (Neoza), Maple, Ash, Celtis, Olive, and Oak, among others.
 - These forests play a vital role in the ecological balance of the region and provide habitat for various species adapted to arid environments.
 - Conservation efforts are crucial to safeguard these unique ecosystems in the Himalayan region.

Alpine Vegetation

- ❑ Alpine vegetation is primarily found in the Himalayan ranges at altitudes ranging from 3,000 metres to 4,000 metres above sea level. This vegetation forms a dense cover in these high-altitude regions.
- ❑ However, as you move from the southern slopes to the northern slopes of the Himalayas, the alpine vegetation transitions into low evergreen shrubs on the southern slopes and dry xerophytic vegetation on the northern slopes.
- ❑ Within the Pir Panjal range, alpine pastures with stunted conifers can be found at altitudes of approximately 2,250 metres to 2,750 metres.
- ❑ These pastures are often used as grazing grounds by the Gujjar nomads who practise transhumance and seasonal migration of livestock in search of better pastures.
- ❑ These high-altitude ecosystems are essential for the biodiversity of the Himalayan region and play a vital role in supporting unique plant and animal species adapted to these extreme conditions.
- ❑ **Flora and vegetation**
 - **Fir, spruce, rhododendron,** etc. are important species.
 - In the Western Himalayas, certain alpine species with white flowers, such as '**Brahmakamal**' and '**Kuth**', are used in perfumery.
 - **The moist alpine scrub** is a low evergreen dense growth of rhododendron, birch etc. which occurs from 3,000 metres and extends upto snowline.

- **The dry alpine scrub** is the uppermost limit of scrub xerophytic, dwarf shrubs, over 3,500 metres above sea level and found in dry zone. Juniper, honeysuckle, artemesia etc. are important species.

Distribution of Alpine Vegetation in India

□ Himalayan Region:

- **Northwestern Himalayas:** Areas in Jammu and Kashmir and Himachal Pradesh. Notable alpine zones can be found in places like **Gulmarg, Sonamarg, and the Great Himalayan National Park**.
- **Central Himalayas:** Regions like Uttarakhand, where areas around **Nanda Devi Biosphere Reserve and Valley of Flowers National Park** showcase diverse alpine flora.
- **Eastern Himalayas:** In Sikkim and Arunachal Pradesh, alpine vegetation is found in higher elevations. **Yumthang Valley in Sikkim** is known for its **alpine meadows** and diverse plant species.

DEFORESTATION IN INDIA

Deforestation is the process of clearing forests for human needs, including agriculture, urbanisation, grazing, mining, and wood use. It has been a practice since ancient times, resulting in habitat loss, biodiversity decline, ecosystem disruption, and carbon emissions. Sustainable land use practices are essential to combat its environmental consequences.

Data-Wise

- **India ranks second after Brazil**, with an average of 668,400 hectares cleared between 2015 and 2020 (rose from 384,000 hectares between 1990 and 2000), as per a Utility Bidder report. According to them, India has recorded the highest increase in deforestation.

Causes of Deforestation

- **Agriculture:** Forests are converted to agricultural land to meet the growing food demand. Shifting cultivators, practising slash-and-burn agriculture, clear substantial forest areas annually, mainly in North-east India, Andhra Pradesh, Bihar, and Madhya Pradesh.
- **Cattle Ranching:** Clearing forests for pastureland is a significant driver in some regions. Across the world, between 2013 and 2019, 70% of tropical forest was cleared illegally for farming and cattle ranching.
- **Commercial Logging:** Logging for global markets, including valuable woods like meranti, teak, mahogany, and ebony, leads to tree loss and opens land for agriculture. Trees are also cut for firewood, construction, and fodder, with grazing animals damaging saplings.
- **Mining:** Mining causes erosion, sinkhole formation, biodiversity loss, and soil, groundwater, and surface

water contamination. Often, additional logging occurs near mines for debris storage, causing pollution and health issues in nearby communities. Since 2000, coal mining has destroyed 2,500+ hectares in Chandrapur district (Maharashtra).

- **Population Growth:** Growing populations increase demands for resources, resulting in permanent forest clearance for agriculture and settlements. From 1930 to 2020, 21.58% of forests were transformed into agriculture and human settlements.
- **Urbanization and Industrialization:** Land needed for urbanisation and industrial growth leads to substantial forest clearance, disrupting ecosystems and causing environmental harm.
- **Dam Reservoir Construction:** Large dams require extensive forest destruction, affecting ecological balance, and increasing the risk of floods, droughts, and landslides. Over 30 years, 14,000 km² of Indian forests were legally cleared for mining, defence, and hydroelectric projects.
- **Forest Fires:** Natural or human-induced fires cause significant forest losses, disrupting ecosystems and biodiversity.

Consequences of Deforestation

- **Land Use Changes:** Deforestation allows for urban development, roads, and agriculture, altering landscapes.
- **Agricultural Challenges:** Deforested areas often degrade in quality, reducing fertility and productivity.
- **Soil Erosion:** Loss of forest cover exposes soil to rain and heat, leading to erosion and reduced arability.
- **Disruption of the Water Cycle:** Forests play a crucial role in regulating the water cycle. Deforestation can lead to altered rainfall patterns, reduced water quality, and increased risk of floods and droughts.
- **Loss of Medicinal Plants:** Many medicinal plants are found in forests. Deforestation can lead to the loss of potential cures and treatments for diseases.
- **Impact on Indigenous Communities:** Indigenous peoples often rely on forests for their livelihoods and cultural practices. Deforestation can displace these communities and disrupt their way of life.
- **Flooding:** Watershed disruption due to deforestation can lead to downstream flooding and coastal sedimentation.
- **Biodiversity Loss:** Deforestation threatens plant and animal species, causing extinctions and disrupting ecosystems.
- **Increased Greenhouse Gas Emissions:** Deforestation contributes to higher levels of greenhouse gases in the atmosphere, exacerbating climate change.
- **Climate Change:** Deforestation contributes to increased CO₂ levels and climate change.
- **Economic Impact:** Flooding and droughts impact economies, and the value of standing forests can surpass that of harvested wood.

Way Forward

❑ Legal Frameworks and Enforcement:

- Establish and enforce robust legal frameworks that regulate and control logging, land conversion, and other activities contributing to deforestation.
 - ❑ **Example:** India has various environmental laws and regulations, such as **the Forest (Conservation) Act, Wildlife Protection Act, and the National Green Tribunal Act**, which provide legal frameworks for the conservation and sustainable management of forests.

❑ Protected Areas and Reserves:

- Promote the sustainable use of forests within protected areas and wildlife reserves by involving local communities in conservation efforts.
 - ❑ **Example:** India has a network of **national parks and wildlife sanctuaries**, designed to protect critical ecosystems and biodiversity by promoting sustainable use of forests.

❑ Afforestation measures:

- Implement afforestation programs to plant trees in areas where they have been previously cut down or degraded in order to restore ecosystems, enhance biodiversity, and mitigate the impacts of deforestation.
 - ❑ **Example:** **The Green India Mission**, part of the **National Action Plan on Climate Change**, aims to increase forest and tree cover, enhance biodiversity, and restore degraded ecosystems.

❑ Agroforestry:

- India promotes agroforestry practices as a sustainable land-use system that integrates trees and shrubs with agricultural crops or livestock.
- The National Agroforestry Policy encourages farmers to adopt agroforestry to enhance biodiversity, mitigate deforestation, and provide alternative sources of income.

❑ Financial Incentives:

- Provide financial incentives for sustainable forest management and conservation, encouraging private landowners and companies to adopt environmentally friendly practices.
- Explore mechanisms such as payments for ecosystem services to reward communities for maintaining healthy and biodiverse forests.
 - ❑ **Example:** **The Compensatory Afforestation Fund Act (CAMPA)** establishes funds for compensatory afforestation and wildlife management. It provides financial incentives for states to undertake afforestation activities and maintain healthy forest ecosystems.

Miyawaki Afforestation - Urban Forestry

- ❑ **Miyawaki afforestation** is a technique named after the Japanese botanist **Akira Miyawaki**.
- ❑ Miyawaki afforestation involves **planting a diverse mix of native plant species in a closely spaced and multilayered pattern, mimicking natural forests**. This technique is particularly effective in urban areas with limited space.
 - Cities like **Bengaluru** and **Pune** have implemented Miyawaki afforestation projects in public spaces, such as parks and school grounds, to increase greenery and biodiversity.
- ❑ Miyawaki afforestation can contribute to **carbon sequestration, improve air and water quality, and create microhabitats for various species**. It also engages local communities and raises awareness about the importance of native flora.



IGNITE YOUR MIND

In Ladakh, there is a mysterious grove of trees near Magnetic Hill that appears to defy gravity. The slope of the hill makes vehicles seemingly roll uphill, and the trees, known as “Magnetic Hill Trees,” grow at unusual angles due to the magnetic properties of the terrain. How can sustainable tourism practices be implemented to preserve these natural wonders while ensuring responsible visitor behavior?

CORAL REEFS

- ❑ Corals are composed of calcareous rocks formed from the skeletons of tiny sea creatures called polyps. Polyps extract calcium salts from seawater to create protective skeletons for their soft bodies, collectively giving rise to corals.



Fig. 7.7: Coral Reefs

- ❑ These coral colonies are anchored to the rocky seafloor, and new generations of polyps develop on the skeletons of deceased polyps.
- ❑ The tubular skeletons grow upwards and outwards, forming a solid, cemented mass of calcareous rock known as corals.

- ❑ These rocky formations create shallow areas called reefs, which can evolve into islands over time.
- ❑ The appearance and colors of corals can vary depending on the salts and constituents they incorporate.
- ❑ The development of corals takes place over time, with diverse forms emerging on the sea surface due to ongoing deposition.
- ❑ Small marine plants like algae also contribute to coral growth by depositing calcium carbonate.

Ideal Conditions

- ❑ Corals flourish in tropical waters, specifically within the latitudinal range of 30 degrees north and 30 degrees south.
- ❑ They prefer depths between 45 metres and 55 metres below the sea surface, where there is ample sunlight available.
- ❑ The water temperature ideally hovers around 20°C to 29°C for optimal coral growth.
- ❑ Clear saltwater with a salinity range between 27 ppm to 30 ppm is conducive to coral development, while both freshwater and excessively saline water can be detrimental to polyp growth.
- ❑ An adequate supply of oxygen and plankton, which are microscopic marine organisms, is crucial for the growth and sustenance of corals. Corals tend to grow more rapidly on the seaward side due to the greater availability of food in these areas.

Location of Coral Reefs in India

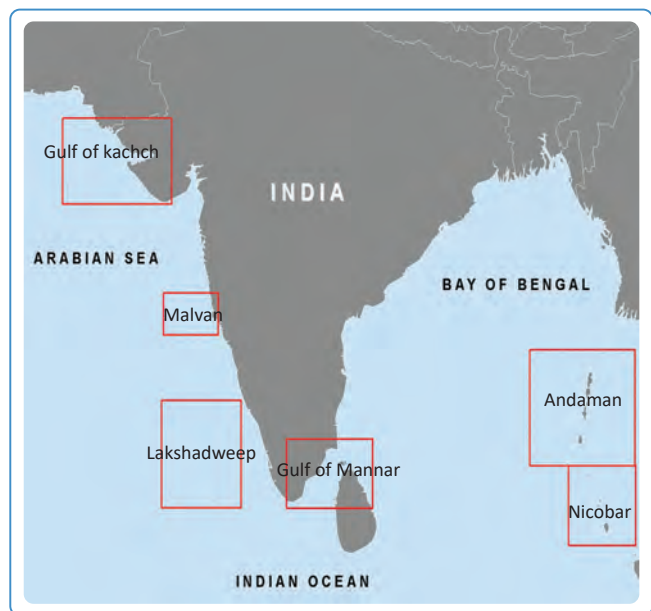


Fig. 7.8: Location of Coral reefs in India

Coral reefs in India are primarily found along the coastline in the tropical and subtropical regions, particularly in the Arabian Sea and the Bay of Bengal. The major coral reef areas in India are as follows:

- ❑ They are mainly restricted to the **Andaman and Nicobar Islands, Gulf of Mannar, Palk Strait, the Lakshadweep islands, Gulf of Kutch and Malvan.**
- ❑ All of these reefs are Fringing reefs, except Lakshadweep which are Atolls.

Coral Bleaching

- ❑ Coral reef ecosystems worldwide have experienced significant deterioration in recent decades, with disturbances arising from both human activities and natural events.
- ❑ The recent accelerated decline of coral reefs is primarily attributed to human-induced impacts, including overexploitation, overfishing, increased sedimentation, and nutrient pollution.
- ❑ Natural disturbances that harm coral reefs involve severe weather events such as storms, floods, extreme temperature fluctuations, and phenomena like El Nino Southern Oscillation (ENSO) events.
- ❑ Coral bleaching occurs when the symbiotic relationship between coral hosts and marine algae, which provide much of the coral's colour, breaks down. This breakdown results in the coral's tissue becoming transparent, revealing the white skeleton.
- ❑ Coral reef bleaching is a common response to stress caused by various disturbances mentioned earlier.
- ❑ When corals bleach, they begin to starve. While some corals can feed themselves, most struggle to survive without their algae. If conditions return to normal, corals can regain their algae, restore their normal colour, and survive.
- ❑ However, this stress often leads to reduced coral growth and reproduction, making the corals more susceptible to diseases.
- ❑ Bleached corals frequently die if the stress persists, and coral reefs experiencing high coral mortality following bleaching may take many years or even decades to recover.

Way Forward

- ❑ **Marine Protected Areas (MPAs):** Establishing marine protected areas helps restrict human activities that may harm coral reefs.
 - **Example: Gulf of Mannar Marine National Park** in Tamil Nadu is a designated Marine Protected Area which encompasses diverse ecosystems, including **coral reefs**, and restricts certain human activities to conserve marine biodiversity.
- ❑ **Sustainable Fishing Practices:** Implementing and enforcing sustainable fishing practices, such as avoiding destructive fishing methods like **blast fishing and cyanide fishing**, helps reduce stress on coral reef ecosystems.

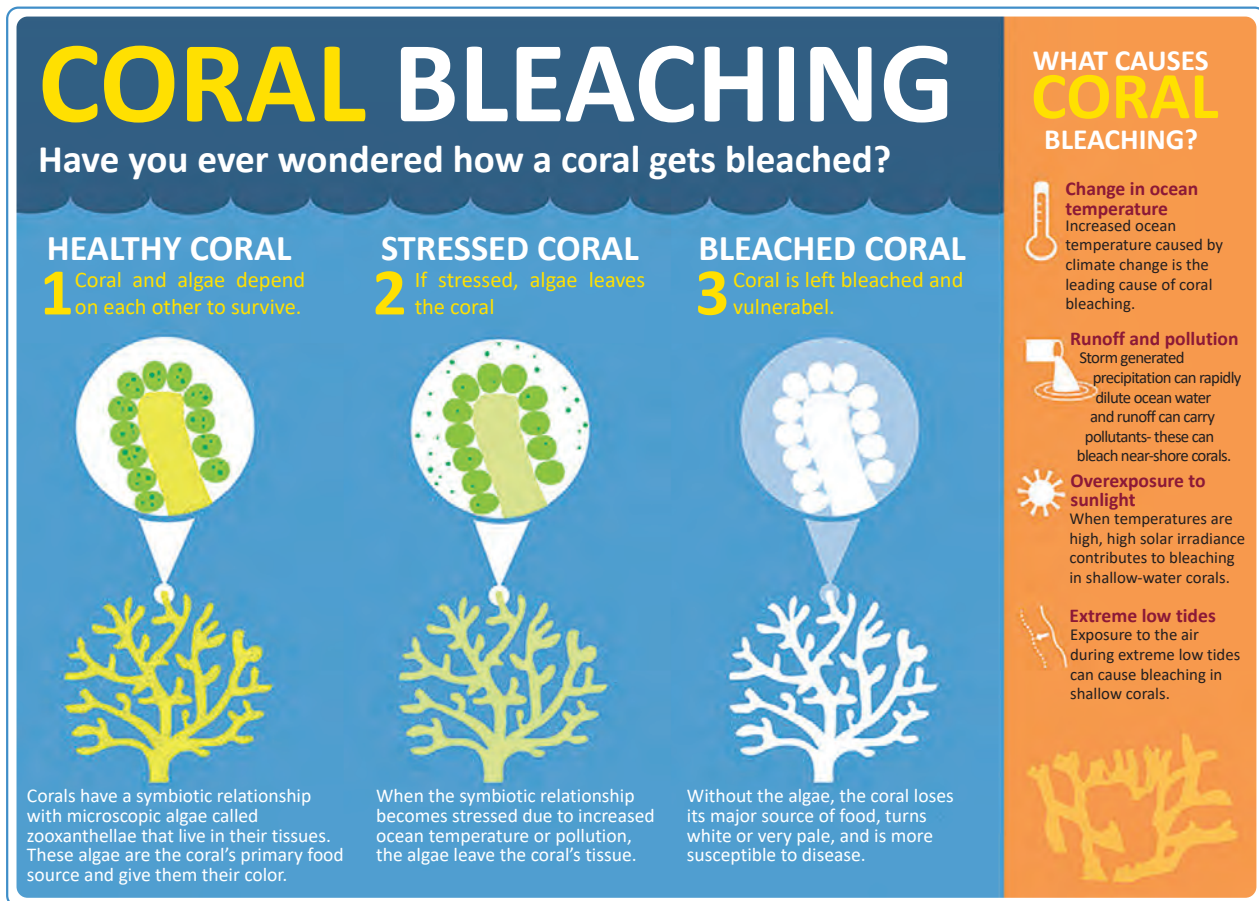


Fig. 7.9: Coral Bleaching

- **Example: Lakshadweep Islands** have implemented sustainable fishing practices, including banning destructive methods like **blast fishing**, which protect coral reef ecosystems from overfishing and habitat degradation.
- ❑ **Coral Reef Restoration Programs:** Initiatives focused on coral reef restoration involve activities like coral transplantation, where healthy coral fragments are grown in nurseries and then replanted onto damaged reef areas.
 - **Example:** Organizations like **ReefWatch Marine Conservation** have undertaken projects to restore damaged coral reefs through coral transplantation and community involvement.
- ❑ **Reducing Pollution:** Minimizing land-based sources of pollution, such as sediment runoff, agricultural runoff, and pollutants from urban areas, is crucial to restore coral health.
 - **Example:** Conservation efforts in **the Netrani Island in Karnataka** involve raising awareness about the impact of pollution on coral reefs and implementing measures to reduce land-based sources of pollution.
- ❑ **Climate Change Mitigation:** Addressing climate change is essential for coral reef conservation. Rising sea temperatures and ocean acidification, linked to climate

change, pose significant threats to coral reefs. Efforts to reduce greenhouse gas emissions can help mitigate these impacts.

- **Example: The Lakshadweep Archipelago** faces threats from rising sea temperatures and climate change. Efforts to reduce the **carbon footprint**, such as promoting **renewable energy** and sustainable tourism practices, contribute to mitigating climate change impacts on coral reefs.

Biorock Technology

Biorock technology, also known as mineral accretion technology, is an innovative approach to coral reef conservation. It involves the use of low-voltage electrical currents to stimulate the deposition of minerals onto metal structures submerged in the ocean. These structures, often made of steel, act as a substrate for coral settlement and growth.

Example: The Gulf of Kutch has seen experimental deployments of Biorock structures to enhance coral growth. The **Zoological Survey of India**, in collaboration with local communities, has explored the potential of Biorock technology as a tool for coral reef restoration and resilience-building in the region.

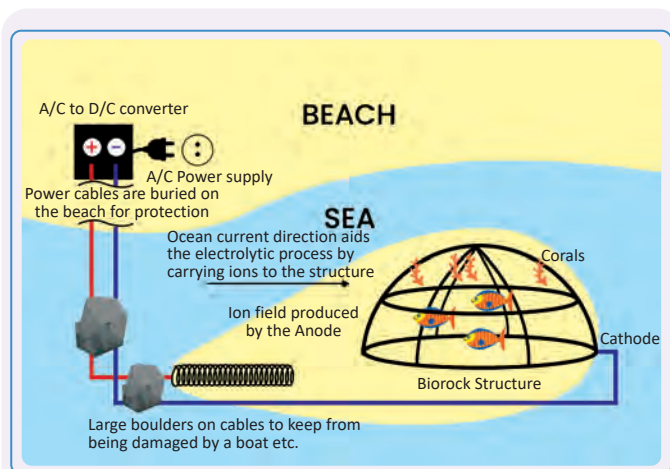


Fig. 7.10: Biorock Technology

MANGROVES

- ❑ Mangrove forests are a dominant presence along the coastlines of tropical and subtropical regions, playing a vital role in the overall forest ecosystem.
- ❑ Mangroves serve as a unique intersection point where the sea, freshwater, and land converge.
- ❑ There are 54 to 75 species of true mangroves that are taxonomically distinct from their terrestrial counterparts and are exclusively found in the intertidal areas of coastlines.
- ❑ Mangrove forests represent one of the world's most intricate ecosystems, thriving in environmental conditions that would rapidly prove fatal to most other types of vegetation.
- ❑ Their ability to endure extremely saline waters and soils is a result of their remarkable adaptability to their surroundings, allowing them to prevent or eliminate salt.

Mangrove Forests have Evolved Specific Growth Patterns to Adapt to the Challenging Conditions of Estuaries



Fig. 7.11: Mangroves

Key Adaptations Include

- ❑ Tolerance of brackish waters and flooding.
- ❑ Ability to withstand waterlogging.
- ❑ Survival in oxygen-deprived (anoxic) soil.
- ❑ Mechanisms for excluding salt.
- ❑ Mangroves are viviparous, which means seeds germinate and develop into seedlings while the seeds are still attached to the parent tree. These seedlings are normally called propagules and they photosynthesize while still attached to the mother tree. The parent tree supplies water and necessary nutrients. They are buoyant and float in the water for sometime before rooting themselves on suitable soil.
- ❑ In some mangrove plants, salt exclusion is achieved through impermeable roots, which act as an ultra-filtration system to prevent sodium salts from entering the rest of the plant, effectively reducing salt content by 90% to 97%. The plant then sheds its old leaves, which concentrate salt and are eventually shed.
- ❑ Other mangroves store salt in cell vacuoles, while certain species have glands on their leaves that actively release salt, leaving visible salt crystals on the leaf's upper surface.
- ❑ Many mangrove trees feature aerial roots known as pneumatophores, which draw oxygen from the air for the roots. This is necessary because the soil in the shallow regions of mangrove forests is often flooded during high tides.
- ❑ Mangrove plants also have a unique mode of reproduction, with mangrove seeds beginning to grow while attached to the parent plant. These seedlings develop roots while still connected to the parent.

Largest Mangrove Areas

The countries with the largest mangrove areas are Indonesia, Brazil, Australia, Mexico and Nigeria.

Mangroves in India

- ❑ According to the **Indian State Forest Report 2021**, Mangrove cover in India is 4975 sq. Km which is 0.15% of the country's total geographical area.
 - India has witnessed a 1.10% increase in its mangrove cover, equivalent to 54 square kilometres, according to the latest assessment.
- ❑ **West Bengal** holds the largest share of India's mangrove cover at 42.45%, followed by **Gujarat** at 23.66%, and the Andaman and Nicobar Islands at 12.39%. Gujarat has experienced the most substantial increase in mangrove forest cover, adding 37 square kilometres.
- ❑ Notably, nearly half of India's mangrove area is situated in the Sundarbans region of West Bengal. Sundarbans in West Bengal are the largest mangrove forest regions in the world. It is listed as a UNESCO World Heritage Site.
- ❑ Other states that have shown significant growth in mangrove cover include West Bengal (2,114 square kilometres), Gujarat (1,140 square kilometres), the Andaman and Nicobar Islands (617 square kilometres),

Andhra Pradesh (404 square kilometres), and Maharashtra (304 square kilometres).

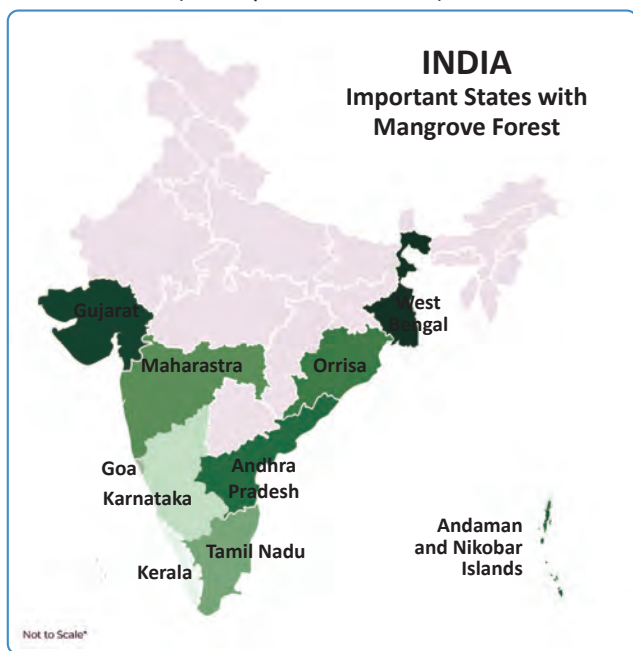


Fig. 7.12: Distribution of Mangrove in India

- ❑ Among the states, Kerala has the smallest mangrove cover at 9 square kilometres, while among the Union Territories, Puducherry has the least, covering 2 square kilometres.
- ❑ For research and development purposes, the Environment Ministry has established the National Mangrove Genetic Resource Centre in Bhitarkanika, Odisha.

Important Mangrove Regions in India

- ❑ **Gujarat:** Mangroves are located in the Gulf of Kutch, Gulf of Khambhat, and areas like Dumas and Ubhrat.
- ❑ **Andhra Pradesh:** Mangrove areas are found in Coringa, East Godavari Delta, and Krishna Delta.
- ❑ **Odisha:** Mangroves are present in Bhitarkanika, Mahanadi, Subarnarekha, Devi-Kauda, Dhamra, and Chilka regions.
- ❑ **West Bengal:** The Sundarbans in West Bengal is a prominent area with mangrove cover.
- ❑ **Andaman & Nicobar:** Mangroves are found in North Andaman and Nicobar regions.
- ❑ **Maharashtra:** Mangrove areas can be found in Achra-Ratnagiri, Devgad-VijayDurg, Veldur, Kundalika-Revdanda, Mumbra-Diva, and Vikhroli.
- ❑ **Goa:** Mangroves exist in Goa.
- ❑ **Karnataka:** Mangrove regions can be located in Coondapur, Dakshin Kannada/Hannavar, Karwar, and the Mangalore Forest Division.
- ❑ **Kerala:** Mangroves are present in Vembanad and in the northern region of Kannur.
- ❑ **Tamil Nadu:** Mangrove areas can be found in Pichavaram, Muthupet, Ramnad, Pulicat, and Kaznuveli.

THE INDIA STATE OF FOREST REPORT 2021

The India State of Forest Report 2021 (ISFR 2021) is a significant biennial evaluation conducted by the Ministry of Environment, Forest and Climate Change (MoEFCC) and marks the 17th iteration. This report provides an extensive overview of India's forests and tree cover, along with several critical findings and assessments.

- ❑ The ISFR 2021 provides the most up-to-date information on the country's 'Forest cover' and 'Tree cover'.
- ❑ It also presents the estimates of growing stock, the extent of trees outside forests, mangrove cover, bamboo resources, and forest carbon stock assessments.
- ❑ The chapter on forest carbon assessment also includes India's progress toward meeting its Nationally Determined Contribution commitments.

New Features in ISFR 2021

- ❑ The report includes a special chapter focusing on "Forest Cover Assessment in Tiger Reserves and Tiger Corridor Areas", a notable addition.
- ❑ It presents the results of two special studies: "Above Ground Biomass Estimation Using Synthetic Aperture Radar Data" (in partnership with ISRO) and "Climate Hot Spots in Forest Areas Studies."
- ❑ A separate chapter highlights new initiatives undertaken by the Forest Survey of India (FSI) in the last two years.
- ❑ For the first time, the ISFR assessed forest cover in tiger reserves, tiger corridors, and the Gir forest, home to the Asiatic lion.

Major Findings of the Report

- ❑ **Area-wise Madhya Pradesh has the largest forest cover in the country**
- ❑ Maximum increase in forest cover witnessed in Andhra Pradesh (647 sq km) followed by Telangana (632 sq km) and Odisha (537 sq km).
- ❑ 17 states/UTs have above 33 percent of the geographical area under forest cover.
- ❑ Total carbon stock in country's forest is estimated to be 7,204 million tonnes, an increase of 79.4 million.
- ❑ Total mangrove cover in the country is 4,992 sq km, an increase of 17 sq Km is observed.
- ❑ **Increase in Total Forest and Tree Cover:** The total forest and tree cover in India have increased by 2,261 sq km since the last assessment in 2019 (ISFR-2019). This increase consists of 1,540 sq km of forest cover and 721 sq km of tree cover. In total, the forest and tree cover now occupy 80.9 million hectares, which is approximately 24.62% of the country's geographical area.

INDIA'S GREEN COVER INCREASES

Change in forest & tree cover

Class	Area (sq km)		% of GA*		Net increase (in 2021 compared to 2019)	% increase
	2021	2019	2021	2019		
Forest Cover	7,13,789	7,12,249	21.71	21.67	1,540	0.22
Tree Cover**	95,748	95,027	2.91	2.89	721	0.76
Total	8,09,537	8,07,276	24.62	24.56	2,261	0.28

*GA - geographical area ; **Tree Cover (patches of size less than 1 hectare outside recorded forest area)

States/UTs showing gain/loss in forest cover in 2021

Gain in sq km		Loss in sq km	
Andhra Pradesh	647	257	Arunachal Pradesh
Telangana	632	249	Manipur
Odisha	537	235	Nagaland
Karnataka	155	186	Mizoram
Jharkhand	110	73	Meghalaya

Forest cover in major mega cities in 2021

Cities	Area (sq km)	Decadal change**
Delhi	194	(+11%)
Mumbai	111	(+9%)
Bengaluru	89	(-5%)
Hyderabad	82	(+15%)
Chennai	23	(+26%)
Ahmedabad	9	(-48%)
Kolkata	2	(-30%)

** Decadal change between 2011 and 2021 in %

Fig. 7.13: Findings of the Report

- ❑ **Top States with Increased Forest Cover:** Andhra Pradesh, Telangana, and Odisha recorded the highest increase in forest cover, with Andhra Pradesh leading with 647 sq km of additional forest cover.
- ❑ **Area-wise Forest Cover:** Madhya Pradesh has the largest forest cover in the country, followed by Arunachal Pradesh, Chhattisgarh, Odisha, and Maharashtra.
- ❑ **Percentage-wise Forest Cover:** Mizoram (84.53%), Arunachal Pradesh (79.33%), Meghalaya (76%), Manipur (74.34%), Nagaland (73.90%) are the top five states with the highest forest cover as a percentage of their total geographical area.
- ❑ **States/UTs with Significant Forest Cover:** The report highlights that 17 states/union territories have forest cover exceeding 33% of their land area. Lakshadweep, Mizoram, Andaman & Nicobar Islands, Arunachal Pradesh, and Meghalaya have over 75% forest cover.
- ❑ **Increase in Mangrove Cover:** The mangrove cover in India has increased by 17 sq km since the previous assessment in 2019, with Odisha (8 sq km), Maharashtra (4 sq km) and Karnataka (3 sq km) leading in the expansion of mangrove cover. The total mangrove cover in the country spans 4,992 square kilometres.
- ❑ **Total Carbon Stock:** India's carbon stock has increased by 79.4 million tonnes since 2019, with an annual carbon store increase of 39.7 million tonnes. The country's forest carbon stock is estimated to be 7,204 million tonnes.
- ❑ **Bamboo Forests:** Bamboo forests have seen remarkable growth, increasing from 13,882 million culms (stems) in 2019 to 53,336 million culms (stems) in 2021.
- ❑ **Forest Fires in India:** The report highlights that forest fires pose a threat to 35.46% of the country's forest cover, with varying degrees of vulnerability. Approximately 2.81% are extremely vulnerable, 7.85% are very highly vulnerable, and 11.51% are highly vulnerable.
- ❑ **Climate Change Hotspots:** The report predicts that climate change and rising temperatures will have a negative impact on 45-64% of Indian forests by 2030. All Indian states, except Assam, Meghalaya, Tripura, and Nagaland, are expected to have highly sensitive climate hotspots. The Union Territory of Ladakh, with forest cover ranging from 0.1-0.2%, is most susceptible.
- ❑ **Forest Cover in Tiger Reserves:** A new chapter assesses forest cover in India's Tiger Reserves, Corridors, and Lion Conservation Areas. While some reserves have shown growth, others have experienced a decline in forest cover over the past decade.

Impact of India's Forest Resources on Climate Change

- ❑ Forests act as a crucial natural stabilizer for the climate, regulating the carbon cycle significantly.
- ❑ Carbon sequestration through photosynthesis is recognized as one of the most potent and cost-efficient strategies for addressing climate change.
 - They provide a unique, secure, and cost-effective method for capturing and storing carbon dioxide, potentially mitigating global CO₂ concentrations.
- ❑ Reduction in forest cover diminishes the land's capacity to absorb carbon dioxide, leading to increased atmospheric concentrations of this greenhouse gas.

- ❑ Forests play a critical role in regulating local and regional climate patterns through evapotranspiration, influencing factors like rainfall distribution and temperature.
- ❑ Deforestation disrupts these climate patterns, causing alterations in rainfall distribution, temperature increases, and shifts in weather patterns.
- ❑ Loss of forest cover also affects biodiversity, with many species relying on forests for habitat and survival, further impacting ecosystem functioning and resilience to climate change.

Therefore, efforts to conserve and enhance the contribution of forest resources to mitigate climate change are essential for India's sustainable development.

FOREST FIRES IN INDIA

Forest fires in India have become a growing concern, with significant implications for biodiversity, ecosystems, and the environment. We will briefly discuss the recent trends, statistics and causes of forest fires

Recent Trends and Statistics

- ❑ The peak fire season in India typically begins in mid-February and lasts for approximately 14 weeks. During the period between August 29, 2022, and August 28, 2023, India experienced an unusually high number of forest fire alerts. There were 14,689 high-confidence fire alerts reported through satellite observations (Visible Infrared Imaging Radiometer Suite).
- ❑ Data analysed by the World Resources Institute (WRI) reveals that from 2001 to 2022, India lost 3.59 lakh hectares of tree cover due to fires. Additionally, 2.15 million hectares were lost due to other drivers of tree cover loss.
- ❑ The year 2008 witnessed the highest tree cover loss due to fires, with 3000 hectares lost, accounting for 3.5% of all tree cover loss for that year.
- ❑ Data from the Forest Survey of India indicates that between November 2021 and June 2022, there were 2,23,333 forest fires in India. This number decreased to 2,12,249 fires between November 2022 and June 2023.
- ❑ Humid primary forests in India have faced significant losses. From 2002 to 2022, India lost 3.93 lakh hectares of humid primary forest, constituting 18% of the total tree cover loss during that period. The area covered by humid primary forests in India decreased by 3.9% during this time.

FOREST CONSERVATION

Forest conservation refers to the sustainable management and protection of forests and wooded areas to ensure their

continued existence and health. It involves various practices and techniques aimed at preserving and maintaining forests for the benefit of both present and future generations. The primary goals of forest conservation are to safeguard biodiversity, mitigate climate change, protect ecosystems, maintain water resources, and provide sustainable livelihoods for communities dependent on forests.

Some of the Key Components and Techniques used for Forest Conservation

- ❑ **Reforestation and Afforestation:** Reforestation aims to restore existing depleted forests, while afforestation involves creating entirely new forests in previously non-forested areas, contributing to increased forest cover, biodiversity, and carbon sequestration.
- ❑ **Selective Logging:** Selective logging is a method of harvesting timber that involves carefully choosing which trees to cut down while preserving the overall health and structure of the forest. This approach minimises the ecological impact of logging activities.
- ❑ **Forest Fire Management:** Forest fires can be destructive and often result from both natural and human causes. Conservation efforts include the prevention and management of forest fires through techniques like controlled burns, firebreaks, and early detection systems.
- ❑ **Sustainable Forest Management:** This approach focuses on balancing the economic, social, and ecological aspects of forest use. It involves carefully planning and monitoring logging activities, ensuring that forest resources are harvested at a rate that allows for natural regeneration.
- ❑ **Wildlife Conservation:** Protecting wildlife habitats within forests is an integral part of forest conservation. It involves creating and maintaining wildlife corridors, protected areas, and habitat restoration initiatives to support diverse species.
- ❑ **Community Engagement:** Involving local communities in forest conservation efforts is essential. Sustainable forest management practices that consider the needs and rights of indigenous and local communities can help ensure the long-term health of forests.
- ❑ **Policy and Regulation:** Governments and international organisations play a critical role in enacting and enforcing policies and regulations that promote forest conservation. These policies can include land-use planning, protected area designations, and restrictions on deforestation and illegal logging.

Sacred Groves

Sacred groves, found in regions like **Rajasthan, Karnataka, Meghalaya, and Maharashtra**, are vital for conserving rare and endemic plant and animal species. These forested areas, protected due to their cultural and spiritual significance, are managed by tribal communities, who strictly prohibit deforestation and contribute to their rich biodiversity. However, sacred groves face modern challenges such as habitat fragmentation and invasive species, requiring community-based conservation efforts to preserve these unique ecosystems.

Constitutional Provisions, Acts and Policies Related to Forest Conservation in India

India has a history of forest policies and laws aimed at protecting and conserving its valuable forest resources:

- ❑ **Constitutional Provisions:** Forests were moved to the Concurrent List in the Constitution, allowing both the central and state governments to legislate on forest-related matters. Article 48A mandates the protection and improvement of the environment, including forests and wildlife. Article 21 of the Constitution includes the right to a healthy environment, ensuring protection from pollution. Article 51A (g) makes it a fundamental duty of every citizen to protect and improve the natural environment, including forests and wildlife.
- ❑ **Indian Forest Policy, 1952:** This policy represented an extension of the colonial-era forest policy. It recognized the need to increase forest cover to one-third (33%) of the total land area and emphasised various measures, including persuading primitive communities to abandon harmful practices like shifting cultivation, strengthening forest administration through adequate forest laws, promoting forest research, and controlling grazing in forestry.
- ❑ **Forest Conservation Act, 1980:** This act mandated central government permission for activities like agro-forestry in forest areas, treating violations as criminal offences. Its primary objectives were to limit deforestation, conserve biodiversity, and protect wildlife. While it held promise for forest conservation, its effectiveness was limited.
- ❑ **Forest Policy of 1988:** This policy marked a significant shift from commercial concerns to focusing on the ecological role of forests and promoting participatory management. Its key goals included preserving and restoring ecological balance, preventing soil erosion and denudation in catchment areas, substantially increasing forest cover through afforestation and social forestry programs, and boosting forest productivity to meet national needs.
- ❑ **Wildlife Protection Act, 1972:** This act facilitated the establishment of wildlife sanctuaries and national

parks and led to the formation of the National Board for Wildlife, which advises the central government on wildlife conservation.

- ❑ **National Green Tribunal Act, 2010:** This act aimed to ensure the effective and expeditious resolution of environmental protection and forest conservation-related cases, including providing relief and compensation for damages to individuals and property.
- ❑ **Compensatory Afforestation Fund Act, 2016:** It is designed to facilitate the efficient and transparent use of funds released in exchange for forest land diverted for non-forest purposes. It also aims to mitigate the impact of forest land diversion through reforestation and afforestation efforts.

Draft National Forest Policy 2018

- ❑ The basic thrust of the **draft National Forest Policy of 2018** is for conservation, protection and management of forests along with addressing other issues associated with forest and forest management.
- ❑ To protect the interest of tribals, the Draft Policy of 2018 proposes to launch **Community Forest Management Mission** for management of community forest resources through a participatory forest management approach.
- ❑ Moreover, specific provisions have been made in the draft policy to ensure that the **interests of local communities are protected** and also provisions are made to consider them as partners in the management of forests.
- ❑ In addition, forest rights of tribals and other traditional forest dwellers are safeguarded under a separate Act, namely, **"The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Right) Act, 2006"** (**Forest Rights Act, 2006**) which recognizes and vests the forest rights of the Scheduled Tribes and other traditional forest dwellers who have been residing in such forests for generations but whose rights could not be vested".

Government Schemes Related to Forest Conservation in India

India has several government schemes and programs aimed at forest conservation and sustainable forest management. These schemes are implemented at both the central and state levels. Some notable government schemes related to forest conservation in India are mentioned below:

- ❑ **National Afforestation Programme (NAP):** Launched in 2000, NAP focuses on afforestation and reforestation of degraded forest lands. It aims to increase forest and tree cover, improve biodiversity, and enhance ecosystem

services. This program is implemented by the Ministry of Environment, Forest and Climate Change (MoEFCC).

- ❑ **National Mission for a Green India (GIM):** Part of India's National Action Plan on Climate Change (NAPCC), GIM launched in 2014 aims to restore and enhance forest and tree cover, improve biodiversity, and mitigate climate change. It focuses on afforestation, reforestation, and sustainable forest management.
- ❑ **Integrated Development of Wildlife Habitats (IDWH):** Implemented by the MoEF&CC, this scheme focuses on the conservation and management of wildlife habitats, both within and outside protected areas. It aims to enhance wildlife protection and improve habitat quality.
- ❑ **National Bamboo Mission (2018-19):** This mission promotes the conservation and sustainable management of bamboo resources, which are essential for rural livelihoods and forest conservation. It includes activities such as bamboo plantation, processing, and market development.
- ❑ **Joint Forest Management (JFM) Program (1988):** Implemented at the state level, the JFM program involves local communities in the protection and management of forest resources. It encourages community participation and sharing of benefits from forest conservation efforts.
- ❑ **Eco-Task Force:** The Indian Army, in collaboration with the MoEFCC, runs the Eco-Task Force program. It focuses on afforestation and ecological restoration in degraded areas, particularly in and around military establishments.

Classification of Forest

As of 2021, the total forest cover in India is 80.9 million hectares, which is 24.62 percent of the total geographical area. Forests are classified in India on various bases:

- ❑ **Based on Administration:**
 - **Reserved Forests:** Areas declared by the state government as reserved for the conservation of forest produce, soil, and water. Access and activities within these areas are restricted.
 - ❑ They cover **54%** of total forest area.
 - **Protected Forests:** Designated for the sustainable use of forest resources, with limited restrictions on activities.
 - ❑ They cover **29%** of total forest area.
 - **Unclassed Forests:** Areas that have not been classified as reserved or protected. They are largely degraded, unproductive and unprofitable forests.
 - ❑ They cover **17%** of total forest area.

{Reserved and Protected forests are permanent forests which are maintained for the regular supply of timber and other forest products as well as for ecological reasons}

❑ Based on Ownership:

- **State Forests:** Include almost all important forest areas of the country and are under full control of the government (state/central).
 - ❑ Cover almost **94%** of the Total Forest Area.
- **Commercial Forests:** Owned and administered by local bodies (municipal corporations, village panchayats, district boards etc.).
 - ❑ Cover **5%** of the Total Forest Area.
- **Private Forests:** Under private ownership.
 - ❑ Cover slightly more than **1%** of the Total Forest Area.

❑ Based on Composition and Types of Leaves:

- **Broad-leaved:** Tropical and subtropical forests that are widely spread and cover ~95% of total forest area. Sal and Teak are the most important species and provide valuable timber.
 - ❑ Other species of broad-leaved forests are **Indian laurel, shistram, garyan and benteak**.
- **Coniferous Forests:** Temperate Forests dominated by cone-bearing trees, adapted to colder climates with a longer winter. They are mainly confined to the Himalayan ranges.
 - ❑ Common species include **deodar, chir, fir, spruce, etc.**

Forest Products in India

The Forest Rights Act (FRA) of 2006 in India is a legislation that recognizes and vests the forest rights and occupation in forestland in forest-dwelling communities and other traditional forest dwellers.

While the FRA primarily focuses on recognizing individual and community rights over land, it also has implications for the collection and use of forest products.

Community Forest Resources (CFR)

- ❑ The FRA grants communities the right to protect, regenerate, and conserve Community Forest Resources (CFR) within traditional or customary boundaries.
- ❑ CFR includes not only land but also the resources in such areas, which can encompass a range of forest products.
- ❑ Communities have the authority to manage and use these resources sustainably.

Forest Produce as Per FRA, 2006

- ❑ **Timber:** India has a variety of timber species, including teak, sal, deodar, and bamboo, which are extensively used in construction, furniture, and paper industries.
- ❑ **Non-Timber Forest Produce (NTFPs):** This category includes a wide range of products such as **gums, resins, medicinal plants, honey, and bamboo shoots**.

- These products contribute significantly to local economies and traditional practices.

❑ **Minor Forest Produce**

- The FRA recognizes the rights of forest-dwelling communities and traditional forest dwellers to access, collect, and use Minor Forest Produce (MFP) from the forests they inhabit.
- As per FRA 2006, “**minor forest produce**” includes **all non-timber forest produce of plant origin** including **bamboo, brush wood, stumps, cane, tussar, cocoons, honey, wax, lac, tendu or kendu leaves, medicinal plants and herbs, roots, tubers and the like.**

❑ **Paper and Pulp:** Wood from various tree species is used in the paper and pulp industry.

- Eucalyptus and bamboo are commonly cultivated for paper production.

Indirect Uses of Forests in India

- ❑ **Biodiversity Conservation:** Forests harbor a vast array of plant and animal species, contributing to the conservation of biodiversity.
- ❑ **Water Regulation:** Forests play a crucial role in regulating water flow, preventing soil erosion, and maintaining watershed health, ensuring a sustainable supply of water for various purposes.
- ❑ **Climate Regulation:** Forests act as carbon sinks, helping regulate the climate by absorbing and storing carbon dioxide. They also influence local climate patterns.
- ❑ **Tourism and Recreation:** Forests attract tourists for activities like trekking, wildlife safaris, and nature walks. This contributes to the economy and promotes environmental awareness.
- ❑ **Cultural and Spiritual Significance:** Many forests in India hold cultural and spiritual importance for local communities, contributing to traditional practices and beliefs.

India's forests provide a wide range of products and services crucial for the well-being of both the environment and society. Sustainable management practices are essential to balance the extraction of forest products with conservation efforts.

Social Forestry in India

- ❑ The management and development of forests with afforestation on barren areas to produce environmental benefit and rural development is known as social forestry.
- ❑ It encompasses activities such as forest management, forest protection, and reforestation of deforested regions with the goal of enhancing rural, environmental, and social development.
- ❑ Unlike other forestry initiatives, the needs of local populations are prioritised in the context of social forestry.

- ❑ The **term “social forestry”** was first used by The **National Commission on Agriculture in 1976**, when the Indian government sought to relieve strain on forests by planting trees on all underutilised and barren lands.

❑ **Objectives of Social Forestry**

- **Enhance Environmental Resilience:** Improve the environment to protect crops from adverse weather conditions.
- **Promote Sustainable Resource Availability:** Augment the supply of home fuelwood, small timber for rural housing, animal feed, and minor forest products for local industries.
- **Beautify the Landscape and Foster Recreation:** Elevate the natural beauty of the terrain by establishing recreational forests, benefiting both rural and urban communities.
- **Generate Employment Opportunities:** Create avenues for unskilled laborers, contributing to economic opportunities and livelihoods.
- **Contribute to Land Restoration:** Impact land restoration initiatives through sustainable forestry practices.
- **Enhance Living Standards:** Improve the living standards and overall quality of life for both rural and urban populations.

❑ **Approach to Social Forestry**

- Social forestry is a new approach to forest management. It is a strategy that balances local needs with multiple external interests.
- Engages communities living in and around forests in the use and management of forests in a sustainable manner.
- Communities are empowered by raising awareness, building capacity, developing policies with local people, and recognising their rights and knowledge systems.
- In exchange for participating in sustainable forest management, the community receives benefits and access to forest resources.

Types of Social Forestry

❑ **Farm Forestry:**

- Farm forestry involves the deliberate management of trees within an agricultural setting, serving specific purposes and offering various benefits.
- **Soil and Water Conservation:** The presence of trees in farm forestry systems helps in maintaining soil health, preventing erosion, and conserving water resources.

- **Shelter and Grazing for Animals:** Trees in farm forestry provide valuable shelter for livestock and can serve as grazing areas, contributing to improved animal well-being.
- **Customization for Domestic Needs:** Individual farmers are encouraged to cultivate trees on their own land tailored to meet the specific domestic requirements of their families.
- **Focus on Non-Commercial Farm Forestry:** While farm forestry can have commercial aspects, the majority of social forestry programs in the nation concentrate on non-commercial objectives, emphasizing the broader societal and environmental benefits.

❑ Community Forestry

- Focuses on the involvement of local communities in the sustainable management and conservation of forest resources.
- Empowers communities to take responsibility for forest protection, regeneration, and equitable distribution of benefits.
- The primary goal of community forestry is to benefit the entire community, providing an opportunity for landless individuals to participate in tree planting and gain access to the benefits typically enjoyed by landowners. Nearly 85,000 joint forest management committees oversee approximately 17.3 million hectares (42.8 million acres) of forestlands in India.

❑ Agro-Forestry

- Agro-forestry involves the integrated cultivation of trees alongside agricultural crops on the same land, including previously unused or wasteland.
- Integrates the cultivation of trees with agricultural practices, providing multiple benefits such as improved soil fertility, increased crop yields, and additional sources of income.
- Promotes sustainable land use by combining agriculture and forestry practices.
- The Food and Agriculture Organization (FAO) has documented 45.4 million hectares of land engaged in agroforestry practices in India.

❑ Extension Forestry

- Extension forestry is becoming more widespread in urban areas and on most residential estates.
- Extension forestry includes planting trees along canals, roadways, and railroads, as well as on wastelands.
- This type of social forestry helps to establish forests on common village lands, panchayat lands, and government wastelands.
- This method of forestry contributes to the expansion of forest borders.

- It is the practice of forestry in regions devoid of tree growth and other flora, located away from traditional forest areas, with the goal of expanding the area under tree growth.

❑ Scientific Forestry or Silviculture

- Scientific forestry is concerned with the replacement and replanting of new crops and trees.
- The many scientific forestry aims are outlined below.
- Afforestation in undeveloped regions.
- Attempting to expand raw materials for forest-based enterprises.
- Expand employment opportunities.
- High-quality timber species production.

Significance of Forestry in India

❑ Environmental Conservation:

- Forestry helps in conserving biodiversity, preventing soil erosion, and maintaining ecological balance.
- Forests act as carbon sinks, mitigating the impacts of climate change.

❑ Economic Contribution:

- Forestry is a significant contributor to India's economy through the timber industry, non-timber forest products, and ecotourism.
- Employment opportunities are created in both rural and urban areas through forestry-related activities.

❑ Social and Cultural Values:

- Forests hold cultural significance for many communities, providing resources for rituals, traditional practices, and spiritual beliefs.
- Forestry initiatives can empower local communities by involving them in decision-making processes and benefit-sharing.

Problems of Indian Forestry and Remedies

Problems of Indian Forestry	Remedies
Deforestation	❑ Strict enforcement of forest conservation laws. ❑ Promotion of sustainable forestry practices. ❑ Afforestation programs.
Encroachment and Illegal Logging	❑ Strengthening forest protection mechanisms. ❑ Community involvement in monitoring. ❑ Awareness programs.
Forest Degradation	❑ Implementing scientifically based forest management practices. ❑ Monitoring ecosystem health. ❑ Promoting afforestation.

Lack of Community Participation	❑ Implementing community forestry programs. ❑ Empowering local communities. ❑ Creating awareness about the importance of sustainable forest management.
Climate Change Impacts	❑ Adopting climate-resilient forestry practices. ❑ Promoting biodiversity conservation. ❑ Integrating climate change considerations into forest management plans.

CONCLUSION

India's diverse natural vegetation plays a crucial role in supporting biodiversity, regulating climate, and providing various ecosystem services. However, many of these ecosystems face threats from human activities, including deforestation, urbanization, and climate change.

Hence, recognizing the value of natural vegetation, understanding the threats it faces, and implementing effective conservation measures are crucial for sustaining India's ecological balance and preserving its rich natural heritage.

MOUNTAIN PEAKS

Mountain Peaks of the Himalayas

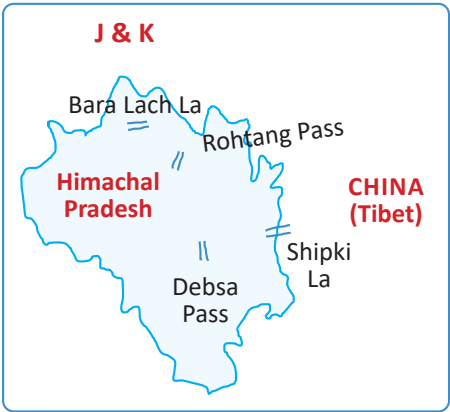
Peak Name	Height (in meters)	Mountain Range	Country
Mount Everest	8,848 m	Himalayas	Nepal
Mt. Godwin Austin (K2)	8,611 m	Karakoram	Pakistan
Kangchenjunga	8,586 m	Himalayas	India, Nepal
Lhotse	8,516 m	Himalayas	Nepal, China
Makalu	8,485 m	Himalayas	Nepal, China
Cho Oyu	8,188 m	Himalayas	Nepal, China
Dhaulagiri I	8,167 m	Himalayas	Nepal
Manaslu	8,163 m	Himalayas	Nepal
Nanga Parbat	8,126 m	Himalayas	India (Pakistan occupied Kashmir Region) (Gilgit-Baltistan Region)
Annapurna I	8,091 m	Himalayas	Nepal
Gasherbrum I (K5)	8,080 m	Karakoram	India, China
Broad peak (K3)	8,051 m	Karakoram	India, China
Gasherbrum II (K4)	8,035 m	Karakoram	India, China
Shishapangma / Gosainthan	8,027 m	Himalayas	China

Hill Ranges of the Peninsular Plateau

Hill Range	Location	Direction	Length	Highest Peak
Aravali	Runs from Delhi in the north to Palanpur in Gujarat	NE-SW	800 km	Guru Sikhar (1,722 m)
Vindhyan	Overlooks the Narmada Valley, stretching from Jobat in Gujarat to Sasaram in Bihar	E-W	1,200 km	Sad-Bhawna Shikhar ("Goodwill Peak") (752 m)
Satpura	Runs south of the Vindhyas, between the Narmada and Tapi rivers	E-W	900 km	Dhupgarh (1,350 m)
Western Ghats (Sahyadri)	Forms the western edge of the Deccan Plateau, from the Tapi Valley to Kanniyakumari	N-S	1,600 km	Anai Mudi (2,695 m)
Eastern Ghats	Runs almost parallel to the east coast of India, from the Mahanadi River in Odisha to the Vagai River in Tamil Nadu	N-S	1131 km	Mahendra Giri (1,501 m)
Nilgiri Hills, Western Ghats	Tamil Nadu, Ooty area	W-NW	50 km	Doddabetta (2,637)

MAJOR PASSES

Major Passes on the Map	Major Passes	Description
Jammu and Kashmir and Ladakh 	Mintaka Pass	Connects Kashmir and China, located at the trijunction of Indo-China and Afghanistan border.
	Zojila Pass	Connects Srinagar, Kashmir to Kargil, Ladakh
	Parpik Pass	The Indo-China border connects Kashmir and China
	Khunjerab Pass	Indo - China border, connects Kashmir and China
	Aghil Pass	Located at 5,000 m above sea level, it connects Ladakh with the Xinjiang Province.
	Banihal Pass	Connects Jammu with Srinagar, located across the Pir Panjal range. It remains snow-covered during the winters.
	Sia La	Near Siachen glacier, Saltoro ridge, Ladakh.
	Changla Pass	Between Leh and Changthang
	Khardung La	Connecting Leh and Nubra. It is the world's highest motorable road that passes through this pass.
	Kongka Pass	Connects Ladakh with Aksai Chin
	Lanak Pass	Connecting Ladakh and Tibet.
	Karakoram Pass	Connects Ladakh with Xinjiang Province of China.
	Saseer La	Connects Nubra Valley with Siachen Glacier
	Spanggur Gap	Connects Tibet with Ladakh
	Gyong La	It is situated in the Siachen glacier
	Bilafond La	It is on the ancient silk route linking the Indian subcontinent and China
	Tanglang La	It is situated adjacent to the Leh-Manali highways.
	Imis Pass	Connects Ladakh with Tibet

Himachal Pradesh 	Baralacha La	Connects Himachal Pradesh and Jammu & Kashmir. The national highway connecting Mandi (Himachal Pradesh) with Leh passes through this pass.
	Kunzum Pass	Connects Lahaul and Spiti valley.
	Debsa Pass	It acts as a link between the Kullu and Spiti districts.
	Rohtang Pass	It links Kullu, Lahaul and Spiti valley. Maintenance and construction of roads in this region are looked at by the Border Road Organisation (BRO)
	Shipki La	Connects Himachal Pradesh with the Tibet region. Sutlej River enters India (originated in the Tibet region) near this pass. Significant pass between Indo-China.
	Chanshal Pass	It is located in the Shimla district and it connects Rohru to the villages of Dodra-Kwar.
	Lamkhaga Pass	It links Harsil in Garhwal to Kinnaur in Himachal Pradesh
	Kimokal Pass	Connects Himachal Pradesh with China.
Uttarakhand 	Lipu Lekh	Located at the trijunction of Uttarakhand (India), Tibet (China) and Nepal.
	Mana Pass	Connects Uttarakhand with Tibet
	Mangsha Dhura	Connects Uttarakhand with Tibet, it is used by pilgrims going to Kailash – Mansarovar.
	Niti La	Connects Uttarakhand with Tibet
	Muling La	Connects Uttarakhand with Tibet
	Sin La	It connects Bidang in Darma Valley to Jolingkong Lake in the Kuthi Yankti valley
	Mayali Pass	It is located on the path from Gangotri to Kedarnath.
	Nama Pass	Nama pass connects the remote villages of Kuti and Sela. It lies in the northern part of Pithoragarh district.
	Dhumdhar Kandi Pass	The pass is situated at an altitude of 5,490 meters above sea level and connects two dhams Yamnotri and Gangotri.
	Auden' Col	It is also an important pass located on the route from Gangotri to Kedarnath.
Sikkim 	Nathu La	Connects Sikkim with Tibet. It was an offshoot of the ancient Silk Route.
	Jelep La	Located on the Sikkim-Bhutan border, it forms an important link between Sikkim and Lhasa.
	Dongkha La	It connects Sikkim with Tibet.
	Goecha La	--

Arunachal Pradesh 	Bom Di La	Connects Arunachal Pradesh with Tibet.
	Dihang Pass	Connects Arunachal Pradesh with Myanmar
	Yong Yap Pass	Connects Arunachal Pradesh with Tibet
	Diphu (Dipher) Pass	Located at the trijunction of India, China, and Myanmar
	Sela Pass	--
	Bum La	--

Western Ghat Passes

Pass	State(s)	Prominent Peaks
Thal Ghat	Maharashtra	Khandala Point, Needle Point, Lingmala Falls
Bhor Ghat	Maharashtra	Khandala Point, Peth Fort
Amba Ghat	Maharashtra	Harishchandragad, Ratangarh Fort
Kasara Ghat	Maharashtra	Khandala Point, Duke's Nose
Chorla Ghat	Maharashtra, Goa	Sahyadri Range, Phonda Mountain
Charmadi Ghat	Karnataka	Mullayanagiri (highest peak in Western Ghats), Bhairava Parvatha
Shiradi Ghat	Karnataka	Kemmanagundi Hills, Baba Budan Giri
Niligiri Pass	Karnataka, Tamil Nadu	Doddabetta Peak, Mukurthi Peak
Palakkad Gap	Kerala, Tamil Nadu	Anamudi Peak (highest peak in Western Ghats), Nilgiri Hills
Munnar Pass	Kerala	Anamudi Peak, Devikulam Lake
Athirappilly Pass	Kerala	Chembra Peak, Vazhachal Falls
Idukki Pass	Kerala	Anamudi Peak, Periyar National Park

IMPORTANT VALLEYS IN THE HIMALAYAS

States	Valleys
Jammu and Kashmir	Kashmir valley, Betab valley, Chenab valley, Nubra valley, Suru valley, Indus valley, Mushkosh valley, Markha valley etc.
Himachal Pradesh	Kullu valley, Kangra valley, Chamba valley, Karsog valley, Spiti valley, Lahaul valley, Sangla valley, etc.
Uttarakhand	Dun valley – Dehradun valley, Bhagirathi valley, Mandakini valley, valley of flowers, etc.
West Bengal	Neora valley
Sikkim	Chumbi valley, Yumthang valley
Assam	Barak valley, Brahmaputra valley
Arunachal Pradesh	Dibang valley
Nagaland	Dzukou valley
Manipur	Kabaw valley and Imphal valley
Nepal	Kathmandu valley

MAJOR GLACIERS IN INDIA

Name	Location/State	Important Feature	River Originated
Siachen	Karakoram (Gilgit-Baltistan (in the Pakistani-administered portion) and Leh in Ladakh)	Largest in Nubra Valley of the Himalayas and the second longest glacier outside the Polar region	Nubra River
Biafo	Karakoram (Baltistan region of Ladakh)	Flows into Shigar River	Shigar River

Chogo Lungma	Karakoram (Gilgit-Baltistan [Pakistan Occupied])	Located in Gilgit Baltistan	Baltoro River
Gangotri	Uttarakhand	Originates below the Chaukhamba Peak and it is also known as 'Gomukh'	Bhagirathi River (a tributary of the Ganges)
Rimo	Siachen Area of Ladakh	Drains in the Shyok River	Shyok River
Hispar	Gilgit-Baltistan		Hispar River
Zemu	Sikkim/Nepal	The largest glacier in Eastern Himalaya it feeds River Teesta	Teesta River
Sonapani	Chandra Valley of Lahaul and Spiti region/Himachal Pradesh	The longest glacier of the Pir Panjal range near the Rohtang Pass	Chandra River
Milam	Uttarakhand	A major source of River Gori Ganga Biggest glacier in Kumaon Himalaya	Gori Ganga River
Diamir (Nanga Parbat)	Kashmir (Pakistan Occupied)	Diamir mountains are known as 'King of Mountains' and are the World's tallest mountains	Diamir River
Bara Shigri	Chandra Valley of Lahaul in Himachal Pradesh	Chandra Valley of Lahaul in Himachal Pradesh	Chandra River
Chong Kumdan	Karakoram/Ladakh	Ladakh; they feed and are a threat to Shyok River and might block it.	Shyok River
Pindari	Kumaon Region of Uttarakhand	Gives rise to River Pindari	Pindari River

Major Waterfalls in India

Name	State	Features	River	Height (m)
Kunchikal Fall	Karnataka	It lies on the Varahi River, it is the highest waterfall in India.	Varahi River	455
Barehipani Fall	Odisha	Located in Mayurbhanj district, it is the 2nd highest waterfall in India.	Budhabalanga River	399
Dudhsagar Fall	Goa	Lies on the Mandovi River, located in Bhagwan Mahaveer Sanctuary and Mollem National Park.	Mandovi River	310
Jog Fall	Karnataka	Lies on the Sharavathi River in the Western Ghats, it is the second-highest waterfall in India. Located on the Uttara Kannada and Sagara border.	Sharavathi River	253
Lodh Fall	Jharkhand	Also known as Budha Ghagh, it is a 2-tiered waterfall in the middle of Palamau Tiger Reserve.	Burha River	145
Shivasamudram Fall	Karnataka	Lies on the Cauvery River, it is a segmented waterfall consisting of two waterfalls – Gangana Chukki and Bhara Chukki.	Cauvery River	122
Hogenakkal Fall	Tamil Nadu	Lies on the Cauvery River.	Cauvery River	120
Palani Fall	Himachal Pradesh	It is one of the highest-surge waterfalls in India, located near Kullu.	Subarnarekha River	105
Kapildhara	Madhya Pradesh	Lies on the Narmada River.	Narmada River	100
Teerathgarh Fall	Chhattisgarh	A block-type waterfall, also known as the Queen of Satpura.	Kanger River	97
Apsara Fall	Madhya Pradesh	Also known as Apsara Vihar or Fairy Pool waterfall, located close to Pachmarhi Hills.	Beehad River	80

Langshiang Fall	Meghalaya	Lies on the Kynshi River, it is the third-highest waterfall in India.	Kynshi River	45
Dhuandhar Fall	Madhya Pradesh	The Marble Rocks and surrounding area support a diverse range of flora and fauna, including rare species like the Indian flying fox and freshwater turtles.	Narmada River	30

IMPORTANT LAKES IN INDIA

- ❑ **Largest Freshwater Lake:** Wular Lake, Jammu and Kashmir
- ❑ **Largest Saline Water Lake:** Chilka Lake, Orissa
- ❑ **Highest Lake (Altitude):** Cholamu Lake, Sikkim
- ❑ **Longest Lake:** Vembanad Lake, Kerala
- ❑ **Largest Artificial Lake:** Govind Vallabh Pant Sagar (Rihand Dam); Dhebar Lake (second-largest artificial lake).

Name	State	Type	Description
Pulicat Lake	Andhra Pradesh and Tamil Nadu	Brackish Water	It encompasses Pulicat Lake Bird Sanctuary; Satish Dhawan Space Centre is located here
Kolleru Lake	Andhra Pradesh	Freshwater	Home to migratory birds
Nagarjuna Sagar	Andhra Pradesh	Freshwater	Artificially constructed on Krishna River
Deepor Beel	Assam	Freshwater	Ramsar Site
Son Beel	Assam	Freshwater (tectonically formed)	Wetland
Kanwar Lake	Bihar	Freshwater (Oxbow)	Asia's largest freshwater oxbow lake
Kankaria lake	Gujarat	Artificial	During the 14th century Muhammed Shah II
Sudarshan lake	Gujarat	Artificial	Built by Pushyagupta, an officer in the province of Chandragupta Maurya. Repaired by Rudradaman
Narayan Sarovar	Gujarat	Artificial freshwater	A pilgrimage site for Hindus
Badkhal Lake	Haryana	Freshwater	Man-made
Brahma Sarovar	Haryana	Freshwater	Ancient water pool sacred to Hinduism
Damdama Lake	Haryana	Freshwater	Constructed by the British government for rainwater harvesting
Karna	Haryana	Freshwater	Connected to the great epic Mahabharat
Chandra Taal	Himachal Pradesh	Sweet water	Ramsar wetland site
Suraj taal	Himachal Pradesh	Freshwater (High altitude)	Bhaga river inflow
Maharana Pratap Sagar	Himachal Pradesh	Freshwater	Ramsar site
Prashar	Himachal Pradesh	Freshwater (Holomitic)	It has a floating island
Renuka	Himachal Pradesh	Freshwater	Ramsar site
Dal lake	J&K	Warm monomictic	Remnants of past glacial period
Pangong Tso	J&K	Endorheic Lake (saline water)	Indo-China Border
Wular lake	J&K	Freshwater (Tectonic lake)	Largest freshwater lake in India
Tso Moriri	J&K	Saltwater	High altitude lake

Agata lake	Karnataka	Freshwater	Located in the southwest part of Bangalore
Pampa Sarovar	Karnataka	Freshwater	On Tungabhadra River
Astamudi Kayal	Kerala	Brackish water	Ramsar site
Vembanad	Kerala	Brackish water	Ramsar site, famous for boat race
Bhojtal	Madhya Pradesh	Freshwater	Ramsar site, the largest artificial lake in India
Salim Ali Lake	Maharashtra	Freshwater	Birdwatching
Shivsagar	Maharashtra	Freshwater	Koyna dam
Lonar lake	Maharashtra	Crater lake	National Geo-Heritage monument
Loktak lake	Manipur	Lenticular freshwater	Ramsar wetland; Phumdis (Floating Islands); Multipurpose project
Chilika lake	Odisha	Brackish water	India's largest brackish water lake, lagoon
Harike	Punjab	Freshwater	Ramsar site
Rupar	Punjab	Freshwater	Man Made riverine lake
Sambhar lake	Rajasthan	Saltwater	Ramsar wetland; largest inland saltwater lake in India
Khecheoplari	Sikkim	Freshwater	Sacred lake for Hindus and Buddhists
Ooty lake	Tamil Nadu	Freshwater	Boathouse
Hussain Sagar	Telangana	Artificial lake	Artificial Gibraltar rock-island
Govind Ballabh Pant Sagar	Uttar Pradesh	Man-made lake	Rihand dam

Lakes in News

Lake	State	Why in News
Chilika Lake	Odisha	Bird Flu outbreak, potential threat to migratory birds
Dal Lake	Jammu & Kashmir	Encroachment on lake area, pollution concerns
Loktak Lake	Manipur	Phumdi (floating islands) shrinking, threats to ecosystem
Lonar Lake	Maharashtra	Mysterious pink color change
Pulicat Lake	Andhra Pradesh & Tamil Nadu	Cyclone damage, impact on coastal communities
Sambhar Lake	Rajasthan	Salt production challenges, environmental concerns
Vembanad Lake	Kerala	Rising water levels, flooding in surrounding areas
Wular Lake	Jammu & Kashmir	Shrinking size, livelihood concerns for fishermen

MAJOR HYDRO PROJECTS IN INDIA

State	Hydroelectric Power Plant	River/Capacity
Andhra Pradesh	Nagarjuna Sagar Dam	Krishna/816MW
	Srisaillam Right Bank Power Station Project	Krishna/770MW
	Shivanasamudra Hydro Power Project	Kaveri/42MW
Assam	Karbi Langpi Hydro Electric Project (KLHEP)	Borpani River/100MW
	Kopili Hydro Electric Project	Kopili River/275MW
Arunachal Pradesh	Kameng Hydro Power Station	Bichom and Tenga Rivers/600MW
	Ranganadi Hydro Power Station	Ranganadi River/405MW
	Subansiri Lower Hydroelectric Project	Subansiri/2000MW (under consideration)
Bihar	Dagmara Hydro-Electric Project	Kosi River/130MW (under consideration)

Chhattisgarh	Hasdeo Bango Dam	Hasdeo River/120MW
Gujarat	Sardar Sarovar Project	Narmada River/1450 MW
Himachal Pradesh	Bhakra Hydroelectric Project	Satluj/1325MW
	Pong Hydroelectric Project	Beas River/396MW
	Dehar Hydroelectric Project	Beas River/990MW
	Nathpa Jhakri Hydro Electric Project	Satluj River/1500MW
	Karcham Wangtoo Hydroelectric Plant	Satluj River/1091MW
	Koldam Hydroelectric Plant	Satluj River/800MW
Jammu & Kashmir	Baglihar Stage- I Hydroelectric Project	Chenab River/450MW
	Salal Hydro Electric Power plant	Chenab River/720MW
	URI-I Uri power station	Jhelum River/480MW
	The Pakal Dul Hydro Electric Project	Marusudar/1000MW
Jharkhand	Subarnarekha Hydel Power Project	Subarnarekha River/130MW
Karnataka	Almatti Hydroelectric Project	Krishna River/290MW
	Sharavathi Hydro Power Plant	Sharavathi River/ 1035MW
Kerala	Pallivasal Hydroelectric Project	Periyar River/37.5MW
	Idukki Hydro Electric Project	Periyar River/780MW
Madhya Pradesh	Indira Sagar Hydroelectric Project	Narmada River/1000MW
	Bansagar Hydroelectric Project	Sone River/435MW
Maharashtra	Koyna (Pophali) Hydroelectric Project	Koyna River/1960MW
Nagaland	Dikhu Hydro Electric Project	River Dikhu and River Yangnyu/186MW
Odisha	Hirakud Hydro Electric Project	Mahanadi/276MW
	Balimela Hydro Electric Project	Sileru River/510MW
Punjab	Anandpur Sahib Hydroelectric Project	Satluj River/134MW
	Ranjit Sagar Dam Hydroelectric Project	Ravi River/600MW
Sikkim	Teesta-VI hydroelectric power project	Teesta River/500MW
Tamil Nadu	Kundah Hydroelectric Project	Bhavani river/585MW
	Kadamparai Hydroelectric Project	Kadamparai River/400MW
Telangana	Srisailem Left Bank Hydroelectric Project	Krishna River/900MW
Uttar Pradesh	Rihand Hydroelectric Project	Rihand River/300MW
Uttarakhand	Tehri Hydroelectric Project	Bhagirathi River/2400MW
	Kishau Dam Hydropower Project	Tons River/660MW
	Kotli Bhel Dam	Ganga/1000MW

Basic Information

Description	Value
Current Capacity of Hydroelectric Plants in India	46,850MW
India's global position in terms of installed hydroelectric power capacity	5th
Classification of Hydroelectric Energy	Renewable source of energy
Administration of Large Hydroelectric Power Projects (Above 25MW)	Ministry of Power
Administration of Small Hydroelectric Power Projects (Up to 25MW)	Ministry of New and Renewable Energy

MAJOR HIGHWAYS

Old National Highway Number	New National Highway Number	Serial No. in NH Act, 1956	States/UTs Through which it Passes
NH 1A and NH 1D	NH 1	1	Jammu & Kashmir
NH 2	NH 19 (Golden Quadrilateral)	63	Bihar, Delhi, Haryana, Jharkhand, Uttar Pradesh, West Bengal
NH 3 NH 50	NH 60	188	Maharashtra
NH 223	NH 4	12	Andaman & Nicobar Islands
NH 4 B	NH 348	161	Maharashtra
NH 5 NH 6 NH 60 NH 217	NH 16 (Golden Quadrilateral)	49	Andhra Pradesh, Odisha, Tamil Nadu, West Bengal
NH 8	NH 48 (Golden Quadrilateral)	157	Delhi, Gujarat, Haryana, Karnataka, Maharashtra, Rajasthan, Tamil Nadu
NH 21 NH 22 NH 95	NH 5	13	Haryana, Chandigarh, Himachal Pradesh, Punjab

RAILWAY ZONES

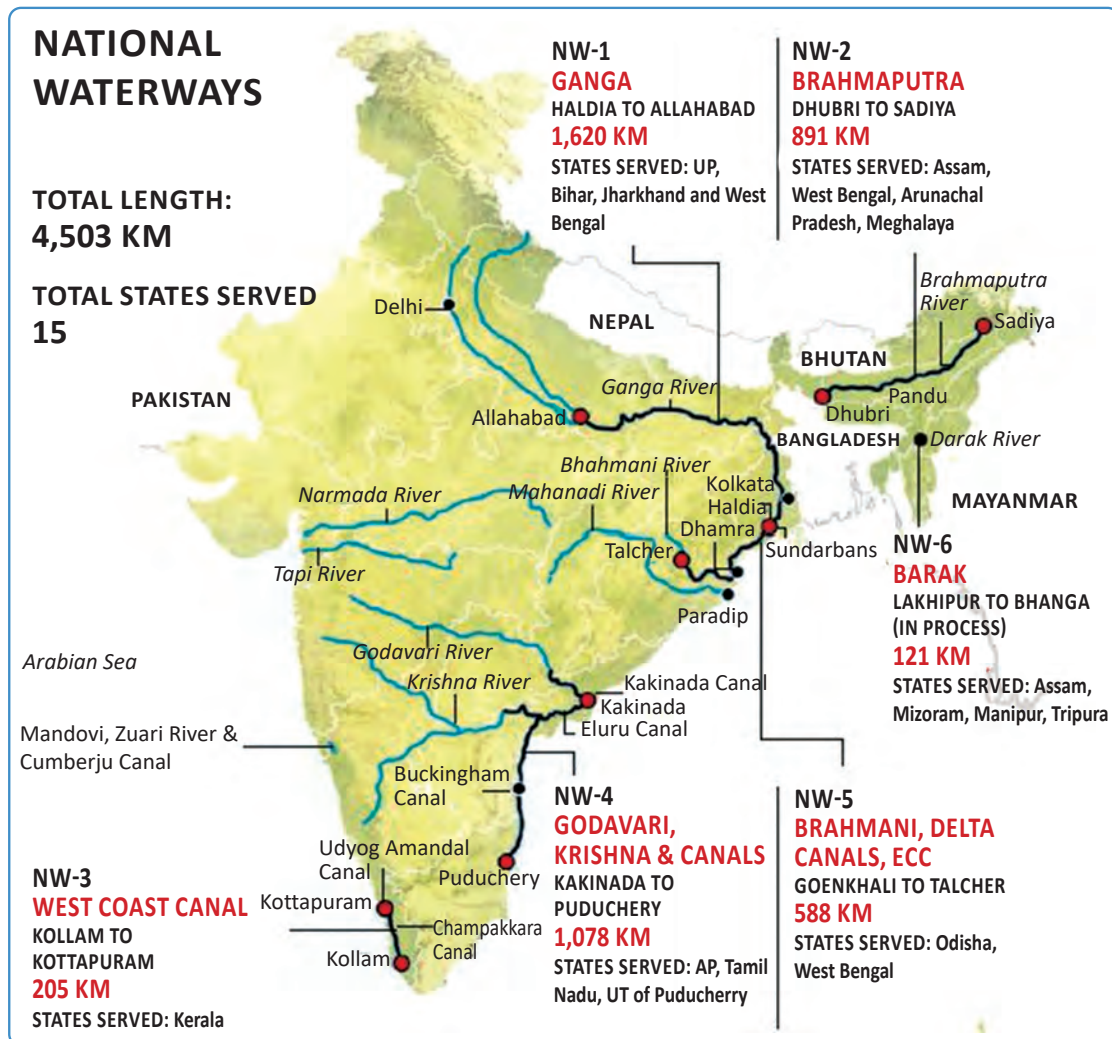
Railway Zone	Division	Zonal Headquarters
Central Railway	❑ Mumbai ❑ Nagpur ❑ Bhusawal ❑ Pune ❑ Solapur	Mumbai
Eastern Railway	❑ Howrah-I ❑ Howrah-II ❑ Sealdah ❑ Malda ❑ Asansol ❑ Chittaranjan ❑ Kolkata Metro	Kolkata
East Central Railway	❑ Danapur ❑ Mugalsarai ❑ Dhanbad ❑ Sonpur ❑ Samastipur	Hajipur
East Coast Railway	❑ Khurda Road ❑ Waltair ❑ Sambalpur	Bhubaneswar
Northern Railway	❑ Delhi-I ❑ Delhi-II ❑ Ambala ❑ Moradabad ❑ Lucknow ❑ Firozpur	Baroda House, New Delhi
North Central Railway	❑ Allahabad ❑ Jhansi ❑ Agra	Allahabad

North Eastern Railway	☐ Izzatnagar ☐ Lucknow ☐ Varanasi ☐ DLW	Gorakhpur
North Frontier Railway	☐ Katihar ☐ Alipurduar ☐ Rangiya ☐ Lumding ☐ Tinsukia	Maligaon, Guwahati
North Western Railway	☐ Jaipur ☐ Jodhpur ☐ Bikaner ☐ Ajmer	Jaipur
Southern Railway	☐ Chennai ☐ Madurai ☐ Palghat ☐ Trichy ☐ Trivandrum	Chennai
South Central Railway	☐ Secunderabad ☐ Hyderabad ☐ Guntakal ☐ Vijayawada ☐ Nanded	Secunderabad
South Eastern Railway	☐ Kharagpur ☐ Adra ☐ Chakradharpur ☐ Ranchi ☐ Shalimar	Garden Reach, Kolkata
South East Central Railway	☐ Bilaspur ☐ Nagpur ☐ Raipur	Bilaspur
South Western Railway	☐ Bangalore ☐ Mysore ☐ Hubli ☐ RWF/YNK	Hubli
Western Railway	☐ BCT ☐ Vadodara ☐ Ahmedabad ☐ Ratlam ☐ Rajkot ☐ Bhavnagar	Mumbai CST
West Central Railway	☐ Jabalpur ☐ Bhopal ☐ Kota	Jabalpur

WATERWAYS IN INDIA

National Waterway Nos.	River System	Route	Location
NW - 1	Ganga-Bhagirathi-Hooghly (Longest waterways 1620 km)	Prayagraj – Haldia	Uttar Pradesh, Bihar, Jharkhand, West Bengal
NW - 2	Brahmaputra	Sadiya-Dhubri	Assam
NW - 3	West Coast Canal, Champakara Canal, and Udyogamandal Canal	Kottapuram – Kollam	Kerala

NW - 4	Krishna and Godavari	Kakinada—A Puducherry stretch of canals, Kaluvelly Tank, Bhadrachalam – Rajahmundry, Waziraba—Vijayawada	Andhra Pradesh, Tamil Nadu, and Puducherry
NW - 73	Narmada River	It traverses through Maharashtra and Gujarat	Gujarat and Maharashtra
NW - 111	Zuari River	Sanvordem Bridge to Marmugao Port	Goa



National Waterways of India

IMPORTANT RIVER

River System	River Name	Origin	Important cities, National Park, Wildlife Sanctuary	Other
Himalayan Rivers	Indus	The northern slope of Kailash in Tibet	Leh	Forms gorge between Ladakh and Zaskar range
	Jhelum	Verinag, foothills of the Pir Panjal range	Srinagar	Flow through Wular Lake
	Chenab	Near Bara Lacha Pass	Doda, Kishtwar, Ramban.	—
	Ravi	Near Rohtang Pass	Chamba, Kathua	—
	Satluj	Rakas Lake, Tibet	Firozpur, Ludhiana	Bhakra-Nangal Project
	Brahmaputra	Chemayungdung Glacier	Guwahati, Dibrugarh, Kaziranga NP, Orang NP, Dibru-Saikhowa NP	—

	Barak	Barail range	Silchar	Tipaimukh Dam in Manipur
	Manas	Rises in Bhutan	Manas Biosphere Reserve	–
	Dhaleshwari	Mizo Hills	–	Passes through tropic of Cancer
	Ganga	Gangotri glacier	Varanasi, Patna, Kanpur, Haridwar	Holiest River of India
	Alaknanda	Satopanth Glacier	Badrinath, Gangotri NP (it joins Bhagirathi River)	–
	Ramganga	Southern slopes of Dudhatoli Hill, Uttarakhand	Bareilly, Jim Corbett NP	–
	Gomati	Gomat Taal	Lucknow, Jaunpur	–
	Sarayu	–	Ayodhya	–
	Buhri Gandak	Western slopes of Sumesar Hills	Muzaffarpur, Valmiki NP	–
	Hooghly	Giria, Murshidabad	Kolkata	–
	Yamuna	Yamunotri Glacier	Delhi, Mathura, Agra, Allahabad (confluence of Yamuna & Ganga)	–
	Banas	Aravalli Range	Ranthambore NP	–
	Kali Sindh	It originates in the Vindhyan ranges	–	Crosses the Tropic of Cancer; Tributary of Chambal
	Ken	Kaimur Hills	Panna NP	–
	Rihand	Ramgarh Hills	–	Passes through Tropic of Cancer
	North Koel	Chotanagpur Plateau	Palamau TR	Passes through Tropic of Cancer
	Subarnarekha	Ranchi Plateau	Ranchi, Jamshedpur	–
	Narmada	Amarkantak Plateau	Jabalpur	Dhuandhar fall, Kapildhara fall
	Tapi	Mahadeo Hills	Surat, Jalgaon, Bhusawal, Burhanpur	Flows parallel to Satpura Range
	Godavari	Trimbak Plateau	Nashik, Trimbakeshwar	–
	Musi	Balaghat plateau	Hyderabad	–
	Krishna	Mahabaleshwar	Vijayawada	–
	Tungabhadra	Baba Budan Hills	Hampi, Kurnool	–
	Cauvery	Taal Cauvery in the Brahmagiri Range	Tiruchirapalli	–
Peninsular Rivers	Lakshman Tirtha	Western Ghats	Nagarhole NP	–
	Kabani	Western Ghats	Wayanad Sanctuary, Nagarhole NP	–
	Pambiyar	Pulachimallai Hills, Western Ghats	Sabarimala Temple	–
	Sharavati	Shimoga district	–	Jog waterfall
	Mandovi	Bhimgad, Karnataka	Panaji, Old Goa	Dudhasagar Falls, Varapoha Falls
	Mahanadi	Sihava Parbat	Sambalpur, Cuttack	Forms Chhattisgarh Plain - 'Rice Bowl of India'
	Sabarmati	Aravalli Hills	Gandhinagar, Ahmedabad	–
	Kshipra	Vindhya range	Ujjain	–

TYPES OF FORESTS IN INDIA

Type	Characteristics	Geographical Distribution	Important Species
Moist Tropical Forest	Rainfall: > 250cm Temperature: 25-27°C Humidity: > 77%	Western sides of Western Ghats, Arunachal Pradesh, Assam, Nagaland, Manipur, Mizoram, Tripura, Andaman & Nicobar Islands	Measua, cedar, bamboo, jamun, canes
Tropical Semi Evergreen Forest	Rainfall: 200-250cm Temperature: 24-27°C Humidity: ~75%	Assam, lower slopes of Eastern Himalayas, Odisha, Andaman	Semul, rosewood, kusum, Indian chesnut, champa
Tropical Moist Deciduous Forest	Rainfall: 100-200cm - Temperature: ~27°C Humidity: 60-75%	Along Western Ghats, Shiwalik range, Odisha, West Bengal, Andaman & Nicobar Islands	Teak, sal, lendi, bamboo
Littoral and Swamp Forests	Can grow in fresh and brackish water Dense mangroves along coasts, tidal creeks, backwaters, salt marshes, mud flats	Deltas of Ganga, Mahanadi, Krishna, Cauvery	Sundri, agar, bhendi, Rhizophora, canes, palms
Dry Tropical Forests	Tropical Dry Evergreen Forests - Rainfall: ~100cm Temperature: ~28°C Humidity: ~75%	Along coasts of Tamil Nadu	Jamun, ritha, tamarind, neem
	Tropical Dry Deciduous Forests - Rainfall: 100-150cm - Shed leaves in dry season - Transitional type between moist deciduous and thorn forests	Along foot of Himalayas to Kanniyakumari, except in Rajasthan, Western Ghats, West Bengal	Teak, tendu, bamboo, sal
	Tropical Thorn Forests - Rainfall: < 75cm - Humidity: < 50% - Temperature: 25-30°C - Trees low and widely scattered	NW parts of India, Rajasthan, SW Punjab, western Haryana, Kuchch, parts of Saurashtra, along leeward side of Western Ghats	Neem, babul, acacias
Montane Sub-Tropical Forests	Sub-Tropical Broad Leaved Hill Forests - Altitude: 1000-2000m - Rainfall: 75-125cm - Temperature: 18-21°C - Humidity: 80%	Eastern Himalayas, Nilgiri, Palni Hills, Mahabaleshwar, Satpura, Maikal range, highlands of Aravali range	Evergreen oaks, chestnuts, ash, beech
	Sub-Tropical Moist Pine Forests - Altitude: 1000-2000m - Rainfall: Varied - Dominated by Chir or Chil - Used for timber and resin/turpentine production	Western Himalayas, Arunachal Pradesh, Manipur, Naga Hills, Khasi Hills	Chir (Chil)
	Sub-Tropical Dry Evergreen Forests - Altitude: Up to 1000m - Rainfall: 50-100cm - Hot summers, cold winters	Bhabar belt, Shiwalik, western Himalayas up to 1000m above sea level	Olive, acacia, pistachio
Montane Temperate Forests	Montane Wet Temperate Forests - Altitude: 1800-3000m - Rainfall: 150-300cm - Temperature: 11-14°C - Humidity: > 80%	Higher hills of Tamil Nadu, Kerala, Eastern Himalayas, West Bengal, Assam, Arunachal Pradesh, Sikkim, Nagaland	Deodar, Indian chesnut, birch, plum, oak, hemlock
	Himalayan Moist Temperate Forests - Altitude: 1500-3300m - Rainfall: 150-250cm	Kashmir, Himachal Pradesh, Uttarakhand, Sikkim	Pines, Cedars, Silver Firs, Spruce
	Himalayan Dry Temperate Forests - - Predominantly coniferous - Found in inner dry ranges of Himalayas - Precipitation: < 100cm	Ladakh, Lahul, Chamba, Kinnaur, Garhwal, Sikkim	Deodar, oak, ash, maple, olive

Alpine Forests	Start at an elevation of above 3000m, up to the region just below the snowline	Main Himalayan regions, Trans Himalaya	Black juniper, drooping juniper, honeysuckle, willow
Low Alpine Grasslands	Vegetation not growing higher than 1.5m - Climatic conditions vary from sub-arctic to arctic	Main Himalayan regions, Trans Himalaya	Juniper, honeysuckle, Artemisia, etc.

Geographical Distribution of Forests in India

Statistic	Information
Percentage of Geographical Area under Forest cover	21.71% (ISFR 2021) (Below world average of 31%)
National Forest Policy Desired Percentage	Around 33% of the total area
State with the Largest Forest Cover	Madhya Pradesh > Arunachal Pradesh > Chhattisgarh > Odisha > Maharashtra
States with highest percentage of Geographical Area under Forests	Mizoram (84.53%), Arunachal Pradesh (79.33%), Meghalaya (76%), Manipur (74.34%), Nagaland (73.90%)

SOIL TYPES IN INDIA

Type of Soil	Formation and Characteristics	Distribution in India	Crops Suitable
Alluvial Soil	Formed from silt deposited by Indo-Gangetic-Brahmaputra rivers. Porous and well-drained.	Indo-Gangetic-Brahmaputra plains, deltas of Mahanadi, Godavari, Krishna, Cauvery, Narmada, Tapi valleys, Northern parts of Gujarat	Rice, wheat, sugarcane, tobacco, cotton, jute, maize, oilseeds, vegetables, fruits
Black Soil	Formed from weathering of basaltic rocks in the Deccan Plateau. Highly argillaceous (clayey) with good moisture retention. Known as regur or black cotton soils.	Deccan Plateau regions of Maharashtra, Madhya Pradesh, Karnataka, Telangana, Andhra Pradesh, Gujarat, Tamil Nadu	Cotton, wheat, jowar, linseed, virginia tobacco, castor, sunflower, rice, sugarcane
Red Soil	Developed on Archean granite, red due to ferric oxides. Well-drained soil with sandy structure. Subtypes based on rainfall.	Peninsula regions from Tamil Nadu to Bundelkhand, Raj Mahal in the east to Kathiawad in the west	Rice, sugarcane, millets, pulses, wheat
Laterite Soil	Developed in regions with laterite rock or structure. Brown in color, rich in iron and aluminum, poor in nitrogen, phosphorous, potash.	Western Ghats, Karnataka, Kerala, Orissa, MP, Gujarat, Jharkhand	Groundnut, cashew nut, coffee, rubber
Mountain Soil	Found on mountains with steeper slopes, high relief. Thin layered, poorly developed profiles. Rich in organic content.	Himalayas, Western Ghats, Nilgiri, Annamalai, Cardamom hills	Rubber, bamboo, tea, coffee, fruits
Desert Soil	Deposited by wind action, found in arid and semi-arid areas. Lacks moisture, low in organic matter.	Rajasthan, West of Aravallis, Northern Gujarat, Saurashtra, Kachchh, Western Haryana, Southern Punjab	Bajra, pulses, fodder, guar
Saline and Alkaline Soil	Contains a large content of NaCl, infertile. Found in Rajasthan, Haryana, Punjab, UP, Bihar, Maharashtra.	Natural in dried lakes, anthropogenic in western UP and Punjab	Leguminous crops
Peaty, Marshy Soil	Found in delta regions. Dominated by clay and mud, high moisture content.	Bengal delta, Alleppey (Kerala), coastal odisha	Jute, rice, spices, rubber

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